CAPITAL MARKETS AND FINANCIAL ADVISORY SERVICES (CMFAS) EXAMINATIONS

STUDY GUIDE

CM-EIP

CAPITAL MARKETS – EXCLUDED INVESTMENT PRODUCTS – SECURITIES, COLLECTIVE INVESTMENT SCHEMES (EIP) AND FOREIGN EXCHANGE



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The Institute of Banking and Finance

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Preface

Capital Markets – Excluded Investment Products – Securities, Collective Investment Schemes (EIP) and Foreign Exchange

The objective of the CMFAS Capital Markets – Excluded Investment Products – Securities, Collective Investment Schemes (EIP) and Foreign Exchange Examination is to test candidates on their knowledge and understanding of Excluded Investment Products (EIPs), as well as the tools and techniques to analyse these products.

Candidates are required to pass the relevant modules of the CMFAS examination pertaining to the regulated activity that they intend to conduct. Once they have passed the relevant CMFAS examination, candidates must ensure that their notification to act as an appointed representative is lodged with the Monetary Authority of Singapore (MAS) on the Public Register via the Representative Notification Framework (RNF) before they can commence any regulated activities. For details, please refer to the MAS Notice on Competency Requirements for Representatives of Holders of Capital Markets Services Licence and Exempt Financial Institutions under the Securities and Futures Act 2001 (SFA 04-N22).

Candidates who have passed the CMFAS Examinations are encouraged to continue their learning journey by attending IBF accredited programmes, attaining the required skills leading to IBF Certification. For more information, please visit www.ibf.org.sg.

Organisation of the Study Guide

The Study Guide consists of 12 Chapters, each devoted to a specific area of the EIPs that the candidate will need to know in order to pass the CM-EIP Examination. Each Chapter begins with a list of learning objectives and ends with a Chapter summary.

The emphasis in each Chapter is to ensure that candidates have a good understanding of the EIPs, as well as their applications. Examples and case studies are used where appropriate to enhance candidate's understanding of key learning points and application of issues discussed.

A summary of each Chapter is provided below:

Chapter 1- 2	**Introduction to Investing, Investments and Financial Markets Risk, Return and Time Value Calculations** Provides an overview of the investment and financial markets and highlights the importance of risk and return which underlies any investment decision.
Chapter 3	**Key Drivers of Market Movements and Asset Values**

Provides an understanding of the key drivers of market movements which underlines any investment decisions.

Chapter 4 – 10

Foreign Exchange
Company Analysis and Understanding Financial Statements
Equity Securities
Deposits and Fixed-Income Securities
Portfolio Management
Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITs), and Insurance
Warrants

Provides a basic understanding of the characteristics, features and valuation concepts and techniques of major EIPs such as foreign exchange, equities, fixed income securities, portfolio management, exchange-traded funds, unit trusts, REITs, insurance and warrants.

Chapter 11

Technical Analysis and Quantitative Analysis

Provides a basic understanding of technical analysis and quantitative analysis which underlines the evaluation of EIPs.

Chapter 12

Case Studies

Provides case studies on various EIPs to demonstrate the application of investment tools and analysis.

We have included a set of Review Questions with the answers highlighted in bold to support the candidate's learning.

A list of essential readings is also provided in Appendix C. Candidates should ensure that they complete the essential readings before attempting the examination.

The Appendices are provided for candidates' reference and to enhance their understanding of the important concepts on EIPs.

Study Guide Updates

The Study Guide is updated at appropriate intervals to reflect changes and developments in the financial industry. Candidates should ensure that they have the latest version of the Study Guide before sitting for the examination. Please refer to the Updates to the Study Guides at www.ibf.org.sg for the latest updates.

Important Notes about the Examination

The CM-EIP Examination is conducted at the IBF Assessment Centre.

The examination includes questions that test candidates' knowledge, understanding and application of EIPs.

The examination comprises 90 multiple-choice questions and multiple response questions, with a duration of 2 hours and 30 minutes. It also includes 3 case studies of 9 questions (3 case studies with 3 questions each). The passing mark is 70%.

Candidates should note the following:

(i) For calculation questions, candidates should assume that:

(a) 365-day count convention applies for Singapore Government Securities (e.g., Singapore government bonds, bills, etc.); and

(b) 360-day count convention applies for all other products (unless otherwise stated).

(ii) Candidates are advised to bring along a non-programmable financial calculator for use during the examination. A formulae sheet will be provided during the examination.

For more information on the examination rules, regulations, and other administrative procedures, please refer to the IBF website at www.ibf.org.sg

Table of Contents

Acknowledgements ..... **ii**
Preface ..... **iii**
Capital Markets – Excluded Investment Products – Securities, Collective Investment Schemes (EIP) and Foreign Exchange iii
Organisation of the Study Guide iii
Study Guide Updates..... iv
Important Notes about the Examination iv
Table of Contents ..... **vi**
Chapter 1: **1**
Introduction to Investing, Investments and Financial Markets **1**
1.1 Introduction 1
1.2 Financial Assets 8
1.3 Financial Markets 11
1.4 Digital Disruption and Fintech 15
1.5 Summary 16
Chapter 2: **18**
Risk, Return and Time Value Calculations **18**
2.1 Risk and Return 18
2.2 Calculating Returns 20
2.3 Measures of Risk 27
2.4 Types of Risk and Risk Management 33
2.5 Time Value of Money 42
2.6 Summary 46
Chapter 3: **48**
Key Drivers of Market Movements and Asset Values **48**
3.1 Introduction to Market Movements and Asset Values 48
3.2 Fundamental Factors and Analysis 50
3.3 Market-Driven Factors and Asset Values 71
3.4 Effect of Investment Behaviour and Emotions on Asset Values 75
3.5 Summary 76
Chapter 4: **79**
Foreign Exchange **79**
4.1 Introduction 79
4.2 Factors Influencing Exchange Rates 80
4.3 Foreign Exchange Markets 84
4.4 Foreign Exchange Rate Quotations 86

4.5 Forward Foreign Exchange Contracts 89
4.6 Risks in Foreign Exchange 94
4.7 Managing Foreign Exchange Risks 94
4.8 Summary 99
Chapter 5: 101
Company Analysis and Understanding Financial Statements 101
5.1 Introduction 101
5.2 Qualitative Analysis 102
5.3 Quantitative Analysis (Financial Statements Analysis) 104
5.4 The Balance Sheet 106
5.5 Assets 108
5.6 Liabilities 115
5.7 Shareholders' Equity 117
5.8 The Income Statement 118
5.9 The Cash Flow Statement 123
5.10 Per Share Calculations – Book Value Per Share (BVPS), Dividends Per Share (DPS), Earnings Per Share (EPS) and Price 125
5.11 Ratio Analysis 126
5.12 Summary 141
Chapter 6: 146
Equity Securities 146
6.1 Introduction 146
6.2 Share Ownership 147
6.3 Choice of Equities 152
6.4 Equity Market Indices 156
6.5 Valuation Approaches 158
6.6 Net Asset Valuation 159
6.7 The Discounted Cash Flow (DCF) Approach 161
6.8 Relative Valuation Ratios 168
6.9 Initial Public Offerings 184
6.10 Valuing Companies in Initial Public Offerings 186
6.11 Depositary Receipts 188
6.12 Summary 189
Chapter 6: Additional Reference Materials 191
Chapter 7: 200
Deposits and Fixed-Income Securities 200

7.1 Introduction 201
7.2 Cash Deposits and Cash-Equivalents 201
7.3 Fixed-Income Securities 205
7.4 Money Market Instruments 206
7.5 Bonds 208
7.6 Price Quotations 210
7.7 Bond Markets 211
7.8 Types of Bonds 214
7.9 Bond Features 216
7.10 Asset and Mortgage-backed Securities 218
7.11 Credit Risk 220
7.12 Other Sources of Risk 229
7.13 Bond Valuation 231
7.14 Relationship between Bond Price and Market Yield 233
7.15 Relationship between Coupon Rate, Market Yield and Price 234
7.16 Relationship between Market Yield and Other Bond Features 234
7.17 Duration and Convexity 235
7.18 Time Path of a Bond 237
7.19 Yield Measures 237
7.20 Term Structure of Interest Rates 240
7.21 Analysis of Convertible Bonds 242
7.22 Summary 246

Chapter 8: 249
Portfolio Management 249
8.1 Introduction 250
8.2 The Efficient Market Hypothesis 251
8.3 Passive versus Active Management Strategies 253
8.4 Portfolio Theory 255
8.5 Portfolio Management Services 270
8.6 Performance Measurement 281
8.7 Portfolio Management and Fund Sales 293
8.8 Summary 294

Chapter 9: 297
Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITs) and Insurance 297
9.1 Introduction 298

9.2 Exchange-Traded Funds 298
9.3 Unit Trusts 305
9.4 Real Estate Investment Trusts 317
9.5 Business Trusts 318
9.6 Stapled Securities 320
9.7 Undertakings for Collective Investment in Transferable Securities 320
9.8 Alternative Investments 321
9.9 Insurance 330
9.10 Annuities 332
9.11 Risk Management for Retirement Funds 334
9.12 Summary 336
Chapter 9: Additional Reference Materials 338
Chapter 10: 341
Warrants 341
10.1 Definition and Terminology 341
10.2 Warrant Valuation 344
10.3 Factors Affecting the Price of a Company Warrant 347
10.4 Summary 348
Chapter 11: 349
Technical Analysis and Quantitative Analysis 349
11.1 Overview 349
11.2 Price-based Technical Indicators 351
11.3 Trends, Support and Resistance Levels 353
11.4 Significance of Volume 354
11.5 Share Price and Volume Techniques 354
11.6 Charts and Chart Patterns 358
11.7 Elliot Wave Theory 368
11.8 Quantitative Analysis 368
11.9 Summary 369
Chapter 12: 371
Case Studies 371
Case Study 12.1 Individual vs Institutional Investor 371
Case Study 12.2 Risk, Return, and Time Value Calculations 373
Case Study 12.3 Key Drivers of Market Movement and Assets 374
Case Study 12.4 Foreign Exchange Carry Trade 376
Case Study 12.5 Equities Analysis 377

Case Study 12.6 Estimating the Fair Value of a Stock 379
Case Study 12.7 Fixed-Income Investments 380
Case Study 12.8 Portfolio Management 382
Case Study 12.9 Warrants and Foreign Exchange 384
Case Study 12.10 Unit Trusts, REITs and Business Trusts 385
Appendix A: 387
Review Questions 387
Appendix B: 399
Formulae Sheet 399
Appendix C: 412
Essential Readings 412
Appendix D: 413
Glossary 413

Chapter 1: Introduction to Investing, Investments and Financial Markets

Learning Objectives

The candidate should be able to:

- ✓ Understand the definition and needs of different types of investors.
- ✓ Describe what makes up the financial markets.
- ✓ Define the term "financial assets".
- ✓ Distinguish between financial assets and real assets.
- ✓ Describe the types of money market securities.
- ✓ Describe the types of capital market securities.
- ✓ Explain the differences between the "primary market" and the "secondary market".
- ✓ Identify the characteristics of a good financial market.
- ✓ Distinguish the money market from the capital market.
- ✓ Discuss broadly the bond markets, the equities markets, and the options and futures markets.
- ✓ Understand the inter-markets relationships and how the economy is sustained by the financial markets.

1.1 Introduction

The current investment environment is seeing incredible innovation that is rapidly changing the landscape of financial markets. Beyond the development of regulatory infrastructure and shift in emphasis on the types of investment products promoted, there have also been significant changes in financial technology giving rise to wider and faster access to investments, as well as the entrance of new players redefining how money, wealth and value is transferred and stored. To stay relevant, representatives of financial institutions must continually expand their breadth of investment knowledge.

Representatives must also have the knowledge and skills to identify suitable products and services that will serve the varying needs, expectations, risk profiles, preferences and limitations of their clients, subject to the regulatory requirements and industry ethical considerations. Understanding the needs of clients forms the basis for all good investment recommendations, yet clients' needs and return expectations will have to match the realities of what the markets offer.

1.1.1 Definition and Needs of Different Types of Investors

Whether a representative is dealing with institutional or retail investors, the first step is to understand the needs of different types of investors before any investment advice or product recommendation is put forth. The Securities and Futures Act 2001 (SFA) Section 4A provides for specific definitions of Accredited Investors (AIs), Expert Investors (EIs), Institutional Investors (IIs), and others excluded under such definitions are categorised as Retail Investors[1].

Institutional Investors

IIs are considered "super-investors" due to their sheer size and impact on the capital markets. Given the massive amount of capital and resources under their control, IIs have timely and wide access to market information and professional research.

Generally, IIs have clearly defined investment goals with prescribed investment directives and explicitly specified requirements (e.g. costs, efficiency in execution, research and reporting support, range for volatility and returns, level of asset allocation, concentration and exposure to asset classes, sectors, and securities). Recognising that IIs are sophisticated investors that act within a structure of an institution, they are less likely to make uneducated investments and therefore are typically permitted access to wider financial products offerings than other investor types.

[1] SFA Section 4A – Specific Classes of Investors:

(i) Retail Investor refers to an investor who is not an accredited investor, institutional investor or expert investor.

(ii) Accredited investor includes:

a) An individual whose net personal assets exceed in value S$2 million, whose financial assets (including banking deposits, investment products and other assets prescribed by regulation, but net of liabilities) exceed S$1 million, or whose income in the preceding 12 months is not less than S$300,000. In determining the value of an individual's net personal assets, the value of the individual's primary residence is net of any outstanding secured credit facility and is capped at S$1 million;

b) A corporation or an entity with net assets exceeding $10 million in value as determined by its most recent audited balance sheet or a balance sheet certified by it as giving a true and fair view of the state of its affairs as of the date of the balance sheet which date must be within the preceding 12 months;

c) The trustee of such trust (including a bare trust) as prescribed by MAS, when acting in that capacity;

d) A partnership (other than a LLP) in which each partner is an accredited investor;

e) A corporation, the entire share capital of which is owned by one or more persons, each of whom is an accredited investor; or

f) A person who holds a joint account with an accredited investor in respect of dealings through that joint account.

(iii) Institutional Investor refers to the Government, a statutory board, an entity wholly and beneficially owned by a central government of a country and whose principal activity is to manage funds and any entity whose funds are so managed, a foreign central bank, government or agency, a prescribed multilateral or international agency, a bank, a merchant bank, a finance company, a company/co-operative society licensed as an insurer in Singapore, a trust company, a capital markets services licence holder for securities dealing, approved exchanges and clearing houses, recognised market operators and clearing houses, licensed trade repositories, an approved holding company, a depository under the SFA, a foreign entity or trust that carries on a financial or regulatory function similar to the Singapore authorities under relevant statutes, a pension fund or collective investment scheme, a person carrying on the business of dealing in bonds with accredited/expert investors, a trustee of a prescribed trust, or any other such person as prescribed by the MAS.

(iv) Expert Investor refers to a person whose business involves the acquisition and disposal, or holding of capital markets products, whether as principal or agent, the trustee of a trust prescribed by the MAS or such other person as may be prescribed by the MAS.

Broadly, IIs can be classified according to the purpose for which money is being invested—pension funds for providing retirement income, insurance funds for meeting expected claims, endowment funds for satisfying philanthropic spending commitments, and sovereign wealth funds for stabilizing the national economy or savings for future generations.

Different types of IIs have differing investment objectives such as the time horizon under which they operate. For example, pension funds may have precise knowledge of future liabilities and will favour long-term investment objectives. A commercial insurer, in contrast, may have lower predictability, as with the climate and weather, and may be inclined towards short-term investment opportunities. In addition, regulations vary considerably from one industry to another. In Singapore, insurers are prohibited from taking uncovered positions in derivatives under the MAS requirements[2].

Accredited Investors

An AI is an individual or a corporate that meets the prescribed net worth threshold[3] under the SFA. Given their wealth, AIs are considered to be financially savvy and have greater ability to understand financial instruments. The accredited status implies that the investor can withstand a higher level of monetary risk than the retail investor. Hence AIs may have access to products that are more complex and riskier, such as unrated bonds and products that require neither a prospectus nor a reasonable basis for recommendation by a representative of a capital market intermediary (CMI)[4].

To enhance regulatory safeguards for AIs, the Monetary Authority of Singapore (MAS) has raised the benchmark for qualifying AIs and introduced an opt-in regime for AIs, which specifies that CMIs can no longer automatically treat anyone who qualifies as an AI unless the investor opts to be treated as such.

Representatives of CMIs should seek to identify factors that influence the investment decisions of AIs and tailor their investment solutions accordingly. The typical factors are set out in Table 1.1.1(a).

Table 1.1.1(a) – Factors Influencing Investment Decisions of AIs

Individual	Corporate
• Demographics (age, gender, education level, etc.)	• Management outlook (growth vs decline)
• Lifestyle (confidence level, no. of dependants, etc.)	• Size of firm
• Risk tolerance level (range of volatility, risk-return spectrum, etc.)	• Corporate governance (ability to maintain balance between vested interests and management control while providing safeguard to the interests of all stakeholders)
• Market trends/forecasts	• Market trends/forecasts

[2] MAS Notice 125 on Investments of Insurers – Paragraph 30 - Prohibitions.
[3] Please refer to footnote 1.
[4] CMI refers to an entity holding a capital markets services licence under the SFA, a fund management company registered under paragraph 5(1)(i) of the Second Schedule to the Securities and Futures (Licensing and Conduct of Business) Regulations (SFR(LCB)) or a person exempted from the requirement to hold such a licence under paragraph 7(1)(b) of the Second Schedule to the SFR(LCB).

Individual	Corporate
• Tax (investors exposed to higher tax brackets prefer investments where the return is tax exempt)	• Tax (fiscal incentives or barriers on asset classes)
• Investment horizon	• Investment horizon
• Liquidity	• Cash flow
• Psychological influences (desires, prejudices and other emotions)	• Corporate culture (aggressive vs passive approach)
• Personal values (social or religious values determining the attitude of the investor towards certain asset classes (e.g., firms profiteering from child labour)	• Corporate values

Expert Investors

An EI is one that trades capital markets products in the ordinary course of business. Like AIs, EIs are afforded less regulatory protection compared to retail investors, albeit for different reasons. Unlike AIs who may have a higher ability to take risk due to their wealth, EIs by virtue of their occupation may have a better understanding of the risks associated with the different products. Their knowledge of investment products and expertise in matters of investment suggest that such investors are more capable of protecting their own interests, even if they may not qualify for II or AI status.

EIs are basically a sub-set of retail investors that meet specific experience requirements as spelt out in the SFA and, thus, their investment decisions are generally influenced by the same factors. Nonetheless, it is important to note that information (e.g., market rumours) that may influence retail investors may not likewise influence EIs. Also, EIs tend to have a clearer idea about their investment objectives and constraints and may prefer greater involvement in the investment process.

Retail Investors

A retail investor is an individual investor who purchases securities for his own personal account rather than on behalf of an organization (i.e., the average working masses). Under the SFA, retail investors are defined as investors who are excluded from the classification of IIs, AIs and EIs.

Retail investors are the polar opposite of IIs. They usually trade in small amounts and tend to trade less frequently. Due to the minute trade size of retail investors, they have little influence over corporate decisions or the capital markets. As retail investors seldom have access to corporate boardrooms or timely market information, they are assumed to be less financially savvy and more risk-averse compared to AIs and IIs.

Given the characteristics of retail investors, they are accorded a higher level of protective regulations as they are deemed to be more vulnerable. When dealing with retail investors in Singapore, representatives of CMIs must be aware of updates to regulations[5] such as:

[5] Revisions to the Code on Collective Investment Schemes on 23 May 2023 and the Securities and Futures (Amendment) Act 2017 on 8 October 2018.

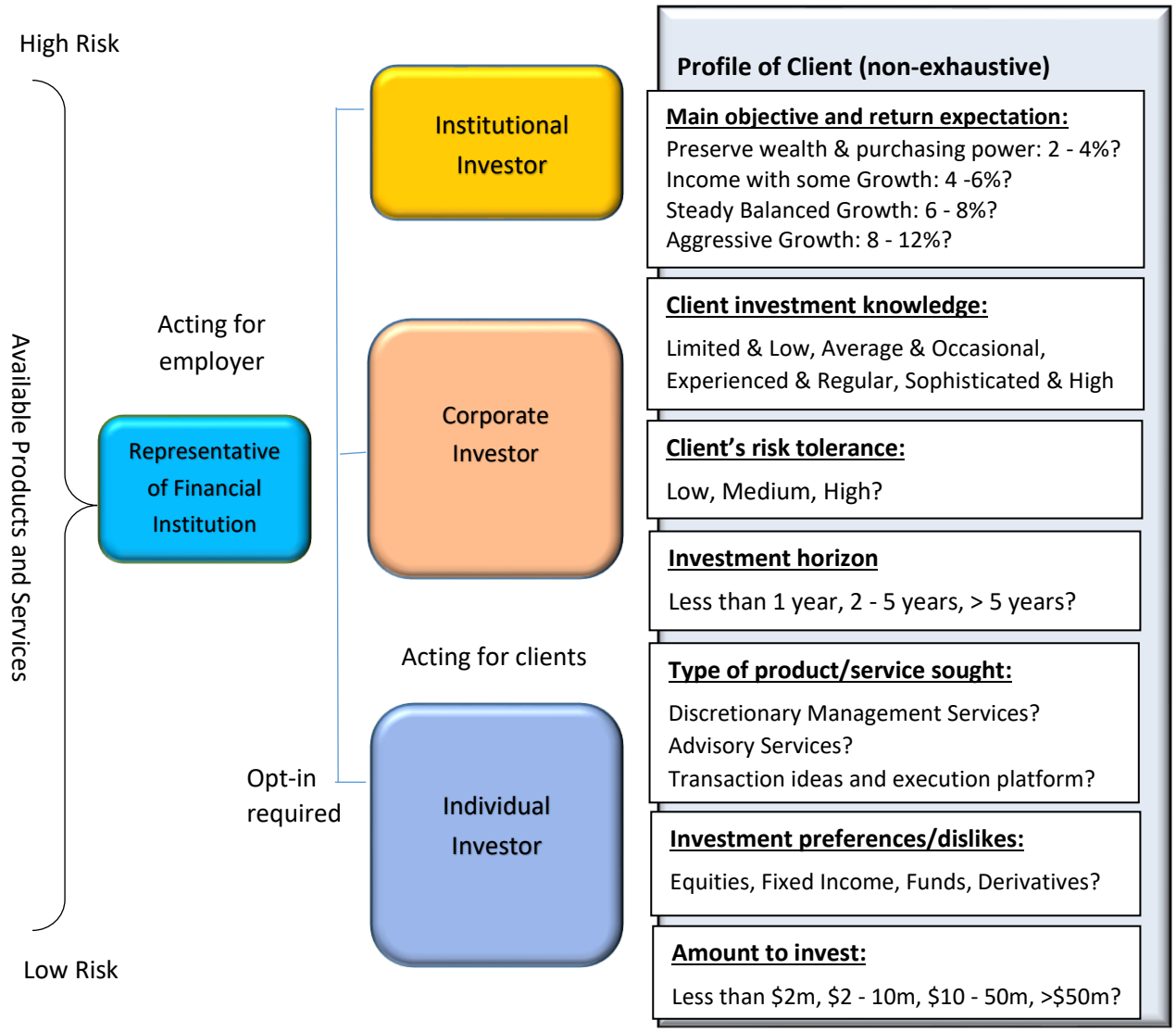
a. More stringent regulations on non-conventional investment products such as precious metal buy-back arrangements and the regulating of collectively managed investment schemes as Collective Investment Schemes (CIS);

b. Refining of investor classifications under the SFA and the Financial Advisers Act 2001 (FAA); and

c. Option for AIs to benefit from the full range of safeguards applicable to retail investors.

Representatives of a CMI should have clear understanding of the legal definition and profile of retail investors prior to recommending any investment solutions or products. Most of the factors influencing the investment decision of the retail investor will coincide with those of the individual AI given that AIs are basically a sub-set of retail investors that meet specific wealth and income requirements as specified in the SFA. Please refer to Figure 1.1.1(b) on the roles of representatives of a CMI.

Figure 1.1.1(b) - Role of Representatives of Financial Institutions



1.1.2 Investment Products and Services

There are a wide range of investment products and services available. Some are actively traded and appeal mainly to IIs whereas others are designed mainly for AIs, high net worth individuals (HNWIs) or retail investors. Most investment products and services crafted from traditional asset classes (e.g., deposits, foreign currency, fixed income, and equity) seek to provide yield and/or potential capital gains. However, with financial innovation, investors can also take a bearish position using products such as put options, futures, and inverse exchange traded funds (ETFs) to generate potentially higher yields.

Financial institutions have provided clients with a wide array of financial instruments which can be used to take advantage of changes in the market. Now clients have access to massive amounts of online information on various financial products, which generates greater client interest in the investment products available. Thus, more than ever, investment professionals must seek to widen their knowledge of financial products so that they can provide the right information and timely advice. Figure 1.1.2 sets out the purpose and benefits of investment products and asset classes for investors.

Figure 1.1.2 - The Purposes and Benefits of Investment Products and Asset Classes for Investors

Purpose or Benefit / Product or Service	Defined Interest Yield or Dividend	Potential of Higher Yield	Potential Capital Gains	Regular Payout Stream	Inverse or Change in Payout (Hedge)	Amplify Gains/Loss (Through Leverage)	Diversify
Traditional Asset Classes							
Local Deposit	X			X			
Foreign Deposit	X		X	X			
Money Market Instruments	X		X	X			
Bonds	X		X	X	X[a]		
Equities	X		X	X			
Convertible Bonds	X		X	X		X	
Derivatives							
Futures			X		X	X	
Swaps	X		X	X	X	X	
Options	X	X	X		X	X	
Warrants			X			X	
Structured Products		X	X		X	X	
Funds							
Portfolio Management		X	X				X
Investment Funds	X		X				X
Exchange-traded Funds/Exchange-traded Notes	X		X			X	X
Alternative Investments							
Hedge Funds			X		X	X	X
Private Equity			X				X[b]
REITS/Property Funds	X		X				X[b]
Commodity Funds							X[b]

Note: a - Inverse floaters; b - Diversification within their asset class only.

The wider range of securities and derivative structures in investment products will allow investors to:

a. Diversify their portfolios;

b. Optimise the risk-return profile of their portfolios;

c. Protect investments or to even take advantage of downward market movement; and

d. Selectively take advantage of investment opportunities through a leverage effect.

1.1.3 Financial Assets and Real Assets from the Individual Investor's Perspective

Beyond financial assets, individual investors may also own real physical assets as well as business assets, which may have values and cash flows that are significantly larger than those of their financial assets.

While we often try to classify assets into neat buckets, in practice, for individuals, the flow of funds from business to physical assets owned or to financial assets invested are very much related (refer to Figure 1.1.3(a)). Individuals who own businesses will have business financing needs and will need corporate finance related products and services as well as cash management services.

Business salaries and profits provide income to business owners, which can be invested carefully in financial assets or ploughed back into business investments. Purchases of real physical assets such as real estate will require financing and can be funded using financial assets. Hence, the investment professional should also be alert to business needs, changes and decisions made about business assets, and physical assets owned, which can have a major impact on financial assets.

Figure 1.1.3(a) – Relationship Between Real Physical Assets, Business Assets and Financial Assets

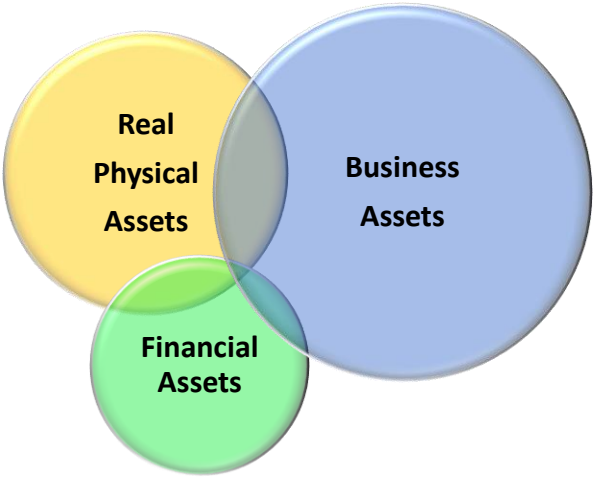


Table 1.1.3(b) lists typical examples of financial assets, business assets and real assets held by an individual investor. Note that this table is not exhaustive.

Table 1.1.3(b): Financial Assets and Real Assets from the Investor's Perspective

FINANCIAL ASSETS	REAL/PHYSICAL & BUSINESS ASSETS
TRADITIONAL ASSET CLASSES • Cash Deposits • Foreign Currencies • Short Term Money Market Instruments • Bonds and Other Fixed Income Instruments	**COMMODITIES** • **Energy** - Crude Oil, LNG, Refined Products, Petrochemicals, Coal, Electricity • **Metal** - Precious, Base, Ferrous • **Agriculture** - Grain, Soft, Livestock, Biofuel, Dairy

<table>
<tr><th>FINANCIAL ASSETS</th><th>REAL/PHYSICAL & BUSINESS ASSETS</th></tr>
<tr><td>• Equities
• Insurance</td><td></td></tr>
<tr><td>DERIVATIVES
• Exchange Traded Derivatives
• Financial Futures, Options, Swaps, Warrants
• Over-the-counter (OTC) Structured Products</td><td>REAL ESTATE
• Commercial, Residential, Industrial, Agriculture</td></tr>
<tr><td>INVESTMENT FUNDS
• Exchange-traded Funds, Private Funds sold OTC
• Pure Asset Class Funds vs Diversified Funds
• Types of Strategy Funds
• Capital Guaranteed Funds</td><td>BUSINESSES
• Privately owned assets - Plant, property, equipment, rights, patents, technology, network, distribution</td></tr>
<tr><td rowspan="2">ALTERNATIVE INVESTMENTS
• Hedge Funds
• Commodity Funds
• REITS and Real Estate Funds
• Private Equity Funds</td><td>PRECIOUS STONES & JEWELRY
• Diamonds, Rubies, Emeralds, etc.</td></tr>
<tr><td>COLLECTIBLES
• Art, Antique pieces, Wine, Limited edition memorabilia, etc.</td></tr>
</table>

1.2 Financial Assets

Financial assets are typically paper (or electronic) claims on the issuer, which may be from the public sector, i.e., government (e.g., Singapore Government) and its agencies (e.g., Housing Development Board), a supranational (e.g., World Bank) or the private sector, i.e., a corporation (e.g., Keppel Corporation). Financial assets may be broadly classified under asset classes such as debt, equity and derivative instruments or securities. The various asset classes are covered in greater detail in the subsequent chapters (Chapters 4 – 10 of this Study Guide).

1.2.1 Local and Foreign Cash Deposits

The most basic asset class is cash. Locally, these would be Singapore dollar deposits placed in current accounts, savings accounts or fixed (time) deposit accounts as well as foreign currency deposit accounts. Given the low interest rate environment for the past years, there has been a high opportunity cost for keeping investible funds in cash and not investing in asset classes offering higher returns to offset inflation (which results in a loss of purchasing power). Investment in foreign currency deposits carry currency exchange risks and have proven to be relatively volatile in the years following the global financial crisis. These risks are further discussed in Chapter 4 - Foreign Exchange.

1.2.2 Fixed Income Securities

Fixed income securities are primarily interest-bearing instruments providing a fixed return. They include treasury bills, fixed-rate bonds, certificates of deposit, commercial papers and debentures. These instruments are covered further in Chapter 7 – Fixed Income Securities. Below is an overview of the different types of fixed income instruments that we will cover in this Study Guide.

a. **Short-Term Money Market Securities**

Money market instruments are relatively low-risk, highly liquid, short-term debt instruments issued by governments, financial institutions and corporations. The minimum denominations of these securities are relatively large, and the size of the transactions can be substantial. The maturities of money market securities range from 1 day to 1 year and are often less than 90 days. Instruments traded in the money market include treasury bills, commercial papers and negotiable certificates of deposits.

Treasury bills are issued by central banks on behalf of their governments to raise short-term funds to finance government expenditures. As they are obligations of the government, they are considered to be free of default. As such, their returns are normally used as a proxy for the risk-free rate of return. They are issued on a discounted basis by tender and payable or redeemable at face value on the maturity date.

Repos or repurchase agreements are key short-term funding arrangements for banks. A repurchase agreement (a form of borrowing) is the sale of a security at a specified price with a simultaneous commitment by the seller (the borrowing party) to buy the security back from the purchaser (the lending party) at a specified price and future date. The effective interest is the difference between the purchase price and the sale price. A **reverse repo** (a form of lending) is the converse transaction of a repo, i.e., purchase of a security (by the lending party) combined with a forward sale of the security (to the borrowing party) at a specified price and date.

Commercial papers refer to short-term unsecured promissory notes issued by a corporation. Like treasury bills, commercial papers are discounted money market instruments.

A banker's acceptance is a time draft drawn on a bank by a customer in which the bank agrees to pay a specified amount on a specified future date. It is created to facilitate commercial trade transactions, particularly for international trade. Banker's acceptances are negotiable instruments that the holder can discount in the money market, i.e., sell it for less than its redemption value.

A certificate of deposit (CD) is a certificate issued by a bank in exchange for a deposit of funds placed with the bank. Yields on CDs are quoted on an interest-bearing basis.

b. **Debt Securities**

Debt securities are long-term fixed borrowings by the issuers. The investor would lend a certain amount of money, called the **principal**, to the issuer and, in return, the issuer agrees to make periodic interest payments and to repay the principal at the maturity date. Debt securities are also called “bonds”.

Singapore has one of the most developed bond markets in Asia. Investors can choose from various types of local currency bonds such as Singapore Government securities, statutory board securities, supranational bonds and corporate bonds.

A Eurobond is an international bond denominated in a currency not native to the country where it is issued. These include Eurodollar bonds, Euroyen bonds and bonds denominated in Euros. A Eurodollar bond is denominated in US dollars and sold outside the United States to non-US investors. For example, a Japanese

company can issue Euroyen bonds for sale in London. **Yankee bonds** are sold in the United States and denominated in US dollars but are issued by foreign corporations or governments. **Domestic bonds** are bonds sold by an issuer within its own country in that country's currency.

1.2.3 Equity Securities

Unlike fixed income securities, equity securities represent an ownership interest in a company. There are two forms of equity:

a. **Ordinary shares** - Ordinary shareholders are entitled to elect the directors of the company and vote on company issues. Each owner is usually allowed to cast votes equal to the number of shares owned; and

b. **Preference shares** - Preference shares are technically an equity security but are also considered a hybrid security because they have limited rights to vote (i.e., they can vote only in limited circumstances) and resemble both equity and fixed income instruments.

In terms of priority of payment or claims over the company's assets, creditors have the highest priority, followed by preference shareholders, then ordinary shareholders, who have residual claim. Equity Securities will be covered in detail in Chapters 5 and 6. Chapter 5 will provide an understanding of financial statements used in fixed income and equity analysis, while Chapter 6 will cover Equities.

1.2.4 Derivatives

Derivatives derive their values from the price of an underlying asset such as a stock, a bond or a market index. Derivatives include options, warrants, futures and convertible securities.

A structured product usually involves bundling a note or deposit with one or more derivatives, providing an investment return that is linked to the performance of one or more underlying assets.

An option gives the buyer the right to buy (call option) or to sell (put option) a specified amount of an underlying asset within a specified period at a specified price.

A **company-issued warrant** provides the right to buy a given number of the company's shares at a specified priced over a given time period (usually over several years).

A **structured warrant** is a financial instrument usually issued by banks that provide investors the opportunity to profit from the upward (call warrant) or downward (put warrant) price performance of an underlying asset (e.g., an index or stock) at a fraction of the price of the underlying asset.

A futures contract is an agreement providing for the future delivery of a particular asset between a buyer and a seller at a predetermined price. A futures contract can be cash settled or physically settled, i.e. by delivery of the asset. Futures contracts are available on commodities and financial instruments, including equity market indices, currencies and fixed income securities. Futures contracts are traded on exchanges and with standardized terms.

A convertible security gives the holder the right to convert the security into a predetermined number of ordinary shares of the issuer within a specified period. It usually pays interest and has a yield that is less than a corporate bond without such a feature. The price difference is the value of the embedded share option.

The different types of derivatives and the strategies used for derivatives trading are covered in further detail in the CMFAS Capital Markets – Specific Investment Products – Derivatives and Collective Investment Schemes (CM-SIP) Examination.

1.2.5 Portfolio Management and Investment Funds

Clients can choose to invest directly in single securities or entrust their funds with portfolio managers. Professional portfolio management can be beneficial for clients who do not have the time, appropriate information, expertise, discipline or emotional temperament to invest actively.

Different asset classes and securities are pooled and managed by professional fund managers. The securities may be pooled together into pure asset funds or they may be diversified under asset allocation strategies based on the risk profile and objectives of clients. Hedge fund managers would also offer alternative ways of investing that seek to generate returns irrespective of the market movement.

Funds are offered on stock exchanges worldwide as **exchange-traded funds (ETFs)** and they are also offered **over-the-counter (OTC)** by financial institutions to investors. The various topics on Portfolio Management are covered in Chapter 8, while the different types of funds are covered in Chapter 9.

1.3 Financial Markets

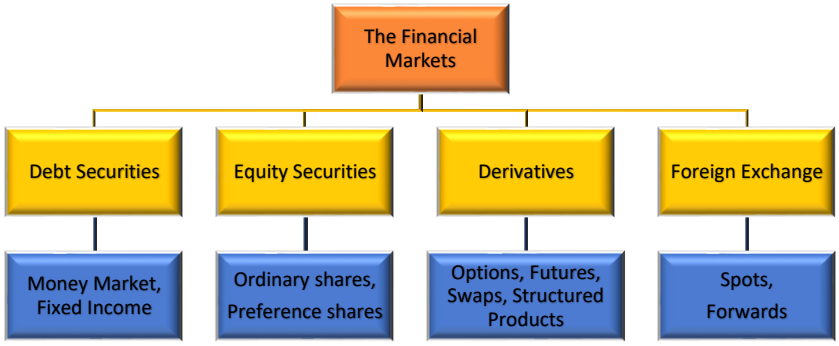
Financial markets are where assets are bought or sold, borrowed or lent, or transacted in various ways among participants such as issuers, investors, speculators, hedgers and intermediaries. Investments can be made in various instruments such as equity securities (e.g., shares and equity unit trusts) and debt securities (e.g., short-term money market papers or long-term fixed income bonds). Sophisticated investors may invest in derivatives such as equity options and futures. Thus, there are many different markets in which assets may be transacted, such as the equities market, money market, fixed income market, commodities market, options and futures market and the foreign exchange market.

Financial markets provide the platform for market participants to carry out their investment and funding decisions, as well as other related activities. These markets have seen extensive liberalization and global consolidation, the impact of information technology which enabled alternative market platforms such as the Electronic Communications Network (e.g., Instinet), and the rapid adoption of online technology which allows professional and retail investors to use brokerage services through computer networks and personal devices such as smartphones.

The rapid advance of technology has also enabled algorithmic trading, which uses a computer program to follow a defined set of instructions to execute a trade and can be used to slice up large trades to minimize market impact. A form of **algorithmic trading** is high frequency trading, which relies on computer speed and power to analyse significant amounts of market information and self-executes trades in fractions of a second.

Figure 1.3(a) provides a general outline of the structure of financial markets in terms of asset classes while Figure 1.3(b) illustrates how financial markets sustain economic growth.

Figure 1.3(a) - Structure of the Financial Markets



Participants in financial markets include:

a. **Investors** – Retail investors, private bank investors, institutional investors (including fund management companies);

b. **Issuers** – Public and private sector companies;

c. **Intermediaries** – Broker/dealer firms and market makers;

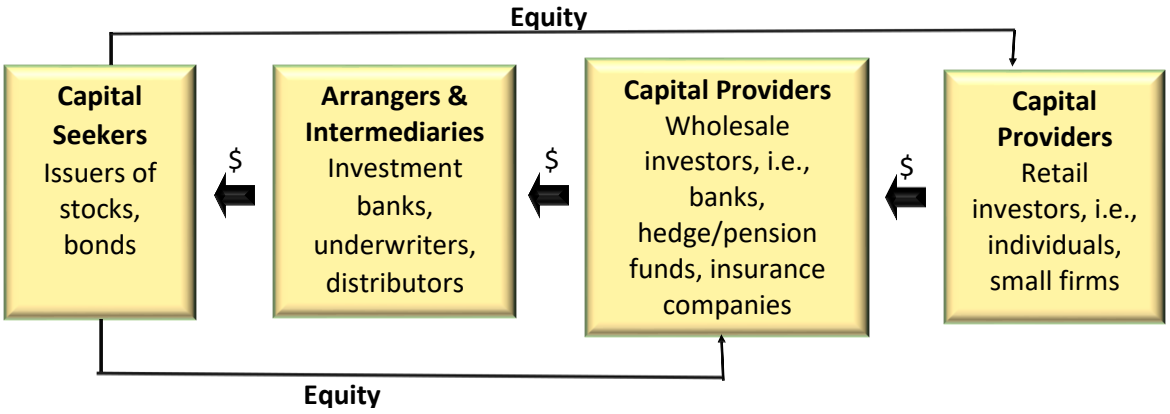
d. **Regulators**;

e. **Electronic exchanges** – For equities, debt, derivatives, and commodities;

f. **Infrastructure providers** – Clearing houses, central depositories, communications networks, trade repositories; and

g. **Financial institutions** - Distribution and underwriting of initial public offerings (IPOs), participation in the secondary markets.

Figure 1.3(b) - How Financial Markets Sustain Economic Growth



Besides regulated exchange-based capital markets, there is an increasing level of activity in the **private equity** or **mezzanine financing** markets, where capital seekers and providers negotiate funding and investments for mergers, acquisitions or buy-outs. In such pre-primary market sectors, capital seekers raise funds from private investors, as opposed to the regulated exchange-based capital markets where investors raise funds from the public. In some forms, this is referred to as “shadow banking”.

1.3.1 Primary Market

The primary market is one where new issues of securities by a company are listed on the stock exchange and sold to the public. New sales of debt or equity securities take place in the primary markets through IPOs (. The issuers of these securities receive cash from the subscribers (buyers) of these new securities, who in turn receive financial claims that previously did not exist.

1.3.2 Secondary Market

After new securities have been sold in the primary market, the securities will be listed on the exchange and begin trading in the **secondary markets**. Secondary markets involve the buying and selling of securities initially sold in the primary market. The proceeds from a sale in the secondary market do not go to the original issuer but to the investor who sold the security. Secondary markets provide liquidity, which is essential to the proper functioning of capital markets.

1.3.3 Characteristics of a Good Financial Market

What are the distinctive features of a good market? Investors should consider the following when evaluating the quality of a financial market:

a. **Availability of information -** To determine the appropriate price at which a security can be quickly sold based on demand and supply, participants need timely and accurate information on the volume and prices of past transactions and on all currently outstanding bids and offers.

b. **Liquidity** - Investors should be able to buy or sell a security quickly at market prices that are not substantially different from the prices for previous transactions, assuming no new information is available. A security's likelihood of being sold quickly is referred to as its **marketability**, which is a necessary but not a sufficient condition for liquidity. Another component of liquidity is **price continuity**, which means that prices do not change much from one transaction to the next, unless new information becomes available. A continuous market requires **depth**, which means there are numerous buyers and sellers willing to trade at current prices, preventing drastic price movements.

c. **Transaction costs** - An efficient market is characterised by low transaction costs, which applies to all aspects of the transaction, including the cost of reaching the market, the actual brokerage cost and the cost of transferring the security. This attribute is referred to as *internal efficiency*.

d. **Bid-offer spread -** This is the spread between the **bid** and **offer** price for a security. In efficient markets, this spread is very small because it reflects wide and deep market-making activities where market makers readily quote two-way prices. Liquid and efficient financial markets usually have reliable market makers, resulting in narrow bid-offer spreads.

e. **Information efficiency** - This means that prices adjust quickly to new information regarding supply or demand, so prices reflect all available information about the security.

f. **Price discovery -** Price discovery is one of the most important functions of any financial market. If all of the above are in place, price discovery will be more efficient, creating a market where the prices of securities being traded are more transparent and responsive to market forces.

1.3.4 Bond Markets

The largest domestic bond market in the world is the US bond market. The US government, through the Treasury Department issues Treasury notes and bonds with maturities greater than 1 year. The US Treasury Department is the largest single issuer of debt globally, making the treasury market the most active and liquid market in the world. Treasury bills, notes and bonds are usually issued through an auction where dealers submit competitive bids.

The Eurobond market remains a very important market for investors who are interested in foreign currency bonds, particularly over the last few years when Quantitative Easing (QE) by central banks has resulted in very low interest rates, leading investors to seek higher returns through long-term bond issues. Due to the worldwide participation of both investors and traders, the dealing and settlement procedures for Eurobonds are governed by the rules of the Association of International Bond Dealers (AIBD), a body comprising professional participants in the international bond markets.

Although a substantial number of corporate bonds are listed on some major exchanges, such as the New York Stock Exchange (NYSE), trading of bonds is mostly done OTC through bond dealers, which are mostly banks and large investment banking firms.

In 2022, while inflation continues to pose as the key macroeconomic challenge, Asian bond markets will also be driven by factors such as the financing needs necessary for Asia's transition to a low carbon future and economic recovery from the reopening of Asia's borders. Singapore's ecosystem of banks, investors, and professional service firms, as well as the digitalisation of bond market infrastructure, will enable us to support fund raising efforts in the region and help channel international capital to support Asia's growth and net-zero transition needs.

1.3.5 Equity Markets

Ordinary shares, preference shares, warrants and ETFs are traded actively on the equity markets. Equity markets in different countries vary in terms of their size, liquidity, degree of regulation, trading and settlement systems, and restrictions on foreigners in buying equities. Today, investors are able to trade on major stock exchanges through internet brokerage firms. With the advent of internet trading, the composition of equity investments for investors has become more international.

According to the World Federation of Exchanges database, the total global equity market capitalization declined about 20% in 2022 compared to the end of 2021 (USD124.5 trillion), with all regions declining significantly. In absolute terms, this represented more than USD25 trillion which have been wiped off worldwide. The high inflation environment, alongside the tightening of monetary policies (including raising interest rates across most economies), cooled down investment in the equity market.

Size-wise, the NYSE and NASDAQ combined account for slightly more than half of the market capitalization for stocks in the world. Singapore is not in the top twenty list of global stock exchanges in terms of market capitalisation. However, the local exchange has sought to distinguish itself in other ways. The Singapore Exchange (SGX) is the world's most liquid offshore market for the benchmark equity indices of China, India, Japan and ASEAN and offers commodity and currency derivative products.

With the globalisation of securities markets, there is a move towards scripless settlement to eliminate inefficiencies from handling voluminous papers. Settlement periods vary amongst securities markets. For example, in China, the settlement period is T+0 (i.e., settlement on the day of the trade date), where the philosophy is that an investor must have the funds on hand to make the purchase. For the US and Singapore,

the settlement periods are T+2[6] (i.e., 2 business days after the trade date), which is the same as that for Hong Kong.

Some countries have one **national stock exchange** while others may have more than one exchange. Japan has 8 stock exchanges with the Tokyo Stock Exchange (TSE) dominating the country's equity markets. Besides the NYSE and the American Stock Exchange (AMEX), the US has several regional exchanges that list small companies with geographic interest.

Securities that are not listed on an organized exchange may be traded in the OTC market. In the US, the OTC market is a negotiated market consisting of a network of dealers or market makers who stand ready to buy and sell securities at market prices. Unlike brokers, dealers have a vested interest in transactions and earn the spread between the bid and offer prices.

1.3.6 Options and Futures Markets

A futures exchange is a marketplace for financial instruments and commodities where the buying and selling of underlying products are contracted for future delivery. The **Chicago Mercantile Exchange (CME) Group** is the largest futures exchange and processes 3 billion contracts of the most diverse derivatives worth approximately $1 quadrillion annually (on average). The second largest futures exchange is the **Intercontinental Exchange (ICE)** which was created out of a period of mergers and acquisitions among exchanges and clearing houses mainly in the US and Europe.

The SGX in Singapore operates the largest derivatives market in Asia with its futures, swaps and options benefitting from its position as the largest foreign exchange centre in Asia. The SGX has elevated its position in Asia by offering investors futures contracts on all the major equities indices in Asia. In Hong Kong, in addition to its stock market and futures exchange, the Hong Kong Exchanges and Clearing Limited also owns the London Metal Exchange.

Futures contracts are traded on electronic futures exchanges, with trades cleared through the clearing house each business day. The clearing house allows participants to easily reverse a position before the futures contracts mature, because the clearing house keeps track of each participant's outstanding contracts.

Options in the US trade on the floor of an exchange, such as the Chicago Board Options Exchange (CBOE), using a system of market makers. The market maker quotes bid and ask prices and floor brokers can trade with the market maker or with other floor brokers. The Options Clearing Corporation (OCC) acts as an intermediary between the brokers representing the buyers and the sellers (writers). Thus, the OCC becomes the buyer for every seller and the seller for every buyer, guaranteeing that all contract obligations will be met.

1.4 Digital Disruption and Fintech

The financial industry, like other industries, has been going through significant technological changes and digital disruption at many levels for the past years. More changes will come as the entire industry goes through the digital revolution. With the accessibility of the internet, the role of financial institutions and capital markets are being challenged by new entrants.

Banks are working with fintech companies to design equivalent internet platforms and solutions to keep pace with technological changes. Table 1.4 provides some examples where digital disruptions have forced adaptation from the financial community.

[6] SGX ST Rule 9.2.1 Settlement Basis and Eligibility for Clearing by CDP. Refers to 2 exchange business days after the trade day.

Table 1.4 - Impact of Digital Developments and Disruptions on Financial Sector

Digital Development	Impact on the Financial Sector
Internet banking	With the growth of internet banking, consumer and branch banking services are being redefined. Banks are identifying tasks that can be carried out directly through the internet as opposed to those that require face-to-face interaction. This has led to a reduction of brick-and-mortar branches for some retail banks, redefining of roles and activities, and an emphasis on getting the banking community more adept at internet banking.
Rise in non-bank internet lending competing with traditional lending	We are witnessing the growth of new non-bank entrants providing direct financing over the internet. With big-data credit analysis, profiling and direct access to individuals, the internet has become a platform for direct lending from new entrants which compete with banks, finance companies and traditional moneylenders for the piece of the lending pie.
Internet business transactions reducing cash float	Cash management is a traditional area of banking where transactions cleared through the banking system may take a few days, providing daily float of cash. This float amount has been reduced because of the greater efficiency of direct internet transactions.
Non-bank digital wallets bypassing banks	Excess cash traditionally held in retail current and savings accounts has started to move into digital wallets on platforms provided by internet retailing (e.g., Tencent, Alibaba). So large is the overall accumulation of cash in digital wallets that the platform providers are now recommending various investment funds and bypassing the retail banks, which offer lower deposit rates. Excess cash may be held on the digital platform rather than in the banking system.
Digital currencies for transactions in cyberspace	Cryptocurrencies, which are not backed by any reserves or central banks, have emerged as a currency for transactions in cyberspace. Authenticated through block-chain technology, cryptocurrencies bypass the banking system and provide some level of anonymity for transactions in cyberspace. Though cryptocurrencies are not recognized as legal tender and only make up about 1% of currency transactions, the lead cryptocurrency Bitcoin has made some headway into mainstream finance with Bitcoin futures being traded on the Chicago Mercantile Exchange (CME) and the Chicago Board Options Exchange (CBOE) in the US.

1.5 Summary

1. Representatives of capital market intermediaries must be able to distinguish between **institutional, accredited, expert and retail investors** in the local context and identify their investment needs.

2. Financial assets may be broadly classified as **debt securities**, **equity securities**, **collective investment schemes**, **funds** or **derivatives**.

3. **Money market instruments** are highly liquid, short-term debt instruments with maturities less than one year.

4. **Fixed income debt securities** are borrowings by issuers who agree to make periodic interest payments and to return the principal to the investor on the maturity date.

5. Equity securities represent ownership interest in a corporation. The two types of equity securities are **ordinary shares** and **preference shares**.

6. Derivative securities derive their values from various underlying assets. **Options, warrants, futures** and **convertible bonds** are examples of derivative securities.

7. Investors have a choice of investing directly in single securities or to entrust their funds with **portfolio managers**.

8. The **primary market** is where new securities are distributed (sold) by the issuer to investors. The **secondary market** is where the securities are traded after they have been distributed in the primary market.

9. In evaluating the quality of a financial market, investors should consider the availability of information, liquidity, transaction cost, the bid-offer spread and informational and price discovery efficiency.

10. The trading of bonds is mostly done "over-the-counter" through bond dealers rather than through organized exchanges.

11. Equity markets in different countries vary in terms of size, liquidity, degree of regulation, trading and settlement systems and restrictions on foreigners in buying equities.

12. The financial sector and capital markets are going through a digital revolution resulting in the upgrade of traditional functions, processes and activities, with new digitally driven processes and capabilities led by new entrants and joint ventures with financial institutions.

Chapter 2:
Risk, Return and Time Value Calculations

Learning Objectives

The candidate should be able to:

- ✓ Explain the concepts of risk and return in the management of investments.
- ✓ Explain the relationship between risk and return in the context of the risk-return trade-off.
- ✓ Identify the components of the total return and compute expected return on an investment.
- ✓ Explain the statistical measures of risk.
- ✓ Compute standard deviation and coefficient of variation as risk measures.
- ✓ Identify the various risks associated with investments.
- ✓ Calculate the present and future value of single cash flow.

2.1 Risk and Return

In economics and finance, the risk-return (or risk-reward) principle tells us that rational investors demand higher potential returns when they take on more risk. When there is lower risk, investors do not demand as high a rate of return.

Depending on the risk profile of the investor, the choice of financial assets to invest in may vary. An investor with a low appetite for risk will be fearful of losing any part of his capital and will consider relatively safe money market instruments, which provide a steady flow of interest income. In contrast, the investor with a higher risk appetite would be prepared to lose some (and, in some cases, all) of his capital and will consider more risky securities such as equities, with the expectation of higher return.

Figure 2.1(a) - General Risk and Return for Different Asset Classes

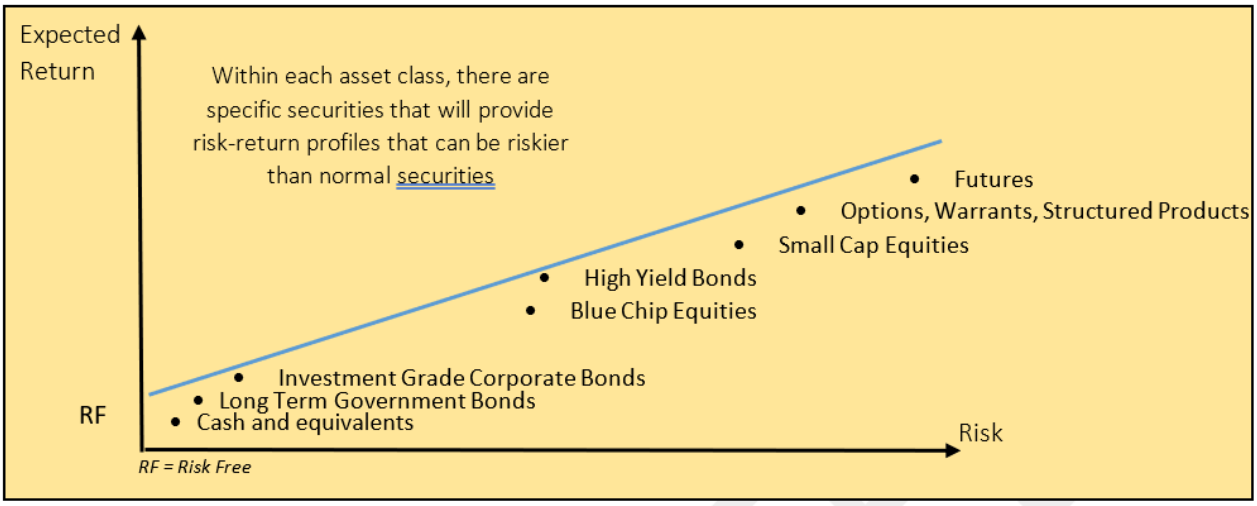


Figure 2.1(a) shows the general perceptions of the risk associated with each asset class relative to the expected returns. The specific risks of each individual investment choice will depend on the selected financial instrument, even within the same asset class.

For example, cash is often considered the least risky due to its role as a medium of exchange and its liquidity. Yet even cash can be volatile when compared to other currencies. Volatility in currency can be high with political changes, currency management policy changes, and fiscal and monetary policy changes. The last few years have seen wide fluctuations in most major world currencies.

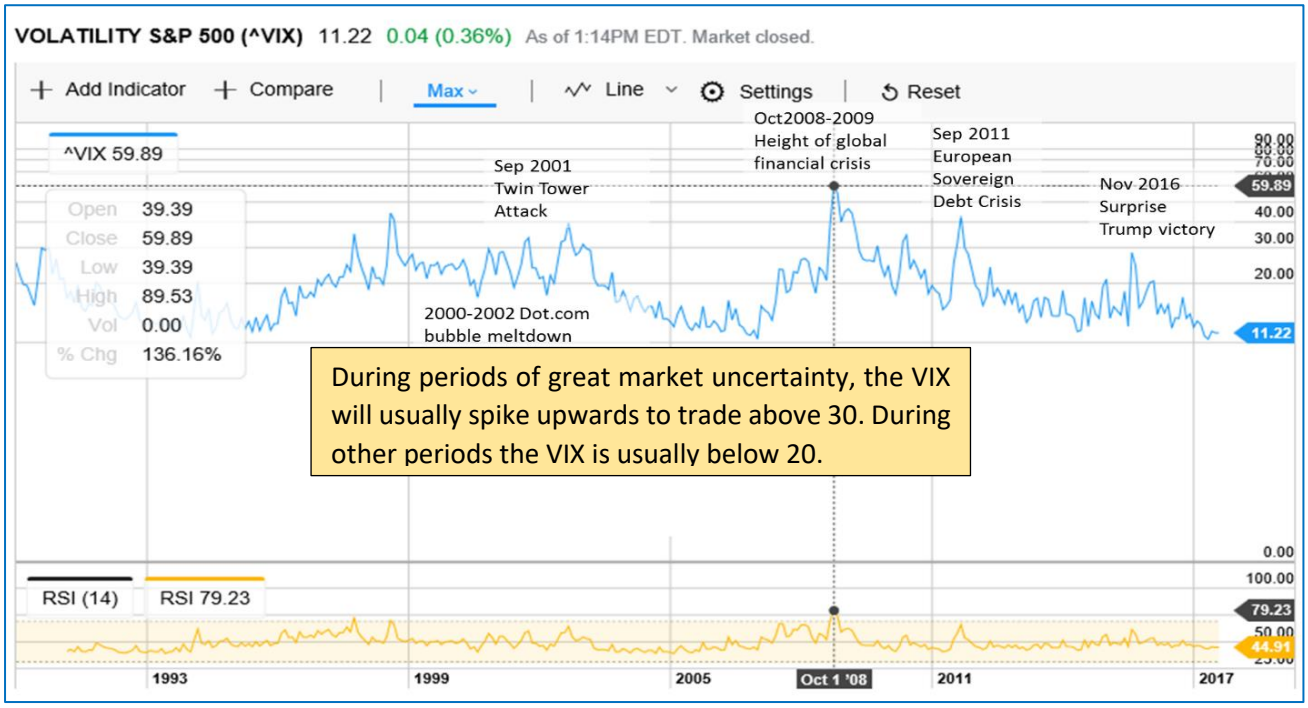
Once unquestionably secure, government bonds appear to carry more counterparty risks in recent years, changing traditional market view about "riskless" sovereign debt (e.g., Portugal, Italy, Greece, Spain, etc.). In general, the risk of a "CCC-rated" bond (junk or speculative grade) will be much higher than that of a blue-chip corporate debt which has "AAA-rating". In fact, the stock of this blue-chip company will probably be considered safer than the "CCC-rated" bond if it has a conservative business and financial profile, even though equities are generally considered to be a riskier asset class than bonds.

Figure 2.1(a) also shows derivatives at the upper end of the risk-return continuum. Since most structured products are constructed with long and/or short embedded options, they are considered risky investments for which investors expect high returns. While this is generally true, the range of structured products is very diverse. Some will have capital guaranteed making them safer than some equities.

From the examples above, it should be clear that representatives should examine the riskiness inherent in the security itself. They will need to understand and assess the risks and payoffs for each product by reviewing the product documents (i.e., Product Highlights Sheet and prospectus) and asking the relevant questions. Only after understanding the product structure, underlying strategy and payoff scenarios should they recommend the investments that are best suited for investors in accordance with their risk-return profiles.

There are periods of greater volatility and uncertainty and period of market confidence and low volatility. One bellwether often quoted as a risk indicator for the US stock market (and, indirectly, for the global equities market due of the importance of the US economy) is the Chicago Board Options Exchange (CBOE) Volatility Index, more popularly known as the "**VIX**".

Figure 2.1(b) - VIX Chart



Source: yahoo.finance.com

The VIX shows the market's expectation of the 30-day S&P 500 volatility and is also referred to as the "investor fear gauge". A higher VIX means more risk and volatility is expected. VIX values of more than 30 generally indicate a high volatility, while values less than 20 generally indicate relative stability. Refer to Figure 2.1(b). At the height of the Global Financial Crisis in October 2008, the VIX shot up to a high of 89.5, while during the European Debt Crisis, it went as high as 48 compared to the more stable periods.

Finally, there is also the overall market consideration as well. Financial assets in mature markets (e.g., the United States) are perceived as less risky compared to those in emerging markets (e.g., China) or frontier markets (e.g., Sri Lanka).

2.2 Calculating Returns

To evaluate the attractiveness of investments, we must be able to measure the rate of return and risk involved in an investment accurately. We focus on the returns in this sub-section.

2.2.1 Total Return

Total return (on a pre-tax basis) on an investment comprises two components:

a. **Current income -** This is income received from the investment during the period in which the investment is held, such as dividends from shares and interest income from fixed income securities or fixed deposits.

b. **Capital gain or loss -** The prices of equity securities and bonds may either increase or decrease, resulting in capital gain or loss, respectively.

Therefore, the total return on a security is:

$$\textbf{Total Return} = \frac{\textbf{Income + Capital Gain or Loss}}{\textbf{Purchase Price}}$$

The capital gain or loss is the beginning (or purchase) price subtracted from the ending (or sale) price. Thus, the equation can also be written as:

$$\textbf{Total Return} = \frac{CF_t + (P_E - P_B)}{P_B}$$

where
CF_t = sum of cash flows during the measurement period *t*
P_E = price at the end of period *t* or sale price
P_B = price at the beginning of period *t* or purchase price

Computation of Total Return

Assume you purchase 1,000 shares of XYZ Company for $5 per share and receive a dividend per share of $0.20. You sold the shares one year later at $6. The rate of return is:

$$\text{Total Return} = \frac{\$0.20 + (\$6 - \$5)}{\$5} = 0.24 \text{ or } 24\%$$

2.2.2 Expected Rate of Return

Previously, we considered the historical rate of return. An investor will be interested in the expected rate of return, i.e., the future rate of return, which may or may not be realised. If the investor expects a return of 15% on his investment, this is referred to as a point estimate. Given that we are dealing with an uncertain future, several possible returns based on different economic outlooks can and will occur.

Probability values can be assigned to all possible returns and the expected rate of return is defined as:

$$E(R) = \sum_{i=1}^{n} r_i p_i$$

where
E(R) = expected return on a security
r_i = i^{th} possible return
p_i = probability of the ith return r_i
n = number of possible returns

The expected rate of return is thus the weighted average of possible returns from a given investment, with the weights being probabilities.

Computation of Expected Rate of Return

Suppose an investor believes an investment can provide different rates of return depending on different possible economic scenarios. The investor might estimate probabilities for each of the economic scenarios based on experience and the current outlook as follows:

Economic Scenarios	Probability	Return
Economic boom	0.20	25%
Recession	0.20	-30%
Stable	0.60	15%
	1.00	

E(R) = (0.20 × 0.25) + (0.20 x -0.30)+(0.60 x 0.15)

= 0.05 - 0.06 + 0.09 = 0.08 or 8%

2.2.3 Annualised Rate of Return

Investment product returns are often presented to investors on an annualised basis. An example is shown below.

Computation of Annualised Rate of Return

An investment product is expected to generate a maximum return of 2.5% over 3 months. For marketing purposes, financial institutions can quote a potential annualised return of 10.38% for the product. The calculation would be as follows:

Annualised Rate of Return $= (1.025)^{12/3} - 1$

$= 1.10381 - 1$

$= 0.104$ or about 10.4%

Another way to look at the concept of an annualised rate of return is to consider the following:

- $1 is invested in an asset which pays an interest of 2.5% for a discrete period of 3 months.
- At the end of 3 months, the $1 grows to: $1 x 1.025 = $1.025.
- This $1.025 is reinvested for the next 3 months at the same rate of 2.5%.
- At the end of 6 months (3 + 3), the original $1 grows to: $1.025 x 1.025 = $1.0506.
- Again, this $1.0506 is reinvested for the next 3 months at the same rate of 2.5%.
- At the end of 9 months (3 + 3 +3), the original $1 grows to: $1.0506 x 1.025 = $1.0769.
- Finally, this $1.0769 is reinvested for the next 3 months at the same rate of 2.5%.
- At the end of 12 months (3 + 3 + 3 + 3), the original $1 grows to: $1.0769 x 1.025 = $1.1038.

Hence, the annual return for the original investment of $1 is: (1.1038-1)/1 = 0.1038 or 10.38%.

This is based on the assumption that the original investment plus interest is reinvested or the return is 'compounded' (i.e.; effect of earning interest on interest). In this example, the original investment of $1 is compounded 4 times to get to an equivalent 1-year return. This is: $(1.025)^4 - 1$, which is $(1.025)^{12/3} - 1$.

Conversely, when asked about the return achievable after 4 months for a product quoting an annualized target return of 12%, the calculation for the 4-month targeted return "r_4" is as follows:

$$(1 + r_4)^{12/4} - 1 = 0.12$$

$$(1 + r_4) = (1 + 0.12)^{4/12}$$

$$r_4 = (1.12)^{1/3} - 1 = 0.0385 \text{ or } 3.85\%.$$

2.2.4 Real Rate of Return (Inflation-Adjusted Return)

The real rate of return takes into consideration the impact of inflation on returns. An example of how the real rate of return is calculated is shown below.

Computation of Real Rate of Return (Inflation-Adjusted Return)

Suppose the inflation rate was 2% for the year. Asset prices in the market would have gone up by 2% on their own. If we consider the annualised rate of return calculated earlier of 10.4%, we can say that part of the return achieved was due to the general inflation. What then is the real rate of return?

The real rate of return is calculated as follows:

[(1 + 0.104) / (1 + 0.02)] - 1 = 0.08235 = 8.24%

Suppose asset values have fallen in value in the country by 1%, a deflation of 1%.

The real rate of return is calculated in the same way as follows:

[(1 + 0.104) / (1 - 0.01)] - 1 = 11.52%

2.2.5 Currency-Adjusted Return

In the accounting and reporting of investments, assets and income generated in foreign currency will have to be restated in the local reference currency, i.e., the base currency in which the investor wishes financial statements to be reflected. The performance of a fund invested in foreign securities denominated in foreign currency will have to be restated in the desired reference currency - usually the local currency.

Computation of Currency-Adjusted Return

A Singapore portfolio manager is looking at the performance of his Japanese portfolio (JPY) and his US portfolio (USD) over a period of 5 years (January 2013 - December 2017). His JPY portfolio, which is pegged to the Nikkei 225 Index is up about 121.88%, while his US portfolio which is pegged to the S&P500 Index was up 87.1%.

Over the 5-year period, SGD/JPY rose from 71.4943 to 85.0038 (up 18.89%) while the USD/SGD rose from 1.2272 to 1.3268 (up 8.12%).

What was the actual performance of the two portfolios over the 5 years when adjusted back to SGD as the reference currency?

The workings are shown below:

	Jan 2013	Dec 2017	+/- %	Currency Adjustment
Nikkei 225	10,688.11	23,714.53	+121.88%	**For the JPY portfolio** In SGD terms, the beginning Nikkei 225 value = 10,688.11/71.4943 = 149.4960 In SGD terms, the ending Nikkei 225 value = 23,714.53/85.0038 = 278.9820 Therefore, the currency-adjusted return = (278.9820/149.4960) – 1 = 0.8662 or 86.62% This is lower than the Nikkei 225 return since the SGD strengthened against the JPY, so the JPY portfolio gains are adjusted down.
SGD/JPY	71.4943	85.0038	+18.89%	
S&P 500	1,466.47	2,743.15	+87.06%	**For the US portfolio** In SGD terms, the beginning S&P 500 value = 1,466.47 x 1.2272 = 1,799.6520 In SGD terms, the ending S&P 500 value = 2,743.15 x 1.3268 = 3,639.6114 Therefore, the currency-adjusted return = (3,639.6114/1,799.6520) – 1 = 1.0224 or 102.24% This is higher than the S&P 500 return since the SGD weakened against the USD, so the US portfolio gains are adjusted up.
USD/SGD	1.2272	1.3268	+8.12%	

Hence, in SGD terms, the US portfolio did better than the Japanese portfolio because the SGD strengthened against the JPY but weakened against the USD. Notice that the currency-adjusted returns can also be calculated by adjusting the portfolio returns with the currency return as follows:

Currency-adjusted return for the Japanese portfolio = (1 + 1.2188) / (1 + 0.1889) – 1 = 0.8662 or 86.62%

Currency-adjusted return for the US portfolio = (1 + 0.8706) x (1 + 0.0812) – 1 = 1.0224 or 102.24%

From the examples above, a simple rule of thumb to follow is to divide (1 + portfolio return) by (1 + currency return) when the reference currency is the first listed currency of the currency pair and to multiply (1 + portfolio return) by (1 + currency return) when the reference currency is the second listed currency of the currency pair.

Some corporate and high net worth (HNW) investors prefer to use a major world currency, e.g., the USD or EUR, as their reference currency because they have an international perspective of their invested assets. Since the US remains the largest capital market and many financial instruments including commodities are still

denominated in USD compared to other currencies, maintaining a USD reference is more convenient and relevant to many such investors.

Some HNW individuals in Asia also prefer to use the USD as their reference currency for their offshore investments, which they keep separate from their domestic investments. Also, in some countries, the local currencies have proven to be more volatile than the global currencies like the USD and EUR. Hence, investors in these countries prefer not to use their local currencies as the reference currency when calculating returns.

2.2.6 Internal Rate of Return

The internal rate of return is a rate which will equate all cash outflows (investments) against all cash inflows (proceeds), evaluated at the same point in time. The internal rate of return (IRR) is used in return calculations for projects and investments with uneven cash flows (e.g., private equity).

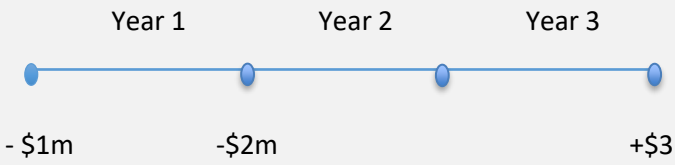
Computation of IRR (1)

Your client has made an investment with the following payout profile:

- The investment requires an initial amount of $1 million ($1m) and a subsequent investment of $2 million ($2m) at the start of the next year; and
- Your client will receive a payback of $3.5 million ($3.5m) only at the end of the 3rd year.

What is the return generated for this investment?

The cash flows are shown below:



The IRR is solved by a trial-and-error process. Evaluated at the end of Year 3, denoting the IRR by r, the equation can be written as:

$$1\,(1+r)^3 + 2\,(1+r)^2 + 0\,(1+r) = \$3.5m$$

The IRR is applied to the first cash flow of $1m for 3 years, the second cash flow of $2m for 2 years, and the third cash flow of $0 for 1 year. The payback of $3.5m comes at the end of 3 years.

To make an educated starting guess for r, note that $3m was invested in total (this is the aggregate outflow) and $3.5m was received (inflow). This implies an estimated return of (3.5/3) - 1 = 16.7% which, when divided by 3 years, would give us 5.55%.

- By applying r = 5.55% to the Left-Hand Side (LHS) expression of the formula, we get $3.4041m, which is lower than $3.5m. Therefore, the rate must be higher.
- By applying r = 7.00% to the LHS expression of the formula, we get $3.5148m, which marginally exceeds $3.5m. Therefore, the rate must be slightly lower than 7%.
- By applying r = 6.80% to the LHS expression of the formula, we get $3.4999m. The accuracy to IRR can be improved by applying 6.81%.

These days, the IRR can be easily calculated using a spreadsheet formula. For example, in Excel, we specify the cell range containing the cash flows in the formula (=IRR (<highlight cell range>, [guess])) and input an estimate under [guess], and the result is calculated through an iterative process.

In the above example, we would specify these cash flows as part of the cell range:

-1	-2	0	3.5

IRR vs Compounded Annualized Rate of Return - The IIR, which is the **money-weighted return** in a portfolio management context, is the actual return generated for the client given the movement of his cash flows and is sensitive to the size and timing of these cash flows.

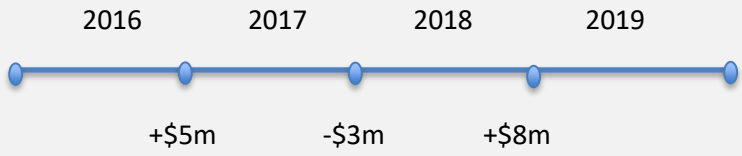
The **compounded annualised rate of return**, which is the **time-weighted return** in a portfolio management context, is unaffected by these cash flows. The difference between the money-weighted and time-weighted returns is illustrated in the next example.

Computation of IRR (2)

A client invested $5 million ($5m) in a fund at the end of 2016, which generated a return of 2% in 2017. He was not too impressed, so he pulled out $3m at the end of the year. In 2018, the fund generated a return of 10%. He was so happy that he added $8m at the start of 2019 but the fund made a loss of 4% for 2019.

(a) What is the value of the fund at the end of 2019?
(b) What is the client's actual return given his cash flows over the 3 years?
(c) What is the annualised compounded return of the fund over the 3 years?

The contributions and withdrawals he made are shown below:



(a) Let us follow the sequence of the cash flows:

Fund value at end of 2017:	$5m × 1.02 = $5.1m
Withdrawal:	- $3.0m
Fund value at end of 2018:	$2.10m × 1.10 = $2.31m
Contribution:	+$8.00m
Fund value at end of 2019:	$10.31m × 0.96 = $9.898m

At the end of 2019, the value of the fund is worth $9.898m.

(b) The client's actual return given his cash flows is the money-weighted return. This is similar to the IRR.

We can begin with a simple estimate of returns. The total negative flows (i.e., investment outlay) is \$5m + \$8m = \$13m, while the total proceeds received is \$3m + \$9.898m = \$12.898m.

Based purely on observation of what went into the investment and what was received, the client made a minor loss. A reasonable starting estimate of the loss would be \$12.898m/\$13m – 1 = 0.007846 or – 0.7846% (we do not divide by the number of years since the contributions and withdrawals are relatively evenly made throughout the period, as opposed to making all contributions early on and all withdrawals later in the period).

Now restating the cash flows into an equation to solve for r:

$$5 (1 + r)^3 - 3 (1+r)^2 + 8 (1+ r) = 9.898$$

Applying trial-and-error in the LHS expression of the formula using negative - 0.7846%:

- $5 (1 - 0.00786)^3 - 3 (1 - 0.00786)^2 + 8 (1 - 0.00786)$ = 9.867 (so rate must be less negative)
- $5 (1 - 0.00700)^3 - 3 (1 - 0.00700)^2 + 8 (1 - 0.00700)$ = 9.881 (so rate must be less negative)
- $5 (1 - 0.00600)^3 - 3 (1 - 0.00600)^2 + 8 (1 - 0.00600)$ = 9.898 (which is close)

Hence, by trial-and-error, we have found the IRR to be negative -0.6%, which is also the money-weighted return when referring to investment portfolios.

(c) The compounded return over 3 years is (1.02 x 1.10 x 0.96) - 1 = 0.0771 = 7.71%.

The annualised compounded rate of return is $(1.0771)^{1/3} - 1 = 2.51\%$.

See following for explanation:

Let R be the annualised compounded rate of return.
$(1 + R)^3 = 1.0771$
$(1 + R) = (1.0771)^{1/3}$
$R = (1.0771)^{1/3} – 1 = 0.0251$ or 2.51%

The compounded annualised return in an investment portfolio context is also the time-weighted return, which is not affected by the size and timing of cash flows moving in and out of the fund.

2.3 Measures of Risk

Risk refers to the uncertainty of returns associated with a given investment. In simple terms, risk is the degree of fluctuation in the value of assets such as stocks. Return and risk are inextricably linked.

2.3.1 Variance and Standard Deviation

The most widely used statistical measures of risk are the **variance** and **standard deviation,** which measure the dispersion of returns from the average return. The smaller the dispersion, the lower the risk of the investment.

The variance of returns (σ2) is measured as follows:

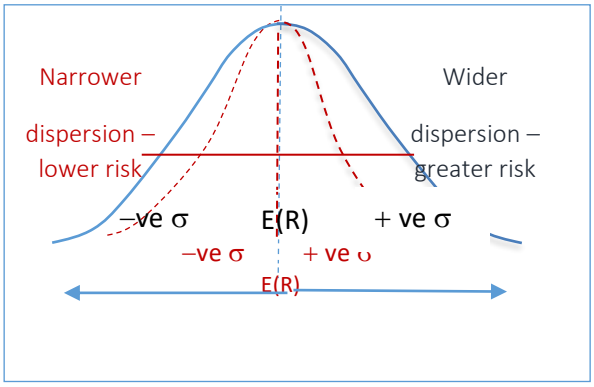
$$\sigma^2 = \sum p_i [r_i - E(R)]^2$$

The standard deviation of returns **(σ)**

is the positive square root of the variance:

$$\sigma = \sqrt{\sum p_i [r_i - E(R)]^2} \quad \textit{or} \quad \sigma = \sqrt{\sigma^2}$$

where all terms are as defined previously.



Computation of Variance and Standard Deviation

Referring to the previous example on the expected rate of return, the variance and standard deviation of the security are as follows:

Possible Return (r_i)	Probability (p_i)	$r_i p_i$	Deviation from Expected Return for Each Return Outcome $r_i - E(R)$	Square of the Deviation for Each Return Outcome $[r_i - E(R)]^2$	Probability-weighted Square of the Deviation for each return outcome $p_i [r_i - E(R)]^2$
0.25	0.20	0.05	0.17	0.0289	0.00578
-0.30	0.20	-0.06	-0.38	0.1444	0.02888
0.15	0.60	0.09	0.07	0.0049	0.00294
	1.00	E(R) = 0.08			0.03760

Variance (σ^2) = 0.03760

Variance is only the first calculation that must be made to get to standard deviation.

Standard deviation (σ) = $\sqrt{0.03760}$ = 0.1939 or 19.39%

Volatility refers to the standard deviation of the security calculated as shown in the example. If a share price has a standard deviation of 19.39%, assuming this is calculated on a daily price change basis, then we say that the share has a daily volatility of 19.39%. This is also known as ***historical volatility***, which is different from ***implied volatility***, a volatility that is implied from the price of a share option.

2.3.2 Coefficient of Variation

Very often, investments have varying rates of return and standard deviations. In evaluating such investments, another measure of risk, the coefficient of variation, can be used. The coefficient of variation (CV) measures the risk per unit of expected return and is defined as:

$$CV = \frac{\text{Standard Deviation of Returns}}{\text{Expected Rate of Return}}$$

Use of Coefficient of Variation

Consider the following two investments A and B:

	Investment A	Investment B
Expected return	0.05	0.15
Standard deviation	0.08	0.20

Based on standard deviation alone, Investment B seems to be less attractive as it has a higher risk than Investment A as indicated by Investment B's higher standard deviation. However, notice that Investment B also has a higher expected return. A calculation of the respective coefficients of variation reveals that Investment B has a lower risk per unit of return as follows:

$CV_A = 0.08/0.05 = 1.60$

$CV_B = 0.20/0.15 = 1.33$

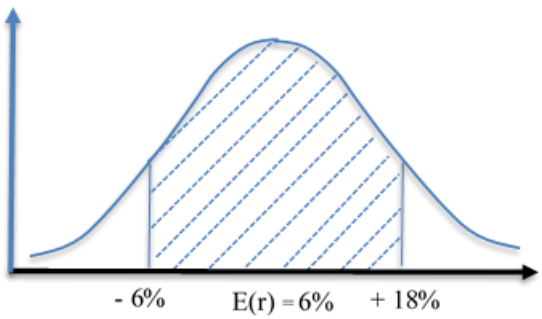
Thus, on a risk per unit of return basis, Investment B is more attractive.

2.3.3 Shortfall Risk - Probability of Losses Using Normal Distribution

In Section 2.3.1, we introduced the concept of standard deviation from an expected return. We can develop that discussion further and look at the probability of observing a range of returns assuming a normal distribution of returns. When we consider risks, we often speak of the likelihood or probability of an event occurring e.g., the probability of a loss occurring.

Please refer to Figure 2.3.3(a) – Distribution of Expected Returns of the Portfolio.

Figure 2.3.3(a) – Distribution of Expected Returns of the Portfolio



Suppose based on historical returns for the last 20 years, a balanced diversified portfolio invested in stocks and bonds has an average return of 6% with an estimated standard deviation of 12%.

What is the return, X, that deviates exactly by +/- 1 standard deviation from the expected return, E(r), if the expected return is equal to the historical average return?

Mathematically:

$X – E(r) = \pm 1\sigma$

$X = 0.06 \pm 0.12 = –0.06$ or 0.18

For any given return, X, we can also calculate something called the Z-score = $\frac{r - E(r)}{\sigma}$

This Z-score can be interpreted as the number of standard deviations that the return, X, deviates from the expected return. Not surprisingly, the Z-scores for the specific X values above must equal $\pm$ 1. That is, the Z-score that corresponds to the –0.06 return is $\frac{-6\% - 6\%}{12\%}$ = -1. Likewise, the Z-score that corresponds to the 0.18 return i$\frac{+18\% - 6\%}{12\%}$ = +1. If we are looking for the probability that the return for a given year would fall between -6% to +18% (within +/- 1 standard deviation of the expected return), we can look up the standard normal distribution table in Figure2.3.3(b). This table gives the probability of observing a return that is equal to or lower than a specified return (i.e.; the area under the standard normal distribution curve to the left of the specified return). From the table, the probability that corresponds to a Z-score of 1.00 is 0.8413, which means there is an 84.13% probability of observing an annual return that is lower than 18%.

Since the total area under the curve is 100%, this means that the probability of observing a return greater than 18% is 1 – 0.84 = 0.16. Also, as the normal distribution is symmetrical (with Z-score of 0 being the mid-line, the left side of the distribution is the mirror image of the right side), this means that the probability of observing a return less than –6% (Z-score of –1.00) is 0.16 as well.

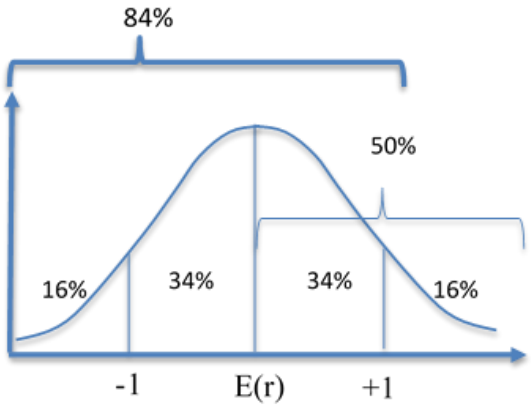
A further explanation:

Note that the standard normal distribution table shown in Figure 2.3.3(b) carries only the cumulative probability (area under the curve) for Z-scores that are greater or equal to zero. It shows the right tail of the distribution. But because it is symmetrical, we can infer the corresponding values for the left tail.

To find the probability that the return for a given year is -6% or worse, we first calculate the Z-score; (-6% - 6%)/12% = -1. But there is no Z-score of -1.00 in the table. Instead look for a Z-score of 1.00 and this corresponds to a probability of 0.8413. Because the normal distribution curve is symmetrical, the probability of X being -6% or worse is therefore 1- 0.8413 = 0.16 approximately.

Hence, the probability of observing a return that is less than 18% but greater than -6% is 0.84 minus 0.16 = 0.68 (see Figure 2.3.3(b).

Figure 2.3.3(b) – Normal Distribution of Expected Returns +/1 Standard Deviation from the Mean



Therefore, there is a 68% chance of returns falling within +/-1 standard deviation from the mean, E(r).

Tables of the Normal Distribution



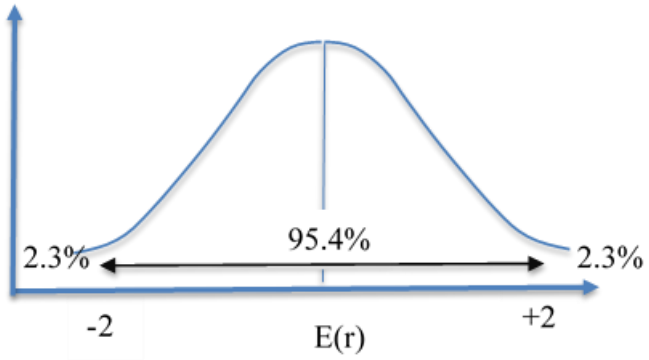
Probability Content from -oo to Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Likewise, if look at +/- 2 standard deviations from the mean, the Z-score (2) from the standard normal distribution table would show 0.9772, or about 97.7% under the curve on one side and 2.3% on the other.

Hence, about 95.4% (i.e. 97.7% minus 2.3%) of returns will fall within +/- 2-standard deviation from the mean. Refer to Figure 2.3.3(c).

Figure 2.3.3(c) – Normal Distribution of Expected Returns with +/- 2 Standard Deviations from the Mean



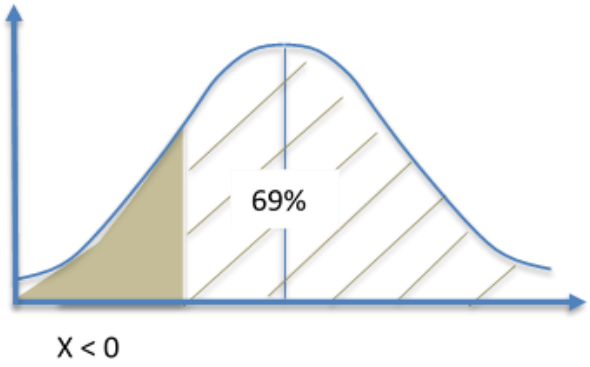
With an understanding of how to use the normal distribution, we can now ask a question that an investor might pose: "What is the probability or likelihood of making a loss, given the expected return and standard deviation of the portfolio as provided by the statistics for the last 20 years?"

So, we are asking what the likelihood is that the return, X, is less than 0.

We can apply the formula to obtain a Z-score, $Z = \frac{X - E(r)}{\sigma} = \frac{0 - 6\%}{12\%} = -0.5$

Looking at the standard normal distribution table, for a Z-score of -0.5, there is a 69% probability of a positive return and a 31% chance of a negative return. Please refer to Figure 2.3.3(d).

Figure 2.3.3(d) – Normal Distribution of Expected Returns with -0.5 Standard Deviations from the Mean



2.3.4 Value at Risk

An extension of the shortfall risk concept is found in **value at risk (VaR),** which is actively used in estimating the risk of an investment portfolio. With VaR, we ask, "What is the smallest proportion of our portfolio value that is at risk 2.5% or 0.5% of the time?"

a. **Balanced Portfolio**

Based on our previous balanced portfolio with a standard deviation of 12% and an expected return of 6%, the Z-score that corresponds to a probability of 0.025 (from the table, we look for a probability of 0.975 i.e. 1 minus 0.025) on the left tail is –1.96. Working back from the formula, $Z = \frac{X - E(r)}{\sigma}$, $X = Z\sigma + E(r) = -1.96 (0.12) + 0.06 = -0.1752$ or approximately 18%.

Next, we find for a Z-score that corresponds to a probability of 0.005. From the table we look for a probability of 0.995 i.e. 1 minus 0.005. Look closely, and you find the values 0.9949 and 0.9951 but no 0.995. Let us say we pick 0.9951, which corresponds to a Z-score of -2.58 (left tail).

Working back from the formula, $X = Z\sigma + E(r) = -2.58\ (0.12) + 0.06 = -0.2496$ or approximately -25%.

Hence, for this balanced portfolio with an expected return of 6% and a standard deviation of 12%, there is a 2.5% chance of the portfolio losing at least 18% and a 0.5% chance of the portfolio losing at least 25% in a given year.

b. **Defensive Portfolio**

Consider a defensive, income-generating portfolio, which is a mix of fixed income and dividend-paying stocks with an expected return of 4% and with a standard deviation of 4% over the last 20 years. The Z-score that corresponds to a probability of 0.025 on the left tail is –1.96.

Working back from the formula, $Z = \frac{X - E(r)}{\sigma}$, $X = Z\sigma + E(r) = -1.96\ (0.04) + 0.04 = -0.0384$ or approximately -4%.

The Z-score that corresponds to a probability of 0.005 on the left tail is around –2.58.

Working back from the formula, $X = Z\sigma + E(r) = -2.58\ (0.04) + 0.04 = -0.0632$ or round up to -7%.

Hence, for the defensive portfolio with an expected return of 4% and a standard deviation of 4%, there is a 2.5% chance of the portfolio losing at least 4% in a given year (i.e., or about 1 in 40 years). Also, there is a 0.5% chance of the portfolio losing at least 7% in a given year.

What is the risk of losses for this defensive portfolio? The Z-score corresponding to a return of 0% is:

$$Z = \frac{X - E(r)}{\sigma} = \frac{0\%-4\%}{4\%} = -1.$$

From the standard normal distribution table, a Z-score of –1 implies that the probability of loss is 16%. 84% of the time, this portfolio should yield positive returns.

The main problem with using historical data as the basis for predicting the probability of returns is that factors and conditions influencing returns for the next few years could be quite different from those influencing past returns. Besides using this traditional standard normal distribution for calculating probability of losses, the financial community also uses the Monte Carlo method[7] and computer-based models to estimate VaR.

2.4 Types of Risk and Risk Management

In our earlier discussion on risk in Section 2.3.1, we defined risk in terms of volatility—a positive or negative deviation from the expected return. However, the average investor is mainly concerned about negative deviation, that is, the risk of losses. In this section, we describe different types of risks and how to manage them.

[7] Instead of using historical data for asset returns and using the point estimates to calculate VaR, Monte Carlo simulation computes VaR using returns that are randomly generated under some distributional assumptions.

2.4.1 Risks in Investments

Risks can be classified in different ways.

In Table 2.4.1 we have grouped risks under the following categories:

a. **Issuer and Counterparty Risks**, which relate mostly to business risk.

b. **Product-Specific Risk**, which is affected by the way a product is designed and structured.

c. **Market Risk**, which tends to affect all assets and may not be diversified away but may be hedged.

d. **Execution and Operational Risks**, which are associated with risks in trade executions as well as internal and external factors affecting investment activities.

e. **Management Risk**, which is associated with the management of investments.

f. **Liquidity Risk**, which refers to the ability to liquidate an investment quickly at a fair price.

Financial losses can materialize from any of these sources at any time during the period of investment.

Table 2.4.1 – Classification of Investment Risks

Issuer & Counterparty Risks	Product-Specific Risk	Market Risk (Systematic)	Execution & Operational Risks	Management Risk	Liquidity Risk
• Business risk • Credit risk • Fraud risk	• Leverage structure • Term and maturity • Product geatures - Option triggers - Strike levels - Pricing - Payoffs - Obligations • Currency Risk	• Market shocks • Country, political, currency • Unexpected economic developments/ changes • Natural and Man-made Disasters, wars, terrorism • Liquidity risk	• Execution error • System failure • Process errors and restrictions • Computer viruses • Data breaches	• Risk Profile Mismatch • Concentration risk (Lack of diversification)	• Lack of market depth
Impact: • Fall in value due to business conditions affecting earnings growth, financial health and causing credit deterioration	**Impact:** • Augmented downside risk from leverage effect in product • Losses due to unfavourable product terms	**Impact:** • Sudden unexpected shocks in the market • Negative impact and volatility on entire market due to economic factors, policy	**Impact:** • Losses incurred on execution errors • Disruption to execution, and sudden corrections triggered by system failure	**Impact:** • Sub-optimal risk-adjusted returns • Failure to diversify, leading to excessive risk exposure, higher	**Impact:** • Trades effected at unfavourable prices that do not reflect fair value, especially when news are released to the market

Issuer & Counterparty Risks	Product-Specific Risk	Market Risk (Systematic)	Execution & Operational Risks	Management Risk	Liquidity Risk
• Failure in meeting financial obligation: - interest - principal - financial ratios - pledge clauses (breach of contract)	• Fall in value due to fall in denominated currency	decisions, rising factor costs, inflationary pressures, and tightened liquidity	• Computer-driven trading augmenting downside risk.	volatility and exposure to larger losses	

a. **Issuer and Counterparty Risks**

Issuer risk refers to the risk of losses arising when issuers of securities (e.g., shares, bonds, commercial papers, certificates of deposit, structured notes/deposits) default on their obligations.

An example of issuer risk was when Lehman Brothers' (Lehman) business of originating mortgaged-backed securities faced difficulties in 2008. At one point, Lehman had close to USD680 billion of assets supported only by USD22.5 billion of the firm's capital. With the weakening of the housing market, mortgage loans began to experience default and this, in turn, led to the value of Lehman's mortgage-backed securities plummeting when it became apparent that the issuer could not support interest payments and principal repayments of its issued securities. Globally (Singapore and Hong Kong included), investors who bought minibonds (structured notes) issued by Lehman suffered major losses when their values collapsed with the fall in Lehman's credit standing.

Some financial instruments are not traded on an exchange but are OTC contracts between counterparties. Hence, there is a risk that the counterparties do not fulfil their contractual obligations.

Counterparty risk describes the risk that one or more parties in a contract fails to fulfil its contractual obligation.

In some instances, the counterparty is also the issuer. When a bank designs a structured investment product and sells it to its clients, the bank is both the counterparty and issuer of the structured product. However, issuer may increase sales through other banks who distribute and market the issuer's product to their clients. The issuer may pay a retrocession fee or a commission to other financial institutions for selling their products.

In the Lehman minibond saga, investors who were affected tried to claim against the banks who sold these bonds to them, but the banks saw themselves merely as distributors and not counterparties to the deal, which was not fully evident to some clients. In such instances, for the sake of preserving reputation and client relationships, some banks may decide to provide "partial compensation" where some responsibility still rests on distributors, even if they are not the issuer, to understand the risks of the products they are selling and to ensure that the products match the risk profile of their clients. The lesson for investment representatives is that the phrase "caveat emptor", along with other typical disclaimers, may not always work, especially when it is unclear whether the clients are novices or sophisticated investors.

Both **issuer risk** and **counterparty risk** can arise from business risk and, in some cases, outright fraud.

i. **Business risk** relates to uncertainty about the ability of a company to meet its financial goals. For equity investors, the downside risks relate to changes in the company's earnings growth or potential, i.e., when there is a risk of:

- Earnings failing to meet market expectations;
- An unexpected slow-down in earnings growth, warranting a downward re-rating of the stock;
- An unexpected loss, which may trigger a sell-off of the investment; or
- Events and developments signalling a deterioration in the potential growth of the business.

Likewise, bond traders and bond fund managers have to be concerned about weakening business conditions for bond issuers (e.g., Singapore's offshore oil and gas engineering industry in 2015-2016). At a minimum, when bond prices fall, bond portfolios will have to be marked-to-market and the bond fund manager will have to report the fall in portfolio value. If business conditions become dire, some bond issuers may default on their bonds and losses will need to be realized.

ii. **Fraud** is relatively uncommon since most financial institutions are supposed to conduct proper due diligence. However, with the advent of outsourcing and the application of open-architecture concepts, many financial institutions (and high net worth individuals) have invested in boutique, hedge and private equity funds. One high-profile fraud case was that of Bernie Madoff, who ran a Ponzi scheme for several years affecting more than 4,800 clients, including some institutions. The fraud case was only discovered during the financial crisis of 2008, when redemptions eventually exposed his Ponzi scheme.

b. **Product-Specific Risk**

We started the Chapter by highlighting that different assets have different levels of risks for the returns generated. Equity shareholders, being part owners in the business, will have higher business risks while bond holders' risks focus on the firm's ability to finance interest and principal repayment over time.

Besides business and credit risks, the risks of investment products also depend on the specific features of the investment product and contractual terms of the agreement. The following are some of the typical features and characteristics that can affect the degree of risks associated with different products:

i. Leverage multiple;

ii. Maturity period;

iii. Payoff structure (e.g., timing, size, sequence, limits, etc);

iv. Sale of put or call options within a structured product; and

v. Currency.

For example, leveraged products (e.g., futures, warrants, daily leveraged certificates, contracts for difference) will have significantly higher volatility, moving multiple times compared to their underlying indices or securities. Some structured products with attractive yields effectively require investors to sell put options to the counterparty (e.g., equity linked notes). For better yield, other structures may require increased accumulation of securities as they fall in value (e.g., accumulators), thereby creating more risk.

Private equity investments will require long gestation periods and will show drawdowns (negative cashflows) in the initial few years before potentially achieving the promised high returns in the later years. Hence,

investors need to be kept fully aware of the nature and design of the products, as well as the risks when market conditions work against their investment positions.

The currency that a product is denominated in also makes a significant impact. The payment flows (e.g., coupons and principal repayment) in a foreign currency can significantly affect the overall returns in the domestic currency. For example, bond yields can be easily wiped out by currency losses when the investment flows are converted back to the local currency.

c. **Market Risk**

Market risk refers to risk that affects most assets in the market. Hence, such risk cannot be fully diversified. All asset classes are affected by market risks in one way or another. However, the impact will be different across the asset classes. There are specific periods where market risk has been especially pronounced and its impact vary significantly across asset classes. Such market shocks can be short- or long-lived.

Table 2.4.1 (c) – Periods of High Impact Market Risk

Financial Market Shocks	When major financial shocks occur, there is usually a ripple effect which affects different markets and different asset classes. For example, during the Global Financial Crisis of 2008-2009, almost all classes of financial assets (except cash and government bonds) fell in tandem. The correlation of the asset classes will tend toward "1" during periods of crisis. Some investment managers may be forced to sell performing asset classes to offset the impact of redemptions. Investors faced with margin calls will sell securities that are less affected to cover losses and margin requirements of other securities. Fear will cause investors to unwind outstanding positions in securities in favour of cash. Some examples of financial market shocks that have spread across different asset classes and markets include: • The Asian Currency Crisis (1997) • The Russian Financial Crisis (1998) • The Long-Term Capital Management (LTCM) Collapse (1998) • The Dot.Com Bubble (2000-2001) • The Global Financial Crisis (2008-2009) • The European Sovereign Debt Crisis (2012)
Political Shocks	Uncertainty due to sudden change in governments can cause uncertainty in the direction of the economy and businesses. Examples include the UK Brexit Referendum (2016) and the US Presidential Election Surprise (2016), both of which saw significant market volatility in the immediate aftermath of the unexpected outcomes.
Unexpected Contractionary Policy Decisions, Unexpected Negative Data	Unexpected contractionary policy decisions and economic numbers can also have a major impact on the market. For example, unexpected interest rate hikes, tax increases, inflation and unemployment figures, rise in factor costs and tighter liquidity can have an impact. Due to the sensitivity of such information, governments and central banks are usually careful in managing expectations so that the investment community is not

	caught off-guard with major policy changes unless it is purposely orchestrated as a surprise (e.g., cooling measures, introduction of capital controls, taxation, purchasing restrictions, etc.).
Natural Disasters, Wars, Terrorism	Natural disasters, wars and acts of terrorism can affect the stability of financial markets domestically and globally, but the impact is usually very short-lived and a rebound usually occurs in the days or weeks after. Some disasters seem not to have an impact at all. So far, those that pose a threat over a longer period (e.g., more than a day) seem to have more impact on the markets. Usually, such events are seen as short-lived and have proven to be buying opportunities. Some examples in the past include: • World Trade Centre Attack in NY, US (Sep 2001) • Severe Acute Respiratory Syndrome, HK/SG (2002-03) • Fukushima Nuclear Disaster, Japan (Mar 2011)

d. **Execution and Operational Risks**

These could arise from a counterparty, the financial exchange, or even from systems and platforms managing the internet trades initiated directly from the investor's laptop.

There are occasional incidences of human error where the wrong trade size or price level entered into the trading system could set off a range of trading activities that may not be fully reversible.

Despite advances in technology, markets are still vulnerable to human error. With the increase in algorithm-based computer trading, high-frequency trading, and an increase in exchange traded funds, trading volatility has been amplified and the momentum of trading has increased significantly. In 2015, a typing error caused Mizuho Securities Co**.** to lose at least 27 billion yen ($225 million) on a stock trade when a trader at Mizuho Securities tried to sell 610,000 shares at 1 yen (less than a penny) a piece. He had actually intended to sell 1 share at 610,000 yen ($5,041).

i. **Execution Risk in Financial Institutions** - Execution risk also occurs within financial institutions when wrong trades are made and must be reversed. While most institutions will usually pay for any loss or costs that are caused by their own error, the potential loss of trust and reputation may cause clients to take their business elsewhere, which is often the greater concern. To ensure that execution mistakes are minimal, the various staff involved in execution are to go through regular training. New systems and process have also been introduced to lower the incidence of errors.

   When there are system upgrades, temporary system glitches, or even compliance-related restrictions implemented at an inopportune time or in an inappropriate way, trading may be frozen for a period and a client could miss opportunities to execute transactions, resulting in a cost of lost opportunity. For example, the computer system could indicate that a client's trading limits or funding limits have been breached while ignoring available flow of funds from transactions that have been completed. These operational and execution problems and risks that exist within financial institutions are being minimised and addressed through improvements in technology and processes.

   The main conflict that can arise between clients and investment representatives is in the communication and confirmation of trade orders. Investment representatives are trained to apply best practices on receiving, confirming, recording, and executing orders so that there are minimal disputes with clients.

   Execution risk is being pushed to the clients these days because the trading platforms are now made directly available to some clients. Internet trading of securities has become very popular. As a result, the

risks of trade execution mistakes are now partly the responsibility of clients as they use the new trading platform applications on their mobile devices and computers that are linked directly to dealing systems. Financial institutions, however, do set limits on client trades, which limit the risk of execution error to the financial institutions and the financial system.

ii. **False Orders and Flash Crash** - Some trading risks arise from market manipulation. In 2010, a software programme was purposely used by a trader to place and cancel large orders within fractions of a second. This led to computer-based trading strategies reacting to the false selling orders by placing real selling orders of their own, causing indices to fall sharply. The Dow Jones Industrial Average Index (DJIA) plunged 9% and recovered its losses within 36 minutes of intraday trading. The trading also caused a spike in futures and options trading and resulted in what we call a flash crash. In February 2018, high frequency trading (HFT) was blamed for another flash crash of a smaller magnitude when the DJIA fell almost 1600 points (4%) in a few minutes.

   Global regulators have yet to fully solve the problem of such flash trades. The response from investment representatives is to be aware of situations in the market that may cause greater volatility and to remind long-term investors not to panic during such events. In most situations, the long-term fundamental factors will reassert their influence and markets will recover. Also, such events may present opportunities to acquire quality investments if clients are willing to hold for the longer term.

iii. **Market Disruption** can be defined as the effects of a large and rapid change in the market prices, causing the market to stop functioning in a regular manner. These are related to systematic risks and market shocks (see Section 2.4.1(c)). Such disruption is usually characterised by price declines, which cause widespread panic and a massive increase in trade execution and volatility. To mitigate this risk, regulators and securities exchanges implement **circuit breakers, shock absorbers** or **limits** (e.g., session or daily price limits), which can be imposed to limit price volatility without slowing or halting trading activity.

   - **Circuit breakers** vary functionally across different exchanges. On the Singapore Exchange Limited (SGX), circuit breakers will be triggered if the price for a stock would move by more than 10% of the traded price 5 minutes prior (the reference price). Hence, a large move can still carry on unimpeded if it is a gradual move over a longer time frame. The circuit breaker only aims to halt trading when the volatility is excessive within a very short time. Once triggered, a 5-minute cooling-off period will follow, where trading can only take place +/- 10% either way from the reference price. Thereafter, trading will resume with new reference prices established during the cooling-off period.

   - **Shock absorbers** are similar to circuit breakers and the terms have sometimes been used interchangeably to mean the same thing. Some use the term “shock absorbers” mainly to refer to the halting of index futures trading when there is a substantial decrease in the futures price. Yet others use the term to refer to the halting of computer-based HFT for specific stocks.

iv. **Technical Disruptions** - Market halts have occurred on different markets due to problems with hardware or software, errors in communication data feed, system updates, etc. Usually, such disruptions last from half an hour to as long as half a day. There are back-up systems which track execution and trade data in most exchanges in the event of major problems with software or hardware.

v. **Computer and Data Protection Breaches** - A major risk and concern in the internet revolution is the risk of viruses, hacking, and data corruption, which could seriously affect the financial markets. Financial institutions and governments have taken steps to build defences to their systems because of the potential problems such events can cause. Investment representatives are expected to take precaution against potential dangers by adhering to best practices in their handling of messages and communications.

e. **Management Risk**

This risk relates to how investments are potentially mismanaged, which can result in excessive risks taken.

i. **Risk Profile Mismatch -** When clients entrust their monies to investment representatives, they expect their investments to be managed professionally. Each client will have a risk profile and return expectation that will consequently determine the type of investment better suited for that client. Investment representatives should help clients make investment and asset allocation decisions based on their risk profiles. Some clients who are more conservative will take on less risky investments and be satisfied with modest returns. Clients who are more aggressive will pick investments with higher expected returns but that come with greater price volatility.

   This process of allocating the right proportion of assets to each client type based on his risk-return expectation is known as **strategic asset allocation.** When the profiling exercise is carried out poorly or only with the intention of fitting the available sales offering, there is the potential of a risk profile mismatch with the recommended products. When this happens, there could be potential case for mis-selling, negligence or mismanagement.

ii. **Concentration Risk or Lack of Diversification -** When clients use professional managers to manage their portfolios, investment representatives who are fund managers will usually ensure that there are proper diversification and minimal concentration risks. When clients manage investments on their own, they may end up with very high concentration of investments in specific securities, industries or markets because there is no formal diversification strategy. Ultimately, it is the client who must decide if he wants to remain concentrated, which can be a deliberate choice. However, investment representatives have the responsibility to advise their clients to diversify their investment portfolios for the longer term, especially if their clients do not have the time, expertise or right emotional temperament to actively manage their own investments.

f. **Liquidity Risk**

Liquidity risk can be examined at two levels. One aspect of liquidity risk relates to the availability of funding of investments. In periods of crisis, a liquidity crunch can occur, resulting in interest rates rising to abnormally high levels.

A second aspect of liquidity risk relates to the ability to liquidate a position quickly at a fair price and depends on the availability of buyers and sellers. An instrument will have more liquidity risk if its trading volume is low or if the available float in the market is small (with major blocks of the security/stock controlled by strategic investors). If there is poor market-price liquidity, i.e., few buyers and sellers in the market, the bid-offer spread will widen, making trading more costly.

Major surprises can cause a drop in liquidity. Suppose negative news is released and sellers are desperately trying to offload the related investment and there are no buyers except at much lower prices, (e.g., 12% lower). When the market sell orders come in, the investment price can suddenly gap downwards, which can happen several times a day as selling pressure mounts with very few buyers.
By similar reasoning, the lack of selling liquidity can create difficult situations for those with short positions. Those with a short position on a derivative instrument may find the price of the underlying asset suddenly rising on good news. If there is an absence of sellers, in this short squeeze situation, traders may be forced to cover their short positions at substantially higher prices because nobody wants to sell unless they are well compensated.

2.4.2 Risk Management

Risk management generally involves conducting due diligence, having timely research and risk management processes, assessing client profile and product suitability, risk diversification, as well as process and system risk management, as discussed below.

a. **Due Diligence Checks**

The security selection process typically comprises both fundamental and technical analysis. Likewise, in the selection of funds managers – be it traditional or hedge funds, it is also important to carry out regular due diligence and audit checks on the fund managers. This goes beyond tracking fund performance. Personnel changes, structural changes, cost pressures and market pressures may cause lapses in management. As such, it is important to ensure that external service providers have the proper checks and balances in place. This is especially important in market segments or jurisdictions where there is lack of regulatory oversight.

b. **Research and Risk Management Processes**

Not every financial institution performs primary research on companies. In most cases, due to cost pressures and efficiency considerations, there is a mix of primary and secondary research (gathering, evaluating and summarising research from other institutions and research providers).

Regardless of the approach, when the research is kept current and there are timely processes in place to review and manage risk exposure, the reaction time to deterioration in an investment or a market situation will be much faster. Timely risk management processes should allow for the underweighting of markets, sectors and companies before conditions worsen.

c. **Client Profile and Product Suitability**[8]

The investment products recommended depend very much on information provided by the client concerning his objectives, investment purpose, risk profile, return expectation, experience and sophistication with regards to investing, desired investment manager engagement and servicing, net worth, quantum or size of investments relative to his overall net worth, financial obligations, limitations, etc. (Refer to Chapter 1, Section 1.1 for details.)

A client could have a different purpose and objective for his investments placed at one bank compared to those placed at another bank. Given the purpose, the selected risk profile for his investments at one bank (e.g., core investments for the family to be kept safe) may be quite different from that for his investments held at another bank (e.g., excess cash for opportunistic investments). In managing risks, we recommend products which are suitable for a client based on his desired risk profile for his investments.

Some corporate clients may have a dedicated purpose with a specific timeline or investment horizon (e.g., monies set aside for a building project starting in 12 months). Thus, the investments selected will have to fit the limits and parameters set by the client.

[8] On 7 October 2023, MAS, in conjunction with the Association of Banks in Singapore (ABS), Association of Financial Advisers (Singapore) (AFAS) and Life Insurance Association (LIA), launched a Basic Financial Planning Guide to help Singaporeans take steps to enhance their financial well-being. The Guide outlines a few rules of thumb, for individuals to start taking proactive steps to address their savings, insurance, and investment needs (Go.gov.sg).

d. **Process and System Risk Management**

To mitigate the problems with operational risk management, most banks have been upgrading their processes and systems to ensure that they can handle the demands of a changing environment.

Over the last few years, the focus has been to ensure that systems and processes help with compliance to regulatory requirements (e.g., documentation, declarations, alerts, and checks). This has slowed down the execution processes and required a re-tooling and re-training of staff.

At the same time, processes and systems are being upgraded to ensure that information is well-protected.

Finally, there are also efforts in upgrading internal processes and systems to ensure "error-proof" execution and efficient proper authorization. Prompts and computer checks are automatically triggered to limit human error. Proper back-ups to computer systems have long been in place for banks to ensure that transaction records can be retrieved in the event of unexpected market disruptions.

2.5 Time Value of Money

Money has time value. This concept is important in financial markets because cash flows can occur at different points in time. The interest received from bonds usually takes place over the life of the bond and, likewise, dividends when declared, are typically paid over the life of the company. Using discounted cash flow techniques to account for the time value of money is necessary to make correct investment analyses and decisions.

The principle behind the time value of money is that a dollar received in the future is worth LESS than a dollar received today. This is because the dollar received today can be immediately invested to make a return. The time value of a sum of money depends on the date of its receipt and the prevailing interest rates.

2.5.1 Simple and Compound Interest

If an investor deposits $10,000 for a fixed period of 1 year at a rate of 2.5%, then the principal is $10,000, the interest period is 1 year, and the interest rate is 2.5%. If this is a deposit in Singapore Dollars, the day-basis will be 365 days i.e., each year is considered to have 365 days for the purpose of computing interest.

Simple interest is calculated with the following formula[9]:

$$\textbf{Simple Interest} = \frac{\textbf{Principal x Interest Rate (in \%) x Interest Period (in days)}}{\textbf{36,500}}$$

Example – Simple Interest

Assume a SGD 1 million deposit with a bank, earning interest at 8% for 91 days.

The simple interest is (SGD 1,000,000 x 8 x 91) / 36,500 = SGD 19,945.21.

[9] The denominator depends on the currency's "day basis". Singapore dollar interest is based on a 365-day basis. US dollar interest is based on a 360-day basis.

Compound interest is best described as the "earning of interest on interest", as seen in the example below.

Example – Compound Interest

Assume the SGD 1 million deposit placed earlier at 8% is now rolled over for another 91 days at 8% again. The compounded interest is:

(SGD 1,019,945.21 x 8 x 91)/36,500 = SGD 20,343.02.

The total interest earned = 19,945.21 + 20,343.02 = SGD 40,288.23.

The annualized compounded interest rate is (SGD 40,288.23 /SGD 1,000,000) x (36,500/182) = 8.08%.

2.5.2 Future Value

With the time value principle, the **future value** (FV) of a sum of money or cash flow can be calculated. The formula[10] to compute the value of an investment, if compounded annually at a rate of k for n years, is:

$$FV_n = PV(1 + k)^n$$

where:
FV_n = future value of the investment at the end of n years
PV = present value or original amount invested at the beginning of the first year
n = number of years during which the compounding occurs
k = annual interest rate

Example – Future Value with Annual Compounding

If $1,000 is placed in a savings account paying 5% interest compounded annually, the amount in the account at the end of 10 years would be:

FV_n = $1,000 $(1 + 0.05)^{10}$

= $1,000 (1.62889)

= $1,628.89

2.5.3 Time Value of Money in Negative Interest Rate Environment

Conceptually, the traditional way of looking at interest rates is that interest rates represent the time value of money determined by the demand and supply of funds. In exchange for having the use of funds, users of funds compensate the suppliers of the funds for the loss of liquidity and opportunity costs. Typically, we would expect interest rates to be positive.

However, the last few years have seen negative interest rates imposed by central banks. In 2017, 10-20 European countries had negative yields on their 2-year sovereign debt, which effectively meant that

[10] Candidates should be familiar with financial / statistical calculators which can perform discounted cash flow and statistical analysis. Excel spreadsheets are also highly recommended for investment computations and analysis.

commercial banks were required to pay for holding excess reserves with the central banks. The idea was to encourage commercial banks to minimise excess cash balances with the central banks so that they could increase productive lending activity with businesses and find better uses for idle funds. Sovereign debt issues may have been issued at a premium (above par) to banks and will, at maturity, return the principal value at par – so there would be a "loss" or implicit interest paid by the banks.

Table 2.5.3 - 2-Year Sovereign Debt Showing Negative Yields (as at 30 August 2017)

Country	Yield (%)	Country	Yield (%)	Country	Yield (%)	Country	Yield (%)
Switzerland	-0.92	Denmark	-0.69	Slovakia	-0.57	Italy	-0.17
Czech Rep	-0.81	Finland	-0.67	France	-0.53	Japan	-0.13
Germany	-0.74	Austria	-0.65	Slovenia	-0.38	Portugal	-0.09
Netherlands	-0.71	Ireland	-0.58	Spain	-0.37	Cyprus	-0.08
Sweden	-0.70	Belgium	-0.58	Bulgaria	-0.33	Lithuania	-0.08

Banks have generally not been able to do the same, i.e., charge negative interest on cash from depositors. If they did, there would be the risk of political backlash from the population against the governments and central banks. Depositors would then likely move money outside of the country and shift into foreign currency deposits. The central bank may not want to spark an outflow of capital into other currency. However, even with positive nominal deposit rates, with the interest rates being so low for the last few years, inflation-adjusted real interest rates were effectively negative. While there has been no formal interest charge by banks for deposits, depositors have already been effectively experiencing negative real interest rates for the periods when inflation rates exceeded deposit rates in their country.

Impact of Negative Rates on Short End of Yield Curve. Although individuals are not formally paying banks to hold deposits, negative central bank rates will drive short term rates on other types of fixed income investments downwards. Initially, there was also some effort by investors to lock in bonds yields at the short end of the yield curve, which drove short-term fixed income investment values up. Hence, in terms of time value of money calculations, one can conclude that negative interest rates will raise the present value of fixed income investments at the short end of the yield curve.

2.5.4 Compounded Annual Growth Rate

The compounded annual growth rate (CAGR) is the effective annual rate that is earned on the initial cash flow. In the above example on Future Value with Annual Compounding, the CAGR can be backed out. If the initial cash outlay is $1,000 and the final cash amount after 10 years is $1,628.89, the CAGR can be computed with the following formula:

$$\text{CAGR} = \left[\frac{\text{FV}}{\text{PV}}\right]^{\frac{1}{n}} - 1$$

where:
CAGR = compounded annual growth rate
FV = future value
PV = present value
n = number of years in the time period

Substituting the FV of $1,628.89, PV of $1,000 and n of 10:

$$\text{CAGR} = \left[\frac{1{,}628.89}{1{,}000}\right]^{\frac{1}{10}} - 1$$

= 0.05 or 5%

So far, it has been assumed that the compounding period is annual although, of course, compounding can be on a daily, monthly, quarterly, semi-annual or annual basis.

If we invest $1,000 for 5 years at 5% interest compounded semi-annually, we are really investing our money for 10 six-month periods, during which we receive 2.5% interest each period. If compounding is quarterly, we receive 1.25% interest per period for 20 three-month periods. This reasoning can be generalized, giving us the following formula for finding the future value of an investment for which interest is compounded more than once in a year:

$$FV_n = PV(1 + k/m)^{mn}$$

where m = number of interest periods in the year

Example – Future Value with Quarterly Compounding

If we place $1,000 for 10 years at 5% interest compounded quarterly, the investment will grow to $1,643.62 at the end of 10 years.

FV_n = $1,000 $(1 + 0.05/4)^{40}$

= $1,000 $(1.0125)^{40}$

= $1,000 (1.64362)

= $1,643.62

2.5.5 Present Value of a Single Cash Flow

Discounting, the opposite of compounding, is the process of finding the present value of a future sum. The process of discounting helps to determine the attractiveness of investment projects. The formula for calculating the present value is as follows:

$$PV = \frac{FV_n}{(1 + k)^n}$$

where:
PV = present value of the future sum of money
FV_n = future value of the investment at the end of n years
n = number of years to the receipt of payment
k = annual discount (or interest) rate

Example – Discounting to Get Present Value

Given a discount rate of 6%, the present value of $1,000 to be received 5 years from today is:

PV = $1,000 / $(1.06)^5$

= $1,000 / 1.3382

= $747.27

The interest rate in present value calculations is commonly known as the **discount rate**. The discount rate can be interpreted as the rate of return that could be earned on alternative similar investments. From the point of view of the firm's security investors, since this is an opportunity cost, the discount rate is also the rate of return expected by holders of the securities.

The discount rate is frequently used to adjust the investment cash flows for risk and, hence, is known as the risk-adjusted discount rate. The actual discount rate is at least the risk-free government bond yield that has the same maturity as the intended holding period of the investment.

2.6 Summary

1. There is a positive relationship between risk and return. Investors who desire higher returns will have to contend with higher risks and vice versa.

2. There are six main measures of return and they are (a) total return, (b) expected rate of return, (c) compounded annualised rate of return, (d) real rate of return, (e) currency-adjusted return, and (f) internal rate of return.

3. Risk from investments can be analysed by looking at the variance and standard deviation of asset returns. The standard deviation is also the return volatility.

4. Risk for investments can also be reviewed in terms of probability of loss and the extent of losses when they do occur. Shortfall Risk and Variance at Risk are two statistical methods for reviewing the probability of losses and the minimal extent of losses.

5. When evaluating investments with varying rates of return and standard deviation, the coefficient of variation, a relative risk measure, is useful. It measures the risk per unit of return.

6. All investments are exposed to risks. The risks that investment representatives and clients have to deal with include:

Issuer & Counterparty Risks	Product-Specific Risk	Market Risk (Systematic)	Execution & Operational Risks	Management Risk	Liquidity Risk
• Business risk • Credit risk • Fraud risk	• Leverage structure • Term and maturity • Product features - Option triggers - Strike levels - Pricing - Payoffs - Obligations • Currency risk	• Market shocks • Country, political, currency • Unexpected economic developments/ changes • Natural and man-made disasters, wars, terrorism • Liquidity risk	• Execution error • System failure • Process errors and restrictions • Computer viruses • Data breaches	• Risk profile mismatch • Concentration risk (Lack of diversification)	• Lack of market depth

7. Risk management generally involves conducting due diligence, having timely research and risk management processes, assessing client profile and product suitability, risk diversification, as well as process and system risk management.

8. Money has time value. The principle behind the time value of money is that a dollar received in the future is worth LESS than a dollar received today.

9. When inflation rates exceed deposit rates, depositors experience negative real interest rates.

10. Compounding is the process of determining the future value of a present sum of money while discounting is the process of finding the present value of a future sum

Chapter 3:
Key Drivers of Market Movements and Asset Values

Learning Objectives

The candidate should be able to:

- ✓ Identify and explain the key drivers of asset values and market movement.
- ✓ Describe and explain how key economic factors affect market movement and asset values.
- ✓ Follow the economic calendar and explain the three main classes of economic indicators - leading, coincident and lagging indicators.
- ✓ Discuss how industry and company-specific factors affect market value.
- ✓ Describe how market factors affect asset prices.
- ✓ Describe how investment behaviour influences investment decisions, market movements, and asset prices.

3.1 Introduction to Market Movements and Asset Values

When assessing asset values, the investment analyst would typically calculate an intrinsic or fair value determined by fundamental analysis based on a set of assumptions and parameters and compare it against the market value. The assessed fair value and accompanying recommendations can often differ across analysts because the assumptions and projections can be quite different. Perception of fair value can also vary among investors because they have different investment horizons and objectives.

Market price, on the other hand, is influenced by a myriad of factors, ultimately translating into the price agreed for the asset given the supply of the security and its demand at a point in time. The market price of the asset is influenced by fundamental factors (economic, industry and company-specific factors affecting its prospects and value), technical factors driven by investment behaviour and sentiment, and market-driven factors such as market structure, trading rules and parameters, type of participants, as well as the flow of funds and assets available in the market at any one time.

The Efficient Market Hypothesis (EMH) postulated by Eugene Fama[11] contends that markets are efficient and asset prices adjust rapidly to reflect any new publicly available information. Therefore, an asset's price is its fair value and no individual may consistently outperform the market. Yet, periods of information inefficiency

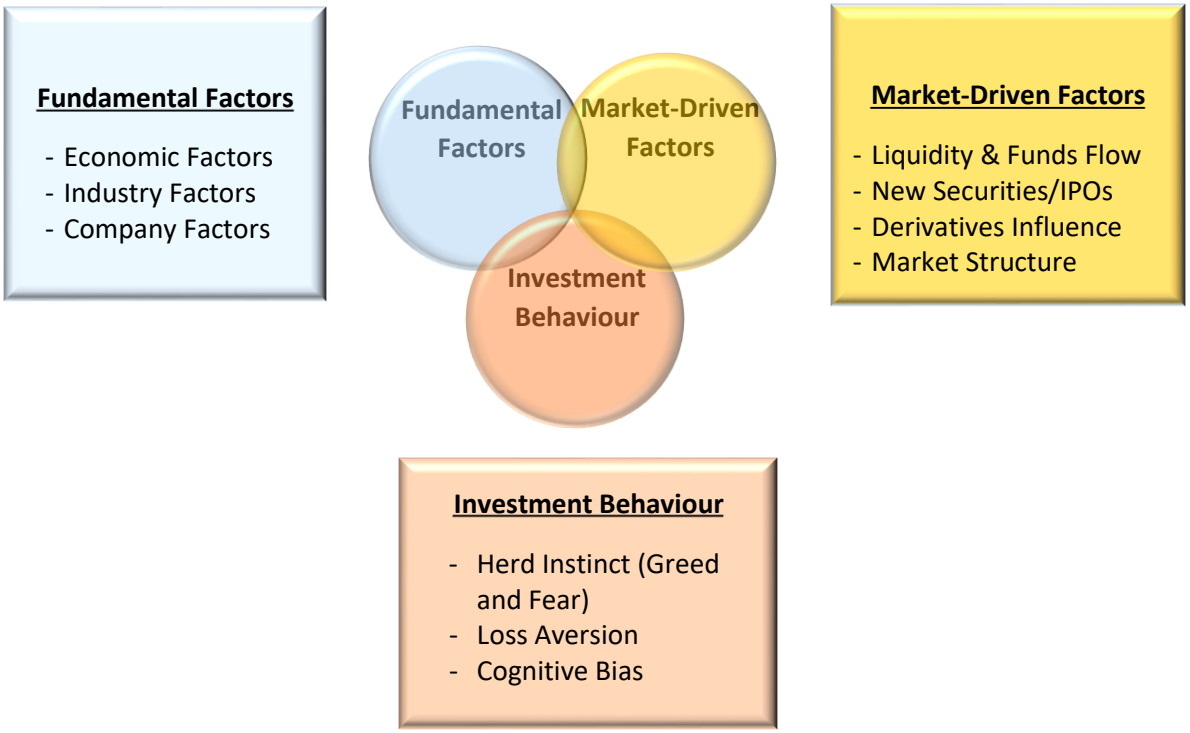
[11] Bergen, Jason Van (17 February 2004). "Efficient Market Hypothesis: Is The Stock Market Efficient?".

appear to exist, leading to under- or over-valuation over extended periods. Investors have also been found to react irrationally due to cognitive biases such as overconfidence, emotional over-reaction, information bias and other predictable human errors in reasoning and information processing. For the more astute and alert investors, there may be opportunities to take advantage of periods of undervaluation and overvaluation.

3.1.1 Factors Driving Asset Values

As set out in Figure 3.1.1, there are 3 primary factors influencing asset values and market movement, namely, fundamental factors, market-driven factors and investment behaviour. Fundamental and market-driven factors provide the rational and logical explanation for how an asset's worth is reflected in its price, while investor behaviour reflects how emotional and market sentiment affects the price movement of the asset. Market participants must be aware of the presence of all 3 factors and incorporate them when analysing and making sound investment decisions.

Figure 3.1.1 - Key Factors Affecting Market Movements and Asset Values



3.2 Fundamental Factors and Analysis

Fundamental analysis attempts to understand and estimate the fair value of an asset by examining macroeconomic factors such as fiscal and monetary policies, industry-specific factors such as the industry life cycle and competitive structure and firm-specific factors such as the business strategy and financial condition of a firm.

3.2.1 Economic Analysis

Expectations about the future health of an economy have direct and significant effects on its financial markets. After all, economic data reflects the composite performance of industries and companies and forms the basis for projections about the future. When investors expect steady economic growth, the stock markets will typically be bullish. Expectations of significant slowdown or the threat of a recession will usually trigger market corrections and bearish market conditions.

Investment instruments are affected by economic fiscal and monetary policies. Such policy changes affect interest rates, currency, inflation, employment, consumption and corporate profits, and expectations of policy changes also influence asset price movements. Hence, the investment community actively tracks leading economic indicators to gauge the significance of policy changes and their impact on different investments and industries. Investors look to invest in the securities that are likely to benefit the most from changes in fiscal and monetary policies.

Economic analysis affects how we allocate investments. In top-down asset allocation, a larger proportion of funds is allocated to the developed economies, growing sectors and industries, and the leading corporate performers within these industries—those that exhibit stronger financials and better earnings growth potential.

3.2.2 Fiscal Policy

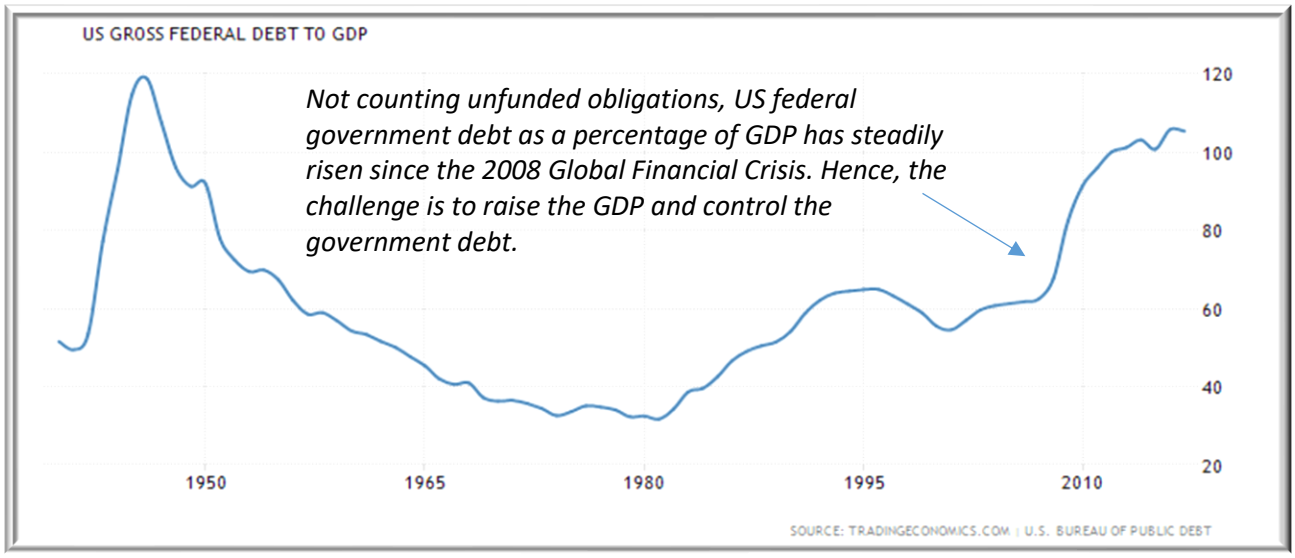
Fiscal policy refers to the use of government expenditure and taxation to influence aggregate demand and supply. Changes in government spending on infrastructure can influence the general economy.

Money from fiscal expenditure passes though the economy and eventually flows to consumers and produces what is called a **multiplier effect**. For example, greater construction and infrastructure spending leads to more demand for construction materials, equipment, services and labour. Wages and profits rise within these industries and higher disposable incomes are created. This triggers further consumption and spending for goods and services, benefitting suppliers of other industries. With the healthier consumption, private capital expenditure rises as well, adding to the expansion. Thus, the initial round of spending ripples through different industries causing the multiplier effect.

Governments can also cut taxes and create higher profits and disposable income, triggering greater consumption and stimulus for the economy. To fund the initial spending, governments may have to run a budget deficit to stimulate the economy.

Excess spending is funded by drawing down on surpluses accumulated from prior years or by issuing government bonds. Most governments had to run higher levels of debt to support fiscal stimulus to their economies immediately after the 2008 Global Financial Crisis. The US government debt as a percentage of GDP was at its highest in 2017 and back to post-World Wall II levels, when heavy government stimulus was needed.

Figure 3.2.2 - US Gross Federal Debt to GDP



3.2.3 Impact of Fiscal Measures on Gross Domestic Product

The Gross Domestic Product (GDP) is the most widely used measure of economic performance. It measures the market value of goods and services produced in the domestic economy during a specific time period, usually one year. In determining the GDP, four basic components need to be considered, namely:

a. **Consumption expenditures** of individuals on goods and services;

b. **Private investments in capital goods**, which consist of capital expenditures for replacement of equipments and for expansion in the supply of capital assets;

c. **Government spendings** that include both investment and consumption. Education, defence, capital investments, infrastructure development and health services are included in the government component; and

d. **Net exports i.e., total exports less total imports**. Net exports may either be positive or negative. When the country sells more to foreigners than they buy, net exports will be positive.

$$\mathbf{GDP = C + I + G + NE}$$

where:
C = Consumption expenditures
I = Private investments
G = Government spendings
NE = Net exports

To evaluate how fiscal policies might affect the economy, we can consider how the policies would affect the components of GDP. For example, with the intent of sustaining a higher GDP growth target of 3-4% compared to prior periods averaging 2.2% (1998-2017), the US government adopted a more aggressive expansionary fiscal policy stance, which included the following measures:

i. Plans for more infrastructure development projects in the US (this raises G);

ii. Introducing tax incentives for US companies to relocate their factory operations back to the US (this raises I);

iii. Lowering the personal income tax rate (this raises C);

iv. Imposing tariffs on selected foreign goods, making the prices of some foreign goods less attractive relative to US-made goods (this is intended to improve NE, which is traditionally negative); and

v. Increasing job creation and retention to encourage higher employment rate (this increases C).

Expectations of the US government's follow-through on the promised fiscal measures led to a major rally in the US market after the November 2016 election and, in 2017, the S&P500 index generated a return of 21%. Successful stimulus in the near term, however, may create new problems that the US government will have to manage in the future (e.g., threat of inflation, higher government budget deficits, etc.)

3.2.4 Monetary Policy

Monetary policy tools are used to stimulate (or cool down) the economy. This is carried out either by increasing (or decreasing) liquidity in the economy and/or by decreasing (or increasing) the cost of funding for short-term liquidity and bank reserves.

All other factors held constant, an expansionary monetary policy seeks to expand the money supply, lower interest rates, and increase aggregate demand. Lower interest rates encourage investments, with less attractive deposit rates, more funds flow into assets such as equity and real estate, which consequentially drive-up asset values. Also, lower interest rates encourage spending via reduced costs of borrowing.

Conversely, a contractionary monetary policy seeks to decrease money supply, raise interest rates and contain inflationary pressures. High interest rates discourage spending and borrowing (through higher costs of borrowing), thereby reducing demand for assets, leading to a fall in asset values. It can also lead to a fall in production and employment, lowering disposable income and further discouraging spending.

For the US economy, an important facet of monetary policy is the rate at which the Federal Reserve (Fed) allows the money supply to grow. Studies on US monetary history have shown that declines in the growth rate of money supply have preceded business contractions, while increases in the growth rate of money supply have preceded economic expansions. The US Fed can influence money supply through:

a. the Open Market Operations (OMOs), where the central bank buys and sells government securities;

b. the raising or lowering of the Fed Funds Rate, which is the rate at which banks lend excess reserves to one another to help those with shortfalls meet their reserve requirements; and

c. the discount window, which is a lending facility that the Fed extends to eligible financial institutions to meet temporary shortfalls in available credit caused by short term illiquidity in the market.

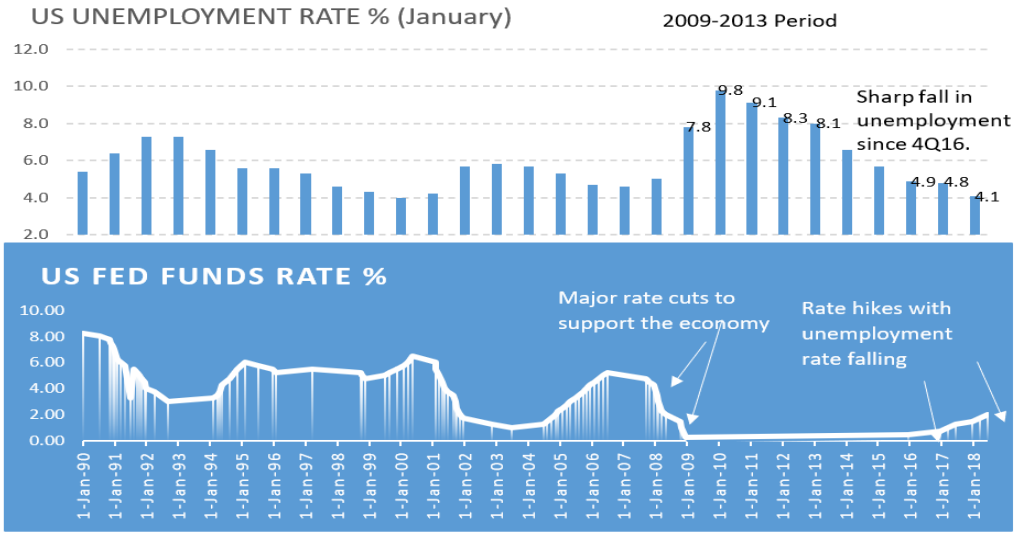
Of the three, the Fed has relied heavily on OMOs as its key method for influencing market liquidity. It was used extensively to pump liquidity into the market in 2008 during the Global Financial Crisis. This type of change in liquidity first affects bond prices, then common stocks and, finally, the aggregate economy. During economic downturns, central banks would adopt an expansionary monetary policy, which means raising liquidity in the market and keeping interest rates low. Funding costs for reserves are kept low and the market is flushed with liquidity to encourage lower cost funding.

The financial crisis of 2008 resulted in the greatest recession since the US Great Depression of the 1930s. To stimulate consumption, employment and growth, the US, the UK and European central banks, embarked on various Quantitative Easing (QE) programs, resulting in a protracted period of very low interest rates (2009-2017). Central banks in Asia also had to keep interest rates low to increase liquidity in their domestic economies.

The Fed openly tied QE to the level of unemployment. Also, interest rates were maintained at a low of 0.25% from Dec 2008 until Dec 2016, when it was gradually raised by 25 basis points each time as employment and job creation numbers improved. With the US unemployment rate reaching a May 2018 figure of 3.8%[12], its lowest in 48 years, and core inflation reaching the Federal target of 2.0%, the Fed e has reiterated its intention to continue to normalize interest rates and to reduce its balance sheet. The Fed Funds Rate was raised to 2.25% in September 2018[13].

When the economy gets overheated, inflation control becomes the focus. Central banks try to contain inflationary pressures and excessive capacity build-up for fear that unsustainable growth and the eventual correction can lead to a hard landing and cause major unemployment, destruction of asset values and social hardships. In such situations, central banks adopt a contractionary policy to try to moderate the pace of growth by reducing money supply in the market and raising the cost of funds, triggering some level of consolidation and engineering a soft landing. Refer to Figure 3.2.4.

Figure 3.2.4 - US Unemployment Rate and Federal Funds Rate (1990-2018)



Source: Data from US Bureau of Labor Statistics and the US Federal Reserve

a. **Singapore's Trade-Weighted Exchange Rate Index**

Because of its open economy, Singapore's monetary regime centres around the management of its exchange rate, as it has a major influence on local inflation and the country's international competitiveness. The Monetary Authority of Singapore (MAS) conducts policy by managing a trade-weighted exchange rate index.

Money supply, on the other hand, is basically driven from within and will only have a limited impact on Singapore's economic activity. It is not possible to set independent policy targets for both exchange rates and

[12] Bureau of Labor Statistics – Labor Force Statistics from the Current Population Survey - https://www.bls.gov/cps/.

[13] Federal Reserve Press Release September 26, 2018 - https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm.

interest rates, as one affects the other. Thus, Singapore does not have independent policy targets for interest rates and money supply.

b. **Singapore's Exchange Rate-Based Monetary Policy**

Below is an extract from an article by the MAS titled "Singapore's Exchange Rate-based Monetary Policy":

"Since 1981, monetary policy in Singapore has been centred on the management of the exchange rate. The primary objective has been to promote price stability as a sound basis for sustainable economic growth. The exchange rate represents an ideal intermediate target of monetary policy in the context of the small and open Singapore economy. It is relatively controllable through direct interventions in the foreign exchange markets and bears a stable and predictable relationship with the price stability as the final target of policy over the medium-term. There are several key features of the exchange rate system in Singapore.

First, the Singapore dollar is managed against a basket of currencies of our major trading partners and competitors. The various currencies are assigned weights in accordance with the importance of the country to Singapore's trading relations with the rest of the world. The composition of the basket is revised periodically to take into account changes in trade patterns.

Second, MAS operates a managed float regime for the Singapore dollar. The trade-weighted exchange rate is allowed to fluctuate within a policy band, the level and direction of which is announced semi-annually to the market. The band provides a mechanism to accommodate short-term fluctuations in the foreign exchange markets and flexibility in managing the exchange rate.

Third, the exchange rate policy band is periodically reviewed to ensure that it remains consistent with underlying fundamentals of the economy. It is important to continually assess the path of the exchange rate in order to avoid a misalignment in the currency value. The length of the policy review cycle is typically six months. A Monetary Policy Statement (MPS) is released after each review, providing information on the recent movements of the exchange rate and explaining the stance of exchange rate policy going forward. An accompanying report, the Macroeconomic Review, provides detailed information on the assessment of macroeconomic developments and trends in the Singapore economy, and is aimed at enhancing market and public understanding of the monetary policy stance.

Fourth, the choice of the exchange rate as the intermediate target of monetary policy implies that MAS gives up control over domestic interest rates (and money supply). In the context of free capital movements, interest rates in Singapore are largely determined by foreign interest rates and investor expectations of the future movements in the Singapore dollar. Domestic interest rates have typically been below US interest rates and reflect market expectations of a trend appreciation of the Singapore dollar over time.

The exchange rate has emerged as an effective anti-inflation tool for the Singapore economy. Over the past twenty years or so since the exchange rate framework has been in place, domestic inflation has been relatively low, averaging 1.9% per annum from 1981 to 2010. As a result of the long record of low inflation, expectations of price stability in Singapore have become more entrenched over the years. The exchange rate system has also helped to mitigate the adverse effects of short-term volatility on the real economy, while at the same time ensuring that the exchange rate remains aligned with economic conditions and fundamentals. The success of the system owes much to the strong economic fundamentals of Singapore. These include prudent fiscal policy, flexible product and factor markets, sound financial system, and robust domestic corporate sector."

c. **Money Supply, Credit Creation and the Changes to Required Reserves**

Traditionally, money supply can also increase from credit creation as a result of the lowering of reserve requirements as seen in Table 3.2.4(a). A bank takes in a new deposit and lends out 90% of it, keeping 10% as statutory reserves. This process makes its rounds through other banks in the system, leading to a 10-fold increase in money supply.

Table 3.2.4(a) – Increasing Money Supply from Credit Creation

Bank Transactions	New Deposit	New Investments	New Reserves
1st round	10,000	9,000	1,000
2nd round	9,000	8,100	900
3rd round	8,100	7,290	810
4th round	7,290	6,561	729
…. subsequent rounds	…..	…..	…..
After all rounds	100,000	90,000	10,000

When the economy is getting overheated, the central bank typically increases the required reserves that banks must hold, indirectly reducing the money supply and liquidity in the system. Likewise, lowering the amount of required reserves will expand the money supply and multiply the effects of lending through the system.

One country that is actively applying this mechanism is China. For example, when the US-China tariff trade-wars were looming and there was an expected slowdown in the China economy, the Chinese central bank lowered the Reserve Requirement Ratio for banks three times in the first half of 2018 to increase liquidity and lending in the China domestic market.

d. **Quantity Theory of Money**

When money supply increases, one of the greatest fears of central banks is that the excessive supply of liquidity in the system will lead to asset inflation. The quantitative theory of money states that there is a direct proportional relationship between the amount of money in the economy and the price of goods and services. Simply stated – too much money in the system causes inflation.

The following formula, known as Fisher's "equation of exchange", shows the relationship between money supply and price levels:

$$M \times V = P \times Q$$

where:

M = Stock of money (currency outside banks plus demand deposits)

V = Velocity of money circulation (number of times per period that an average unit of money changes hands/accounts)

P = General price level or a price index

Q = The number of transactions made in an economy during a year

The significance of this equation is that when it is fully operative, the central bank will not be able to increase output by increasing money supply because, as money M increases, the price level P will increase (more money chasing goods, causing inflation), and the velocity of money V (which is socially determined i.e., decided by consumers) will then determine the output Q.

e. **Inflation, Interest Rates and Asset Values**

Interest Rates and Bond Prices - Mathematically, lower interest rates also mean lower discount rates for valuation calculations, giving higher fair value targets. When interest rates rise, the higher discount rates used to discount bond-related cash flows will mean lower intrinsic bond values. Since interest rate inflection points tend to coincide with business cycle peaks, the highest bond yields tend to occur close to the peak of the business cycle at the tail end of a series of interest rate hikes. Conversely, the lowest yields tend to be offered at the bottom of a cycle before economic recovery.

Interest Rates and Stock Prices - The relationship between interest rates and stock prices is more complex. The stock market can and have risen through periods of rising interest rates when the global economies are strong, even with rising inflation. While expectations and announcements of interest rate hikes can bring short-term stock market corrections, the markets can continue upward despite several rate hikes if the economy continues to grow. However, at some point in the cycle, the markets will recognise that further interest rates hikes to control inflation will no longer be accompanied by strong corporate earnings and economic performance. Any further rate hike will then be viewed more as a negative trigger to an overvalued market at that point in time and a deeper correction typically ensues.

Consider the following simplified scenarios:

i. **Positive scenario**: Interest rates rise due to an increase in the rate of inflation, and strong corporate earnings. Cash flows increase because firms are able to pass on the cost increases to consumers in the form of higher prices because employment is high. In this case, stock prices may remain fairly stable, because the increase in required rate of return (k) is offset by the increase in earnings growth (g).

ii. **Slightly negative scenario**: Interest rates rise but there is little change in expected earnings because firms are unable to increase prices to match higher costs. This will result in a fall in stock prices, as k increases but g remains constant.

iii. **Very negative scenario**: Interest rates rise while cash flows decline because the rise in interest rates result in a fall in sales revenue and earnings. Companies unable to pass on the higher interest costs in the market identify other ways to reduce cost, e.g., by cutting wages and labour. Short term lay-offs and retrenchments occur, increasing unemployment, making it even more difficult to grow earnings. Stock prices will decline significantly because k will increase while g declines, causing a large increase in the k-g spread.

Rising interest rates is not negative for the entire market. Certain industries or segments of the economy may have earnings and dividends that react positively to inflation and interest rate hikes. For example, when interest rates rise, during the initial expansion, the banking sector and banking stocks will tend to benefit from their ability to widen the net interest spread (raising lending cost faster than deposit rates) while still enjoying the demand in loan growth, creating net interest income expansion for banks. Hence, banks typically benefit from the gradual increase of interest rates.

f. **Effectiveness of Fiscal and Monetary Policies**

The efficacy of fiscal and monetary policies relies on several factors.

i. When fiscal and monetary policies run in opposite directions, they can cancel each other out. In the US, fiscal policies are under the purview of the US government whereas monetary policies are implemented

by the US Fed . There are times when the fiscal policy priorities, which can be politically motivated, run contrary to the monetary policies managed by the Fed. For example, job creation and employment may be foremost on a government's agenda at a point when maintaining a contractionary money policy to stave off inflation may be more important to the US central bank. Such misalignment could lead to some offsetting effects. However, in most countries, the central bank policies are generally aligned with those of the ruling government.

ii. In some instances, fiscal policies may be less effective. In Asian markets, where there is a culturally high propensity to save (e.g., Japan), the increase in disposable income from fiscal tax breaks may lead to a smaller than expected impact on consumption because of the high savings rate.

iii. In very open markets and economies, where there is excessive dependence on major foreign companies and labour involved in construction, government fiscal stimulus on infrastructure development will lead to some leaks or outflows of funds, which means a lower impact from fiscal stimulus.

iv. Both fiscal and monetary policies need time to take effect. The deeper an economic recession, the longer it may take for the stimulus to flow through the economy and have a positive effect.

v. Finally, the strength and stability of the government plays an important part in determining the effectiveness of the policies. Fiscal policies rely on the ability of the government to continue funding them. Monetary policies depend on the belief in the ability of the government to control money supply. When there is political corruption, social unrest, lack of taxable income, and governments do not appear to have the means to continue spending or effectively control the currency value (e.g., hyperinflation), both fiscal and monetary stimuli may not work.

g. **Negative Interest Rates as a Monetary Policy Tool**

In the aftermath of the Global Financial Crisis, a few central banks (Sweden, Denmark, Switzerland and Japan) adopted a new approach of setting negative rates on commercial banks that hold excess funds as reserve deposits with the central bank. In effect, banks are paying to keep excess funds held on deposits at the central bank.

The aim of the Eurozone-related central banks was to stimulate economic growth and push their economies out of deflation. Exacting negative interest rates on banks was aimed at encouraging banks, instead of retaining substantial levels of cash for a fee, to put money to more productive use by providing more loans and encouraging businesses and individuals to invest or spend.

h. **Money Supply from Investment Flows**

So far, we have discussed money flows driven by the central bank. Money supply can also come from external sources. External money flows can come from flights to quality as investors reallocate funds to markets with stronger currency, better economic outlook, healthier corporate earnings growth and potentially higher investment returns. For example, there was strong demand for the US dollar, as money flowed out of emerging market investments to the US market for most of 2018 due to expectations of economic slowdown amongst emerging markets caused by US trade tariffs.

3.2.5 Business Cycles and Market Movement

Business cycles reflect the rise and fall in the general level of economic activity. While no single business cycle is identical, they do have some common traits. There is a business peak, a slowdown or contraction phase, a recessionary trough followed by a recovery and expansion phase.

Another common feature is that the stock market tends to be predictive of economic conditions. Stock market peaks and troughs have typically preceded turning points of general economic business activity. In other words, stock prices tend to turn ahead of the economy. Before GDP growth numbers reflecting two quarters of contraction (i.e., a technical recession) are announced, the equity markets would have typically corrected ahead of the announcements. Equity market investors would have read the trends from leading indicators to make appropriate adjustments to their investments. While economic conditions are still weak, the market will be poised and waiting for leading investment strategists to declare that the worst is over, so as to participate in an equity rally ahead of reported economic improvements.

Figure 3.2.5 – The Business Cycle and Financial Market

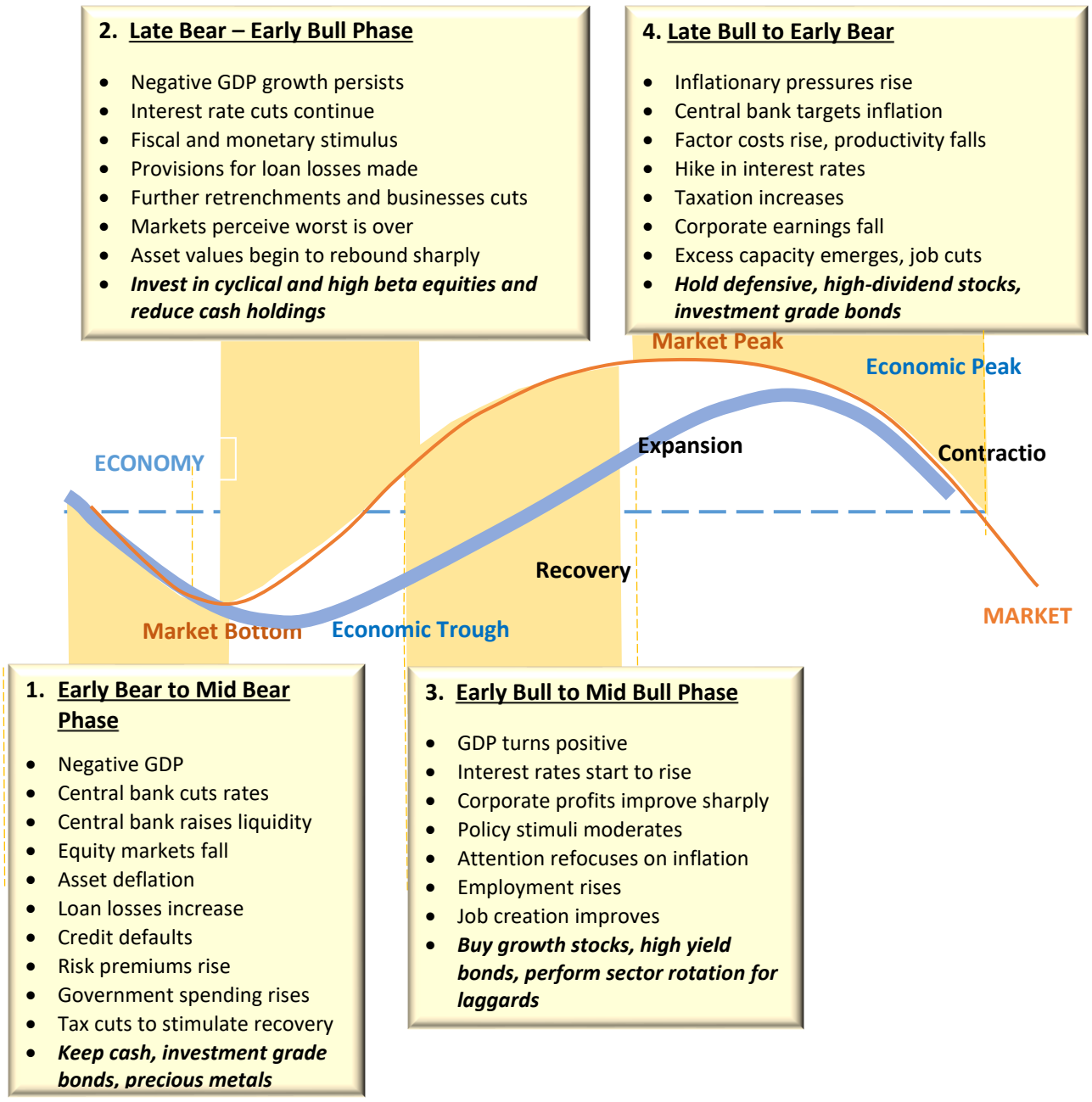


Figure 3.2.5 illustrates how financial markets typically lead the real economy business cycle. Financial assets tend to decline ahead of a slowing economy (as indicated by shrinking/negative GDP growth) and rise ahead of the trough/bottom of a business cycle. Each of the four boxes represents the four phases of the business

cycle and within it are the typical observations and policy actions accompanying each of the phases. The relevant investment strategy for each of the four phases of the business cycle is also highlighted in each box (last bullet point).

Intuitively, the equity market reflects the broad range of companies and industries in the economy. Hence, it follows that the projected forecast of earnings of the companies when aggregated would be representative of the growth and performance of the economy. Given the strong correlation between business cycles, the stock market, and the overall economy, having insight to the various phases of the economy within the business cycle enhances one's ability to forecast major turning points in stock prices.

The bond market tends to be predictive of expected interest rate trends. Hence, with expectations of interest rate hikes, yields will start rising ahead of the actual central bank rate increases. The bond market also reflects the market demand for credit. Typically, when the economy is doing well, there should be credit expansion – expected demand for credit should then be higher in the longer term on top of the liquidity premium so yields would be upward sloping. However, when longer-term credit demand appears weak, long-maturity bond rates may not rise as much, even with Fed fund rate hikes. This results in a flattening of the yield curve, which is considered as one of the early predictors of an upcoming recession.

3.2.6 Economic Calendars

Figure 3.2.6 is an extract of the Singapore economic calendar for September 2023. Similar calendars for other countries documenting key economic data release dates are published by various financial information providers. Investors closely monitor releases of new economic data to evaluate the impact of such data on the financial markets.

Figure 3.2.6 - Singapore Economic Calendar (September 2023)

Wednesday September 06 2023	Actual	Previous	Consensus	Forecast
08:30 AM SG S&P Global PMI AUG	53.6	51.3		51
Thursday September 07 2023	**Actual**	**Previous**	**Consensus**	**Forecast**
05:00 PM SG Foreign Exchange Reserves AUG	S$455.5B	S$452.5B		S$447.0B
Tuesday September 12 2023	**Actual**	**Previous**	**Consensus**	**Forecast**
01:00 PM SG MAS 12-Week Bill Auction	4.05%	4.0%		
01:00 PM SG MAS 4-Week Bill Auction	4.06%	4.04%		
Thursday September 14 2023	**Actual**	**Previous**	**Consensus**	**Forecast**
10:30 AM SG Unemployment Rate Final Q2	1.9%	1.8%		1.9%
01:30 PM SG 6-Month T-Bill Auction	3.73%	3.7%		
Monday September 18 2023	**Actual**	**Previous**	**Consensus**	**Forecast**
08:30 AM SG Non-Oil Exports MoM AUG	-3.8%	-3.5% ®	5.5%	
08:30 AM SG Non-Oil Exports YoY AUG	-20.1%	-20.3% ®	-15.8%	
09:30 AM SG Balance of Trade AUG	$3.584B	$6.509B ®	$5.787B	$7.0B
Tuesday September 19 2023	**Actual**	**Previous**	**Consensus**	**Forecast**
01:00 PM SG MAS 12-Week Bill Auction	4.06%	4.05%		

Source: https://tradingeconomics.com/Singapore/calendar

3.2.7 Economic Indicators

Economic indicators are key statistics that provide cues and signals about the direction of the economy. Analysts interpret them to make forecasts of future performance. The three broad categories of economic indicators most widely used are **leading indicators**, **coincident indicators** and **lagging indicators**.

a. **Leading Indicators**

Leading indicators are the economic data with peaks and troughs that precede the economy and are useful as short-term predictors of the economy. In the US, a popular leading indicator is the **Composite Leading Economic Index (CLEI)** published by The Conference Board[14]. The CLEI comprises 10 indicators used to predict the direction of the economy. Please refer to Table 3.2.7(a). A downward spiralling of the CLEI components typically signals a forthcoming recession whereas an upturn indicates an upcoming expansion.

Composite leading indicators (CLI) have been compiled by entities such as the Organization for Economic Co-operation and Development (OECD) to forecast growth and contraction in an economy.

While such composite indicators are useful as short-term indicators of turning points in the economy, they do not tell investors anything about the magnitude of these shifts.

Table 3.2.7 (a) - Leading Indicators in the Composite Leading Economic Index

No	Leading Indicator	Source and Description
1	Average Weekly Hours (manufacturing)	Monthly average hourly earnings and average work week hours from the US Department of Labor.
2	Unemployment Claims	Weekly report on first-time claims for state unemployment insurance from the US Department of Labor
3	Manufacturers' Shipments, Inventories, and Orders	Monthly preliminary report on Manufacturers' Shipments, Inventories, and Orders from the monthly consumer goods and materials report gathered by the United States Census Bureau.
4	Manufacturers' Shipments, Inventories, and Orders for Non-Defence Capital Goods	Monthly preliminary report on Manufacturers' Shipments, Inventories, and Orders from the monthly non-defence capital goods report gathered by the United States Census Bureau.
5	Housing Starts and Building Permits	Monthly report on housing starts and building permits from the United States Census Bureau.
6	10-year T-bond spread over the Fed Rate	The spread between the interest rate of the 10-year US Treasury Note and the Fed Funds Rate.
7	Adjusted M2 Money Supply	The Fed 's inflation-adjusted measure of M2 Money Supply.

[14] The Conference Board is an objective, non-profit, independent source of economic and business knowledge with one agenda: to help member companies understand and deal with the most critical issues of today. (www.conference-board.org).

No	Leading Indicator	Source and Description
8	ISM Index of Manufacturing (new production orders)	Monthly ISM Index of Manufacturing consisting of supplier deliveries, imports, production, inventories, new orders, new export orders, order backlogs, prices and employment.
9	The S&P 500 Index	Free-float capitalization-weighted index of 500 companies with common stock listed on the New York Stock Exchange (NYSE), the NASDAQ stock exchange, or the Cboe BZX Exchange
10	Consumer Sentiment	The University of Michigan Consumer Sentiment Index, which is a survey of consumer sentiment.

For Singapore, there is likewise a set of CLI with components selected for their leading cyclical properties, economic significance, timeliness, periodicity and data quality.

Singapore's CLI (published by the Singapore Department of Statistics) are:

• Total New Companies Formed	• Purchasing Managers' Index (Manufacturing)
• Money Supply (M2)	• Total Non-Oil Seaborne Cargo Handled
• Stock Exchange of Singapore Indices	• Domestic Liquidity Indicator
• Business Expectations for Wholesale Trade	• Total Non-Oil Retained Imports
• Business Expectations for Stock of Finished Goods (Manufacturing)	

The Purchasing Managers' Index (PMI) is a widely followed leading indicator. The PMI measures the activity level of purchasing managers for the manufacturing, construction and services sectors. In the US, the PMI is compiled by the Institute of Supply Management (ISM) on a monthly basis.

In general, a reading of 50 or higher suggests expansion in the manufacturing sector whilst a reading below 50 indicates a contraction. PMI reading is actively followed particularly for countries like China and US, where there is a large manufacturing base contributing to the growth of the economy. The PMIs of the US and China are particularly important because they are the two largest economies. China is known as the factory of the world. Expansion or contraction in the US or Chinese manufacturing sector will have significant ripple effects on other economies.

In Singapore, the PMI is compiled and published by the Singapore Institute of Purchasing and Material Management. It is a key barometer of the health of the Singapore manufacturing sector.

b. **Coincident Indicators**

Coincident indicators are statistical data with peaks and troughs that generally align with those of business cycles and move in tandem with the overall economy. The coincident indicator changes simultaneously with general economic conditions, reflecting the current state of the economy.

Examples of coincident indicators include sales and personal income which peak or bottom as the economy does, illustrating a direct correlation. In the US, the Coincident Economic Indicators (CEI) comprise 4 economic data series components, as set out in Table 3.2.7(b).

Table 3.2.7(b) - Economic Statistics to the Index of Coincident Economic Indicators

No	Coincident Indicator	Source and Description
1	Nonfarm Payroll Employment	Nonfarm payroll employment which excludes farm workers, private household employees, or non-profit organization employees compiled by the US Bureau of Labor Statistics.
2	Industrial Production Index	Measure of inflation-adjusted production output from the manufacturing, mining and utilities sector
3	Manufacturing and Trade Sales Index	A monthly, inflation-adjusted manufacturing and trade industries sales index constructed by the Federal Reserve Bank of St. Louis.
4	Personal Income and Consumption Expenditure	Calculation and analysis of the change in personal income reported by the Bureau of Economic Analysis, showing disposable personal income (DPI) and personal consumption expenditures (PCE).

In Singapore, the Composite Coincident Index (CCI) is the equivalent of the CEI. The CCI is an aggregation of economic indicators that show coincident relationships with the growth cycles of the economy. Singapore's CCI, as published by the Singapore Department of Statistics, are:

- Gross Domestic Product (GDP) at Constant Prices;
- Index of Industrial Production
- Non-Oil Domestic Exports at Constant Prices;
- Total Employment; and
- Retail Sales Index at Constant Prices, excluding motor vehicles.

c. **Lagging Indicators**

Lagging indicators are statistics that experience their peaks and troughs after those of the aggregate economy. They tend to change only after the economy has begun to form a pattern and are useful for confirming turning points in the business cycle. Typically, these indicators lag the economy by a few quarters.

The index of Lagging Indicators is also published monthly by The Conference Board and comprises 7 components:

- Average Duration of Employment;
- The Manufacturing and Trade Inventory to Sales Ratio;
- The Change in Labour Cost per Unit of Manufacturing Output ;
- Average Prime Rate Charged by Banks;
- The Value of Outstanding Commercial and Industrial Loans;
- The Consumer Credit Outstanding to Personal Income Ratio; and
- The Change in the Consumer Price Index for Services.

3.2.8 Other Economic Factors Affecting Asset Values and Market Movements

In addition to the various indicators discussed above, market observers also assess the impact of several other economic indicators on the financial markets. They include (but are not limited to):

a. **Foreign Currency Reserves and Balance of Payments Deficit** - Countries with a positive balance of payments (BoP) from year to year will generate a BoP surplus. Whether this comes from the trade account (exports exceed imports) or the capital account (capital inflows exceed capital outflows), there will be an increase in foreign exchange reserves. A country with a widening trade deficit can become the target of speculative attacks because this deficit makes the country a "debtor nation". It is generally believed that such weaknesses led to the Asian Currency Crisis in 1997, when many of the then-weaker Asian currencies came under speculative attack, plummeting sharply, in some cases by more than 50%.

   Singapore and Hong Kong were able to defend their currencies during the crisis. Singapore has one of the largest foreign reserves in the world totalling USD337 billion as at July 2023[15], ranking 11st globally. Countries with no natural resources such as Singapore should maintain high levels of foreign reserves to help it weather through financial storms and global economic downturns.

   A country's foreign currency exchange rate also has a significant impact on asset values. If the market expects the country's currency value to fall sharply, all securities denominated in that particular currency will be affected. Such a sentiment will result in an outflow from investments in financial products denominated in the specific currency.

b. **Inflation and the Consumer Price Index (CPI)** – The CPI measures the price level of goods and services purchased by households as an indication of inflation. Most central banks are wary of inflation as it is difficult to control once it gathers momentum.

   The US Fed uses Core Personal Consumption Expenditure (Core PCE) data or core inflation as its primary measure and has traditionally set 2% as the long-run inflation rate. The Fed has indicated that it will attempt to stave off inflation by raising interest rates when there are signs of inflationary pressure. However, 2017 was different because the Fed decided to raise interest rates even though Core PCE remained below 2%.

   Inflation is a major concern because:

   i. it erodes purchasing power;

   ii. it pressures consumers to buy ahead for fear of escalating prices (especially for big-ticket items like housing);

   iii. it encourages businesses to build in buffer pricing thereby aggravating the expected price increase;

   iv. it forces central banks to raise borrowing costs to try to control inflation; and

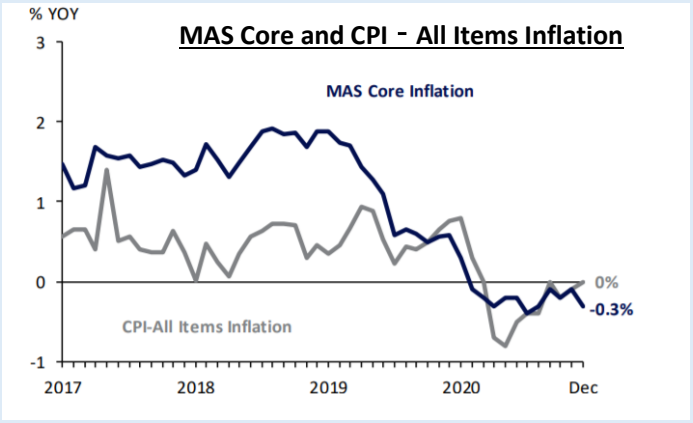
   v. rising costs affect the quality of life for majority of the population and can create social unrest.

In Singapore, the Department of Statistics rebased the CPI to 2019 in February 2020. For 2020 as a whole, both MAS Core Inflation and CPI - All Items inflation came in at negative 0.2%, lower than their respective outturns of 1.0% and 0.6% in 2019. Please refer to Figure 3.2.8(a) which sets out the Singapore historical CPI data for 2017-2020.

Despite low reported inflation rates, the Economist Intelligence Unit (EIU) ranked Singapore as the costliest city in the world from 2014-2018. Whilst the 2020 EIU ranking placed Singapore at fourth place, it is still one of the most expensive cities in the world. One reason for this inconsistency lies in the strong Singapore Dollar, which has kept costs lower for the local population compared to the expatriate population. Another reason is the difference in the basket of goods used in Singapore's CPI measurement compared to EIU's measurement. For example, the EIU includes the cost of buying a car, which is not part of Singapore's measurement of what the average consumer using public transport would need.

[15] Refer to https://www.mas.gov.sg/Statistics/Reserve-Statistics/Official-Foreign-Reserves.aspx.

Figure 3.2.8(a) - Singapore Historical CPI Data (2017-2020)



Composition (2019)	Percent
Food	21%
Clothing & Footwear	2%
Housing & Utilities	25%
Household Durables & Services	5%
Health Care	7%
Transport	17%
Communication	4%
Recreation & Culture	8%
Education	7%
Miscellaneous Goods & Services	5%
All	**100%**

Source: Department of Statistics, Singapore - https://www.mas.gov.sg/-/media/MAS/EPG/CPD/2020/Inflation202012.pdf and https://www.singstat.gov.sg/find-data/search-by-theme/economy/prices-and-price-indices/related-info.

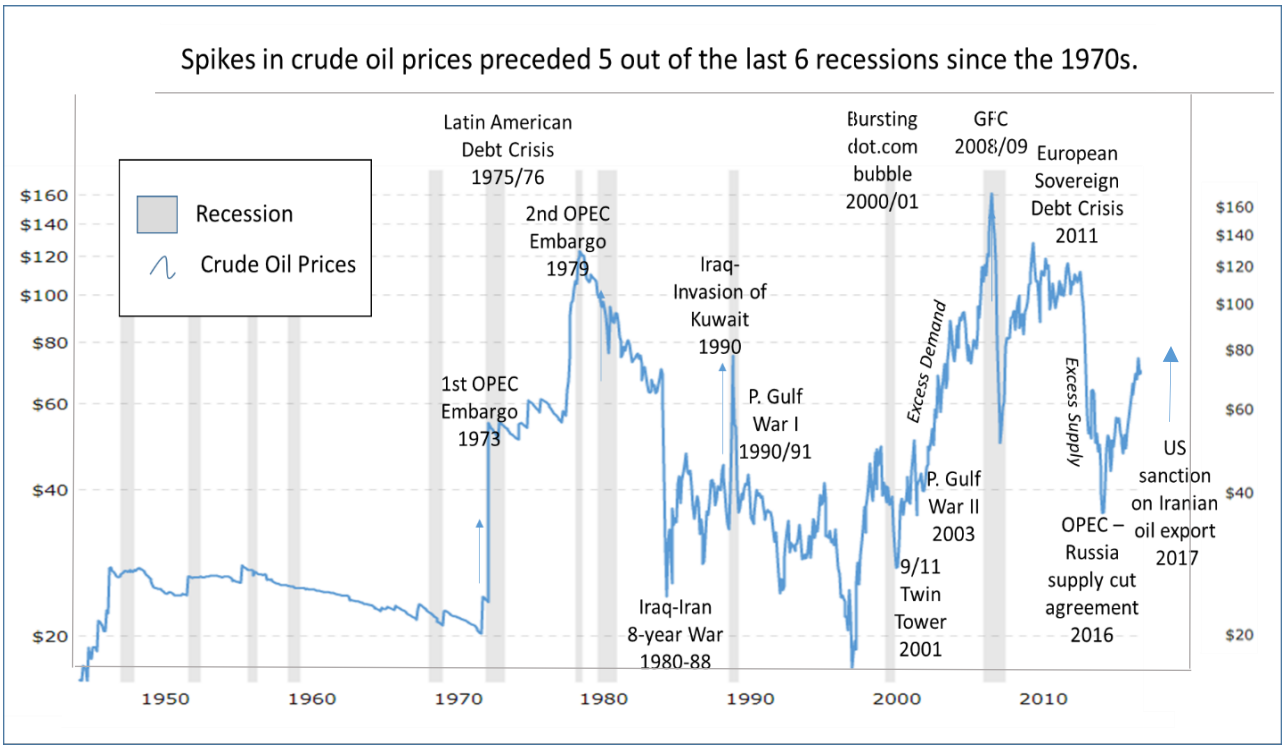
a. **Persistent Deflation** – Deflation is the fall in the real value of assets (which differs from a reduction of inflation or disinflation). Falling prices shift spending into the future and this can trigger a fall in consumption and investments and create an unwarranted economic contraction and negative economic growth.

Persistent deflation occurred after the Japanese stock and real estate markets imploded in 1989, causing Japan's central bank to defer part of the huge corporate losses by allowing outstanding debts to be restructured into low-interest, longer-term government bonds – effectively a bailout. These bonds were, in turn, purchased by local corporates, banks, public entities, and the Japanese people. In deferring part of the pain from the crash, the Japanese economy had to endure 12 years of deflation, low and negative GDP growth (1989-2014), and anaemic consumption. The efforts of the European and US central banks are aimed at avoiding a similar fate of long-term deflation.

b. **Factor Cost Changes** - Crude oil is a key factor cost that has made a dramatic impact on global economic growth over the last century. Before fracking technology and the development of energy alternatives, the supply of crude oil was easily affected by output from the Organization of the Petroleum Exporting Countries (OPEC). Interestingly, five out of the last six US recessions since the 1970s (1973, 1975, 1979, 1980/81, 1990, 2008/09) were preceded by spikes in crude oil prices. Please refer to Figure 3.2.8(b).

From 2015-2017, excess capacity and supply from new oil producers reversed the situation and crude oil prices plunged to USD28 per barrel at one point. The lower crude price reduced factor costs and inflation globally. While lower crude oil prices may have a negative impact for oil-producing countries and oil companies, the rest of the world benefitted from cost savings, which supported the global economic recovery. US oil and refining companies have emerged to become a major producer and dominant player. Amidst agreement from the OPEC member nations and Russia to cap output in 2017, the excess supply was eventually absorbed, and prices began to stabilise in 2017-2018.

Figure 3.2.8(b) – Crude Oil Prices and Recession (1950 -2018)



Sources: Adapted from Macrotrends – Crude Oil Price History Chart; http://www.macrotrends.net/1369/crude-oil-price-history-chart
Research: http://econweb.ucsd.edu/~jhamilton/oil_history.pdf.

Other major factor costs include rent and labour. High land prices, rental and labour costs can significantly raise the cost of doing business, stifle new business creation, and skew the composition of viable industries - all of which will have a negative long-run impact on economic growth.

3.2.9 Political Factors Affecting Markets and Asset Values

Country and Political Risk – To assess country risk, markets look beyond monetary and fiscal policy actions to examine the legal framework, foreign exchange reserves, political stability and credibility of countries.

Dedicated research organizations like BMI Group, Economist Intelligence Unit, and Country Risk Solutions have their in-house assessment criteria for assessing country and political risks. Other organizations such as Euromoney provide reports for surveys conducted to gauge the market's view of country risk. Countries with stable governments that have a good track record and consistency in driving their economic agenda and national initiatives, maintaining social peace, and abiding by proper legal and accounting framework in doing business are ranked highly. Such countries attract investors due to less return uncertainty. Asset valuations will be lower for countries with higher risks as higher yields will be expected to compensate for the higher country risks borne by investors.

In the sphere of credit analysis, Moody's, Standard & Poor's, and Fitch Ratings are renowned rating agencies for sovereign debt credit ratings. The topic on credit analysis is covered in Chapter 7 – Deposits and Fixed Income Securities.

Political Shocks and Uncertainty - Brexit was a major surprise in 2016 that caused massive turmoil for investors in Great Britain. The surprise 51.9% affirmation vote for Britain to leave the European Union resulted in panic selling of the GBP in June 2016. It fell 10.5% in 4 days and continued to drift downwards to a low of

GBP/USD 1.21714 (down 18%) by October after the referendum because the investment and business community regarded the exit decision to be negative for Great Britain. Likewise, during the night of the US Elections, when it became clear that the unexpected win of Republican President Trump was imminent, the Dow Jones Industry Average Index (DJIA) plunged 800 points. However, markets recovered soon after when the President highlighted that his plans included fiscal stimuli such as tax cuts and infrastructure spending.

Continuity and Political Will - Regardless of the type of government, be it Democratic, Socialist or Communist, investors desire strong and capable country leaders that can uphold their legal framework and help drive the economy for the good of their people, businesses, and industries. When there is uncertainty with regard to succession and continuity, or a lack of political will to follow through, risk rises and there tends to be investment outflows.

Sectors and Companies that Fall In and Out of Favour - When new governments come into power, their political and social agenda will benefit some companies while threatening others. For example, a government with a socially conscious agenda may seek cost reductions in healthcare and curb excessive prices for drug remedies. Governments who are pro-defence spending will raise investor interest in the defence industry. Governments that receive huge political support from oil-rich or coal-producing states may be less inclined towards developing renewable energy as compared to governments that seek energy self-sufficiency. Some governments will welcome the gaming industry for the taxable income while others will seek to curb it for its social ills. Thus, changes in government can lead to changes in the fortunes of businesses.

3.2.10 Industry Analysis

Some factors affect certain industries more favourably than others. At times, these industry-related factors have a greater influence than economic factors and can make certain industries more resilient to economic or financial crisis, which is usually reflected in share prices. On the other hand, other factors (e.g., structural changes and disruptive technology) may cause the rapid decline of a particular industry.

In this section, three approaches to industry analysis will be considered. First, the industry life cycle and the potential risk and return at each stage of the cycle are examined. Next, Michael Porter's five competitive forces, which determine the profitability of an industry, are discussed. Lastly, business cycle analysis, which explains the performance of companies in relation to the economy, is covered.

a. **Importance of Industry Analysis**

Industry analysis is important for the following reasons:

i. The risks and rates of return among different industries vary widely during a given time period because of the different mix of products and services and the different business cycle sensitivities.

ii. Industries display different rates of return over time, so past industry performance cannot simply be extrapolated into the future.

b. **Industry Life Cycle**

A useful framework for analysing industry trends is the industry life cycle concept. The purpose is to evaluate the general health and current stage of the industry.

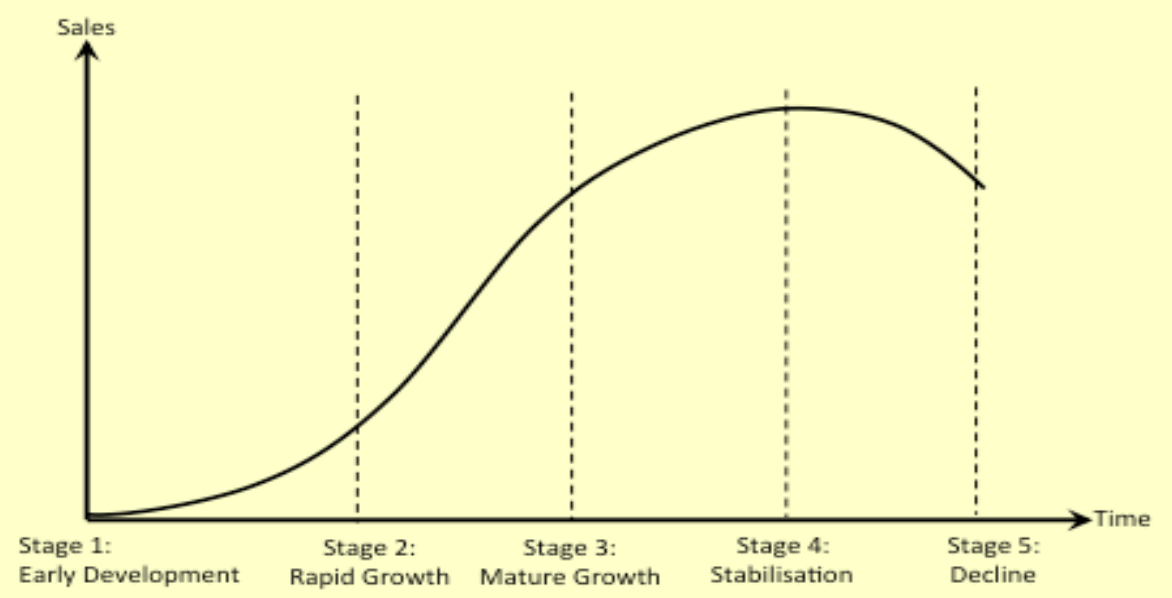
Over time, the development of an industry can be divided into five stages:

i. Early development;
ii. Rapid expansion;

iii. Mature growth;
iv. Stabilisation and market maturity; and
v. Deceleration of growth and decline.

Figure 3.2.10(a) shows the growth path of sales during each stage of the life cycle. Besides sales estimates, profit margins and earnings growth can also be analysed using the industry life cycle framework.

Figure 3.2.10 (a) – Industry Life Cycle



i. **Early Development -** This is also referred to as the pioneering or start-up stage. It is often characterised by the introduction of a new technology and a new product that did not previously exist. For example, currently, we have the advent of renewable energy (e.g., wind and solar technology) as well as the launch of battery-driven electric cars. Companies involved face high start-up research and development costs and will struggle to find ways to make their products commercially viable. Sales, if any, tends to be modest and initial profit margins are either very small or negative. The demand for the industry's product or service is low.

   For example, Tesla Inc., the maker of electric cars, reported 10 years of losses leading up to 2018. Many pharmaceutical and biotech companies have no sales, with high R&D costs burning through the equity capital in this phase. Many start-up firms may be attracted to the new industry, but few survive this stage as competitive pressures build up and losses are incurred due to high start-up costs and low levels of initial sales.

   Investors' risk in an unproven company is high but so are expected returns should the company succeed. At this stage, it may be difficult, albeit very rewarding, to identify who the likely "winners" are.

ii. **Rapid Growth -** In the second stage of the life cycle, as awareness of the product or service grows, a market develops, and demand picks up significantly. The industry expands its capacity to meet rising demand. The limited number of firms who had survived the early development stage will face less competition and individual firms can experience substantial backlogs. They typically become the industry leaders. Rising sales growth, coupled with greater efficiency, result in accelerating profit growth. During this phase, profits can more than double from the low earnings base. The profit margins are also very high.

Examples of rapid revenue and profit growth include internet marketing companies Amazon (2014-2018) and Alibaba (2011-2018), social media giant Facebook (2010-2017), and Google from (2002-2012). Earnings for such companies jump in multiples initially and continue to grow for several years at a compounded pace (e.g., 30-50%) well above the average corporate earnings growth. Other examples of industries in this stage of the life cycle are industrial robotics in China, where only 20% of industries currently use any form of industrial robots. As such, the growth potential has been described as phenomenal. However, the rapid pace of growth will naturally slow as firm size increases, with revenues and profits.

iii. **Mature Growth -** After the explosive growth in the previous stage, the industry enters the mature growth stage. Demand for the industry product or service is largely satisfied. Sales growth stays above average but no longer accelerates. The larger the revenues and profits, the more likely the pace of growth slows as the base expands each year. In this phase, industry sales can still run between 12-20%. Performance of the companies in the industry will begin to vary more significantly from year to year and in comparison, with each other because of differences in business models. Attracted by the rapid growth of sales and high historic profit margins, new competition will also emerge, resulting in profit margins declining. Established companies with excess cash may also venture into the profitable space, adding to the competition. Companies that have dominant market share and an established brand will be able to continue growing.

iv. **Stabilisation -** Probably the longest phase of the life cycle, the industry growth rate tends to be in line with the growth rate of the overall economy. Profit growth of individual companies may vary, depending on the ability of management to control costs. Products become more standardised and less innovative. The marketplace is full of competitors, resulting in tight profit margins and rates of return on capital eventually becoming equal to or slightly below the competitive level.

v. **Decline -** In the last phase of the life cycle, the industry's sales growth declines because of shifts in demand or emergence of substitutes. The growth of the industry is unable to keep pace with economic growth. Profit margins continue to be squeezed, and some firms experience low profits or even losses. Firms that remain profitable may show very low rates of return on capital and investors begin to consider alternative uses for the capital tied up in this industry.

   An example of an industry that is in the decline stage is the music record industry, which retails audio CDs or tapes. With the advent of YouTube and music streaming over the internet, sales in the traditional music record industry have almost come to a halt, replaced by the likes of the iTunes store and other on-line sales distributors.

An understanding of the growth pattern of an industry and of the various stages of growth helps investors to assess the growth potential of companies in the industry. While the average investor may want to consider industries that are in the rapid growth or mature growth stages of the industry life cycle, aggressive investors may consider industries in the early development stage, which offer the highest potential returns but also the greatest risks. Industries that are in the decline stage should generally be avoided. Comparing an industry's sales to that of firms in the overall economy should help investors identify the industry's stage within the industry life cycle.

c. **Competitive Forces and Industry Structure**

Another framework for sector/industry analysis is the model proposed by Michael Porter, who believes that the competitive environment of an industry and the intensity of competition among the firms in that industry determine a firm's ability to sustain above-average rates of return on invested capital.

According to Porter, there are 5 basic competitive forces that determine industry profitability:

i. Rivalry among existing competitors;
ii. Threat of new entrants;
iii. Threat of substitute products;
iv. Bargaining power of buyers; and
v. Bargaining power of suppliers

In industries where the 5 forces are favourable, many companies earn attractive returns. In industries where pressure from one or more of the forces is intense, it is more difficult to command high returns despite management efforts.

i. **Rivalry among the Existing Competitors -** The intensity of rivalry influences prices as well as the costs of competing in areas such as product development, advertising, and sales force. Rivalry is more intense when there are many firms of relatively equal size competing in the industry. Both local and foreign competitors must be considered. Slow growth causes competitors to fight for market share. The desire to sell at full capacity to cover fixed costs results in price cutting and greater competition in some industries. Exit barriers, such as specialized facilities or labour agreements, can keep firms in the industry despite having below average or negative rates of return.

ii. **Threat of New Entrants -** The threat of entry affects pricing power and shapes the investment required to deter entrants. Although the number of existing competitors in the industry may be few, one must assess the possibility of new entrants and increasing competition. High barriers to entry include low current prices relative to costs, large capital requirements, substantial economies of scale or the need for extensive distribution channels to compete. Similarly, high costs of switching products or brands, such as costs incurred to change a computer or telephone system, keep competition low. In addition, government policy can limit the number of competitors by imposing licensing requirements or limiting access to materials. Without some of these barriers, it might be very easy for competitors to enter an industry, increasing the competition and potentially driving down rates of return.

iii. **Threat of Substitute Products -** Substitute products limit the profit potential of an industry because they limit the prices that firms can charge. Although almost everything has a substitute, one must determine how close the substitute is in price and function to the product in the industry. With rapid technological progress, many businesses and products and even industries will become obsolete and replaced by new products. For example, the emergence of digital photography and imaging resulted in the demise of camera film, bankrupting film-producing giant, Kodak.

iv. **Bargaining Power of Buyers -** Buyers can influence the profitability of an industry because, by bargaining among competitors, they can bid down prices or demand higher quality or more services. Buyers become powerful when they purchase large volumes relative to the sales of a supplier. The most vulnerable firm is one that supplies a single large customer, as is common for auto parts manufacturer suppliers or software developers. Buyers will be more sensitive if the cost of an item represents a significant percentage of their firm's total costs, or if they are feeling cost pressures from their own customers. Also, buyers with intimate knowledge of pricing of supplier costs will bargain more intensely, such as when the buying firm partially supplies its own needs. Examples of buyers with strong bargaining power are the US government (for defence contractors and manufacturers of military aircrafts) and Apple (as the key customer for many of its contractors).

v. **Bargaining Power of Suppliers -** Suppliers can alter future industry returns if they increase prices or reduce the quality or quantity of services that they provide. Suppliers are more powerful if there are fewer of them and if they supply critical inputs for which few, if any, substitutes exist. If so, we essentially have a sellers' market, in which a few sellers have the power to dictate their terms of sales and service to the buyers. Workers that have the support of strong labour unions have greater

bargaining power. Also, monopolistic and oligopolistic suppliers generally have strong bargaining power. For example, utilities producers in some countries have great bargaining power over consumers who need the electricity that they provide, and pharmaceutical and biotech companies often have stronger bargaining power due to the specialty nature of the drugs that they supply.

By examining these competitive forces, the intensity of competition and the long-run profit potential of an industry can be assessed. As the competitive structure of an industry is not static, regular review of an industry's competitive environment is necessary.

d. **Business Cycle Analysis**

Another way to analyse industries is to look at their performance in relation to the overall economy. The earnings performances of many companies tend to move in line with business cycles, and some industries are able to cope with the recession reasonably well, while others perform badly during a recession. Investors have to pay attention to these relationships.

The sales and earnings of companies in **cyclical industries** are greatly affected by the business environment. These companies do exceptionally well during economic expansions and rather badly during economic contractions. Durable goods are cyclical in nature. When times are good, demand for durable goods such as cars and refrigerators will rise. During a recession, purchases of durable goods may be postponed as consumers continue to use existing goods.

Least affected by recession and economic adversity are the **defensive industries**. The food and public utilities industries are considered to be defensive. People need food for subsistence regardless of the health of the economy. Even so, there are certain segments of the food industry that are not necessarily defensive, e.g., specialist coffee chains in Singapore.

Many investors seek **growth industries**, where earnings are expected to be significantly above the average of all industries, and such growth may occur notwithstanding setbacks in the economy. Cell phones have been a growth industry in the past, as have audio CDs and computers. Current growth industries include mobile devices game applications and music purchased via internet stores. Not surprisingly, a primary goal of security analysis is to identify growth industries.

Interest-sensitive industries are sensitive to changes in interest rate expectations. Financial services and real estate are examples of interest-sensitive industries.

What are the implications of these classifications to investors?

To predict the performance of an industry, investors should carefully analyse the stage of the business cycle and its likely responses to macroeconomic changes, which may be as diverse as the balance of payments, the trade account, inflation and unemployment, just to name a few. If the economy is heading into a recession, cyclical industries are likely to be affected more than other industries, whereas defensive industries are least likely to be affected. The outcome of industry analysis will enable the investor to select the industry that better fits his risk and return requirements.

3.2.11 Firm-Specific Factors and Analysis

Just as industry factors affect the performance of firms across industries, unique characteristics of firms even within a particular industry affect their attractiveness to investors.

By applying a bottom-up approach to stock-picking, firms with specific competitive advantages distinguish themselves from others. For instance, what makes Alibaba and Tencent dominant compared to other internet

sales platforms in China? Why did Nvidia Corporation trade at a significantly higher valuation in 2017 compared to other semi-conductor chip makers in the US? The following are some examples of firm-specific factors that have allowed firms to stay ahead of the competition.

Firm-Specific Factors			
1	Dominance and control of technology/research	5	Cost efficiency, lowest cost producer
2	Control over resource supply	6	Successful focus on target niche clients
3	Superior brand differentiation	7	Established diversified core business
4	Dominance in sales distribution/marketing	8	Superior leadership, management and talent retention

There will be further discussion of firm-specific factors alongside financial statements analysis in Chapter 5. Firm-specific factors are critical for making precise sales and earnings forecast, which increases accuracy in the valuation of share prices.

3.3 Market-Driven Factors and Asset Values

Market-driven factors refer to physical market conditions that affect short-term demand and supply and, consequently, the pricing of investment products.

3.3.1 Liquidity and Money Flow

Excessive Liquidity - Typically, during periods of uncertainty, investors will sell off investments and keep more cash. During such periods, interest rates are also usually low as the central bank attempts to stimulate the economy. At the onset of a market rally, aggressive investors and money managers will commence investing. The market will continue to rise as long as there is sufficient liquidity and money flows to keep driving the market upward.

Liquidity Shortage - At some point in time, the bulk of funds is almost fully invested in the market. When there are insufficient funds to sustain the upward drive, the market will start to plateau, with funds rotating to industry sectors and assets that have been lagging in relative performance. Typically, this is when parts of the market become overvalued. With the lack of fresh funds to sustain a bull market, the market is then poised for a factor to trigger correction and profit-taking.

Some analysts track the flow of funds, the excess cash balances of mutual funds, and the deposit levels in banks to predict the amount of funds available for investing. When there is a build-up of liquidity and deposit rates are low, money will eventually flow back into the stock market as investors seek higher returns, so there is more upside to the stock market. When the available or excess funds dwindle, investors are increasingly invested and there is less liquidity to push the market up further, which would be a signal to sell.

3.3.2 New Securities Issues

Excessive Equities Issued - A significant supply of securities in the market can lead to a drain of funds for a particular period. For example, if there are several major initial public offerings (IPO) of securities in the same

market during a particular period, funds will be absorbed by IPOs (which, in some cases, may be deliberately priced at a discount to ensure demand). This flow of monies to the IPOs can come at the expense of similar stocks already trading in the market. Some investors will look to take profit on existing stocks to participate in the IPOs for potential capital gains. Thus, the number of IPOs per year can have an impact on demand for other similar securities. Additionally, when there are too many IPOs in a year, demand for some IPOs might be affected if funds are already committed for other earlier issues.

Excessive Treasury Bills/Bonds (T-bills) Issued - The same phenomena can happen in the fixed-income market. Excessive issuances of government T-bills and bonds of a particular maturity can lead to an oversupply situation for that maturity tranche. As a result, bond prices may briefly dip slightly to provide better yields for that particular maturity tranche. The same situation can happen if there is a sudden reduction of interest by regular buyers of US Treasuries. If China and/or Japan were to decide to buy fewer US Treasury bonds, this can suddenly cause an excess supply situation.

3.3.3 Trading Influence from Derivatives and Other Asset Classes

a. **Influence from Structured Products**

The general belief is that markets are usually quite efficient and difficult to influence by any one investor or speculator. However, with the creation of structured products, there may be a collective incentive for proprietary traders of financial institutions to move the price of certain stocks to reach strike price targets for stocks or indices. This is because these financial institutions are usually the counterparty of the structured products and the profitability of their positions depend on the stock and index prices moving in a favourable direction.

On days of thin trading or during specific time periods of the day, there may be attempts by certain parties to push prices in a specific direction. When more than one financial institution is selling similar structured products with relatively similar target price ranges, the collective ability to move the underlying price of the securities and indices can be quite significant.

While proving collusion and the illegality of such actions is difficult, such actions are nonetheless unethical because the financial institution is profiting from actively working against investors to whom it has sold an investment product. Some institutions have argued that their proprietary trading desks have no knowledge of the position of structured products held by other investors, and they receive no instruction to act in one way or the other, but it should be clear that financial institutions and investment professionals should not tolerate practices that profit at the expense of clients.

b. **Influence from Futures and Options**

When the markets are bearish, the sale of futures contracts and put options can add significant pressure on the underlying assets e.g., the equities or bonds. Derivatives, such as futures and options, are leveraged financial instruments. They are traded on margins i.e.; an investor need only put forth a small fraction of capital to own the full notional amount of the underlying asset. Because of the leverage, returns to a derivative investor are magnified vis-à-vis the underlying asset returns. As the underlying assets fall in value, they in turn cause further reduction to futures and options prices, with the losses magnified. Derivatives investors and traders may be forced to reduce their holdings to meet margin requirements thereby further driving down the market. This leads to the risk of the derivative and underlying asset markets feeding off each other in a downward spiral. The reverse can also occur but is less worrying, at least initially since the markets will be rising.

c. **Major Common Expiration Dates for Futures and Options**

There are a large number of financial instruments that mature and have to be rolled over towards the end of quarterly periods. As such, volume and volatility can pick up towards the last two weeks of March, June, September and December. In the case of the US market, on the third Friday of every March, June, September and December, the last hour of the stock market trading session (3.00 - 4.00pm New York local time) is called the **triple witching hour.** Three kinds of securities mature on this day, namely, stock market index futures, stock market index options, and stock options. During this last hour, the expirations generally increase the trading volume of options, futures and the underlying stocks and, occasionally, the price volatility of related securities as well.

3.3.4 Other Trade Flows

Hot Money - Global money from hedge and other large funds that seek to exploit temporary inefficiencies and arbitrage opportunities may flow into a market, a sector or a particular set of stocks and temporarily push up volume, interest and prices. In some other cases, such hot money can be orchestrated with the dissemination of additional positive information to generate interest in the asset class, only for the hot money to exit once desired return targets have been achieved. This is essentially the "pump-and-dump" trading activity occasionally seen in stock markets, but with more finesse and sophistication and taken at a global level.

Computer-driven Momentum-based Trading – Computer-driven trading follows a defined set of instructions to make high-speed decisions and transactions in many financial markets (e.g., commodity futures, stocks, currency, and money market) to generate potential profits at a frequency that is impossible for human traders to match. Such trading can be based on algorithms that are predominantly influenced by technical market-related indicators that may dictate continued buying as long as the uptick and specific price and volume trends are in place (regardless of valuation) and continued selling based on downward momentum trends. As a result of the widespread use of such trading strategies, prices of securities and assets have become more volatile.

3.3.5 Market Microstructure Influence on Price and Asset Values

In the short term, prices and values of assets are also influenced by market structure. Market structure reflects the trading and execution processes, rules, liquidity, and strength of participants, all of which can have significant influence over pricing.

a. **Differences in Price Discovery Process**

For some markets, the price discovery process for buyers simply involves looking at the price quote provided by the issuer because the issuer has control over the pricing of the security. When there is relatively high demand for the investment, there is typically no room for negotiation, and the buyer is simply a price taker. However, for some OTC financial instruments, the prices of trades are negotiated over the phone with OTC dealers or market-makers.

In stock markets, the fair market price for securities can be established through a rule-based process with bid and ask price quotes provided continuously throughout the trading day by different parties in the system. Over automated platforms and exchanges, this pricing process first prioritizes matching the highest bid price to the lowest ask price, followed by the timing of the quotes. In continuous markets, buyers and sellers can gauge the trends and sentiment throughout the day and decide if they wish to adjust or close out their positions.

In certain European bourses (e.g., France), a call market approach is used instead. This happens when all the bids are submitted and are batched for simultaneous execution once or twice a day when the market is

"called". Price, but not timing, prioritization is practised. The traders can submit limit and/or market orders but will not be able to ascertain the likelihood of the trade being executed until after the market is "called". The IPO process follows a somewhat similar pricing process where bids are submitted, but there is an added criteria of bid size to determine whether an allocation is made.

Finally, if an auction is called for a pool of stocks held, the bids are all submitted, and the auctioneer will take the highest bids available to match the number of stocks offered. However, this process is rare, even in distress situations. In most distress situations, the creditors or parties taking over will undergo a negotiation process to agree on a price.

Hence, the types of players in each market structure and the specific systems and rules applied will impact the pricing process in any given market.

b. **Impact of Computer-driven, Momentum-based Trading on Pricing Process**

Globally, one of the current and most commonly applied computer-driven, momentum-based trading strategies is **algorithmic trading**. **High frequency trading (HFT),** the primary form of algorithmic trading, is the rapid execution of extremely short duration trades, generating profit by arbitraging small differences in security prices across different exchanges.

The success of HFT strategies is mainly due to the capability to simultaneously process information and execute trades without the time and emotional burden of human decisions and interactions. With new technologies creating capabilities that no human trader can offer, computer-driven trading that is more sophisticated will be developed and assimilated into the global market structure. Though the use of HFT has improved market liquidity and produced higher price efficiency through synchronised statistical arbitrage, it also poses the risks of increased volatility and potential predatory trading at the expense of less sophisticated market participants, destabilising the market.

c. **Inventory Costs and Liquidity**

Some research studies indicate that the bid-ask spread quoted can be influenced by inventory costs as well as the amount of liquidity certain traders or investors possess. Hence, the pricing of securities by a dealer and market-maker while holding an inventory of stocks (and incurring costs) can be different in comparison to the pricing by those who do not hold inventory. Other studies have suggested that parties with greater cash liquidity can have a greater influence over pricing.

d. **Dark Pools and Price Efficiency**

Dark pools are networks such as private financial forums or exchanges for the trading of securities, derivatives, and other financial instruments. The bulk of dark pool trades consist of large trades conducted by financial institutions, outside of public exchanges, maintaining confidentiality as to the trade details. Dark pools are not new as a medium since block trades and private transactions have long existed. However, there has been a growth in the pools of entities facilitating and financing the private trading process, away from the prying eyes of the exchange, until the transactions are completed and brought into the open.

Although it generally assumed that the standard price discovery process ensures that, at any given time, the price of publicly traded assets is relatively ''fair'', in the case of dark pools, internally agreed trade prices, along with the size and lack of transparency of identities of dark pool participants, put less sophisticated market participants at a disadvantage.

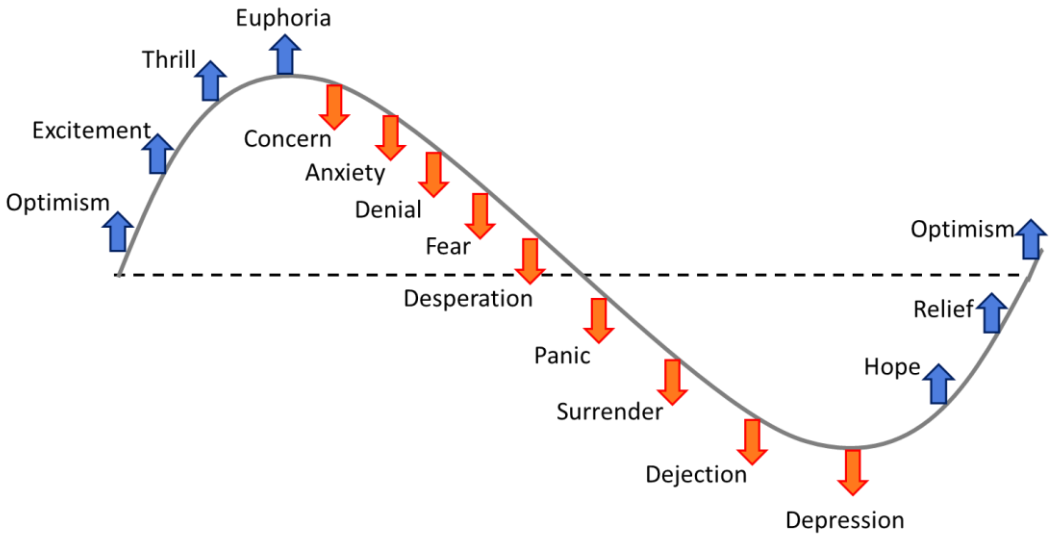
3.4 Effect of Investment Behaviour and Emotions on Asset Values

3.4.1 Herd Instinct – Greed and Fear

In the financial markets, a herd instinct is an emotional phenomenon characterised by the lack of independent decision-making ability for individuals in a crowd that is acting without centralised direction. Such instincts may be attributed to the emotions of greed and fear and can drive asset values way above or below the levels justified by fundamental indicators. Behavioural studies show that the herd instinct kicks in especially when information is scarce, and threats seem imminent.

As the market declines and reaches a trough or turning point, market participants are inclined to begin accumulating investments when they believe that the worst is over. When the trough is confirmed, confidence gradually returns with more market participants and funds returning into the market. Retail investors start to pile back into the market and greed starts taking over as market participants lose sight of valuations and push the market towards new heights. Generally, an investment bubble occurs when the market is driven by investors overlooking the fundamentals and overcome with optimism about asset values, perpetuated by exuberant market behaviour. Please refer to Figure 3.4.1 on the different market emotions cycle.

Figure 3.4.1 - Cycle of Market Emotions



An example of mass euphoria when fundamental considerations were thrown out of the window was the 1997-2000 US Internet Dot.com Bubble. Greed took over as investors bought into overpriced companies (some that are not even generating revenues, much less profits). NASDAQ rose sharply from 1000 to 5000, with a market price-earnings (P/E) ratio of 200x. Then Fed chairman, Alan Greenspan, warned the market of overvaluation by coining a term “Irrational Exuberance” in Dec 1996, but the bubble continued until 2001 when it finally burst and the NASDAQ collapsed by 75%. Under the influence of emotion, markets can become irrational, inefficient and overvalued.

Conversely, the initial disappointment when markets first correct quickly turns into panic when markets start to free fall and gap downwards. Fear of huge losses will cause investors to dump their stocks. As selling picks up, investors disregard valuations as the entire market is falling from panic selling. Somewhere between the point of surrender and depression, some investors will start nibbling on the blue-chip counters which they believe will be among the first to rebound when the market recovers. However, the bulk of the market is still nursing its losses and suffering from depression.

3.4.2 Loss Aversion in Portfolios

Loss aversion is associated with prospect theory[16]. An implication of prospect theory is that individuals prefer less uncertainty when it comes to gains and more uncertainty when it comes to losses. When the market is falling, there may be higher level of fund redemptions and securities sold. Investors and investment managers may be unconsciously influenced by loss aversion to sell off their profitable holdings to lock in positive returns (i.e., lowers uncertainty), while still holding on to their book losses on poorer performing securities in the hope that the securities rebound (i.e., raises uncertainty).

The profitable securities sold may actually have more upside potential but, because of loss aversion, investors have sold these off first and have instead retained the ones making losses that may fall deeper into the red because they are trying to avoid realising losses. Thus, this loss aversion investment behaviour potentially has a double negative impact.

3.4.3 Cognitive Bias

Cognitive bias refers to the tendency to make conclusions based on one's past experiences, preferences or predetermined beliefs and conclusions.

An example of a cognitive bias is the confirmation bias, which refers to how we, as investors, tend to look for views and recommendations that are aligned with our own. For instance, investors who have committed their funds into an investment will be more prone to listen to and agree with research that calls for upward appreciation than research that calls for investors to sell. We then reinforce the view by committing even more funds to the same investment. An investor who is less influenced by such a cognitive bias would actively look out for research opinions that are different from his own and to check if there are any merits to the alternative views.

3.5 Summary

1. There are 3 primary factors affecting market movements and asset values – (i) fundamental factors, (ii) market-driven factors, and (iii) investment behavioural factors.

2. Fundamental analysis revolves around the understanding and analysis of economic, political, industry, and firm-specific factors.

3. Fiscal and monetary policies, business cycles and interest rates form the main areas of interest of study in economic analysis.

4. Expansionary fiscal and monetary policies are implemented by governments and the central bank within a country to stimulate growth in the economy. Contractionary fiscal and monetary policy actions are deployed to control inflation and cool the economy. Expectations and announcements of policy actions can dramatically affect asset values and move markets.

5. Business cycles reflect the general level of economic activity. Given that stock market peaks and troughs have typically preceded turning points of general business activity, there is a strong correlation between business cycles, the stock market, and the overall economy. Hence, having insight to the various phases

[16] According to the prospect theory in finance, investors will prefer certain gains over uncertain gains which are, on average, larger; also, investors will prefer uncertain losses which are, on average, larger over a certain, smaller loss.

of the economy within the business cycle enhances one's ability to forecast major turning points in stock prices.

6. The investment community closely monitors the release of key information to determine if the actual data are in line with market estimates and to determine if forecast revisions are needed. Knowing the dates and times of the releases of key information is useful because the price volatility of affected financial assets tends to increase during the period immediately before and after the information release.

7. Some economic indicators are used to give early warnings of changes in economic activity. The three categories of economic indicators are the leading indicators, the coincident indicators, and the lagging indicators.

8. Other economic indicators that contribute to market movements and changes in asset values include, but are not limited to, foreign currency reserve level, balance of payments, inflation/deflation and changes in factor costs.

9. Generally, market participants favour investing in countries with stable, capable, and honest governments that can help drive the economy for the good of their people, businesses, and industries. Investors should be aware of political developments such as elections, changes in political leadership, dissent in a particular party, and stated party policies and guidelines to mitigate return uncertainty.

10. Industry analysis is important because the risks and rates of return among different industries vary widely. The profitability of industries does not remain constant and, hence, needs to be updated over time. At times, industry-specific factors are more dominant than economic factors in terms of the impact on the prospect of companies and their securities.

11. A useful framework for analysing industries is to examine where they are in the industry life cycle. The development of an industry can be divided into five stages: early development, rapid expansion, mature growth, stabilisation and decline.

12. Another framework for industry analysis is to consider Porter's 5 competitive forces that determine the intensity of competition within an industry which, in turn, affects industry profitability. The five competitive forces are rivalry among existing competitors, threat of new entrants, threat of substitute products, bargaining power of buyers, and bargaining power of suppliers.

13. The business cycle of the industry should also be looked at, by evaluating the earnings performance of companies within the industry in relation to that for the overall economy.

14. Firm-specific factors are those that uniquely distinguish one firm within a particular industry from another. Analysis of firm-specific factors provide market participants more accurate earnings and cash flows forecasts for asset valuation purposes.

15. Market-driven factors refer to physical market conditions that affect short-term demand and supply and, consequently, the pricing of investment products.

16. Major market-driven factors include liquidity flows, new issues of securities, trading influences arising from derivatives and other asset classes, hot-money flows, computer-driven and momentum-based trading. These influences must not be underestimated as they are increasingly dominant over fundamentals in effecting changes in asset values.

17. Other factors that impact asset values include market microstructure influences such as varying pricing processes, inventory costs, and the existence of dark pool liquidity.

18. The investment behaviour of investors also plays a critical role as psychological and emotional influences of fear, greed, loss aversion, and cognitive biases can have significant impact on asset values and market movements.

Chapter 4:
Foreign Exchange

Learning Objectives

The candidate should be able to:

- ✓ Describe the features of the foreign exchange market.
- ✓ Understand how foreign exchange rates are quoted.
- ✓ Understand the difference between direct and indirect quotations and cross quotations.
- ✓ Calculate cross rates from indirect quotes in USD against other currencies.
- ✓ Understand the bid-offer spread.
- ✓ Distinguish between a spot and a forward transaction.
- ✓ Explain how the forward foreign exchange market works, the concept of forward premium and discount and their relationship with the interest rates of the two currencies being quoted.
- ✓ Calculate the forward exchange rate.
- ✓ Use the interest rate parity theorem to explain movements in the forward exchange rates, in particular the concept of covered interest arbitrage.
- ✓ Explain the quotations of forward rates.
- ✓ Demonstrate how a carry trade works.

4.1 Introduction

The foreign exchange market today is the largest decentralised global market, with an average of USD7.5 trillion[17] worth of foreign exchange contracts traded daily. Spot trades account for about a third of the trades with currency derivative trades accounting for the remainder.

Despite intensified regulation and the corresponding decline in investment banking activities which has curtailed derivatives trading, the underlying foreign exchange market growth remains strong. The foreign exchange market is borderless, with financial flows having diminishing regard for national boundaries. It is effectively linked by global communication networks and integrated with overlapping market segments, i.e., money markets, equity markets, derivatives and commodity markets.

[17] According to the BIS Triennial Central Bank Survey of foreign exchange and OTC derivatives market in 2022, trading in foreign exchange markets averaged $7.5 trillion per day in April 2022 (https://www.bis.org/statistics/rpfx22_fx.htm).

When we consider investments in foreign currencies, we have to contend with foreign exchange risk. The values of foreign currencies fluctuate daily and can affect international investments significantly. Market participants must be aware of these fluctuations, which can either enhance investment returns through exchange gains or depress investment returns through exchange losses. It is therefore important to understand how foreign exchange rates are determined, how the foreign exchange market operates, and how best to use it in financial transactions.

In addition, investing in a foreign currency will always involve another investment decision: which currency should be used to fund the investment? Should the investor borrow in a foreign currency or should he convert funds from his base currency? These choices can result in differing foreign exchange and interest rate risk exposures.

4.1.1 Definition and Terminology

A foreign exchange transaction involves the purchase of one currency against the sale of another currency for settlement or delivery on a specified date. The exchange rate is the price per unit of one currency expressed in units of another currency.

Usually, foreign exchange transactions are **spot** deals i.e., made for delivery in two working days (**value spot**), but there is also a market for same day delivery (**value today**) and next day delivery (**value tomorrow,** or **value tom**).

For investors, it is important to decide what the **base currency** is. Currencies are quoted in pairs, e.g., USD/SGD 1.2520. The first listed currency is called the base currency, and the second listed currency is called the counter currency or term currency. The base currency is the operating currency, otherwise known as the reference currency for the investor, and all his profits and losses will be measured in this currency. Holding investments in any currency other than the base currency is a source of foreign exchange risk, which has to be managed.

Foreign exchange risk positions are described as **long** or **short**. In foreign exchange trading, to go long means to buy with the expectation that your purchase will rise in value. You are betting the base currency will strengthen against the term currency. To short a currency means to sell the base currency in hope that its price will go down in the future. For example, if an investor borrowed USD to fund a SGD investment, he would have sold USD spot for SGD. His foreign exchange position will be described as short USD and long SGD. To close out this foreign exchange position, he enters into the opposite transaction i.e., sell spot SGD and buy USD. After doing this, we say he has "squared off" his position as he no longer has any foreign exchange risk exposure.

4.2 Factors Influencing Exchange Rates

Besides interest rates and inflation, the exchange rate is one of the most important determinants of a country's overall economic health. As the exchange rate plays a vital role to the economic stability of a country, it is important to understand exchange rates and how they are analysed. There are numerous factors affecting exchange rates. In this section, we examine some of the leading factors that can explain exchange rate fluctuations, which include:

- Interest rates;
- Inflation rates;
- Political stability;
- Foreign currency reserves;

- Trade account balances and terms of trade; and
- Government/Public debt.

4.2.1 Interest Rates

Interest rates, inflation and exchange rates are highly correlated. Governments and central banks vary interest rates to manage inflation and currency values. Higher interest rates offer lenders in an economy with higher returns on investment relative to other countries. Consequently, higher interest rates attract inflow of foreign capital, which causes the exchange rate to rise. However, the impact of higher interest rates may be moderated by higher inflation rate in that country compared to other countries, or by other economic and political factors.

4.2.2 Inflation Rates

Typically, a country with a consistently lower inflation rate than other countries will see an appreciation in the value of its currency. The opposite relationship exists for countries with high inflation – that is, high inflation tends to see depreciation in the country's currency.

Inflation can have a very severe impact on currency value. The IMF data for 2017-2018 showed that Venezuela, South Sudan, and the Democratic Republic of the Congo have the 3 highest inflation rates – all three countries were experiencing hyperinflation in 2018. The IMF indicated that Venezuela's inflation would have hit 1,000,000% in 2018. Long before, Venezuelans were already converting to the USD, gold, and even bitcoin as better stores of value. The Venezuelan Bolivar was considered virtually worthless. Despite launching the New Venezuelan Bolivar in late August 2018, the new currency fell by 95% in value the very day of the launch.

In the developed economies of the world, inflation rates were negligible for most of the decade after the Global Financial Crisis (GFC) of 2007/2008. Some of the major developed economies were more concerned about the ills of deflation in the years immediately after the GFC. However, with a decade of monetary and fiscal stimulus in US, Europe, and many other economies, and the recovery of oil prices, inflation looks to be finally returning to the fore for developed economies.

Table 4.2.2 shows the major global currencies against the USD, interest rates and their mid-year CPI (Year-on-Year percentage change).

Table 4.2.2 List of Major Global Currencies, Interest Rates and CPI

Countries	Exchange Rate September 2018	12-mth LIBOR interbank	June 2018 CPI
United States	-	+2.88713	2.55%
Europe	EUR/USD 1.1756	- 0.16700	2.00%
United Kingdom	GBP/USD 1.3238	+1.06000	2.37%
Switzerland	USD/CHF 0.9614	- 0.51860	0.94%
Japan	USD/JPY 112.5000	+0.13900	0.96%
Canada	USD/CAD 1.2914	+1.78650	3.00%
Australia	AUD/USD 0.7278	+3.33800	2.10%

4.2.3 Political Stability

Investors seek stable countries with strong economies to make investments. Countries with stable government executing sound financial and trade policies will attract investment. Currencies tend to appreciate when there are strong inflows of capital and demand for domestic investments. The opposite happens when there is corruption in government, social unrest, frequent changes in government, and inconsistencies in policies. These often lead to loss of investors' confidence and lower demand for the currency. Political uncertainty and social unrest will also lead to outflows of money from residents who feel it is safer to park their domestically acquired wealth in foreign currencies that are more stable.

4.2.4 Foreign Currency Reserves

A larger foreign currency reserve does not necessary lead to appreciation in the domestic currency – some governments may not want their currency to strengthen. However, countries with strong foreign currency reserves are generally better positioned to defend and manage their currency levels compared to those with weak reserves. For example, during the Asian Currency Crisis, hedge funds and other speculators were able to exploit the vulnerability of some Asian currencies that had weaker foreign currency reserves but countries with strong currency reserves (e.g., Hong Kong, Singapore, and Taiwan) successfully defended their currencies against the speculators. Countries with the strongest foreign currency reserves[18] as of June 2022 are shown in Table 4.2.4.

Table 4.2.4 – Countries with Strongest Foreign Currency Reserve (as of June 2022)

Rank	Country	Amount (in USD billions)	Figures Obtained as at		Rank	Country	Amount (in USD billions)	Figures Obtained as at
1	China	3,480	June 2022		6	Taiwan	548	Dec 2021
2	Japan	1,376	June 2022		7	Hong Kong	504	June 2022
3	Switzerland	1,033	June 2022		8	Saudi Arabia	451	June 2022
4	Russia	630	June 2022		9	South Korea	449	June 2022
5	India	599	June 2022		10	Singapore	365	June 2022

4.2.5 Trade Account Balance and Terms of Trade

The trade account balance of a country is another important factor influencing exchange rates.

When a country has a trade deficit, it means that the country is spending more on imports than receiving through exports. In other words, the country needs to acquire more foreign currency to pay for the imports, resulting in higher demand for foreign currency. Conversely, a trade surplus is generated by having higher export receipts than expenditures on imports, leading to a net increase in demand for domestic currency. All

[18] https://data.imf.org/regular.aspx?key=61280813.

other things held constant, a trade deficit should lead to weaker demand for local currency and a trade surplus should lead to stronger demand for local currency.

The ratio of export prices to import prices is a measure of the terms of trade. The terms of trade for a country improve when its export prices rise by a greater rate than its import prices. This in turn implies higher demand for the country's currency and accompanying currency appreciation. If the ratio of export price change is proportionally smaller than that of import price change, it is an indication of lower demand for the domestic currency and accompanying currency depreciation.

Export-oriented countries might find it difficult to manage the upward pressure on the local currency. A massive inflow of money from export-driven surpluses could also encourage inflation in the domestic economy. To offset the impact, a government can take money out of the system by investing the money abroad, thereby moderating the impact on its currency. China, for example, has been investing in US Treasuries and the Chinese governments and its companies have been investing abroad to allow the Chinese Yuan (CNY) to strengthen more gradually. Another example is India. The Indian Rupee (INR) fell by about 14% in the first 9 months of 2018 (refer to Figure 4.2.5(a))). India, which has consistently been running a trade deficit for the last 20 years, saw a further widening of its trade deficit in 2018 as well. (refer to Figure 4.2.5(b)).

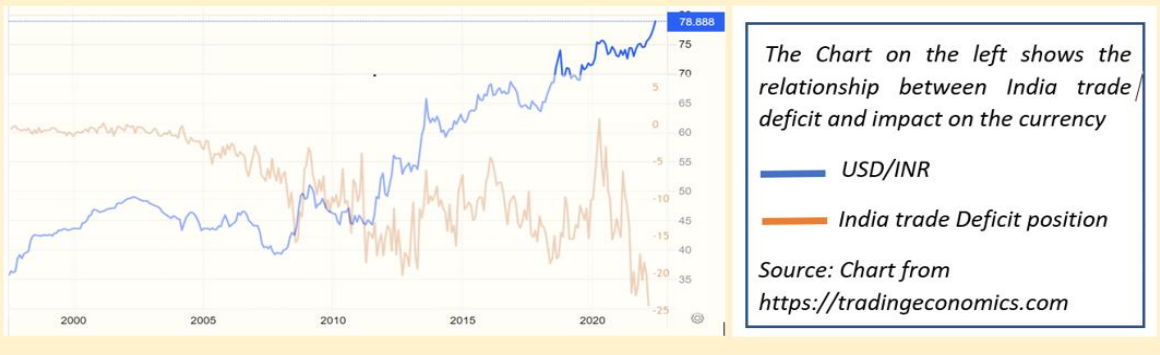
Figure 4.2.5(a) USD has strengthened against INR



US-China Trade and Foreign Exchange

In 2018, the US government under President Trump accused its major trading partners, and in particular China, of "ripping off the US" and enjoying hundreds of billions of surpluses in USD each year over the last decade. The argument was that, by keeping the CNY value low and by not having tariff barriers, China artificially made its exports more competitive while limiting imports. With the intent of reversing the trend to protect US jobs, the US government started imposing larger tariff barriers on a wider range of Chinese goods.

Figure 4.2.5 (b)Relationship between India trade balance and INR currency movements



US-China Trade and Foreign Exchange

In 2018, the US government under President Trump accused its major trading partners, and in particular China, of "ripping off the US" and enjoying hundreds of billions of surpluses in USD each year over the last decade. The argument was that, by keeping the CNY value low and by not having tariff barriers, China artificially made its exports more competitive while limiting imports. With the intent of reversing the trend to protect US jobs, the US government started imposing larger tariff barriers on a wider range of Chinese goods.

Figure 4.2.5(c) USD has actually weakened against the CNY over time



Contrary to the US political rhetoric of the CNY being artificially kept low, the CNY has actually appreciated from 2002 to 2014 by about 25% and China's cost competitiveness has been falling over the years as its labour and land costs have risen quite significantly.

Chart Source: Reuters 27/9/2018

4.2.6 Government Debt

The level of government debt has significant impact on interest rates and currency value. Nations with excessive government debt, also known as public debt, relative to their income generating capabilities and GDP are usually less attractive to foreign investors. The large amount of debt is often needed to increase liquidity and stimulate the economy as income from taxes and other sources are insufficient to support it.

When the government debt is denominated in USD, a fall in the value of the local currency will make the debt more difficult to repay. If public debt is repaid through the printing of money, then the country could experience massive inflation that inevitably further devalues its currency, making further financing through local currency government debt even more difficult.

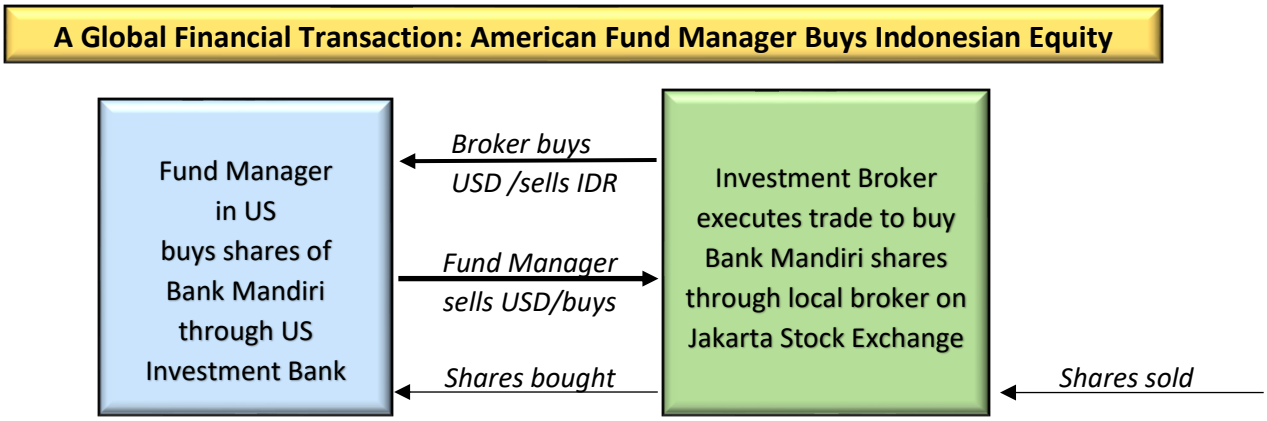
Sometimes, the printing of money to pay off public debt is not even an option. A case in point is the Greek public debt crisis, which started after its Prime Minister admitted in 2009 that its budget deficit figures had been understated for years and could be double those originally reported, triggering a wave of rating downgrades on its debt. As Greece did not have monetary policy independence because it is a member of the Eurozone, it could not simply inflate its way out of debt. Since default appeared imminent, many foreign investors sold their Greek bonds to eliminate that vulnerability. Foreign investors will be less willing to acquire securities or investments in a country where its government is perceived to have high default risk of its debt obligations. To date, Greece has needed three bail-out programmes, with amounts totalling 300 billion euros.

4.3 Foreign Exchange Markets

In the foreign exchange market, financial institutions, commercial banks, business firms, governments or their central banks, tourists, speculators, and investors buy and sell foreign currencies. Unlike the stock market or futures market, the foreign exchange market is not confined to any specific geographical location, such as a

stock exchange. The transactions are made through a complex communication network of computer terminals, telephones, faxes, telexes, cable and communication satellites, and even personal digital devices such as smartphones. The foreign exchange markets work round the clock, and although markets open and close throughout a 24-hour trading day, there are operations with 24-hour dealing, starting the day in Sydney, then Tokyo, Singapore/Hong Kong, London, New York and San Francisco. Figure 4.3 is an illustration of the role of the foreign exchange market in global investment flows.

Figure 4.3 – The Role of Foreign Exchange in Global Investments



"The 2019 Triennial Central Bank Survey of turnover in the over-the-counter (OTC) Foreign Exchange (FX) markets reflects growth in FX derivatives trading, especially in FX Swaps, which outpaced that of spot trading. Trading in FX markets averaged $6.6 trillion per day in April 2019. This is up from $5.1 trillion in April 2016. Though spot trading rose to $2 trillion per day in April 2019 from $1.7 trillion in April 2016, it declined as a share in global FX activity. By contrast, activity in FX swaps rose to $3.2 trillion per day in the latest survey and accounted for almost half of global FX trading. Trading in outright forwards rose by a notable 43% to $999 billion per day.

The US dollar remained the world's dominant vehicle currency - it was on one side of 88% of all trades in April 2019. The Euro remained the second most actively traded currency, edged up to 32% in April 2019, compared to 31% in April 2016. In contrast, turnover for the Yen stagnated, with the JPY's share in global turnover falling by 5 percentage points, to 17% in 2019. The global share of emerging market currencies rose to 25% of the total FX turnover in April 2019, continuing the trend observed in previous surveys. Renminbi trading increased in line with aggregate market growth with $284 billion in turnover and retained its position as the world's eighth most traded currency."

(Source: Bank for International Settlements Triennial Central Bank Survey 16 September 2019.)

Based on IMF data on Currency Composition of Official Foreign Exchange Reserves (COFER), USD-denominated assets among foreign exchange reserves continue to dominate in the second quarter of 2019, accounting for US$6.8 trillion or about 62% of allocated reserves.

According to September 2019 BIS Triennial Central Bank Survey data, currency pairs within the Group of Seven[19] (G-7) countries continue to dominate the global market. Underlying economic, business and financial investments from governments and financial institutions drive the bulk of daily currency trades.

[19] The G7 countries include Canada, France, Germany, Italy, Japan, UK and the US.

The 8 most traded currency pairs for 2019[20] have been:

- EUR/USD
- USD/JPY
- GBP/USD
- AUD/USD
- USD/CAD
- USD/CNY
- USD/CHF
- USD/HKD

The Swiss Franc (CHF) (and, occasionally, the Japanese Yen (JPY)) are considered low-volatility, risk-off currencies that the market tends to gravitate towards when volatility and risks rise in the US Dollar (USD) and Eurodollar (EUR). This role as the safe-haven currency can be quite onerous at times since it can lead to the currency being driven up excessively during period of higher risk, along with gold. In the Southeast Asian region, the Singapore Dollar (SGD) is also seen as the safe haven currency when risk rises in the region.

The Australian Dollar (AUD) and South African Rand (ZAR) are considered commodity currencies and tend to be particularly strong during periods of higher global economic growth or fiscal stimulus where demand for metals and other raw materials leads to strong demand for both currencies. When commodity demand is projected to fall, the commodity currencies tend to weaken.

4.4 Foreign Exchange Rate Quotations

FX rates can be quoted on a direct, indirect, or cross basis, and are usually quoted to the fourth decimal place. The last digit of the quote is called a pip (= 0.0001). A pip is the smallest unit of price movement or quotation for foreign exchange just as a basis point is to interest rates and fixed income instruments.

Currencies are quoted in pairs, e.g., USD/SGD 1.2520. The first listed currency is called the base currency, and the second listed currency is called the counter currency or term currency. In the previous example, the USD is the base currency and the SGD is the counter currency. The quotation refers to the number of units of counter currency that equals one unit of base currency. In the same example, USD 1 = SGD 1.2520.

4.4.1 Bid-Offer Spread

A bank or an authorized FX market maker will always provide two prices when asked for a quote. The first price is the one at which the bank/FX market maker will buy a currency (the **bid**) and the second is the price at which the bank/FX market maker will sell a currency (the **offer**).

For example, USD/SGD 1.2520-25 means that the bank dealer will pay SGD 1.2520 to buy 1 USD. He will sell 1 USD for SGD 1.2525. The difference between 1.2525 and 1.2520 is 0.0005, also known as 5 basis points, which is the spread or profits earned by the bank. The spread varies according to market conditions and the value of the transaction involved. The greater the uncertainty and volatility in the market, the wider the spread. The party asking for a quote will always buy at the offer rate and sell at the bid rate. In market parlance, we say that when asking for a quote, the price is always against the price taker i.e., the party looking for a quote.

4.4.2 Direct Quotes/European Convention

A **direct quote** expresses the price of one unit of foreign currency in number of units of domestic currency. In other words, the foreign currency is the base currency and the domestic currency is the counter currency, i.e., foreign/domestic. As there are many foreign currencies to choose from, the USD is often used as the base currency because it is the most widely accepted, so most direct quotes involve the USD. For example:

[20] BIS Triennial Central Bank Survey 16th September 2019 – Annex Table 3 OTC foreign exchange turnover by currency pair.

- USD/CNY[21] 6.6186 (means 1 USD = CNY 6.6186)
- USD/JPY 101.86 (means 1 USD = JPY 101.86)
- USD/SGD 1.2520 (means 1 USD = SGD 1.2520)

However, there are a few exceptions, as discussed in the next section.

4.4.3 Indirect Quotes/American Convention

Unlike other currencies, currencies in the colonial or old 'sterling bloc' (comprising the UK Pound Sterling (GBP), AUD and the New Zealand Dollar (NZD) are the base currency when quoted against the USD, which is the counter currency. The EUR is also quoted in this manner. For example:

- GBP/USD 1.6816 (means 1 GBP = USD 1.6816)
- AUD/USD 0.9235 (means 1 AUD = USD 0.9235)
- NZD/USD 0.8630 (means 1 NZD = USD 0.8630)
- EUR/USD 1.3622 (means 1 EUR = USD 1.3622)

An **indirect quote** expresses the price of one unit of domestic currency in number of units of foreign currency. In other words, the domestic currency is the base currency and the foreign currency is the counter currency, i.e. domestic/foreign. For example, the way the GBP/USD is typically quoted in the FX markets would be considered an indirect quote in the UK.

Therefore, whether a given quote is direct or indirect really depends on the context; specifically, which is the domestic currency and which is the foreign currency. Also, to note, the indirect quote is just the reciprocal of the direct quote. For example, in Singapore, if the direct quote is USD/SGD 1.2520, then the corresponding indirect quote is SGD/USD 0.7987 (i.e. the reciprocal, 1/1.2520).

4.4.4 Cross Quotes

Although most FX transactions involve the USD as one of the currencies, the increasing importance of other major currencies in international transactions has resulted in direct quotation of two non-USD currencies. The cross-currency rate or cross rate for short is commonly used to refer to currency pairs that do not involve the USD. The idea of cross rates implies two exchange rates with a common currency enabling the calculation of exchange rate between the remaining two currencies.

Cross Rate for SGD/JPY – Deriving a Cross Rate from 2 Direct Quotes

The general rule is to divide the term currency on the same side of the spread by the base currency on the opposite side of the spread. More specifically, to get the SGD/JPY bid, divide the USD/JPY bid by the USD/SGD ask. Similarly, to get the SGD/JPY ask, divide the USD/JPY ask by the USD/SGD bid.

[21] Technically, CNY is the onshore Renminbi, while CNH is the offshore Renminbi.

Currency pair	Bid	Ask
USD/SGD	1.3362	1.3382
USD/JPY	112.6530	112.7330
SGD/JPY	84.1825	84.3684

USD is base, SGD is term

USD is base, JPY is term

SGD/JPY **Bid Rate** = 112.6530 / 1.3382 = 84.1825
This is the rate at which the market **buys** SGD and **sells** JPY.
SGD/JPY **Ask Rate** = 112.7330 / 1.3362 = 84.3684
This is the rate which the market **sells** SGD and **buys** JPY.

Cross Rate for GBP/SGD – Deriving a Cross Rate from a Direct Quote and an Indirect Quote

The general rule is to multiply on the same side of the spread. More specifically, to get the GBP/SGD bid, multiply the GBP/USD bid by the USD/SGD bid. Similarly, to get the GBP/SGD ask, multiply the GBP/USD ask by the USD/SGD ask.

Currency pair	Bid	Ask
GBP/USD	1.3501	1.3510
USD/SGD	1.3362	1.3382
GBP/SGD	1.8040	1.8079

GBP is base, USD is term

USD is base, SGD is term

GBP/SGD **Bid Rate** = 1.3501 x 1.3662 = 1.8040
This is the rate at which the market **buys** GBP and **sells** SGD.

GBP/SGD **Ask Rate** = 1.3510 x 1.3382 = 1.8079
This is the rate at which the market **sells** GBP and **buys** SGD.

4.4.5 Settlement Dates

The date when both parties of a FX transaction deliver their respective currencies is known as the value date or settlement date.

A **spot transaction** is a contract in which one currency is exchanged for another for settlement usually in 2 business days. For example, a foreign exchange deal transacted on Monday, 4 April 2022 will be settled on Wednesday, 6 April 2022. The 2-day period gives ample time for both parties to exchange the currencies. As Saturdays and Sundays are not considered working days, a spot deal done on Thursday or Friday will be settled on the following Monday and Tuesday, respectively. Public holidays are also considered non-working days.

Market participants transact spot FX to cover FX exposures, pay in foreign currencies, switch liabilities or assets for yield reasons, for hedging purposes, or simply to trade in the currency.

Forward transactions are contracts to exchange one currency for another at some future date (later than spot) at a rate of exchange that is determined on the trade date. The purpose of forward deals is to establish and fix the cost of exchange for a known foreign exchange commitment at some future date.

4.5 Forward Foreign Exchange Contracts

There are at least two reasons why an investor will enter into a forward exchange contract:

a. to hedge a currency exposure, or

b. to speculate by taking a position in the currency.

An investor who has bought or sold foreign securities may want to cover his foreign currency exposure by fixing a forward exchange rate. By covering his exposure, through locking-in a known forward FX rate, the investor also forfeits potential gains from favourable movements in market rates. These transactions, where there is only a single forward transaction, are normally referred to as **'outrights'**.

Some investors or traders may not have an initial currency exposure but hope to profit by taking a view on the direction of the foreign currency. They can buy the forward position on the foreign currency if it is expected to appreciate and vice versa.

4.5.1 Forward Rates

A forward transaction is made up of two components:

a. Spot exchange rate; and

b. Interest differential expressed in FX rate terms.

The forward rate is therefore dependent on the movement of the spot exchange rate of the currency pair and the respective interest rates for each of the two currencies. As an example, the 3-month forward rate for USD/SGD depends on the spot rate for USD/SGD, the 3-month interest rate for USD and the 3-month interest rate for SGD.

The interest differential between the two currencies is converted into **forward** (or swap) **points,** which are either added to or subtracted from the current spot rate. Hence,

Forward exchange rate = Spot rate ± Forward points

When forward points are added to the spot rate, we say that the forward rate is at a premium. When they are subtracted from the spot rate, the forward rate is at a discount. Generally, the currency of the country with the higher interest rate will trade at a **forward discount** (forward rate is cheaper than the spot rate). The currency with the lower interest rate will trade at a **forward premium** (forward rate is more expensive than the spot rate).

4.5.2 Calculating Forward Points

The general formula for calculating forward points is as follows:

$$\textbf{Forward Points} = \frac{S \times (R_c - R_b) \times n/360}{1 + (R_b \times n/360)}$$

where:
S = spot rate
R_c = annualised interest rate of counter currency
R_b = annualised interest rate of base currency
n = number of days from spot date to the forward date

Sometimes, supply or demand for outright forwards can be so great that the forward rate will not reflect the interest differential. When this happens, there is room for arbitrage between the money market and the FX market.

There is also proprietary trading of the forward foreign exchange market because banks take trading positions based on their expectations of the swap points movements, i.e., the relative movements between the interest rates of the two currencies. This is also known as **forex gap trading**.

It is important to note that the forward points are only a reflection of the relative interest rates of the currencies. They are not an indication of what the future spot rate of the currencies will actually be.

How to Calculate Forward Points

Suppose the prevailing market rates are:

Spot USD/SGD = 1.2520

$R_b = R_{US}$ = 1.75% p.a.

$R_c = R_{SGD}$ = 0.75% p.a.

n = 30 days

USD/SGD forward points

$$= \frac{1.2520 \times (0.0075 - 0.0175) \times 30/360}{1 + (0.0175 \times 30/360)}$$

= -0.0010 / 1.0015

= -SGD 0.0010 or -10 points (forward USD/SGD at a discount)

The 30-day USD/SGD forward rate will be quoted at a discount, which is USD/SGD 1.2510

4.5.3 Interest Rate Parity

The value of the premium or discount can be theoretically computed from the **interest rate parity theorem**. This theory states that (except for the effects of small transaction costs) the forward premium or discount should be equal to the difference in the national interest rates for securities of the same maturity. If the

interest rate parity relationship is violated, arbitrageurs will be able to make risk-free profits in FX markets. Their actions will force the forward and spot exchange rates back into alignment.

According to the interest rate parity theorem, the relationship between the spot rate and forward rate is:

$$F = S\left[\frac{1 + R_c(\frac{n}{360})}{1 + R_b(\frac{n}{360})}\right]$$

where:
F = forward rate
S = spot rate
R_c = annualised interest rate of counter currency
R_b = annualised interest rate of base currency
n = number of days

We can illustrate the interest rate parity theorem by using two currencies, the USD (counter currency) and GBP (base currency).

Covered Interest Arbitrage - Interest Rate Parity with USD and GBP

If R_{USD} = 0.06 and R_{GBP} = 0.05 annually, while GBP spot is USD 1.50, then the forward price for a 1-year contract would be:

F = USD 1.50 (1.06/1.05) = USD 1.5143 per GBP

What if the interest rate parity relationship is violated? For example, suppose the forward price is USD 1.5100 instead of USD 1.5143.

The following strategy can be taken to reap arbitrage profits.

- Borrow GBP 1,000,000 for 1 year at 5% per annum. After one year, the arbitrageur will need to repay GBP 1,050,000.
- Sell the borrowed GBP into USD at the spot exchange rate of USD 1.50. He receives GBP 1,000,000 x 1.50 = USD 1,500,000.
- Invest the USD for 1 year at 6% per annum. He will receive USD 1,500,000 x 1.06 = USD 1,590,000 at the end of one year.
- Enter into a 1-year forward contract to buy GBP 1,050,000 at forward price of USD 1.5100. At the end of 1 year, the arbitrageur will need to pay GBP 1,050,000 x 1.5100 = USD 1,585,500.

After 1 year, the arbitrage position is closed out as follows:

- Receive USD 1,590,000 from the USD deposit.
- Pay USD 1,585,500 to obtain GBP 1,050,000 at an exchange rate of USD 1.5100.
- Pay GBP 1,050,000 toward the principal and interest on the GBP 1,000,000 loan.

At the end of 1 year, the arbitrageur's net position is zero because all the transactions are closed out. However, he makes a profit of (USD 1,590,000 – USD 1,585,500) = USD 4,500. Effectively, the arbitrageur has sold GBP position at 1.5143 (USD 1,590,000 / GBP 1,050,000) and bought the GBP in the forward market at the lower rate of 1.5100. This arbitrage was accomplished by simultaneously borrowing in one currency, investing in another currency, and covering the exchange position in the forward exchange market.

The entire process is known as **covered interest arbitrage**.

4.5.4 Forward Rate Quotes

Most market information services show forward rates not as outright rates of exchange but as margins, in points, above or below the spot rate.

For example:

	Spot	**1-mth**	**2-mth**
USD/JPY	119.22/25	52/51	106/104
AUD/USD	0.6475/80	17/21	36/43

Sometimes, the forward points do not have a sign in front of them. Similar to spot rates, market makers quote forward points on a bid and offer spread. For currencies that are traded on a forward discount, the bid (left hand side) is higher than the offer (right hand side) i.e. the market maker is buying at a deeper discount and selling at a lesser discount. For currencies that are traded on a forward premium, the bid (left hand side) is lower than the offer (right hand side) i.e. the market maker is buying at a lesser premium and selling at a higher premium.

To derive the outright rates, the convention is to:

a. Subtract the points from the corresponding spot rates if the forward points are descending i.e. the lefthand side is higher than the right-hand side (e.g., 52/51). For example:

	Spot	**1-mth**	**Outright**
USD/JPY	119.22/25	-52/-51	119.2148/119.2449

b. Add the points to the spot rates if the forward points are ascending i.e. the left-hand side is lower than the right-hand side (e.g., 17/21). For example:

	Spot	**1-mth**	**Outright**
AUD/USD	0.6475/80	+17/+21	0.6492/0.6501

On an intuitive basis, for a market maker to maintain a bid-offer spread, the lower (in value) forward point is always quoted first, followed by the higher (in value).

This can only be true if 52/51 are negative i.e., -52/-51.

4.5.5 Non-Deliverable Forwards

Some countries have tight capital and exchange controls to reduce the outflow of currencies and prevent exchange rate volatility. Often, the currency market in these countries has low liquidity and inaccessibility due to capital control restrictions. A Non-Deliverable Forward (NDF) is essentially a financial instrument commonly used to hedge exposure to currencies offshore that are not internationally traded or whose trade is limited or legally restricted to within the domestic market. The NDF market is an OTC market that developed over the last 2 decades with the growth of emerging economies such as China, Taiwan, Vietnam, and Indonesia.

The concept of an NDF is relatively similar to that of a regular FX forward. The key difference is that there is no physical delivery or exchange of the domestic currency in the offshore market. An NDF is a cash-settled, short–term forward contract settled at an agreed forward rate at maturity in a freely traded currency. Typically, NDFs are quoted with USD as the reference currency. The only exchange of cash flows or calculation of profits or losses are based on the difference between the agreed rate (the contracted NDF rate) and the spot rate prevailing at the time of settlement with no exchange of the notional amount. The normal tenor of the NDF contract ranges from 1 month to a year.

Besides hedging currency risks of investments in countries with rigid capital control policies, the mechanisms of NDF also provide arbitrage opportunities for market participants and allow speculative positions to be held.

Non-Deliverable Forwards for Hedging Risks

You have a client based in Singapore importing goods from China. The supplier quotes the cost in CNY but requires payment in USD at the prevailing rate. Your supplier's invoice shows the USD equivalent of CNY10 million, to be paid 3 months later. Assume that the current USD/CNY spot rate is 6.5845.

USD/CNY	**Current Rate @ 6.5845**	**If USD/CNY rises 5% to 6.9137**	**If USD/CNY falls 5% to 6.2553**
CNY 10 million equivalent in USD	USD 1,518,718	USD 1,446,404	USD 1,598,644
FX Gain (Loss)	-	FX Saving: USD 72,314	FX Loss: USD 79,926

If your client does nothing, what kind of exchange rate risk does he face?

If nothing is done, he faces the risk of the USD depreciating against the CNY. If the USD/CNY falls to 6.2553, then the CNY 10 million amount will require an additional USD 79,926 in 3 months' time.

What can your client do if he wishes to be more certain about the USD/CNY exchange rate?

Since your client faces the risk of the USD depreciating, to offset this risk, he can enter into a USD/CNY NDF contract set for 3 months to sell USD (buy CNY), fixing the USD/CNY rate at 6.5845 for the notional balance of CNY 10 million. If the USD/CNY level falls (i.e., CNY rises) by about 5% to USD/CNY 6.2553, then the bank will pay your client about USD79,926, which will serve to offset the USD79,926 more he has to pay his Chinese supplier.

Of course, if the CNY falls and USD/CNY actually rises to USD/CNY 6.9137, he will have to pay the bank USD 72,314 for the notional loss, but this is offset by the USD72,314 less that he needs to pay his supplier. The NDF contract allows your client to fix the rate at a level he is comfortable with to remove the uncertainty.

4.6 Risks in Foreign Exchange

Like any asset class, FX has its own risks. Compared to the other asset classes, most FX trading is very short-term, with proprietary trading making profits and losses over periods that may be as brief as minutes. Participants in the FX market have to manage the following risks:

a. **Market Risk** – movement and volatility of FX rates.

b. **Price Liquidity** – in less traded currencies, and even in major currencies when there is an extreme market event e.g., a major power failure.

c. **Credit Risk** – when counterparty fails to honour the deal. In forex transactions, credit risk is but a fraction of the total transaction amount because the transaction is an exchange (buy and sell), and not an outright extension of credit.

d. **Settlement/Delivery Risk** – counterparty fails to pay on time (one has to view settlement risk as a specific counterparty credit risk which takes place in the period just before settlement. In FX, this is 2 days before payment). Continuous Linked Settlement[22] (CLS) has reduced delivery risk significantly.

4.7 Managing Foreign Exchange Risks

A number of FX risk mitigating strategies may be applied to reduce the level of exchange rate risk. Certain exchange rate management strategies involving complex mechanisms such as currency intervention and controls may only be administered by governments or central banks.

Besides the use of money market instruments and FX derivatives such as forward and futures contracts quoted on exchanges, firms and individuals use options and swaps to hedge their risks and to take advantage of perceived opportunities. Currency forward contracts is covered in this Chapter whilst other derivative instruments such as futures, options, and swaps are covered in the CMFAS CM-SIP Study Guide. Apart from derivatives, there are also alternative strategies to hedge exchange risks and to take advantage of opportunities, which will be discussed below.

In this section, we will first discuss how governments manage currency, before discussing how firms and individuals can manage currency risk.

4.7.1 Exchange Rate Regimes

The policy framework adopted by a country's central bank to manage its currency in relation to other currencies and the FX market is called an exchange rate regime. Each country is free to adopt the exchange

[22] https://www.mas.gov.sg/news/media-releases/2003/sgd-joins-continuous-linked-settlements-system.

rate regime that is aligned with its monetary policies or fiscal policies. In most cases, countries will either choose some form of fixed rate pegged to a dominant world currency, a floating exchange rate regime, or some variant in between like a managed trade-weighted float.

A country may also choose not to manage their own currency but to simply adopt another nation's currency as legal tender. This exchange rate regime is known as currency substitution or dollarization. With the circulation of the currency of another country as the sole legal tender, the adopting country inherits the credibility of that currency, but not that other country's creditworthiness. A monetary union (also known as a currency union) is formed when several countries share a common currency (e.g., EUR). Such monetary unions are difficult because some countries and governments may be more extravagant than others. A currency board is an institutional arrangement based on the full convertibility of a local currency into a reserve one backed by foreign currency reserves (e.g., HKD).

The SGD is managed as a trade-weighted currency band based on Singapore's major trading partners and competitors. The Monetary Authority of Singapore (MAS) manages the SGD on a managed float within an undisclosed currency band, which provides some flexibility to handle short term market fluctuations. The exchange rate band is also reviewed from time to time against the fundamentals of the economy. Under a Currency Interchangeability Agreement in 1967, the Brunei Dollar is interchangeable with the Singapore Dollar at par. The Brunei Dollar is legal tender in Singapore and vice versa.

In general, there are main four methods in which exchange rates can be managed (refer to Table 4.7.1):

- The fixed peg;
- The managed soft peg;
- The trade-weighted band; and
- The free float (or managed free float).

Table 4.7.1 – The Different Exchange Rate Regimes

Exchange Rate Regime	Description
Fixed or Pegged Exchange Rate	In the fixed peg, the central bank or government simply manages its currency against a major international currency. Many currencies are directly pegged either to the USD or the EUR.
Managed Peg (Soft Peg) Exchange Rate	In a managed soft peg, there is no official fixing of a currency to another, but the government will use that other currency as a reference point. A government or central bank can allow for a wider drift up or down against the currency but keep within a pre-defined limit.
Trade-weighted Band (Trade weighted Peg or Float)	Such currencies are managed within a trading band based on a trade-weighted basket of currencies. For countries preferring a tighter band (e.g., China), more central bank intervention will take place and the process becomes more like a peg against the trade-weighted currency benchmark. For currencies where a wider trading band is allowed (e.g., SGD), the currency will behave closer to a managed float.

Exchange Rate Regime	Description
Floating Exchange Rate	The currency is allowed to float freely and adjust based on the forces of demand and supply in the market with little government interference. Many central banks claim to adopt such policies but in reality, when the currency trend interferes with the economy, governments do often step in by buying or selling foreign reserves to weaken or strengthen the currency.

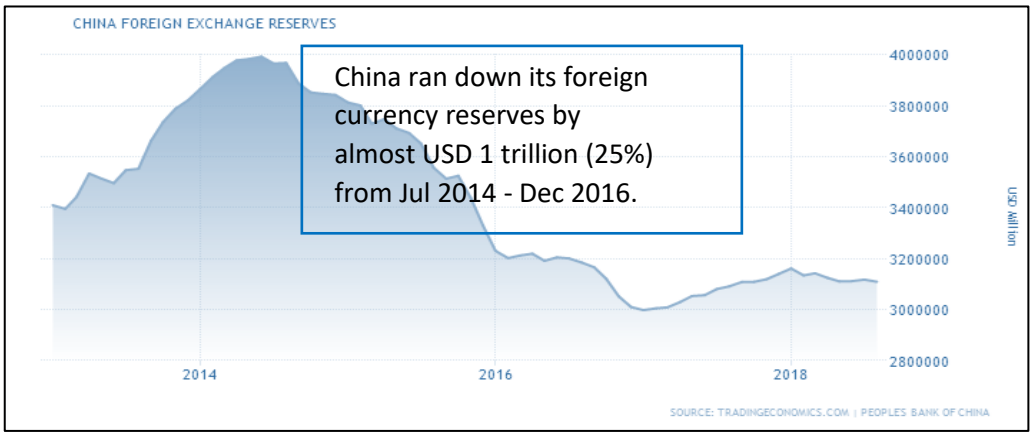
At times, it can be costly for central banks to attain a desired exchange rate level against another currency.

The level of foreign currency reserves, the amount of capital outflows, economic fundamentals, and market conditions facing a country can make it challenging for the central bank to support its desired exchange rate. An example would be when the Chinese government decided to move towards a managed trade-weighted approach in managing its currency in late 2015, which resulted in major currency outflows and pressure on the CNY. The Chinese government had to deplete some of its foreign reserves to defend and avert depreciation of the CNY against the USD (refer to Figures 4.7.1(a) and 4.7.1(b)).

Figure 4.7.1(a) – Keeping a strong CNY was costly from Jan 2014 - Dec 2016



Figure 4.7.1(b): China's Foreign Currency Reserves



4.7.2 Hedging Currency Risk by Borrowing in Similar Currency

Investors and entrepreneurs often borrow in the currency of the investment, to mitigate any fall in the currency. For example, Singapore investors in UK property will establish their borrowings in GBP. In SGD, if the GBP weakens, the GBP borrowings to fund the investment would be a natural hedge against GBP exposure. Since they did not pay in full and did not fund the investment with SGD, the impact from the GBP fall will have a less damaging impact.

To illustrate this, suppose a Singaporean firm would like to buy a GBP 1 million property in the UK. The firm needs to borrow either SGD or GBP to fund 80% of the property value. The current GBP/SGD rate is 1.8060 but the GBP depreciates by 5% to 1.7157 in a year's time. Ignoring interest and transaction costs, and assuming the firm sells off the property in a year at the same price that it purchased it, would the firm have been better off selling SGD- or GBP-denominated bonds?

If the firm had borrowed in SGD,

Down payment in SGD = 0.2 x 1 million x 1.8060 = SGD 361,200

SGD loan amount = 0.8 x 1 million x 1.8060 = SGD 1,444,800

Property sales proceeds after 1 year in SGD = 1 million x 1.7157 = SGD 1,715,700

Total Profit/(Loss) in SGD = 1,715,700 - 361,200 - 1,444,800 = - SGD 90,300

In this case, initially, the firm had to borrow SGD 1,444,800 and come up with another SGD 361,200 for down payment, for a total of SGD 1,806,000, then convert it into GBP 1 million at the GBP/SGD rate of 1.8060 to buy the house. After a year, the house is sold for GBP 1 million, converted into SGD 1,715,700 at the GBP/SGD rate of 1.7157, and pay off the loan of SGD 1,444,800.

If the firm had borrowed in GBP,

Down payment in SGD = 0.2 x 1 million x 1.8060 = SGD 361,200

Net property sales proceeds, after loan repayment in 1 year = 1 million – (0.8 x 1 million) = GBP 200,000 = SGD (200,000 x 1.7157) = SGD 343,140

Total Profit/(Loss) in SGD = 343,140 - 361,200 = - SGD 18,060

In this case, initially, the firm had to come up with SGD 361,200, which is then converted into GBP 200,000 at the GBP/SGD rate of 1.8060. Together with the GBP 800,000 that the firm borrows, it can then buy the GBP 1 million house. After a year, the house is sold for GBP 1 million, and the GBP 800,000 loan is paid off. The firm then converts the remaining GBP 200,000 into SGD 343,140 at the GBP/SGD rate of 1.7157.

Therefore, the firm would be better off selling the GBP-denominated bonds as it would have resulted in a smaller loss of SGD18,060.

4.7.3 Carry Trades

Another common strategy used by investors in the FX market to manage exchange rate risk is carry trade. It is a strategy in which an investor borrows at a low interest rate to invest in an asset that is likely to provide a higher return. A currency carry trade entails the sale of a currency with relatively low interest rate and the use

of the funds to purchase a different currency with higher yields. While paying low interest on the currency borrowed or sold, the investor attempts to capture the interest rate differential. This strategy relies on relative stability in currency prices as an adverse exchange rate movement can wipe out the returns from the underlying interest rate differential.

Where possible, carry trades should be structured on stable currency pairs or when one is more confident of weakness in the borrowing currency. If there is weakness in the invested currency, an investor stands to lose from both the additional interest incurred and FX losses. Investments become highly speculative when we use currencies facing high inflation and weak economies with a history of currency weakness and devaluation.

One popular foreign exchange trade over the last few years among Singapore and other Asian investors was to borrow in JPY at the low rates and invest in AUD products offering reasonable returns, as illustrated in the example below.

Borrowing in JPY and Investing in AUD - Carry Trade from 2008-2016

For example, a Singapore investor borrows JPY 100 million every year from 2008-2016 and invests in Australian corporate bonds creating a 350-400+ basis point spread annually over the Japanese loan rate. For most of years, the strategy worked, but for 2008, 2011 and 2015 the strategy failed because the AUD (the invested currency) fell sharply against the borrowed currency (JPY).

The table shows the potential benefits of borrowing cheaply in one currency, while investing in products in a currency with higher interest rates. Naturally, the higher the differential between the AUD investment and JPY borrowing cost, the better for the investor. As long as the borrowed currency stayed flat or depreciated against the currency invested, the carry trade strategy will be effective.

CARRY TRADE	2008	2009	2010	2011	2012	2013	2014	2015	2016
Yield on Australian Bond	7.00	5.50	6.00	6.60	6.00	5.50	5.15	4.75	4.50
Avg JPY Loans Rate	2.25	2.00	1.65	1.50	1.25	1.25	1.20	1.00	0.95
AUD/SGD Start of Year	1.2611	1.0178	1.2636	1.3142	1.3286	1.2831	1.1284	1.0523	1.0233
AUD/SGD End of Year	1.0178	1.2636	1.3142	1.3286	1.2831	1.1284	1.0523	1.0233	1.0641
SGD/JPY Start of Year	77.7445	63.3266	66.1700	63.1530	59.2623	71.3046	82.9152	90.4936	83.7575
SGD/JPY End of Year	63.3266	66.1700	63.1530	59.2623	71.3046	82.9152	90.4936	83.7575	82.8787
AUD/JPY Start of Year	98.0466	64.4513	83.6150	82.9938	78.7352	91.4938	93.5631	95.2237	85.7065
Borrowing in JPY100m	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000	100,000,000
Interest Cost (JPY)	2,250,000	2,000,000	1,650,000	1,500,000	1,250,000	1,250,000	1,200,000	1,000,000	950,000
Total Repayment Due (JPY)	102,250,000	102,000,000	101,650,000	101,500,000	101,250,000	101,250,000	101,200,000	101,000,000	100,950,000
Conversion into SGD									
Initial Borrowed (SGD)	1,286,265	1,579,115	1,511,260	1,583,457	1,687,415	1,402,434	1,206,052	1,105,051	1,193,923
Japanese Loan Yr End (SGD)	1,579,115	1,511,260	1,583,457	1,687,415	1,402,434	1,206,052	1,105,051	1,193,923	1,206,583
Japanese Int. Yr End (SGD)	35,530	30,225	26,127	25,311	17,530	15,076	13,261	11,939	11,463
Japanese Loan + Int. (SGD)	1,614,645	1,541,485	1,609,584	1,712,726	1,419,964	1,221,128	1,118,311	1,205,862	1,218,045
Initial AUD Investment (AUD)	1,019,923	1,551,559	1,195,958	1,204,910	1,270,079	1,092,970	1,068,797	1,050,159	1,166,772
AUD Investment Return	71,395	85,336	71,757	79,524	76,205	60,113	55,043	49,883	52,505
Total Proceeds in AUD (AUD)	1,091,317	1,636,895	1,267,715	1,284,434	1,346,284	1,153,084	1,123,840	1,100,041	1,219,277
				1,706,486					
				1,712,726					
AUD Proceeds Yr End (SGD)	1,110,699	2,068,445	1,665,993		1,727,471	1,301,163	1,182,583	1,125,639	1,297,432
JPY Borrowing + Interest	1,614,645	1,541,485	1,609,584	-(6,240)	1,419,964	1,221,128	1,118,311	1,205,862	1,218,045
Net Gain (Loss)	-(503,946)	526,960	56,409	-(6,240)	307,507	80,035	64,272	-(80,223)	79,387

In 2008, JPY 100 million was borrowed at the prevailing rate of 2.25%. The JPY 100 million was converted into AUD 1.020 million (SGD 1.286 million) and was invested in an AUD investment paying 7%. Unfortunately, over 2008, the AUD fell 34.2% against the JPY and 19% against the SGD. The total Australian proceeds (invested amount and returns), when converted back into SGD terms at the end of the year, was SGD 1.111 million, which was lower than the –SGD 1.615 million equivalent of the JPY loan principal repayment and interest cost. Hence, the carry trade resulted in a loss due to the weakening of the invested

currency and the strengthening of the borrowed currency. Carry trades are performed on different currency pairs depending on interest rate differentials and expected currency movement.

4.8 Summary

1. The leading factors that can explain exchange rate fluctuations include:
   - Interest rates;
   - Inflation rates;
   - Political stability;
   - Foreign currency reserves
   - Trade account balances and terms of trade; and
   - Government/Public debt.

2. Higher interest rates offer lenders in an economy higher return on investment relative to other countries. Consequently, higher interest rates attract inflow of foreign capital which causes the exchange rate to rise.

3. Typically, a country with a consistently lower inflation rate than other countries will see an appreciation in the value of its currency. High inflation tends to see depreciation in the country's currency.

4. Countries with stable government executing sound financial and trade policies will attract investments. Currencies tend to appreciate when there are strong inflows of capital and demand for domestic investments. The opposite happens for countries with corruption in government, social unrest, frequent changes in government, and inconsistencies in policies.

5. Countries with strong foreign currency reserves are generally better positioned to defend and manage their currency levels when compared to those with weak reserves.

6. If public debt is paid through the printing of money, then the country could experience massive inflation that inevitably further devalues its currency, making further financing through local currency government debt even more difficult.

7. A foreign exchange transaction involves the purchase of one currency with the simultaneous sale of another currency for settlement or delivery on a specified date. Foreign exchange rates can be quoted on a direct, indirect or cross basis.

8. A spot transaction is a contract in which one currency is exchanged for another for settlement usually in two working days. A forward transaction is a contract to exchange one currency for another at some future date (later than spot) at a rate of exchange which is determined on the trade date.

9. An investor may enter into a forward exchange contract to hedge a currency exposure or to speculate by taking a currency position.

10. The major factor determining the forward rate of one currency against another is the difference in interest rates in the two countries over the specific forward period. The interest differential can be converted into forward or swap points, which are either added to or subtracted from the current spot rate.

11. Generally, the currency of the country with the higher interest rate will trade at a forward discount; the currency with the lower interest rate will trade at a forward premium.

12. According to the interest rate parity theorem, the relationship between the spot rate and forward rate is:

$$F = S\left[\frac{1 + R_c(\frac{n}{360})}{1 + R_b(\frac{n}{360})}\right]$$

13. Currencies can be managed using:

    - A fixed peg, where the central bank or government simply manages their currency against a major international currency.

    - A managed soft peg, where there is no official fixing of a currency against another but the government will use another currency as a reference point and can allow for a wider drift up or down against that currency but will keep within a pre-defined limit.

    - A trading band based on a trade-weighted basket of currencies.

    - A free float approach allows the currency to move and find its value based on demand and supply for that currency.

14. A carry-trade is a strategy that seeks to exploit the interest rate differential between two currencies by borrowing in the currency with low interest rate and investing in investments denominated in the currency with high interest rate. (For example, borrowing in JPY at less than 1% and investing in AUD products offering 4.5% return).

Chapter 5: Company Analysis and Understanding Financial Statements

Learning Objectives

The candidate should be able to:

- ✓ Explain company analysis in the context of fundamental analysis.
- ✓ Read and interpret the three major financial statements – the balance sheet, the income statement and the statement of changes in cash flow.
- ✓ Analyse a company's financial performance using financial ratios.
- ✓ Explain the liquidity of a company, using the current, quick and cash ratios.
- ✓ Explain financial risk and leverage in a company, using the debt-to-equity ratio, total debt to total assets ratio, interest coverage and total fixed charge coverage.
- ✓ Analyse a company's operating efficiency using total asset turnover, inventory turnover, days to sell inventory, receivables turnover and the collection period for accounts receivables.
- ✓ Analyse profitability in relation to sales or capital using gross profit margin, operating profit margin, net profit margin, return on assets, return on equity and earnings per share.
- ✓ Understand the use of valuation ratios in estimating the value of a share.
- ✓ Understand what good quality of financial statements and good quality of earnings mean.
- ✓ Explain the importance of comparative and trend analysis.
- ✓ Identify the limitations in the use of financial ratios to analyse a company's performance.

5.1 Introduction

In Chapter 3, we discussed several aspects of fundamental analysis, covering economic, industry, market, and company analysis. We also highlighted the impact of economic and industry factors on asset values and market movements. In this Chapter, we will cover the company analysis portion of fundamental analysis in greater detail. Company analysis does not only relate to equity analysis, but it is also relevant for credit

analysis of bond investments and other securities whose cashflow payments are dependent on the ongoing viability of the issuing company.

Company analysis can be divided into two parts:

- Qualitative Analysis; and
- Quantitative Analysis (Financial Statements Analysis).

When performing company analysis, there are a few principles to remember:

a. **Asset price movements are influenced by the future prospects of a company**. Historical financial statements provide us with a picture of the profitability of the business in the recent past and the way in which the company has been managed. We use the historical numbers, ratios and relationships to help make projections and forecasts given the outlook and prospects of the company. Asset prices in the market will move when there are changes to the expectations about the future performance of a company.

b. **A company's performance is driven by qualitative factors.** Asset price movements are first driven by news and information which have a bearing on the prospects of the company. It is more important to focus on the qualitative factors and conditions that allow for the company to be successful than to simply rely on ratios and numbers as these qualitative factors drive and precede the quantitative outcomes. These qualitative factors are discussed in Section 5.2.

c. **Excellent companies with sustainable profit growth potential can appear expensive in the near term but they usually remain good investments for the longer term.** Companies with strong business models and excellent earnings growth which have attracted investors' interest may have its market price run ahead of its earnings in the near term. However, if the positive qualitative factors continue to provide competitive advantages, and the companies are able to deliver steady and sustainable earnings growth over time, the earnings will catch up with the valuations, eventually driving up market prices. In contrast, buying into weaker-performing companies or loss-making companies that have been massively sold down for a period will, at best, bring short-term trading opportunities. At worst, such companies could trap investors if prospects continue to deteriorate.

5.2 Qualitative Analysis

In the qualitative analysis of companies, investors aim to identify characteristics and factors that allow a company to outperform its peers. A suggested list of competitive advantages is reflected in Table 5.2:

Table 5.2 - Examples of Companies with Competitive Advantages

	Competitive Advantage	Companies Exhibiting Competitive Advantage
1	Dominance and control of key technology, research, and patents	Apple, Nvidia, Pfizer, Johnson & Johnson, Novartis (and other top technology, pharmaceutical and biotech companies)
2	Control over key resource supply	De Beers, Bayer-Monsanto
3	Superior brand differentiation	Apple, Coca-Cola, Microsoft, BMW, Mercedes, Samsung, Toyota, Facebook, SIA, Disney, Google, Nestle, Nike

	Competitive Advantage	Companies Exhibiting Competitive Advantage
4	Control of distribution channels	Alibaba, Amazon, Tencent
5	Cost efficiency, lowest-cost producer	Xiaomi, TSMC, McDonald's, IKEA
6	Successful focus on niche clients	Harley Davidson, Swiss private banks (e.g., UBS, Credit Suisse, Julius Baer), LVMH, Prada
7	Superior management, talent, connections	Apple, Alibaba, Microsoft, Berkshire Hathaway, Cheung Kong-Hutchinson Holdings
8	Diversified core businesses with shared cost	General Electric, 3M, Honeywell

Dominance and control of key technology, research, and patents. Companies with control over technology are able to maintain higher prices and margins while growing their revenue. With patented technology, some companies are able exert an almost monopolistic influence as the sole supplier in the high-end range of the product or service. This allows them to establish significant price premiums over their competitors. For example, Apple continues to enjoy a price premium when compared to other smart phone producers. Nvidia, the top-end chipmaker that designs graphic processing units for the gaming, mobile computing, and automotive industries enjoys higher profit margins than its competitors because it focuses on top-end chips.

Dominance over supply. Companies that have substantial control over the raw materials and resources required for production will have an advantage over other companies. Despite giving up its almost monopolistic control of diamond supply and production for several decades, DeBeers continues to control about 30-35% of the world's diamond supply. Monsanto, a leading producer of genetically modified seeds (GMO), was acquired by Bayer, which is also another leading producer in the same industry. With the acquisition, Bayer should have captured more than half of the GMO seed crop market.

Superior brand differentiation. By creating a strong brand image, a company's products and services can command higher pricing, better sales, and better profit margins than its competitors. Luxury companies are especially conscious about their brand image as they differentiate themselves through branding. Some brands have developed solely within a specific range of products (e.g., Japanese and German car manufacturers such as Mercedes, BMW, Honda, and Toyota) and are known mainly for their precision, quality, and dependability. Other brands are known for their superior standards of service (e.g., Singapore Airlines).

Controlling the distribution channel. Companies that dominate the sales channel usually gain a critical advantage. The need for costly physical infrastructure to develop distribution and sales networks is now less of a barrier to entry due to the ongoing internet revolution. The internet has redefined the way company market, communicate, collect payments, and deliver products. Traditional industries with established brick-and-mortar retail chains and distribution networks have been forced to adopt a blended approach to interact with their clientele. As the world becomes more internet savvy, companies with a first-mover advantage on internet marketing sales platforms (e.g., Alibaba, Amazon, Tencent) are taking a larger share of sales commissions and advertising revenue on top of selling their own products.

Cost efficiency, lowest-cost producer. Some companies compete by being the most efficient producers. Often, they will drive sales volume on very low margins due to their efficiency and economies of scale. Some companies have become the outsourcing partners of manufacturers who no longer want to bear the cost of manufacturing and the risk of having to retool and restructure when technologies advance. These outsourcing partners have lower cost and scalability; they typically operate on very thin profit margins but can offer economies by aggregating orders from different companies, creating high sales volumes. The cost of setting

up the infrastructure in order to have scale advantages becomes a natural barrier to entry. Examples of companies that have done very well using such an approach are Taiwan Semiconductor Manufacturing, Foxconn and Samsung, being major subcontractors for other technology companies.

Successful focus on niche clients or niche products. Some companies have successfully focused on a niche product and clientele. They become specialized in serving a particular client group and often command premium fees and pricing because of their specialized offering and customised focus. An example of a listed company in this area is Harley Davidson, which caters to a certain high-end niche of motorcycle enthusiasts. Swiss private banks and luxury goods manufacturers have long focused on high net worth clients.

Superior management, talent, relationships. The strength of strong management and founders with foresight, expertise, or relationships and contacts will be an advantage that is not easy for other companies to emulate. Entrepreneurial and business acumen and creative genius cannot be easily transferred, which has allowed some companies with the right mix of management expertise to dominate their respective fields. Other companies have benefited from the strength of relationships that management possesses with the market and governments. Examples of such companies are Steve Jobs and Apple, Bill Gates and Microsoft, Warren Buffet and Berkshire Hathaway, Jack Ma and Alibaba, and Lee Kah Shing and Cheung Kong Hutchinson Holdings.

Diversified core businesses with shared costs. Some companies regularly identify new core businesses. Such companies will typically have 3-4 core businesses where they have a competitive advantage and lead in market share. Resources would mainly be focused on their core businesses, with a smaller portion allocated to the development of new businesses. By sharing common costs (e.g., administrative, marketing, human resources, etc.), businesses will be able to compete more aggressively in their pricing and leverage on shared capabilities across divisions. Also, when one business goes through a cyclical downturn, the company's earnings can be supported by the other core businesses until it recovers. Typically, one or two of the core businesses will become cash cows for funding new business growth. When profit margins for some businesses narrow as industries mature and product life cycles reach their tail-end, such companies will seek to convert some of their promising peripheral business into new core businesses. Examples of such companies are General Electric, Honeywell, and 3M.

5.3 Quantitative Analysis (Financial Statements Analysis)

Financial statements provide information on the resources available to management, how these resources are financed, and what the company has accomplished with them. The financial statements are typically prepared for use by many parties, e.g., creditors, shareholders and investors, tax authorities, industry regulators, senior managers, board of directors, and company employees.

For this Chapter, we will be focusing on the analysis of three major financial statements, namely:

a. The Balance Sheet;
b. The Income Statement; and
c. The Cash Flow Statement.

5.3.1 Historical Financial Statements

As mentioned in Section 5.1, the current value of a company depends on its potential earnings and future cash flow performance rather than historical data. Why, then, do investors look at historical financial statements, financial ratios and data?

While financial statements present the historical performance and financial health of companies, for most companies, the most recent historical performance data still provides the best reference point for making near term forecasts. While we are generally familiar with the disclaimer that "history is not necessarily a good predictor of the future", we may be able to tilt the odds in our favour by investing in companies that have a track record of solid earnings performance and financials. Likewise, companies facing difficulties in the recent past are generally more likely to continue encountering problems unless business conditions change significantly.

Analysing past growth and financial ratios can help investors and analysts pick out the leading companies with the right strategies and competitive advantages. Analysts can draw on research about future demand and supply conditions within the industry to augment the reference that historical financial statements provide in order to better project future earnings.

5.3.2 Presenting Financial Statements

Financial statements have to be audited. The investment community expects auditors to make the necessary qualifying comments if financial statements are not well-prepared. Qualified reports are provided when the auditors are not completely satisfied with how the company's financial statements are prepared.

In the US, the norm is to provide quarterly financial statements to shareholders and the investment community. In some other markets, companies are only required to provide semi-annual and annual financial statements. In the interest of transparency, most companies in Singapore currently provide quarterly reporting.

Listed US companies follow the **US Generally Accepted Accounting Principles (US GAAP)** adopted by the US Securities and Exchange Commission. All listed European companies follow the **International Financial Reporting Standards (IFRS)**, as does Singapore. As there are variations between the US GAAP and the IFRS, adjustments and restatements must be made in order to properly compare companies across geographical markets.

5.3.3 Reviewing Financial Statements

When reviewing financial statements, we will:

a. Examine key ratios over time and explain the reasons behind the company's performance and financial health;

b. Forecast the likely future performance of the company based on the research outlook about the economy and industry, key performance drivers, as well as interviews with management and industry experts;

c. Compare the financial health and projected future performance of the company with other companies in the same industry (or league); and

d. Estimate the fair value of the company given its future earnings potential relative to other similar companies and general market conditions.

5.4 The Balance Sheet

The **balance sheet** shows the financial condition of a company at a point in time, typically at the company's financial year-end. There are three parts to a balance sheet: **assets**, **liabilities** and **shareholders' equity**. Please refer to Table 5.4.

a. **Assets** represent the resources owned by the company. They include cash, accounts receivables, inventories, motor vehicles, property, plant and equipment, among others.

b. **Liabilities** are amounts owed by the company to banks, suppliers, vendors, and other non-bank parties outside of the firm. These obligations may be accounts payables, bank loans or overdrafts.

c. **Shareholders' equity** is equivalent to total assets minus total liabilities and represents the amount that the company has financed using ordinary and preferred shares. It is also equivalent to the company's share capital plus retained earnings minus treasury shares[23] (refer to Section 5.7.2).

Table 5.4 – Balance Sheet Equations

Jurisdiction	Definition
US companies have Assets on one side with Liabilities and Shareholders' Equity summing up to the same total on other side.	**Assets = Liabilities + Shareholders' Equity**
For British companies, the practice is to present Assets net of Liabilities (i.e., Net Assets on one side), which then equates to Shareholders Equity on the other side.	**Assets – Liabilities = Shareholders' Equity (i.e., Net Assets)**

For the rest of this Section, the publicly available financial statements of Singapore Airlines (SIA) are used to illustrate the analysis of financial statements. SIA's financial year starts from 1 April and ends on 31 March of the following calendar year. The financial statements used here are for financial years: 2020/2021, 2019/2020 and 2018/2019. The competitor airlines used for comparison have a financial year which is the same as a calendar year i.e., covering the period from 1 January to 31 December.

Note also that the analysis covers the period of the COVID pandemic, which came to global prominence in Q1 2020. The global pandemic brought on unprecedented policy actions by governments globally to curb its spread, principally through border controls and movements of people. As a result, the airline industry was severely hit and it would have likely been fatal if not for intervention from government actions. The effects of the unprecedented industry downturn as well as policy assistance can be seen in the financials of the global airlines. It is still too early to tell if the pandemic will have long term implications on the airline industry or what we are seeing today is just a short-term aberration. In the intervening period, however, the policy assistance received differs among the global airlines. This further complicates trend analysis and peer comparison.

Nevertheless, it is still useful and educational to feature a global brand such as SIA in the case study.

We have re-casted the **Statement of Financial Position** of SIA into a standard US balance sheet format in Figure 5.4(a):

[23] Treasury shares or reacquired stock represents the capitalisation of the company that is financed by ordinary and preferred shares.

Figure 5.4(a) – Balance Sheet of Singapore Airlines Ltd

Statement of Financial Position As at 31 March 2021 (in $ million)	2021	2020	2019
Equity attributable to owners of the company			
Share capital	7,180.2	1,856.1	1,856.1
Mandatory convertible bonds	3,496.1	-	-
Treasury shares	(133.2)	(156.0)	(171.5)
Other reserves	5,362.8	7,614.0	11,602.2
	15,905.9	9,314.1	13,286.8
Non-controlling Interests	372.2	418.6	396.4
Total Equity	16,278.1	9,732.7	13,683.2
Long-term liabilities			
Deferred Account	41.0	33.3	83.9
Deferred Taxation	1,032.5	1,335.3	2,040.3
Long-term Lease Liabilities	2,373.6	1,467.6	-
Borrowings	10,564.8	7,162.9	6,423.3
Other long-term Liabilities	506.4	1,875.7	89.1
Provisions	965.1	990.7	702.5
Defined Benefit Plans	106.6	112.5	104.5
Total long-term liabilties	15,590.0	12,978.0	9,443.6
Current liabilities			
Borrowings	907.1	2,661.0	231.1
Lease liabilities	491.4	493.0	-
Current tax payable	95.4	68.5	87.3
Trade and other creditors	2,117.2	3,016.0	3,163.6
Amounts owing to subsidiaries	-	-	-
Sales in advance of carriage	568.1	2,041.4	2,715.4
Deferred revenue	957.8	755.8	610.9
Deferred account	16.9	31.6	44.9
Derivative liabilities	130.4	1,570.8	89.5
Provisions	428.9	364.0	435.7
Total current liabilities	5,713.2	11,002.1	7,378.4
Total liabilities and shareholder equity	37,581.3	33,712.8	30,505.2

	2021	2020	2019
Long-term assets			
Property, plant and equipment	23,483.3	25,485.8	22,176.3
Right-of-use-assets	2,395.7	1,477.7	-
Intanglible assets	301.1	487.0	451.3
Subsidiary companies	-	-	-
Associated companies	833.1	817.0	1,104.5
Joint venture companies	200.2	191.5	171.7
Long-term investments	49.9	65.2	343.9
Other long-term assets	646.0	345.7	713.7
Deferred account	-	-	44.1
Total long-term assets	27,909.3	28,869.9	25,005.5
Current assets			
Deferred account	-	-	8.9
Derivative assets	156.8	169.3	371.4
Inventories	194.9	239.3	229.9
Trade debtors	939.5	820.5	1,527.2
Amounts owing by subsidiaries	-	-	-
Deposits and other debtors	117.3	330.8	93.8
Prepayments	80.7	121.1	164.9
Other short-term assets	29.4	38.9	42.8
Investments	271.8	423.5	116.8
Cash and bank balances	7,783.0	2,685.3	2,944.0
Assets held for sale	98.6	14.2	-
Total current assets	9,672.0	4,842.9	5,499.7
Total assets	37,581.3	33,712.8	30,505.2

The **Common Size Statement** is shown in Figure 5.4(b). The items on the **common size statement numbers** below are reflected as a percentage of total assets (which is 100%). The common size statement allows for an easier comparison of the balance sheet items across different companies.

Figure 5.4(b) – Common Size Statement of Singapore Airlines Ltd

Common Size Statement of Financial Position As at 31 March 2021 (in %)	2021	2020	2019		2021	2020	2019
Equity attributable to owners of the company							
Share capital	19.1	5.5	6.1				
Mandatory convertible bonds	9.3	-	-				
Treasury shares	(0.4)	(0.5)	(0.6)				
Other reserves	14.3	22.6	38.0				
	42.3	27.6	43.6				
Non-controlling Interests	1.0	1.2	1.3				
Total Equity	43.3	28.9	44.9				
Long-term liabilities				**Long-term assets**			
Deferred Account	0.1	0.1	0.3	Property, plant and equipment	62.5	75.6	72.7
Deferred Taxation	2.7	4.0	6.7	Right-of-use-assets	6.4	4.4	-
Long-term Lease Liabilities	6.3	4.4	-	Intanglible assets	0.8	1.4	1.5
Borrowings	28.1	21.2	21.1	Subsidiary companies	-	-	-
Other long-term Liabilities	1.3	5.6	0.3	Associated companies	2.2	2.4	3.6
Provisions	2.6	2.9	2.3	Joint venture companies	0.5	0.6	0.6
Defined Benefit Plans	0.3	0.3	0.3	Long-term investments	0.1	0.2	1.1
				Other long-term assets	1.7	1.0	2.3
Total long-term liabilties	41.5	38.5	31.0	Deferred account	-	-	0.1
				Total long-term assets	74.3	85.6	82.0
Current liabilities							
Borrowings	2.4	7.9	0.8				
Lease liabilities	1.3	1.5	-	**Current assets**	-	-	-
Current tax payable	0.3	0.2	0.3	Deferred account	-	-	0.0
Trade and other creditors	5.6	8.9	10.4	Derivative assets	0.4	0.5	1.2
Amounts owing to subsidiaries	-	-	-	Inventories	0.5	0.7	0.8
Sales in advance of carriage	1.5	6.1	8.9	Trade debtors	2.5	2.4	5.0
Deferred revenue	2.5	2.2	2.0	Amounts owing by subsidiaries	-	-	-
Deferred account	0.0	0.1	0.1	Deposits and other debtors	0.3	1.0	0.3
Derivative liabilities	0.3	4.7	0.3	Prepayments	0.2	0.4	0.5
Provisions	1.1	1.1	1.4	Other short-term assets	0.1	0.1	0.1
				Investments	0.7	1.3	0.4
Total current liabilities	15.2	32.6	24.2	Cash and bank balances	20.7	8.0	9.7
				Assets held for sale	0.3	0.0	-
				Total current assets	25.7	14.4	18.0
Total liabilities and shareholder equity	100.0	100.0	100.0	**Total assets**	100.0	100.0	100.0

When we examine the balance sheet of companies, we should focus on:

a. the largest items on both sides of the balance sheet;

b. the relationship between assets and liabilities; and

c. significant trends in the accounts over time.

5.5 Assets

Assets that can be converted to cash within a year are classified as **current assets**, with the rest being classified as **long-term assets or non-current assets**. The amount of **current assets** relative to **current liabilities** provide

a reflection of the short-term liquidity of the company. When we calculate ratios, we can compare current assets against current liabilities, overall liabilities, and as a proportion of total assets.

We are generally interested in how assets are allocated for productive use to generate earnings and whether asset values will change over time. We are also interested in the funding from debt and equity used to support these assets.

5.5.1 Cash and Bank Balances

Some companies keep a large portion of their earnings in cash because of the liquidity, risk and operational needs of their industry. In cyclical downturns, having a large cash balance accumulated from past earnings allows a company to continue funding its business and meet short-term obligations. The excess cash in such times can also allow a company to capitalise on the lower price of raw materials and to make direct investments in technology, property, equipment, and businesses without having to rely on additional debt or asset liquidations. Liquidity ratios will be further discussed later in this Chapter.

Please refer to the cash ratios of selected airlines set out in Table 5.5.1(a).

Table 5.5.1(a) – Cash/Total Assets Ratio of Selected Airlines

Cash/Total Assets
(December 2020)

Airline	Percentage
IAG	19.1%
Cathay Pacific	9.5%
American Airlines	0.4%
SIA (Mar' 2021)	20.7%

In the earlier example of SIA, the company maintained for several years a cash war chest that was much higher than that of most airlines. More recently, the proceeds from the capital raised (rights issue and mandatory convertible bonds) at the height of the pandemic in mid-2020, is kept mostly in cash, presumably as a war chest; to conserve liquidity should the downtown become protracted or for future investment in plant and machinery when the market recovers.

Note that in the case of IAG (International Airlines Group, the parent of British Airways and Iberia), the 'high' cash ratio is a result of a reduced balance sheet (denominator) brought about largely by a decline/depletion in shareholders' equity (from losses and reduced equity reserves) exacerbated by the pandemic.

Holding high cash balances has its disadvantages as well. When not invested in the business, cash balances represent underutilized assets. Keeping a perpetually large cash balance would also mean sacrificing potential returns that could have been generated from investing in other financial assets. In some countries, large cash balances can make companies as targets for acquisitions.

High cash balances and cash generation capabilities also suggest potential for high dividend payouts. Management can choose to set aside cash for larger dividend payouts over time if it does not see the need to make additional investments. Companies in cash-based businesses like gaming and mature industries like tobacco (e.g., Genting and Las Vegas Sands from the gaming industry and Altria, BATS, and Swedish Match

from the tobacco industry) will naturally run a high cash balance on their balance sheets and provide high dividend payouts. Please refer to Table 5.5.1(b).

Table 5.5.1(b) – Cash balances of Various Companies in the Gaming and Tobacco Industries

	Cash as a Percent of Current Assets %					Cash as Percent of Total Assets %					Dividend Payout % (DPS/EPS)				
Company	2016	2015	2014	2013	2012	2016	2015	2014	2013	2012	2016	2015	2014	2013	2012
Altria	62.9	48.2	48.3	48.2	45.9	9.9	7.5	9.6	9.1	8.2	32.3	81.3	78.1	81.4	82.5
BAT	17.8	20.0	19.9	22.1	23.1	5.5	6.2	6.9	7.8	7.6	68.0	66.9	63.4	65.7	68.4
Swedish Match	48.4	35.0	35.2	51.0	47.8	21.9	11.7	14.0	21.3	19.7	111.2	230.6	65.0	65.2	59.3
Genting	85.0	85.5	58.6	58.6	70.4	43.4	41.6	29.2	27.8	33.8	135.7	161.3	23.8	20.8	20.8
Las Vegas Sands	68.7	60.4	67.6	65.4	56.2	10.4	10.4	15.7	15.9	11.4	137.1	105.3	56.7	50.2	202.7

5.5.2 Inventories

Inventories do not feature significantly on most airlines (usually 2-3% of all assets). However, inventory features significantly in luxury/specialty retail, food and beverage, and manufacturing companies. During periods of economic downturn, rapid technological advancement, and change in tastes, older products may become obsolete and more difficult to sell. As a result, keeping too much inventory can result in major price discounts in order to move the goods and there could be major write-downs in value if the inventory becomes totally obsolete, outdated, or goes bad (as is the case for grocers and supermarkets).

Table 5.5.2 – Inventory to Assets Ratio of Various Industry Sectors

Inventory/Total Assets (2017)

Jewellers	Percent
The Hour Glass	51.3
Chow Tai Fook	59.2
Sookee Group (2016)	44.1
Tiffany & Co (2016)	42.3

Inventory/Total Assets (2017)

Food Companies	Percent
Tung Lok Group	5.5
Domino's Pizza (USA)	5.6
Dairy Farm Group	19.0

Inventory/Total Assets (2017)

Tech Companies	Percent
Apple (2016)	< 1
Microsoft (2016)	1.1
Foxconn (2015-2016)	14.9 – 18.3

As set out in Table 5.5.2, the inventory of jewellers has very high value. As their inventory can retain its value for a very long time, jewellers can afford to have very high inventory balances. In some years, some of the inventory they hold can actually appreciate in value (e.g., gold, diamonds). These companies will also keep high cash balances to maintain enough liquidity in order to weather cyclical downturns.

In contrast, the inventory balances for food companies are maintained at a lower level to minimize the risk of spoilage. Inventory as a percentage of total assets at restaurant chains are kept low. For grocers and supermarkets, where frozen food is carried, inventory balances tend to range from 18 - 22%.

Technology companies are concerned with obsolescence due to the speed of technological development. As such, some technology companies have outsourced part or most of their manufacturing activities to sub-contractors. By doing so, they do not have to deal directly with the rapid obsolescence of inventory and the need to upgrade plants and factories when technologies change, and new products have to be created. Instead, the sub-contractors will be the ones that have to deal with these risks.

When we carry out quarterly reviews of inventory balances for retailers, one warning signal to watch for is the sudden build-up in inventories amidst falling sales numbers. If products (e.g., sports shoes) are staying longer on the shelves for brick-and-mortar retailers, it may mean that the company is struggling with sales.

However, an inventory build-up in the accounts can be due to positive reasons as well, such as:

a. an increase in the finished goods balance in anticipation of greater demand in the next period;

b. an increase in the raw material inventory ahead of rising prices or an opportunity to buy raw materials at very low-price levels; and

c. a change in the mix of products being manufactured.

Hence, it is important to investigate the underlying reasons for the build-up, instead of forming conclusions from the numbers alone.

5.5.3 Account Receivables

Account receivables are payments owed to a company by its customers. These payments are for products or services delivered on credit terms. If a company is a cash-based business (such as supermarkets and restaurants), accounts receivables should not be high. For instance, airline passenger ticket sales are largely a cash-based business (usually 95% or more), hence accounts receivables usually make up less than 5% of an airline's assets. On the other hand, manufacturing companies that sell products on credit to other companies will have to be more concerned about receivables, which may become bad debts.

We should pay attention to the quality of accounts receivables during major market downturns. If companies are highly dependent on a few buyers, then the accounts receivables may unexpectedly form bad debts if one or more of the main buyers become insolvent. By knowing a company's main buyers or clients, analysing the portion of sales to them, and tracking their performance, we may be able to spot early signals of potential problems with sales and accounts receivables. If there are falling sales and a build-up in accounts receivables during economic recessions, it could signal problems with collection and that some accounts receivables may turn into bad debts in subsequent quarters.

If there is a rise in accounts receivables from quarter to quarter in conjunction with rising sales, it could mean that the company is using a strategy of offering more credit terms to distributors and retailers so that its goods will have an advantage over competing products. If there are low cash sales, it could suggest that there are more credit sales and hire purchase sales. Hire purchase sales generate additional interest earnings for the company and can be a very lucrative strategy (e.g., Courts furniture shop in Singapore, car dealers offering credit sales alongside banks in some countries, etc.).

A reduction in accounts receivables (for example, by shortening the receivables collection period) raises liquidity. This may be an indication that the company is encountering liquidity challenges, or it may simply be an initiative by the company for a more efficient collection system.

5.5.4 Investments

Some companies hold sizeable amounts of investments. Investment holding companies listed on the exchange are in the business of managing investments, one of the most famous with investments featuring prominently being Warren Buffet's Berkshire Hathaway.

When businesses generate strong cashflows, managers may choose to invest some excess cash in financial assets instead of reinvesting entirely back into the business. Some will also park cash in short-term securities

until they can find a better use for the additional funds. Over time, the size of the investment portfolio can become quite large. In the example of SIA, short and long-term investments combined amounted to no more than $500 million for each of the last 3 years.

For some companies, their investment income streams may be even larger than the income from their core business. When the size of investments becomes too large, they should be outsourced to different fund managers to ensure that there is diversification, regular professional monitoring, and proper accountability. Companies can take on the task of self-managing their investment portfolios as long as there are sufficient checks and balances to ensure that best practises are adopted.

Investing in relatively low-risk, yield-oriented investments will provide a steady investment income stream. Holding riskier investments can lead to significant mark-to-market (MTM) losses when the markets turn negative.

Accounting best practices dictate that we reflect investments at market value. However, in some countries, more latitude is given to the accounting for unrealised gains and losses. For example, in some cases, companies are allowed to re-classify short-term investments into long-term investments at cost (or at book) with no immediate requirement to MTM. Investment professionals and representatives should be aware of such potential accounting manipulation when reviewing short and long-term investments.

5.5.5 Plant, Property and Equipment

Returning to the SIA example, the largest item on SIA's balance sheet is its Aircraft, Parts and Engines account. This, together with Plant, Property and Other Equipment, account for 62.5% of overall assets. Other airlines may classify both items under **Plant, Property and Equipment (PPE)**. The large balances of PPE are characteristic of capital-intensive industries (i.e., shipping, airlines, railway, steel manufacturing, plantations and mining). Table 5.5.5 sets out the PPE/Total Assets ratio for various airlines.

Table 5.5.5 – Aircraft, Plant, Property & Equipment as a Percentage of Total Assets for Various Airlines

Aircraft, plant, property & equipment as a percentage of total assets (December 2020)

Airline	Percentage
IAG	57.9%
Cathay Pacific	64.5%
American Airlines	51.1%
SIA (Mar' 2021)	62.5%

Some considerations when reviewing the PPE of a company are:

a. the age of the plants, equipment (aircraft, vessels, etc.);

b. whether the technology employed is obsolete;

c. the estimated cost of upgrading; and

d. how resources are deployed to fund upgrades.

In the case of SIA, the renewal and upgrade of its aircraft fleet and equipment are largely supported by equity capital and reserves, and SIA carries out this renewal process diligently as part of its branding and value proposition to customers. For manufacturing, a major concern with large commitments to plants and factories is that they can become obsolete quite quickly due to the pace of technological advancement and disruption.

When there are competing technologies with eventual victors, technological obsolescence must be managed. If the choice of technology becomes obsolete too quickly, there could be massive write-downs of existing plant and equipment and the need to spend on new replacement technology. Also, industries that go through such major changes will have to retrain staff.

As mentioned, some companies have chosen to pass on some of the obsolescence risk to its sub-contractors in order to avoid this problem. This allows them to keep their spending on PPE manageable despite the rapidly changing environment they face. For example, in Apple's 2015/16 accounts, PPE accounts for only 7-8% of overall assets.

5.5.6 Property

In Singapore, Hong Kong and most of Asia, property companies (developers and owners of commercial and retail properties, and real estate investment trusts (REITS)) are blue-chip stock investments[24]. Naturally, property holdings form the bulk of the value of these companies (typically 55-70%) and function both as an earnings generator and as a long-term "store-of-value". Once we factor in holdings through associates and joint ventures, the property holdings of these companies account for an even larger proportion (70-85%) of overall asset value. Please refer to Table 5.5.6.

Table 5.5.6 – Property as a Percentage of Total Assets for Various Real Estate companies (2016)

Property Companies	Investment Pty., Landbank & Development Pty. (Percent)	Current Development Property (Percent)	Total Property Holdings (Percent)
City Development	37.8	26.3	64.1
CapitaLand	43.2	10.6	53.8
UOL	47.3	10.2	57.5
Guoco Land	41.1	31.2	72.3
Note: All figures have yet to include property holdings through associate companies and joint ventures.			

Properties developed and completed for sale in the current period are "inventory" for property companies and are reflected as part of current assets. Correspondingly, the undeveloped land bank, developments beyond a year, and investment property held for rental income (e.g., commercial, retail) are classified under long-term assets.

Property companies and their values can be significantly affected by economic slowdowns when inventory and asset values are revised downwards. Though rare, write-downs in the value of properties have occurred in the past during major economic downturns. Upward revisions in property value, however, are more common due to re-zoning, reclassification of the use of land and property, and the general price appreciation of land due to economic growth in Asia. For example, companies in Singapore and Malaysia that initially acquired industrial and agricultural properties as part of their natural course of business later found the opportunity to redevelop these properties for residential use in the 1990s. Land values in Singapore have risen about 14x or about 6.8% per year over the last 40 years (1976-2016)[25].

[24] A blue-chip stock is a huge company with an excellent reputation. These are typically large, well-established, and financially sound companies that have operated for many years and that have dependable earnings.
[25] https://tradingeconomics.com/singapore/housing-index, URA.

Due to the sizeable value of property holdings, significant value is locked in physical property. As this can lower the productive use of capital, many property companies have sought different ways to unlock the capital. One way of unlocking capital is the injection of non-core property holdings into REITS for listing. The capital freed can then be used for new investments that have the potential to generate higher returns.

During economic downturns with high unemployment, investors should pay close attention to the level of available funding support for property holdings with large development commitments. Residential properties under development may stay on the books for a much longer time and incur more interest costs and other charges. Small property developers face a higher risk of insufficient liquidity. During downturns, lines of credit from bankers will also tighten or become costlier, especially for the smaller developers. If they have insufficient cash reserves and financing in relation to development commitments, the over-commitment may result in a liquidity crunch. In previous property downturns, several small developers were either forced to liquidate choice properties at the bottom of the market cycle or were acquired. Investors looking at the balance sheets of smaller property companies should therefore consider the size of their development commitments in relation to funding and liquidity.

5.5.7 Intangible Assets

Intangible assets of value often include goodwill, patents, copyrights, trademarks, and licences, which have a long predetermined legal life. Many intangible assets (goodwill is a notable exception) are amortized over time.

For companies that have gone through a few mergers and acquisitions, a common intangible asset booked on the balance sheet is goodwill. Goodwill refers to the premium paid by an acquiring company that is above the book value of the target company. The acquiring company is willing to pay a premium because of certain qualities that the target company possesses such as brand value, proprietary technologies, or superior management expertise. Amongst the intangible assets, goodwill represents a value premium over the fair market value of tangible assets. As such, there are many instances of large write-downs of goodwill. For example, Energy companies wrote down more goodwill than businesses in many other industries due to the sharp drop in oil prices caused by the Covid-19 pandemic. So far, the largest impairment in 2020 was Baker Hughes Co. ’s $14.77 billion write-down. The goodwill is associated with an oil-field services and equipment business, most of which Baker Hughes acquired in its merger with General Electric Company’s oil and gas division in 2017.

Many companies make the effort to write-down goodwill even when there appears to be no urgent need to do so. A goodwill write-down is typically accompanied by a direct charge against equity reserves. As unproductive asset value is also removed from total assets, but nothing much really changes operationally or businesswise, the outcome from the goodwill write-down is that the Return on Assets (ROA) and Return on Equity (ROE) figures in subsequent years would be higher as a result of the lower denominators. Some bank Chief Executive Officers (CEOs) who are remunerated on the basis of improvements to ROA and ROE will be able to deliver “better performance” over their tenure in subsequent years, but if you really do the homework, you will realise the improvements are due more to the systematic write-downs of asset and equities that appear as better performance in later periods.

5.6 Liabilities

5.6.1 Debt and Bank Borrowings

When looking at the liabilities side of the balance sheet, we pay particular attention to bank borrowings and other forms of debt financing (e.g., bond issues). We are interested in the:

a. size of debt used to finance assets and relative to equity;

b. funding mix for the debt obligations;

c. currency denomination of the debt; and

d. security and terms and conditions of the debt.

a. **Size of Debt and Bank Borrowing**

The most obvious consideration relates to the size of debt obligations that are financing the company's assets. The higher the debt level used to fund the asset base, the greater the financial risk of the company. Over reliance on debt in low interest rate environments can cause difficulty in servicing that debt if interest rates start to rise significantly over time. As such, there should be an appropriate mix between debt and equity financing and a company's financing approach should be compared against the norm for the industry.

Companies with larger debt and borrowings tend to be those that:

i. are regularly engaged in major infrastructure or property projects (e.g., construction and engineering companies, property developers, etc.);

ii. require large asset commitments (e.g., steel and iron foundries, power generation plants, purchase of ships and aircraft, mining, oil-drilling, etc.); and

iii. are involved in leveraged mergers and acquisitions.

While perpetual refinancing is an approach adopted by some, such refinancing can suddenly become very difficult and costly during economic downturns or if there are project overruns due to miscalculations.

Having sufficient cash flows to meet immediate obligations and the ability to mobilize liquid assets to retire debt commitments are especially important considerations for bond investors. These considerations will be expanded upon when we look at solvency ratios and liquidity ratios later in this Chapter.

b. **Funding Mix for Debt Obligations**

Ideally, companies should seek to maintain an appropriate mix between:

i. Short-term and long-term financing;
ii. Funding from different banks and the capital markets; and
iii. Funding from shareholders.

Excessive funding of long-term assets with short-term financing may lead to liquidity problems because of insufficient short-term assets covering short-term obligations.

During downturns, long-term borrowings can suddenly become short-term obligations if banks decide to deny credit extensions. There are often enough clauses in the documentation that will allow banks to shorten the credit term. Hence, it is important for a company to have an appropriate mix of funding sources, so it is not overly dependent on just one or two banks. For companies that are large enough, bond issues through the

capital markets during low interest rate cycles can provide an additional source of cheap financing beyond bank borrowings.

c. **Currency Denomination of Debt**

An unhedged foreign currency debt obligation can be precarious. Companies borrowing in a foreign currency because of low interest rates (e.g., Japan) run the risk of the overall debt obligation suddenly becoming larger if that foreign currency appreciates.

In 1997, many companies in Southeast Asia financed their business using Eurobonds; specifically, borrowings denominated in USD. These borrowings provided lower rates compared to the much higher Asian rates in local currencies. Carry trade was exploited by borrowing in USD and investing in the local debt. When the Rupiah and the Baht fell by more than 75% during the Asian Currency Crisis, the USD Eurobond borrowings suddenly quadrupled in value (in local currency terms) and many companies found that their local currency cash flows and liquid assets could not cover the USD-denominated debt. Several companies had to declare bankruptcy due to their USD-denominated borrowings.

5.6.2 Accounts Payables

Accounts payables show the short-term credit extended by a company to its suppliers for various goods and services purchased. The longer the period before payment, the longer the effective financing received. To improve their operating cash flow position, a company can try to stretch the payment period for their accounts payables while shortening the collection period for their accounts receivables.

If there are signs in the quarterly statements that a company is stretching payables as indicated by a build-up of payables, it could suggest that the company is trying to increase liquidity. A negative scenario could be that the company is facing difficulty in making payments, which may be signalled by a build-up of payables each quarter accompanied by falling sales.

5.6.3 Deferred Taxes

The amount of deferred taxes expected to be paid in the current period is classified as current liabilities while portions that are payable in future years are booked as long-term liabilities. When a company's earnings consistently rise year-on-year and the company has a policy of deferring a portion of tax, the amount of deferred taxes tends to rise. The constant deferring of taxes provides a form of free capital. However, should earnings fall sharply or turn negative, the amount of deferred taxes will fall.

5.6.4 Pension Obligations

Less popular in Singapore, pension obligations are liabilities that feature in countries without a national retirement fund. Instead, a pension is set aside as a retirement benefit for employees who have a long-term career with a company. Some international companies (especially those with exposure to the US) have pension funds for their international staff.

In markets like the US, where pension obligations are monitored under the requirements of the Employee Retirement Income Security Act 174, pension obligations can become a major problem for companies if they are not properly managed and the funding set aside for them is insufficient.

5.7 Shareholders' Equity

Shareholders' equity reflects the accumulated net worth (or accounting book value) of a company and shows the value of what shareholders own after all the company's liabilities have been paid off. It consists of the following:

- Share capital;
- Reserves;
- Provisions; and
- Foreign exchange adjustments.

5.7.1 Share Capital

Share capital refers to the original amount of capital provided by **common shareholders** and **preferred shareholders**. Preferred shareholders receive a fixed amount of dividend payment and receive dividends before common shareholders. However, they do not benefit from future growth in earnings.

In contrast, the dividends of common shareholders have the potential to increase with earnings over time, but they will only be paid after the preferred shareholders have received their fixed payments. Common shareholders are therefore residual owners of the company and earn the profits that remain after all other obligations have been paid. The share capital is the amount of capital issued out of the total share capital that the company has been authorized to issue.

5.7.2 Treasury Stock

This is the portion of stock that has been bought back and "retired" by the company. When a company buys back stock, the remaining shareholders will own a larger share of the earnings and dividend payouts. Performing stock buybacks will create a treasury stock account and will effectively benefit the remaining common shareholders.

5.7.3 Reserves and Hidden Reserves

A company's accumulated net earnings (or losses) over the years (after paying off dividends) are reported in reserves or capital reserves. Write-downs in the value of long-term assets/investments and provisions for losses are sometimes charged directly against reserves, instead of flowing through the profit and loss statement. This is because some of these write-downs and provisioning charges are not seen as part of the company's normal operating performance.

For example, banks make huge provisions for loan losses during economic downturns, but the actual losses may not actually materialize. When economies subsequently recover and customers start repaying what was previously classified as bad loans, some banks may be slow in reversing their provisioning. This creates hidden reserves. Similarly, hidden reserves can also be created by excessive write-downs of assets like real estate and by provisions for extraordinary losses.

Having reserves means that a company is probably worth more than what is stated in its accounts. Some companies will choose to write back their provisions under their profits or extraordinary profits to smooth or lift earnings performance in subsequent years.

5.8 The Income Statement

Amongst the different financial statements, the **income statement** or **profit and loss statement** remains the main focus for most investors since it reflects the sales and earnings performance. It is also accompanied by comments reflecting the outlook for future earnings. The income statement can be recast as a common size statement based on sales revenue.

The income statement of Singapore Airlines Limited ("the Group") in Figure 5.8 is used to illustrate the review of the income statement.

Figure 5.8 – Income Statement and Common Size Statement for Singapore Airlines Limited

Singapore Airlines Income Statement for financial year ended March 2021 (in $ million)				Common Size Statement (%)		
	2021	2020	2019	2021	2020	2019
Revenue	3,815.9	15,975.9	16,323.2	100.0	100.0	100.0
Expenditure						
Staff costs	1,160.5	2,563.6	2,816.9	30.4	16.0	17.3
Fuel costs	1,015.5	4,636.5	4,587.1	26.6	29.0	28.1
Fuel hedging ineffectiveness	214.0	709.8	-	5.6	4.4	-
Depreciation	2,075.9	2,134.2	1,327.9	54.4	13.4	8.1
Impairment of property, plant & equipment	2.0	14.2	-	0.1	0.1	-
Amorisation of intangible assets	65.8	57.5	61.9	1.7	0.4	0.4
Aircraft maintenance and overhaul costs	446.4	835.4	899.2	11.7	5.2	5.5
Commission and incentives	(13.6)	489.8	472.5	(0.4)	3.1	2.9
Landing, parking and overflying charges	219.0	886.4	884.0	5.7	5.5	5.4
Handling charges	394.1	1,276.7	1,315.0	10.3	8.0	8.1
Rentals on leased aircraft	9.8	79.2	679.7	0.3	0.5	4.2
In-flight meals	10.7	539.0	550.4	0.3	3.4	3.4
Advertising and sales costs	12.6	334.2	331.8	0.3	2.1	2.0
Company accomodation and utilities	35.7	50.2	93.5	0.9	0.3	0.6
Other passnger costs	31.1	187.7	188.0	0.8	1.2	1.2
Crew expenses	15.0	172.4	168.1	0.4	1.1	1.0
Impairment of amount owing by joint venture company	12.6	61.6	-	0.3	0.4	-
Other operating expenses	621.3	888.4	880.1	16.3	5.6	5.4
	6,328.4	15,916.8	15,256.1	165.8	99.6	93.5
Operating (Loss)/Profit	(2,512.5)	59.1	1,067.1	(65.8)	0.4	6.5
Finance charges	(267.9)	(220.9)	(116.1)	(7.0)	(1.4)	(0.7)
Interest income	35.4	42.1	41.9	0.9	0.3	0.3
Impairment of aircraft	(1,734.3)	-	-	(45.4)	-	-
Impairment of base maintenance assets	(36.9)	-	-	(1.0)	-	-
Impairment of goodwill	(170.4)	-	-	(4.5)	-	-
Impairment of intangible assets	(11.4)	-	-	(0.3)	-	-
(Loss)/Surplus on disposal of aircraft, spares and spare engines	(27.0)	6.9	(5.8)	(0.7)	0.0	(0.0)
Dividends from long-term investments	8.4	3.2	3.1	0.2	0.0	0.0
Other non-operating items	(127.8)	(31.9)	(47.4)	(3.3)	(0.2)	(0.3)
Share of profits of joint venture companies	14.0	46.4	23.2	0.4	0.3	0.1
Share of losses of associated companies	(126.8)	(125.1)	(97.4)	(3.3)	(0.8)	(0.6)
				-	-	-
Loss before taxation	(4,957.2)	(220.2)	868.6	(129.9)	(1.4)	5.3
Taxation	673.8	50.8	(147.0)	17.7	0.3	(0.9)
(Loss)/Profit for the financial year	(4,283.4)	(169.4)	721.6	(112.3)	(1.1)	4.4
(Loss)/Profit attributable to:						
Owners of the company	(4,270.7)	(212.0)	682.7			
Non-controlling interests	(12.7)	42.6	38.9			
	(4,283.4)	(169.4)	721.6			
Loss per share (cents)	(115.6)	11.2	57.7			
Diluted loss per share (cents)	(115.6)	11.3	57.4			

From the income statement, we can identify the major factors affecting a company's performance. What are some possible conclusions and inferences we can make from trends in the income statement above?

Some possible conclusions and inferences are:

a. As a result of the pandemic, top-line revenue for financial year 2020/2021 declined by a massive 76% when compared to financial year 2019/2020. However, given the more inelastic nature of costs, the cuts to operating expenses ($9.5 billion) were not enough to offset the loss in revenue ($12 billion). This resulted in an operating loss of some $2.5 billion.

b. Fuel costs, being a variable cost item, came down by more than 75% to roughly $1 billion. Fuel costs still accounted for some 26% of a reduced revenue base.

c. While the Group's operating loss came in at $2.5 billion, the pre-tax loss was bigger at $4.9 billion. This is due to non-operating losses/expenses, the significant items were impairment of aircraft ($1.7 billion), impairment of goodwill ($170 million), losses of associated companies ($127 million) and others ($128 million). The impact of these non-operating items resulted in the decline of the final results appearing more severe.

d. The common size statement also makes the nature of the business clearer - we can see that the Group was a volume-driven business with small profit margins. Operating profits only made up 6.5% of sales for the financial year 2018/2019, which was considered a good year before the onset of the pandemic. After taking in non-operating expenses, including financing charges, net margin before tax was 5.3% of sales. Like other airlines, changes in key factors like fuel prices or intensified price competition on airfares can dramatically affect profitability.

Aside from the key trends observed in the financial statements, investors can also evaluate the accompanying qualitative comments made during press and analysts' briefings. Such comments will also reveal management's outlook for the business and the key variables they are focusing on. These, along with other research inputs, will lead to a forecast of the company's future sales and earnings.

5.8.1 Sales and Revenue

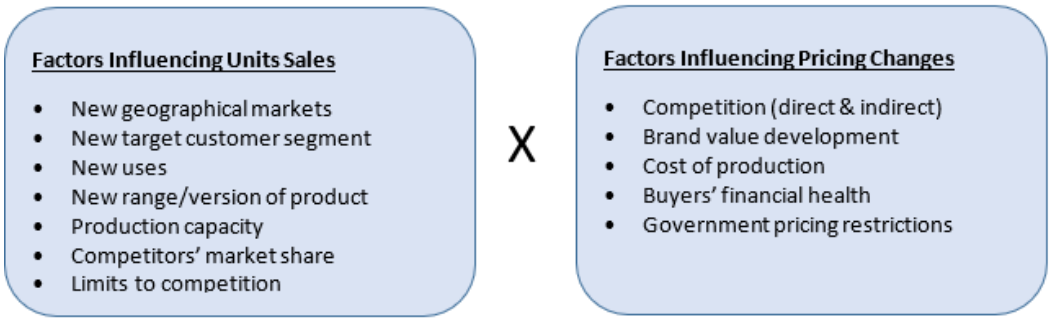
a. **Drivers of Sales Revenue**

When analysing the key drivers of sales growth, we should break the sales revenue component into two parts:

i. Unit sales growth; and
ii. Pricing changes.

Whenever a company raises either of the two, sales revenue will increase. Hence, in reviewing and forecasting revenue, we should pay close attention to factors that can drive unit sales growth and cause price changes as shown in Figure 5.8.1(a).

Figure 5.8.1(a) - Factors Driving Revenue Growth



Returning to the example of airlines, the ability to secure new lucrative routes would lead to new ticket sales (more unit sales). Alternatively, the successful implementation of service quality and branding allows an

airline price at a premium (better pricing). Conversely, new players with deep financial pockets could lead to strong competition and negatively affect sales revenue (less unit sales).

b. **Segmental Breakdown of Revenues and Earnings**

To gain a clearer picture on the contribution from various lines of businesses, we can break down and review the revenues and profits by segments. Common segmental breakdowns are by:

i. Geographical markets;
ii. Businesses divisions;
iii. Products and services; and
iv. Customer segments.

In the case of SIA, revenue is first broken down into its various business divisions/customer brands. The revenue is then further segmented by geographical origin.

The impact of the major downturn caused by the Covid-19 pandemic appeared to have a systemic impact across both customer segments and geographies, as shown in Figure 5.8.1(b).

Figure 5.8.1(b) - Segmental Breakdown of Revenue and Profits for Singapore Airlines

Singapore Airlines - Segmental Breakdown of Revenue and Profits ($ mn)

	Revenue				Net Profit		
	2021	2020	2019		2021	2020	2019
SIA	3,422.1	12,918.0	13,054.2		(3,066.8)	(22.2)	635.6
SilkAir	30.1	888.2	1,011.1		(342.4)	(103.8)	(52.3)
Budget Aviation	125.7	1,624.8	1,710.8		(847.0)	(252.1)	(59.2)
SIAEC	163.4	444.9	485.7		(19.6)	190.5	161.0
Others	74.6	100.0	61.4		(13.1)	9.2	13.2
Total Revenue	3,815.9	15,975.9	16,323.2		(4,288.9)	(178.4)	698.3
				Eliminations	5.5	9.0	23.3
				Net Profit	(4,283.4)	(169.4)	721.6

Segmental Revenue - Revenue by Origin of Sale ($ mn)

	2021	2020	2019
East Asia	2,253.5	8,485.1	8,966.2
Europe	530.1	2,169.8	2,283.0
South West Pacific	302.5	2,114.5	2,091.8
Americas	120.7	970.2	889.0
West Asia and Africa	187.6	1,095.3	1,028.8
Systemwide	3,394.4	14,834.9	15,258.8
Non-scheduled services and incidental revenue	303.2	764.4	696.4
Total Revenue	3,697.6	15,599.3	15,955.2

Research analysts in brokerages, banks and fund houses will do a further detailed breakdown by different airline services, and also carry out some model forecasting given the estimated fleet, load factor and average ticketing fee for each airline. This allows them to come up with a forecast of sales revenue and earnings, before applying a discounted cashflow model (see Chapter 6) to come up with a fair target price for the stock based on the projections.

The segmental breakdown and model forecasts help to highlight the main sources of revenue and income. By tracking news related to the sales and performance of key businesses and new products and services from each division, we would be able to better anticipate the impact on overall growth in revenues and profits. Companies also provide segmental breakdown of assets to show resource allocation, i.e., how assets are deployed in the business.

5.8.2 Interest and Investment Income

For cash-rich companies that have generated strong cash flows over several years, excess cash usually finds its way into financial investments. This usually leads to large interest and dividend flows, which may not really be part of a company's core business activity. In the case of SIA, interest income amounted to $35 million, while dividend income was $8.4 million for financial year 2020/2021.

While many successful companies with excess cash have made investments that generate steady income contributions, there have also been cases where companies made speculative investments and suffered losses that wiped out the profits from their core businesses. It is therefore important to review the accounting notes to better understand the type of investments that the company has made. This will help investors gain a better picture of the likely income volatility associated with these investments.

5.8.3 Costs and Expenses

When reviewing costs, close attention should be to potential changes in the major cost items, measured as a percentage of sales.

a. **Major Cost Items**

For manufacturing companies, production costs (which includes labour costs, raw materials costs, and utility costs) are factored into the cost of goods sold. Raw material costs typically feature significantly in manufacturing. If raw material prices fall, and the manufacturing companies can keep finished goods prices and sales constant, then there will be increased profit margins. Some manufacturing companies reflect costs relating to production as separate from the costs of marketing, distribution and administration. For such companies, costs would be broken down into the **cost of goods sold** (which focuses on production cost) and **selling and administrative costs** (which focuses on marketing, distribution and administration costs). Research and development costs are another significant cost that does not go directly into production.

For distribution and service-oriented companies, the practice is to reflect **cost of sales** or **operating cost** before operating profit.

When reviewing costs and expenses, we should take note of:

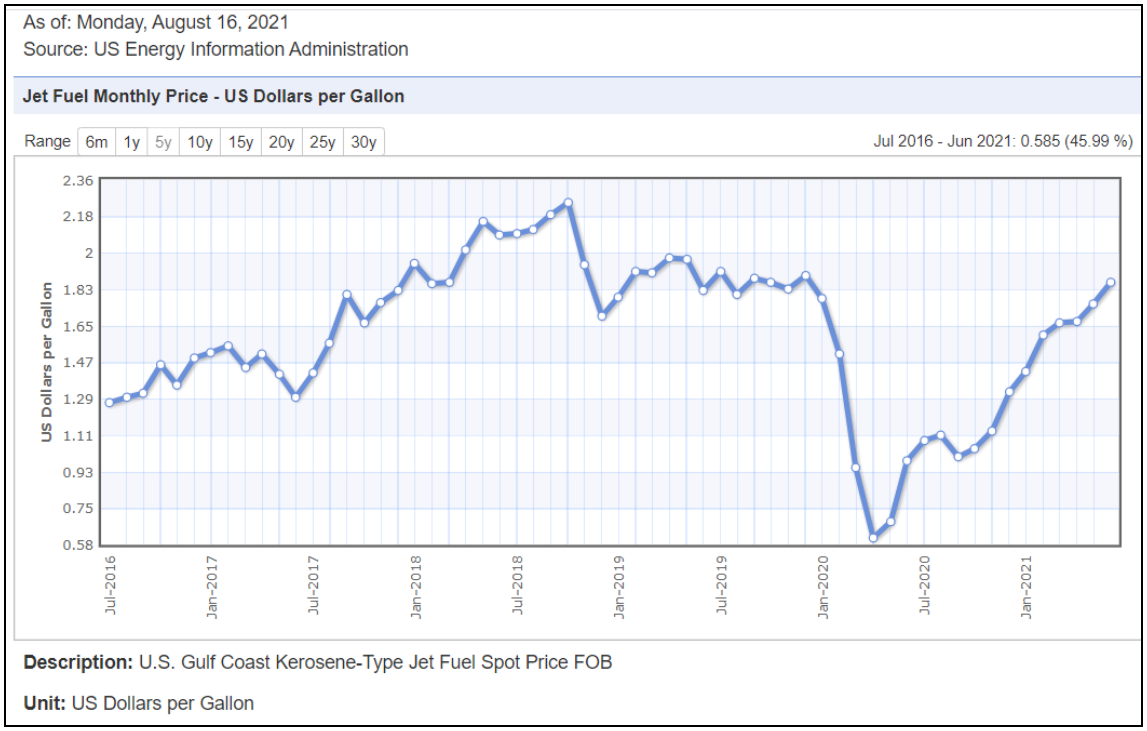
i. Major cost items measured as a percentage of sales;

ii. Significant rising or falling trends in cost (e.g., raw materials, fuel costs); and

iii. Cost items that are non-cash accounting costs (which do not affect cash flows).

As expected, major cost items differ from industry to industry. In service industries like banking and hospitality, the largest costs tend to be staff costs. In the transportation industry, fuel costs, staff costs and depreciation expenses will tend to be the main cost items.

Figure 5.8.3(a) - Jet fuel prices



For SIA, the major cost items were staff costs (30.4% of revenue in 2020/2021 and 16% in 2019/2020) and fuel costs (26.6% of revenue in 2020/2021 and 29% in 2019/2020).

From Figure 5.8.3(a), we can see that jet fuel prices can be highly volatile. After the precipitous fall in H1 2020, triggered by the pandemic, the recovery of jet fuel price was equally stunning. As it is often difficult to pass on higher fuel costs to customers through higher ticket prices, higher fuel price would typically cut into an airline's profitability. Airlines hedge their fuel costs by taking forward and futures position for jet fuel. Hedging may also bring its own unique problems. While the purpose of hedging is to reduce cost uncertainty, airlines that bought jet fuel futures/forwards for delivery in H1 2020 may find that they become 'over-hedged' when demand for air travel collapsed. Hence the 'unused' long hedges had to be unwound at a loss when jet fuel prices declined. Even if demand for air travel did not collapse, airlines that have hedged their fuel may find themselves, having locked in at forward prices, uncompetitive to those airlines that have not hedged when jet fuel prices declined unexpectedly.

b. **Depreciation**

Another major cost item is **depreciation expense**. For capital-intensive companies, depreciation expense can be significant. Depreciation for SIA in 2020/2021, albeit under atypical market conditions, accounted for more than 50% of sales and roughly 8% for 2018/2019 and yet depreciation is not a real cash outflow but is only a method of accounting. For transport-related companies, which must make large capital commitments on aircraft or ships on an ongoing basis, depreciation expense will always be quite large.

Depreciation expense can be a source of abuse. When a company is not doing well, one cosmetic approach is to change its accounting practice by stretching depreciation expense over more years. For example, if a taxi fleet operator that usually depreciates its taxis over a 5-year period suddenly decides to depreciate them over 10 years, it will dramatically lower depreciation expense. This, in turn, will have a positive cosmetic effect on net profit in the accounts. However, nothing has changed in terms of cash flow.

5.9 The Cash Flow Statement

The statement of changes in cash flows is useful for investors since it provides a picture of the real cash earnings generated from operations. Items like depreciation, amortization, and losses from associates are among some of the major adjustments that need to be made in order to reflect cash flow from operating activity. There are 3 parts to the cash flow statement:

a. Cash flow from operating activities;
b. Cash flow from financing activities; and
c. Cash flow from investing activities.

Figure 5.9(a) – Consolidated and Common Size Statement of Cash Flow for Singapore Airlines Limited

Singapore Airlines - Consolidated Statement of Cash Flows
For the financial year ended 31 Mar 2021 (in $ million)

				Common Size Statement (%)		
	2021	2020	2019	2021	2020	2019
Cash Flow from Operating Activities						
(Loss)/Profit before taxation	(4,957.2)	(220.2)	868.6	150.6	(8.1)	31.0
Adjustments for:						
Depreciation	2,075.9	2,134.2	1,327.9	(63.1)	78.1	47.4
Impairment of aircraft	1,734.3	-	-	(52.7)	-	-
Impairment of property, plant and equipment	2.0	14.2	-	(0.1)	0.5	-
Impairment of base maintenance assets	36.9	-	-	(1.1)	-	-
Impairment of goodwill	170.4	-	-	(5.2)	-	-
Impairment of intangible assets	11.4	-	-	(0.3)	-	-
Amortisation of intangible assets	65.8	57.5	61.9	(2.0)	2.1	2.2
Imparment of trade debtors	4.6	7.5	5.7	(0.1)	0.3	0.2
Impairment of amount owing by a joint venture company	12.6	61.6	-	(0.4)	2.3	-
Writedown of inventories	12.4	8.8	5.4	(0.4)	0.3	0.2
Income from short-term investments	(1.2)	(1.4)	(1.1)	0.0	(0.1)	(0.0)
Provisions	207.0	180.0	302.2	(6.3)	6.6	10.8
Share-based compensation scheme	13.2	20.5	21.0	(0.4)	0.8	0.7
Exchange differences	45.5	(11.0)	(2.4)	(1.4)	(0.4)	(0.1)
Net (gain)/loss on financial assets mandatorily measured at fair value through profit ir loss ("FVTPL")	(2.6)	0.5	(0.7)	0.1	0.0	(0.0)
Fuel hedging ineffectiveness	214.0	709.8	-	(6.5)	26.0	-
Foreign currency hedging ineffectiveness	12.6	(30.0)	-	(0.4)	(1.1)	-
Amortisation of deferred loss on sale and operating leaseback transactions	-	-	4.3	-	-	0.2
Finance charges	267.9	220.9	116.1	(8.1)	8.1	4.1
Interest income	(35.4)	(42.1)	(41.9)	1.1	(1.5)	(1.5)
Loss/(Surplus) on disposal of aircraft, spares and spare engines	27.0	(6.9)	5.8	(0.8)	(0.3)	0.2
Dividends from long-term investments	(8.4)	(3.2)	(3.1)	0.3	(0.1)	(0.1)
Other non-operating items	127.8	31.9	47.4	(3.9)	1.2	1.7
Share of profits of joint venture companies	(14.0)	(46.4)	(23.2)	0.4	(1.7)	(0.8)
Share of losses of associted companies	126.8	125.1	97.4	(3.9)	4.6	3.5
Operating cashflow before working capital charges	149.3	3,211.3	2,791.3	(4.5)	117.5	99.7
Decrease in trade and other creditors	(2,621.5)	(346.2)	(28.1)	79.6	(12.7)	(1.0)
Decrease in sales in advance of carriage	(1,473.3)	(674.0)	273.3	44.7	(24.7)	9.8
Decrease in trade debtors	225.8	660.3	(206.1)	(6.9)	24.2	(7.4)
Decrease/(Increase) in deposits and other debtors	215.6	(237.0)	(10.9)	(6.5)	(8.7)	(0.4)
Decrease in prepayments	40.4	10.6	19.7	(1.2)	0.4	0.7
Increase in inventories	(13.1)	(18.2)	(66.6)	0.4	(0.7)	(2.4)
Increase in deferred revenues	202.0	144.9	54.8	(6.1)	5.3	2.0
Cash (used in)/generated from operations	(3,274.8)	2,751.7	2,827.4	99.5	100.7	100.9
Income taxes paid	(17.6)	(19.8)	(26.3)	0.5	(0.7)	(0.9)
Net Cash (used in)/provided by operating activities	(3,292.4)	2,731.9	2,801.1	100.0	100.0	100.0

For SIA (refer to Figure 5.9(a), depreciation expense accounted for 63% (or $2 billion) of the (negative) operating cash flow for financial year 2020/2021. When added back to the pre-tax loss, the net impact is significant. It greatly reduced the negative operating cashflow. Under more benign market conditions of

2018/2019, while profit before tax was $868 million, SIA generated $2.8 billion in cash flow from operations, which is more than three times larger.

Figure 5.9(b) – Consolidated Statement of Cash Flow for Singapore Airlines Limited (continued)

Singapore Airlines - Consolidated Statement of Cash Flows
For the financial year ended 31 Mar 2021 (in $ million)

	2021	2020	2019
Cash Flow from Investing Activities			
Capital expenditure	(2,695.5)	(5,103.5)	(5,562.3)
Purchase of intangible assets	(74.1)	(91.6)	(105.4)
Proceeds from disposal of aircraft and other property, plant and equipment	156.4	124.3	180.2
Proceeds from sale and leaseback transactions	1,230.5	117.2	-
Purchase of long-term investments	-	-	(5.0)
Proceeds from disposal of long-term investments	30.3	41.5	157.6
Purchase of short-term investments	(584.6)	(689.1)	(798.8)
Proceeds from disposal of short-term investments	739.7	661.7	850.6
Dividends received from associated and joint venture companies	39.5	84.0	108.2
Dividends received from investments	8.4	5.2	5.2
Interest received from investments and deposits	33.7	42.2	47.8
Proceeds from finance leases	8.1	9.2	-
Proceeds from disposal of interest in a subsidiary company, net of cash disposed	5.3	-	1.6
Investments in associated companies	(212.0)	(141.0)	(205.6)
Investments in joint venture companies	-	(11.9)	(40.5)
Loan to a joint venture company	-	(18.1)	-
Proceeds from disposal of associated companies	-	-	4.0
Proceeds from liquidation of an assoicated company	-	5.1	-
Net Cash Used In Investing Activities	(1,314.3)	(4,964.8)	(5,362.4)

e.g., Direct investment purchases of Boeing or Airbus aircraft are cash outflows, while proceeds from sale of aircraft are cash

	2021	2020	2019
Cash Flow from Financing Activities			
Dividends paid	-	(355.5)	(449.8)
Dividends paid by subsidiary companies to non-controlling interests	(14.3)	(30.7)	(34.4)
Acquisition of non-controlling interests without a change in control	(4.1)	-	(4.7)
Issuance of share capital by subsidiary companies	-	1.0	19.5
Interest paid	(232.4)	(206.0)	(124.5)
Proceeds from issuance of shares and mandatory convertible bonds	8,829.2	-	1,350.0
Paymment of transaction costs related to shares issued and mandatory convertible bonds	(9.0)	-	-
Repayment of bonds	(500.0)	-	-
Proceeds from issuance of bonds	2,013.0	-	-
Payment of transaction costs from issuance of bonds	(10.9)	-	-
Proceeds from borrowings	4,579.0	3,985.2	2,280.4
Repayment of borrowings	(4,344.1)	(878.3)	(93.7)
Repayment of lease liabilities	(551.6)	(575.9)	-
Payment of transaction costs related to borrowings	(17.8)	(4.2)	(11.1)
Net Cash Provided by Financing Activities	9,737.0	1,935.6	2,931.7
Net Cash Inflow/(Outflow)	5,130.3	(297.3)	370.4
Cash and Cash Equivalents at Beginning of the Year	2,685.3	2,944.0	2,568.3
Effect of exchange rate changes	(32.6)	38.6	5.3
Cash and Cash Equivalents at End of the Year	7,783.0	2,685.3	2,944.0

e.g., Proceeds from capital raising and borrowings are cash inflows while dividends and repayment of borrowings are cash

For capital-intensive industries, a company's net earnings may not provide an accurate picture about its ability to support short-term liabilities and borrowings. As such, we need to examine the company's cash flow statement, which adds back depreciation and other non-cash items, to really assess the ability of the company

to support financing and investing activities. Hence, it is important to analyse a firm's cash flows to better understand the quality and sustainability of the business and not rely on the profit and loss alone. For most investors, especially bond holders, a major concern is when the cashflows from operations are consistently negative. Sometimes, a poor overall net cashflow position can be temporarily masked by the selling of assets and businesses, creating positive cashflows from investment activities. Sometimes, this begins with the selling of all peripheral and poorly performing businesses and excess assets but, when a company becomes desperate, it starts to sell assets and businesses that would fuel future growth or assets it can temporarily do without. If there is no recovery in the core businesses, these efforts to temporarily improve liquidity will come to nought.

In the case of SIA (refer to Table 5.9(b), the negative cash flow from operations sustained in 2020/2021 was largely funded by the issuance of shares (rights issue) and mandatory convertible bonds totalling $8.8 billion. Analysing the cash flow from investing activities, we may also infer that, for 2020/2021, certain large capital investments (e.g., aircraft purchase) were deferred. To further conserve cash flow, no dividends were paid out in 2020/2021. With the reduced investing activities and the additional capital raised, SIA ended 2020/2021 with a net cash and cash equivalent position of $7.8 billion.

5.10 Per Share Calculations – Book Value Per Share (BVPS), Dividends per Share (DPS), Earnings per Share (EPS) and Price

For per-share computations using numbers on the balance sheet, we use the number of outstanding shares at the end of the financial year. Since the balance sheet reflects a snapshot of the financial position of the company, for calculations like **BVPS,** the total number of outstanding shares at that same point is used.

When we consider dividends paid to shareholders, we use the total number of outstanding shares as at the ex-dividend date in order to calculate **DPS.**

When we calculate **EPS**, we use the weighted average number of shares outstanding because the earnings were generated over an entire year. The shares outstanding can fluctuate throughout a year with new share issuances and repurchases, among others, so the weighted average number of shares is used in the calculation of EPS. The EPS computation should also be on a fully diluted basis, which assumes all derivatives and convertibles outstanding are exercised and converted to shares. The fully diluted basis is particularly relevant for EPS projections.

Per Share Calculations

Book Value Per Share (BVPS)	= Shareholders' Equity / Total Number of Shares Outstanding
Earnings Per Share (EPS)	= Net Earnings/Weighted Average Number of Shares
Fully Diluted Earnings Per Share	= Net Earnings/Fully Diluted Number of Shares
Dividend Per Share (DPS)	= Dividends Paid/Total Number of Shares Outstanding

The share price of a company is already on a per-share basis, and it is frequently compared to the other computed per-share numbers. For example, the forward **price-earning (PE) ratio (**is the price paid for future EPS and is a popular method of making comparisons between companies. Similarly, the **price-to-book value**

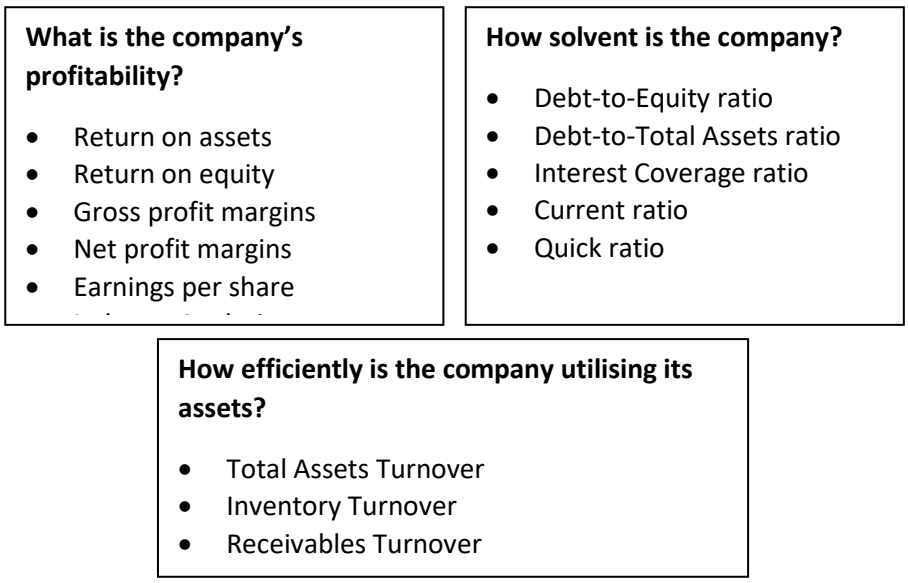
ratio is another measure that is commonly used. We will revisit these ratios later in Chapter 6 on equities analysis. To ensure comparability, we should calculate all future EPS forecasts on a fully diluted basis.

5.11 Ratio Analysis

Financial ratio analysis is an established and popular method to interpret companies' financial accounts. Based on a company's financial accounts, one can evaluate its financial strength, the "quality" of its earnings, and its performance using simple mathematical ratios (refer to Figure 5.11).

However, used in isolation, financial ratios are quite meaningless. For example, an interest coverage ratio of two times may be low for some industries, but very high for others. Hence, it would be more meaningful if the findings are compared and benchmarked against other companies in the same industry. When properly applied and interpreted, these ratios can help to diagnose the financial health of the company.

Figure 5.11 – Important Questions Answered by Financial Ratios



Key financial ratios in the following categories will be examined:

- Liquidity ratios;
- Leverage ratios;
- Operating efficiency / activity ratios; and
- Profitability ratios.

5.11.1 Liquidity Ratios

Liquidity may be defined as the availability of funds at any given time and at a reasonable cost. Liquidity ratios indicate the ability of the company to meet its short-term financial obligations.

Banks are governed by liquidity ratios levels specified by their respective local authorities. One of the lessons from the Global Financial Crisis in 2008 is the importance of liquidity, not just for banks but also for companies. While it is important to maintain strong liquidity, doing so can be expensive in times of high interest rates.

Common liquidity ratios used in the financial community are shown in Table 5.11.1.

Table 5.11.1 – Common Liquidity Ratios

Current Ratio	**:**	Current Assets / Current Liabilities
Quick Ratio	**:**	(Cash + Marketable Securities + Accounts Receivable) / Current Liabilities
Cash Ratio	**:**	(Cash + Near-Cash) / Current Liabilities
Cash flow Ratio	**:**	Cash Flow from Operating Activities / Current Liabilities

a. **Current Ratio**

The most common liquidity ratio is the current ratio, which measures the degree to which current assets cover current liabilities. Although conventional wisdom suggests a current ratio of 2 or more, the cost to having too much liquidity is the unproductive use of assets, since current assets tend to generate low returns. Companies are now more efficient in managing their working capital, resulting in generally lower current ratios.

b. **Quick Ratio**

Also known as the **acid test ratio**, it has the same denominator as the current ratio, but its numerator excludes inventories because these are considered less liquid. This ratio indicates the amount of highly liquid assets that are available to cover current liabilities, and would typically include cash, marketable securities, and accounts receivables in the numerator.

Quick ratios vary across industries. For example, the quick ratio of a supermarket may be as low as 0.2 because it has substantial inventory, but it may not be a liquidity concern because a supermarket's current assets are "fast moving consumer goods" that are financed by short-term supplier credit. Quick ratios are not as meaningful for non-retail/merchandising companies such as banks and other financial institutions, which do not have inventory.

c. **Cash Ratio**

A more stringent test of liquidity is the **cash ratio**, which looks at cash and cash-equivalents. Typically, the numerator will only include cash and short-term marketable securities that can be converted quickly to cash. Any assets that cannot immediately be converted to cash will be omitted.

d. **Operating Cash Flow Ratio**

This ratio provides a picture of the cash flow from operating activities that is available to cover the current liabilities. The current ratio of some companies may look weak, with barely enough current assets to cover current liabilities, but these companies are actually generating cash flows well in excess of their current liabilities. This is typically true of cash-based, service-type businesses, which primarily cover their short-term liabilities not by selling inventory on credit but by rendering services for cash.

Conversely, there are also cases where the current ratio looks fine, but the operating cash flow ratio looks worrying. A company in dire straits and overcommitted in loss-making projects may liquidate peripheral long-

term assets to shore up their cash position to meet near-term obligations. However, if operating cash flows and incomes remain negative, this strategy is non-sustainable, and the company may end up defaulting on its obligations later. A look at some of the defaults on Singapore corporate bonds and perpetual securities will show that most of the defaulting companies had negative operating cashflows prior to their defaults.

e. **Market Adjustments to Liquidity Ratios in Financial Crisis Situations**

During difficult market conditions, creditors reviewing a company's accounts will prefer to value the company's marketable securities at the lower of book or market. When securities have clearly fallen in value, it makes little sense to continue to use the book value when measuring liquidity.

In situations where it is clear that sales are falling and there is an inventory overhang, credit analysts may also write-down part of the inventory value by assuming that a portion cannot be sold or that inventories have to be sold off at a discount. When there is a financial crisis, even accounts receivables can become uncollectable, so creditors may discount part of the value of receivables. Additionally, if qualitative assessments are made, it is important to apply them consistently with justifications across the companies being assessed.

Finally, if current liabilities include foreign currency-denominated borrowings, when assessing liquidity, these debt obligations should be reported at prevailing market exchange rates in order to properly reflect the market value of the debt obligations. The liquidity ratios for SIA are reflected in Table 5.11.1(a) as follows:

Table 5.11.1(a) - Liquidity Ratios for Singapore Airlines (2019-2021)

	2021	2020	2019
Current Ratio			
Current Assets / Current Liabilities	9672 / 5713	4843 / 11002	5500 / 7378
	1.69	0.44	0.75
Quick Ratio			
Cash + Mkt Sec + A/C Receivables / Current Liabilities	7783 + 272 + 940 / 5713	2685 + 424 +821 / 11002	2944 + 117 + 1527 / 7378
	1.57	0.36	0.62
Cash Ratio			
Cash + Near Cash / Current Liabilities	7783 + 272 / 5713	2685 + 424 / 11002	2944 + 117 / 7378
	1.41	0.28	0.41
Operating Cash Flow Ratio			
Operating Cash Flow / Current Liabilities	-3292 / 5713	2732 / 11002	2801 / 7378
	-0.58	0.25	0.38

From Table 5.11.1(a), we can see there were significant increases in liquidity ratios, with the exception of the operating cash flow ratio, for financial year 2020/2021 from the previous year. This was largely due to the fact that most of the funding raised through share issue and mandatory convertible bonds (in 2020) was kept as cash (numerator). At the same time, a significant portion of the short-term borrowings was paid off, presumably, using funds from the capital raising. Also given the industry downturn, sales in advance of carriage were also greatly reduced. Together they had an effect on reducing total current liabilities (denominator). As for the operating cash flow ratio, the negative operating cash flow is a result of the significant trading loss sustained in 2020/2021, hence turning the operating cash flow ratio from 0.25 (2019/2020) to -0.58 in 2020/2021.

It is clear from the review that the funding raised was used to maintain a highly 'liquid' and 'defensive' balance sheet, presumably to weather a protracted downturn and also to provide a war chest for future investments, when the market recovers. Table 5.11.1(b) sets out the current ratios of various airlines.

Table 5.11.1(b) - Comparing SIA's Current Ratio with Competitors

Current Ratio	2020	2019
Cathay Pacific	0.59	0.48
IAG	0.68	0.89
American Airlines	0.67	0.45
Singapore Airlines (Mar' 2021/2020)	1.69	0.44

SIA ratios are based on end March 2021 & 2020 respectively

Current and quick ratios at financial reporting dates are susceptible to manipulation, otherwise known as "window dressing". For example, toward the financial year-end, companies can collect receivables earlier, reduce inventory below normal levels, and delay purchases. Proceeds from speeding up collections can be used to pay off current liabilities.

One should also note that an equal dollar increases in both current assets and current liabilities does not mean that the current ratio remains unchanged; an equal increase for the numerator and the denominator will lead to a lower ratio if it was previously more than 1 and to a higher ratio if it was previously less than 1.

5.11.2 Leverage Ratios - Debt and Solvency Ratios

Leverage ratios provide an understanding of a firm's financing mix between debt and equity. They also help to determine a firm's amount of financial risk and to measure its capacity to meet principal and interest repayments. These ratios are of primary interest for bondholders, whose main risk is default on principal and interest payments by the firm. For other stakeholders of the firm, these ratios are more useful when analysed together with other financial ratios of the firm.

One of the principal considerations in managing a firm's capital structure is the extent of leverage. While a higher leverage will increase the return on equity and reduce any potential dilution on earnings, it might lower the firm's credit rating and raise its cost of debt.

a. **Debt-to-Equity Ratio (Gearing Ratio)**

The debt-to-equity ratio measures the relationship between total debt and shareholders' equity. Debt includes all short- and long-term borrowings, including subordinated convertible bonds. Deferred taxes may or may not be included. Operating leases, especially if these are extensive, have to be included as an estimate of the present values of the lease payments.

By stating total debt in relation to total equity, we can see how much the firm is reliant on external borrowings, i.e.; debt in its capital structure. The greater the debt, the higher the leverage, and the greater its impact on the firm's **return on equity (ROE)**. Also, the greater the debt, the harder it is for the firm to generate sufficient income to repay the debt. The ratio is calculated as follows:

$$\textbf{Debt/Equity Ratio} = \frac{\textbf{Total Debt}}{\textbf{Total Shareholders' Equity}}$$

b. **Debt-to-Total Assets Ratio**

This measures the relationship between debt and total assets. It is an overall indicator of the portion of a firms' assets that are financed by debt. The debt includes all borrowings, both long-term and short-term, and the assets includes intangibles. Note that debts are carried at their face values on the balance sheet and are not revalued. The ratio is calculated as follows:

$$\textbf{Debt-to-Total Assets Ratio} = \frac{\textbf{Total Debt}}{\textbf{Total Assets}}$$

When we review the Total Debt-to-Equity or Total Debt-to-Assets, it is important to check for foreign currency-denominated debt. If there is appreciation in the foreign currency against the local currency, the market value of the debt will be much larger. Companies with foreign currency-denominated debt (bonds) in countries with weak currencies are especially susceptible to a ballooning of their debts due to unfavourable exchange rate movements.

c. **Total Liabilities-to-Total Assets**

This shows how much of a firm's assets is funded by its liabilities. The Total liabilities-to-Total assets ratio is calculated as follows:

$$\textbf{Total Liabilities-to-Total Assets Ratio} = \frac{\textbf{Total Liabilities}}{\textbf{Total Assets}}$$

d. **Interpreting Leverage Ratios**

The leverage ratios for SIA are reflected in Table 5.11.2(a).

Table 5.11.2(a) - Leverage Ratios for Singapore Airlines (2019-2021)

	2021	2020	2019
Debt/Equity (i.e.; Gearing)			
Total Borrowings / Total Shareholders' Equity	10565 + 907 / 16278	7163 + 2661 / 9733	6423 +231 / 13683
	0.70	1.01	0.49
Debt/Asset			
Total Borrowings / Total Assets	10565 + 907 / 37581	7163 + 2661 / 33713	6423 +231 / 30505
	0.31	0.29	0.22
Liabilities/Assets			
Total Liabilities / Total Assets	21303.2 / 37581	23979.8 / 33713	16822 / 30505
	0.57	0.71	0.55

The higher the leverage ratio, the greater the financial risk and leverage effect. SIA's improvement in the gearing ratio (debt/equity) for 2020/2021 from previous year, was driven principally by an increase in the shareholders' equity (denominator) through the rights issue and mandatory convertible bonds. The company has always been very conservative with its borrowings. Hence, as can be seen in Table 5.11.2(b), SIA's gearing is the lowest among airlines in the industry.

Table 5.11.2(b) - Comparing SIA's Gearing Ratio (Debt to Equity) with Competitors

Gearing Ratio	2020	2019
Cathay Pacific	1.27	1.55
IAG	11.91	2.09
United Airlines	4.49	1.26
Singapore Airlines (Mar' 2021/2020)	0.70	1.01

SIA ratios are based on end March 2021 & 2020 respectively

5.11.3 Coverage Ratios

a. **Interest Coverage**

The interest coverage ratio or **times-interest-earned ratio** indicates how many times the fixed interest charges are covered by the earnings of the company. A higher ratio of earnings relative to interest charges indicates lower financial risk. The interest coverage ratio is computed as follows:

$$\text{Interest Coverage} = \frac{\text{Earnings Before Interest and Taxes}}{\text{Interest Expense}}$$

How safe an interest coverage ratio depends on the earnings volatility of the firm. In a slowing economy, earnings may not be sustainable, but interest obligations remain, leading to a higher financial risk for the firm. This ratio is especially relevant for bond investors, whose primary risk is the issuer's continued ability to meet its interest and principal payments on time.

b. **Earnings before Interest, Taxes, Depreciation, and Amortisation (EBITDA) Interest Coverage**

For companies with very high depreciation and amortisation expenses, the cash flow generated is actually much higher than the earnings before interest and taxes suggests. Hence, when a company has a very low interest coverage ratio, a secondary measure of interest coverage, EBITDA/Interest Expense, can be used to really see how well interest is covered by cash earnings. A third measure, Operating Cash Flow/Interest Expense, can also be used but EBITDA is usually a sufficient measure of how well cash flows cover interest expense.

$$\text{EBITDA Interest Coverage} = \frac{\text{EBITDA}}{\text{Interest Expense}}$$

c. **Interpreting Interest Coverage Ratios**

Table 5.11.3(a) - Interest Coverage Ratios for Singapore Airlines (2019-2021)

	2021	2020	2019
Interest Coverage			
$\frac{\text{EBIT}}{\text{Interest Expense}}$	$\frac{(4957) + 268}{268}$	$\frac{(220) + 221}{221}$	$\frac{869 + 116}{116}$
	-17.5	0.003	8.5
EBITDA Int. Coverage			
$\frac{\text{EBITDA}}{\text{Interest Expense}}$	$\frac{(4957) + 268 + 2076 + 66}{268}$	$\frac{(220) + 221 + 2134 + 58}{221}$	$\frac{869 + 116 + 1328 + 62}{116}$
	-9.5	9.9	20.5

SIA's interest coverage ratios for 2019-2021 are set out in Table 5.11.3(a). The negative coverage ratios were the result of trading losses sustained in 2020/2021. While a higher positive coverage ratio is a good thing (indicates lower financial risk), a negative coverage ratio indicates the opposite (the loss as a multiple of financing charges i.e.; the opposite of coverage). Note that the size of negative coverage ratio is also determined by the absolute amount of interest/financing charges. For example, a firm with an interest coverage ratio of -1 does not necessarily indicate lower financial risk when compared to a firm with a ratio of -3. To be sure, both firms are incurring trading losses, but the lower ratio of -1 is due to higher financing charges (denominator), which is not a good thing, even though both firms made the same trading loss.

Table 5.11.3(b) - Comparing SIA's Interest Coverage Ratios with Competitors

EBIT/Interest	2020	2019
Cathay Pacific	-6.3	1.7
IAG	-10.7	4.7
United Airlines	-7.3	6.4
Singapore Airlines (Mar' 2021/2020) SIA ratios are based on end March 2021 & 2020 respectively	-17.5	0.003

EBITDA/Interest	2020	2019
Cathay Pacific	-1.5	6.2
IAG	-6.2	8.2
United Airlines	-5.0	9.5
Singapore Airlines (Mar' 2021/2020) SIA ratios are based on end March 2021 & 2020 respectively	-9.5	9.9

All the airlines in Table 5.11.3(b) incurred net losses in 2020. Hence, they exhibit negative coverage ratios. We have to be careful not to over interpret the meaning of negative coverage ratios when assessing financial risk. As explained earlier, the denominator (interest/financing charges) has a direct effect on the ratio. It is therefore imperative for the analyst to look for the reasons behind the ratios.

d. **Total Fixed Charge Coverage**

This ratio measures how well earnings cover total fixed financial charges, including fixed lease payments and preferred dividends paid out of earnings after taxes.

The ratio is expressed as:

$$\textbf{Fixed Charge Coverage} = \frac{\textbf{Earnings Before Interest, Taxes and Lease Payments}}{\textbf{Interest Expense + Lease Payments + Preferred Dividend/(1 - Tax Rate)}}$$

5.11.4 Operating Efficiency / Activity Ratios

Efficiency ratios measure how productively management uses company assets and capital to generate sales. Some of these ratios measure the speed at which a company converts its inventory to sales and its receivables to cash and are part of the inputs for the **cash conversion cycle**, one of several measures of management effectiveness. The cash conversion cycle measures the time period between when a firm pays its suppliers for the inventory that it purchased and when the firm collects from its customers for the inventory that it sold; thus, the cash conversion cycle represents the period for which short-term financing is needed.

a. **Total Asset Turnover**

This ratio indicates the overall effectiveness of a firm's use of its assets. The higher the total asset turnover ratio, the greater the efficiency of the company in using its assets to generate sales.

The ratio is computed as follows:

$$\textbf{Total Asset Turnover} = \frac{\textbf{Sales}}{\textbf{Average Total Assets}}$$

Note that in the calculation of this ratio, we use **average total assets** instead of the year-end total asset balance. Conceptually, since sales revenue is generated by possibly different amounts of assets over the period, it follows that we should take the average of the company's total asset balance at the start and end of the year. Please refer to the computation of SIA's total asset turnover ratios for period from 2019-2021 in Table 5.11.4(a)

This ratio should be compared with that of other firms in the industry because it varies substantially across industries. Total assets turnover ratios range from 1 for capital-intensive industries to over 10 for some

retailing operations. This ratio can be affected by factors other than a firm's efficiency. For example, a high ratio relative to the industry might imply too few assets for the potential business (i.e., foregone sales) or the use of outdated, fully depreciated assets.

Table 5.11.4(a) – Total Asset Turnover Ratios for Singapore Airlines (2019-2021)

	2021	2020	2019
Asset Turnover Ratio			
Sales / Total Average Assets	3816 / (37582 + 33713)/2	15976 / (33713 + 30505)/2	16323 / (30505 + 25893)/2
	0.11	0.50	0.58

Table 5.11.4(b) - Comparing SIA's Total Asset Turnover Ratio with Competitors

Sales/Assets	2020	2019
Cathay Pacific	0.22	0.53
IAG	0.24	0.80
United Airlines	0.27	0.85
Singapore Airlines (Mar' 2021/2020)	0.11	0.50

SIA ratios are based on end March 2021 & 2020 respectively

The pandemic induced industry downturn in 2020 meant that many commercial airliners were left to 'idle' instead of generating revenues. As can be seen from Table 5.11.4(b), there was a significant decline in the total asset turnover ratio for the major airlines.

SIA maintains a lower total asset turnover than its competitors, so there appears to be some room for improvement in terms of the use of its assets. While keeping older fleets for longer periods will increase the total asset turnover, this should not be done at the expense of safety and customer experience.

b. **Inventory Turnover**

This ratio measures the number of times in a year that inventory is sold and restocked. A high inventory turnover ratio means a rapid rate of sales and, all else equal, higher profits. It is calculated as follows:

$$\textbf{Inventory Turnover} = \frac{\textbf{Sales}}{\textbf{Average Inventory}}$$

Some analysts prefer to use the cost of goods sold because inventories are stated at cost while sales, on the other hand, includes a profit component. However, as the cost of goods sold figure is not normally available, the sales figure is often used as a substitute.

Inventory turnover ratios vary widely. It is more applicable to manufacturing and the retail sales industry and is less useful in service-oriented industries. A high ratio is normally desired as it indicates efficiency. On the other hand, it can also indicate inadequate inventory to meet current demand, which can lead to shortages,

back orders, and eventually lost sales. Too low an inventory turnover ratio relative to the industry average may mean excess inventory and, possibly, obsolete physical stock.

c. **Days to Sell Inventory**

Another way to express the Inventory Turnover ratio is as the required number of days to sell inventory.

$$\textbf{Days to Sell Inventory} = \frac{\textbf{365 days}}{\textbf{Inventory Turnover}} \quad \textbf{or} \quad \frac{\textbf{Average Inventory}}{\textbf{Sales / 365}}$$

d. **Receivables Turnover**

This ratio indicates how many times, on average, a firm collects on its receivables and resold on credit during the year. It is computed as follows:

$$\textbf{Receivables Turnover Ratio} = \frac{\textbf{Sales}}{\textbf{Average Accounts Receivables}}$$

Note again that an average is calculated for the denominator because the numerator is measured over the period and, so, the denominator should be an average of the two year-end balances.

e. **Collection Period for Accounts Receivables**

Given the receivables turnover ratio, the number of days it takes (on average) to collect accounts receivables can be determined as follows:

$$\textbf{Collection Period} = \frac{\textbf{365}}{\textbf{Average Accounts Receivables Turnover}}$$

$$\text{or} = \frac{\textbf{Average Accounts Receivables}}{\textbf{Sales / 365}}$$

To determine whether a given collection period is good or bad, it should be compared to the firm's credit policy and the industry average. If the industry average is 60 days, a collection period of 70 days would indicate slow-paying customers, which ties up capital for a longer time in receivables and increases the possibility of bad debts. A figure substantially below the norm may indicate overly stringent credit terms relative to competitors, which could be detrimental to sales.

5.11.5 Profitability Ratios

Profitability may be analysed in relation to sales or capital.

a. **Gross Profit Margin**

Gross profit is the profit after deducting cost of goods sold from sales, but before deducting selling, general and administrative expenses. The gross profit margin is:

$$\textbf{Gross Profit Margin} = \frac{\textbf{Gross Profit}}{\textbf{Sales}}$$

Gross profit margins are useful for making comparisons between manufacturing companies, distributorships, supermarkets, and other companies that have goods for sale.

b. **Operating Profit Margin**

Operating profit refers to gross profit minus selling, general, and administrative expenses and reflect earnings before the financing decision and income taxes have been accounted for.

The operating profit margin is expressed as:

$$\textbf{Operating Profit Margin} = \frac{\textbf{Earnings Before Interest and Tax}}{\textbf{Sales}}$$

The variability of the operating profit margin over time is an indicator of the firm's business risk. Business risk is the uncertainty of income that is caused by the firm's industry. In turn, this uncertainty can be attributed to the firm's sales variability due to fluctuating demand for its products, changing preferences of its customers, and the way the firm produces its products.

In some instances, investors add back depreciation expense and compute an EBITDA profit margin. This alternative approach reflects the cash earnings of the company since depreciation and amortization are non-cash items and can provide insights into the profit performance of heavy manufacturing firms with large depreciation charges.

Please refer to Table 5.11.5(a) for SIA's profit margin ratios for period of 2019-2021.

Table 5.11.5(a) – Profit Margin Ratios for Singapore Airlines (2019-2021)

	2021	2020	2019
Operating Profit Margin			
EBIT / Sales	(4957) + 268 / 3816	(220) + 221 / 15976	869 +116 / 16323
	-122.89%	0.00%	6.03%
Return on Equity			
Net Profit / Average Shareholders' Equity	-4283 / (16278 + 9733)/2	-169 / (9733 + 13683)/2	722 / (13683 + 13228)/2
	-32.94%	-1.45%	5.36%
Pre-tax ROA			
Pre-tax Income + Interest / Average Total Assets	(4957) + 268 / (37582 + 33713)/2	(220) + 221 / (33713 + 30505)/2	869 +116 / (30505 + 25893)/2
	-13.15%	0.00%	3.49%
Basic EPS			
Net Profit/(Loss) / Weighted Average Shares	(115.6) cents	(11.2) cents	57.7 cents
	As per annual report	As per annual report	As per annual report
Fully Diluted EPS			
Net Profit/(Loss) / Fully Diluted Weighted Average Shares	(115.6) cents	(11.3) cents	57.4 cents
	As per annual report	As per annual report	As per annual report

c. **Net Profit Margin**

Net income is profits after taxes but before dividends on preference and ordinary shares. The net profit margin measures net income as a percentage of sales:

$$\textbf{Net Profit Margin} = \frac{\textbf{Net Income}}{\textbf{Sales}}$$

All of the above margin calculations are calculated against sales revenue. This approach is similar to the common-size income statement, which also shows the various line items as a percentage of sales. The Net Profit Margin calculation for SIA can be found in its Common Size Statement.

d. **Return on Equity (ROE)**

ROE indicates the rate of return on the equity capital provided by the shareholders.

$$\textbf{Return on Equity} = \frac{\textbf{Net Income}}{\textbf{Average Common Equity}}$$

Like all financial ratios, ROE numbers need to be reviewed carefully to see if the strong performance is indeed due to better earnings, instead of being a depletion of the equity base in prior years. For example, the ROE numbers in Table 5.11.5(b) below seem to suggest that International Airlines Group (IAG) and United Airlines produced superior ROE in 2019, prior to the onset of the pandemic. However, the numbers, taken at a snapshot, do not tell the full picture.

Table 5.11.5(b) - Comparing SIA's ROE Ratio with Competitors

ROE	2020	2019
Cathay Pacific	-37.5%	2.7%
IAG	-170.2%	25.3%
United Airlines	-80.8%	27.9%
Singapore Airlines (Mar' 2021/2020)	-33.9%	-1.9%

SIA ratios are based on end March 2021 & 2020 respectively

IAG is a merger between British Airways and Iberia Airline Group. In 2012, IAG substantially reduced its shareholders' equity balance, with losses of EUR0.9 billion charged to reserves. In 2013, it charged EUR2 billion for restatement of merger value to reserves. In 2014, it also charged EUR1.2 billion in losses under net value in changes of hedges. Hence, IAG has effectively been reducing shareholders' equity over the years with losses and charges. Likewise, United Airlines charged its losses to retained earnings in 2011 and 2012, which substantially reduced its equity base.

Subsequent years of ROE for both IAG and United Airlines look high due to the reduction of the value of shareholders' equity with losses in prior years. As such, simply interpreting ROE numbers for two years without reviewing the past may lead to an incomplete picture.

e. **Return on Assets (ROA)**

ROA measures the overall efficiency of the firm in managing its investment in assets to generate profits. It is normally expressed before interest but could be before or after tax. Profit before interest is used to separate operating decisions from financing decisions and to enable comparisons of efficiency for companies with different capital structures.

Table 5.11.5(c) - Comparing SIA's ROA Ratio with Competitors

ROA	2020	2019
Cathay Pacific	-10.3%	0.8%
IAG	-21.1%	5.4%
United Airlines	-12.6%	5.9%
Singapore Airlines (Mar' 2021/2020)	-12.0%	-0.5%

SIA ratios are based on end March 2021 & 2020 respectively

Note: ROA is calculated on a post tax basis

$$\textbf{Pre-tax Return on Assets} = \frac{\textbf{Pre Tax Income + Interest Expense}}{\textbf{Average Total Assets}}$$

$$\textbf{After Tax Return on Assets} = \frac{\textbf{Net Income + Interest Expense (1-Tax Rate)}}{\textbf{Average Total Assets}}$$

The after-tax ROA is the more popular calculation and is the commonly understood definition of ROA among the investment community. However, to be clearer, we will refer to the former as Pre-tax ROA and the latter as After-tax ROA.

Note that the comparative ROA figures in Table 5.11.5(c) are based on the following:

$$\textbf{Return on Assets} = \frac{\textbf{Net Income}}{\textbf{Average Total Assets}}$$

This is also another popular calculation used in some annual reports.

f. **Earnings Per Share (EPS)**

EPS is an important measure for assessing the profitability of a firm.

$$\textbf{Earnings Per Share} = \frac{\textbf{Net Income Available to Common Shareholders}}{\textbf{Average Number of Common Shares Outstanding}}$$

If a company issued convertible bonds, share options or other instruments where there is an option to purchase new shares, then there is a need to consider a 'worst-case' scenario and use the number of shares that would be in existence if all of the options were exercised. The resulting EPS is referred to as the 'fully diluted' earnings per share.

5.11.6 Quality of Financial Statements

Analysts sometimes refer to the "quality" of a company's earnings or its balance sheet. This typically is a reference to the reliability of the financial statements. A quality financial statement should reflect the economic and financial reality of the company's financial health and prospects. Also, it should not be reliant on biased or self-serving accounting treatment and interpretations that make the company appear more profitable and attractive as an investment than it actually is.

Accounts of such companies are usually conservatively stated and are relatively transparent on revenues and costs. Most telling is not how the management reports numbers when business is rosy, but rather the level of transparency and treatment of accounts and losses when market conditions are poor and when the industry is at its cyclical trough.

Signs of strong and high-quality financial statements include:

a. Limited or lower use of debt relative to the industry, which reduces or minimizes financial risk. Lower debt also implies that the company has sufficient credit lines to draw from if needed.

b. Higher levels of liquidity than competitors than the industry.

c. Assets that are generally carried below market values.

d. A preference not to reflect intangible assets[26] such as patents, goodwill, trademarks or valuable proprietary research, with costs expensed rather than deferred into amortisation schedules for accounting purposes.

e. The setting aside of funds for replacement and upgrades of plants and factories, with scheduled plans for renewal and technological upgrades. Companies using obsolete or inferior technology are overstating the values of these assets because the asset market values are likely to be below original costs.

f. Conservative revenue recognition policies and principles that do not change just to reflect higher sales, conservative accounting of unrealized income by deferring its recognition until the income is realized, and generally adequate provisioning for losses and expenses.

g. Absence of off-balance sheet activities. When a company has off-balance sheet liabilities, it is important to account for them when estimating the true financial position of the company.

h. Discipline in keeping to activities that are part of the company's core business and avoidance of speculative financial activities.

i. Realistic recognition of bad debts and adequate provisioning for potential bad debt and losses during economic or cyclical downturns. A rising average collection period usually means that bad debts may increase in the future. Also, as an example, a bank with an overly optimistic view of its credit quality will understate the volume of its non-performing loans.

j. Steady sustainable revenue and earnings growth from existing businesses. A company with good quality earnings has earnings that are realistic and repeatable.

From the review of SIA's financial statements, we can infer that in response to the severe industry downturn caused by the onset of the pandemic in 2020, the firm took decisive actions to build a more resilient business.

The key actions taken by SIA and their outcomes that can be seen on the financial statements include:

i. Raising of a significant amount of equity capital ($8.8 billion) through rights issue and mandatory convertible bonds.
ii. Conserving cash flow by withholding dividends and deferring major capital investments.
iii. The capital raised was partly used to reduce borrowings and hence lowered leverage.
iv. Keeping most of the funding raised in cash, presumably as a war chest, to see the firm through a protracted downturn.
v. As a result, the balance sheet is highly liquid and solvent, with reduced leverage and has a substantial war-chest that can be deployed in a market recovery.
vi. This level of financial resiliency was achieved at the expense of short-term profitability and asset efficiency.

[26] In December 2012, a Muddy Waters' report on Olam's treatment of impairment of its intangible assets caused Olam's stock price to fall by more than 30%.

5.11.7 Making Comparisons and Trend Analysis

Besides comparing with industry averages, trend analysis, which involves studying the past performance of the company over time, can provide valuable insights into the financial health of the company.

Business cycles affect the performance of all companies. Thus, to have a better understanding of the companies' performance, it is necessary to cover at least one cycle and to track the overall performance over a cycle. Otherwise, the analysis could be misleading and indicate that a company is doing well when it is simply at the crest of a cycle that is about to turn down.

Most companies provide either a 5-year or 10-year summary of key statistics as part of their annual reporting. Table 5.117.7(a) shows, in addition to financial statistics, some key operating statistics for SIA that some analysts will use to help forecast and assess future performance.

Table 5.11.7(a) - Key Operating Statistics for Singapore Airlines Ltd (2011 – 2021)

		2020/21	2019/20	2018/19	2017/18 [R1]	2016/17	2015/16	2014/15	2013/14	2012/13	2011/12
SILKAIR											
Passengers carried	('000)	57	4,440	4,902	4,687	4,106	3,836	3,553	3,411	3,295	3,032
Revenue passenger-km	(million)	87.4	8,195.4	8,940.3	8,343.5	7,138.0	6,516.2	5,864.9	5,516.1	5,223.1	4,469.4
Available seat-km	(million)	239.3	10,599.6	11,731.8	11,365.9	10,086.3	9,117.8	8,355.2	7,926.9	7,096.3	5,904.8
Passenger load factor	(%)	36.5	77.3	76.2	73.4	70.8	71.5	70.2	69.6	73.6	75.7
Passenger yield	(cents/pkm)	28.6	10.7	10.9	11.5	13.0	13.5	13.9	13.7	14.1	14.5
Revenue per available seat-km	(cents/ask)	10.4	8.2	8.3	8.4	9.2	9.6	9.8	9.5	10.4	11.0
Passenger unit cost	(cents/ask)	48.3	8.5	8.5	8.4	8.6	9.0	9.7	9.8	9.9	10.1
SCOOT [R4]											
Passengers carried	('000)	82	10,454	10,455	9,467	8,503	7,540	7,018	-	-	-
Revenue passenger-km	(million)	221.6	28,668.5	29,325.9	25,599.8	22,083.8	18,225.0	16,415.0	-	-	-
Available seat-km	(million)	2,228.2	33,445.8	34,388.6	29,888.4	26,792.8	21,732.8	19,983.0	-	-	-
Passenger load factor	(%)	9.9	85.7	85.3	85.7	82.4	83.9	82.1	-	-	-
Passenger yield	(cents/pkm)	55.0	5.6	5.7	5.8	5.9	6.3	6.3	-	-	-
Revenue per available seat-km	(cents/ask)	5.5	4.8	4.9	5.0	4.8	5.3	5.2	-	-	-
Cost per available seat-km	(cents/ask)	10.9	5.4	5.2	4.9	4.8	5.3	5.9	-	-	-
GROUP AIRLINES (PASSENGERS)											
Passengers carried	('000)	596	35,800	36,095	33,659	31,599	30,405	29,308	22,039	21,505	20,187
Revenue passenger-km	(million)	2,890.6	140,998.5	140,838.1	129,798.3	122,135.6	119,008.6	116,489.1	100,580.4	98,988.7	92,293.4
Available seat-km	(million)	21,721.2	171,211.2	169,606.6	159,381.0	154,541.4	149,217.1	148,339.0	128,429.7	125,360.7	119,314.5
Passenger load factor	(%)	13.3	82.4	83.0	81.4	79.0	79.8	78.5	78.3	79.0	77.4

[R1] *Operating statistics for FY2017/18 have been adjusted for adoption of IFRS and re-integration of SIA Cargo into the Parent Airline Company. Overall statistics for Singapore Airlines only shown with effect from FY2017/18.*

[R2] *Capital disbursements comprised capital expenditure in property, plant and equipment, intangible assets, investments in subsidiaries, associated companies and joint venture companies, and additional long-term equity investments.*

[R3] *Capacity per employee is defined as passenger and cargo capacity production (in tonnes) divided by Singapore Airlines' average staff strength.*

[R4] *Operating statistics for Scoot only shown with effect from FY2014/15.*

Source: Singapore Airlines 2020-2021 Annual Report

5.11.8 Limitations of Ratio Analysis

Ratio analysis has limitations and one must keep the following in mind when using ratios to analyse companies:

a. Each company may have different accounting policies, and adjustments are required to make the figures comparable.

b. Consolidated accounts are used for groups of companies; in such cases, it is often difficult to determine the performance of one company within the group.

c. The figures stated in the accounts may not be representative of the figures for the rest of the year, e.g., year-end inventory may be drawn down to abnormally low levels to give a misleading perception of a strong financial position.

d. The analyst should be alert to special situations that may depress profitability, such as research and development costs for new products.

e. When doing trend analysis, it is important to note the possible effects of inflation and general economic conditions.

f. It is important to read the footnotes of financial statements as these contain additional information on the company's financial situation.

g. When doing company comparisons, the companies used should be similar in terms of industry sector and size.

h. Ratios can be artificially manipulated (e.g., high return on equity from reducing equity in prior years). Analysts should look at accounts from prior years to better understand the genesis of the numbers.

i. Ratios can only be meaningful when reviewed over time and in comparison, with sector peers or industry benchmarks. Companies in emerging sectors (e.g., emerging technology/business model) do not lend themselves easily to ratio or, for that matter, financial analysis. The lack of comparable and reliable financial information greatly limits ratio analysis.

j. In addition to (i) , ratio analysis can be challenging for emerging business models with little or no 'hard' balance sheets (of operating assets), and where the value of the enterprise may rest on its brand perception and/or its business founder.

5.12 Summary

1. Company analysis can be divided into two parts:
    a. Qualitative analysis; and
    b. Quantitative analysis (Financial statements analysis).

2. Companies establish competitive advantage over other companies by/through:
    a. Dominating and controlling key technology, research, and patents.
    b. Controlling or dominating a key resource (e.g., raw material) supply.
    c. Becoming very cost efficient and becoming the lowest-cost producer.
    d. Establishing superior brand differentiation.
    e. Controlling the distribution channel.
    f. Adopting successful focus on niche clients.
    g. Developing superior management, talent, and relationships.
    h. Establishing diversified core businesses with shared costs.

3. When performing company analysis, there are several things to remember:
    a. Current asset values and future price movements are driven by expected future sales and earnings, not by historical data.
    b. A company's performance is driven by qualitative factors.
    c. Excellent companies with sustainable profit growth potential can appear expensive in the near term but they usually remain good investments for the longer term.

4. The current value of a company depends on its earnings and cash flow performance in the future, but we continue to look at historical financial data because:
    a. for most companies, the recent historical performance data provides the best predictor of the likely performance in the near future;
    b. historical financial data provides a good foundation and reference point for making forecasts.

5. The three major financial statements are:
    a. the balance sheet, which shows the financial condition of a company at a point in time;
    b. the income statement, which is a summary of the revenue and expenses of a firm during some period of time; and
    c. the statement of changes in cash flow, which shows the changes in the firm's cash position over a year.

6. When we examine the balance sheet of companies, we focus on:
    a. the largest items on both sides of the balance sheet;
    b. the relationship between assets and liabilities; and
    c. significant trends appearing in the accounts over time.

7. Some companies keep a large cash holding because of the liquidity, risk and operational needs of their industry. However, when not reinvested in the business, cash balances represent underutilised assets. Keeping a perpetually large cash balance would mean sacrificing potential returns that could have been generated from investing in other business or financial assets.

8. During periods of economic downturn, rapid technological advancement, and changes in tastes, older inventory may become obsolete and more difficult to sell. Keeping too much inventory can require major price discounts in order to move the goods and there could be major write-downs in value if the inventory becomes obsolete, outdated, or goes bad. A steady build-up of inventories accompanied by falling sales may indicate a problem.

9. Likewise, for manufacturing operations, because of the speed of technological advancement and disruption, a major concern with large commitments to plants and factories is the possibility of obsolescence in a short amount of time.

10. To reduce the risk of obsolescence and the need for plant retooling, as well as the impact on inventory, some technology companies have outsourced part or most of the manufacturing activity to subcontractors while retaining ownership of the software, intellectual property and research for the technology.

11. If there is falling sales and a build-up in accounts receivable during economic recessions, it could signal that there are problems with collection and that some accounts receivables may turn into bad debt in subsequent quarters.

12. If there is a rise in accounts receivables from quarter to quarter in conjunction with rising sales, it could be that:

    a. the company is using a strategy of offering more credit terms to distributors and retailers so that its goods will have an advantage over competing products; or

    b. if there are low cash sales, it could suggest that there are more credit sales and hire purchase sales. Hire purchase sales generate additional interest earnings for the company and can be a very lucrative strategy (e.g., Courts furniture shop in Singapore, car dealers offering credit sales alongside banks in some countries, etc.).

13. Best practices dictate that we reflect both short- and long-term financial investments at market value.

14. When we examine property holdings with large property development commitments, we should examine the funding supporting the commitments. During major economic downturns with high unemployment, properties under development may stay on the books for a long time. A slowdown in sales and longer holding periods can result in a drain on liquidity.

15. Due to the sizeable value of property holdings, significant value is locked in physical property, which can lower the productive use of capital. Many property companies have sought different ways to unlock the capital. One way is the injection of non-core property holdings into REITS for listing.

16. Amongst the intangible assets, goodwill represents a value premium over the fair market value of tangible assets. Write-downs of goodwill have a direct impact on equity and regular reduction of equity through such charges can artificially make ROE look better than they otherwise would in subsequent periods.

17. When looking at the liabilities side of the balance sheet, we pay particular attention to bank borrowings and other forms of debt financing (e.g., bond issues). We are interested in the:

    - size of debt used to finance assets and relative to equity;
    - funding mix for the debt obligations;
    - currency denomination of the debt; and
    - security and terms and conditions of the debt.

18. Companies with larger debt and borrowings tend to be those that:

    a. are regularly engaged in major infrastructure or property projects (e.g., construction and engineering companies, property developers, etc.);

    b. require large asset commitments (e.g., steel and iron foundries, power generation plants, purchase of ships and aircraft, mining, oil-drilling, etc.); and

    c. are involved in leveraged mergers and acquisitions.

19. When analysing the key drivers of sales growth, we should break the sales revenue component into two parts: (i) Unit sales growth and (ii) Pricing changes. Whenever a company is able to raise either of the two, sales revenue will increase.

20. Factors influencing unit sales growth include:
    - New geographical markets;
    - New target customer segment;
    - New uses;

- New range/version of product;
- Production capacity;
- Competitors' market share; and
- Limits to competition.

21. Factors influencing pricing changes include:
    - Competition (direct and indirect);
    - Brand value development;
    - Cost of production;
    - Buyers' financial health; and
    - Government pricing restrictions.

22. The segmental breakdown and model forecasts help to highlight the main sources of revenue and income. By tracking news related to the sales and performance of key businesses and new products and services from each division, we would be able to better anticipate the impact on overall growth in revenues and profits. Companies also provide segmental breakdown of assets to show resource allocation, i.e., how assets are deployed in the business.

23. When reviewing costs and expenses, we should take note of:

    - Major cost items as a percentage of sales;
    - Significant rising or falling trends in cost (e.g., raw materials, fuel costs); and
    - Cost items that are non-cash accounting costs (which do not affect cash flows).

24. The statement of changes in cash flows is useful for investors since it provides a picture of the real cash earnings generated from operations and the level of liquidity in the business. There are 3 parts to the cash flow statement:

    a. Cash flow from operating activities;
    b. Cash flow from financing activities; and
    c. Cash flow from investing activities.

25. Financial ratios can be grouped into the four categories:

    a. Liquidity ratios (which include the current ratio, quick ratio and cash ratio) measure the ability of the firm to meet future short-term financial obligations.

    b. Leverage ratios indicate the firm's dependency on external sources of debt financing and the earnings available to pay fixed financial charges. Measures of leverage include the debt-to-equity ratio, the debt-to-total assets ratio, the interest coverage ratio, and the fixed charge coverage ratio.

    c. Operating efficiency ratios measure how efficient management uses its assets and capital to generate sales;

    d. Profitability ratios analyse profitability in relation to sales (e.g., gross margin, operating profit margin and net profit margin) or in relation to capital (e.g., ROE and ROA).

26. The quality of a firm's financial statements and earnings depend on good accounting practices, conservative revenue recognition principles, limited reliance on external debt, understating rather than overstating asset values and inclusion of all financial liabilities (including off-balance sheet items).

27. The analysis of the financial statements of a company should always be carried out using data over several (say 5) years so that underlying trends can be identified.

28. Ratios can be artificially manipulated (e.g., high ROE from reducing equity in prior years). Analysts should look at accounts from prior years to better understand the genesis of the numbers.

29. Financial ratios should be analysed and reviewed against the performance of the industry, the firm's competitors, and the firm's historical performance.

30. Financial ratios can only be meaningful when reviewed over time and in comparison, with sector peers or industry benchmarks. Companies in emerging sectors do not lend themselves easily to ratio or financial analysis. The lack of comparable and reliable financial information greatly limits ratio analysis.

Chapter 6:
Equity Securities

Learning Objectives

The candidate should be able to:

- ✓ Describe the characteristics of equity securities, namely ordinary shares and preference shares.
- ✓ Distinguish between blue-chip, growth-oriented, income-oriented, cyclical, and defensive equity investments as well as those with high net tangible assets (hard asset) backing.
- ✓ Understand the differences between absolute and relative valuation approaches.
- ✓ Understand how to apply relative price valuation ratios, firm valuation ratios and yield-oriented ratios used in equity analysis, such as dividend yield, price-earnings ratio, price-to-book value ratio, price-to-sales ratio, price-to-cash flow ratio and Enterprise Value-to- earnings before interest, taxes, depreciation and amortisation (EBITDA) ratio, and their relevance to different types of companies.
- ✓ Describe the composition and computation of equity market indices.
- ✓ Perform calculations relating to rights and bonus issues.
- ✓ Explain the use and computation of Discounted Cash Flow (in the estimation of fair market value (intrinsic value of a stock).
- ✓ Apply relative valuation ratios (e.g., Price/Book Value, Price/Earnings, EV/EBITDA) in comparing stocks and explain how the ratios are calculated and used.
- ✓ Understand how prospectuses for Initial Public Offerings are read and analysed.
- ✓ Describe depositary receipts as a means of investing in foreign markets.

6.1 Introduction

Equity is possibly the most popular asset class among investors in Asia. Despite the volatility associated with equity investments, the potential for substantial capital gains remains a major draw for investors. This Chapter covers the following topics on equity securities:

a. Share ownership;
b. Characteristics of equities;
c. Strategies for investing in equities;
d. Relative pricing and valuation using cash flow models; and
e. Composition and application of equity indices.

This Chapter should also be read together with Chapter 5 on Company Analysis and Understanding Financial Statements.

Shares represent ownership rights in a corporation and allows investors to participate in the performance of the company. When companies perform well, with managerial discretion, they can pay out dividends and benefit investors on top of the general price appreciation (capital gains) of the stock.

However, if the businesses run by the company perform poorly, not only are lesser or no dividends paid, but the shareholder will also face the risk of capital loss as the stock falls in value. In the worst-case scenario, stock prices can go to zero. There have been many stocks on exchanges all over the world (including those in Singapore) with prices that have fallen either to zero or close to zero.

Hence, as investment representatives for banks and brokerage firms, it is important to provide the right advice to clients to help them select the stocks of better companies and avoid lemons.

6.2 Share Ownership

6.2.1 Why share the ownership of a great business?

Why would the owners of a booming company decide to give other investors the chance to participate in their company's successful performance? Aside from altruism, here are some possible reasons why the original owners of companies may decide to share the ownership of their companies by listing them on an exchange:

- Monetising Business Value
- Redeployment of Capital
- Raising Capital for Growth
- Risk Reduction
- Lack of Family Succession
- Satisfy Differing Ownership Agenda
- Professionalize Management
- Raise Corporate Profile

As shown above, original owners may want to share their risk of ownership, realise and monetise the value of the company, and redeploy funds to other businesses for even greater gains. Usually, after growth has started to slow from the high double digits, owners of privately held companies will seek a way to monetise part of their ownership of the still profitable business.

For example, the owners of a 100% family-owned business could reduce their stake to 51% by selling their 49% stake to common shareholders through an Initial Public Offering (IPO) and placement. Families may not have the right successors and may prefer to professionalise the management of the business. Different family members could also have different ownership agenda and motives for cashing out.

Through a well-planned IPO, ownership control can still be retained in the family, the business' value can be partially monetised, additional capital can be raised from new shareholders, management is professionalised, and the risks and returns associated with the future of the business is shared. It should also be noted that if the share value of a publicly listed company falls below its perceived intrinsic value given its prospects, the company is sometimes taken private by its major shareholders.

Finally, owners of new growth businesses (e.g., high-tech, biotech, internet, companies with disruptive technology) with creative ideas and prospects often require capital to help fund growth. These founding owners may have some initial support from private equity partners looking to realise and monetise the potential of the business and raise sizeable capital through an IPO to fuel its growth potential.

6.2.2 Types of Shares

Share ownership carries certain rights and privileges. However, not all types of shares carry the same rights and privileges. Before listing the company, the original owners would usually have to decide how much ownership to give up. This is important because in most instances, majority share ownership will dictate the stance on major decisions.

In Singapore, there are traditionally two types of listed shares:

- **Common or ordinary shares** with voting rights; and
- **Preference shares** with non-voting rights.

Typically, one common share carries one vote, and this is still predominantly the case. However, since June 2018, the Singapore Exchange Limited (SGX) has allowed the listing of companies with dual class shares i.e., in addition to ordinary voting shares, companies can have multiple vote shares. The intent of this initiative is to encourage founders who need substantial funding to rapidly grow the business but would like to retain majority control to list with the SGX. Safeguards are also put in place to protect non-controlling shareholders e.g., to prevent expropriation, extra votes cannot be exercised on important matters where there may be conflicts of interest between the founders and minority shareholders; to prevent founder entrenchment, each multiple vote share can only carry a maximum of 10 votes.

In the US and Europe, there are also classes of shares with different voting rights. For example, Berkshire Hathaway[27] has two classes of common stock, designated as:

a. Class A common stock (priced at USD284,570/share as at 24 October 2017); and

b. Class B common stock (priced at USD189.39/share as at 24 October 2017).

Each share of Class A common stock is convertible, at the option of the holder, into 1,500 shares of Class B common stock. Conversely, Class B common stock is non-convertible. In addition, Class B shares have 1/200th the voting power of Class A shares.

Another example in the US of a company with different share classes is Google (Alphabet Inc), which has 3 share classes:

a. Class A shares with 1 vote per share;

b. Class B shares with 10 votes per share; and

c. Class C shares with no voting interest.

Class A shares trade at a premium over Class C shares. However, Class B shares of Google are not traded. Instead, they are internally distributed for the founders and insiders.

In Europe, there are also a large number of companies controlled by founding families who have different share classes, with each share class carrying a different number of votes.

[27] Berkshire Hathaway is one of the most successful equity investments holding companies in the United States. Berkshire's gain in net worth during 2016 was $27.5 billion, which increased the per-share book value of both Class A and Class B stock by 10.7%. Over the last 52 years (that is, since present management took over), the per-share book value has grown from $19 to $172,108, a rate of 19% compounded annually. Warren Buffett is the CEO and Chairman and owns 36.8% of the company.

6.2.3 Ordinary Shareholders

In Singapore, companies mainly issue ordinary or common shares for ownership interest in the company. Holders of ordinary shares are granted the following:

a. Shareholders stand to gain when the company makes profits, when the share price rises, and when management decides to declare a dividend to shareholders. Ordinary share dividends are decided at the discretion of management and depends on the performance of the company. Common shareholders do not have a right to dividend payouts and such pay-outs are not guaranteed.

b. Shareholders are legally entitled to vote on major policy decisions, such as electing directors and issuing new shares. The more shares one owns, the greater one's voting power. Minority shareholders, on the other hand, may not have sufficient votes to exert pressure on the management of the corporation.

c. Shareholders have a right to be informed of major events and decisions affecting the company and its value.

d. Shareholders are residual claimants. In the event of the company's bankruptcy or liquidation, they will only receive their share of the remaining value of assets after all other creditors and claimants have been paid (including debt owed to bondholders, monies owed to suppliers, other creditors, employees and any deferred taxes).

e. Shareholders have limited liability in relation to the company. Limited liability means that in the event of a corporation failing, the most shareholders can lose is their original investment. Creditors have recourse only to the assets of the corporation and its shareholders are not personally liable for the obligations of the company.

6.2.4 Preference Shares

Although technically as an equity security, a preference share is considered a hybrid security because it resembles both fixed-income and equity instruments. As an equity security, a preference share has an infinite life and pays dividends. Although most preference shares are callable by the issuer, there is typically a period of protection against such calls. Usually, the issuer will not call in the preference share within the first 5 years of its life. Although not as prevalent, non-callable preference shares may also be issued. These carry a lower rate of return than the callable ones because the issuer has no option to redeem the shares.

A preference share resembles a fixed-income security with the dividend fixed and specified in advance. In effect, a preference share can provide a stream of income very similar to that of a bond. The difference is that the income stream continues indefinitely unless the preference shares are called or otherwise retired. Given its stipulated payments, preference shares are viewed by most investors as somewhat equivalent to a fixed-income security and are therefore considered as an alternative to bonds.

Preference share dividends are specified as an annual dollar amount or as a percent of par value. Unlike a bond, however, the issuer can decide not to pay the preference dividend if earnings are insufficient and failure to pay the specified preference dividend does not result in a default.

Investors regard preference shares as less risky than ordinary shares because preference dividends are specified and must be paid before ordinary share dividends can be paid. However, preference shares are riskier than bonds because bondholders will be paid first in the event of liquidation. Preference shareholders do not have voting rights except in the event of financial distress. Companies may issue several classes of preference

shares, with each class having different benefits. The main distinctions between the classes would be the method and amount of dividend payment and the special privileges associated with each class.

In summary, preference shares are "preferred" because of the benefits they confer. Firstly, they have redemption priority over ordinary shares in the event of liquidation. Secondly, preferred dividends are also given a priority to be paid over ordinary dividends.

Features of Preference Shares

Cumulative Feature - Once common, most preference shares today do not have this feature. The cumulative feature requires all past unpaid preference share dividends to be paid out before any ordinary share dividends are declared. This provides some degree of protection for the preference shareholder because preference share dividends do not enjoy the same legal right as the interest that is paid to bondholders. The cumulative feature secures the right for preference shareholders to receive their share of dividends before the ordinary shareholders can receive any.

Callable Feature - Most preference shares are callable by the issuer at a stated redemption price, usually only after several years from its issue. The call feature allows the issuing firm to replace its preference shares if interest rates fall. Callable preference shares normally carry a higher yield compared to similar preference shares without the call feature. The initial redemption price is usually set at a premium above the par value. By setting the initial call price above par, the investor is protected from an early call that carries no premium.

Participation Feature - The participation feature allows the preference shareholder to receive extra dividends when there are sufficient earnings. Although attractive to investors, this is not a common feature in preference shares.

Convertibility Feature - Although not common, this feature gives a preference shareholder the right to exchange his preference share for a specified number of ordinary shares. This convertibility feature is attractive to the investor and reduces the cost of the preference shares to the issuer. Essentially, the issuer has sold an option to the investor, and the cost of the option premium to the investor is reflected in the lower yield on the preference shares. Investors will generally exercise the conversion from preference to ordinary shares when ordinary share prices appreciate.

6.2.5 Rights Issue, Bonus Issue, Stock Split and Reverse Stock Split

What is the difference between a rights issue, a bonus issue, and a stock split?

a. **Rights Issue**

A rights issue is an injection of new capital into the company via the issuance of new shares. Rights to subscribe for the new shares are first offered to existing shareholders, who have the option to purchase a stated number of new shares at a specified price. This prevents a dilution of their share of profit and voting interest in the company.

These rights may be renounceable or non-renounceable. Renounceable rights can be traded while non-renounceable rights are not transferable and cannot be traded. Companies are now allowed to issue pro-rata renounceable rights shares of up to 100% of share capital, which is an increase from the previous limit of 50%. If an existing shareholder declines to exercise the rights, the rights may be taken up by other investors in the market.

Rights Issue - Stock Price Impact

It is September 2022 now. Of the authorised capital of 500 million (500m) shares, a company Dragonfly Ltd has 300m shares currently outstanding. Its stock was trading at $2.00 per share at the end of September 2022. Management announced a rights issue of 100m shares at $1.90 per share to raise capital, mainly for its acquisition of a company involved in the development of parts for the navigation and tracking technology used in commercial drones. Part of the funds will be used for refinancing. Dragonfly has a forecast earnings per share (EPS) of $0.11 for financial year ending December 2022.

(1) What is the likely reaction of the stock price when the stock begins trading?

When an announcement to raise capital through a rights issue is made, the price reaction in the subsequent days will depend on the market's perception on the purpose and benefits of the capital-raising exercise and the prospects for the company afterwards.

If the purpose of the capital-raising exercise is seen to be positive for its business and perceptions are that the venture, acquisition, or project relating to the capital raised adds to the future revenue, earnings and value of the company, the stock price may actually rise after the announcement. However, if the purpose is seen as risky with a potential negative impact, or if the market believes that the reasons for the capital-raising exercise are not clear, the stock price may even fall below the issue price.

(2) What is the theoretical ex-rights price of Dragonfly Ltd?

Before rights issue, Dragonfly Ltd shares outstanding is 300m shares at $2.00 per share, which implies a market capitalisation of $600m.

After a 1-for-3 rights issue, there will be 400m shares.

The capital raised from the new 100m shares is $190m (100m x $1.90), so the new theoretical market capitalisation is $790m ($600m + $190m).

The theoretical ex-rights market price is $1.975 ($790m/400m).

(3) What is weighted-average number of shares used in the EPS calculation for Dragonfly Ltd? What is its revised EPS forecast assuming there is no change in earnings for the financial year 2022?

Weighted-average number of shares = (300m x 9 + 400m x 3)/12 = 325m

Revised EPS forecast for 2022 = ($0.11 x 300m)/ 325m = $0.1015/share

b. **Bonus Issue**

In a bonus issue, there is no additional capital raised. It is usually carried out as a signalling exercise when the company has done well and expects its stock to rise with an announcement of good news. For example, a 1-for-3 bonus issue for Dragonfly Ltd leads to a total number of 400m shares, but the share price will be adjusted downwards more sharply because there is no new capital injected.

Hence, based on the same initial stock price of $2.00/share and 300m shares outstanding, a 1-for-3 bonus issue will give us a theoretical ex-bonus issue stock price of $600m/400m = $1.50/share, which is a sharply lower stock price compared to the rights issue price calculated earlier.

c. **Stock-Split**

A stock split occurs when the company wants to make its shares more accessible to retail shareholders. Several companies in Singapore have done this in the past. For example, OCBC made a 2-for-1 stock split in 2005 when its stock price was above $13. After the stock split and another rights issue, its stock as of October 2017 has risen to $11.55.

Hence, when stock prices reach a certain level where the composition of shareholders is excessively skewed towards institutional clients, a company may wish to reduce the stock price through a stock split to make the unit price of the stock more affordable. The stock split also gives the majority holders more convenience in selling off units of their existing shares. Stock splits do not raise capital.

d. **Reverse Stock Split**

A reverse stock split occurs when a number of shares is combined into one (or fewer) shares. For example, Citibank carried out a 1-for-10 reverse stock split in 2011. During the Global Financial Crisis, banks like Citigroup that were previously trading above $60 found themselves trading below $5/share (the lowest for Citigroup during the crisis was $0.97/share).

Certain institutional investors are governed by rules which discourage them from investing in "penny stocks". In the US, the Securities and Exchange Commission (SEC) defines "penny stocks" to be stocks priced below $5/share[28]. This could have made it difficult for institutional investors to own Citigroup shares.

Hence, its management called for a 1-for-10 reverse stock split, which brought Citigroup shares from $4.52 (just before the reverse split) to $44.16 (at the close of the first trading day after the reverse split). As of mid-2022, Citigroup shares are trading at $47.10 per share.

6.3 Choice of Equities

When giving advice and recommendations for equity investments, we can adopt different strategies depending on the potential return sought by the client and the client's risk threshold. Naturally, the basic strategy is to search for undervalued stocks, which will be discussed in Section 6.5. This Section looks mainly at the different types of stocks available for investing.

6.3.1 Blue-Chip Companies

A relatively simple approach for long-term investors is to invest in the shares of blue-chip companies, which are major conglomerates with great financial strength and well-established, stable and mature businesses. As these companies have a long and consistent record of earnings, they generally can pay steady dividends over a long period of time during both good and bad years.

Blue chips hold leading positions in their respective industries and the economy in which they operate. They are usually pacesetters and frequently determine the standards by which other companies in their fields are measured. The management of these companies have shown foresight in growing the companies without

[28] US Securities and Exchange Commission: "The term "penny stock" generally refers to a security issued by a very small company that trades at less than $5 per share. Penny stocks generally are quoted over-the-counter, such as on the OTC Bulletin Board (which is a facility of FINRA) or OTC Link LLC (which is owned by OTC Markets Group, Inc., formerly known as Pink OTC Markets Inc.); penny stocks may, however, also trade on securities exchanges, including foreign securities exchanges. In addition, the definition of penny stock can include the securities of certain private companies with no active trading market." Link: https://www.sec.gov/fast-answers

jeopardising current earnings. Such companies typically have the advantage of size and resources for sustainable earnings. Investors who seek safety, stability, and are conservative in their investment approach usually turn to blue chip shares.

Although blue chips would have had some periods of high earnings growth during their early years, growth would have slowed over time due to their sheer size. Many mega-growth stocks dominating the world today are likely to become blue chips in the years to come. Due to their financial strength, strong connections, and operating capabilities, many blue chips in Asia have diversified into new businesses or ventured into new geographical markets to sustain their growth. In smaller markets, blue chips would also account for a large share of the stock market's capitalisation.

Singapore companies that are considered blue chips would include Keppel Corporation, SembCorp Industries, City Development, CapitaLand, DBS, OCBC, UOB, and SIA. For US stocks, typical blue chips would be 3M, Procter & Gamble, General Electric, Walmart, IBM, Hewlett Packard, Apple, and Microsoft.

However, it is still important not to be complacent when investing in blue chips, because the conditions fundamental to the success of all companies can always change, especially with technological disruption.

6.3.2 High-Growth Companies

High-growth companies usually come from industries and businesses in the new economy. These are typically companies at the forefront of research and development for new technologies, new business processes, renewable energy, and other new ways of conducting business. Some of the products and services of these companies may disrupt traditional ways of conducting business.

When the investment community talks about growth companies, they are usually referring to companies that can generate high double-digit sales and earnings growth over a certain number of years (e.g., > 20% growth compounded annually over several years). There are two stages of high growth:

a. High growth in sales, but still reporting negative profits; and
b. Companies with high growth in both sales and earnings.

a. **High Growth, Negative Profits**

On the NASDAQ, there are numerous concept companies that have strong future sales growth potential but continue to incur even higher costs that prevent them from breaking even. There are numerous technology and biotech companies on the NASDAQ that are bought for their potential breakthrough in research.

For many, the "burn-rate" from research and development activities eventually catch up before they can even show profit, and they become bankrupt. On the other hand, there are also companies that persist on with their research and marketing, thanks to financial support from key investors who continue to fund the companies' activities until they reach the point of profitability.

There are a few well-known examples of such companies. Tesla, Inc**.,** a manufacturer of electric cars, solar panels and energy storage devices reported many years of losses but was still heavily supported by investors in the automotive industry until it made its first quarterly profit in 2017. Social business network company Twitter reported almost 12 years of losses until the 4th quarter of 2017.

When it becomes clearer that the high sales growth will translate to profitability, the rest of the investment community will be become more interested in the company. In some cases, such companies become acquisition targets.

b. **High Growth, High Profits**

The second group of high-growth companies would have both excellent sales and earnings growth. Part of earnings growth would be reflected in the stock price. The key is to identify companies with sustainable high earnings growth to justify continued price appreciation on the stock. These companies are usually aggressive with strong research teams and normally pay relatively small dividends as most of their earnings are retained for expansion purposes.

Key characteristics of such growth companies include a solidly entrenched position in an expanding market, distinctive or unique products, and high profitability in most cases by pre-empting a special part of the market. Examples of companies that demonstrated phenomenal earnings growth from 2014-2018 are Amazon, Alibaba, Tencent, Nvidia, and Facebook.

c. **Characteristics of High Growth Companies**

High-growth companies typically carry high Price-to-Earnings multiples because investors are willing to pay a premium for the higher growth in earnings compared to similar companies in the industry. The pace of growth cannot last forever as growth companies get bigger. Once earnings growth starts to slow, the premiums will narrow, and the growth stock will be re-rated with lower ratios.

These stocks are usually high-beta stocks with greater sensitivity to the market. When there is a general sell-off on the market, these high-beta, growth stocks also tend to correct much faster. High growth stock investors must be able to cope with the higher volatility associated with such stocks.

6.3.3 Companies with High Dividend Yields (Income Stocks)

Investing in companies that pay higher average dividend yields is attractive to retirees and other investors who buy shares for current income. Such income stocks are often sought by trust funds, pension funds, endowment funds, and charitable and educational foundations. Companies that pay higher dividends include mature firms that can generate large cash flows. As these companies are at a mature stage of growth, they may not require cash flows for expansion purposes and can afford to pay higher dividends.

A benefit of high dividend-paying shares is that when markets correct, stocks with high dividend yields will usually find price support on the strength of their yields, as long as there is confidence that the same dividends will be paid out. For example, if the fixed deposit rate is 3% for annual deposits, stocks with dependable dividend payouts are unlikely to fall below a dividend yield of 4 - 5% before tax (i.e., the price will be supported at the point where Dividend/Price = 4%).

Companies with high dividends do come with some drawbacks. For example, the share may be that of a lacklustre company in a socially unpopular industry (e.g., tobacco and cigarettes). With low growth potential, the company provides high dividend payouts. In some companies, the high dividends may be offset by the company's tendency to call for rights issues, which can be potentially dilutive (e.g., some real estate investment trusts (REITS)). Finally, the high dividends will only be present if earnings can be sustained. Companies facing difficult conditions have been forced to cut dividends.

6.3.4 Cyclical Stocks

The stock market rises and falls with the general economy and business conditions. However, there are certain stocks that are expected to benefit more than others in a booming economy. These same stocks will also do

worse than other stocks in an economic downturn. We call such stocks that are more sensitive to the economic conditions, cyclical stocks.

Traditionally, for economies driven by domestic consumption, cyclical stocks are amongst the first to benefit from increased consumer and business spending during periods of economic growth and low unemployment. For example, stocks of automobile manufacturers (e.g., GM, Ford) are representative of consumer-based cyclical stocks. The earnings and share prices of these companies either fluctuate with the business cycle or are accentuated by it. In good business conditions, such companies would enjoy strong earnings growth and ordinary share price appreciation. When business conditions deteriorate, these companies would be the first to be affected by the slowdown in spending, and their share prices drop as earnings growth declines.

Due to the difference in the composition of economies, the stocks that are cyclical can be quite different in different markets. Some markets are driven very much by high-tech industries and electronic manufacturing; hence, for Taiwan and South Korea, stocks in the technology and electronic manufacturing sectors are cyclical in nature. For Singapore and Hong Kong, the cyclical stocks will also come from the property and banking sectors because they are usually amongst the first sectors to be affected in an economic downturn.

6.3.5 Defensive Stocks

The earnings of defensive companies have a higher likelihood of withstanding economic downturns or uncertainties. Defensive companies tend to be those that provide consumers with necessities, such as pharmaceuticals, food, utilities, and transportation, and shares of these companies tend to be less volatile than cyclical shares and growth shares. For example, shares of companies like Nestle and Coca-Cola, would be typical defensive stocks. If a recession is expected, there would be a strong interest in the shares of 'recession-proof' companies, especially if they pay a reasonable dividend yield.

6.3.6 International Equities

With the expansion of internet brokers, and local brokers offering execution and research for stocks outside of Singapore, investing in stocks internationally is now quite common even for retail investors. The need for diversification and exposure to growth companies ultimately forces investors to search for good opportunities internationally.

Participation in foreign shares can take place via:

a. **Direct Investments -** Investors can purchase foreign shares directly on the respective share exchanges through their local brokers. This diversifies away risks that the home market is exposed to. However, direct investment may be complicated by withholding taxes, significant transaction costs, and regulations that limit foreign ownership.

b. **Depositary Receipts (DRs) -** DRs represent indirect ownership of shares of foreign companies and enable investors to buy and hold foreign securities without going to the foreign share exchange directly. They provide ready access to foreign markets that may have barriers to foreign ownership and capital flows.

c. **Exchange-Traded Funds (ETFs) -** ETFs replicate a broad-based index like the Straits Times Index (STI) or the Standard and Poor's (S&P) 500. Investors who buy an ETF effectively assume the market risk of the foreign index's underlying shares. ETFs are covered in greater detail in Chapter 9..

d. **Foreign Shares with Dual Listings -** Some shares are listed both in their home country and another jurisdiction (e.g., Singapore). Dual listing refers to the practice of listing a security on a foreign share

exchange in addition to its local exchange. For the investor, such listings enhance liquidity and allow a share to continue trading on the other exchange after one exchange has closed.

6.4 Equity Market Indices

The general movement of the share market is usually measured by an equity market average or index, consisting of a group of securities that is supposed to reflect the entire market. The composition of the index, the weight given to each security, and the method of calculation are all important considerations in the construction of an equity market index or average. There are three main weighting schemes used:

- Value-weighted series;
- Price-weighted series; and
- Un-weighted or equal-weighted series.

6.4.1 Value-Weighted Series

A **value-weighted series** is generated by deriving the initial total market value of all shares in the series (market value = number of shares outstanding x current market price). This figure is typically established as the base and assigned an index value (normally 100).

Subsequently, a new market value is computed for all securities in the index, and this new value is compared to the initial 'base' value to determine the percent change. This percent change is, in turn, applied to the beginning index value to compute a new index value.

Value-Weighted Share Index

Day 1

Share	No. of Shares	Share Price ($)	Market Value ($)
A	1,000,000	5.00	5,000,000
B	3,000,000	3.00	9,000,000
C	5,000,000	10.00	50,000,000
			Total: 64,000,000

Base value equal to an index of 100.

Day 2

Share	No. of Shares	Share Price ($)	Market Value ($)
A	1,000,000	5.20	5,200,000
B	3,000,000	3.10	9,300,000
C	5,000,000	10.50	52,500,000
			Total: 67,000,000

$$\text{New Index Value} = \frac{\text{Current Market Value}}{\text{Base Value}} \times \text{Beginning Index Value}$$

$$= \frac{67{,}000{,}000}{64{,}000{,}000} \times 100$$

$$= 1.05 \times 100 = 105$$

An advantage of the value-weighted index is that it automatically adjusts for bonus and rights issues, as well as other capital changes. This is because these corporate actions typically do not affect market capitalisation, i.e., a decrease in share price is offset by an increase in the number of shares outstanding. In a value-weighted index, the importance of a constituent share is dependent on the market value of the share. Price changes for shares with high capitalisation have a greater impact on the index value than those for shares with low capitalisation.

The S&P 500 is a value-weighted index that comprises the top 500 companies, based principally on market capitalisation. It accounts for around 75% of the total U.S. stock market capitalisation. In Singapore, the STI is also weighted by market capitalisation. The STI replaced the previous market index, the Straits Times Industrial Index, which was price-weighted.

6.4.2 Price-Weighted Series

A **price-weighted series** uses an arithmetic average of current prices. Share prices are added together and divided by the number of securities. The divisor must be adjusted when there are capital changes (such as rights and bonus issues) and changes in the sample over time. With the series being price-weighted, a high-priced share carries more weight than a low-priced share. As such, a stock split will reduce the weight of the share in the index even though the market capitalisation of the company remains unchanged.

The Dow Jones Industrial Average (DJIA) is a price-weighted average that comprises 30 large, well-known industrial shares. The DJIA has been criticized because it uses only 30 non-randomly selected blue-chip shares, which are not considered to be representative of the shares listed on the New York Share Exchange (NYSE).

6.4.3 Un-Weighted or Equal-Weighted Series

In an **un-weighted index** or **equal-weighted index**, all shares carry equal weight regardless of their price and/or market capitalisation. A $2 share is as important as a $10 share, and the total market value of the company is not taken into consideration. Changes in the index may be based on the arithmetic or geometric average of the percentage price changes for the shares in the index.

The Value Line Composite Average is an equal-weighted geometric average of share prices. More than 80 percent of the shares making up the Value Line Average are listed on the NYSE.

Equal-Weighted Share Index

Share	Share Price ($)		1 + Percentage Change
	Day 1	Day 2	
A	5.00	5.20	1.04
B	3.00	3.10	1.03
C	10.00	10.50	1.05

Geometric Average

$= (1.04 \times 1.03 \times 1.05)^{1/3}$

$= 1.040$

Index Value (Day 2) = Index value (Day 1) x 1.00

6.5 Valuation Approaches

The time value discussion in Chapter 2 provides much of the foundation needed for calculations in this section. There are essentially three different ways of valuing a listed company and its businesses:

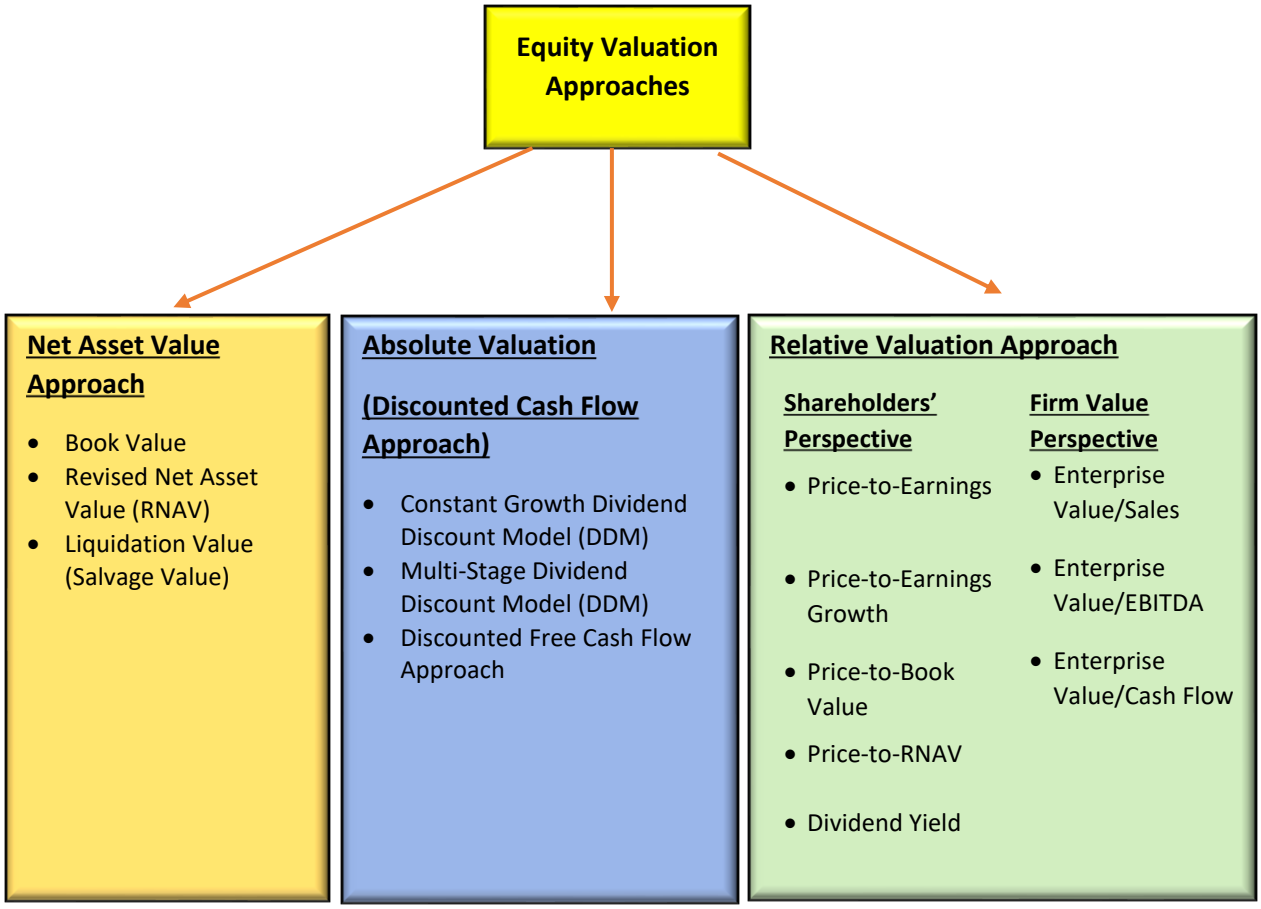
a. **Net Asset Valuation,** which includes Book Value, Liquidation Value, and Revised Net Asset Value;

b. **Discounting of Future Cash Flows** (which is also called the **Absolute Valuation Approach)**; and

c. **Relative Valuation Approach,** where comparisons are made against historical trends and other companies in the industry.

Please see Figure 6.5 – Different equity valuation approaches.

Figure 6.5 - Equity Valuation Approaches



6.6 Net Asset Valuation

The **Net Asset Valuation Approach** is a basic balance sheet approach of valuing companies. Under this approach, net assets are valued at book value, and are also revised to market value or liquidation value depending on the circumstances.

The net asset value approach is particularly relevant for companies with values that are predominantly determined by the values of the tangible assets (e.g., loans of banks, premiums of insurance companies, and real estate holdings of property companies) on their balance sheets.

6.6.1 Book Value

The book value of equity reflects the total value of the company's assets as stated on its balance sheet, net of all of its liabilities. It is one of the key numbers that investors and acquiring companies will examine. The book value is divided by the outstanding number of shares to arrive at **book value per share.**

When economic and stock market conditions are severe (e.g., when there is a market crash) or when a company is making losses with a challenging earnings outlook, the book value is one way of assessing the value of a company's equity.

The book value also represents the accumulation of retained earnings over the years. During bad times, if shareholder's equity has not been wiped out by losses, the book value often represents the floor value.

The problem with assessing book value during crisis situations lies with assessing the deterioration in asset values. This deterioration can often be understated. When this happens, the book value needs to be revised downwards.

6.6.2 Liquidation Value (Salvage Value)

This value assumes that the company will be discontinued and that viable business operations will be sold off to other companies. Assets and existing businesses are usually priced for ease of sale to recover monies to pay off creditors quickly.

Hence, the price obtained may be much lower than the asset's market value or book value, leading to a residual liquidation value that can also be much lower than the equity book value. While not usual, there have been cases of listed companies that have encountered severe business failures and unexpected losses, which resulted in eventual liquidation.

6.6.3 Revised Net Asset Value

$$\text{BVPS} = \frac{\text{Book Value of Assets - Book Value of Liabilities}}{\text{Total Outstanding Number of Shares}}$$

$$\text{RNAVPS} = \frac{\text{Revised Asset Value – Market Value of Liabilities}}{\text{Total Outstanding Number of Shares}}$$

When the value of a company's assets or liabilities have changed significantly before the end of the financial reporting period, the book value of these assets/liabilities may be revised to reflect the actual value of the assets/liabilities. From this, the revised net asset value per share (RNAVPS) can be calculated. The RNAVPS is derived in a similar manner as the book value per share (BVPS).

One example of the revised net asset value approach can be found in the real estate industry. For property companies with substantial land bank and property holdings, professional real estate consultants are asked to provide a valuation in order to mark-to-market the property asset values on the balance sheet.

However, real estate prices could appreciate significantly or fall rapidly within a financial year. Before the close of a given financial year, investment analysts and specialists may have to revise the net asset value to reflect major asset appreciation or correction.

When we revise the asset values to their respective market values, we should also check whether there is a need to revise debt and other liabilities to market value as well. For example, the market value of USD-denominated debt may have appreciated in the interim. If so, the market value of the company's liabilities should be revised as well.

6.7 The Discounted Cash Flow (DCF) Approach

For CM-EIP examination, investment representatives are only expected to know how to explain the process of valuing stocks and the variables used but will not be asked to compute and derive the fair value.

The investment community mostly uses the DCF Method to derive the fair value of a company so, for the purposes of CM-EIP Module, we will focus on the DCF approach. If you would like to review the dividend discount model (DDM) for valuation, which is only used occasionally for valuing companies with high dividend payouts (DPS/EPS of more than 80%), please refer to the step-by-step explanation in Section 6.13 – Additional Reference Materials which will take you through the different models.

6.7.1 Introduction to the DCF Approach

The key principle behind the DCF approach is the time value of money. As covered earlier in Chapter 2, this simply means that money received in the future is worth less than money received today. Under the DCF approach, a company's current fair value can be estimated by 'discounting' the company's future free cash flows to their present values.

There are several different methods of conducting a DCF valuation, depending on the assumptions made. As such, the end results of the different DCF approaches can vary significantly.

For this module, we will focus on the **Free Cash Flow to Firm (FCFF)** approach to DCF valuation, using the **Weighted Average Cost of Capital (WACC)** as the **Discount Rate**.

a. **Free Cash Flow to Firm (FCFF)** – This is a commonly used measure of a company's projected cash flows. First, we derive the after-tax cash flow generated from the company's operations. Next, we deduct cash set aside for capital expenditures and to sustain the company's business activities. Lastly, we add back depreciation and other non-cash expenses to reflect the **free** cash flows available to the company.

b. **Terminal Value** – We can project the future FCFF over a reasonable period (e.g., 10 years). Beyond that, it makes more sense to assume a terminal value at the end of the 10th year that encompasses all future cashflows thereafter (i.e., indefinitely if the company is assumed to be a going concern). The terminal value 10 years from now will also have to be discounted to a present value.

c. **Discount Rate** – The discount rate to be applied to the future cashflows is dependent on how the company is financed – by either equity or debt (or both). Hence, the discount rate is a **WACC** of both the cost of equity and cost of debt used to generate future cashflows.

d. **Enterprise Value** – By discounting the future FCFF and Terminal Value using the WACC, we arrive at the current value of the company, often referred to as the Enterprise Value. In other words, the Enterprise Value is the sum of all the discounted future FCFF and the discounted Terminal Value.

If the current market value of the stock is below the intrinsic share value by a significant percentage (i.e., undervalued), we should buy the stock. Conversely, if the current market value of the stock is above the intrinsic share value by a significant percentage (i.e., overvalued), we should sell the stock.

To demonstrate these steps, we apply the DCF model analysis to Procter and Gamble (P&G).

6.7.2 Calculating the Weighted Average Cost of Capital

The FCFF is discounted to the present value using the WACC. This is because the firm's value and future cash flows are generated with the help of funds from both debtholders and shareholders. The returns that these investors demand represent the Cost of Capital to the company. Simply put, it is the cost that the company has to incur when it raises funds through debt (e.g., bonds) and/or equity (e.g., share issues). As such, the WACC is simply a weighted average of both the cost of equity and cost of debt.

a. **Calculating Cost of Equity**

We apply the **Capital Asset Pricing Model (CAPM)** formula to calculate the cost of equity for P&G (refer to Figure 6.7.2(a)). The variables in the CAPM formula are further explained below:

Figure 6.7.2(a) – The CAPM formula

$$R_{P\&G} = R_f + \beta\,(R_m - R_f)$$

$R_{P\&G}$ is the cost of equity for P&G.

R_f is the risk-free rate.

R_m is the market return.

Rm-Rf is the equity market risk premium.

β is the sensitivity of the stock price to the market.

The following are the typical steps taken in computing firm and equity values:

(a) Calculate the WACC.

(b) Estimate the FCFF generated over a more predictable period – typically 10 years.

(c) Estimate the long-run Terminal Value (TV) at the end of the estimation period (i.e.; price at Year 10).

(d) Sum the present value of the FCFFs and TV in order to obtain the company's Enterprise Value.

(e) Subtract the market value of net debt and preferred shares from the Enterprise Value to get the estimated market value of equity (or fair value of equity), which is the estimated market value of the common shares.

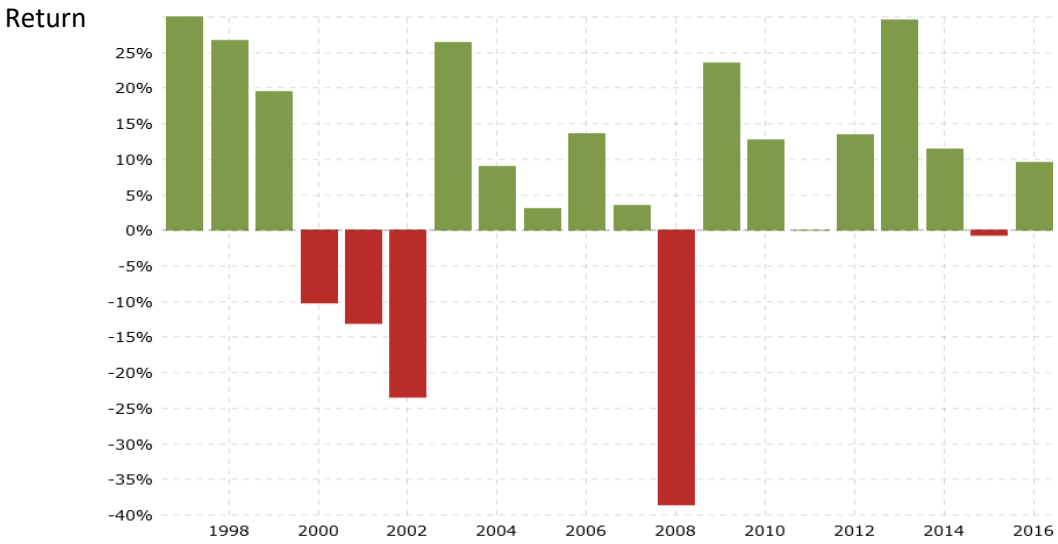
(f) Divide the estimated market value of equity by the number of shares outstanding to get the intrinsic share value (or fair value per share).

i. **Risk Free Rate (R_f)**: The US 10-year Treasury bill rate was 2.33% at the time of calculation, so the risk-free rate for P&G stock (which is a US stock) is R_f = 2.33%. (Note: For the valuation of Singapore stocks, the 10-year SGS rate can be used as the risk-free rate.)

ii. **Required Return of the Market (R_m)**: The required return of the market is needed to calculate the market risk premium (R_m-R_f). The market risk premium represents the additional returns that investors required to be compensated with to invest in the equity market instead of a risk-free security.

How do we go about estimating the required return of stocks in, say, the US market? We can begin by first looking at how the general market performed in the US over the last 10 or 20 years (refer to Figure 6.7.2(b)).

Figure 6.7.2(b) – Historical US Market Returns



The average price return of the S&P 500 over the last 20 years (including the global financial crisis and dot-com bubble) was +7.36%. The annual dividend yield on the S&P500 is about 1.92%. Thus, the total return of the S&P500 would be about 9.28%. Hence, we can conclude that the required market return is about **R_m = 9.28%.**

Beta of P&G (β): Beta measures the sensitivity of the stock to market movements. One way to derive beta is to plot the returns of the market (e.g., S&P 500 weekly or monthly returns for a few years) against P&G's weekly or monthly returns. We then estimate the change in P&G return for a unit change in the S&P 500 return. The gradient of the line would be equal to beta. We can also use Excel to graph the relationship and calculate beta.

Stock betas are provided by Bloomberg, Reuters, and other financial information services. In the case of P&G, beta is quoted as being between 0.65-0.68. We use the more conservative estimate of 0.68.

Finally, we apply all of the variables to the CAPM formula:

$R_{P\&G}$ $= R_f + \beta (R_m - R_f)$

$= 2.33 + 0.68(9.28 - 2.33)$

$= 7.056\%$

$\approx 7\%$

where:
R_f = 2.33%
R_m = 9.28%
B = 0.68

This is the cost of equity for P&G. It represents the rate of return that P&G must offer on its shares to entice shareholders to finance the business.

b. **Cost of Debt**

Cost of debt refers to the effective interest rate that a company has to pay on its debt. From the balance sheet, P&G's effective interest cost (calculated using Interest Expense / Borrowings) was 1.4% in 2016 and about 1.1% for the first nine months of 2017.

Another way to estimate the cost of debt is to identify the interest cost associated with each debt item from the notes to the accounts. Analysts will typically gather information about the costs of borrowing from the financial controller.

$$WACC = (1 - t)\, R_{Debt} \frac{Debt}{(Debt + Equity)} + R_{Equity} \frac{Equity}{(Debt + Equity)}$$

t is the effective tax

Taking this conservative approach, we will use the higher 1.4% rate as the cost of debt for our valuation, since interest rates are expected to rise with the Federal Reserve (Fed) hikes.

c. **Weighting**

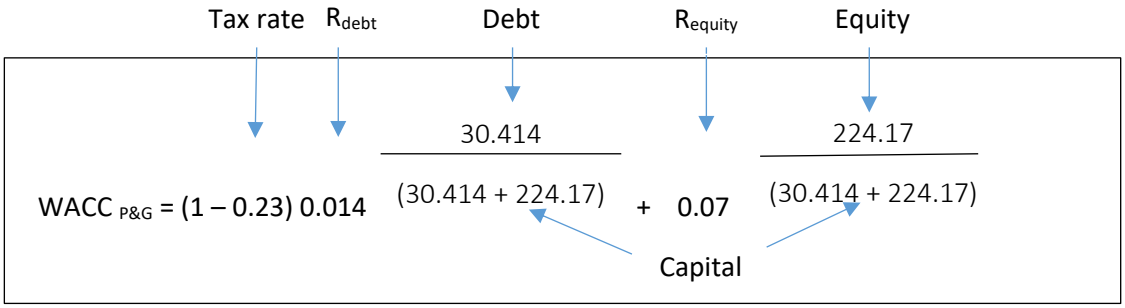
To derive the WACC for P&G (WACCP&G), we need to combine the cost of debt (R_{Debt}) and cost of equity (R_{Equity}). We also need the following inputs:

i. **Current Market Value of Debt (Debt)** - Based on the quarterly information as at 8 July 2017, the total book value of P&G's debt is USD30.414 billion. If there are no significant issues with the company's credit and if the debt is in the local currency, the book value of debt is a reasonable estimate of the market value of debt. Hence, we assume that **Debt is USD30.414 billion.**

ii. **Current Market Value of Equity (Equity)** - For common equity, we should always use the market value of equity. This market value can either be obtained from financial information providers or calculated by multiplying the total shares outstanding by the current price per share. Suppose P&G's stock price at the time of this analysis is $87.65 and the company has USD2.5576 billion shares outstanding. Then the market capitalisation of P&G would be **USD 224.17** billion (or $87.65 x 2.5576 billion shares)

iii. **Effective Tax Rate (t)** - The effective tax rate of P&G for the last nine months was 22.6%. In 2016, it was 25%. We will use an effective tax rate of 23%.

iv. **Total Capital (Debt + Equity)** - The total capital comprises the market value of debt and the market value of equity (i.e.; market capitalisation).



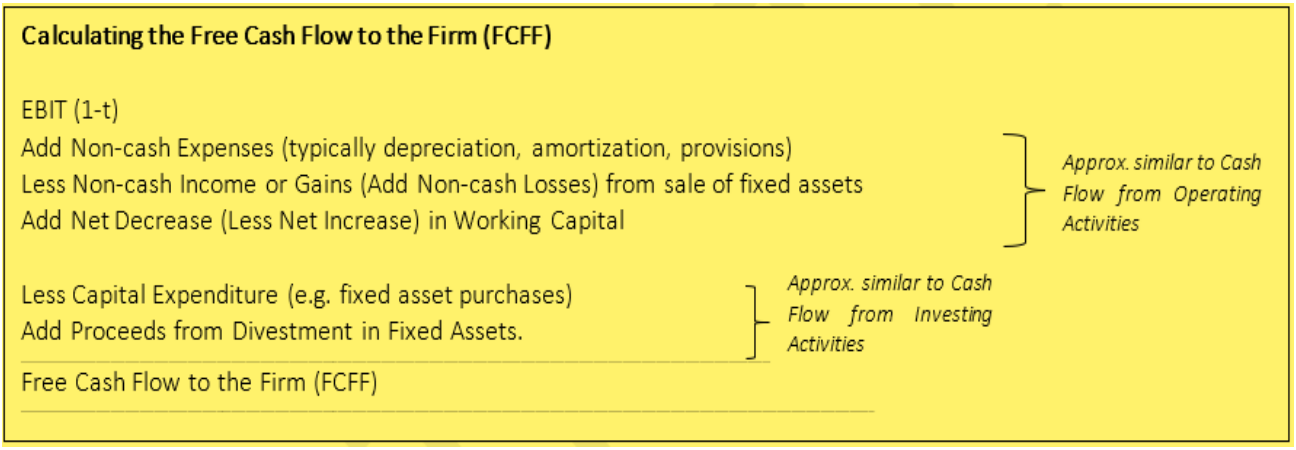
Based on the information, WACC $_{P\&G}$ = (0.0108 x 0.1195) + (0.07 x 0.8805) = 0.0629 = 6.29%.

This will be the discount rate we will use for discounting all future cash flows of the company. With the low interest rate environment, P&G has locked in some long-term debt that are priced very cheaply when compared to the cost of equity.

6.7.3 Estimating Free Cash Flows to the Firm

In order to estimate the FCFF, most analysts would build projections of the pro-forma income statement[29], balance sheet and cash flow statement. Projections would be made for about 10 years. When performing primary research to make future FCFF projections, an analyst would typically interview the company's management to discuss prospects and expenditure needs. The earnings estimates will come from projections of sales, other income, and cost items (refer to Figure 6.7.3)

Figure 6.7.3 – Calculating the Free Cash Flow to the Firm



Analysts will also rely on historical ratios and relationships to make projections for the years ahead. They may review past major expenditure patterns and ask management about the likely major capital outlays over the next few years.

Alternatively, the FCFF is approximately the sum of the cash flow from operating activities and the cash flow from investment activities, since the underlying numbers are often the same.

6.7.4 Calculating the Enterprise Value

To derive the Enterprise Value (or Firm Value), we discount the estimated FCFF and the TV of the company over the chosen time period (10 years):

$$\text{Present Value of Free Cashflows} = \frac{E(FCFF_1)}{(1+WACC)^1} + \frac{E(FCFF_2)}{(1+WACC)^2} + \frac{E(FCFF_3)}{(1+WACC)^3} + \ldots + \frac{E(FCFF_{10})+TV}{(1+WACC)^{10}}$$

[29] A pro-forma income statement is an income statement that contains figures that are estimated based on certain assumptions.

In order to do this, we need an estimate of the TV of the company in the 10th year. The TV is calculated using the estimated FCFF in the 11th year, divided by (WACC – g_c), where g_c refers to the **terminal growth rate, the rate** at which the company's FCFF are expected to grow indefinitely.

The TV gives us the present value of all future cash flows as at the end of year 10. This lump sum value is then discounted back over 10 years to obtain the present value of the terminal value.

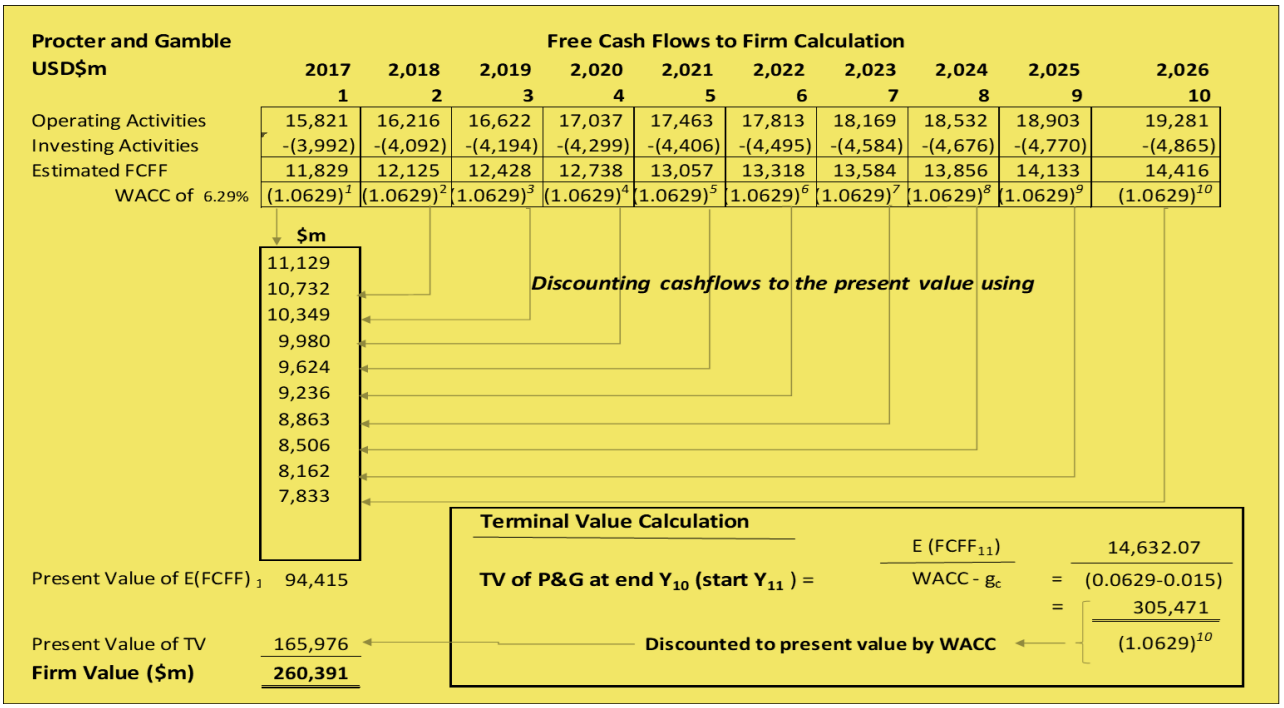
$$\text{Present Value of Terminal Value (TV)} = \frac{\left(\frac{E(FCFF_{11})}{(WACC-g_c)}\right)}{(1+WACC)^{10}}$$

We have taken P&G's Cash Flows from Operating Activities and Investing Activities in 2016 as the base for the projection of future FCFF. P&G's cash flows grew at an average of 1.5% over the last 10 years (2006-2016) and by about 3% over the last 5 years.

For our DCF analysis below, we have applied a growth rate of 2.5% for the cash flows in the first 5 years (2017-2021), slowing to 2% in the next 5 years (2022-2026), and eventually settling at a terminal growth rate of 1.5%. This is consistent with P&G's past growth pattern. P&G's WACC is 6.29% (as calculated earlier in Section 6.7.2(b)).

The Firm Value of P&G is USD 260,391m, as calculated in Figure 6.7.4.

Figure 6.7.4 - Calculating the Firm Value (Enterprise Value) of P&G



6.7.5 Calculating the Fair Value of the Stock

As the enterprise value represents the value of the company as estimated by the DCF model, it is a sum of the **estimated**:

a. Market value of common shareholders;

b. Market value of preferred shareholders; and
c. Market value of debt.

To derive the fair value of the common stock, we first subtract the estimated market value of preferred shareholders and debt from the enterprise value. This leaves us with the estimated market value of common shareholders (or **fair value of common equity**).

We then divide this by the number of shares outstanding to get the estimated value per share. This is also known as the **fair value per share** or the **intrinsic share value**.

$$\text{Est. Common Shareholders' Value} = \text{Firm Value} - \text{Market Value of Debt} - \text{Preference Share Value}$$

$$\text{Est. Intrinsic Share Value} = \frac{(\text{Firm Value} - \text{Market Value of Debt} - \text{Preference Share Value})}{\text{Total Number of Common Shares Outstanding}}$$

Continuing with our example on P&G, we have:

Enterprise Value of P&G	= 260,391
Less Market Value of Debt	= (30,514) information obtained from balance sheet
Less Value of Preference Shares	= (1,038) information obtained from balance sheet
Estimated Market Value of Share Capital	= 228,839
P&G's Estimated Intrinsic Share Value or Price	= (228,839 / 2557.6m shares)
	= USD 89.47

Compared with P&G's price of USD87.65 on 7 July, the estimated value indicates an upside of about 2%. If we believe in our valuation estimate, we can conclude that P&G is fairly valued. This estimated intrinsic share value can be used as the target share price or fair value accompanying the qualitative research report on P&G.

6.7.6 Criticism of the Discounted Cash Flow Model in Estimating Fair Value

The DCF model involves making several assumptions and estimates in order to arrive at the projected cash flows used in the calculation of the firm value and, eventually, the intrinsic (or fair) value of P&G.

Different research teams and analysts will come up with different target price levels. The process of determining fair or intrinsic value using the DCF model is not an exact science, but only serves to provide an estimate that is based on certain assumptions.

Cynical observers may also suggest that the estimate is open to a fair bit of "fudging" until the assumptions chosen enables a price range that is aligned with the qualitative story being presented. The DCF model is still heavily dependent on the calculation of the TV, which can vary significantly if different assumptions are made.

6.7.7 Actual Market Practice for Cash Flow Models

The industry applies the discounting of a company's Free Cash Flows to estimate an intrinsic value or target price for the company's stock. The model usually calls for the discounting cash flows estimated over 10 years back to the present value and the discounting of the terminal value in the 10th year back to the present as well. This DCF model is used in conjunction with relative valuation approaches to come up with a more accurate view of the company's stock value (i.e.; whether it is overvalued or undervalued).

The Multi-stage DDM (Dividend Discount Model – Refer to Section 6.13 – Additional reference material) is seldom applied for valuation on a day-to-day basis. On occasion, the Multi-stage DDM may be used for companies with extremely high dividend payouts (e.g., tobacco stocks such as British American Tobacco).

6.8 Relative Valuation Ratios

The DCF model highlighted earlier is an example of an **absolute valuation approach**, whereby the company's intrinsic value is derived from its future cashflow projections.

In contrast, a relative valuation approach aims to derive a company's value by comparison against companies from the same industry.

With different ratios, relative valuation of companies can be made from either:

- a shareholder's perspective (e.g., Price-Earnings (P/E) Ratio ; or
- a firm-wide perspective (e.g., Enterprise Value to Earnings before Interest, Taxes, Depreciation and Amortisation (EV/EBITDA) ratio), which takes both debtholders and shareholders into consideration.

In practice, multiple relative valuation ratios are used together to make a more holistic assessment and comparison between companies.

6.8.1 Price-Earnings (P/E) Ratio

The P/E ratio an indication of how much investors are willing to pay for a dollar of the company's earnings. It is calculated as the ratio of the company's current price per share to its EPS.

$$\textbf{P/E Ratio} = \frac{\textbf{Market Price Per Share}}{\textbf{Earnings Per Share (EPS)}}$$

A P/E ratio can be in 'trailing' or 'leading' form.

A trailing P/E, also known as the "current P/E", is the current market price divided by the most recent four quarters of EPS. The trailing P/E is what is typically published in newspaper share listings.

A leading (or prospective) P/E is obtained by dividing the current market price by the consensus forecasted earnings for the next year, as projected by research analysts.

When applying P/E ratios, common questions to ask would be:

- How much are you paying for the future earnings generated by a company?

- How does this compare against the company's historical P/E ratio in past years?
- How does this compare against the P/E ratio of other similar companies?
- Are there qualitative reasons for this company to remain at a premium (or discount) compared to other companies?

a. **Making P/E Ratio Comparisons with Historical P/E Ratios and Against Competing Companies**

Suppose we are comparing semiconductor companies: Intel (INTC), Taiwan Semiconductor Manufacturing (TSMC), and United Microelectronics (UMC). Assume the following scenario:

INTC closed at a price of USD34.68. For FY 2016, INTC's EPS was USD2.72. It has a consensus forecasted EPS of USD2.86 and USD2.97 for FY2017 and FY2018, respectively.

Given the information above, INTC's trailing and prospective P/E Ratios are as follows:

Trailing P/E Ratio for 2016	=	34.68/2.72
	=	12.8x
Prospective P/E Ratio for 2017	=	34.68/2.86
	=	12.1x
Prospective P/E Ratio for 2018	=	34.68/2.97
	=	11.7x

We have carried out the same calculations for TSMC and UMC, as shown in Figure 6.8.1(a).

Figure 6.8.1(a) – P/E Ratios for INTC, TSMC, UMC

			CONSENSUS FORECAST			Trailing	Prospective		
	14-Jul-17	EPS	EPS	EPS	EPS	PE	PE	PE	PE
Company	Price	2016A	2017F	2018F	2019F	2016A	2017F	2018F	2019F
INTC	34.68	2.72	2.86	2.97	3.05	12.8	12.1	11.7	11.4
TSMC	36.25	2.00	2.17	2.44	2.65	18.1	16.7	14.9	13.7
UMC	2.43	0.10	0.11	0.13	0.15	24.3	22.1	18.7	16.6

The basic interpretation for P/E would be that an investor will pay less for a dollar of INTC's earnings, as compared to its other two competitors. For example, an investor will have to pay just USD12.1 for a dollar of FY2017 earnings from INTC, as compared to USD16.7 for TMSC and USD22.1 for UMC for an equivalent dollar of earnings.

INTC's lower leading P/E for all 3 projected years implies that an investor will pay less for INTC's future earnings when compared to TSMC and UMC.

However, it is not as straightforward as simply buying companies with the lowest P/E ratio and selling those with the highest.

Figure 6.8.1(b): P/E Ratio for Semi-Conductor Companies (2012-2017)

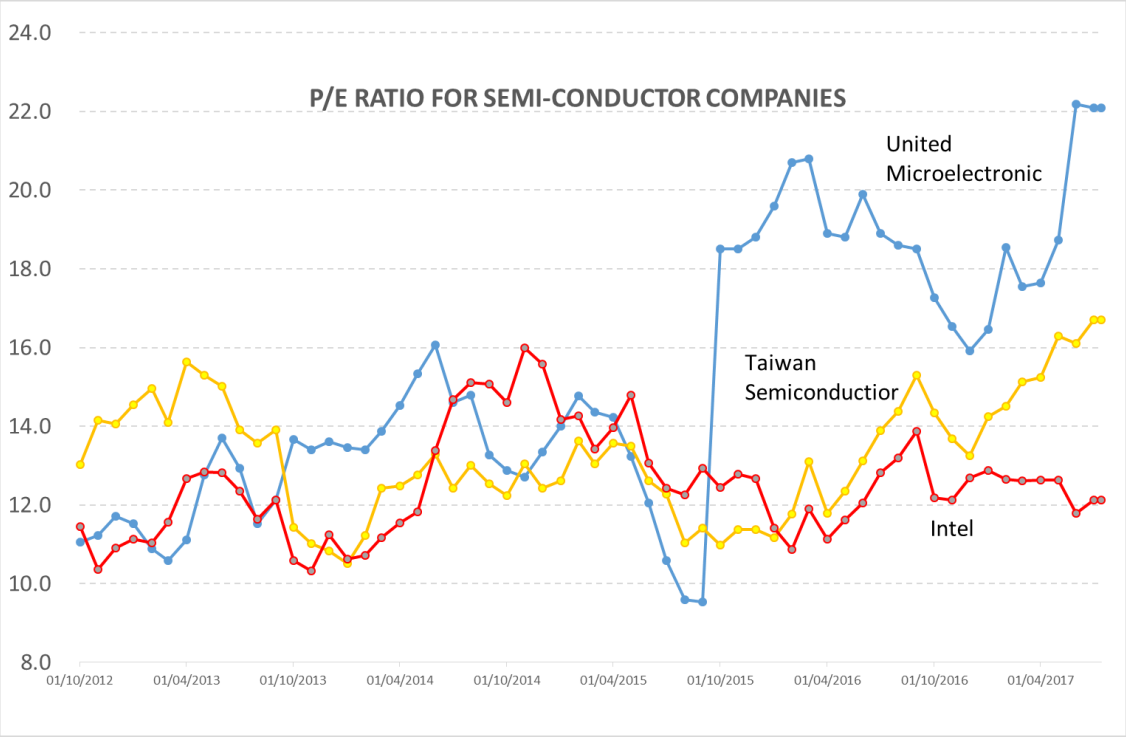


Figure 6.8.1(b) shows the historical P/E trading range of the three companies. The P/E chart is constructed by taking the 12 months of prices, with the last month being three months before the financial year end. Hence, for example, the historical PE chart uses Oct 2012-Sep 2013 prices divided by the actual 2013 EPS. All 3 companies have December financial year ends. Hence, if expected earnings are close to actual earnings, the PE chart reflects what the market decided to trade at, given its expectations of future earnings at that point in time.

The chart shows that the P/E of INTC trades very closely to that of TSMC, while UMC has been trading at a higher P/E in the past. Note that INTC usually traded at a premium to TSMC in the past. Until recently when the roles reversed. UMC, on the other hand, after its recent P/E expansion, usually trades a premium over its peers.

An example of how to compare the P/E ratios between companies and with historical values follows:

INTC: How does INTC's prospective P/E Ratio for FY2018 (11.7x) compare with its historical P/E range?

Given the EPS forecast is accurate, the prospective FY2018 P/E is at the lower end of INTC's normal P/E trading range (10.3x to 16.0x). Comparing with the historical P/E trends of its competitors, we can see that INTC has been trading in tandem with TSMC over the last few periods. However, from 2016 onwards, INTC has been trading at a larger discount to TSMC. We should check if such a discount is justified going forward.

Suppose that we believe that INTC should have a prospective P/E of 14x. This implies an upside potential of 18-20%, which represents a possible BUY signal for INTC.

TSMC: For TSMC to continue sustaining its P/E premium over INTC, it must maintain a stronger EPS growth than INTC. The consensus estimate from analyst forecasts shows that it does indeed have a higher projected

EPS growth than INTC. This market expectation that TSMC will perform better than INTC justifies its P/E premium. However, if TSMC does not deliver, its price will fall relative to INTC.

At the current price level, TSMC's prospective 2017 P/E is 16.7x, which is beyond its normal trading range. For FY2018, TSMC's P/E of 14.9x is also at the high end of its historical trading range.

Is there justification for TSMC's high prospective P/E? The answer is yes (for the moment), because TSMC is vying with Samsung to be the main supplier for Apple's A11 chips for the new Apple iPhone. Hence, TSMC's earnings success for FY2018 will be partly tied to the success of the new Apple iPhone. Combining the P/E analysis with what we know qualitatively at this point, TSMC's premium over INTC appears to be justified.

Most analysts continue to recommend buying TSMC but based purely on the prospective FY2018 P/E alone, TSMC appears to be fairly valued at current price levels. TSMC's price has already risen based on news of its positive exposure to the new Apple iPhone. If the new phone takes off, TSMC's EPS forecast may be revised upwards and there may be further upside potential in its stock price. Conversely, if the take up and sales are lower than expected, then TSMC's EPS forecast for FY2018 may be revised downwards.

UMC: At prospective FY2017 and FY2018 P/E ratios of 22.1x and 18.7x, respectively, UMC appears expensive compared to its historical trading range of 11-15x. While its EPS growth is higher than TSMC's and INTC's, it is trading at P/E multiples well beyond its normal trading range. Moreover, its EPS growth is coming from a much lower absolute EPS base. Qualitatively, there has been no major information that will suggest a more sustainable earnings growth except for the recovery of the US economy. If there is no known justification for UMC's share price to trade at such high multiples, UMC's share price may weaken.

b. **Factors affecting P/E Ratios**

From the analysis above, it is clear that investment is not simply about buying companies with low P/E and selling those with high P/E. Instead, we must examine the underlying qualitative rationale for the ratings. Companies will remain at a perpetual premium (discount) to others so long as they have stronger (weaker) earnings potential compared to their peers.

Other than projected earnings, some other factors that can affect P/E ratios are:

i. **Low transparency and predictability of earnings.** When the risks are high, and transparency and predictability of earnings is low, even with high earnings growth, the lower predictability increases risks for investors; hence, a lower P/E multiple is justified.

ii. **Impending events that may affect earnings**, such as the upcoming renewal of major contracts or pending legal decisions. In such scenarios, analysts will have to make major assumptions and the P/E ratio may be lower because of the uncertainty.

iii. **Uneven growth trajectory.** For companies with potential exponential growth in the long term but very low earnings in their short term, then prospective P/E ratios in the first few years may be very high in anticipation of growth in the longer term. The Price/Earnings-to-Growth (PEG) ratio will help in this assessment.

iv. **Market inefficiency.** In some markets, when the stock is less liquid and controlled by major shareholders, the markets may not be able to price the stock efficiently. There could therefore be a distorted P/E due to illiquidity.

c. **Growth Companies vs Mature Companies**

Stocks of growth companies (growth stocks) would typically command higher P/E ratios as compared to the market average. Because of their higher expected earnings growth, investors are willing to pay more for growth stocks, thus driving up their P/E ratios. While high-growth industries will vary from country to country, in general, companies in the tech, biotech, and disruptive technology industries are generally considered growth companies and command higher P/E ratios. In contrast, defensive, mature companies with low but steady earnings growth will tend to have lower P/E ratios (e.g., utilities, tobacco, and diversified food).

d. **Bear and Bull Market P/E Ratios**

P/E ratios are also affected by market psychology and market cycles. In bear market conditions, market sentiment is typically poorer, and the prospective outlook can be more pessimistic. During such times, P/E ratios will generally be lower. Likewise, during bullish markets, P/E ratios will generally be higher. Hence, it is useful to compare P/E ratios during similar market periods. Also, we can use the P/E multiples under bear market conditions as the likely floor for P/E levels.

e. **Shiller's Inflation-Adjusted P/E ratio - Predicting Overvaluation**

Robert Shiller[30] used inflation-adjusted P/E ratios to try to track periods when the S&P500 was overvalued. According to Shiller, the S&P500's P/E ratio will pass the 30x mark during periods of overvaluation (otherwise termed as irrational exuberance by former US Federal Reserve Chairman, Alan Greenspan). Although we have been flirting with that level in recent years, there were only two other times in the past – the Great Depression of 1929 and the 2000 Dot.com bubble – when the inflation-adjusted S&P500 P/E ratio was above 30x.

f. **P/E Ratios and Interest Rates**

The P/E ratio is related to the required rate of return, which is in turn related to interest rates. As interest rates increase, required rates of return on all securities generally increase.

When interest rates rise, the fall in price valuations will be greater than the fall in earnings. Hence, P/E ratios will fall. Conversely, when interest rates fall, price valuations tend to be more significantly impacted than interest savings. Hence, P/E ratios will tend to rise. Therefore, all else being equal, P/E ratios should decline (rise) if interest rates rise (decline).

Since the 2008 Global Financial Crisis, we have been experiencing a low interest rate environment. As such, P/E ratios have tended to rise and will continue to do so until more significant rate hikes are carried out by central banks around the world.

g. **Advantages and Disadvantages of P/E Ratios**

Advantages	Disadvantages
1. The P/E ratio concept is intuitive and easy to understand. Investors are paying for future earnings and want to pay less for future earnings.	1. P/E ratios do not explicitly reflect the amount of debt used to drive earnings, as it looks at a company mainly from a shareholder's perspective. 2. P/E ratios cannot be used when there are negative earnings. 3. P/E ratios can be influenced by changing the capital structure.

[30] Author of the book "Irrational Exuberance".

Advantages	Disadvantages
2. P/E ratios can be easily used for comparison across companies and sectors.	4. Accounting differences can affect earnings and P/E ratios. 5. Earnings can be volatile or lumpy, making P/E calculations difficult to interpret and apply. 6. Earnings forecasts can be quite inaccurate. 7. P/E ratios do not distinguish between valuation and expected growth differences.

6.8.2 Price-Earnings Growth (PEG) Ratio

One criticism about the P/E ratio is that it does not account for a company's expected future growth. Specifically, a company with a low P/E can be undervalued or simply have poorer expected future growth. Conversely, a company with a high P/E can be overvalued or simply have stronger expected future growth. Strictly speaking, P/E ratios should only be compared for companies with similar expected future growth in order to make conclusions about relative valuation. To address this criticism, analysts use the Price-Earnings Growth (PEG) ratio.

The PEG ratio is simply the P/E ratio divided by the expected annual earnings growth (in %), as measured by EPS:

$$\text{PEG} = \frac{\text{P/E}}{\text{\% Growth in EPS}}$$

$$= \frac{\left(\frac{P_0}{EPS_{Y1}}\right)}{\left(\left(\frac{EPS_{Y4}}{EPS_{Y1}}\right)^{\frac{1}{3}} - 1\right) * 100}$$

assuming the % prospective EPS growth is compounded over 3 years.

By controlling for the expected growth in earnings, the PEG ratio becomes an indicator of whether the current share price is understated or overstated when compared to that for another company, which may have different earnings growth prospects.

PEG is most widely used for high-growth companies and industries. On occasion, when the growth potential is further into the future, the PEG may be calculated with a 5-year projected growth rate for comparison purposes.

A low PEG ratio indicates an undervalued share. As a rule of thumb, a PEG ratio below 1 would be attractive but typically very hard to find. Such stocks could be relatively new and therefore not actively covered and may also carry a higher level of risk (e.g., the earnings growth may not actually materialize). These companies are typically involved in high-tech, biotech, or disruptive technologies that would be in high demand. Once noticed, more investors will be interested these stocks due to the future growth potential, raising the share price and causing the PEG ratio to rise above 1.

The example below illustrates how the PEG ratio could provide a more comprehensive picture of a company's prospects, especially if the companies involved are growth companies with highly skewed P/E ratios:

Figure 6.8.2 – P/E Ratios of High Growth Companies

Company	14-Jul-17 Price	Actual EPS 2016A	Concensus Forecast EPS 2017F	EPS 2018F	EPS 2019F	EPS 2020F	Trailing PE 2016A	Prospective PE 2017F	PE 2018F	PE 2019F	PE 2020F	Growth* 2017-2020	PEG
Alphabet	955.99	34.73	35.00	40.44	45.27	50.70	27.5	27.30	23.70	21.10	18.90	13%	2.1
Facebook	159.97	4.23	4.87	6.03	7.24	8.54	37.8	32.80	26.50	22.10	18.70	21%	1.6
Alibaba	151.83	2.60	3.41	4.49	5.75	7.18	58.4	44.50	33.80	26.40	21.10	28%	1.6
Baidu	188.55	5.53	5.78	7.87	9.84	12.10	34.1	32.60	24.00	19.20	15.60	28%	1.2
**Annualised Compound Growth*													

Figure 6.8.2 shows some global high-growth companies in the tech sector. Note that, while Alibaba has the highest prospective P/E ratios of 44.5 in 2017 and 33.8 in 2018, respectively, its PEG ratio is amongst the lowest at 1.6, reflecting its high (28%) projected annual growth rate over the next few years.

The ability of such companies to generate high double-digit growth makes them very lucrative investments, especially if you can identify them early enough. Finding the next Apple, Google, Facebook or Alibaba is very difficult. Some companies on the NASDAQ and the secondary bourses in other markets may also have double-digit growth potential but will naturally have commensurate business risks.

A note of caution with projections is in order: When markets are bullish, and the economy is strong, the analyst community tends to overestimate the sustainability of high earnings growth.

6.8.3 Price-to-Book Value (P/BV) Ratio

We calculated book value per share and revised net asset value per share earlier in the Chapter. In this section we compare the net asset value-related calculation with the current stock price. This is the P/BV ratio:

$$\textbf{P/BV Ratio} = \frac{\textbf{Market Price Per Share}}{\textbf{Book Value Per Share}}$$

The P/BV ratio is best suited for companies whose value is tied directly to their asset holdings. For example, book value is more important for investment companies, banks, insurance companies and property companies, as many of the assets held by such companies are reflected in the books and can be liquidated.

a. **Price-to-Book Ratio for Financial Institutions**

As the calculation of the P/BV ratio is based on book and not market value, the ratio can be misleading if the current market values of the assets are substantially different from their book values. For example, if a bank's loans have become non-performing, the market value of these loans may be near to zero, even though the book value of the loans remains the same.

Since the Global Financial Crisis, several banks have seen their book values eroded by losses from proprietary trading, excessive risk-taking on distressed debts, and credit derivative losses, amongst other reasons. If adequate provisions are made and proper recapitalisation is carried out, the book value can be accurately adjusted. However, if provisions are inadequate, additional write-downs have to be made over time.

The lack of transparency concerning the quality of loan books and investment positions is a problem when applying the P/BV ratio to evaluate banks. Besides this, the P/BV ratio has also become less dependable and more difficult to apply because banks continue to encounter credit losses even after the crisis – the latest being provisioning for non-performing loans to oil-related companies and commodities trading-related losses (refer to Figure 6.8.3(a)).

Figure 6.8.3 (a) – Price/BV Ratio of Selected International Banks

Some International Banks	14-Jul-16 Price/BV
JP Morgan Chase	1.43
Goldman Sachs	1.24
Morgan Stanley	1.21
UBS	1.16
Standard Chartered	1.10
HSBC	1.06
Bank of America	0.99
Citigroup	0.88
Credit Suisse	0.73
RBS	0.62

b. **Price-to-Book Ratio for Property Companies**

The P/BV ratio is also actively applied to property companies because the bulk of its asset values are well-reflected on the balance sheet. Property companies and REITs with sizeable commercial, office and residential property holdings may enjoy significant appreciation in book value when the property market rises. Conversely, they may also suffer a substantial fall in book value in a property downturn. Accountants rely on independent professional real estate consultants to revalue property assets annually.

During major stock market and economic downturns, stock prices can fall much more sharply than book value. In such situations, some companies with diffused ownership concentration can became good takeover targets. With the stock price being lower than book value, control of the assets can theoretically be acquired cheaply. Investors with deep pockets can potentially acquire distressed property companies with hard assets.

In the aftermath of the Asian Currency Crisis (1997), some small, listed property developers were caught in a liquidity crunch. These companies were eventually acquired by larger property developers at stock price levels below book values.

c. **Hidden Reserves in the Book Value**

There are also cases where the market values of property land bank and properties are significantly different from their book values. This can happen when land is being rezoned for a different use (e.g., from agriculture land to commercial or residential usage) or when land acquired become more accessible due to better road and rail links. This can happen in many large countries.

Although such land may remain on the balance sheet at book value, they are actually worth a lot more when revised to market value. When accounting limits allow for such land to remain at book value, a hidden reserve is created. Hidden reserves are also created when provisioning made in prior years are no longer needed but are not written back in the accounts.

In such situations, the book value is understated, and the true P/BV ratio would be lower than what is calculated, making it more likely that the company is undervalued.

d. **Price-to-Revised Net Asset Value (P/RNAV)**

When asset values are significantly different from book values, equity analysts will revise the book values of assets to their estimated market values (which may be lower or higher). This happens in between reporting periods when the balance sheet has yet to be revised to reflect market prices.

Where possible, the same revisions will be made for liabilities as well, resulting in a revised net asset value (RNAV). The current price is then compared against the RNAV per share. Just as book value is the net asset value at book, the RNAV reflects the net asset value at market.

$$\text{P/RNAV Ratio} = \frac{\text{Market Price Per Share}}{\text{(Market Value of Assets - Liabilities)/Shares Outstanding}}$$

This P/RNAV ratio is then applied across similar companies that are being compared.

e. **Application of Price/Book Value and Price/RNAV Ratio**

When we use these ratios, we are asking how much we are paying for the net asset value of the company either at P/BV or at P/RNAV. Like the P/E ratio, when applying these two ratios, it not just a matter of picking the lowest number. Instead, we will look at how a company's existing ratio compares to:

i. its historical ratios; and

ii. the ratios of similar companies.

Often, companies will trade within a certain level or band.

Some companies will trade almost perpetually at a premium over others because of the quality of their assets (e.g., property holdings). Other companies will trade at a discount because of their higher debt levels. However, if the premium or discount compared to other companies narrows or widens significantly, like P/E ratios, it is important to review the underlying qualitative factors that influence the ratio.

f. **Advantages and Disadvantages of Price-to-Book Value Ratio**

Advantages	Disadvantages
1. Book value is cumulative and is usually positive, unlike EPS. 2. Book value is usually more stable than EPS, which can be volatile. 3. P/BV is well suited for companies whose value is closely tied to its tangible asset holdings (banks, insurance companies, real-estate companies).	1. Companies with intangible assets, such as patents and proprietary research knowledge, cannot be fairly valued using the P/BV ratio. 2. The validity and effectiveness of the ratio is dependent on the transparency and proper treatment of provisions and charges against book value. 3. Book value may be understated or overstated when assets rise or fall in value. This affects P/BV calculations.

6.8.4 Price-to-Sales (P/Sales) Ratio and Enterprise Value-to-Sales (EV/Sales) Ratio

a. **Price-to-Sales Ratio**

P/Sales ratio is a feasible measure of comparison for companies who generate very low, zero, or negative earnings. It indicates the number of times the market is willing to pay for a firm's sales revenue per share.

$$\text{P/Sales Ratio} = \frac{\text{Current Market Price Per Share}}{\text{Sales Per Share}}$$

This is calculated by dividing the current share price by its projected sales per share, i.e., the annual sales divided by the number of ordinary shares outstanding.

P/Sales ratio is useful in situations of strong sales but where earnings are lacking or negative. Some high-tech and biotech companies may be generating sales but remain unprofitable. Proponents of this ratio hold the view that consistent sales are indicative of a strong company and that sales are a fundamental requirement for earnings.

This ratio should, however, be used with caution. It does not present a complete picture because it does not consider the expenses and liabilities of a company. Besides, a lower P/Sales ratio is not necessarily a positive indicator because it may simply indicate that the company is unprofitable.

Conceptually, the P/Sales ratio is also flawed. The numerator (Price) relates to shareholders only, even though sales are generated from funds supplied by both shareholders and debtholders. Hence, the numerator and denominator are inconsistent.

b. **Enterprise Value-to-Sales Ratio**

Some investors believe that the EV/Sales ratio is a more appropriate measure than the P/Sales ratio, since both the numerator (EV) and denominator (Sales) are consistent because both debt and equity financing are used to generate sales.

Enterprise Value became more widely adopted among the investment community in the 1980s and is a total firm value concept. It is calculated as follows in Figure 6.8.4(b):

Figure 6.8.4(b) – Derivation of Enterprise Value

	Market Value of Equity (i.e. Market Capitalisation)
+	Market Value of Debt (usually book value is sufficient)
+	Market Value of Minority Interest (if any)
+	Market Value of Preferred Shares (if any)
+	Market Value of Unfunded Pensions (if any)
-	Less Market Value of Associate Companies
-	Less Cash Equivalents
	Enterprise Value

c. **Price-to-Sales Ratio vs Enterprise Value-to-Sales Ratio**

The P/Sales ratio is easier and more convenient to use than the EV/Sales ratio. The former also intuitively poses to investors the question: "How much am I paying for the projected or current sales of the company?"

However, the P/Sales Ratio does not consider risks that may arise from debt financing, especially if large amounts of debt is used to generate sales. It is therefore possible for a company with high debt levels to appear inexpensive due to a low P/Sales ratio. As Enterprise Value encompasses both debt and equity, the EV/Sales ratio accounts for the value of debt and is more appropriate when making comparisons across companies.

d. **Advantages and Disadvantages of Enterprise Value-to-Sales Ratio**

Advantages	Disadvantages
1. Generally, sales figures are less subject to manipulation compared to earnings. 2. Sales are positive, unlike EPS, which can be negative. This is useful for start-ups and companies that have no earnings yet. 3. Sales are more stable than EPS, which can be quite volatile. 4. Applicable for shares of almost all companies. 5. EV/Sales directly reflects the impact of higher debt on the relative valuation comparison (but P/Sales does not). 6. EV/Sales is conceptually correct since both the numerator and the denominator relate to the total enterprise.	1. High sales revenues do not necessarily mean that the company is profitable. 2. Sales revenue recognition practices, even without manipulation, can vary from one company to another and may make comparisons difficult. 3. As expenses are not included, the comparison may be incomplete or misleading.

6.8.5 Enterprise Value-to-Cash Flow (EV/CF) Ratio

The EV/CF ratio is used as a comparison between companies because of concerns over the potential manipulation of earnings. Cash flows are much less prone to manipulation and, usually, operating cash flow (or free cash flow) is used. Operating cash flow refers to the total amount of cash generated by a company through its core operations. It can be found in the Statement of Cash Flows under "Cash generated from operations".

Assume that the market price of Fulton Corporation is $18 per share and it has 1 billion shares outstanding as at 31 December 2018. It has $10 million of preference shares and borrowings worth $400 million. Assume there are no associates, minority interest, and unfunded pension obligations. Please refer to the Statement of Cash Flows for Fulton Corporation Ltd at Table 6.8.5.

The EV/CF ratio for Fulton Corporation is as follows:

$$\textbf{Enterprise Value-to-Cash Value Ratio} = \frac{\text{Market Cap + Debt + Preferred Shares - Cash}}{\text{Cashflow from Operations}}$$

$$= \frac{(\$18 \times \$1000\text{m}) + \$400\text{m} + \$10\text{m} - \$8000\text{m}}{\$2210\text{m}}$$

$$= 4.71$$

When comparing between companies, a lower EV/CF ratio is desirable as it implies that the company is generating more cash flows from the capital supplied.

Table 6.8.5 – Statement of Cash Flows for the year ended 31 December 2018 for Fulton Corporation Ltd

	Notes	2018 $m	2017 $m
Profit before taxation			
Cash generated from operations	20	2,210	2,000
Interest paid	22	(1,500)	(1,200)
Income tax paid	23	(1,200)	(750)
Net cash from operating activities		(490)	50
Cash flows from investing activities			
Purchase of property, plant and equipment	21	(1,820)	(2,300)
Purchase of intangible assets	21	(750)	(1,200)
Proceeds from sale of equipment	18	150	120
Proceeds from sale of intangible assets	18	180	250
Interest received	18	1,500	1,050
Net cash used in investing activities		(740)	(2,080)
Cash flows from financing activities			
Proceeds from issue of share capital	4	3,000	-
Proceeds from long term borrowings	6	3,030	6,030
Dividend paid	21	(500)	(500)
Net cash used in financing activities		5,530	5,530
Net increase in cash and cash equivalents		4,300	3,500
Cash & cash equivalents at start of the period		3,700	200
Cash & cash equivalents at end of the period		8,000	3,700

6.8.5.1 **Advantages and Disadvantages of Enterprise Value-to-Cash Flow from Operations**

Advantages	Disadvantages
1. Cash flows are less likely to be manipulated, compared to earnings. 2. Cash flows are independent of accounting methods. 3. EV/CF considers the impact of debt since debt funding is also used to generate operating cash flows.	1. Cash flows can be from operating, investing or financing activities, and the choice of cash flow used will affect the valuation comparison among companies.

6.8.6 Enterprise Value-to-Earnings before Interest, Taxes, Depreciation and Amortisation (EV/EBITDA) Ratio

The EV/EBITDA ratio is also referred to as the **enterprise multiple or the earnings before interest, taxes, depreciation and amortisation (EBITDA) multiple** and is actively used in the relative valuation of stocks.

EV/EBITDA is a more popular valuation ratio than EV/Sales or EV/Operating Cash Flow.

a. **EBITDA vs Net Earnings and Operating Cash Flow**

There are several reasons why EBITDA is preferred over net earnings and operating cash flow in valuation:

i. EBITDA is preferred over net earnings because it is closer to cash-based earnings. Unlike net earnings, EBITDA does not subtract non-cash expenses like Depreciation and Amortisation, which are accounting adjustments. By adding back depreciation and amortisation to EBIT, we get the bulk of cash-related earnings.

ii. EBITDA is a firm value concept, adding back interest cost because it considers the total earnings that are available to be paid to creditors and shareholders. Net earnings are a shareholder concept as interest expense has been subtracted.

iii. EBITDA is preferred over operating cashflows because the latter can experience wide swings due to changes in working capital. Depending on the industry, working capital adjustments may not really reflect business performance, as they are influenced by decisions on inventory as well as payables and receivables management.

iv. It is easier to just forecast EBITDA instead of operating cash flow.

b. **Why Enterprise Value is Preferred over Price**

EV considers the impact of debt and other capital on the total value of the firm while price relates only to equity value. Hence, EV/EBITDA is preferred over P/E. EV/EBITDA became actively used as another measure for comparing the relative value of companies in the 1980s because it was recognized that P/E did not directly reflect the value of debt in its ratio. Unlike the P/E, the EV/EBITDA ratio cannot be manipulated by changes in capital structure, which makes possible fair comparison of companies with different capital structures.

$$\frac{\text{Market Cap + Debt + Preferred Shares + Minorities + Pension – Associates – Cash \& Near Cash}}{\text{Earnings Before and Taxes and Depreciation and Amortization}}$$

c. **Application of the EV/EBITDA Ratio**

When applying the EV/EBITDA ratio, we typically use the current EV in the numerator and the projected annual EBITDA for one to three years into the future. By doing so, we are asking the question: "What is the total capital deployed, net of cash, to generate the cash earnings for a given year for the company?"

All else being equal, a lower EV/EBITDA ratio will mean that less capital (EV) is deployed to generate cash earnings (EBITDA). However, as with the other ratios, some companies will have perpetual EV/EBITDA discounts or premiums of over others because of the various reasons highlighted in Section 6.8.1(b).

The EV/EBITDA ratio is reflected in Bloomberg and most equity research reports from investment brokerage firms.

d. **Advantages and Disadvantages of Enterprise Value-to-EBITDA**

Advantages	Disadvantages
1. Numerator (EV) captures the impact of debt used to generate earnings and enables comparison across companies with different amounts of debt used. 2. Denominator (EBITDA) reflects cash-based earnings that are unaffected by swings from changes in working capital.	1. Less intuitive for retail investors, who prefer and can identify with share price comparisons. 2. EV is more tedious to measure as it requires the additional estimation of the current market value of debt, associates, preference shares, and minority interest.

6.8.7 Absolute Valuation vs Relative Valuation

Absolute and relative valuation approaches have the following advantages and disadvantages:

	Advantages	Disadvantages
Absolute (Discounted Cash Flow) Method	1. Conceptually relates the intrinsic value to the earnings and cash flows due to the firm over time. 2. Discounted cash flow analysis is a widely accepted way of determining current value. 3. Flexibility to incorporate changing growth rates over time in cash flows.	1. Discounted cash flow models depend heavily on the calculation of terminal values. Terminal values can vary significantly with different assumptions for WACC and growth rates. 2. The calculation of terminal value always requires the discount rate to be higher than the growth rate.

	Advantages	Disadvantages
Relative Valuation	1. Allows for easy comparison across companies. 2. Intuitive and easy to implement.	1. Market sentiment and cycles can influence relative valuation. 2. Need to have a comparable set of entities to allow proper relative valuation.

6.8.8 Dividend Yield

A key consideration, especially for income-oriented stock investors, is dividend yield. Companies with stable, non-cyclical businesses that pay a constant stream of healthy dividends have downside support for their share prices.

The dividend yield is the income component of a share's return and is stated as a percentage of the current share price.

For example, if the price of a share is $2 and the annual gross dividend is $0.10 per share, the dividend yield is 5%:

$$\text{Dividend Yield} = \frac{\text{Dividend Per Share}}{\text{Market Price Per Share}} \times 100$$

$$=(0.10/2.00)*100$$

$$=5\%$$

The dividend payout ratio is the ratio of dividends to earnings. It indicates the percentage of a firm's earnings that is paid out as cash dividends to its shareholders. For example, if a company earns $0.40 per share and pays out a dividend of $0.10 per share, the dividend payout ratio is 25%. The retention ratio is equal to 1 minus the dividend payout ratio and indicates the percentage of a firm's current earnings that are retained for reinvestment purposes.

Expected Dividend Yield

KW Corp is an engineering, property and manufacturing company. Stock-broking company, ABC & Partners (ABC), expected KW Corp to pay a 1.5 cent dividend in 2017, with the dividend expected to rise to 3.0 cents in 2018.

A 3-cent dividend on KW Corp, whose share price was 31 cents at the time of ABC's research report, would have translated into a high dividend yield of 9.7%.

However, ABC's analyst expects some "yield compression", which means that as the share price rises, the dividend yield on the share will go down. If we have a dividend yield expectation of 7.5% on a dividend of 3 cents, the implied share price will be 3 cents / 7.5% (i.e., 40 cents).

Real estate investment trusts (REITs) can be valued using a dividend discount model as they are required to pay out 90% of their distributable income and their earnings are relatively predictable.

6.8.9 Cash Per Share and Valuations

Related to dividend is the cash per share. When a company has a significant proportion of cash holdings, there is a potential for large dividend payments. Such companies are also potential targets for acquisition.

Consider two companies with the same P/E ratio of 10x earnings:

	Company A (Cash Rich)	Company B (Cash Poor)
Price	2.50	3.00
EPS	0.25	0.30
Net cash per share	0.50	Net debt

Company A is cash-rich and has excess idle cash of $0.50 per share even if it pays off its debt. In contrast, Company B is cash poor as it is in a net debt situation.

In this scenario, Company A is undervalued. If Company A is acquired at 2.50 and we net out the idle cash of $0.50 per share, the buyer is buying the company at a P/E of only 2.00/0.25 or 8x, which is lower than the 10x P/E of Company B.

6.8.10 Strategic Value and Break-up Value of a Listed Company

In cases of corporate acquisition, it is important to consider the strategic and break-up value of the company. When performing an acquisition, the acquiring company naturally hopes that the acquisition will result in more earnings and a higher overall value to the combined entity than what is reflected in the offer price for the acquisition.

Strategic Value: When the perceived strategic value is higher than the offer price (which is, in turn, usually higher or equal to the market price), there is potential for the share prices of both the acquiring and target companies to rise. If the perceived strategic value is less than the offer price, it is likely that the stock price of the acquiring company will fall.

Break-Up Value: There are also acquisitions in which the sum of the parts sold is expected to be higher than the target company's value as an ongoing concern (i.e., the company is currently undervalued because of how it is managed).

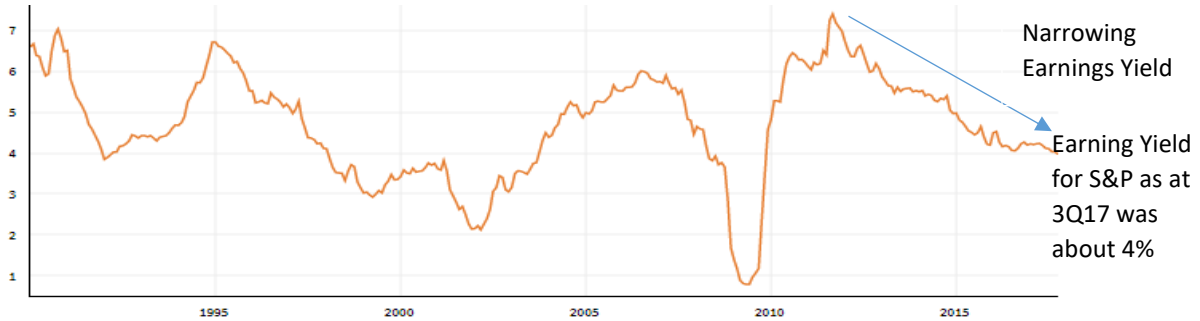
During the seventies in the US, there were a host of hostile takeovers carried out with the expectation to sell off parts of the businesses and assets at a higher overall value than the takeover market value (share price x shares outstanding).

6.8.11 Earnings Yield

The Earnings-to-Price (E/P) ratio measures a company's earnings against its share price. When expressed as a percentage, it is also known as the earnings yield.

Figure 6.8.11 (a) shows the monthly earnings yield of the S&P500 from 1990 to 2017(3Q). Following the Global Financial Crisis in 2007/08, there was a sharp drop in earning yields. However, the extremely low interest rates that has continued for the bulk of 2009-2016 provided support for earnings recovery and earnings yields rebounded to as high as 6.0-7.0% immediately after the Global Financial Crisis.

Figure 6.8.11(a): Monthly S&P500 Earnings Yield Chart



Source: www.Quandl.com

As the term suggests, the E/P ratio is the inverse of the P/E ratio. Earnings yields are usually compared against bond yields to assess whether share prices are over- or under-valued. The difference between these yields is referred to as the yield gap. If the S&P500 earnings yield is still 1.5-2.0% above the US Treasury bond yields, there is still room for the S&P to rise. As shown in Figure 6.8.11 (b), the US Treasury bond yield as of Oct 2017 was 2.45%, compared to the S&P500 earnings yield of about 4% during the same period.

Figure 6.8.11(b): US 10-Year Treasury Yields



6.9 Initial Public Offerings

In Singapore, listing on the SGX Mainboard requires the fulfilment of listing requirements spelt out by the exchange. However, listing on Catalist (SGX's secondary board) is supervised by a "sponsor" (the equivalent of Issue Manager for a Mainboard listing) and no quantitative requirements are needed. Sponsors decide on the suitability of the candidate company seeking a listing. Sponsors are qualified professional firms with experience in corporate finance and mergers and acquisitions.

In any Initial Public Offerings (IPOs) process, the issuer will make all necessary preparations in order to obtain approval from SGX to list the issuer's share either on the Mainboard or the secondary board. To do this, the issuer will have to engage an investment bank or a member company of SGX to be its lead manager or sponsor. Underwriters such as brokerages and other financial institutions will also be involved. The lead manager earns management fees, while the underwriters earn underwriting fees from the issuer. Lawyers will also be needed to draft the IPO prospectus and related documents, while public accountants will be needed to prepare the audited financial reports.

Once the SGX approves the draft prospectus for the IPO, it can be lodged on the MAS Offers and Prospectuses Electronic Repository and Access (OPERA) website. The issuer will distribute its prospectus to institutional and retail investors. However, if there is pre-placement activity, institutional investors may view the offering before retail investors. IPOs are usually distributed in two tranches – a private placement tranche and a public retail tranche. The private tranche is typically placed with institutional investors and accredited investors, usually preferred customers of brokerages and banks, while the public tranche comprises retail investors. Usually, the private placement tranche comprises a small group of investors and takes up a major portion of the offering, while the retail tranche is smaller and much more widely distributed.

The commencement of retail or public offerings begins from the date of registration of the prospectus with the MAS and typically open for at least 2 markets days (excluding the date of commencement of offer). The share applications received will then be allocated or balloted, if necessary, because of over-subscription, within a few days. Trading begins the next business day after allocation and the shares commence trading on a "ready" basis i.e., bought or sold for settlement 2 days after the transaction date (T+2).

When deciding whether to subscribe to an IPO, potential investors should be aware of the following risks:

a. **Vendor share sales** i.e., the sale of the issuer's own holdings, rather than new shares, in the market. The obvious risk here is that the issuer may be offloading its own shares to the public due to its knowledge of a poor outlook for the company. Potential investors must watch out for this and for the presence of a lock-up option. The lock-up option places a moratorium on the sale of the issuer's shares for a stipulated period. In a way, this is the issuer's expression of confidence in the company that is being brought public with the IPO.

b. **Deferred shares** or share options that have been awarded before the IPO may potentially dilute the earnings of the new IPO investors, because the EPS will be depressed once these deferred shares or share options are exercised.

c. The **over-allotment option** (also known as the Greenshoe option) gives the issuer the right to increase its offering through an over-allotment of shares. IPO underwriters can use this option to stabilise the price of the IPO shares if they perform poorly on initial trading by buying back the additional shares sold from the market. However, not all IPOs carry this over-allotment option.

d. Potential IPO investors should also look out for the issuer's right to "**claw-back**". This term refers to the re-allocation of shares between institutional and retail investors. For example, if the institutional take-up is poorer than expected, the "claw-back" provision may be invoked, and retail investors will get a higher allocation of a share that is probably avoided by institutional investors for good reason.

In addition to the specific IPO risks above, investors have to carry out their own due diligence on IPOs and evaluate these by studying the prospectus carefully. IPO prospectuses can be found in the "OPERA Public Portal" on the MAS website (www.mas.gov.sg).

When evaluating the prospectus, the following should be noted:

i. **Underwriters** - The underwriters of the IPO would generally purchase any remaining unsubscribed IPO shares.

ii. **Business Model** - The business model, strategy and industry competition should be examined. This is usually discussed in sections on Management's Discussion, Business Overview, and Competition.

iii. **Use of Proceeds** – If the proceeds are to be used to pay off a large amount of debt, there may not be enough left for the expansion of operations. Also, if the proceeds will be used to buy out existing shareholders, this may be a bail-out at the expense of new investors.

iv. **Management** - The experience and background of the company's management, especially those of key appointment holders such as the Chief Executive Officer and Chief Financial Officer etc., should be examined to validate their experience and track records.

v. **Financial Analysis** - A financial analysis of the balance sheet, income statement, and statement of cash flows should be conducted, using the financial ratios and equity valuation ratios covered earlier in this Chapter. Special attention should be paid to:

a. overly rapid expansion of sales;

b. inventory growing faster than sales;

c. unusual fall in free cash flows; or

d. aggressive acquisitions.

vi. **Timing** – The IPO's "lockup period", which refers to the moratorium period during which pre-IPO investors and promoters are not allowed to sell their shares, should be noted.

vii. There should be **no conflicts of interest** between the existing owners and management.

viii. After an IPO, shares of a listed company can subsequently be issued through **secondary offerings**. A secondary offering is typically carried out as a financing exercise for the company. This could be in the form of a rights issue, which can be renounceable or non-renounceable. Rights issues are basically an offer to existing shareholders to purchase additional shares in proportion to their holdings. In order to prevent dilution of their earnings, existing shareholders would have to take up these rights, but it also means they have to put up more money for the additional shares. In a rights issue, the company will lodge an Offer Information Statement (OIS) instead of a prospectus with the MAS.

ix. In addition, a secondary offering can also be a **compliance placement** made to meet regulatory requirements in regard to the distribution of shares.

6.10 Valuing Companies in Initial Public Offerings

Both the DCF and relative valuation methods are used in identifying a target price or price range for IPOs. The use of the DCF method in IPO valuation is akin to what was laid out earlier in Section 6.7. We would:

a. Discount the free cash flow of the company to derive the firm's value;,
b. Subtract debt to obtain the equity value; and
c. Divide it by the total number of shares outstanding to come up with the target IPO share price.

There are alternative ways to arrive at an overall value of the company. Below is an example of how we can use relative valuation with enterprise value-to-sales, EBITDA, and assets to determine a target share price range of Q-Star Pharmaceuticals.

Valuing the IPO of Q-Star Pharmaceuticals

A Sovereign Wealth Fund owns a pharmaceutical and biotech company, Q-Star Pharmaceuticals. Q-Star Pharmaceuticals is now going through a preparatory period of 2-3 years before seeking a major listing on the NASDAQ or the NYSE with a secondary listing on an Asian exchange. Q-Star's owners would like to get an estimate of the possible current value of the company.

IPO Company - Q-Star Pharmaceuticals

Sales = USD 7.65 billion (bn)
EBITDA = USD 2.20bn
Assets = USD18.30bn
Shares post-issue = 750 million (m)
Debt = USD11.0bn

Table 6.10 shows the information we have selected to conduct a relative valuation of Q-Star. The ratios used in the relative valuation would be the Enterprise Value (EV)-to-Sales, EV-to-EBITDA, and EV-to-Asset ratios of the top 5 pharmaceutical companies. To be conservative, the lowest multiples are selected from each category of the EV ratios. On top of that, we will apply a 20% discount on the chosen multiple, to account for the fact that Q-Star Pharmaceuticals does not have the same pipeline of patents and brand image as compared to the top pharmaceutical companies.

Table 6.10 - Key Measures of Top 5 Pharmaceutical Companies

18-Jul-17	MRK	JNJ	PFE	LLY	GSK
Sales ($Bn)	39.9	72.2	52.8	21.2	37.8
EBITDA ($Bn)	15.2	25.1	19.5	5.6	9.5
Assets ($Bn)	95.4	141.2	171.6	38.8	59.1
Shares (Bn)	2.740	2.690	5.970	1.050	4.890
Share Price ($)	62.61	132.15	33.25	83.64	42.26
Market Cap ($Bn)	171.6	355.5	198.5	87.8	206.7
Preference Shares & Minority Interest ($Bn)	0.2	0	0.3	0.1	0
Market Value of Debt ($Bn)	28.5	46.5	44.2	10.3	33.4
Cash ($Bn)	6.5	19.0	2.6	4.6	4.9
Enterprise Value ($Bn)	**193.8**	**383.0**	**240.4**	**93.6**	**235.2**
Enterprise Value/Sales	4.86	5.30	4.55	4.42	6.22
Enterprise Value/EBITDA	12.75	15.26	12.33	16.72	24.75
Enterprise Value/Assets	2.03	2.71	1.40	2.41	3.98

We multiply Q-Star Pharmaceuticals' Sales, EBITDA, and Assets against the discounted multiple to arrive at the possible Enterprise Value of Q-Star Pharmaceuticals. Netting off the debt of USD11bn from Q-Star Pharmaceuticals' Enterprise Value, we are left with the Market Capitalization. Dividing this by the number of shares outstanding after the IPO (750m) gives us a target share price range of $12.67-$21.33 per share.

Selected Valuation Ratio	Lowest (times)	20% Disc. (times)		Q Star $bn	Firm value $bn	Mkt Cap $bn	Price USD
Enterprise Value/ Sales	4.42	3.53	Sales	7.65	27.00	16.00	21.33
Enterprise Value/ EBITDA	12.33	9.86	EBITDA	2.20	21.69	10.69	14.25
Enterprise Value/ Assets	1.40	1.12	Assets	18.30	20.50	9.50	12.67

Some prefer to apply the discounting to the target price range rather than to the multiples. This alternative approach will result in a higher target price range.

In the valuation example above, we used Enterprise Value as our relative valuation multiple. A more common approach is to compare the P/E and P/BV of similar companies and to use a selected average or discounted multiple to come up with an appropriate valuation range. The value of the firm will be discussed with the owners that are seeking the listing and the eventual price level decided upon will be a function of what the market conditions allow and the expectations of the owners.

6.11 Depositary Receipts

6.11.1 Depositary Receipts (DRs)

DRs are equity securities that trade in an investors' home market and represent an underlying security in the foreign issuers' home market. They offer a simple and convenient way to overcome the obstacles to issuing and investing in overseas markets. Such obstacles can include inefficient cross-border settlements, imperfect information flow, and unfamiliar market practices.

For issuers, DRs offer a few advantages such as:

a. broader and more diversified investor exposure, which may improve liquidity;
b. enhanced visibility for the company's products or services in an overseas market; and
c. a flexible mechanism for raising capital.

A variant is the Non-Voting Depositary Receipt (NVDR), which is similar to a DR in the sense that it facilitates foreign investment in an issuer's home market. The main difference between investing indirectly through an NVDR and directly in the issuer's shares is that NVDRs do not come with voting rights and, thus, do not allow foreign investors to participate in corporate decisions. This feature, however, enables foreign investment in firms that would otherwise be prohibited due to regulatory restrictions on foreign ownership.

6.11.2 American Depositary Receipts (ADRs)

ADRs are the most common type of DRs and trade on the NYSE, the NASDAQ or the Over-the-Counter (OTC) markets in the US. Hence, an ADR is a non-US company's equity security that trades in the US as though it were a US equity security.

There are 3 levels of ADRs:

a. **Level 1 ADRs**, which trade in the traditional US OTC market and are not listed on any US securities exchange such as the NYSE or the NASDAQ.

b. **Level 2 ADRs**, which are listed on a US securities exchange (such as the NYSE or the NASDAQ).

c. **Level 3 ADRs**, in which the issuer uses a listed ADR program to offer new shares to US investors in a capital-raising exercise.

Sponsored Level 2 and 3 ADRs are issued by a depositary bank on behalf of, and under a deposit agreement with, the non-US company. Level 2 ADRs do not involve capital raising, but Level 3 ADRs allow for the raising of new capital. Unsponsored ADRs are issued by depositary banks, but without a formal agreement between the depositary bank and the non-US company.

Global Depositary Receipts (GDRs) are generally issued offshore from the US. GDRs are commonly listed on the London Stock Exchange and the Luxembourg Stock Exchange.

6.11.3 Risks of Investing in DRs

a. DRs carry currency risks because of the currency in which the ADR is denominated, the currency in which the share is denominated, and the investor's home currency.

b. DRs also carry price risk, which is essentially the risk association with fluctuations in the price of the underlying share. However, ADRs often trade at a premium to their underlying shares because of regulatory restrictions in the issuer's home market on foreign holdings of the shares.

c. A potential counterparty risk is the financial or operational failure of the depositary bank or its custodian in their processing of DR obligations.

d. Surrendering single-listed ADRs—ADRs whose underlying shares are not listed in their home markets—will result in investors holding common shares that are not traded and have no transactional liquidity. It is not advisable to convert ADRs into unlisted shares.

e. Not all DRs have two-way fungibility, which means that the DRs can be converted into the underlying shares and the underlying shares can be converted into the DRs. All DRs have underlying shares backing them. The DRs are issued and traded outside of the issuer's country, but the underlying shares backing the receipts are lodged in custody with a custodian in the issuer's country.

6.12 Summary

1. When we conduct valuation, we can use the various relative valuation and discount cash flow methods to come up with an estimated value (or range of values).

2. The two most important characteristics of an ordinary share are its residual claim and limited liability features. Residual claim means that, in the event of liquidation, ordinary shareholders are the last to be paid. Limited liability means that the losses of shareholders are limited to their original investment.

3. Legally, a preference share is an equity security. However, it is often viewed as a hybrid security because it has both the characteristics of a fixed income security and an equity security. There are different classes of preference shares, which are distinguished by the method and amount of dividend payment, and the associated special benefits.

4. There are different types of equity investments available to clients depending on their return expectation and risk appetite. An investor can invest in the stocks of Blue-Chip companies, High-Growth companies, Income Stocks (Companies with High Dividend Yield), Defensive Stocks, Cyclical Stocks, and stocks with strong net tangible assets backing (i.e., stocks with hard assets).

5. The general movement of the equity market is usually measured by an equity market average or index consisting of a group of securities that is supposed to reflect the entire market. A price-weighted series is an arithmetic average of the share prices, while a value-weighted series considers the market capitalization of the shares in the computation of the equity market index.

6. In the construction of an unweighted index, all shares carry equal weight regardless of their price and/or market capitalisation.

7. The two broad approaches to equity valuation are the absolute valuation approach and the relative valuation approach.

8. The major weakness of the absolute, discounted cash flow method of valuation is the wide variation in the terminal value resulting from small changes in the weighted average cost of capital and the terminal growth rate.

9. In absolute valuation, the intrinsic value that is determined for the share through discounted cash flow analysis, is compared to its current market value. If the intrinsic value is greater (lesser) than the market value, the share is undervalued (overvalued) by the market i.e., the market is valuing the share lower (higher) than what it is intrinsically worth.

10. The most widely applied version of absolute valuation is the discounted free cash flow to the firm approach, using the weighted average cost of capital.

11. In relative valuation, a comparison is made between price multiples (Price-Earnings ratio, Price-to-Book Value ratio, etc.) of shares. When comparing the ratios, it is important to consider the qualitative factors underlying the premium or discount of a company's valuation relative to that of other companies.

12. In practice, multiple relative valuation ratios are used together to make a more holistic assessment.

13. To find the fair value of a company, we would have to discount all future free cash flows of the company and its terminal value to the present using the weighted average cost of capital. The sum of all these cash flows gives us the Enterprise Value (Firm Value or Company Value). Subtracting the market value of debt from the Enterprise Value will give us the fair market value of the total equity. Dividing this value by the total shares outstanding will give us the fair or intrinsic value per share.

14. The price-earnings ratio is an indication of the number of times that the market is willing to pay for the earnings of a company.

15. There is an inverse relationship between the price-earnings ratio and interest rate.

16. The price-to-book ratio is a multiple that the marketplaces on a company's book value.

17. The price-to-sales ratio indicates the number of times the market is willing to pay for a firm's sales revenue.

18. The enterprise value-to-cash flow ratio is deemed to be more reliable than the price-earnings ratio because cash flow is less subject to manipulation than earnings.

19. Enterprise value-to-EBITDA is a good ratio to use alongside the Price-to-Earnings ratio because the former takes the impact of debt into consideration for comparing companies whereas the latter does not.

20. Dividend yield measures the income component of a share's return.

21. Earnings yield is the reciprocal of the Price/Earnings ratio i.e., it is the Earnings/Price ratio, expressed as a percentage.

22. Depositary receipts are equity securities that trade in an investor's home market and represent an underlying security in another domicile i.e., the issuer's home market. For example, American depositary receipts are certificates of ownership issued by US banks or brokers and represent claims to a certain number of shares of a specific foreign company. They provide a means for US investors to invest in non-US companies in a form that is most familiar and convenient to them.

Chapter 6: Additional Reference Materials

The material below will help you improve understanding of this Chapter.

The absolute valuation approach or discounted cash flow approach is used to determine the intrinsic value (fair value) of a listed company. In absolute valuation, this estimated intrinsic value of the share is compared to its current market value. If the estimated intrinsic value is greater than the price, then the stock is undervalued by the market i.e., the valuation model suggests that the market is valuing the share lower than what it is intrinsically worth. On the other hand, if the intrinsic value is less than the price, then the share is overvalued by the market i.e., the market is valuing the share higher than what the valuation model suggests it is intrinsically worth.

6.13.1 Dividend Discount Model

The intrinsic value of an asset can be estimated using the dividend discount approach to determine the present value of a security. According to the dividend discount model (DDM), the intrinsic value of a common share is the present value of all future dividends:

$$P_o = \frac{D_1}{(1+R)} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \ldots + \frac{D_\infty}{(1+R)^\infty}$$

where:
P_0 = intrinsic value or theoretical value of the share today
D_t = dividends expected to be received at the end of period *t*
R = required rate of return (discount rate) on the share

The main difficulty with implementing the above equation as a valuation model is that investors must forecast and discount an infinite stream of dividends, both insurmountable tasks.

The DDM can be modified to account for a subsequent sale of the common shares as follows:

$$P_o = \frac{D_1}{(1+R)} + \frac{D_2}{(1+R)^2} + \ldots + \frac{D_n}{(1+R)^n} + \frac{P_n}{(1+R)^n}$$

where:
n = holding period
P_n = expected selling price of the share at the end of period *n*.

Dividend Discount Model

Assume a holding period of 3 years and that the estimated dividend payments for shares of Company A at the end of each year are:

Year 1	$1.00 per share
Year 2	$1.05 per share
Year 3	$1.10 per share

Suppose the expected selling price of the share at the end of 3 years is $10. Given a 12% required rate of return, the value of the share of Company A is:

$$P_o = \frac{1.00}{(1.12)} + \frac{1.05}{(1.12)^2} + \frac{1.10}{(1.12)^3} + \frac{10.00}{(1.12)^3}$$

$$= 0.8929 + 0.8371 + 0.7830 + 7.1178$$

$$= \$9.63$$

To make an investment decision, one would compare this estimated share value to its price to determine whether one should buy the stock.

6.13.2 Constant Growth Dividend Discount Model

When dividends are expected to grow at a constant rate (g) over time, the model can be generalized as follows:

$$P_0 = \frac{D_0(1+g)}{(1+R)} + \frac{D_0(1+g)^2}{(1+R)^2} + \frac{D_0(1+g)^3}{(1+R)^3} + \ldots + \frac{D_0(1+g)^{\infty}}{(1+R)^{\infty}}$$

where:
P_0 = intrinsic value or theoretical value of the share today
D_0 = current dividend payment
g = constant growth rate of dividends
R= required rate of return (discount rate) on the share

The above equation can be simplified to:

$$P_0 = \frac{D_1}{R - g}$$

where D_1 = dividend expected to be received at the end of period 1

This model is more suitable for the valuation of very large and broadly diversified companies.

The constant-growth model is often referred to as the Gordon model (named after Myron J. Gordon, who played a large part in its development and use).

The constant-growth model makes the following assumptions:

i. Dividends grow at a constant rate;.

ii. The constant growth will continue indefinitely.

iii. The required rate of return (R) is greater than the constant growth rate (g). If this assumption is not true, the model does not apply.

Based on the constant-growth model formula, the required rate of return can also be derived:

$$\textbf{Required Rate of Return (R)} = \frac{D_1}{P_0} + g$$

Constant Growth Model

Assume Alpine Limited is currently paying dividends per share of $0.20 and investors expect dividends to grow at a rate of 5% a year for the foreseeable future. For investments of similar risk, investors require a return of 12% a year.

The estimated price of Alpine is:

$$P_0 = \frac{D_1}{R - g}$$

$$= \frac{\$0.20\ (1.05)}{0.12 - 0.05} = \$3.00$$

Note that the current dividend (D_0) must be compounded for one period because the constant growth model formula uses the dividend expected one period from now (D_1) as an input.

Note, however, that a small change in any of the estimates (namely R and g) will have a significant impact on the value of the share as depicted in the following examples:

a. Assume R increases from 0.12 to 0.13:

g = 0.05

R= 0.13

Then D_1 = $0.20 (1.05) = $0.21

$$P_0 = \frac{\$0.21}{0.13 - 0.05} = \$2.63$$

b. Assume g increases from 0.05 to 0.06:

g = 0.06

R = 0.12

D_1 = $0.20 (1.06) = $0.212

$$P_0 = \frac{\$0.212}{0.12 - 0.06} = \$3.53$$

These examples show that a small change of 1% in either g or R produces a large difference in the estimated value of the share. The crucial relationship that determines the value of the share is the spread between the required rate of return (R) and the expected growth rate (g). Anything that causes a decline in the spread will cause an increase in the estimated value, whereas any increase in the spread will decrease the estimated value.

6.13.3 Estimating Expected Growth Rate of Dividends

A starting point for estimating the growth rate in dividends is to use the company's sustainable growth rate, which is the maximum growth rate a company can achieve while maintaining a constant debt-equity ratio and without relying on external equity financing. Specifically, the growth rate is:

g = Retention Rate × Return on Equity = RR × ROE

A firm can increase its sustainable growth rate by increasing its retention rate (reducing its dividend payout ratio) and reinvesting these additional funds at its historical ROE. Alternatively, the firm can maintain its retention rate but increase its ROE to increase sustainable growth.

Calculation of Growth Rate

If a firm retains 60% of its earnings and has a ROE of 15%, its estimated growth rate is:

g = RR x ROE

= 0.6 x 0.15 = 0.09 or 9%

Applying the Constant Growth Model for Valuing a Stock

Now let us see how we can obtain the variables used in the Constant Growth Model in practice and if the model can really be applied in a credible and dependable way to value stocks. First, we need to obtain the variables used in the Constant Growth Model to estimate the fair value or intrinsic value of a stock. We have selected a relatively well-known US stock, Procter and Gamble (P&G), on which to apply this model.

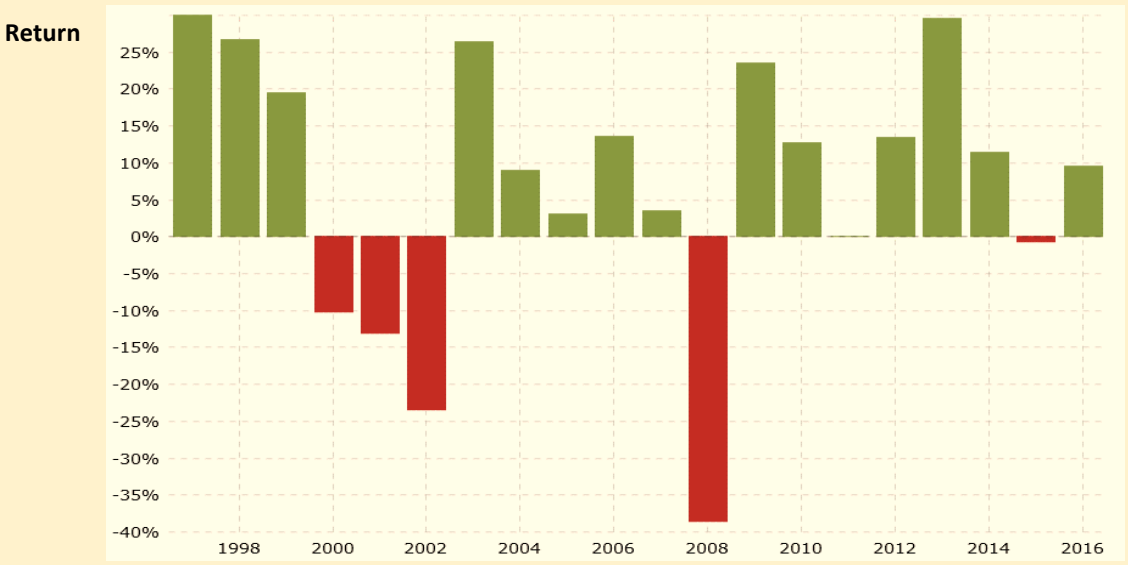
For 2016, P&G's EPS was 3.80 and its DPS was 2.66.
We need the required return of the stock and dividend growth rate.

$$P_0 = \frac{D_0(1+g)}{R_{stock} - g}$$

Estimating the Required Return of the Equity Market

How do we go about really estimating the required return of stocks, say, in the US market? We can begin by first looking at how the general market performed in the US over the last 10 or 20 years to estimate the required return of the market.

If we consider the average price return of the S&P 500 for the last 20 years including the global financial crisis and the dot-com bubble, the return was +7.36%. The annual dividend yield on the S&P500 is about 1.92%. Thus, the total return of the S&P500 would be about 9.28%. Hence, we can conclude that the required market return is about **R_m = 9.28%.**



1. Calculating Required Return of P&G Stock using the CAPM

Now let us apply the CAPM formula to calculate P&G's cost of equity. The company pays a large dividend that has been growing steadily.

$$R_{P\&G} = Rf + \beta\,(R_m - Rf)$$

Rm-Rf is the equity market risk premium.

β is the sensitivity of the stock price to the market.

The US 10-year Treasury bill rate was 2.33% at the time of calculation, so we use a risk-free rate of 2.33%. P&G has a beta of 0.65-0.68 (over the last 5 years according to popular providers of financial statistics (Bloomberg, Reuters). Let us use 0.68, i.e.; assuming more risk sensitivity to the market.

Applying the CAPM model, the required return for P&G would be:

$R_{P\&G}$ = 2.33 + 0.68 (9.28-2.33) = 7.056%.

This is the cost of equity required by for shareholders to finance the business.

2. Estimating the Growth Rate

Estimated earnings growth is based on the formula: g = ROE (1 – dividend payout).

Average ROE for P&G in 2016 = Net Income/ Average Shareholder's Equity = 10508/60516.5 = 17.36%.

For 2016, P&G's EPS was 3.80, its DPS was 2.66 and, therefore, the payout was DPS/EPS = 0.70.

Using the formula, **Growth $_{P\&G}$** = 0.1736 x (1 – 0.70) = **0.0521 or 5.21%**.

(To check for consistency, the annual DPS growth for the last 5 years is $(2.66/1.97)^{1/5}$ -1 = 0.0619.)

3. Applying the Constant Growth Model

$$P_0 = \frac{D_0(1+g)}{R_{stock} - g} \qquad P_0 = \frac{2.66(1.0521)}{0.07056 - 0.0521}$$

From the previous calculation for the required return of P&G, $R_{P\&G}$ = 7.056%.

Using the Constant Growth Model, the estimated value of P&G is 2.7986/0.0185 = $151.28, which is almost twice the value of P&G at the close of 7th July 2017, $87.65.

The wide value difference highlights the problem with the Constant Growth Model - the estimated value based on the model will swing widely with small changes to the value of "R" and "g". The table below shows the range of estimated fair value prices of P&G by making small changes to "R", the required rate of return, and "g", the estimated growth rate.

	Required Rate of Return %					
Growth %	7	8	9	10	11	12
2	54.26	45.22	38.76	33.92	30.15	27.13
3	68.50	54.80	45.66	39.14	34.25	30.44
4	92.21	69.16	55.33	46.11	39.52	34.58
5	139.65	93.10	69.83	55.86	46.55	39.90
6	281.96	140.98	93.99	70.49	56.39	46.99
7	infinity	284.62	142.31	94.87	71.16	56.92

When R is equal to g, the value goes to infinity. The model becomes invalidated when the required rate of return for equity is less than the growth rate.

The Constant Growth Dividend Discount Model can possibly be improved if we are able to account for some of the more predictable dividend flows for the early years, which leads us to the Multi-Stage Dividend Discount Growth Model.

6.13.4 Multi-Stage Dividend Discount Growth Model

Multi-stage growth refers to a situation in which the expected future dividend growth must be described using two or more growth rates. A two-stage growth model assumes near-term growth at a given rate for some period (typically 2 to 10 years) followed by a different long-term growth rate that is sustainable.

a. **Two-Stage Dividend Discount Model**

Many companies grow at a rapid rate for several years and then their growth slows down to a stable rate. Other companies pay no dividends for several years, often during their early-growth period. It is inappropriate to use the constant-growth model to value the stocks of these firms because their growth is not constant throughout.

In the two-stage dividend discount model, we improve on the original model by forecasting a dividend growth rate for the near-term, when dividends are more predictable, and assuming a growth rate that is close to the long-run GDP growth of the economy or possibly 1-2% points above for the long-term, when dividends are less predictable. The equation for a two-period growth model can be written as:

$$P_0 = \sum_{t=1}^{n} \frac{D_0(1+g_1)^t}{(1+R)^t} + \frac{D_n(1+g_c)^t}{R - g_c} \frac{1}{(1+R)^n}$$

where:
P_0 = intrinsic value of the share today
D_0 = current dividend
g_1 = growth rate when dividends are more predictable
g_c = constant-growth rate when dividends are less predictable
R = required rate of return
n = number of periods of supernormal (or subnormal) growth
D_n = dividend at the end of the abnormal growth period

In the equation, the first term on the right side applies to a dividend stream for the first n periods, when dividends grow at a high (or low) growth rate of g1, by discounting these dividends at the required rate of return, R. The second term on the right side is the constant-growth version discussed earlier, which divides the dividend expected at the end of period n+1 by the difference between R and g; since the resulting figure is a present value at the end of period n, it must be discounted back to time period zero for the corresponding number of periods.

Two-Stage Dividend Discount Model

Let us apply a simplified version of the same data on P&G to the Two-Stage Model. We assume that we are more confident of projecting dividends for the first five years and that a long-run constant growth will apply after year 5, when dividend growth becomes less predictable.

D_0 = \$2.66, R = 7%, g_1 = 5% g_c = 3.5% (higher than the US long-run GDP growth of 2%)

The first step in the valuation process is to project dividends over the period when we are more confident about dividend growth:

D_0 = \$2.66

D_1 = \$2.66 (1.05) = \$2.7930

D_2 = \$2.66 $(1.05)^2$ = \$2.9327

D_3 = \$2.66 $(1.05)^3$ = \$3.0793

D_4 = \$2.66 $(1.05)^4$ = \$3.2333

D_5 = \$2.66 $(1.05)^5$ = \$3.3950

To determine the expected price of the share at the end of year 5, when the long-run constant growth of 3.5% begins, the constant-growth model is used:

D_6 = 3.3950 (1.035) = 3.5138

$P_5 = D_6 / (R - g_c)$

P_5 = 3.5138 / (0.07 – 0.035) = 100.3943

The last step is to determine the sum of the present values of the stream of dividends and the present value of P_5.

$$P_0 = \frac{2.7930}{(1.07)} + \frac{2.9327}{(1.07)^2} + \frac{3.0793}{(1.07)^3} + \frac{3.2333}{(1.07)^4} + \frac{3.3950}{(1.07)^5} + \frac{100.3943}{(1.07)^5}$$

$$= 2.6103 + 2.5615 + 2.5136 + 2.4667 + 2.4206 + 71.5797$$

$$= \$84.15$$

With this two-stage model, we happen to come close to the current price level of P&G and, if we believe in our model, then we would say that P&G is currently overvalued.

However, this two-stage model still suffers from the same sensitivity issues as the Constant-Growth Model. If we use only 3% for long-run constant growth rate instead of 3.5%, the valuation will reduce by about 10% to $74.90. Since 85% of the value is determined by the terminal value (i.e., 71.5797/84.15), changes that affect the terminal value will still affect the model significantly.

b. **Three-Stage Dividend Discount Model (DDM)**

We can also stretch our model to a 3-stage DDM by assuming a different growth rate for Years 6 to 10. The three-stage model is quite useful to gradually taper dividend growth rates to the eventual terminal growth rate. For example, one might assume that Stage 1 (Years 1-5) has supernormal dividend growth, Stage 2 (Years 6-10) has a gradual slowdown in dividend growth, and Stage 3 (Year 11 onwards) is when dividend growth rate reaches a long-run sustainable level.

c. **Practical Application of Multi-stage DDM Growth Model**

The DDM is best applied when the dividend payouts are much higher and when we can shift more of our estimate to be based on the more predictable early years of dividend flows. If the early years can account for a larger proportion of the estimated fair value, we become less dependent on the terminal constant-growth assumption. The accuracy and effectiveness of the dividend valuation model is better for stocks of companies that are in the mature phase of their life cycles and that have very high dividend payouts (e.g., 70-120% of EPS (DPS can be higher than EPS if a company pays out from retained earnings and has sufficient cash)). Hence, the DDM model tends to be more useful for stocks of gaming companies, tobacco companies, and other select mature companies with huge dividend payouts.

d. **Intrinsic Value or Estimated Fair Value**

After obtaining the estimated fair value or intrinsic value of the share, this value is compared with the prevailing share price. If the intrinsic value is greater than the price, the share is considered undervalued and should be purchased. On the other hand, if the price is greater than the intrinsic value, the share is considered overvalued and should be avoided or sold.

Figure 6.13.4 - Stock Price Trend for Proctor & Gamble



Source: Yahoo Finance

6.13.5 Zero-Growth Model

The owner of a preference share receives a fixed dividend for an infinite period. Since dividends of a preference share is a perpetuity, the share value is simply the stated annual dividend divided by the required rate of return as follows:

$$P_0 = \frac{\text{Dividend}}{\text{Discount Rate}}$$

Zero-Growth Model

A preference share has a $100 par value and pays a dividend of $7 a year. Given a required rate of return of 8%, what is the intrinsic value of the preference share?

Value of preference share = $7 / 0.08 = $87.50

Chapter 7: Deposits and Fixed-Income Securities

Learning Objectives

The candidate should be able to:

- ✓ Discuss the different ways we use cash.
- ✓ Describe various basic money market instruments.
- ✓ Describe the characteristics of a fixed-income instrument such as a bond.
- ✓ Define and explain the various terminology used in fixed-income securities.
- ✓ Explain the credit risk in holding bonds.
- ✓ Describe the issuers and their motivation in issuing bonds.
- ✓ Describe the primary and secondary bond markets.
- ✓ Explain how prices are quoted in the trading of bonds.
- ✓ Discuss various special features in bond issuance and explain their significance to investors.
- ✓ Describe high yield bonds and their implied bond rating.
- ✓ Discuss asset-backed fixed-income instruments.
- ✓ Describe features of the internal and external bond markets, and the Eurobond market.
- ✓ Explain the differences among the various grades of debt rating by rating agencies.
- ✓ State the various risks that a fixed-income investor is exposed to.
- ✓ Calculate the value of a bond.
- ✓ Calculate the bond equivalent yield for bonds that pay interest more often than once a year.
- ✓ Explain the relationship between coupon rate, market yield, required yield, and bond price.
- ✓ Explain why bond prices converge to par on maturity.
- ✓ Explain the concepts of duration and convexity and the implications for a bond's price risk.
- ✓ Understand what is meant by price value of a basis point and how this is used to monitor the price risk in a bond portfolio.
- ✓ Calculate various yield measures such as current yield and yield to maturity to make investment decisions.
- ✓ Explain reinvestment risk and how it is affected by the coupon rate and the maturity of the bond.

- ✓ Discuss the term structure of interest rates in relation to the expectations theory and the liquidity preference theory.
- ✓ Analyse convertible bonds for investment purposes, including the conversion parity price and conversion premium.

7.1 Introduction

This Chapter covers the basic interest-bearing assets classes:

a. Cash;
b. Money market instruments; and
c. Fixed income instruments.

Cash is an interesting asset class because the position of physical cash as a medium of exchange is gradually being replaced by other digital forms of storage and disbursement. This basic asset class is often classified as a non-volatile, safe asset class. However, for that reason, cash pays the lowest return. Keeping excessive cash from 2008-2017, for example, could be non-optimal because of the opportunity costs and loss of purchasing power.

Beyond cash, money market instruments provide better returns than cash, followed by fixed-income instruments of various credit risks and tenures. Most money market instruments are usually short-term instruments of 3-6 months. For wider range of tenures, risks, returns and pay-off profiles, investors can invest in fixed-income instruments, from sovereign government notes and bonds to corporate bonds, and from 'plain vanilla' bonds to bonds with embedded options and other structured elements.

7.2 Cash Deposits and Cash-Equivalents

All investments begin with, and can be liquidated to, cash. For example, during economic downturns and financial crises, cash deposit balances rise as other financial assets are sold down. Cash can either be denominated in local or foreign currency. Cash deposits generally provide some form of interest earnings through savings or term deposits. As such, they can be considered as the most basic form of fixed-income investment.

Since cash provides the lowest return, institutional investors and fund managers will generally seek to minimise cash balances by holding only what is necessary to satisfy redemption requirements. However, if a major market downturn is anticipated, institutional investors and fund managers may increase their cash holdings. This would allow them to have readily available funds to take advantage of opportunities during market downturns.

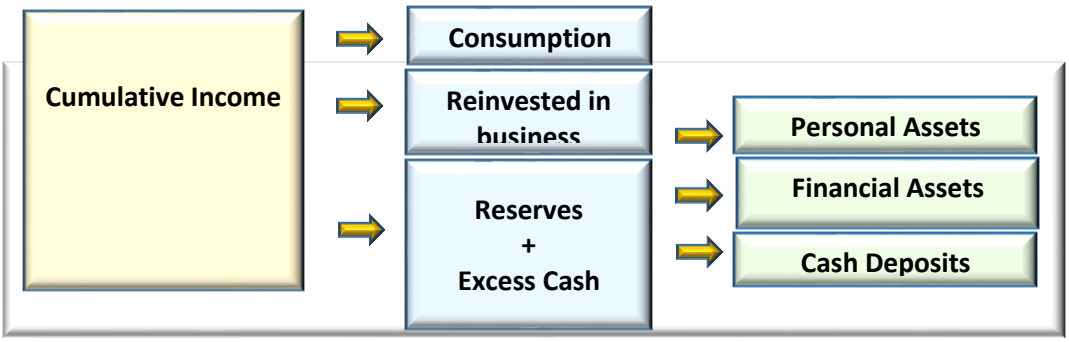
Similarly, corporate entities also have to manage excess cash and liquidity requirements. Excess cash needs to be invested to optimize overall returns. For very large conglomerates, even the short-term available cash float needs to generate a reasonable return before they are remitted out of the firm’s accounts. For some banks, cash management services for such corporate entities have become a lucrative activity.

For individuals, income and monies received can usually be categorised as:

a. money for consumption;
b. reinvestment into the business (in the case of entrepreneurs);
c. reserves set aside for family needs, life's risks, and unfortunate eventualities; and
d. excess cash.

Reserves and excess cash can either be invested in personal assets (e.g., property) and financial assets or held as cash and cash-equivalents for liquidity purposes. Please refer to Figure 7.2.

Figure 7.2 – Allocation of Cash and Income

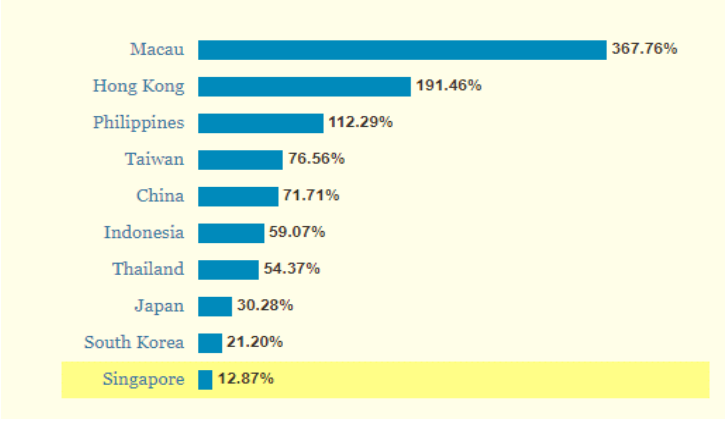


7.2.1 Currency Deposits

a. **Local Currency Deposits**

The most widely known and used type of cash-equivalent investments are savings and time deposits at banks and finance companies. Time deposits are deposits held for a specified period of time, such as 3, 6, 12 months or even longer (such as 36 months). The longer the deposit period, the higher the interest. However, a portion of interest earnings may be forfeited as a penalty for any premature withdrawals made. Even though they are low-risk instruments, the disadvantage of such deposits is that they provide low yield. Over the period of July 2009 to July 2017, money kept in savings or fixed deposits would have accumulated returns of at most 5-6%. In comparison, the S&P500 index has appreciated by more than 133% over the same period. Similarly, average housing prices in Asia rose significantly over the same period as reflected in the Figure7.2.1(a)[31].

Figure 7.2.1(a) – Home Price Changes in Asia – Jul 2009 to Jul 2017



[31] Source: https://www.globalpropertyguide.com/.

As such, time deposits, as a financial investment, do not provide a good inflation hedge and can be "costly" to investors by forgoing other forms of investment.

b. **Foreign Currency Deposits**

Foreign currency deposits placed with banks in Singapore can offer potentially higher interest, but they come with potential exchange rate risk. In Singapore, foreign currency deposit accounts are not covered by the Deposit Insurance (DI) Scheme, which is further discussed in Section 7.2.1(c).

Though local currency deposits reduce an individual's direct exposure to foreign currency fluctuations, an investor may be better off investing in foreign currencies when the local currency continually depreciates in value relative to other currencies. For example, if the domestic economy has a poor outlook, coupled with weak Gross Domestic Product (GDP) growth and worsening political stability, local currency weakness can be expected to follow. In order to mitigate sustained local currency depreciation, investors can include a portion of deposits in a stable and relatively stronger currency within their investment portfolios. For growing companies and wealthy individuals, maintaining a portion of assets in foreign currency has become an integral part of the diversification and risk management process.

c. **Deposit Insurance (DI) in Singapore**

DI Scheme was introduced in Singapore on 1 April 2006. The Scheme was reviewed and enhanced by the MAS and the Singapore Deposit Insurance Corporation (SDIC) in May 2011. All full banks and finance companies in Singapore are members of the DI Scheme, except for those exempted by the MAS. The DI Scheme has been set up to protect small depositors in the event that a DI Scheme member fails.

The coverage limit has been increased from S$50,000 to S$75,000 on 1 April 2019, and further revised to S$100,000[32] effective 1 April 2024. The scope of coverage includes individuals and other non-bank depositors such as sole proprietorships, partnerships, companies, and other unincorporated entities, such as charities, associations, and societies.

This Scheme is only valid for Singapore dollar-denominated deposits placed with a DI member. These deposits include:

i. a deposit held in a savings account;

ii. a deposit held in a fixed deposit account;

iii. a deposit held in a current account;

iv. any monies placed under the CPF Investment Scheme (CPFIS);

v. any monies placed under the CPF Retirement Sum Scheme (CPFRS);

vi. any monies placed under the CPF Supplementary Retirement Scheme; and

vii. Murabaha, as prescribed by MAS.

The type of financial products that are not insured by the SDIC include:

i. foreign currency deposits;

[32] https://www.mas.gov.sg/-/media/mas-media-library/publications/consultations/ppd/2023/response-to-feedback-received-on-proposed-enhancements-to-the-di-scheme-in-singapore-part-1.pdf.

ii. structured deposits; and

iii. investment products such as unit trusts, shares, and other securities.

Monies and deposits held under the CPFIS and CPFRS are aggregated and separately insured up toS$100,000.00[33].

A deposit insurance payout may be made based on decision by the MAS given that:

i. a court order has been made to wind up a DI Scheme member; or

ii. it is so determined by the MAS that a DI Scheme member is insolvent, unable or likely to become unable to meet its obligations, or about to suspend payments.

The following examples illustrate how the SDIC computes the insured amount under some common scenarios (based on the coverage of S$100,000).

Example 1 – Deposit insurance coverage for individual accounts all with one bank

An individual with S$650,000 places his money in different accounts, but all with a single bank.

Type of Balance	Account Balance in S$	Insured	Not Insured
1. Individual Current Account 2. Individual Savings Account 3. Individual Fixed Deposit Account	10,000 10,000 280,000	100,000	200,000
4. CPF deposits – CPFIS5. Individual USD Currency Investment	100,000 250,000	100,000 -	 250,000
Total	650,000	200,000	450,000

As shown above, in the case of accounts with a single bank, the DI scheme will only cover up to S$100,000 of the aggregate balance in the different accounts. CPF deposits invested with the bank also receive insurance cover of up to S$100,000. Hence in this example, although the individual has $650,000 worth of deposits, he is only insured up to S$200,000.

For individuals with deposits above S$100,000, the DI scheme currently allows for some deposit insurance benefits of having joint accounts and for having multiple accounts. Partners in a joint account will be separately insured for their share of the account (see Example 2).

Example 2 – Deposit insurance coverage for accounts with different banks and joint accounts

A man and his wife hold accounts with 3 banks. In Bank A, he has $250,000 in USD deposits, S$30,000 in an individual current account, $$40,000 in an individual savings account, and $$80,000 in a joint account with his wife. In Bank B, he has $$100,000 in a joint fixed deposit account with his wife and $$50,000 in a CPFIS

[33] https://www.mas.gov.sg/-/media/mas-media-library/publications/consultations/ppd/2023/response-to-feedback-received-on-proposed-enhancements-to-the-di-scheme-in-singapore-part-1.pdf.

deposit. In Bank C, he has another S$100,000 in an individual account. If all 3 banks are members of the SDIC, how much of the couple's deposits are insured?

Type of Balance	Account Balance in S$	Insured	Not Insured
Balance in Bank A – Under SDIC 1. Individual Current Account 2. Individual Savings Account 3. Man's Share of Joint Fixed Deposit (80,000/2) 4. Wife's Share of Joint Fixed Deposit (80,000/2) 5. CPF deposits with Bank A – CPFIS 6. Individual USD Currency Investment	30,000 40,000 40,000 40,000 50,000 250,000	 100,000 40,000 50,000	 10,000 250,000
Balance in Bank B – Under SDIC 7. Man's Share of Joint Fixed Deposit (100,000/2) 8. Wife's Share of Joint Fixed Deposit (100,000/2) 9. CPF deposits with Bank B – CPFIS	 50,000 50,000 50,000	 50,000 50,000 50,000	 - - -
Balance in Bank C – Under SDIC 10. Man's Individual Fixed Account (100,000)	100,000	100,000	-
Total	700,000	440,000	260,000

7.3 Fixed-Income Securities

Fixed-income securities are financial instruments that represent debt obligations. Technically, the term "fixed income" covers any security where the issuer is obligated to pay the lender fixed payments at agreed set intervals. Cash deposits can provide interest earnings at set times as well, but there is no issuance of tradable securities.

Fixed-income securities are issued by governments and their linked entities, financial institutions, and large corporations to raise funds. These securities can be issued publicly or privately. Governments issue debt to manage money supply and to partly fund their budget spending. Corporates issue debt to fund capital expenditure, the acquisition of other businesses, and the repayment of maturing debt obligations. Corporate bonds held by investors are fixed income debt instruments that bear a claim on the assets of the issuing entity. The issuers are essentially borrowing money from investors and, in exchange, investors receive periodic interest payments followed by the repayment of the borrowed amount, the principal, at maturity. The yield investors expect to receive from investments in fixed-income instruments is normally expressed as an interest rate. Investors take on issuer risk, which is the risk of the issuer defaulting on interest payments or principal repayment. Table 7.3 lists the different types of fixed-income instruments.

Table 7.3 - Types of Fixed-Income Instruments

Money Market Instruments	Sovereign Debts	Corporate Bonds	Asset-Backed Securities	Other Instruments
• US Treasury Bills • Certificates of Deposit • Commercial Papers • Repurchase Agreements • Banker's Acceptances • Trade Bills	• Govt. Bonds - US Treasury - European - Japanese - Other Govts • Agency Debts • Municipal & Provincial Govt Debts	• Corporate Bonds	• Mortgage-Backed Securities – Fannie Mae – Ginnie Mae – Freddie Mac • Other Asset-Backed Securities – Auto Loans – Student Loans – Credit-card Debts • Collateralised Debt Obligations (CDOs)	• Non-convertible Preferred Stocks • Convertible Bonds

One major way to classify fixed-income instruments is by credit risk, which splits the range of securities into investment-grade bonds and high-yield bonds. Investors in short-term money market instruments look for low-risk instruments; hence, the majority of these instruments are of investment-grade quality.

As bonds are the most common form of fixed income securities, there will be a more comprehensive coverage on bonds in the following sections.

7.4 Money Market Instruments

Money market instruments are debt instruments issued by governments, financial institutions, and corporations. They have maturities of less than one year. Owing to their relatively short maturity, money market instruments have relatively low risks. However, the minimum denomination of money market instruments is usually relatively large, at S$250,000 and above. Some examples of money market instruments are **Treasury bills (T-bills), negotiable certificates of deposit (CDs)** or **banker's acceptances**, **commercial paper**, **repurchase agreements,** and **bills of exchange**.

For institutional portfolios, a portion of the funds can be placed in money market instruments for liquidity purposes. Liquidity usually refers to the ability to convert an investment into cash quickly with little or no loss in value. Due to their status as near-cash instruments, the primary function of money market instruments is to provide a pool of reserves that can be readily utilised for emergencies or to accumulate funds that can be readily made available for some specific purposes.

Investment in money market instruments generally increases when investors are uncertain about the prospects of other investment classes. Investors will also increase their holdings of money market instruments when stocks and bonds are expected to perform poorly. Conversely, they will hold less cash when they perceive a bull (rising) stock/bond market.

In general, professional fund managers try not to allocate too much of their funds in excess of liquidity requirements to money market instruments due to the low yield. Although money market instruments carry almost no risk of principal loss, they are subject to reinvestment risks.

Reinvestment risk refers to the risk that an investor may not be able to reinvest the proceeds of his maturing investments at equivalent rates, possibly because of declines in interest rates. Money market instruments are subject to more reinvestment risk because they are short-term instruments, requiring the investor to reinvest his investment proceeds each time the investment matures.

7.4.1 Treasury Bills

These are short-term government securities (with maturities of one year or less) issued by governments to the investing public in order to fund government expenditure. Investing in a T-bill is equivalent to lending money to a sovereign government.

T-bills issued by a creditworthy government are the safest investments and are generally considered to be risk-free. When the term "risk-free investment" is used, it usually refers to T-bills. In this context, "risk-free" means that there is no risk of defaulting on interest or principal payments when due.

T-bills are sold on a discount basis – that is, investors pay an amount less than the face value to purchase them and will receive the face value at maturity. The difference between the purchase price and the face value represents the interest earned on T-bills and is used to calculate the yield on such securities.

The yield on T-bills is usually used as a benchmark for risk-free rates. They can be readily sold and converted to cash before maturity at a modest cost to meet client needs.

7.4.2 Banker's Acceptances and Certificates of Deposits

Banker's acceptances and certificates of deposits (CDs) are money market instruments issued by financial institutions.

A **banker's acceptance** is issued to facilitate international commercial trade transactions as it represents a claim on the issuing bank for a specific amount on a specific date. A banker's acceptance is a negotiable security that is also issued on a discounted basis.

A **CD** is typically issued by a bank that indicates that a specific sum of money has been deposited with the bank. This certificate carries a maturity date and a specific interest rate, generally on a compounding basis. The interest amount is payable upon maturity.

7.4.3 Commercial Papers

Commercial papers are short-dated unsecured promissory notes issued by a corporation. Due to their unsecured nature, only corporations with very strong credit ratings are able to issue commercial papers. Similar to T-bills and banker's acceptances, commercial papers are issued at a discount to face value.

7.4.4 Repurchase Agreements

Secondary trading in money market instruments can take place through a repurchase agreement (commonly known as a repo). A repo is the sale of a money market instrument, with a commitment by the seller to buy the security back from the buyer at a specified price on a future date. Effectively, it is a collateralised short-term loan, where the collateral is the money market instrument. The yield on a repo can be calculated using the difference between the sale and purchase prices.

7.4.5 Bills of Exchange

Bills of exchange are widely used to finance international trade. They play an important role because of the time lag in trade transactions and the distance between the buyers and sellers. A bill may be drawn payable on sight or demand. Alternatively, a bill may be drawn to be paid after a period of time, commonly 91 days; such a bill is known as a term bill. Bills of exchange are useful because they are negotiable and are generally transferable by endorsement and delivery.

7.5 Bonds

A coupon bond is defined as a debt security on which the issuer is obligated to make regular interest payments (called coupons) to the bondholder on specified dates and repays the principal to the bondholder on the maturity date.

Usually, the coupon rate that is payable on the bond is fixed for the life of the bond, giving rise to the term "fixed income". The debt securities market may be divided into three maturity segments based on the issue's original maturity:

a. Money market instruments which are short-term issues that mature within one year;

b. Medium-term notes with maturities of more than one year and up to 10 years; and

c. Long-term bonds with maturities longer than 10 years.

7.5.1 Benefits and Drawbacks of Investing in Bonds

a. **Benefits**

Bonds and other forms of fixed-income securities can be good investments for investors interested in:

i. principal preservation; and

ii. receiving a steady stream of periodic income.

Fixed-income securities have long been regarded as good vehicles for those seeking current income. Under normal circumstances, an investor in fixed-income securities will receive cash flows with reasonable certainty.

For investors interested in capital gains, the value of bonds can appreciate from improvements to their issuer's credit quality or when market yields fall. Investment professionals who actively engage in bond trading look for undervalued bonds that are likely to enjoy credit rating improvements and receive credit rating upgrades.

b. **Drawbacks**

A major disadvantage of investing in fixed-income securities with fixed coupons and principal repayment is that these securities do not adjust for inflation. Not only can inflation erode the purchasing power of cash flows from the investment, it can also lead to swings in interest rates. This can produce fluctuations in the prices of fixed-income securities, which can cause substantial capital losses if the fixed-income securities are not held to maturity.

Another disadvantage of fixed-income securities is the lack of activity and liquidity in the smaller secondary markets; that is, those other than the major bond markets of the US, Europe and Japan. This poses an added challenge for investors of fixed-income securities as they have to be confident of holding the relatively illiquid bonds until maturity.

7.5.2 Terminology

The **par value** (face value) or **principal** represents the redemption value of the bond. This is the amount that will be repaid to the investor at maturity. For most bonds, the par value is $1,000. A bond's par value is not the same as its market value, which may rise or fall depending on market factors that impact its required return, such as interest rate movements.

The **coupon** of a bond represents the periodic interest payments made to the bondholders over the life (or holding period) of the bond. The coupon rate, when multiplied by the principal or par value of the bond, provides the dollar value of the coupon payment. Coupon payments are typically paid on an annual or semi-annual basis.

In Singapore, the bonds issued by the Government are scripless, and ownership of Singapore Government Securities is reflected as electronic book entries. Otherwise, bonds may be issued as **bearer** bonds or **registered** bonds, but since most markets have progressed to electronic trading and settlement platforms, few bonds are issued as bearer bonds today (given, also, the high security risk of such ownership).

An important feature of a bond is its **term-to-maturity**, which indicates the number of years remaining to the maturity of the bond. There are two different types of maturity - most corporate bonds are typically **term bonds**, which have a single maturity date. In contrast, **serial bonds** have a series of maturity dates and are essentially bundles of bonds issued at the same time, but with different maturities. The longer the maturity, the more volatile the bond price will be.

Conservative investors tend to choose securities with shorter maturities in order to reduce their exposure to interest rate and default risks. Note that price volatility is not a concern if the investor intends to hold the bonds until maturity. However, it would be a concern if investors are forced to sell their bonds before maturity due to emergencies or other unforeseen reasons.

7.5.3 Issuers

Bond issues can be classified as follows:

a. Government;

b. Government-related bodies; and

c. Corporations.

The government issues bonds to pay for various government projects such as infrastructure development and the provision of essential services, such as healthcare and education. In some countries, such as the US, government agencies are quite active in issuing bonds to finance their programmes. In Singapore, government-linked corporations tap the bond market for funds rather than sourcing directly from the government.

Besides raising funds through equity securities or bank loans, corporations can also raise funds through the issue of bonds to the public. The corporate sector provides the most diverse issues in terms of type and quality.

7.6 Price Quotations

For pricing purposes, fixed-income securities can be classified as discount or coupon securities. Fixed-income securities with maturities of 1 year or less, such as T-bills or commercial papers, are discount securities while fixed-income securities with maturities of more than one year are coupon securities.

7.6.1 Discount Basis

T-bills are quoted on a bank discount basis. The yield on a bank discount basis is calculated as follows:

$$Y_d = \frac{D}{F} \times \frac{360}{t}$$

where:
Y_d = annualised yield on a bank discount basis
D = dollar discount, which is equal to the difference between the face value and price
F = face value
t = number of days remaining to maturity[34]

Calculation of Yield on a Discount Basis

A T-bill that has 90 days to maturity, a face value of $100,000, and is selling for $98,000 would be quoted at 8.0% on a bank discount basis.

D = $100,000 - $98,000 = $2,000

$$Y_d = \frac{\$2000}{\$100,000} \times \frac{360}{90} = 8.0\%$$

Though it is widely used, the quoted yield on a bank discount basis is not a meaningful measure of return from holding a T-bill because it is calculated based on the face value rather than on the actual dollar amount invested.

7.6.2 Price Basis

The prices of fixed-income securities of more than one year are quoted as a percentage of par value. Assuming a par value of $1,000, a quote of 98.50 represents $985 and a quote of 95¾ represents $957.50.

US Treasury bonds and notes are quoted in thirty-seconds (1/32) of a percentage point, whereas corporate bonds are quoted in eighths (1/8) of a percentage point. Specifically, a quote of 90-24 for a Treasury bond means 90 24/32.

All fixed-income securities trade on an accrued interest basis. This means that the buyer of a bond must pay the seller the price of the bond plus the interest that has been earned on the bond since the last interest

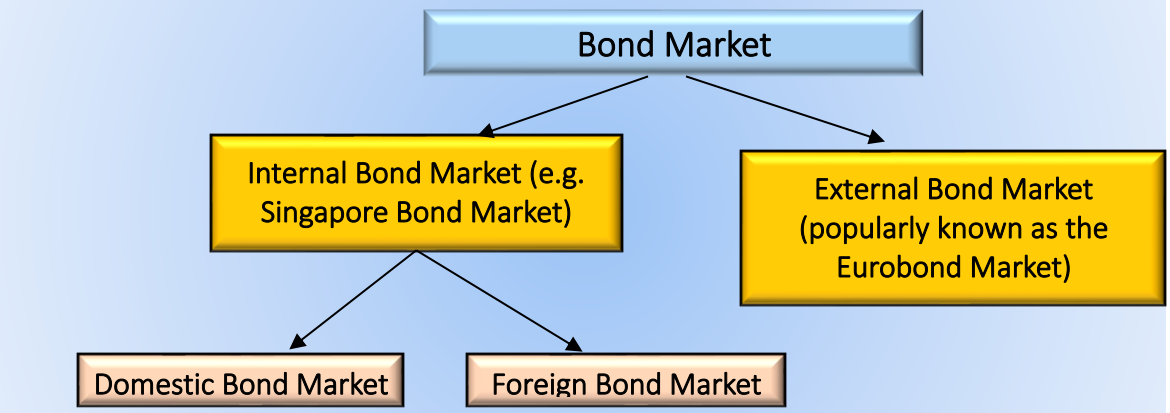
[34] The numerator of 360 as well as the denominator t will follow the Day Count Convention of the respective market. Day count convention refers to the practice in each market to use either 360 or 365 days to represent a year, and whether it should be the actual number of days or 30 days for each calendar month.

payment date. Assume an 8% bond that pays semi-annual interest on 1 June and 1 December. If the bond is sold on 1 September, the seller is entitled to accrued interest of $20 ($80 x 3/12) for the three months that the seller has held the bond since the last coupon payment.

7.7 Bond Markets

Before discussing the types of bonds, we should look at the different bond sectors. Essentially, the bond market can be divided into the internal bond market and the external bond market.

Figure 7.7 – Bond Market Sectors



As illustrated in Figure 7.7, in the context of the Singapore bond market, the domestic bond market comprises Singapore Government Securities (SGS) and corporate bonds issued by Singapore companies. The foreign bond market refers to foreign issuers who have issued their bonds in the Singapore bond market.

Global bond portfolios invest 40% or more of their assets in foreign bonds. Some global bond portfolios follow a conservative approach, favouring high-quality bonds from developed markets. Others are more adventurous and own some lower-quality bonds from developed or emerging markets. Some portfolios invest exclusively outside of the US, while others invest regularly in both US and non-US bonds.

7.7.1 The Singapore Bond Market

The bond market in Singapore comprises Government securities (T-bills and bonds), as well as corporate bonds issued by quasi-government statutory organizations, local companies, and foreign entities.

a. **Singapore Government Securities**

SGS were initially issued to meet banks' needs for a risk-free asset in their liquid asset portfolios. The Singapore Government does not need to finance its expenditures through the issuance of government bonds as it operates on a balanced budget policy and often enjoys budget surpluses. Hence, the issuance of SGS serves primarily to:

i. build a liquid SGS market to establish a benchmark yield curve for the pricing of Singapore dollar private debt securities;

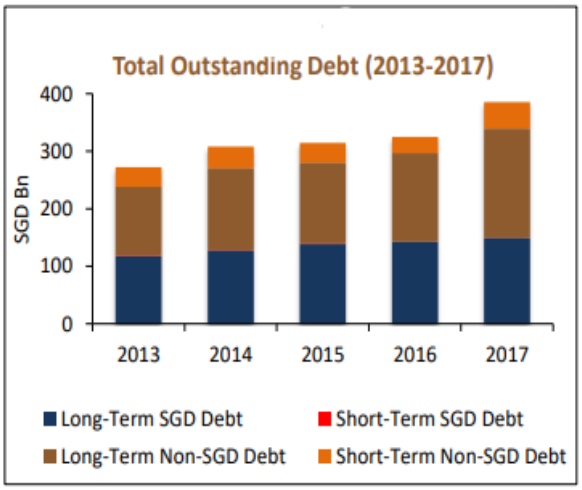
ii. encourage the growth of an active secondary market, both for cash transactions and derivatives, for the purposes of risk management; and

iii. attract domestic and international issuers and investors to participate in the Singapore bond market.

SGS auctions of T-Bills and bonds are conducted on a regular basis according to an annual calendar. MAS introduced ''mini-auctions" of SGS bonds in 2015 as a regular feature in the issuance calendar to address unexpected instances of strong demand for bonds outside of the issuance calendar. Singapore T-bills and bonds are issued with standard characteristics:

i. Short-term T-bills are issued as zero-coupon discount bonds with typical issue size ranging between SGD 2-4 billion in minimum denomination of SGD1,000.

ii. Government bonds have fixed semi-annual coupon payments with tenors ranging from 2, 5, 10, 15, 20 to 30 years. Issue sizes are typically between SGD2-3 billion (for benchmark issues) in minimum denomination of SGD1,000.



Investors keen to invest in SGS must submit a bid through the approved SGS primary dealers. As at the end of December 2022, outstanding SGS bonds and T-bills totalled SGD233.7 billion[35].

b. **Singapore Savings Bonds (SSB)**

MAS issued the first SSB on 1 October 2015, with regular issues of new savings bonds every month thereafter. SSBs are a new type of government bonds that offer individual investors a safe and flexible way to meet their long-term savings and investment needs.

SSBs are fully backed by the Singapore government and investors can redeem the invested amount before maturity without any loss of capital. The SSBs promote long-term savings with interest that increases over time. In addition, they offer flexibility to investors by allowing investors to exit their investment early without penalties. The minimum investible amount per bond is SGD500 with a maximum individual holding of SGD200,000. The investment sum has to be in multiples of SGD500.

The return on SSBs depends on how long investors hold them for. Investors receive less interest at the start, but the interest increases or "steps up" over time so that they receive a higher effective return when they invest in the SSBs over a longer period. At the beginning of each month, MAS will announce the interest rates for the entire 10-year term of that particular month's SSB issue. These interest rates will be locked-in at the point of subscription.

The interest rates for each SSB issue are based on the average yields of SGS the month before applications open for that issue. However, the rates may be adjusted to maintain the "step-up" feature if market conditions do not otherwise allow it. Nonetheless, if investors were to hold an SSB for the full 10 years, their return will match the average 10-year SGS yield for the month prior to investment.

Investors can redeem their SSBs in any given month before maturity and there is no penalty for exiting the investment early. However, early redemption of SSBs will result in a lower return. Redemption amounts must be in multiples of SGD500 (up to the amount invested) and accrued interest on the redemption amount will be paid. Interest income from SSBs is exempt from tax.

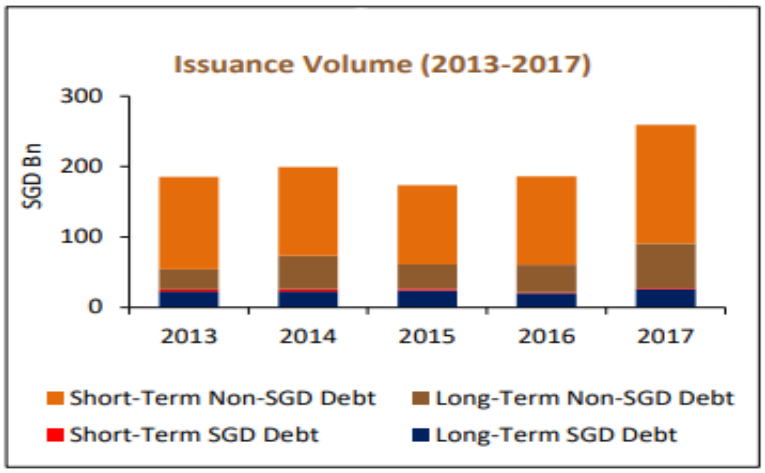
[35] https://www.mas.gov.sg/bonds-and-bills/outstanding-sgs.

c. **Singapore Corporate Bond Market**

Despite challenges in the offshore energy market, which triggered some defaults, and the slowdown of global economies, which led to lesser demand for capital, overall corporate bond issuance rose in 2017 (refer to Figure 7.7.1(a).

Figure 7.7.1 (a) Singapore Corporate Debt[36]

Corporate debt issued totalled SGD259 billion while total debt outstanding rose to SGD386 billion in 2017.



Several new bond issues arose from the launch of green bonds, which provide funds to projects that deliver environmental benefits. Also, with the recovery in crude oil prices bringing relief to the offshore oil industry, conditions in the corporate bond market are expected to improve.

7.7.2 Eurobond Market

The Eurobond market is an international bond market. A Eurobond can be defined as a bond that is:

a. underwritten by an international syndicate;
b. offered at issuance simultaneously to investors in a number of countries; and
c. issued outside of the jurisdiction of any single country.

Eurobonds derive their name from the fact that they were originally bond issues sold in Europe (typically London). The Eurobond market is divided into the different currencies in which the issues are denominated. For example, when Eurobonds are denominated in USD, they are referred to as **Eurodollar bonds**, while Eurobonds denominated in Japanese Yen are referred to as **Euroyen bonds**.

7.7.3 Primary and Secondary Bond Markets

The primary and secondary bond markets each serve a different function for issuers and investors.

a. **Primary Bond Market**

Bonds are issued to the public in the primary bond market. SGS are sold at auctions through Primary Dealers. When a company decides to raise funds with a bond issue, it will usually appoint an investment bank to help

[36] https://www.mas.gov.sg/bonds-and-bills/understanding-singapores-bond-market/sgd-corporate-dept-market.

with the management, underwriting, and distribution of the bonds. For its effort, the investment bank (or a syndicate of underwriters) will be paid management and underwriting fees, generally based on the amount of bonds that are issued.

Bonds and other debt securities that will be sold to non-retail investors are exempted from prospectus requirements, but an offering memorandum is used. Issuers of bonds and similar debt securities that will be sold to retail investors are required to prepare a prospectus unless the issuer is already listed on SGX, in which case only an Offer Information Statement is needed.

When evaluating a bond issue, the following factors should be noted:

i. Credit rating and debt structure of the issuer.

ii. Will there be a reliable secondary market for the bond?

iii. Are there cross default clauses? The presence of cross default means that the issuer may be considered insolvent if one of its related entities becomes insolvent.

iv. How the bonds rank against the other liabilities of the issuer?

b. **Secondary Bond Market**

After the bonds have been sold in the primary market, they can be traded in the secondary market. The secondary market may be the exchange where the bond is listed (for retail bond issues) or the over-the-counter (OTC) market. Other than being a place where bonds can be bought and sold, the secondary market also serves the important market function of benchmarking the market values and yields of the traded bonds. Such information is necessary for the valuation of bond portfolios and for establishing benchmark yield curves of bonds with different credit ratings. Of interest is the spread over the yield on Government issues of similar maturity, i.e., the difference between the corporate bond yield and the Government bond yield.

7.8 Types of Bonds

7.8.1 Floating Rate Bonds

In contrast to a coupon rate that is fixed for the entire life of the bond, the coupon rate on floating rate securities is periodically reset based on some predetermined benchmark e.g., SIBOR (Singapore Interbank Offered Rate). For example, the coupon rate may be reset every six months to be equal to a spread of 100 basis points over the six-month Treasury bill rate. Floating rate securities are attractive to some institutional investors because the securities allow investors to buy an asset with an income stream that more closely matches the floating nature of some of their liabilities.

7.8.2 Zero-Coupon Bonds

Zero-coupon bonds do not make any periodic coupon payments. They are sold at prices far below their par values, i.e., they are sold at deep discounts. At maturity of the bond, the investor will receive the par value. The difference between the maturity value and the purchase price generates an effective interest rate of return.

Perhaps the most attractive feature of zero-coupon bonds is that there is no reinvestment risk within the life of the bond because there is no coupon to reinvest, so the rate of return is locked in when they are held to maturity.

However, there are potential disadvantages associated with zero-coupon bonds. Firstly, holders must be alert to zero-coupon bonds with call features. The call feature potentially limits capital appreciation when interest rates decline because the issuers are likely to redeem the bonds. Secondly, zero-coupon bonds experience high price volatility—a rise in interest rates will negatively impact the price of a zero-coupon bond much more than that of a coupon bond. If the bonds are held to maturity, then the interim volatility will not be a concern. Lastly, liquidity may be limited for such bonds.

7.8.3 Convertible Bonds

Convertible bonds are bonds that can be converted, at the holder's option, into ordinary shares of the issuing company within a specified period. Yields on convertible bonds are generally lower than those on comparable straight bonds because of the conversion privilege, which the bondholder effectively pays for through the lower yield. There is potential capital appreciation from increases in the underlying share price (through the option), a fall in interest rates or an improvement in the credit rating of the issuer.

Convertible bonds are advantageous to the issuer because they allow the issuer to refinance at a lower rate and defer potential dilution of earnings. This is because, at the time of issue, the conversion rate is set such that the effective paid per share from the conversion is above the current share price, making it unattractive to convert. If the share price appreciates, it may be worthwhile to convert; at this point, the amount of conversion from bonds to shares will determine the extent of earnings dilution.

7.8.4 Covered Bonds

Generally, a covered bond is a highly rated paper, backed by bank assets, and typically costs less to issue than unsecured bonds. Covered bonds have a long history in Europe that dates to 1769. Though many banks have since come and gone, these bonds have never experienced a default and, partly because of this appeal and as a response to the Global Financial Crisis of 2007-2008, covered bonds are now issued in many countries around the world. The MAS released guidelines in December 2013 to allow local banks to sell covered bonds. A bank in acquisition mode might consider a covered bond offering as an additional, and more attractive, way to raise funds.

7.8.5 High-Yield Bonds

Bonds with a credit rating below triple B (BBB) are called high-yield bonds or junk bonds. High-yield bonds are securities that trade primarily on their creditworthiness as opposed to the level of interest rates. Their high yields reflect a variety of factors such as the small size of the company or a lack of credit history.

There are complex junk bond structures, particularly for bonds issued for leveraged buyout (LBO) financing and recapitalizations. To reduce the heavy interest payment, firms involved in LBOs and recapitalizations will issue bonds with deferred coupon structures that permit the issuer to avoid using cash to make interest payments for a period of three to seven years.

7.8.6 Deferred-Coupon Bonds

There are three types of deferred-coupon structures:

a. **Deferred-interest bonds** are the most common type of deferred-coupon structure. These bonds sell at a deep discount and do not pay interest for an initial period, typically from three to seven years.

b. **Step-up bonds** pay coupon interest. However, the coupon rate is low for an initial period and then increases ("steps up") to a higher coupon rate thereafter.

c. **Payment-in-kind bonds** give the issuer an option to pay cash at a coupon payment date or give the bondholder a similar bond (i.e., a bond with the same coupon and a par value equal to the amount of the coupon payment that would have been paid). The period within which the issuer can make this choice varies from five to ten years.

d. **Perpetual bonds** have no maturity. Such bonds pay a higher coupon rate to compensate the investor for an indefinite holding period and the correspondingly higher interest rate risk. In a rising interest rate environment, perpetual bonds will usually suffer one of the highest declines in value. Although issuers of perpetual bond are not legally obligated to redeem these bonds, there is usually an option for the bonds to be redeemed by the issuer on specific dates (usually three to five years after the bond issuance). Depending on the prevailing cost of funds and financing needs during the specific redemption dates, the issuer can decide to redeem the bond. Perpetual bonds rank before equities in the order of discharge in the event of default.

7.9 Bond Features

Unlike a 'plain vanilla' or 'straight' bond, which is a basic fixed-income security with no special features, some bond issues have special features that can augment the riskiness of the bonds.

7.9.1 Call Provisions

A bond with a **call feature or call provision** gives the issuer the right to retire the debt, fully or partially, before the maturity date. This is in contrast to a non-callable bond, for which the issuer cannot retire the bond before maturity.

The call provision is attractive to issuers as it gives them the flexibility to control financing costs if and when interest rates move downward. The call feature can also be useful for corporations that may want to circumvent restrictions in the bond indenture.

The call provision is, however, unattractive to investors who run the risk of losing a high-coupon bond when interest rates decline. To protect the investor from an early redemption, there may be a provision limiting the issuer's right to call the bond for a certain period of time after the date of issue. Another disadvantage for the investor is that the prospect of a call limits the upside price potential of the bond in a falling interest rate environment. Due to these disadvantages, bonds with call features carry higher yields than equivalent non-callable bonds.

The **call price**, i.e., the price at which the bond may be called, is normally higher than the par value of the bond. The difference between the call price and par value is the **call premium**, which tends to be higher in the early life of the bond and declines gradually towards par as the bond approaches maturity.

7.9.2 Sinking-Fund Provision

Corporate bond indentures may require the issuer to retire a certain amount of the outstanding debt each year. If so, the issuer may purchase the amount of bonds to be retired in the open market if their price is below par, or the company may make payments to the trustee who will then call the bonds for redemption using a lottery. The purpose of a sinking-fund provision is to reduce credit risk.

Some sinking-fund provisions may have embedded options that allow the issuer the option to accelerate the repayment of the principal. An example is the doubling option, which grants the issuer the right to retire twice the amount of debt the sinking fund requires, supposedly to reduce the credit risk of an issuer.

7.9.3 Put Provisions

Puttable bonds give the bondholder the right to sell the bonds back to the issuer at par value on designated dates. A put provision helps to limit the downside risk of the bond. When interest rates rise, resulting in a decline in the bond price, the investor can "put" or sell the bond to the issuer at par. As with a convertible bond, the investor has effectively bought a put option from the issuer and will pay for it with a lower yield.

7.9.4 Security for Bonds

The promises of a corporate bond issuer and the rights of the bond investors are specified in a contract called a **bond trust deed**. A trustee is legally appointed to look after the interests of the bondholders and to ensure that all the provisions under the deed are met, including the timely payments of interest and principal.

Failure to pay either the principal or interest due constitutes legal default, and court proceedings can be instituted to enforce the contract. Bondholders, as creditors, have a priority claim over ordinary and preferred shareholders on both the income and assets of a corporation for the principal and interest due to them.

Secured (senior) bonds are backed by a legal claim on some specified property of the issuer in the event of default. Such bonds are secured by collateral – similar to collateral for a loan. In the event of a default, the bondholders will have first rights over the asset. Secured bonds can have different types of collateral. For example, mortgage bonds are secured by real estate assets. Secured bonds can also be secured via a specific dedicated revenue stream that comes from a particular business activity or project (such as tolls from toll roads or utility fees from a power generation project).

When an asset **is pledged**, the title of the asset is held by the lender until the loan is paid. Although the borrower still owns and can still use the asset, the title (documents of ownership) will be held by the lender. Secured bonds will often involve the pledging of assets as collateral.

The term **debenture** refers to any debenture stock, bond, note and any other debt securities as described in the Securities and Futures Act 2001.

Subordinated (junior) bondholders have a claim on the income and assets of the issuer that is subordinated to secured debt and after the main creditors have made their claims.

Income bonds are junior debt as the issuing firm can omit or delay the payment of interest if its earnings are too low.

When things go bad, creditors will be aggressively seeking recovery and recourse. The ranking of claims helps in some way to make the process more orderly.

7.9.5 Guarantees and Credit Insurance

Some loans and bonds that are issued by subsidiaries will carry guarantees from their parent company. Such guarantees would add to the credit standing of the bond. Where this is lacking, the company could also try to improve the credit quality of its bond issue by securing credit insurance for a fee.

7.10 Asset and Mortgage-backed Securities

Asset and mortgage-backed securities are typically securities created for sale to investors by lenders from pooling and securitising illiquid asset cash flows. The primary difference between asset-backed securities (ABS) and mortgage–backed securities (MBS) is the underlying asset that is being securitised. MBS are created from the pooling of mortgages whereas ABS evolved from MBS and are created from the pooling of non-mortgage assets.

ABS represent an interest in a wide range of pooled assets, which include (amongst others):

a. Auto loans and leases;
b. Consumer loans;
c. Credit card receivables; and
d. Royalty payments.

Prior to the 2007 sub-prime crisis in the US, the Federal Home Loan Mortgage Corporation (Freddie Mac) and the Federal National Mortgage Association (Fannie Mae) issued large amounts of MBS. Numerous mortgages were actively promoted and approved despite their poor credit quality (i.e., sub-prime loans) because loan officers were rewarded on a commission basis. Such lending was carried out amidst the growing property bubble which had been building from 2001-2007. Freddie Mac and Fannie Mae purchased and invested in huge numbers of mortgages and MBS with their funding advantage, as they had lower capital requirements than other regulated financial institutions and banks.

Once the property bubble burst, mortgage defaults began and the MBS started falling in value. Several financial institutions that purchased these loans suffered massive losses and became insolvent. Their credit default and demise triggered further losses and subsequent defaults in other investments, which put other banks globally at risk. Defaulting MBS were at the root of the sub-prime crisis in 2007 and the global financial crisis that ensued.

The asset securitisation market in Singapore was active when it commenced in 2006 but became very quiet following the Global Financial Crisis in 2007. Examples of MBS issues in Singapore include the commercial MBS structured and sold by DBS Bank, which dealt with pre-sales of properties under construction. The progress payments on the property were securitised and provided construction finance.

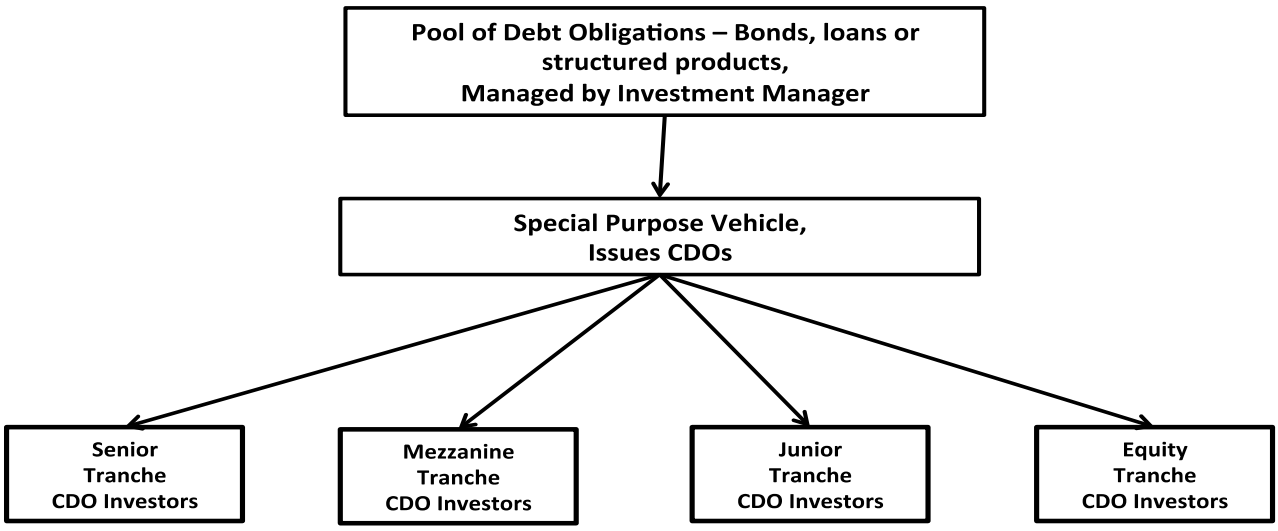
Another deal, which was structured and lead-managed by HSBC for Courts Asia, used two distinct special purpose vehicles (SPVs) – one in Malaysia and one in Singapore – to structure a multi-currency, multi-jurisdiction transaction. Courts' loan receivables were used as assets for the SPVs.

7.10.1 Collateralised Debt Obligations (CDOs)

A CDO is structured by an investment manager or a bank, which pools together cash flow-generating assets (e.g., bonds or loans) that are then repackaged and sold to investors. The investors purchase the CDOs, which are typically issued by a SPV and secured and backed by the asset acquired i.e., bonds or loans.

Please refer to Figure 7.10.1 for the basic CDO structure.

Figure 7.10.1 - Basic CDO Structure



CDOs that are backed by a pool of bonds are also known as collateralised bond obligations (CBOs), while those that are backed by a pool of loans are known as collateralised loan obligations (CLOs). CDOs are effectively a type of asset-backed security.

Therefore, with a CDO structure, a fund manager will look after a portfolio of assets (i.e., bonds or loans that have been purchased), and the proceeds from the issuance of the CDOs effectively finances the acquisition of these assets.

Hence, a CDO is effectively a vehicle for an investment manager to gather assets to generate management fees or to carry out a form of arbitrage in the market, earning a yield between the return on the assets and the rate paid to the CDO investors.

During the sub-prime crisis, 40-50% of CDOs in the US were made up of MBS. Such MBS were mixed with other investment grade securities and, with the help of insurance purchased on them, were re-sold as CDOs with higher credit ratings (as given by credit rating agencies).

Some banks selling CDOs included Credit Default Swaps (CDS) to protect and insure the buyers against default. An insurance company thought the premiums on CDS to be attractive and began selling CDS to different parties even for a given MBS – essentially guaranteeing payment to several different parties if the same MBS defaulted.

As long as the CDOs and other MBS did not default, the insurance company would be able to earn premiums from different investors. Hence, there would be several premiums for the same asset insured. However, in the event of default, the insurance company would be obligated to pay different parties who bought the CDS. When the defaults occurred in 2007/8, the insurance company had to pay many times over.

7.11 Credit Risk

Credit risk or default risk refers to the risk that the issuer of the bond will be unable to make timely principal and interest payments as agreed in the terms of issue. Most corporate bonds pay a higher yield than comparable US Treasury securities, which are regarded as free of credit risk.

Normally, institutional investors that trade bonds are concerned about both:

a. changes in the perceived credit risk; and

b. the actual event of default.

However, it is important to note that even though the likelihood of an actual default may be small, a change in the perceived credit risk of the issuer can have an immediate and significant impact on the value of the bond, which will affect the value of the portfolio in which it is a constituent asset.

Individual investors who tend to hold investments until maturity focus on credit risk. As such investors usually have fewer means to conduct their own research, they rely on the advice of bond specialists from banks to help select suitable credit instruments. Such investors will typically look at the rating agencies' commentaries to get any negative comments that could affect ratings or some forewarning of any impending credit rating downgrades.

As with all investments, there is no free lunch. Investors seeking higher yielding bonds will have to take on credit risks. Moreover, investors usually have greater difficulty unwinding deteriorating bond investment positions since liquidity usually vanishes quickly once credit conditions become dire. This was encountered by investors in the Singapore bond market when they faced a series of bond defaults from five Singapore firms in 2016 due to the contraction in the offshore oil industry. In hindsight, the right time to cut losses would have been more than a year before, when oil prices began their decline.

Hence, investors need to be vigilant of key factors that will affect the viability of companies issuing bonds. Unfortunately, in such situations, the bond default will still have to eventually be borne by an investor.

7.11.1 Credit Risk Rating

The credit risk of bonds is gauged by the quality ratings assigned to them by commercial rating companies. The three largest credit-rating agencies are **Moody's Investors Service (Moodys), Fitch Ratings (Fitch), and Standard & Poor's Global (S&P)**. These agencies aim to provide objective assessments of the credit risk of the issuers and their securities.

S&P will be discussed for illustrative purposes. S&P's bond ratings consist of letters ranging from **AAA, AA, A, BBB, and so on, to D**. The first four categories, AAA through BBB, represent investment grade bonds. Issues that carry a rating below the top four categories, i.e., BB and below, are referred to as non-investment grade bonds or, more popularly, as high-yield bonds or junk bonds. Ratings from AA to CCC may be modified by the addition of a plus (+) or minus (-) sign to show relative standing within the rating categories.

Issuer credit ratings can either be long-term or short-term. Short-term ratings are generally assigned to those obligations that are considered short-term in the relevant market. In the US, this would mean obligations with an original maturity of no more than 365 days. Short-term ratings are also used to indicate the

creditworthiness of an obligor with respect to put features[37] on long-term obligations, leading to a dual rating, in which the short-term rating addresses the put features, in addition to the usual long-term rating. Medium-term notes are assigned long-term ratings. Please refer to Tables 7.11.1(a) and 7.11.1(b) on S&P's long-term and short-term issuer credit ratings respectively.

Table 7.11.1(a) – Long-Term Issuer Credit Ratings of S&P [38]

Category	Definition
AAA	An obligor rated 'AAA' has extremely strong capacity to meet its financial commitments. 'AAA' is the highest issuer credit rating assigned by S&P.
AA	An obligor rated 'AA' has very strong capacity to meet its financial commitments. It differs from the highest-rated obligors only to a small degree.
A	An obligor rated 'A' has strong capacity to meet its financial commitments but is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than obligors in higher-rated categories.
BBB	An obligor rated 'BBB' has adequate capacity to meet its financial commitments. However, adverse economic conditions or changing circumstances are more likely to weaken the obligor's capacity to meet its financial commitments.
BB, B, CCC, and CC **BB**	Obligors rated 'BB', 'B', 'CCC', 'CC' and 'C' are regarded as having significant speculative characteristics. 'BB' indicates the least degree of speculation and 'CC' the highest. While such obligors will likely have some quality and protective characteristics, these may be outweighed by large uncertainties or major exposure to adverse conditions. An obligor rated 'BB' is less vulnerable in the near term than other lower-rated obligors. However, it faces major ongoing uncertainties and exposure to adverse business, financial, or economic conditions that could lead to the obligor's inadequate capacity to meet its financial commitments.
B	An obligor rated 'B' is more vulnerable than the obligors rated 'BB', but the obligor currently has the capacity to meet its financial commitments. Adverse business, financial, or economic conditions will likely impair the obligor's capacity or willingness to meet its financial commitments.
CCC	An obligor rated 'CCC' is currently vulnerable, and is dependent upon favourable business, financial, and economic conditions to meet its financial commitments.
CC	An obligor rated 'CC' is currently highly vulnerable. The 'CC' rating is used when a default has not yet occurred, but S&P expects default to be a virtual certainty, regardless of the anticipated time to default.
R	An obligor rated 'R' is under regulatory supervision owing to its financial condition. During the pendency of the regulatory supervision, the regulators may have the power to favour one class of obligations over others or pay some obligations and not others.
SD and D	An obligor rated 'SD' (selective default), or 'D' is in default on one or more of its financial obligations including rated and unrated obligations but excluding hybrid instruments classified as regulatory capital or in non-payment according to terms. An obligor is considered in default unless S&P believes that such payments will be made within five business days of the due date

[37] A bond with a put feature (or a puttable bond) gives the bondholder the right, but not the obligation, to force the issuer to redeem or repurchase the bond on one or more specified dates before maturity.
[38] Source: https://www.spglobal.com/ratings/en/about/intro-to-credit-ratings.

Category	Definition
	in the absence of a stated grace period or within the earlier of the stated grace period or 30 calendar days. A 'D' rating is assigned when S&P believes that the default will be a general default and that the obligor will fail to pay all or substantially all of its obligations as they come due. An 'SD' rating is assigned when S&P believes that the obligor has selectively defaulted on a specific issue or class of obligations, but it will continue to meet its payment obligations on other issues or classes of obligations in a timely manner. An obligor's rating is lowered to 'D' or 'SD' if it is conducting a distressed exchange offer.

Table 7.11.1(b) – Short-Term Issuer Credit Ratings of S&P [39]

Category	Definition
A-1	An obligor rated 'A-1' has strong capacity to meet its financial commitments. It is rated in the highest category by S&P. Within this category, certain obligors are designated with a plus sign (+). This indicates that the obligor's capacity to meet its financial commitments is extremely strong.
A-2	An obligor rated 'A-2' has satisfactory capacity to meet its financial commitments. However, it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than obligors in the highest rating category.
A-3	An obligor rated 'A-3' has adequate capacity to meet its financial obligations. However, adverse economic conditions or changing circumstances are more likely to lead to weaken the obligor's capacity to meet its financial commitments.
B	An obligor rated 'B' is regarded as vulnerable and has significant speculative characteristics. The obligor currently has the capacity to meet its financial commitments; however, it faces major ongoing uncertainties which could lead to the obligor's inadequate capacity to meet its financial commitments.
C	An obligor rated 'C' is currently vulnerable to non-payment that would result in a 'SD' or 'D' issuer rating, and is dependent upon favourable business, financial, and economic conditions for it to meet its financial commitments.
R	An obligor rated 'R' is under regulatory supervision owing to its financial condition. During the pendency of the regulatory supervision the regulators may have the power to favour one class of obligations over others or pay some obligations and not others.
SD & D	An obligor rated 'SD' (selective default), or 'D' has failed to pay one or more of its financial obligations (rated or unrated), excluding hybrid instruments classified as regulatory capital or in non-payment according to terms, when it came due. An obligor is considered in default unless S&P believes that such payments will be made within any stated grace period. However, any stated grace period longer than five business days will be treated as five business days. A 'D' rating is assigned when S&P believes that the default will be a general default and that the obligor will fail to pay all or substantially all of its obligations as they come due. An 'SD' rating is assigned when S&P believes that the obligor has selectively defaulted on a specific issue or class of obligations, excluding hybrid instruments classified as regulatory capital, but it will continue to meet its payment obligations on other issues or classes of obligations in a timely manner. An obligor's rating is lowered to 'D' or 'SD' if it is conducting a distressed exchange offer.

The credit rating categories and notations applied by Moodys and Fitch are similar to those of S&P.

[39] Source: https://www.spratings.com/understanding-ratings.

Table 7.11.1(c) outlines the common rating notations and definitions used by Moodys, S&P and Fitch.

Table 7.11.1(c) – Comparing Ratings of Moodys, S&P, and Fitch

Moodys	S&P	Fitch	Credit Worthiness
Aaa	AAA	AAA	An obligor has EXTREMELY STRONG capacity to meet its financial commitments.
Aa1	AA+	AA+	An obligor has VERY STRONG capacity to meet its financial commitments. It differs from the highest-rated obligors only to a small degree.
Aa2	AA	AA	
Aa3	AA−	AA−	
A1	A+	A+	An obligor has STRONG capacity to meet its financial commitments but is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than obligors in higher-rated categories.
A2	A	A	
A3	A−	A−	
Baa1	BBB+	BBB+	An obligor has ADEQUATE capacity to meet its financial commitments. However, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity of the obligor to meet its financial commitments.
Baa2	BBB	BBB	
Baa3	BBB−	BBB−	
Ba1	BB+	BB+	An obligor is LESS VULNERABLE in the near term than other lower-rated obligors. However, it faces major ongoing uncertainties and exposure to adverse business, financial, or economic conditions which could lead to the obligor's inadequate capacity to meet its financial commitments.
Ba2	BB	BB	
Ba3	BB−	BB−	
B1	B+	B+	An obligor is MORE VULNERABLE than the obligors rated 'BB', but the obligor currently has the capacity to meet its financial commitments. Adverse business, financial, or economic conditions will likely impair the obligor's capacity or willingness to meet its financial commitments.
B2	B	B	
B3	B−	B−	
Caa	CCC	CCC	An obligor is CURRENTLY VULNERABLE, and is dependent upon favourable business, financial, and economic conditions to meet its financial commitments.
Ca	CC	CC	An obligor is CURRENTLY HIGHLY-VULNERABLE.
	C	C	The obligor is CURRENTLY HIGHLY-VULNERABLE to nonpayment. May be used where a bankruptcy petition has been filed.
C	D	D	An obligor has failed to pay one or more of its financial obligations (rated or unrated) when it became due.
	SD	RD	This rating is assigned when the agency believes that the obligor has selectively defaulted on a specific issue or class of obligations but it will continue to meet its payment obligations on other issues or classes of obligations in a timely manner.
NR	NR	NR	No rating has been requested, or there is insufficient information on which to base a rating.
e, p	pr	Expected	Preliminary ratings may be assigned to obligations pending receipt of final documentation and legal opinions. The final rating may differ from the preliminary rating.
WR			Rating withdrawn for reasons including: debt maturity, calls, puts, conversions, etc., or business reasons (e.g. change in the size of a debt issue), or the issuer defaults.
Unsolicited	Unsolicited		This rating was initiated by the ratings agency and not requested by the issuer.

[40]

7.11.2 Credit Spreads

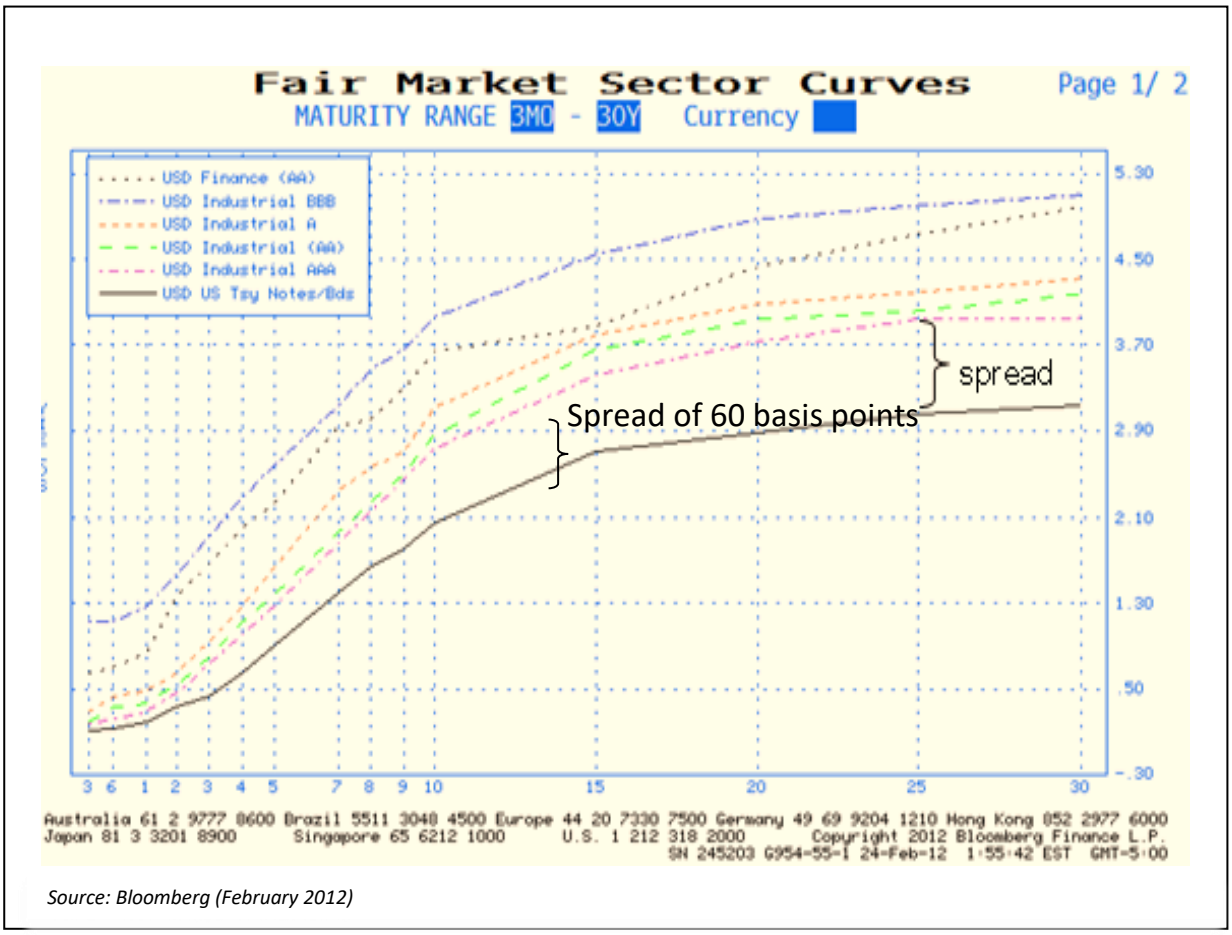
A credit spread is the rate difference (usually quoted in basis points) between two bonds of the same maturity but of differing credit quality. It is a direct measure of the cost of risk, or the extra yield (or rate of return)

[40] https://en.wikipedia.org/wiki/Bond_credit_rating#cite_note-MRSD-3.

investors demand to take on a bond with a certain level of risk of defaulting. As the issuer's default risk increases, its bond price falls and its bond spread widens.

To illustrate, if a 10-year Treasury bond is trading at a yield of 2% and a 10-year corporate bond is trading at a yield of 4%, then the corporate bond is said to offer a 200-basis point spread over the Treasury. Figure 7.11.2(a) reflects an approximate 60-basis point spread between the AAA (investment grade) Industrial Bond over the USD Treasury Note at the 10-year mark. Comparisons can be made between any bond type (e.g., High Yield bond) and a benchmark bond yield, with the spread monitored over time.

Figure 7.11.2(a) - Bond Yield Curves Showing Spread over US Treasury Note/Bonds



Source: Bloomberg (February 2012)

Figure 7.11.2(a) also shows that the US Financials Bond rated AA offer a higher yield than a US Industrial bond rated A. Since the Global Financial Crisis and the European Sovereign Debt Crisis, credit ratings of financial institutions remain weak. Credit spreads will widen against the benchmark during crises and narrow when market conditions, and investors' risk appetites, improve.

Figure 7.11.2(b) illustrates the high yield bond spreads for period of 2007 to 2017.

Figure 7.11.2(b) High-Yield Bond Spreads 2007-2017



Source: *Markit iBoxx / Credit Market Daily.com*

7.11.3 Non-Rated/Unrated Bonds

Some companies in Asia have chosen not to be rated by rating agencies. They do not see a need for it because they believe the main investors who would buy their bonds know their business and history. For example, an Indonesian corporate would be seeking funding mainly from other local institutions and corporates who are familiar with the company. An unrated company may not necessarily be of poor credit or afraid of being rated – they are simply "not rated".

Regulators are generally concerned about such unrated debt because of the lower transparency and higher risk to investors. Local and regional investors are attracted to buying unrated bonds because of their higher yields, which reflect the higher risks associated with such unrated bonds.
While some local investors rely on the strength of the company's brand name or the company's association with the government or government entities, others depend on the credit analysis carried out by the financial institutions underwriting and marketing the bonds. Financial institutions that wish to underwrite and market such unrated/non-rated bonds have to carry out due diligence and ensure that there are no conflicts of interest in the entire process.

7.11.4 Mitigating Poor Credit Quality of the Bonds

Poor credit quality can adversely impact the cost of financing for companies. Hence, companies seeking to improve their credit quality of their bonds can do the following:

a. Create a dedicated sinking fund which will set aside income earned for the purposes of sustaining interest payments and eventually paying off the debt;

b. Offer specified assets (e.g., property holdings) as collateral or security for bond owners;

c. Offer parental guarantees where the debt issue is further backed by the credit standing of the parent company or major equity holder;

d. Work with a financial institution to structure and repackage the bonds of lower credit quality with other bonds of higher credit quality. Pooling them together will form the underlying that will fund the issuance of a security of higher credit rating (see Collateralised Debt Obligations - section 7.10.1); or

e. Get a financial institution to offer credit insurance or credit default swaps where the financial institution will "guarantee" to cover losses of the bond buyer if the bond issuer defaults.

7.11.5 Credit Default Swaps

In this section, we will only provide a brief description of CDS for completeness. Further details on CDS will be covered in the CMFAS CM-SIP Study Guide.

A CDS is the most commonly utilised form of credit derivative. In simple terms, a CDS is a contract of insurance that guarantees a buyer of debt protection against the risk of debt default. CDS are meant to help investors exchange the risk of default for a premium. The buyer of a CDS pays a premium to be insured against a debt default. On the other hand, the seller of a CDS receives regular payments from the buyer to indemnify the buyer with a lump-sum payment (i.e., principal) in the event of a debt default (refer to Figure 7.11.5).

CDS were conceived to provide financial institutions with the means to transfer credit exposure and free up regulatory capital. Apart from protecting against default of high-risk bonds, sovereign debts and corporate debts, investors also use CDS to hedge the credit risk of MBS and CDOs. In the context of bonds, CDS contracts mitigate the risk of bond investing by transferring a given risk from one party to another without transferring the underlying bond.

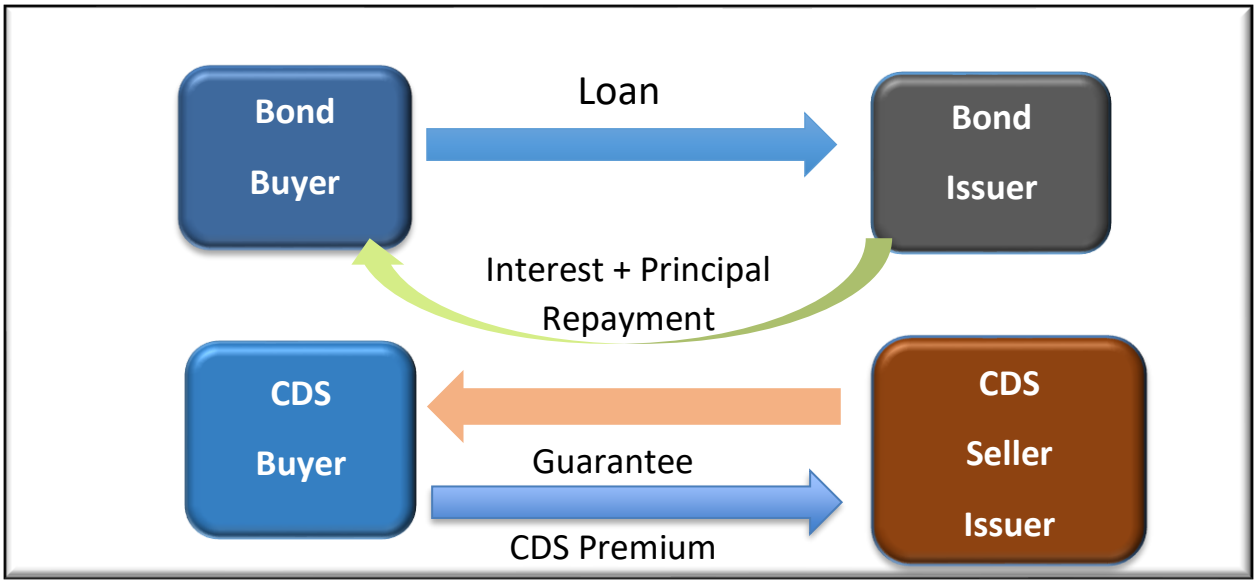
The derivatives market was largely unregulated and lacked transparency until 2009, after the Global Financial Crisis. Issuers of CDS saw the lucrativeness of the CDS market given the lack of regulation to ensure that the seller of a CDS had the money to pay the holder (buyer) if the bond defaulted. There was also no clarity on which sellers ultimately held the exposure.

Many financial institutions, including insurers, were excessively selling CDS for premiums even though most were heavily undercapitalised and unhedged against the possibility that the debt might decline in value. CDS provided a false sense of security to investors (buyers), who rationalised that CDS offer them protection against any losses. Hence, they bought riskier debts with the CDS providing the "protection". The system worked for a while since there was no default on most of the debts and there was no expectation of all the debts coming due at the same time.

Unfortunately, when US mortgagees started defaulting, the MBS were unable to make their payments and defaulted. Consequently, the financial institutions that purchased the securities came under distress and some eventually went into bankruptcy (e.g., bankruptcy of Lehman Brothers). Insurers that sold multiple CDS on the MBS found themselves unable to provide the protection that they sold (e.g., bailout of AIG). The CDS market fell apart overnight.

Today, CDS have become the engine that drives the credit derivatives market with enhanced regulatory structure, enabling investors to effectively apply it as a tool to hedge or gain exposure to credit risk. CDS also allow for arbitrage opportunities.

Figure 7.11.5 - Credit Default Swaps (CDS)



7.11.6 Red Flags for Credit Analysis and Distressed Debts

Earlier in Chapter 5 on Company Analysis and Understanding Financial Statements, we discussed various ratios. The following types of ratios are particularly relevant to credit analysis:

Liquidity	Current; Quick; Cash; Operation Cash Flow Ratios
Debt & Solvency	Gearing; Debt/Asset; Total Liabilities/Total Assets Ratios
Interest Coverage	EBIT/Interest Expense; EBITDA/Interest Expense

Please refer to Chapter 5 for a detailed discussion on financial ratios.

The same ratios are used by credit rating agencies to assess credit quality and identify significant deterioration in credit quality. However, these are still at best derived from historical financial statements, which are often released too late, and only serve to confirm perceptions and estimates. Fluctuations to the value of credit assets still tend to precede the release of the financials and typically put investors in a reactive position in which losses or lower profits may still occur.

To proactively manage a portfolio of debt (or bonds), one should focus on a few key variables that will impact both current/future revenues and costs before they reflect in the accounting numbers.

Projected Revenue Falls

a. Sudden, significant, and sustained falls in sales and/or prices due to oversupply or structural changes in the industry (e.g., fall in crude prices or freight rates);

b. Sudden or gradual fall in product demand due to technological disruptions that result in new competitors having the lead over traditional players (e.g., Kodak, Nokia);

c. Deferment of major contracts that will affect revenue numbers, especially in cases where the downturn of one industry (e.g., oil) affects another industry more severely (e.g., rig building); or

d. Non-diversified revenue stream coupled with the loss of major clients.

Projected Cost Rises/Stickiness

a. When key cost items rise and causes operating costs to spike; or

b. When revenues are falling due to industry changes, but the company is unable to reduce cost as quickly due to inefficiencies or inertia.

Regulatory Restrictions

a. Where new regulation and laws result in a restriction to old forms of revenue creation; or

b. Where new regulation and laws drastically raise the costs of doing business, forcing some to exit the industry.

Projected Cashflow from Operation Falls

If current and projected cashflows from operations are expected to stay negative over time, the only way companies can service debt obligations is through refinancing or the sale of assets. Quarterly cashflow tracking should reveal some challenges the company is facing but investors would probably want to act on cashflow projections if they are significantly negative. Cash flow liquidity concerns would be the immediate focus for most bond and debt holders of a company.

Departure of Financial Controllers, Accountants, Chief Executive Officers (CEOs), Chief Operating Officer (COO), Chief Financial Officer (CFO), Chief Investment Officer (CIO) and Board of Directors

Regular changes in Financial Controllers, Accountants, CEOs, COOs, CFOs, CIOs and Board of Directors can be a signal that there are matters of disagreement or concern on how the accounts are reflected. They may also indicate concerns over the performance of the company.

Another red flag could be the window-dressing of financials. However, this typically does not allow for timely mitigation of possible losses incurred by investors. Adopting a “wait-and- see” approach to assess the impact of the above red flags on revenue and cost to the point where a company has difficulty meeting its debt and interest obligations would be too risky for most investors. The primary objective for any investor should be to minimize as much risk as possible in their portfolio whilst still achieving a desired rate of return.

7.11.7 Investing in Distressed Debts

A company facing distress may not default. When there is credit deterioration, the debt (or bonds) may be priced below par but that does not mean that the company will ultimately fail as an ongoing concern.

At this point, the projected cash flows from operations and the disposal of assets should be re-examined to assess whether the company can continue to meet its immediate obligations and the availability of assets to fulfil maturing debt payments. If it is able to do so with sufficient buffer, the debt (or bond) could actually be a worthwhile investment since it is priced at a discount i.e., with higher yield.

Some hedge funds, private equity firms, and investors who have a higher risk threshold do look at distressed debt as opportunities. There is potential for recovery and substantial gains from purchasing distressed debt,

especially if the company's challenges are related more to a short-term credit squeeze or cyclical downturn with the potential for recovery when key variables (sales price or factor costs) normalise. For example, a distressed bond priced at $0.30 - $0.40 on the dollar may provide for potential to recover a significant portion of its value when key variables normalise.

If, however, the likelihood of revenue recovery is low (e.g., structural changes in industry), the market values of peripheral assets are either not available or are insufficient to support the debt, and there is insufficient cash flow to tide over the period, then investors should try to salvage whatever is possible from the debt (or bond) investments. Several major financial institutions have sustained massive losses from their investments and trades of distressed debt during the Global Financial Crisis in 2008.

7.12 Other Sources of Risk

Apart from credit risk, there are several other sources of risk associated with investments in fixed-income securities.

7.12.1 Interest Rate Risk

Bond prices are most vulnerable to interest rate changes. There is an inverse relationship between interest rates and bond prices. As interest rates rise (fall), the prices of bonds fall (rise). This is also referred to as **delta risk**. For an investor who intends to hold a fixed-income security to maturity, the intermediate price volatility is not of concern. However, if the investor plans to sell the fixed-income security before maturity, an increase in interest rates will translate into capital loss.

7.12.2 Reinvestment Risk

The interest income received from a fixed-income security is usually reinvested. However, in an environment where interest rates are ever-changing, one is never sure of the rate at which future interest coupons can be reinvested. This uncertainty is referred to as reinvestment risk. There is the risk that the interest rate at which interim cash flows can be reinvested will decline, resulting in a lower eventual return. Reinvestment risk is greater for higher coupon bonds and longer holding periods.

Interest rate risk and reinvestment risk make opposite predictions about the impact of interest rate movements on bond price. For example, a rise in interest rates will have, on the one hand, the effect of pushing down the bond price (since the bond price must fall to give investors the higher yield, because the coupon is fixed); on the other hand, the higher rate at which interest received from the coupons can be reinvested will have the effect of pushing up the bond price. However, these are not fully offsetting and, generally, the effect of the reinvestment risk will only slightly mitigate the effect of interest rate risk.

7.12.3 Timing or Call Risk

Callable bonds are subject to the risk of bond redemption by the issuer before the maturity date. From the investor's viewpoint, there are three disadvantages associated with the call feature. First, issuers are likely to redeem the bonds when interest rates decline, thus exposing the investor to the reinvestment risk from having to invest the proceeds of the redeemed bonds in bonds offering lower yields. Second, the potential for capital appreciation is limited, as the market price of a callable bond will not rise much above the call price. Lastly, the promised cash flow pattern of a callable bond is not known with certainty.

7.12.4 Inflation or Purchasing Power Risk

As the promised cash flows from fixed-income securities do not change, as general price levels rise, the value of the promised cash flows in real terms declines. Inflation risk refers to the uncertainty regarding how much lower the real return will be compared to the nominal (dollar) return.

Inflation risk is actually concerned with **unanticipated inflation**. If inflation is anticipated and has been factored into the price of the bond, then investors are adequately compensated for anticipated inflation.

7.12.5 Marketability or Liquidity Risk

Marketability or liquidity risk refers to the ease with which a security can be sold at or near its fair value in the secondary market. An indication of the marketability of a security is the size of the spread between the bid and the offer quoted by the dealer. The greater the spread, the greater the marketability risk. For an investor who plans to hold the bond until maturity date, marketability risk is less important.

7.12.6 Exchange Rate or Currency Risk

The SGD cash flows, of a foreign currency-denominated bond, are dependent on the prevailing foreign exchange rates at the times the payments are received. If the foreign currency depreciates relative to the SGD, then fewer SGD will be received. This is referred to as exchange rate or currency risk. On the other hand, if the foreign currency appreciates relative to the SGD, then the investor will enjoy currency gains.

7.12.7 Volatility Risk

The price of a bond that has an embedded option depends on the level of interest rates and other factors that influence the value of that embedded option. One of the factors is the expected volatility of interest rates, which is also referred to as vega risk. Specifically, the value of an option rises when expected interest-rate volatility increases. The risk that a change in volatility will adversely affect the price of a security is called volatility risk.

7.12.8 Political Risk

Sometimes, the government can declare withholding taxes on a bond or declare a tax-exempt bond taxable. There may also be capital controls imposed on the remittance of funds. These actions can adversely impact the value of the security. The possibility of any political or legal actions adversely affecting the value of a security is known as political or legal risk.

7.12.9 Event Risk

Occasionally, the ability of the issuer to make interest and principal payments may be seriously impaired because of a natural catastrophe, some regulatory change, or a takeover or corporate restructuring. These risks are referred to as event risks and will result in a downgrading of the issuer by rating agencies.

7.12.10 Sector Risk

Sector risk is a risk that applies only to a particular industry, or sector. In sector risk, the stocks of many of the companies in one sector (like healthcare or technology) will fall in price at the same time because of an event that affects the entire industry. Investors can reduce sector risk by diversifying their portfolios.

7.13 Bond Valuation

The value of a bond equals the present value of its expected cash flows. The cash flows from a bond are the periodic interest payments to the bondholder and the repayment of principal at its maturity. Therefore, the current value of a bond is the total present value of the interest payments plus the present value of the principal repayment, where the discount rate is the required rate of return on the bond. We can express bond value using the following present value formula:

$$P = \frac{C_1}{1+k} + \frac{C_2}{(1+k)^2} + \frac{C_3}{(1+k)^3} + \ldots + \frac{C_n}{(1+k)^n} + \frac{M}{(1+k)^n}$$

$$= \sum_{t=1}^{n} \frac{C_t}{(1+k)^t} + \frac{M}{(1+k)^n}$$

The above equation applies to bonds that pay annual interest. In the case of bonds that pay interest semi-annually, the equation is:

$$P = \sum_{t=1}^{2n} \frac{C_t}{(1+k/2)^t} + \frac{M}{(1+k/2)^{2n}}$$

where
P = present value of the bond today (t = 0)
C_t = coupon payments for period t
M = maturity value
n = number of years to maturity
k = required market yield i.e.; the yield that is currently available in the market for a similar security of this maturity

The discount rate (k) that an investor wants from investing in a bond is called the **required rate of return**. The required yield is the yield offered by comparable bonds with the same maturity and of the same credit quality.

Calculation of Bond Price

Consider a 10-year bond with a par value of $1,000 and a coupon rate of 8%. Given a required market yield of 6% per year, what price would you pay for the bond?

Compute the bond value assuming:

a. Annual coupon payments; and

b. Semi-annual coupon payments.

a. Assuming annual coupon payments

$$\text{Price} = \frac{80}{1.06} + \frac{80}{1.06^2} + \frac{80}{1.06^3} + \ldots + \frac{80}{1.06^{10}} + \frac{1{,}000}{1.06^{10}}$$

This can be calculated with a financial calculator (See Figure 7.13(a)) with the following calculator inputs:

- N = number of interest rate periods, in this example, N = 10
- PMT = the coupon payment each year, PMT = 80
- FV = future value at the end of 10 years, FV = 1,000
- I/Y = rate of discount, I/Y = 6

Figure 7.13(a) – Financial Calculator



Press CPT followed by PV to obtain the present value of the cash flows.

PV = $1,147.20

b. Assuming semi-annual coupon payments

$$\text{Price} = \frac{40}{1.03} + \frac{40}{1.03^2} + \frac{40}{1.03^3} + \ldots + \frac{40}{1.03^{20}} + \frac{1{,}000}{1.03^{20}}$$

= $1,148.77

Note that in the computation, the number of periods must be doubled and, hence, the discount rate must be divided by two.

7.13.1 Valuation of a Zero-Coupon Bond

Zero-coupon bonds do not make any periodic coupon payments. The only cash flow is its maturity value. Therefore, the price of a zero-coupon bond is simply the present value of the maturity amount.

$$\mathbf{P = \frac{M}{(1 + k/2)^{2n}}}$$

Calculation of Bond Price for a Zero-Coupon Bond

Given a required rate of return of 9%, compute the price of a zero-coupon bond that matures in 15 years with a par value of $1,000.

$$P = \frac{1,000}{(1.045)^{30}}$$

= $267.00

Even though there are no coupons, the semi-annual rate of 4.5% and the interest period of half a year are used to make the valuation consistent with that of semi-annual coupon bonds.

To use the financial calculator, input the following:

FV = 1,000

N = 30 (for 30 half-years)

I/Y = 4.5% (annual return of 9% ÷ 2)

Press the CPT, followed by the PV button to obtain:

PV = $267.00

7.14 Relationship between Bond Price and Market Yield

There is an inverse relationship between bond price and market yield i.e., bond prices move in the opposite direction of market yields. This is because the price of the bond is the present value of the cash flows and as market yields increase, the present value of the cash flows decreases with the higher discount rate, leading to a decrease in bond price. Given the expectations of a series of yield adjustments through monetary policy such as the Federal Reserve rate hikes, then existing bond prices and bond portfolio will fall in value as market yields increase. The reverse is also true. When market yields decrease, the present value of the cash flows increases and, therefore, the price of the bond rises.

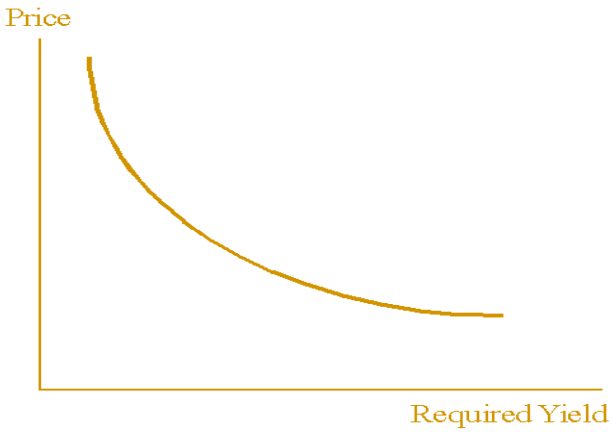
Table 7.14(a) shows the bond value (as a percentage of par) for a 10-year maturity, 12% semi-annual coupon bond for market yields of 10%, 12%, and 14%. Note that as the required yield rises, the bond value declines.

Table 7.14(a) – Price-Yield Relationship for a 10-year, 12% Semi-annual Coupon Bond

Required Yield	Bond Value
10%	112.46%
12%	100.00%
14%	89.41%

This inverse relationship between bond price and required yield is illustrated in Figure 7.14(b):

Figure 7.14(b) - Relationship between Bond Price and Required Yield



7.15 Relationship between Coupon Rate, Market Yield and Price

Generally, when a bond is issued, the coupon rate is set at approximately the prevailing yield in the market. When the market yield is equal to the coupon rate, the price of the bond is its par value i.e., 100% of par value. For instance, if you bought an 8% coupon bond at par, the yield on the bond would be 8%.

Market yields are always changing. If market yields rise above the coupon rate to 9%, the investor will have to be compensated for the higher yield through an adjustment in the bond price. This is accomplished when the bond price falls below its par value. When a bond trades below its par value, it is said to be selling at a **discount**.

If, instead, the market yield falls below the coupon rate to 7%, the price of the bond will move up to adjust for the lower yield. When a bond trades above its par value, it is said to be selling at a **premium**.

The relationship between coupon rate, market yield, and price can be summarised as follows:

a. When the coupon rate equals the market yield, the bond will trade at its par value.

b. When the coupon rate is lower than the market yield, the bond will trade at a discount.

c. When the coupon rate is higher than the market yield, the bond will trade at a premium.

7.16 Relationship between Market Yield and Other Bond Features

7.16.1 Yield and maturity

The longer the maturity of a bond, the greater the impact of a change in interest rate on the price of the bond because longer term cash flows are more sensitive to changes in interest rates. When yields are rising, the longer the maturity, the greater the downward price impact on bonds. When yields are falling, prices of bonds with longer maturities will rise more significantly.

An investor whose investment horizon matches a bond's maturity and who is also unconcerned about marking price fluctuations to market would be indifferent to interest rate risk. Such an investor would also have the

opportunity of capital gains should yields fall. For bond fund managers and traders, however, their bond portfolios will reflect gains or losses each year with changes in market yields. Hence, active management will entail adjusting the maturity and duration of these portfolios.

7.16.2 Yield and coupons

Current yield is the interest rate of return for the bond given its current price whereas coupon rate is the rate of return for the bond based on the par value, which is usually the price of the bond when it was first issued. A bond with smaller coupons generally has a larger percentage of its value that is attributable to its long-term principal repayment component. Since longer term cash flows are more susceptible to changes in interest rates, bonds with smaller coupons will be more severely affected by a rise in yield compared to bonds with larger coupons. For two bonds with equal maturity, the bond with the larger coupon will be less affected by rising market yields and will be priced slightly higher compared to the bond with the smaller coupon.

7.16.3 Yield and callable features

A call feature gives the issuer the right to redeem the bond when there are cheaper refinancing options. When interest rates fall, the call feature becomes an advantage to the issuer as it has the ability to refinance the bond at the lower interest rate without having to pay market price to call its bonds in. From the perspective of the buyer (bond investor), it is a disadvantage as the buyer is exposed to reinvestment risk should the bond be called when interest rates fall.

7.17 Duration and Convexity

7.17.1 Duration

Duration and convexity are concepts that are important to understand in the risk management of a bond portfolio.

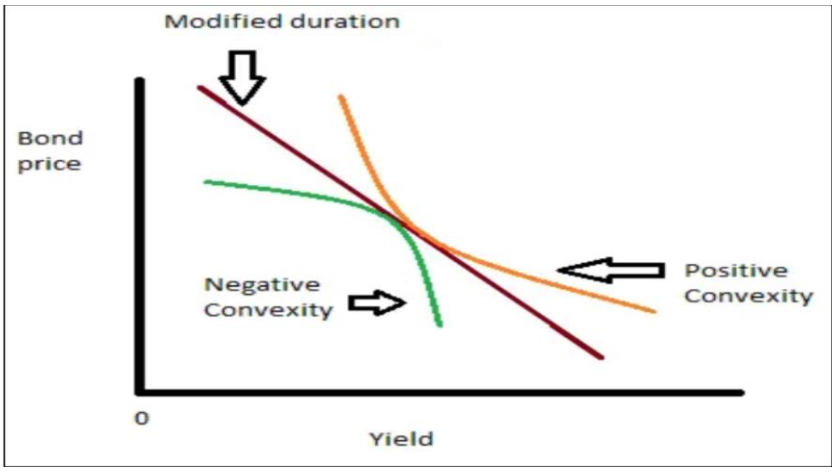
The duration of a bond is affected by its coupon rate, yield, and remaining time to maturity. It is a linear measure of the approximate price sensitivity of a bond to small interest rate changes, i.e., not more than 1% or 100 basis points. All else equal, duration is higher for bonds with lower coupons, lower yields, and longer maturities.

Modified duration is used to calculate changes in bond prices from small changes in yields. Typically, a bond portfolio's overall duration is calculated to monitor the profit and loss sensitivity of the portfolio to small changes in yields. The portfolio duration is the weighted-average duration of all bonds in the portfolio.

7.17.2 Convexity

Duration is an imperfect way of measuring a bond's price change; the price sensitivity is linearly approximated when in fact the price-yield graph exhibits a "convex" shape. With positive convexity, duration increases as yields fall. This means that, for large changes in interest rates, the amount of price appreciation will be greater than the amount of price depreciation. Since these favours the investor, a higher convexity bond will have a higher price than a lower convexity bond. With negative convexity, the amount of price appreciation is less than the amount of price depreciation for large interest rate changes. See Figure 7.17.2.

Figure 7.17.2 – Duration and Convexity



Convexity is directly related to the coupon rate and the bond's remaining term to maturity. The lower the coupon, the higher the convexity. Zero coupon bonds have the highest convexity.

7.17.3 Duration and Convexity: The Price-Yield Relationship

Duration and convexity are two metrics used to help investors assess how the price of a bond will be affected by changes in interest rates. They are the most prevalent measures for quantifying market risk for bonds in different interest rate environments.

Duration is particularly important when attempting to capitalise on the sensitivity of a bond portfolio to interest rates. A higher duration implies a greater price movement upwards (downwards) should rates fall (rise). Following this rationale, with the anticipation of falling interest rates, bond prices are anticipated to rise, and a bond portfolio's duration should be increased to maximise the sensitivity of the portfolio. Conversely, when there is expectation of rising interest rates, bond prices are expected to fall. Bonds with shorter duration and maturity will be less affected and those with longer duration will be more sensitive to the interest rate rise. The duration of a bond portfolio should be decreased to minimise the sensitivity of the portfolio in this case.

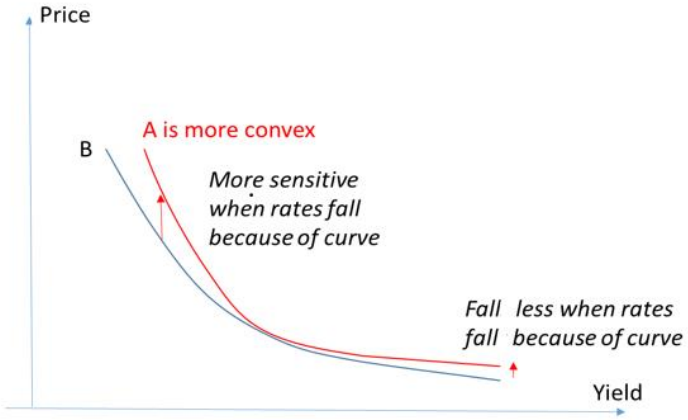
Convexity is the second derivative of the price function and measures the rate of change in the duration – effectively showing the curvature of price–yield relationship. Regardless of whether interest rates are rising or falling, investors should seek bonds with greater convexity. When interest rates rise, the price of a more convex bond will not fall as much. When interest rates fall, the price of a more convex bond will rise much faster. Please refer to Table 7.17.3(a).

Table 7.17.3(a) – Relationship between Duration, Convexity and Interest Rates

Interest Rate	More Duration	More Convexity
Rising	Price falls more	Price falls slower
Falling	Price rises more	Price rises faster

Figure 7.17.3(b) shows that Bond A will outperform Bond B when rates rise and when rates fall because of its greater convexity.

Figure 7.17.3(b) –Convexity



7.17.4 Present Value of a Basis Point (PVBP or PV01)

The PVBP or PV01 is used as a risk management tool for a bond portfolio. It is the change in the present value of the entire bond portfolio for a one basis point movement in interest rates. Typically, a limit is set for the total PVBP effect, given parallel movements in interest rates of 100 bps, 200 bps, and 300 bps.

7.18 Time Path of a Bond

Over time, holding all other factors constant, a bond's price must converge to its par value as the maturity date approaches because the bond will be redeemed at par. Therefore, over time, the price of a discount bond will increase, and the price of a premium bond will decline, holding all other factors constant. For a bond selling at par, its price will remain at par as the bond moves toward the maturity date.

7.19 Yield Measures

The most common measures of bond yield quoted by bond dealers are the current yield and yield-to-maturity.

7.19.1 Current Yield

The current yield is simply the ratio of the interest payment to the current price of the security. Current yield measures the rate of return from the annual interest payment of the bond and is expressed as a percentage of its market price.

$$\textbf{Current Yield} = \frac{\textbf{Annual Interest Payment}}{\textbf{Bond Price}}$$

For example, a 10-year, 6% coupon bond with a par value of $1,000 and a price of 115% would provide a current yield of 5.22%.

$$\text{Current Yield} = \frac{\$60}{\$1,150} = 5.22\%$$

The current yield considers only the annual interest payment and does not account for any potential capital gain or loss that the investor may realize in a year.

7.19.2 Yield-to-Maturity (YTM)

YTM is a measure of the rate of return that will be earned on a bond if it is purchased at a given price and held until maturity. Thus, the YTM is the discount rate that will make the present value of the bond's cash flows equal to the current market value of the bond.

Calculation of Yield-to-Maturity of a Bond

Given a price of $1148.80, determine the YTM of a 10-year bond with a par value of $1,000 and a coupon rate of 8%. Interest payment is on a semi-annual basis.

The YTM (k) for the bond can be determined by solving the following equation:

$$P = \frac{C_1}{1 + k/2} + \frac{C_2}{(1 + k/2)^2} + \frac{C_3}{(1 + k/2)^3} + \ldots + \frac{C_{2n}}{(1 + k/2)^{2n}} + \frac{M}{(1 + k/2)^{2n}}$$

$$1{,}148.80 = \frac{40}{1 + k/2} + \frac{40}{(1 + k/2)^2} + \frac{40}{(1 + k/2)^3} + \ldots + \frac{40}{(1 + k/2)^{20}} + \frac{1{,}000}{(1 + k/2)^{20}}$$

The YTM is determined using a trial-and-error process that equates the left-hand side of the equation with the right-hand side. In practice, a financial calculator can be used to determine the YTM.

Input the following in the financial calculator:

N = 20
PV = -1,148.80
PMT = 40
FV = 1000

Press CPT followed by I/Y to get a semi-annual rate of 3.00%.

The market convention is to annualise the semi-annual yield by doubling its value. The YTM computed on the basis of this market convention, 6%, is also called the bond-equivalent yield.

Market convention aside, the proper way to annualise the semi-annual yield is to apply the following formula:

$$\textbf{Effective annual yield} = (1 + \text{Semi-annual Interest Rate})^2 - 1$$

Therefore, the effective annual yield is $(1.03)^2 - 1 = 6.09\%$

An approximation formula exists to calculate the YTM. This formula relates the net annual effective cash flow to the average amount of money invested in the bond during the holding period. Effectively, the investor is assumed to have an average investment of this amount in the bond as the price gradually converges to $1,000, as it must by the maturity date.

$$\textbf{Approximate YTM} = \frac{\text{Annual Interest Payment} \pm \text{Amortised Gain or Loss}}{(\text{Current Market Price} + \text{Par Value})/2}$$

Consider the bond above in the example of "Calculation of Yield-to-Maturity of a Bond". Since the current bond price is $1,148.80 and it is expected to decrease to $1,000 over 10 years, the amortised loss per year is (1,148.80 – 1000)/10 = $148.80/10 and the:

$$\text{Approximate YTM} = \frac{80 - 148.80/10}{(1{,}148.80 + 1{,}000)/2}$$

= 65.12/1,074.40

= 6.06%

It is easier to compute the YTM for a zero-coupon bond. To find the YTM, we substitute zero for the coupon payments into the present value model and solve for k.

$$\frac{k}{2} = \left[\frac{\text{Maturity Value}}{\text{Price}}\right]^{1/2n} - 1$$

For example, for a 15-year zero coupon bond with a maturity value of $1,000, selling for $300:

$$\frac{k}{2} = \left[\frac{\$1{,}000}{\$300}\right]^{1/30} - 1$$

= 0.0409 or 4.1%

Note that the number of periods is equal to 30 semi-annual periods, which is double the number of years. The bond-equivalent yield is 8.2%.

The YTM is a more meaningful measure of return compared to the current yield as it considers both coupon income and any capital gain or loss, which the investor will realise when holding the bond to maturity. Also, the timing of the cash flows is also accounted for in the YTM.

The investor will only realise the YTM stated at the time of purchase if:

- the coupon payments can be consistently reinvested at the YTM; and
- the bond is held to maturity.

With regard to the first assumption, the risk that an investor faces is that future reinvestment rates may be different from the YTM at the time the bond is purchased. This is the reinvestment risk, which was covered earlier in this Chapter. If the bond is not held to maturity, the bond may have to be sold at a price that is different from par value and at a timing that is different from the maturity date, resulting in a return that is different from the YTM. The risk that a bond will have to be sold at a loss because yields have risen (resulting in a lower bond price), is referred to as interest rate risk or price risk.

7.19.3 Reinvestment Risk

The maturity and coupon rate of a bond affect the degree of reinvestment risk. First, for a given coupon rate, the longer the maturity, the more the bond's total dollar return is affected by the compounding process to realise the YTM at the time of purchase. In other words, the longer the maturity, the greater the reinvestment risk.

Second, for a given maturity, the higher the coupon rate, the more dependent the bond's total dollar return will be on the reinvestment of the coupon payments in order to produce the YTM expected at the time of purchase. In other words, higher coupon rate bonds carry higher reinvestment risk because there is more to reinvest. The converse is also true. So, a zero-coupon bond has no reinvestment risk if it is held to maturity because there is nothing to reinvest.

7.19.4 Taxable Equivalent Yield (TEY)

Some bonds are exempted from tax. Accordingly, the stated rate on these bonds will be lower than that on comparable non-exempt bonds. To make the return on these bonds comparable to those of taxable bonds, the TEY can be calculated. The formula of TEY is:

$$\textbf{Taxable Equivalent Yield} = \frac{\textbf{Tax-Exempt Yield}}{\textbf{1 - Marginal Tax Rate}}$$

An investor in the 28% marginal tax bracket who invests in a 5-year a tax-exempt bond with a YTM of 7% would have a TEY of:

$$\frac{0.07}{1 - 0.28} = 9.7\%$$

7.20 Term Structure of Interest Rates

The relationship between yield-to-maturity and term to maturity is known as the maturity structure or term structure of interest rates. It is also referred to as the yield curve.

7.20.1 Types of Yield Curves

The rising yield curve is the most common. It is observed when yields on short-term issues are low and rise consistently with longer maturities. A rising yield curve is a signal that short-term interest rates are expected to rise in the future i.e., the entire yield curve will move upwards.

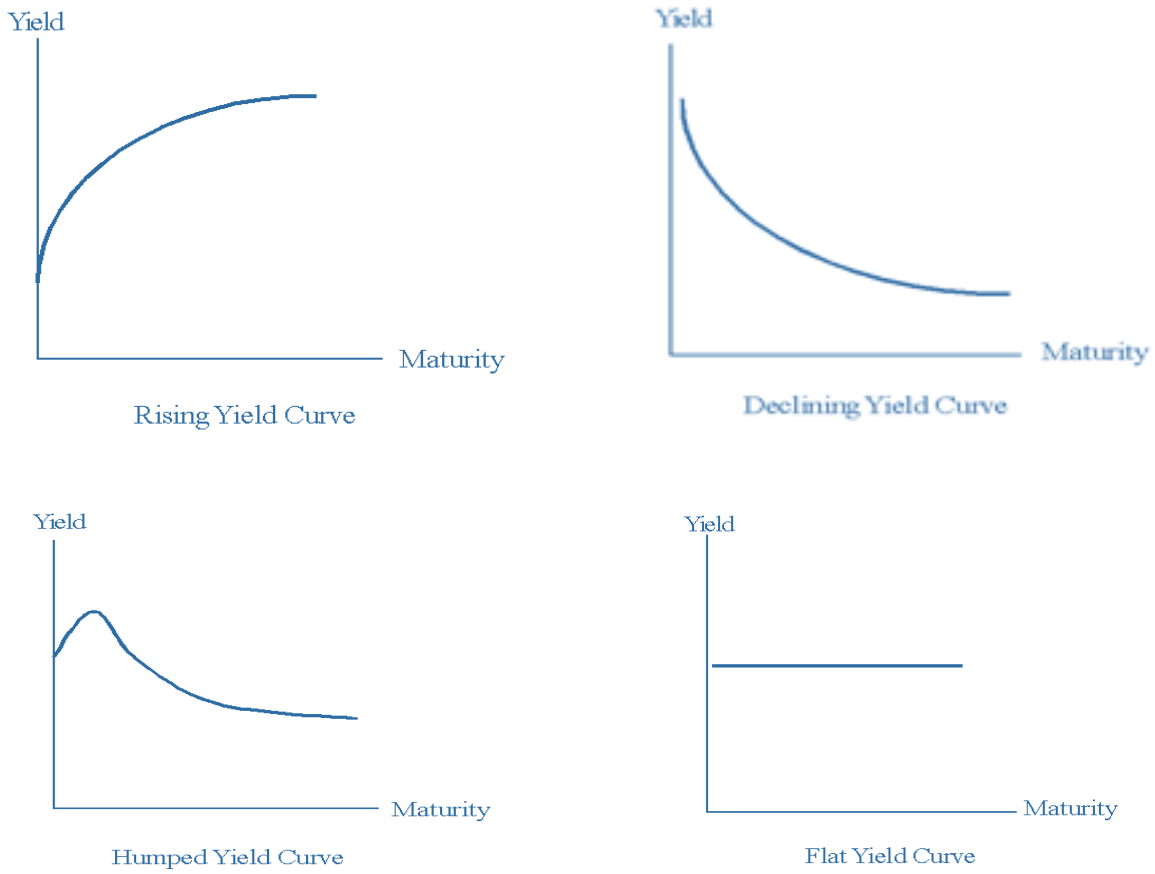
The declining yield curve is seen when the yields on short-term issues are higher than the yields on longer-maturity issues. This generally indicates lower short-term interest rates in the future.

A humped yield curve is formed when yields on intermediate-term issues are higher than yields on short-term issues, and rates of long-term issues decline to levels below those of the short-term issues before levelling out.

When the yields on short-term and long-term issues are the same, this will result in a flat yield curve.

Please see Figure 7.20.1 that shows the four different types of yield curves – Rising, Declining, Humped and Flat.

Figure 7.20.1 - Types of Yield Curves



7.20.2 Theories of Interest Rates

Three major theories of interest rates are used to explain the different types of yield curves. They are the expectations theory, liquidity preference theory and market segmentation theory.

a. **Expectations Theory**

The expectations theory asserts that the shape of the yield curve is determined by interest rate expectations of investors. When the market expects interest rates to rise, lenders will prefer to lend for shorter periods, forcing short-term interest rates down. Conversely, borrowers will prefer to borrow for longer periods, forcing long-term interest rates up. These opposing pressures create an upward-sloping yield curve.

On the other hand, if the market expects interest rates to fall, lenders will prefer to lend at the long end of the curve (forcing interest rates down) and borrowers will prefer to borrow at the short end (forcing interest rates up), resulting in a downward-sloping yield curve.

When the market is ambivalent about the direction of interest rates, the yield curve will tend to be flatter.

Suppose the interest rate on a 1-year T-bill is 6%. After one year, the yield on a new 1-year T-bill is expected to rise to 8%. There are two options for an investor with a 2-year time frame today.

He can invest in one-year T-bills on a rolling basis, which will provide him an average yield of 7%:

Average annual rate of return= $[(1.06)(1.08)]^{1/2} - 1 = 0.07$

Alternatively, he can invest in a 2-year security. He will require a yield of 7%, the average of the 6% he should realise in the first year and the 8% expected in the second year.

b. **Liquidity Preference Theory**

The theory of liquidity preference argues that the shape of the yield curve should always be upward sloping and that any other shape would be a temporary aberration. Because of the greater price volatility of long-maturity securities (compared to shorter-maturity ones), investors will require a higher return to hold long-term securities relative to short-term securities. Since short-term securities are more easily converted into cash without the risk of large price movements, investors are willing to pay a higher price for short-term securities and thus receive a lower yield. From a lender's perspective, higher yields will be required to induce a lender to lend long-term since liquidity is sacrificed for a longer period of time. This again justifies an upward-sloping yield curve.

c. **Market Segmentation Theory**

The market segmentation theory presumes that investors have their own maturity preferences, which are dictated by the nature of their liabilities. As an example, life insurance companies with long-term liabilities prefer long-term bonds while banks tend to prefer short-term liquid securities to match the nature of their deposits. The yield for each maturity is determined solely by the supply and demand in each "segment". The theory holds that activities of long-term borrowers and lenders determine rates on long-term bonds. Similarly, short-term trades set short-term rates.

7.21 Analysis of Convertible Bonds

A convertible bond is a fixed-income security that gives the holder the right to convert the bond into a predetermined number of ordinary shares of the issuer within a specified time period. Exchangeable bonds grant the bondholder the right to exchange the bonds for the ordinary shares of a firm other than the issuer of the bond. A major reason for adding a convertible feature to the debt security is to increase the attractiveness of the issue, thereby ensuring a market for it.

Convertible bonds are issued by companies as part of their fundraising activities. A convertible feature is attached to the bond as a sweetener, thereby allowing the company to issue the bonds at lower coupon rates than comparable straight bonds, thus lowering the company's financing costs.

Analysis of a Convertible Bond

The following hypothetical bond, Genesis convertible bond, can be used to illustrate the analysis of a convertible bond:

Maturity	= 10 years
Coupon rate	= 4%, paid semi-annually
Conversion ratio	= 40 (i.e., 1 bond can be converted to 40 common shares)
Par value	= $1,000

Market price of Genesis convertible bond = $1,100

Market price of Genesis common share = $24 per share

Annual dividend = $0.35 per share

7.21.1 Minimum Value of a Convertible Bond

A convertible bond can be considered as a "double" security, consisting of a straight bond and a call option to purchase common shares of the issuing company. Therefore, to determine the value of a convertible bond, it is necessary to consider both aspects of the bond. First, we want to consider the value of the convertible as a straight bond. Second, we want to consider the conversion value, the value of the shares when conversion takes place.

a. **Investment Value**

The bond value, also referred to as the straight value or investment value, is the value of the convertible bond without the conversion feature. It is calculated based on the cash flows of the convertible as if there is no conversion option. To determine the bond value, we need to know the market yield on a comparable straight bond. Given the market yield, we will discount the bond's cash flows, thus deriving the bond value.

Suppose the market yield on comparable straight bonds is 8%, the value of the Genesis convertible bond is $728.

$$\text{Bond Value} = \sum_{t=1}^{2n} \frac{C_t}{(1 + k/2)^t} + \frac{\text{Par Value}}{(1 + k/2)^{2n}} = \$728.19$$

b. **Conversion Value**

The conversion value of a convertible bond is the total market value of the ordinary shares that the bond can be exchanged for. For the Genesis convertible bond:

Conversion value = Conversion Ratio x Market Share Price
= 40 x $24
= $960

The minimum value of a convertible bond is the greater of its:

- Conversion value, or
- Investment value.

Given a conversion value of $960 and an investment value of $728, the minimum value of the Genesis convertible bond is $960. If the bond sells below $960, say at $728, arbitrageurs will buy the bond for $728, convert it, and sell the shares at $24 per share. This would result in a profit of $232 per bond. If the investment value is higher than the conversion value, the convertible bond will trade at its investment value. If it trades below its investment value, the yield would be greater than those of comparable straight bonds. Investors would buy the convertible bond, thus bidding up the price until the yields are equal.

7.21.2 Conversion Ratio and Conversion Price

The conversion ratio represents the number of ordinary shares for which the convertible bond (CB) can be exchanged.

$$\textbf{Conversion Ratio} = \frac{\textbf{Par Value of CB}}{\textbf{Conversion Price}}$$

In the case of the Genesis convertible bond, the conversion ratio is 40, which means that a $1,000 par value convertible bond is exchangeable for 40 ordinary shares.

Given the conversion ratio, we can derive the conversion price. The conversion price for the Genesis convertible bond is:

$$\text{Conversion Price} = \frac{\text{Par Value of CB}}{\text{Conversion Ratio}}$$

$$= \frac{\$1{,}000}{40}$$

$$= \$25$$

7.21.3 Conversion Parity Price

The conversion parity price is what an investor effectively pays for an ordinary share if the convertible bond is purchased and then converted into ordinary shares. The price is calculated as follows:

$$\textbf{Conversion Parity Price} = \frac{\textbf{Market Price of CB}}{\textbf{Conversion Ratio}}$$

For the Genesis convertible bond, the conversion parity price is:

$$\text{Conversion Parity Price} = \frac{\$1{,}100}{40}$$

$$= \$27.50$$

If the investor purchases the Genesis convertible bond and converts it to ordinary shares, he is effectively paying $27.50 per share. The conversion parity price can be considered as a share price breakeven point. Any further increase in the share price beyond this point will result in the same percentage increase in the value of the convertible bond.

7.21.4 Conversion Premium

The investor who purchases the Genesis convertible bond instead of the underlying share is paying a premium of $3.50 ($27.50 - $24) or 14.6% over the current share price. This is the conversion premium, which is the difference between the convertible's market price and its conversion value. It can be expressed as an absolute value or as a percentage.

For the Genesis convertible bond, the conversion premium is calculated as follows:

Conversion premium ($) = Market Value of CB – Conversion Value = $1,100 - $960 = $140

$$\text{Conversion Premium (\%)} = \frac{\text{Market Value of CB - Conversion Value}}{\text{Conversion Value}}$$

$$= \frac{\$140}{960}$$

$$= 14.6\%$$

Normally, a convertible bond trades at a premium to its straight value as well. This is because the convertible bond has an embedded option equivalent to a call option on an ordinary share at a strike price of $25 (i.e., 1 bond of $1,000 can be converted to 40 shares). The price differential between the convertible bond and its straight value is therefore effectively, the value of the embedded call option. An investor is willing to paying the higher price for the convertible because of the potential for significant capital gains when the share price appreciates. At the same time, his downside risk is limited to the straight value of the bond.

7.21.5 Breakeven Time

Convertible bonds are often issued with a current yield that is higher than the dividend yield one would receive from converting the bond to the ordinary shares. Investors compare the attractiveness of convertible bonds by looking at how fast it takes to recover the conversion premium from the income advantage of holding the convertible bond.

By comparing the conversion premium and the income differential, we can calculate the breakeven time or payback period, which is:

$$\text{Breakeven Time/Payback} = \frac{\text{Conversion Premium}}{\text{Bond Income - Dividend Income}}$$

For the Genesis convertible bond, the conversion premium is $140.

Bond income = 0.04 x $1,000 = $40

Dividend income = Conversion Ratio x Dividend per Share
= 40 x $0.35 = $14

Differential between bond income and dividend income
= $40 - $14 = $26

Breakeven time = $140/$26 = 5.38 years

The breakeven time does not consider the time value of money. However, it is a useful indicator of the relative attractiveness of a convertible bond.

7.21.6 Downside Risk

The investment value is usually used as a measure of the floor price of a convertible bond. To measure the downside risk, we compare the market price of the convertible bond with the investment value as follows:

$$\text{Premium Over Straight Value (\%)} = \frac{\text{Market Price of CB}}{\text{Investment Value}} - 1$$

The downside risk for the Genesis convertible bond is:

Premium over straight value = $1,100/$728 - 1 = 51%

Despite its use in practice, this measure of downside risk is flawed because the straight value changes as the discount rate changes.

7.21.7 Investing in Convertible Bonds

Convertible bonds offer investors a unique combination of an income stream from a fixed-income security and an opportunity to participate in the capital appreciation of the underlying ordinary share. They offer upside potential and downside protection. The yield on a convertible bond is usually higher than the dividend yield of the underlying ordinary share but lower than the yield of a comparable 'plain vanilla' bond.

When the share price rises above its conversion “strike” price, the convertible bond price will mainly be influenced by that of the underlying share. If the share price for the convertible bond in the example above rises to $50, the bond will have a minimum value of $2,000, and the investor will make a clear profit from conversion. When the convertible is trading at a reasonable conversion premium, its value will be influenced by interest rates, the price of the underlying share, and the share’s price volatility. When the convertible is trading near the investment value, the embedded option is almost worthless because the underlying share price is substantially below the conversion “strike” price; in this scenario, the convertible trades like a straight bond.

7.22 Summary

1. For individuals, income and monies received can be categorised in one of four ways – (1) consumption, (2) business reinvestment, (3) reserves set aside for family needs, life’s risks, and unfortunate eventualities, and (4) excess cash.

2. All full-banks and finance companies in Singapore are members of the Deposit Insurance (DI) Scheme, except for those exempted by the MAS. The DI Scheme protects depositors in the event that a DI Scheme member fails. Effective April 2024, the coverage limit will be increased to S$100,000 and the scope of coverage includes individuals, charities, sole proprietorships, partnerships, companies, and other unincorporated entities, such as associations and societies.

3. Money market instruments are debt instruments issued by governments, financial institutions, and corporations, with maturities of less than one year. Examples of money market instruments are **Treasury bills (T-bills), negotiable certificates of deposit (CDs)** or **banker’s acceptances, commercial paper, repurchase agreements,** and **bills of exchange**.

4. A fixed-income security is a debt instrument that provides periodic payments at an interest rate known in advance by the holder with eventual return of the principal at maturity to the holder.

5. A bond is a fixed-income security on which the issuer makes regular interest payments to the bondholder on specified dates and repays the principal to the bondholder on the maturity date.

6. Essentially, the bond market can be divided into the internal bond market and the external bond market.

7. In the context of the Singapore bond market, the domestic bond market comprises Singapore Government Securities and corporate bonds issued by Singapore companies. The foreign bond market refers to foreign issuers who have issued their bonds in the Singapore bond market.

8. The Eurobond market is divided into different sectors based on the currency in which the issue is denominated. Eurobonds can be denominated in any currency, with the most popular issues being USD denominated, called Eurodollar bonds.

9. The secondary market also serves an important market function of benchmarking the market values and yields of the traded bonds.

10. Special bond types include callable bonds, puttable bonds, convertible bonds, bonds with sinking-fund provisions, zero-coupon bonds, floating rate bonds, deferred-coupon bonds, and perpetual bonds.

11. Convertible bonds are advantageous to the issuer because they allow the issuer to finance at a lower rate and defer potential dilution of earnings.

12. Bonds with a quality rating below triple B (BBB) are called high-yield bonds or junk bonds.

13. Asset-backed securities (ABS) may be secured against auto loans and leases, consumer loans, commercial assets, and credit card receivables.

14. A collateralised debt obligation (CDO) is structured by an investment manager or a bank, which pools together cash flow-generating assets that are then repackaged and sold to investors.

15. The credit risk of bonds is rated primarily by Standards & Poor's Global, Fitch Ratings, and Moody's Investors Service. Ratings range from AAA (the highest) to D (bonds in default).

16. Deferred-interest bonds, step-up bonds, and payment-in-kind bonds have deferred-coupon structures.

17. The credit spread is a direct measure of the cost of risk, or the extra yield (or rate of return) investors demand to take on a bond with a certain level of risk of defaulting. As the issuer's default risk increases, its bond price falls, and its bond spread widens.

18. A credit default swap (CDS) is a contract that guarantees against bond defaults. It is designed to transfer credit exposure. The buyer of a CDS pays a premium to be insured against a debt default and receives a lump-sum payment in the event of a debt default. The seller of a CDS receives regular payments from the buyer. In the event of a debt default, the seller has to pay the insured amount to the buyer.

19. Apart from financial ratios, investors should also rely on other key variables such as swings in projected revenues, costs, and cashflows, as well as regulatory policies and changes to the company's key management personnel for an informed investment opinion of a firm's debt.

20. Bonds are most vulnerable to credit and interest rate risk. Other sources of risk include reinvestment risk, call risk, inflation risk, liquidity risk, currency risk, volatility risk, political risk, event risk, and sector risk.

21. The value of a bond is the present value of its future cash flows. For a coupon bond, the cash flows comprise periodic interest payments and the principal. For a zero-coupon bond, the value is equal to the present value of its principal.

22. There is an inverse relationship between bond prices and the required market yield.

23. Interest rate risk and reinvestment risk make opposite predictions about the impact of interest rate movements on bond price.

24. When the coupon rate is equal to the yield-to-maturity (YTM), the bond will trade at its par value. When the coupon rate is less than the YTM, the bond will trade at a discount. When the coupon rate is greater than the YTM, the bond will trade at a premium.

25. The duration of a bond is basically affected by its coupon rate, yield, and remaining time to maturity. It is a linear measure of the approximate price sensitivity of a bond to small interest rate changes, i.e., not more than 1% or 100 basis points.

26. Positive convexity means that, for large changes in interest rates, the amount of price appreciation will be greater than the amount of price depreciation.

27. The present value of a basis point (PVBP or PV01) is used as a risk management tool for a bond portfolio. It is the change in the present value of the entire bond portfolio for a one basis point movement in interest rates.

28. Over time, holding all other factors constant, the price of a bond will converge to its par value on the maturity date.

29. There are two conventional bond yield measures: the current yield and YTM. The current yield considers only the annual interest payment but fails to account for capital gain or loss. The YTM is a measure of the average rate of return that will be earned on a bond if it is purchased at a given price and held until maturity.

30. The relationship between yield and maturity is known as the term structure of interest rates. There are four types of term structures or yield curves: the rising yield curve, the declining yield curve, the humped yield curve, and the flat yield curve.

31. There are three major theories of interest rates to explain the shapes of the yield curve: the expectations theory, liquidity preference theory, and market segmentation theory.

32. A convertible bond is a fixed-income security that gives the holder the right to convert the bond into a predetermined number of the issuer's ordinary shares within a specified time period.

33. Analysis of a convertible bond requires calculation of the investment value, conversion value, conversion parity price, conversion premium, and breakeven time. The minimum price of a convertible bond is the greater of its conversion value or its investment value.

Chapter 8
Portfolio Management

Learning Objectives

The candidate should be able to:

Efficiency Market Hypothesis (EMH)

- ✓ State the different levels of market efficiency under EMH in the context of the current market environment.
- ✓ Explain how active and passive portfolio management can be applied by investors.

Portfolio Theory

- ✓ Appreciate how investment portfolios are constructed and managed.
- ✓ Explain the investment management principles behind diversification and portfolio risk.
- ✓ Distinguish between systematic and non-systematic risks, particularly in the hedging of each of these risks.
- ✓ Compute the correlation coefficient and using it to determine the extent of diversification of risks in a portfolio.
- ✓ Calculate portfolio return and risk.
- ✓ Show how portfolio risks can be diversified internationally.
- ✓ Understand and explain the Capital Asset Pricing Model (CAPM) and the risk-free rate.
- ✓ Explain the Capital Market Line for the portfolio's risk-return trade off.
- ✓ Explain the Security Market Line for the portfolio's risk-return trade off.
- ✓ Use the Security Market Line to determine overvalued and undervalued securities.
- ✓ Explain the Arbitrage Pricing Theory, and how it compares with the CAPM.

Portfolio Management Process

- ✓ Describe and explain the portfolio management process.
- ✓ Explain how investment policies are established.
- ✓ List and explain the investment objectives in the investment policy statement.
- ✓ Describe the contents of a typical investment policy statement.

- ✓ Discuss and explain the investment constraints in portfolio management.
- ✓ Explain asset allocation and the difference between strategic and tactical asset allocation.

Performance Measurement

- ✓ Compute and measure investment returns across different cash flow assumptions and movements in the portfolio.
- ✓ Use Dollar-weighted rate of return to measure performance.
- ✓ Use Time-weighted rate of return to measure performance.
- ✓ Compare the advantages/disadvantages of Dollar-weighted versus Time-weighted rates of return.
- ✓ Describe the Modified Dietz Method, in assuming a constant rate of return on the portfolio over the measurement period.
- ✓ Appreciate the Global Investment Performance Standards for investment management performance.
- ✓ Explain the Risk-adjusted Measures of Performance such as the Sharpe Ratio, Treynor Ratio and Jensen's Alpha.
- ✓ Describe Performance Attribution as a means to measure a portfolio against an established benchmark.

8.1 Introduction

This Chapter is divided into four parts. The first part begins a description of the Efficient Market Hypothesis (EMH), an examination of the variants of the EMH and a review on the relevance of the hypothesis to portfolio management. Additionally, we will also discuss portfolio management strategies in light of the EMH.

The second part covers Harry Markowitz's **Portfolio Theory**[41], which forms the basis and foundation for portfolio management. We will discuss diversification and correlation, systematic and non-systematic risks, and William F. Sharpe's Capital Asset Pricing Model (CAPM)[42], which describes the relationship between risk and return by evaluating investments in the context of a portfolio.

The third part deals with the **portfolio management process**, which covers the construction of an investment portfolio. Specifically, we will examine the determinants of investment policy - the investor's objectives and constraints. In addition, we will also discuss strategic and tactical asset allocations for a portfolio.

The final part deals with the performance of the portfolio, covering how **portfolio returns** are measured and how to use risk-adjusted performance measures, such as the Sharpe Ratio, Treynor Ratio and Jensen's Alpha. Lastly, we look at performance attribution analysis and the various contributors to investment performance.

[41] "Portfolio Selection" by Harry Markowitz; The Journal of Finance, Vol. 7, No. 1. (March 1952).
[42] "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk" by William F. Sharpe; The Journal of Finance, Vol. 19, No. 3 (Sep 1964).

8.2 The Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH[43]) states that an efficient market is one where security prices always fully reflect available information. As mentioned in Chapter 3, the EMH assumes that market participants act collectively, with stock prices adjusting quickly, reacting to all known information about the firm. The direct implication is that stocks always trade at their fair values and it is impossible to outperform the overall market consistently on a risk-adjusted basis through expert stock selection or market timing. This is because, in an efficient market, all available information is already factored into the stock prices and market participants cannot earn a risk-adjusted excess return.

After more than half a century of studies and research on the EMH, its validity is still inconclusive. Possibly, this is because there are varying degrees of market efficiency, which allows for a minority who have an advantage to exploit various pockets of informational inefficiency to make abnormal gains over certain periods.

There are three forms of the EMH: Weak, Semi-strong and Strong. They are further elaborated upon in the following sections.

8.2.1 Weak Form of the EMH

Under the weak form of the EMH, investors cannot consistently make abnormal profits using historical price and volume information. As markets are efficient, all past price information would be fully reflected in the current stock price. Future stock prices cannot be predicted by analysing past prices and, thus, will follow a random walk (unpredictable) pattern without exhibiting any serial dependency on past information. This implies that investors will not be able to consistently achieve excess returns using technical analysis.

However, some financial market participants do not behave as if the markets are weak form efficient. Technical analysis remains used by market participants to detect unusual volume increases and price trends that signal buying or selling interest and to determine whether to buy or sell with the momentum and direction of the price.

Extensive application of computer-based high-frequency trading (HFT) seeking to exploit price trends and momentum is again observed behaviour that is inconsistent with the weak form of the EMH. Financial institutions operating alternative trading systems such as dark pools (refer to Chapter 3 for more information) obtain usage of confidential client trading information which could provide an unfair advantage, as exemplified by enforcement action taken by the US Securities and Exchange Commission (SEC) against two major banks that supplied dark pool trading information to external high-frequency traders[44].

In spite of the behaviour of these market participants, the empirical evidence indicates that technical analysis generally does not yield abnormal returns and is supportive of the weak form of the EMH.

[43] Harry Roberts (Roberts, 1967) coined the term efficient markets hypothesis and made the distinction between weak and strong form tests, which became the classic taxonomy in Fama (1970).
[41] US Securities and Exchange Commission Press Release: "Barclays, Credit Suisse Charged with Dark Pool Violations Firms. Collectively paying more than $150m to settle cases". https://www.sec.gov/news/pressrelease/2016-16.html.

8.2.2 Semi-Strong Form of the EMH

Under the semi-strong form of the EMH, all publicly available information would also be fully reflected in the current stock price. In addition, prices would react rapidly to new public information (such as earnings announcements, management changes, etc.). This implies that neither fundamental analysis nor technical analysis will allow investors to consistently beat the market.

However, as is the case for weak-form efficiency, many financial market participants do not behave as if the markets are semi-strong form efficient. There are many anecdotes of value-oriented investors that have a consistently successful track record although it has been argued that this can occur by pure chance given the sheer number of investors and that the observed excess returns may simply be compensation for an unidentified risk.

8.2.3 Strong Form of the EMH

Under the strong form of the EMH, all available information, be it public or private, is fully reflected in the current stock price. This implies that investors will not be able to make consistent abnormal profits on any type of information (including insider or private information). As such, even insider trading would not give investors any advantage.

The evidence suggests real-world financial markets are not strong form efficient. The history of insider trading in the US and numerous high-profile cases in recent years continue to suggest that having inside information does provide an information advantage to generate abnormal gains.

Hence, there is a need for regulatory oversight provided by the US SEC and equivalent bodies worldwide. In less developed financial markets with fewer enforcement and monitoring capabilities, the potential advantage that inside information confers could be even higher.

8.2.4 Interpreting and Applying the EMH

Supporters of market efficiency highlight that the average fund manager does not outperform the market. Despite all the expertise and capability most fund managers claim to have, the majority underperform their respective benchmarks even if fees are excluded from consideration. Even for managers that outperform their benchmark, the outperformance does not generally persist.

Thus, markets are probably quite efficient, making it difficult to consistently beat the benchmark. However, there may be periods of inefficiency in which irrational behaviour and uneven distribution of relevant price-sensitive information can allow investment professionals that are more astute to outperform the market.

For those who believe in efficient markets, it makes sense to adopt a passive investment management strategy by looking for an efficient low-cost fund that aims to replicate the performance of a benchmark index.

For those who believe that markets can be inefficient, they can:

a. try to outperform the market by trading actively;

b. adopt a longer-term investment approach by selecting and holding investments with strong fundamentals; or

c. identify and choose an active portfolio manager to manage their investments.

8.3 Passive versus Active Management Strategies

Institutional and corporate investors can choose between passive or active management investment strategies. For individuals, the choice of strategy adopted will depend on a few factors in addition to their belief in the efficiency of the market.

Additional key considerations in choosing a strategy include:

a. Funds available for investing (size of investment);

b. Desire for personal involvement in trading or investment;

c. Amount of time available to manage investments;

d. Investment expertise; and

e. Market structure (please refer to Chapter 3, section 3.3.5 for more details).

8.3.1 Passive Strategy

As covered earlier, the EMH implies that it is not possible to outperform the market through active share selection and analysis. Hence, for investors who believe in efficient markets, it does not make sense to trade frequently as this generates large brokerage commissions without enhancing expected performance. Passive management, such as a **buy-and-hold** strategy, is therefore recommended for these investors. Passive investors believe that such a strategy will, over the long term, produce results that are just as good as active management strategies because of market efficiency.

Alternatively, such investors can choose to invest in an **equity index fund** or portfolio. Index funds are designed to track the performance of a selected market index. A manager of an equity index portfolio is judged on how well he tracks the target index. Expenses, including research costs, management fees, and brokerage commissions, are kept to a minimum.

On a practical basis, investors who:

a. are only able to invest in small amounts,

b. lack the time and / or expertise to monitor and manage their investments, and

c. do not wish to pay fees,

should probably adopt a passive investment approach and rely on index funds to gain some diversification in their portfolio.

8.3.2 Active Strategy

The existence of numerous asset management firms suggests that most investors do not subscribe to the EMH. Instead, they favour an active approach to equity investing.

The use of valuation models and the analysis of the economy, industries, and individual companies are all elements of active management. The objective of active management is to outperform the market on a risk-adjusted basis.

Active investors believe they have some advantages over other market participants. Such advantages could include superior analytical skills or access to information that is not available to other investors. Security selection, sector rotation and market timing are examples of active strategies.

a. **Security Selection**

Value-oriented share specialists believe they can outperform the market by selecting securities that they believe are undervalued. Such shares are typically selected using fundamental analysis of securities and valuation models such as the dividend discount or price earning (P/E) ratio model. These specialists tend to look for low P/E shares or shares that are trading below their intrinsic values.

b. **Sector Rotation**

This strategy involves shifting sector weights in the portfolio to take advantage of sectors that are expected to outperform while avoiding or underweighting sectors that are expected to underperform the market average.

Under the Industry Classification Benchmarks, the market is classified into 10 industries, which are further sub-divided into super sectors and sectors. A sector is the smallest subset of an industry. It is quite common in sector analysis (i.e., industry analysis) to categorise ordinary shares into broad sectors, such as interest-sensitive shares, cyclical shares, and defensive shares.

Each of these sectors is expected to perform differently during the various phases of the business cycle:

i. **Interest-sensitive shares** are expected to underperform during periods of high interest rates.

ii. **Cyclical shares** are expected to do well in an economic upturn and badly in an economic downturn.

iii. **Defensive shares** are not hurt as badly during the downturn of a business cycle.

Effective strategies involving sector rotation depend heavily on accurate assessment of current economic conditions.

c. **Market Timing**

When equities are expected to do well, market timers try to achieve superior performance by shifting their asset holdings from cash-equivalents to equities. Conversely, when equities are expected to do poorly, the opposite occurs and equity holdings are liquidated. Market timing strategies, thus, involve active asset allocation.

8.4 Portfolio Theory

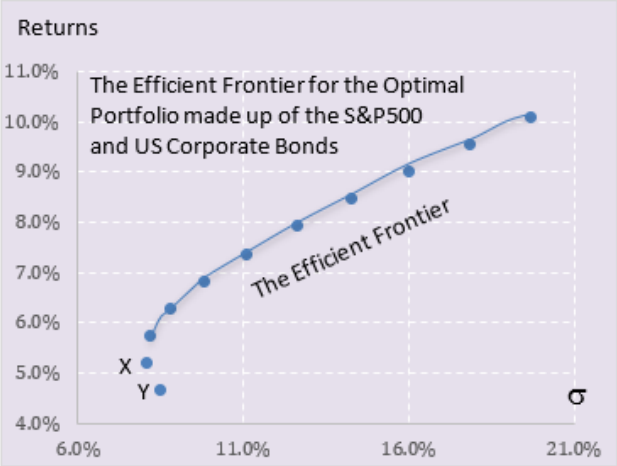
8.4.1 Introduction to Portfolio Theory

Portfolio theory discusses how investors can create **optimal portfolios** that maximize expected returns while minimizing risks. To illustrate this, we have created a simple optimal portfolio based on two asset classes – Equities on the Standards & Poor's (S&P) 500 and US Treasury Bonds.

Looking at returns and standard deviation (our measure of volatility) over the last 90 years, the portfolio returns for different percentage weights of equity and bonds were plotted against the calculated portfolio standard deviation.

Although correlation between stocks and bonds over the last decade was mostly negative (averaging around ρ_{EB} = - 0.20), since August 2017, correlation has turned positive again. Hence, for calculation purposes in the optimal portfolio example set out in Figure 8.4.1(a), a positive correlation value of ρ_{EB} = 0.1 is applied.

Figure 8.4.1(a) Creating the Efficient Frontier



Return R_E =	10.1%	σ_E =	19.7%
Return R_B =	4.7%	σ_B =	8.5%
ρ_{EB}	0.1		

Weight in Asset E	Weight in Asset B	Volatility of Portfolio EB	Return for Portfolio EB
0.0%	100%	8.5%	4.7%
10.0%	90%	8.1%	5.2%
20.0%	80.0%	8.2%	5.7%
30.0%	70.0%	8.8%	6.3%
40.0%	60.0%	9.8%	6.8%
50.0%	50.0%	11.1%	7.4%
60.0%	40.0%	12.6%	7.9%
70.0%	30.0%	14.3%	8.5%
80.0%	20.0%	16.0%	9.0%
90.0%	10.0%	17.8%	9.5%
100.0%	0.0%	19.7%	10.1%

The Efficient Frontier represents all possible combinations of the two asset classes (equity and bonds) that would generate the highest return at a given level of risk. As such, all the different combinations of equities and bonds that lie on the Efficient Frontier represent optimal portfolios, but for clients with different risk profiles.

For example, as equities have higher returns and volatility (i.e., risk), portfolios with higher weights on equities would be optimal for clients with higher risk appetites, with the highest-risk portfolio being a 100% equity-weighted portfolio. Likewise, portfolios with higher weights on bonds would be optimal for clients with lower risk appetites.

It is important to understand that investors should not be picking portfolios such as Portfolio X (5.2% return, 8.1% volatility) and Portfolio Y (4.7% return, 8.5% volatility) that lie below the Efficient Frontier because they are not efficient portfolios. This means that, at the specified level of risk (8.1% for Portfolio X and 8.5% for

Portfolio Y), there are better portfolio choices that can provide higher returns for the same amount of risk taken.

Most banks offering diversified portfolios will present the potential optimal allocations diagrammatically on the efficient frontier. Lower risk (and return) portfolios will be at one end, while higher risk (and return) portfolios will be at the other end. The frontier is a curve instead of a straight line because the returns of the two assets are not perfectly correlated.

In practical application, if the addition of alternative investment (e.g., hedge funds, derivatives, private equity, commodity funds, and real estate funds) can either:

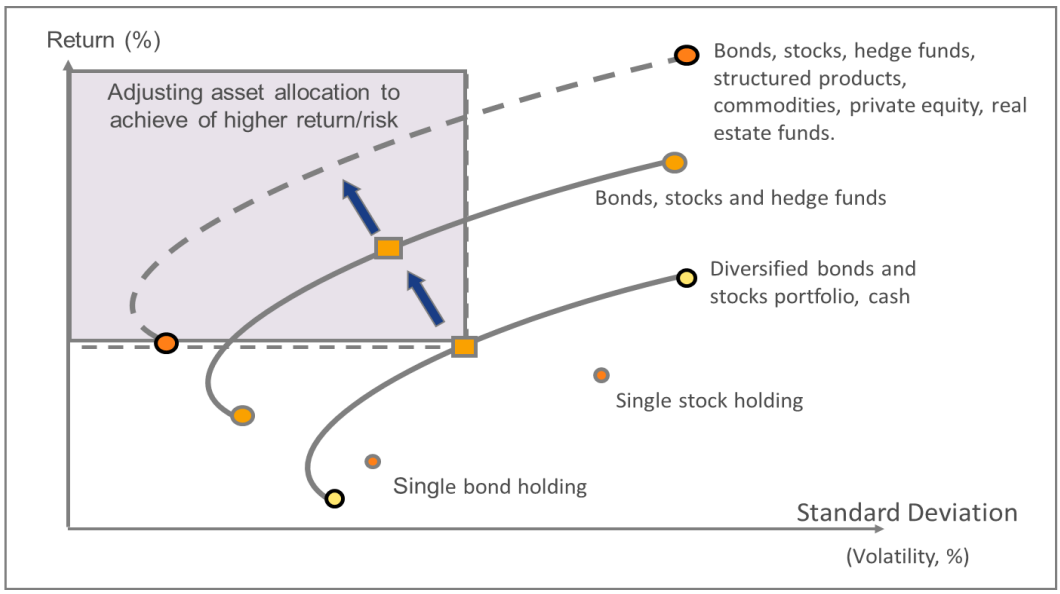
a. add to the return per unit risk, or

b. lower the risk per unit return,

then they are beneficial in further optimizing the portfolio. This would move the efficient frontier towards the upper left corner of the graph. Conceptually, this means raising the return per unit risk or lowering the risk per unit return.

Figure 8.4.1(b) illustrates the concept of optimizing the efficient frontier by adding less-correlated assets into the portfolio.

Figure 8.4.1(b) Optimizing the Efficient Frontier by Adding Less-Correlated Assets



However, simply adding new assets to the portfolio will not necessarily increase return per unit risk. There are some assets that will increase portfolio risk levels without adequately improving portfolio returns.

As such, while the concept appears to work and most investment professionals selling alternative investments will show evidence of improved risk-adjusted returns, one should review the impact on expected portfolio volatility and returns associated with the proposed additions. By doing so, investors can evaluate and confirm whether these additions can indeed improve portfolio return per unit risk. Professional portfolio managers

are actively engaged in the process of maximizing the expected returns of the portfolio while minimizing the corresponding portfolio risks.

8.4.2 Portfolio Diversification

Diversification is the financial equivalent of the old saying "Don't put all your eggs in one basket." Studies have shown that by increasing the number of securities from different industries in a portfolio, portfolio risk can be reduced. Further diversification can be achieved by investing in different asset classes and selectively investing internationally.

Diversifying an investment portfolio across different asset classes is recommended as a better risk-adjusted return can be achieved. However, the diversification process should not be carried out in an ad-hoc or blanket fashion. For example, buying into assets with a lower return per unit risk and high correlation with an existing portfolio is unlikely to improve portfolio diversification. To identify assets that have the potential to generate higher return per unit risk requires a well-thought-out process of reviewing the markets, sectors, and companies to overweight or underweight, either as part of a long-term allocation or tactically during specific periods.

Example of Diversification Within an Asset Class

Consider an investor's portfolio that only consists of one security (e.g., Apple). This portfolio will face three broad levels of risk:

- Firstly, there is the risk arising from changes in the general economic environment, such as changes in business cycles, interest rates, and currency movements.
- Secondly, there are industry-specific risks, which are those related to the industry that Apple operates in (IT/consumer electronics), such as changes in consumer preferences or technological obsolescence. Such risks exist below the economic level.
- Thirdly, there are firm-specific risks, which are those that relate only to Apple. These include risks arising from new product launches, research and marketing successes on new products, and Apple's management. These risks are unique to the company.

Now, consider the most basic level of diversification: Adding another stock (e.g., Monsanto, a company that produces genetically modified grains) into the portfolio. What will happen to portfolio risk then?

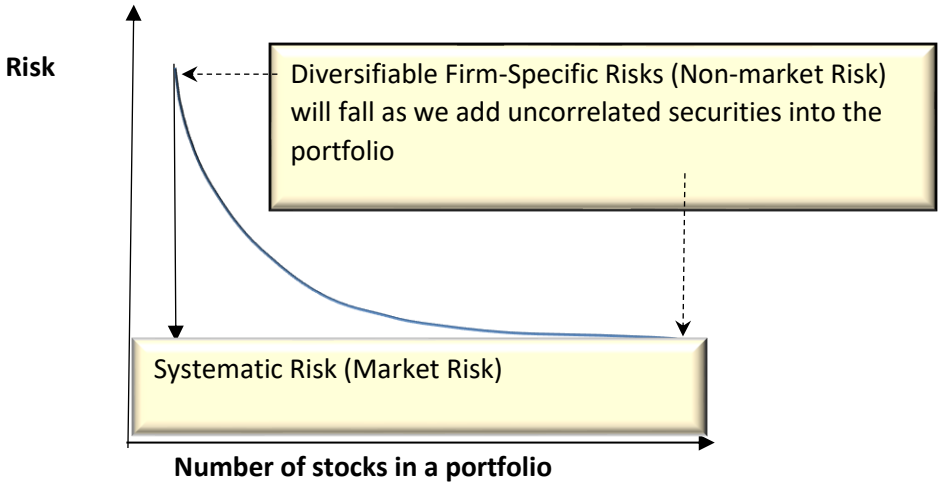
To the extent that firm-specific influences on the two shares differ, portfolio risk will be reduced. It does not matter whether factors affecting Monsanto are diametrically opposed to factors influencing Apple. All that matters is that there is **less than perfect positive correlation** between the factors driving Monsanto's price movement and Apple's price movement. As long as correlation is not 1, the addition of Monsanto will have the potential to diversify risk in a portfolio that previously consisted solely of Apple stock.

As more securities are added to the portfolio, the effect of firm-specific factors on volatility is further limited. As such, portfolio volatility should continue to decline, but at a slower rate as more shares are added. Studies have shown that around half of portfolio standard deviation is eliminated by the time that there are 10

securities in the portfolio. Unfortunately, the benefits of diversification diminish (i.e., the marginal risk reduction is small) as more securities are added.

Theoretically, a portfolio that contains all the shares in a given market (i.e., a stock market index) will have completely diversified away all non-systematic risks, with only systematic risk remaining (risks common to, and that affects all, securities). For example, as all shares are affected by the business cycle to varying degrees, it is impossible to avoid exposure to business cycle risk no matter how diversified the portfolio is.

Figure 8.4.2 - Portfolio Risk as a Function of the Number of Shares in the Portfolio



The risk that remains even after extensive diversification is called **market risk** or **systematic risk**. This is the risk that is attributed to market-wide sources. In contrast, the risk that can be eliminated by diversification is called **firm-specific risk** or **non-systematic risk**. See Figure 8.4.2.

8.4.3 Systematic and Non-systematic Risks

As mentioned above, the total risk in a portfolio can be divided into two general types:

Total risk = Systematic risk + Non-systematic risk

Systematic risk (also known as undiversifiable risk) results from factors that affect all firms, such as interest rate risk and inflation risk. It is caused by factors such as changes in the economic, political, and sociological environments, which affect the prices of all marketable securities. As such, systematic risk cannot be eliminated by diversification.

Non-systematic risk (also known as firm-specific risk) can be mitigated through diversification. It is that portion of total risk that is peculiar to a firm or industry. Non-systematic risk arises from factors that impact a company's earnings potential such as company strategies, management errors, new product developments and innovations. As such, these factors cause non-systematic variability of returns in a firm. Since non-systematic risk is caused by factors that affect specific firms or industries, it can be minimized by investing in

a broad number of different firms or industries (i.e., diversification). Non-systematic risk can theoretically be reduced to zero by having a well-diversified portfolio containing shares that are uncorrelated with each other.

Some further points to note about systematic and non-systematic risks are:

a. Systematic risk cannot be reduced through diversification. However, depending on the level of exposure and asset class invested, investors can partly or almost fully offset systematic risk through hedging (e.g., the use of options and futures).

b. Non-systematic risk can be reduced by having a diversified portfolio as an individual asset faces risks peculiar to it. The larger the number of assets in the portfolio, the lower the overall level of asset-specific risk the investor is exposed to i.e., the greater the diversification of specific risks.

c. Systematic risk is reflected in the beta (β) of an investment portfolio. This is the sensitivity of the portfolio to market movement. β will be covered later in the Chapter.

d. The realized Total Risk of a portfolio is captured by the realized standard deviation of the portfolio. Total realized portfolio risk includes systematic risk, non-systematic risk, and an error term that captures the deviation between predicted and actual standard deviation.

8.4.4 Correlation Coefficient

The correlation coefficient ρ (pronounced as "rho") measures the extent to which the returns of any two securities move together. However, it is important to note that correlation only denotes association and not causation.

The value of ρ can range from +1.0 (indicating perfect positive correlation) to -1.0 (indicating perfect negative correlation). Perfect **positive correlation** implies that the returns have an exactly direct linear relation. This means that the two securities have identical return patterns. Conversely, perfect **negative correlation** implies that the returns have an exactly inverse linear relation; as such, when one security's return is high, the other security's return would be negative. If the value of ρ is zero, then the returns of the securities do not have a linear relation.

For efficient diversification, we should include securities with returns that are not highly correlated to each another. If returns of securities are highly correlated, then risk cannot be reduced significantly.

Specifically, it should be noted that:

a. Combining securities that are perfectly positively correlated with each other ($\rho = 1$) provides no reduction in portfolio risk. The risk of the resulting portfolio is simply a weighted average of the individual risks of the securities.

b. Combining two securities with zero correlation ($\rho = 0$) will bring about some reduction to the overall risk of the portfolio. If more securities with lower return correlation with the portfolio are added, significant risk reduction can be achieved.

c. Combining two securities with perfect negative correlation ($\rho = -1$) with each other can eliminate portfolio risk altogether. This is the principle behind hedging strategies.

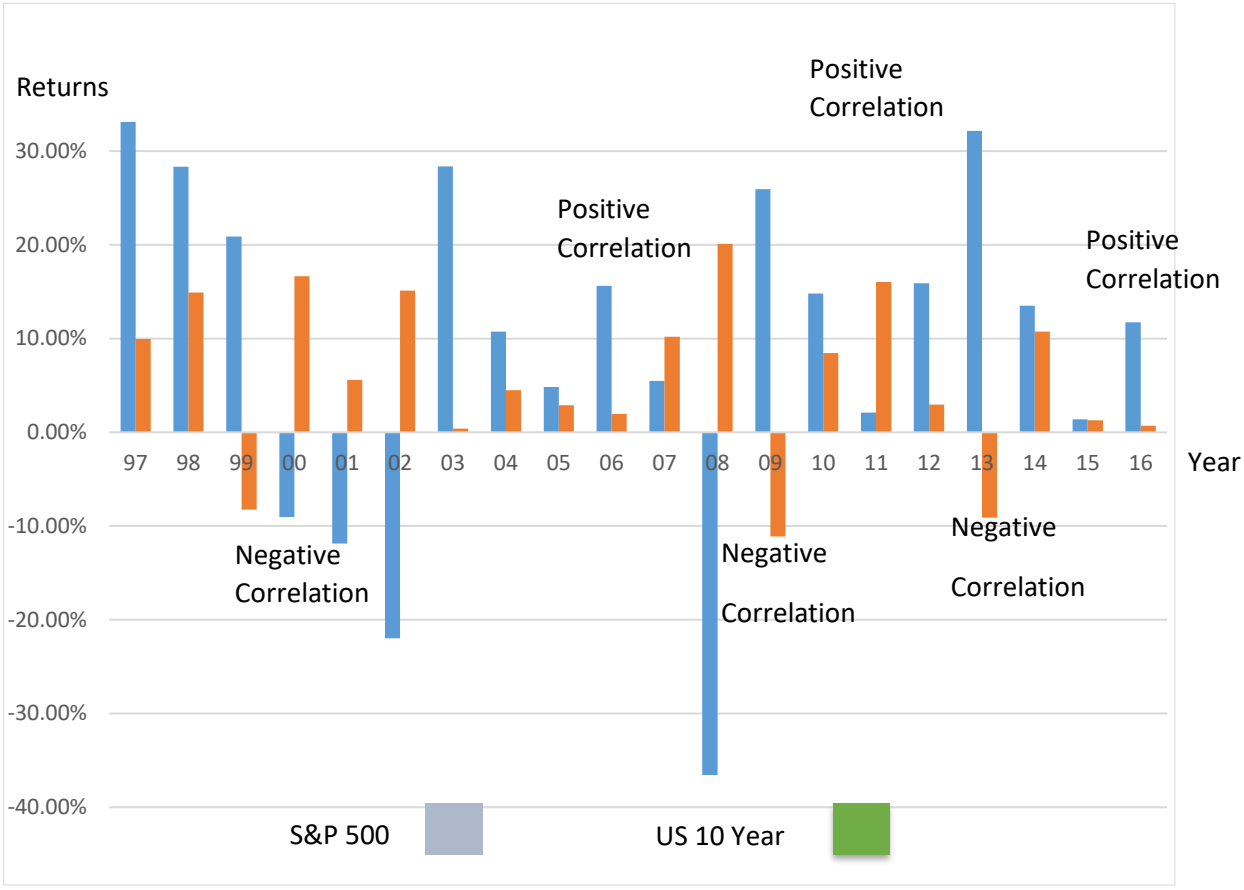
As mentioned earlier, as long as correlation is less than perfectly positive, risk can be reduced. Typically, returns of securities have some correlation with each other but are not perfectly positive. Thus, although portfolio risks can almost always be reduced, it usually cannot be completely eliminated.

8.4.5 Correlation is Not Stable

For portfolio management, assumptions on correlation must be adjusted from time to time. This is because the correlations between asset groups change over time.

Figure 8.4.5 shows that the correlations between S&P 500 Index returns and US 10-Year Treasury yields have fluctuated between negative and positive territory from 1997-2016.

Figure 8.4.5 – S&P 500 Returns vs US 10-Year Treasury Yields



8.4.6 Correlation Rises During Crisis

Example of Extreme Correlations

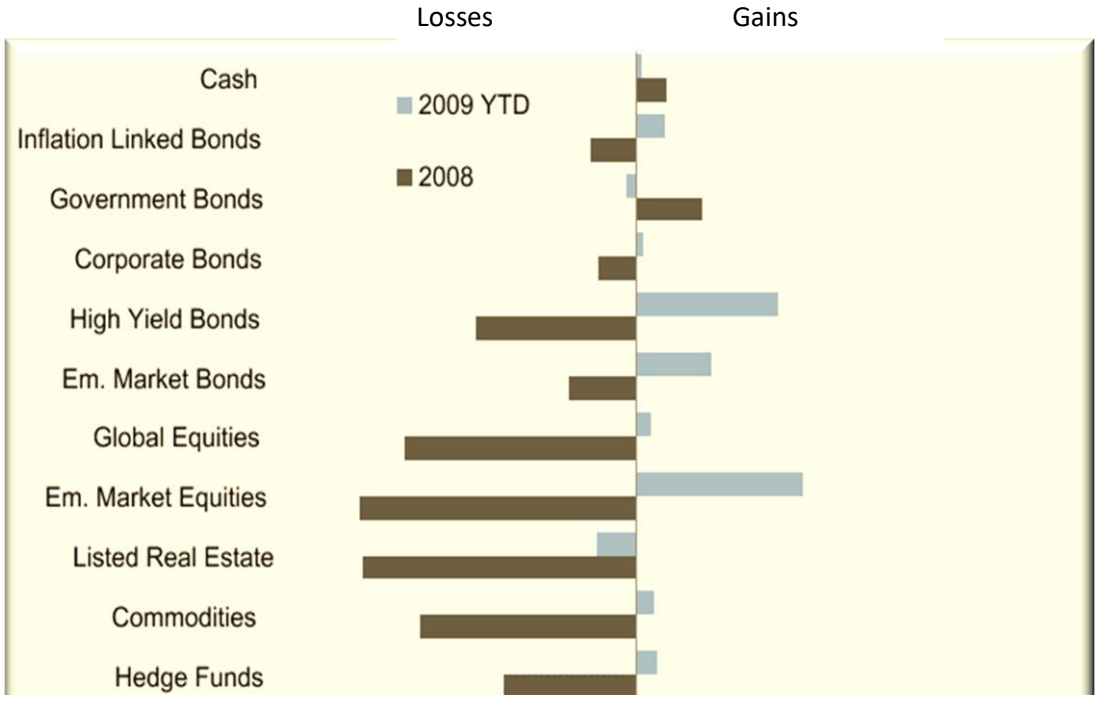
One concern in relation to diversification is the potential for asset correlations to move towards 1 during major financial crises. Though rare, extremely high correlation occurred amidst the 2008 Global Financial Crisis.

Systematic risk was higher because financial institutions and the entire banking sector were threatened. As credit dried up and redemptions became rampant, fund managers had to sell off assets to meet redemption demands. Even companies that were less related to the crisis had to sell off strong-performing assets to cover margin positions on assets that were directly affected.

The contagion effect rippled through the global financial market because counterparty risk was at an all-time high when financial institutions had to be bailed out. As a result, almost all assets corrected downwards. Under such conditions, correlation between assets gravitate towards 1. The silver lining was that the correlation remained highly positive during the rebound in 2009.

Figure 8.4.6: Asset Classes Move in Sync in 2008/09

(Except for Cash & Government Bonds)



Source: Bloomberg, Thomson Financial, as of 22 May 2009

There was nowhere for funds to "hide" in 2008 except in cash and government bonds since the contagion risk in the financial system spread to individual asset classes. The reverse happened in 2009, with recovery for most asset classes except for listed real estate, which took a bit longer to recover. See Figure 8.4.6.

8.4.7 Calculating Portfolio Return and Risk

The expected return of a portfolio is a weighted average of the expected returns of the individual securities in the portfolio and can be calculated as follows:

$$R_p = \sum_{i=1}^{n} w_i \; r_i$$

where

R_p = expected return on the portfolio
w_i = portfolio weight for the ith security
r_i = expected return on i^{th} security
n = number of different securities in the portfolio
$\sum w_i$ = 1.0

Although the expected portfolio return is a weighted-average of its constituent assets' expected returns, portfolio risk (measured by the variance or standard deviation) is not a weighted-average of the risk of the individual securities in the portfolio.

For a two-asset case, the formula for calculating **portfolio risk** is as follows:

$$\textbf{Portfolio risk } (\sigma_p) = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2}$$

where

w_1 and w_2 = portfolio weights of security 1 and security 2
σ_1 and σ_2 = standard deviation of security 1 and security 2
$\rho_{1,2}$ = correlation coefficient of security 1 and 2

Calculating Portfolio Return and Risk

Consider the following portfolio which consists of two securities, A and B.

Security	Expected Return	Standard Deviation	Weight in Portfolio
A	18%	20%	40%
B	9%	10%	60%

The correlation coefficient between Security A and Security B is **0.8**.

Expected return on the portfolio

= (0.4)(0.18) + (0.6)(0.09)

= 0.072 + 0.054 = 12.6%

Portfolio risk

$= \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2}$

$= \sqrt{0.4^2*0.2^2+0.6^2*0.1^2+2(0.4)(0.6)(0.8)(0.2)(0.1)}$

$= \sqrt{0.0064 + 0.0036 + 0.00768}$

$= \sqrt{0.01768}$

= 0.1330 or 13.30%

If ρ = 1, then σ_p = 14%

If ρ = 0, then σ_p = 10%

If ρ = -1, then σ_p = 2%

These calculations show the impact of combining securities with less than perfect positive correlation has on portfolio risk. The risk of the portfolio decreases steadily from 14% to 2% as the correlation coefficient declines from +1.0 to -1.0.

8.4.8 International Diversification

Portfolio risk reduction can also be achieved through international diversification, as studies have shown that movements in share prices in different countries are fairly uncorrelated.

Inherent in international diversification is the risk of fluctuating exchange rates of foreign currency investments (called foreign exchange risk). However, one way to remove foreign exchange risk from international portfolio investments is to hedge the risk using forward exchange contracts.

Additionally, international diversification does not always lead to higher risk-adjusted returns. It makes sense to add securities from new markets only if the additions have the potential to raise the risk-adjusted return of the portfolio. Sometimes, adding the wrong securities and markets can actually lower the risk-adjusted return.

Top-down fundamental research on the prospects of the economy, industry and company would be important. Reviews of the political and currency risks would also be useful. Back-testing simulations might also help provide information on how securities or markets performed during different periods, including with high stress. Investment managers may apply such analytics for appropriate portfolio allocation.

8.4.9 The Capital Asset Pricing Model

Derived by Sharpe, Lintner, and Mossin in the mid-1960s, CAPM describes the equilibrium relation between the systematic risk and the expected return of assets.

The CAPM is based on the following assumptions:

1. Investors evaluate portfolios on the basis of risk and return.
2. Investors are rational and maximise return for a given level of risk and minimise risk for a given level of return.
3. Individual assets are infinitely divisible, which means that it is possible to buy or sell fractional shares.
4. Investors can borrow or lend at the risk-free rate of return.
5. There are no taxes or transaction costs involved in buying or selling assets.
6. There is no inflation or any change in interest rates or inflation is fully anticipated.
7. All investors have the same one-period time horizon.
8. Information is freely and instantly available to all investors.
9. Investors have homogenous expectations of risk, return, and covariances of securities.
10. Capital markets are in equilibrium.

Based on the CAPM assumptions above, since all investors have the same expectations, they will therefore arrive at the same optimal portfolio of risky assets, which is referred to as the **Market Portfolio** (M). As the market portfolio is completely diversified, it comprises only systematic risk, which cannot be eliminated because it is the result of macroeconomic factors that affect the value of all securities and the market as a whole. All assets are included in portfolio M in proportion to their market value.

In theory, the market portfolio should include all risky assets worldwide, both financial (bonds, options, futures, etc.) and real (gold, real estate, etc.). However, in practice, we often just use the portfolio of all ordinary shares on the S&P 500 to represent the universe and the market index such as the S&P 500 Composite Index as the proxy.

The CAPM introduces another asset to the universe of assets under consideration; namely, the **risk-free asset**. As the name implies, the risk-free asset has no risk and provides a risk-free rate of return (R_f). The risk-free asset's return variance is zero and its return covariance with any risky asset is zero as well.

Given the analysis above, we can now derive two important relations between expected returns and risk, expressed as:

- The **capital market line**, which specifies the equilibrium relation between expected return and **total risk** for efficiently diversified portfolios.
- The **security market line**, which specifies the equilibrium relation between expected return and **systematic risk**. It applies to individual securities as well as portfolios.

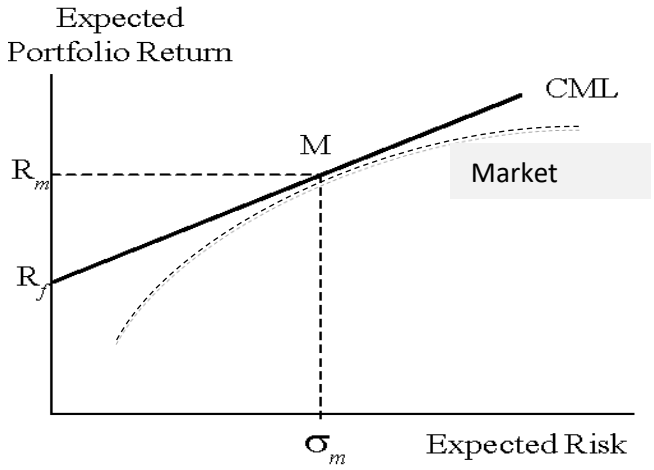
a. **Uses of the CAPM**

The CAPM is used to evaluate investments using the market benchmark risk-return relation to see if an investment provides a fair return given its systematic risk. In spite of its many shortcomings, the CAPM is still used in the industry to offer insights into risk-return relation in the market and to provide a basis for determining the cost of capital for securities.

b. **The Capital Market Line**

The capital market line (CML) in Figure 8.4.9(b) shows the risk-return trade-off for efficient portfolios. It is essentially the efficient frontier formed from different combinations of the market portfolio with the risk-free asset. In equilibrium, all investors should hold a portfolio somewhere on the CML.

Figure 8.4.9 (b) – Capital Market Line



The equation for the CML is:

$$E(R_p) = R_f + \frac{E(R_m) - R_f}{\sigma_m} \sigma_p$$

Where:

$E(R_p)$ = expected return on any efficient portfolio, P, on the CML

R_f = rate of return on the risk-free asset

$E(R_m)$ = expected return on market portfolio, M

σ_m = standard deviation of the market portfolio returns

σ_p = standard deviation of efficient portfolio P's returns

$E(R_m)–Rf$ represents the market risk premium

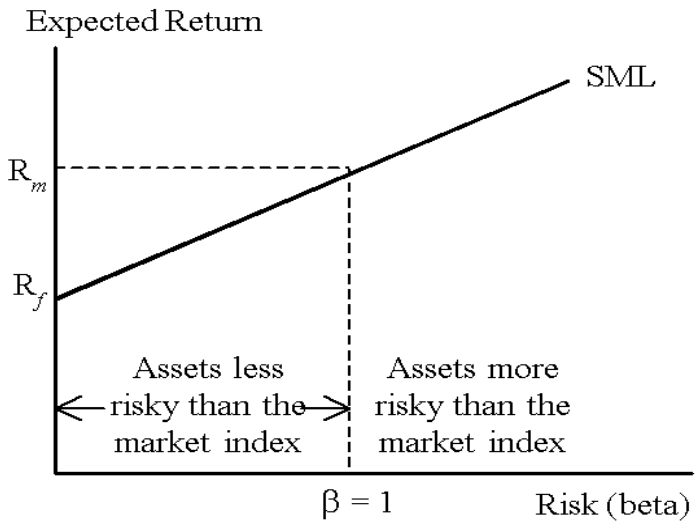
The slope of the CML, $[E(R_m) - R_f]/ \sigma_m$ is the **market price of risk** for efficient portfolios. It indicates the additional return that the market requires for a 1% increase in the portfolio's risk, i.e., its standard deviation of returns.

Based on the equation for the CML, the expected return for any portfolio on the CML is equal to the risk-free rate plus the product of the market price of risk and the quantity of portfolio risk (i.e., standard deviation).

c. **The Security Market Line**

While the CML applies only to efficient portfolios, the security market line (SML) depicts the risk-return trade-off for all assets, including individual securities and portfolios, both efficient and inefficient. Figure 8.4.9(c) shows the SML.

Figure 8.4.9 I – The Security Market Line



The relative systematic risk of a security or portfolio is the volatility of its return relative to the volatility of the market portfolio's return, multiplied by the correlation coefficient. This relative measure of systematic risk is referred to as beta (β). The beta of the market portfolio is 1. If a security or portfolio has a beta greater than 1, this means that it contains proportionately higher systematic risk than the market portfolio and should therefore be compensated with a higher expected return. A beta less than 1 means that the security or portfolio has systematic risk lower than the market portfolio, which would also mean lower expected return.

For example, a security with beta of 1.5 indicates that, on average, the security has 1.5 times as much systematic risk as the market. A risk-free asset has a beta of zero but an asset with a beta of zero need not be risk-free (i.e., it can have unsystematic risk); either way, both should yield the risk-free rate of return.

The CAPM relates the expected rate of return for any security or portfolio with the relative systematic risk measure, β, as follows:

$$\textbf{E}(R_i) = \text{Risk-free Rate} + \text{Risk Premium for i} = R_f + \beta_i (R_m - R_f)$$

where
$E(R_i)$ = expected return on security or portfolio i
R_f = risk-free rate
R_m = expected return on the market portfolio
β_i = systematic risk of security or portfolio i relative to the market portfolio
$R_m - R_f$ = market risk premium

This equation shows that securities with betas greater than the market beta of 1 should have larger risk premiums than the market portfolio and therefore require higher rates of return. This is what investors expect since beta is a measure of undiversifiable risk and greater systematic risk should be accompanied by greater expected return. Conversely, securities with betas less than that of the market are less risky and should have required rates of return that are lower than that of the market portfolio.

Using the CAPM to calculate Expected Rate of Return on an Equity Growth Portfolio

Assume that the beta for Growth Portfolio A is 1.20. Given a risk-free rate of 2.5% and an expected market return of 14%, the expected rate of return for Growth Portfolio A is:

$$E(r) = r_f + \beta(r_m - r_f) = 2.5 + 1.2(14 - 2.5) = 16.3\%$$

d. **Generating Positive Alpha**

In portfolio performance measurement, we often speak of identifying portfolios that generate positive alpha.

What, then, is alpha in the context of portfolio management? Alpha refers to the actual outperformance of the portfolio above what is predicted.

Hence, if the actual return is higher than the predicted CAPM return for the portfolio, we can say that the fund manager is generating positive alpha. Alpha is the fund manager's ability to outperform the predicted CAPM return (or whatever model return a fund is using as the benchmark).

Based on the example above, if Growth Portfolio A gives an actual return of 17%, its implied alpha would be 17 – 16.3 = 0.7%. At 15%, the portfolio's alpha would be negative at -1.3%.

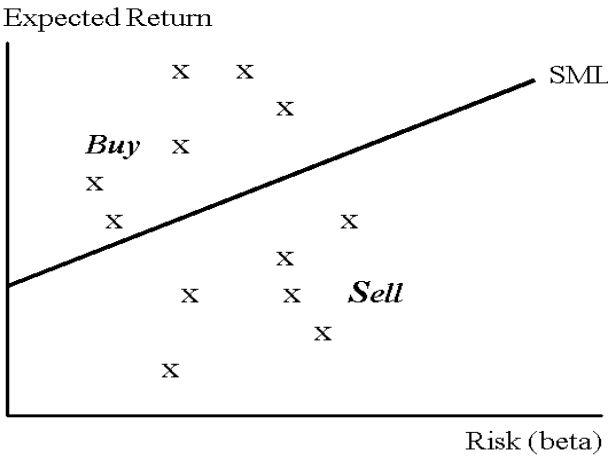
Investors should buy portfolios generating positive alpha and sell or short sell those generating negative alpha. Investor demand for a positive alpha will increase its price. As the fund/portfolio price rises, other things being equal, expected return falls, reducing and ultimately eliminating the alpha that first created the excess demand. Conversely, a drop in demand for the negative alpha fund/portfolio will reduce its price and increase its expected return, pushing its alpha back up to zero.

Ultimately, such buying and selling pressure should leave most securities with zero alpha most of the time and positive/negative alphas are unlikely to persist for many securities. Hence, unless the analysis shows otherwise, alpha should be assumed to be zero.

e. **Overvalued and Undervalued Securities**

As seen in Figure 8.4.9(e) the SML may be used to identify securities that are overvalued or undervalued.

Figure 8.4.9(e) - Overvalued and Undervalued Securities Using the SML



The SML may be used to identify securities that are overvalued or undervalued. Securities that lie above the SML provide above average expected returns given their respective betas. Such securities are considered to be under-priced. Conversely, securities that lie below the SML are considered to be overpriced as they offer lower returns than predicted by the SML for a given level of risk.

However, it should be noted that in a perfectly efficient market, all securities should lie on the SML. If they do not, the market mechanism would force it back onto the SML. For example, if a security lies above the SML, demand for the security would push the price of this security up, thereby reducing the expected return of the security.

Similarly, if the security lies below the SML, pressure to sell this security would then push its price down. With lower prices, the expected return of this security would correspondingly increase. Thus, the buying and selling of attractive and unattractive shares, respectively, would continue until equilibrium is reached; that is, when all securities lie on the SML.

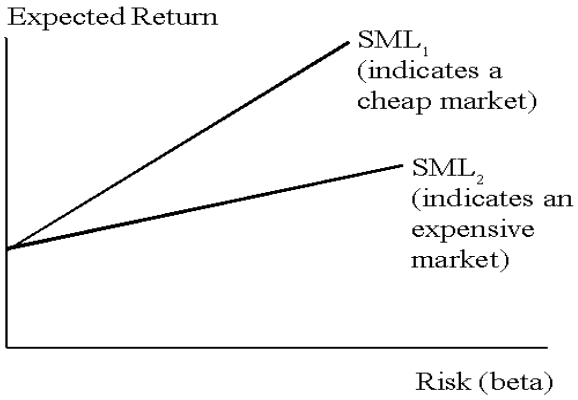
f. **Expensive or Cheap Markets**

The SML is also useful in reflecting the value of markets at any point in time, as shown in Figure 8.4.9(f).

A steep SML indicates that a large incremental return can be earned in the market with a small increase in systematic risk. As such, the market may be excessively risk-averse and, therefore, cheap.

On the other hand, a flat SML indicates that there is little incremental return in the market for taking additional systematic risk. Since the market appears to be ignoring risk, such a market may be overvalued.

Figure 8.4.9(f) - Expensive and Cheap Markets Using the SML



8.4.10 Arbitrage Pricing Theory

One of the main criticisms of the CAPM is the inability to observe or properly measure the return on the market portfolio, which consists of all risky assets. Moreover, the underlying assumptions of the CAPM are largely unrealistic. The CAPM is essentially a single factor model where fluctuations in security prices are only explained by the market factor.

In response to these criticisms, Stephen Ross developed an alternative asset pricing model in the early 1970s - the Arbitrage Pricing Theory (APT). The APT is a multifactor equilibrium pricing model.

The APT, like the CAPM, has the following assumptions:

a. Investors have homogeneous beliefs.

b. Investors are risk-averse utility maximisers.

c. Markets are perfect, and frictions like transactions costs are irrelevant.

d. Returns are generated by a factor model.

However, in contrast to the CAPM, the APT does not assume the following:

a. A single-period investment horizon.

b. Absence of taxes.

c. Borrowing and lending at the risk-free rate.

d. Investors' selection of portfolios on the basis of expected return and variance of return.

A factor model takes the view that there are underlying risk factors that affect the realised and expected returns of securities. These risk factors are not firm-specific factors and represent broad economic forces with three characteristics:

a. The risk factor must have a pervasive influence on share returns. Firm-specific events are not APT risk factors.

b. The risk factor must influence expected return.

c. At the beginning of each period, the risk factor must be unpredictable by the market as a whole. The rate of inflation is not an APT risk factor because it is partially predictable. In contrast, unexpected inflation (which is the difference between actual and expected inflation) is an APT risk factor.

Economic factors might include real economic growth, unexpected inflation, changes in interest rates, and political uncertainties. The APT assumes that there are several underlying factors that generate returns for the security:

$$R_i = E(R_i) + b_{i1}f_1 + b_{i2}f_2 + \ldots + b_{in}f_n + e_i$$

where:

R_i	=	actual rate of return on security i in a given period
$E(R_i)$	=	expected return on security i in a given period
f	=	deviation of a systematic factor f from its expected value
b_i	=	sensitivity of security i's returns to a factor
e_i	=	random error term, unique to security i

The actual return for security i will be equal to its expected return if the factors are at their expected levels (i.e., the factors do not deviate from their expected values) and the random error term is zero.

The problem with the APT is that the factors are not well-specified, at least for the future. Most empirical studies suggest that there are three to five factors influencing security returns. Chen, Roll and Ross (1986) identified four systematic factors:

i. Unanticipated changes in inflation;
ii. Unanticipated changes in industrial production;
iii. Unanticipated changes in the default-risk premium; and
iv. Unanticipated changes in the term structure of interest rates.

Investors can better understand return by identifying the risk factors affecting asset returns. An investment manager can structure a portfolio to increase exposure to one or more of these risk factors or he can hedge against them.

8.5 Portfolio Management Services

Portfolio management services of varying service levels are available, depending on the client type.

Dedicated and personalized portfolio management services are made available to institutional funds, sovereign wealth funds, corporate funds, as well as funds of accredited investors individuals. With the provision of dedicated services, these funds (e.g., USD50m AUM) may be separately managed by the fund manager from other funds. Additionally, the fund manager will provide regular monthly updates and one-to-one reviews of the portfolio's performance with the client.

For retail investors, who generally have smaller amounts to invest, available options are unit trust funds traded on different global exchanges and private funds sold by banks. One notable example of a listed investment company that is solely involved in managing investments is Berkshire Hathaway. Some funds are country-specific, while others offer regional or global diversification; there are also sector-specific and thematic funds. By pooling investor monies together, retail investors are therefore also able to benefit from the effects of diversification through economies of scale and by leveraging on common resources.

8.5.1 Investment Approach and Style

Portfolio management services are not suitable for all investors, due to differing investor preferences and objectives. For example, some investors do not trust anyone else to manage their funds and prefer a hands-on approach in their portfolio security selection. There are also investors who do not wish to incur investment management fees, which can range from 1.0% - 1.5%, on average.

Some investors prefer a hybrid approach, instructing their fund managers to manage their core investments conservatively, while directly investing a portion of their portfolio themselves.

Nonetheless, regardless of the preferences of the investor, both individual and institutional investors would benefit from adopting a disciplined portfolio management approach.

Figure 8.5.1(a) – Investment Approaches of Clients

Clients who self-manage their own portfolio	Clients who have both self-managed and professionally managed portfolios	Clients who prefer professional portfolio management
• Likes trading ideas and strategies • Wants good research • Believes in their own ability to manage investments • Wants a cost effective and efficient execution platform • Has time to manage own portflio • Can choose to trade listed diversified funds and thematic funds	• Manages some investments directly and trades occasionally • Has a proportion of funds to be managed externally • May use external fund managers as well as trade and invest directly • Long term-oriented but will also participate in short-term opportunities	• No time for dedicated focus on investment portfolio • Desire for investments to be professionally managed in a portfolio approach • Wants to minimise emotional influence in the investing process • Willing to trust external professionals to manage investments over longer term

a. **Advisory Portfolio Management**

For investors who wish to remain actively engaged in investment decisions, an advisory portfolio management approach is needed. However, banks will only provide such services for affluent (i.e., accredited investors) clients who are willing to place significant amounts of money (e.g., USD10 million) with the bank, so as to justify the added deployment of expertise and resources. Less affluent clients can rely on their bank's

relationship manager or client internet portal for research, advice, and asset allocation recommendations. Finally, at the lowest end of the spectrum, retail banking clients may not receive any support services at all when they purchase funds.

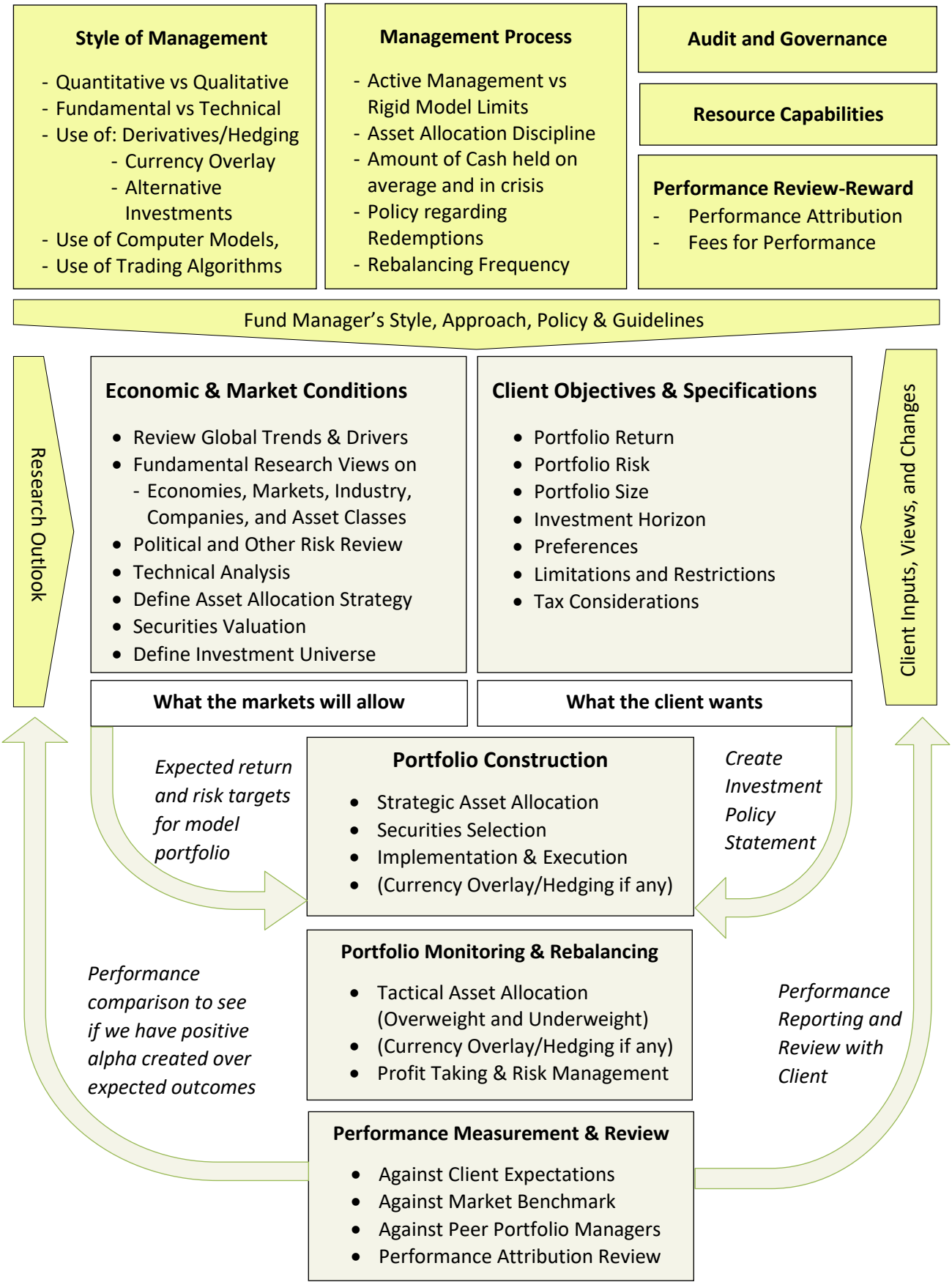
b. **Discretionary Portfolio Management**

For most investors buying into portfolio investment mandates, the general practice is to give the fund manager the discretion to manage the funds purchased.

Under the discretionary portfolio management approach, the fund manager is able to decide on asset allocation and securities selection without having to obtain permission from the client for each transaction. The fund manager's decisions would be guided by the client's investment profile and policy statement.

Please refer to Figure 8.5.1(b) for an overview of Portfolio Management Process.

Figure 8.5.1(b) Overview of Portfolio Management Process



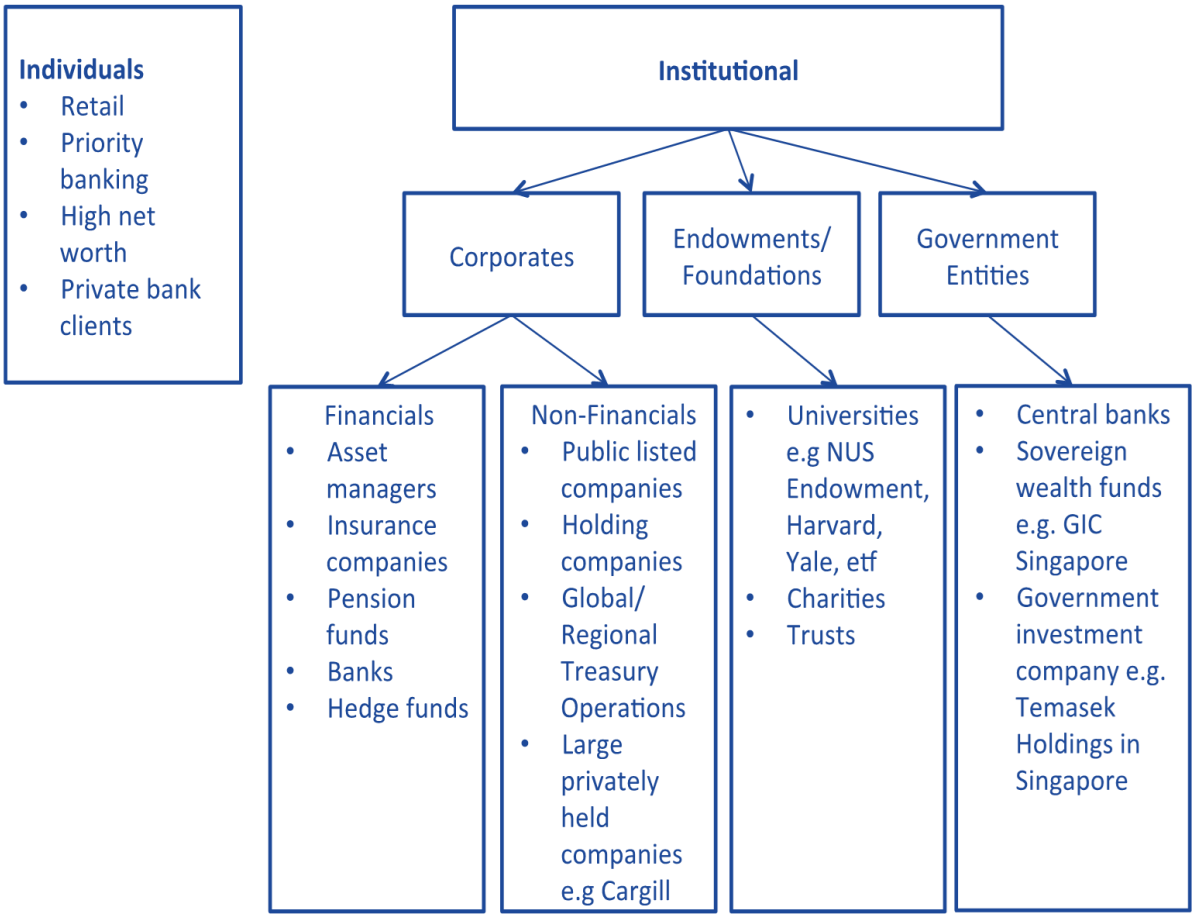
8.5.2 Portfolio Management Process for Retail Investors

Portfolio management involves the evaluation of investment risk and returns to construct an optimal portfolio for the investor. When doing so, the investor's investment objectives, risk profile, and any constraints and circumstances that are unique to the particular investor must be taken into account.

It would be useful to understand an investor's personal goals, as these help the investment representative to determine investment objectives, identify a suitable fund or portfolio management service that meets the client's needs, and recommend investment strategies. For example, conservative investment strategies may make sense for those setting aside monies for retirement or their children's education. Generally, individual (retail) clients have shorter time horizons than institutional clients but, for an investor setting aside dedicated funds (e.g., through a trust) for their grandchildren, the long investment horizon may allow for the higher volatility inherent in a growth-oriented fund.

Figure 8.5.2 shows the different types of client categories.

Figure 8.5.2(a) Different Client Categories



In this section, we will focus on portfolio management for individuals; particularly, retail investors. As illustrated in Figure 8.5.2(b), the portfolio management process for individual investors is dynamic, continuous, and systematic, with the following elements:

a. **Setting investment objectives**. The investor's objectives, requirements, preferences, and constraints are analysed. The aim is to define a level of acceptable risk that the investor can take and the expected return that is reasonable for that level of risk.

b. **Developing and implementing strategies**. This involves setting guidelines to determine the allocation of assets and defining the circumstances in which this allocation may vary.

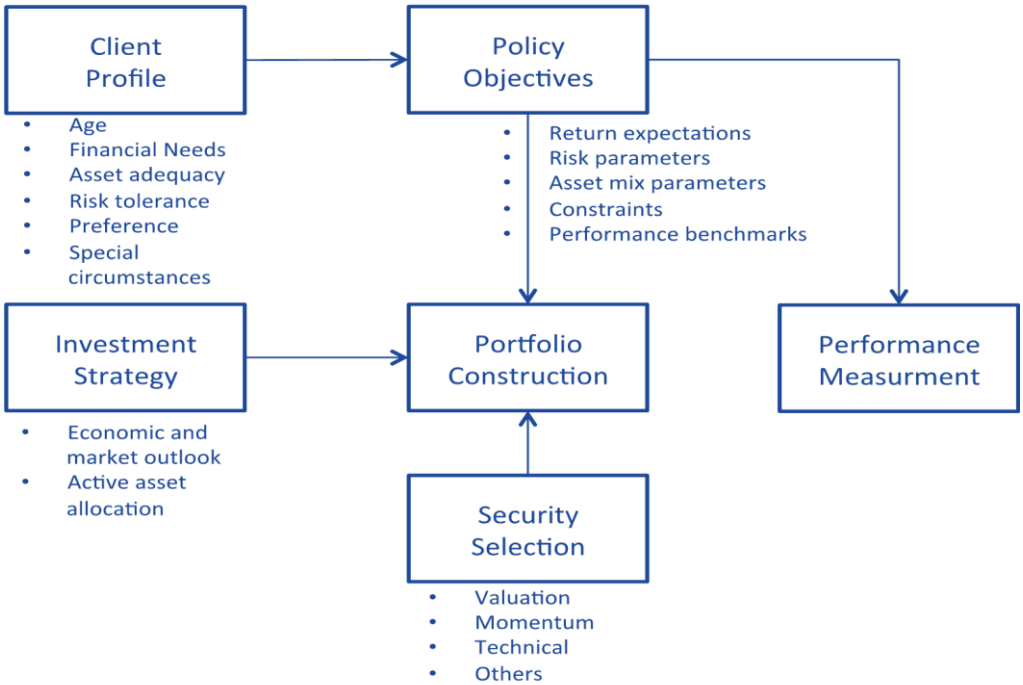
c. **Monitoring market conditions**. Market conditions and changing relative values of various asset classes and securities are closely monitored. In addition, the investor's needs, circumstances, and objectives are re-evaluated periodically.

d. **Reviewing and adjusting the portfolio**. Portfolio adjustments are made at appropriate times to reflect changes in the market or the investor's requirements.

e. **Measuring investment portfolio performance**. This step involves measuring the performance of the portfolio and then evaluating that performance relative to a benchmark.

The portfolio objectives, constraints, and strategy should always be stated clearly in a written document. A well-drafted investment policy provides the discipline required for efficient management of the portfolio, minimising whipsaw reactions to temporary price swings or even protracted volatility.

Figure 8.5.2 (b) - The Portfolio Management Process for Retail Investors



8.5.3 Determinants of the Investment Policy Statement (IPS)

The IPS or "investment mandate" belongs to the client and not to the portfolio manager. It is, therefore, essential that the investment manager spends sufficient time with the client to define the client's investment goals and acceptable levels of risk, and to note any constraints or preferences. This analysis, coupled with the manager's understanding of the capital markets, would aid in creating the investment policy statement.

A typical IPS will include the following:

a. **Introduction** – describes the investor;

b. **Statement of purpose** – expresses the purpose of the IPS;

c. **Statement of duties and responsibilities** – lays down the respective duties and responsibilities of the client, the custodian of the client's assets, and the investment manager;

d. **Procedures** – sets out the processes for maintaining and updating the IPS;

e. **Investment objectives** – e.g., capital appreciation, income generation, or wealth preservation;

f. **Investment constraints** – e.g., liquidity requirements, time horizon, tax considerations, regulatory and legal requirements, and circumstances peculiar to the investor;

g. **Investment guidelines** – provides guidance for the execution of the policy (e.g., leverage permissibility, loss limits, asset size limits, non-permissible activities or markets, etc.);

h. **Evaluation and review** – describes how evaluation and review of the portfolio will be carried out under the IPS;

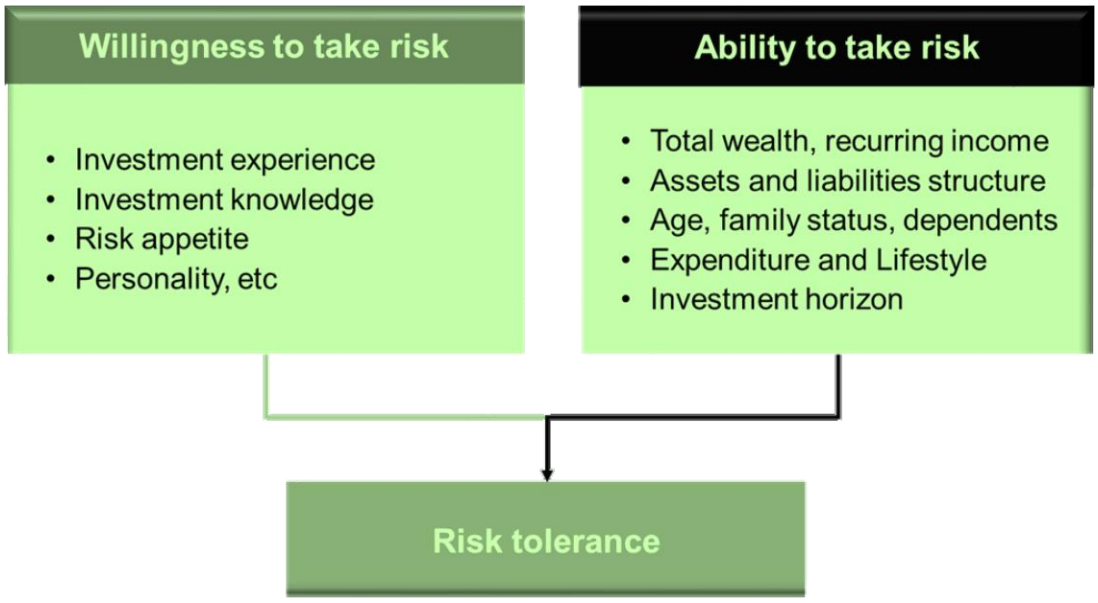
i. **Appendices on strategic asset allocation, rebalancing policy** – investors often specify a policy portfolio (e.g., conservative portfolio, balanced portfolio, or aggressive portfolio) as part of their investment objectives.

8.5.4 Investment Portfolio Objectives - Risk and Return Requirements

Return Requirements and Risk Tolerance - In defining the investment objectives, emphasis should be placed on determining the client's sensitivity to risk. Determining the individual's return requirement and risk tolerance level is usually carried out with an interview and discussion session, and through a risk profiling questionnaire.

While client profiling and risk tolerance are covered extensively in the CMFAS Rules and Regulations Modules, the main point to note with regard to understanding the client's profile (including the client's risk and return requirements) is that it goes beyond fulfilment of regulations and embodies taking a genuine interest in the client's needs and financial well-being. Factors affecting a client's risk tolerance is set out in Figure 8.5.4.

Figure 8.5.4 – Factors affecting Risk Tolerance



A useful framework for analysing the risk appetite of individual investors is the life cycle approach.

Life Cycle Approach - An individual's risk and return preferences are often related to the stage of the life cycle the individual is at:

a. 1st stage **(early career)** – The individual's priorities at this stage would typically include saving for liquidity purposes, buying a home, obtaining life insurance, and then investments. Given the need to service a large housing loan and possibly a car loan, assets tend to be small relative to liabilities. However, because the individual is young and has a long time horizon with a growing stream of discretionary income, he can undertake high return, high risk investments that are growth-oriented.

b. 2nd stage **(mid-career)** - The individual can establish a serious investment program, noting that liabilities should be substantially lower by this stage. While the time horizon remains quite long, it is not so long that capital preservation is unimportant. The investor can continue to undertake high risk, high return growth-oriented investments, but should reduce overall risk exposure.

c. 3rd stage **(late career)** – At this stage, the individual is near retirement, and there is a greater emphasis on income needs and capital preservation. Assets significantly exceed liabilities. Time horizon is shorter and the investor's portfolio is typically shifted to significantly lower return, lower risk assets with large dividend or interest income components and relatively secure asset values.

While the life cycle approach is generally used to portray the risk and return preferences of individuals, each individual's risk tolerance is unique. The risk profile of an individual is affected by other factors such the investor's net worth, health, family situation, future financial commitments, familiarity with investments, job stability, and emotional temperament.

8.5.5 Investment Constraints

a. **Liquidity Requirements**

Liquidity requirements vary considerably from one investor to another and must therefore be clearly communicated. Investors must carefully assess their own financial situation to determine their liquidity needs. If one is investing funds to be used for the next mortgage payment or the coming semester's tuition, then immediate liquidity is essential, and financial assets would be preferred. To meet such liquidity needs, funds should be invested in high quality, short-term debt instruments which can be easily sold, such as Treasury bills and commercial paper. If the assets held have poor marketability, such as real estate or shares in a closely held business, then liquidity needs should be monitored carefully.

In practice, few investment companies actually hold a liquidity reserve, given that most financial assets are highly liquid. Shares and bonds can generally be sold within a matter of minutes at a price close to their market values. Shortfalls in liquidity are normally met through the sale of financial assets.

b. **Time Horizon**

Time horizon is defined as the investment period of the investor. The investment time frame of the client is important as it affects the nature of investments selected and the investment strategies to be employed.

Investors with a long investment horizon can absorb and smooth out the ups and downs of risky assets like ordinary share portfolios. These investors can also build portfolios with investment strategies that are riskier.

Conversely, investors who have a shorter investment horizon tend to have fewer volatile portfolios, typically consisting more of bonds than shares, with the former having shorter duration and the latter having lower volatility.

c. **Tax Considerations**

To determine investment returns, the fund manager must consider the tax implications for the client as well. Consideration must be given to:

- the client's income tax bracket;
- different tax treatments from different assets (tax-exempt or not); and
- different components of total return (dividend or interest income versus capital gains).

Investors in high tax brackets should consider tax-exempt fixed-income securities if the security's taxable equivalent return exceeds the return on comparable taxable bond issues. For investors in high tax brackets, equities with a large capital gain component may be more attractive compared to high dividend income-yielding shares.

d. **Regulatory and Legal Constraints**

Institutional investors such as insurance companies are regulated with regard to their investments. Generally, institutional investors are more regulated than private individual investors. An individual using his personal

savings has greater freedom in what he wants to invest. However, if he is using his CPF[45] money to invest, his investment will be governed by the CPF Investment Scheme, which specifies the types and quantum of investments permissible.

e. **Clients' Unique Needs or Preferences**

It is the fiduciary duty of fund managers to ensure that clients' instructions are adhered to. It is important to understand the client's preferences or restrictions. Some clients may specify a limit on equity investments and/or foreign currency exposure. These preferences and instructions from the clients must be clearly spelt out in the investment policy and be in writing in order to avoid potential misunderstandings in the future.

8.5.6 Asset Allocation

The purpose of the asset allocation process is to maximize portfolio return at a level of risk consistent with the investor's objective. It has always been vigorously debated whether asset allocation is more important than securities selection when it comes to portfolio performance. A study by Brinson, Hood and Beebower[46] showed that 91.5% or more of a portfolio's return comes from asset allocation and, therefore, less than 10% can be attributed to security selection. Furthermore, it was found that asset allocation has a much greater impact on reducing total exposure to risk than picking an investment in any single asset category. Clearly, asset allocation is an important aspect of portfolio management.

At the asset allocation stage, economists and market strategists will provide input regarding their expectations on macroeconomic factors and market trends, because these will affect the returns of the various asset classes.

There are two types of asset allocation:

a. Strategic asset allocation (SAA); and

b. Tactical asset allocation (TAA).

a. **Strategic Asset Allocation**

SAA refers to how a portfolio's funds are divided among the various asset classes, given the client's risk profile, risk and return requirements, the portfolio manager's long-term forecasts of expected returns, variances and co-variances of the various asset classes.

SAA constitutes the "policy" or "long-run" asset mix. This asset mix is normally expressed in terms of the percentage of total portfolio value invested in each asset class. For example, SAA may be:

[45] The Central Provident Fund (CPF) is a social security scheme for Singapore Citizens and Permanent Residents, and encompasses retirement, healthcare and home ownership.

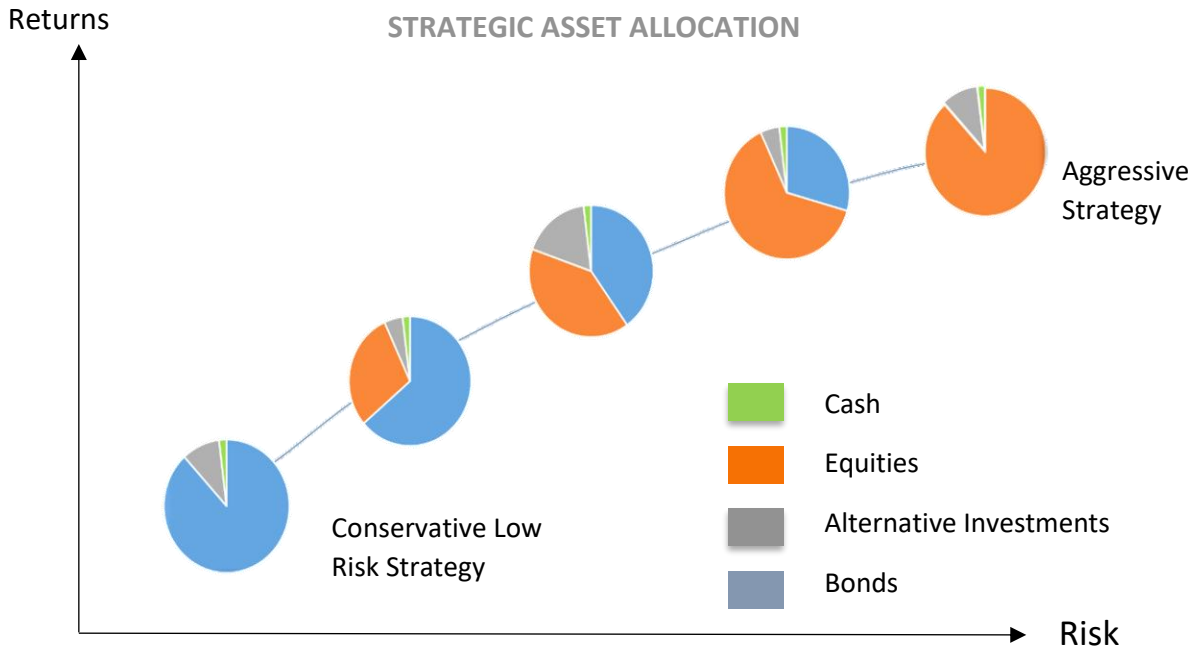
[46] "Determinants of Portfolio Performance" by Gary P: Brinson, L. Randolph Hood, and Gilbert L. Beebower, Financial Analysts Journal (January-February 1995).

Asset Classes	Allocation Range
Equities	45-55%
Bonds	35-45%
Cash-equivalents	5-15%
Total Portfolio	**100%**

Such an approach can be termed as a **constant-mix strategy**.

Maximum and minimum limits can also be allocated to each asset class to accommodate different risk profiles and market conditions. The available and permissible asset classes have to be clearly defined by the nature of the assets (e.g., cash, bonds, shares) and also by sector, industry, and geographical region or country. There may be sub-categories for each of these.

Figure 8.5.6 – Client's Risk and Return Objectives Help Define Strategic Asset Allocation



The optimal asset mix depends on the investor's return objectives and risk tolerance. The lower the tolerance for risk, the more conservative the asset mix. A conservative asset allocation (low-return/low-risk) relies heavily on bonds and short-term securities to provide stable returns, and there is more emphasis on current income and capital preservation.

Conversely, an aggressive portfolio that seeks capital gains might have the highest exposure to ordinary shares and less to bonds and short-term securities, thereby increasing the expected portfolio return and risk. Note that the representation of optimal asset allocation is similar to Figures 8.4.1(a) and 8.4.1(b).

The SAA is not expected to change very often. However, for the individual, there are situations when this long-run attitude toward investment risk changes. For example, there could be changes in a client's family circumstances, health conditions, business circumstances, or an overall negative impact or experience on his

wealth that may cause the client to become much more risk averse. In such cases, there is a change in the client's profile and risk tolerance. His strategic asset allocation should then be adjusted to reflect the change. When annual reviews are carried out, it is important to note whether there are changes in the client's view towards risk-taking.

b. **Tactical Asset Allocation**

For a given SAA mix, the investment manager will typically be allowed to reduce or increase the allocation to a given asset class to take advantage of market opportunities, as long as the allocation stays within a certain range. The decision to adjust the SAA slightly to exploit market opportunities is called TAA.

TAA refers to actions taken to overweight or underweight asset classes over the short run-in response to changing economic and market conditions, so as to optimize returns given the perceived risks. The process also seeks to take advantage of relative inefficiencies in the security prices across different asset classes.

For example, if the outlook for equities is favourable, the allocation mix can be changed as follows:

Asset Classes	SAA Range	Original TAA	New TAA
Equities	45-55%	45%	55%
Bonds	35-45%	50%	40%
Cash equivalents	5-15%	5%	5%
Total portfolio	**100%**	**100%**	**100%**

TAA models can be divided into three groups: valuation approaches, cyclical considerations, and a combination of the two.

The **valuation approach** seeks to identify the relative value of an asset class. In general, asset classes that are considered expensive are sold, and those considered cheap are bought. The most common valuation approach is the **risk premium approach** or **spread approach**. Basically, if the expected return on equities relative to bonds is above average, a higher allocation is made to equities. However, if the expected return on equities relative to bonds is below average, a lower allocation is made to equities. This is also referred to as the equity-bond yield gap.

The **cyclical consideration approach** is based on the view that cycles in share and bond prices are closely related because of macroeconomic factors. For example, equity market movements typically precede changes in economic activity. Managers adopting this approach will attempt to identify one or more components of general economic activity that are believed to be good predictors of market movements.

8.6 Performance Measurement

The major components of the investing process have been discussed. One important issue that remains is the "bottom line" of the investing process: evaluating the performance of a portfolio. In evaluating performance, we want to answer the following questions:

- How well (or how badly) is the manager performing, relative to a benchmark?
- What were the factors that contributed to the success or failure of the portfolio?

In this section, we will examine the types of composite performance measures, which consider both risk and return. We will also look at performance attribution analysis, which helps to identify the reasons for good or bad performance.

8.6.1 Measuring Portfolio Returns

A proper measure of total return captures both the income component and the capital (gain or loss) component of return. Performance is typically measured over a regular time period (e.g., monthly or quarterly). Return measurements are relatively simple if there are no cash flow movements into or out of the portfolio during the measurement period but become more complicated when there are movements in cash flows (i.e., deposits or withdrawals by the client). In this section, we will look at the methods of measuring investment returns based on different cash flow assumptions.

a. **No deposits or withdrawals**

In the simplest case, where there are no withdrawals or deposits by the client, the market value of a portfolio can be measured at the beginning and end of a period, and the rate of return can be calculated as:

$$\textbf{Return on the Portfolio } R_p = \frac{V_E - V_B}{V_B}$$

where:
V_E = end value of the portfolio
V_B = beginning value of the portfolio

Example 8.6.1(a) - Portfolio Return (No Movement of Cash Flows)

If a portfolio has a market value of $30 million at the beginning of the quarter and a market value of $35 million at the end of the quarter, then the return on the portfolio over the quarter is:

$$R_p = \frac{\$35m - \$30m}{\$30m}$$

$$= 16.7\%$$

b. **Deposits or withdrawals at the end of the measurement period**

Measurement of portfolio returns becomes more complicated when the client either adds or withdraws money from the portfolio.

Example 8.6.1(b) - Portfolio Return (Cash Flow Movements at End of Period)

Consider a portfolio with a market value of $50 million at the beginning of the quarter. Suppose the client deposits $5 million into the portfolio just before the end of the quarter. Subsequently, the market value of the portfolio at the end of the quarter is $53 million. When calculating the portfolio return, the deposit of $5 million has to be deducted from the ending value first. This results in a return of -4%.

$$R_p = \frac{(\$53m - \$5m) - \$50m}{\$50m}$$
$$= -4.0\%$$

In general, if a deposit or withdrawal occurs just ***before*** the end of the period, the return on the portfolio should be calculated by adjusting the ending market value of the portfolio as follows:

$$\mathbf{R_p = \frac{(V_\varepsilon + Withdrawals - Deposits) - V_B}{V_B}}$$

c. **Deposits or withdrawals at the beginning of the measurement period**

If a deposit or withdrawal occurs just after the start of the period, then the return on the portfolio should be calculated by adjusting the beginning market value of the portfolio. In the case of a deposit, the beginning value should be increased by the deposited amount. In the case of a withdrawal, the beginning value should be decreased by the withdrawn amount.

$$\mathbf{R_p = \frac{V_\varepsilon - (V_B + Deposits - Withdrawals)}{V_B + Deposits - Withdrawals}}$$

Example 8.6.1(c) - Portfolio Return (Cash Flow Movements at Start of Period)

If the $5 million deposit in the previous example was received at the beginning of the quarter, then the return should be -3.6%.

$$R_p = \frac{\$53m - (\$50m + \$5m)}{\$50m + \$5m}$$
$$= -3.6\%$$

d. **Dollar-Weighted Rate of Return (DWR)**

When deposits or withdrawals occur sometime between the beginning and end of the period, the DWR can be used to calculate the return on the portfolio. The DWR is equivalent to the internal rate of return. It equates the present value of all cashflows, including the ending market value, with the beginning market value of the portfolio.

Example 8.6.1(d) - Portfolio Return (DWR)

If the $5 million deposit in the previous example was made in the middle of the quarter, the DWR is calculated by solving the following equation for r:

$$\$50m = \frac{-\$5m}{(1 + r)} + \frac{\$53m}{(1 + r)^2}$$

r = -1.92% (semi-quarterly return)

The quarterly return is $(1 - 0.0192)^2 - 1 = -3.8\%$.

This is a loss. In the first half of the quarter, the loss was 1.92%, resulting in a portfolio value that is 0.9808 of the original value. In the second half of the quarter, there is another 1.92% loss on the 0.9808. Hence, $(1 – 0.0192)^2$ or (0.9808) x (0.9808) of the original portfolio value remains.

e. **Time-Weighted Rate of Return (TWR)**

An alternative method of computing a portfolio's performance when cash flows occur between the beginning and end of the period is the TWR. Calculating the TWR requires information about the value of the portfolio before each cash flow occurs.

Example 8.6.1(e) - Portfolio Return (Time-Weighted)

Consider the following information:	
Portfolio value at beginning of quarter	$50 million
Portfolio value just before deposit	$49 million
Deposit in middle of quarter	$5 million
Portfolio value immediately after deposit	$54 million
Portfolio value at end of quarter	$53 million

Return of portfolio for first half of the quarter is:

$$R_p = \frac{\$49m - \$50m}{\$50m}$$

$= -2.0\%$

Return of portfolio for second half of quarter is:

$$R_p = \frac{\$53m - \$54m}{\$54m}$$

$= -1.85\%$

The quarterly return is computed by linking the two rates of return as follows:

$R_P = [(1 - 0.02)(1 - 0.0185)] - 1$

$= -3.81\%$

If there are frequent cash flows, many calculations may be necessary in order to derive the TWR.

f. **Annualizing Returns**

Based on the quarterly returns, the annualized return can be calculated as follows:

$$\textbf{Annual Return} = [(1+ r_1)(1+ r_2)(1+ r_3)(1+ r_4)] - 1$$

where r_1, r_2, r_3, r_4 are the quarterly returns.

When the measurement period is over several years, the annualized geometric mean rate of return (also known as the compound annual growth rate or CAGR) is calculated. The following example illustrates the computation of the CAGR.

Example 8.6.1(f) - Calculating Annualized Returns (Compound Annual Growth Rate)

A portfolio grew from $100 to $250 in 5 years. The annualized geometric mean return, which is the compound annual growth rate (r), is given by:

$250 = 100 \times (1 + r)^5$

$(1 + r)^5 = 2.5$

$r = 20.1\%$

g. **Dollar-Weighted Return versus Time-Weighted Return**

The two methods of calculating returns, the DWR and the TWR, can produce different results. At times, these differences can be substantial. In general, the DWR is regarded as inappropriate for the purpose of evaluating the performance of a portfolio manager. This is because the DWR is strongly influenced by the size and timing of the cash flows, over which the investment manager has no control.

For evaluating the performance of a portfolio manager, the TWR is more appropriate as the objective is to measure the performance of the portfolio manager independent of the actions of the client. Fund managers are evaluated using the TWR because they usually have little or no control over the cash inflows and outflows, which are decided by the client.

h. **Modified Dietz Method**

An essential part of calculating a true TWR is the valuation of the portfolio just before each cash flow occurs. In practice, cash flows often occur intra-month when portfolio valuations may not be available. The Dietz method overcomes the need to know the valuation of the portfolio on the date of each cash flow by assuming a constant rate of return on the portfolio during the period. The original Dietz method assumes that all cash flows occur at the midpoint of the period.

The modified Dietz method weights each cash flow by the amount of time it is held in the portfolio. The formula for estimating the TWR using the modified Dietz method is:

$$R_p = \frac{V_E - V_B - CF}{V_B + WCF}$$

where:
CF = net cash flow over the period
WCF = weighted cash flow

The formula for WCF is:

$$WCF = \sum_i CF_i \frac{n - t_i}{n}$$

where:
CF_i = cash flow on day t_i
t_i = number of days from beginning of period until CF_i
n = number of days in the period

The numerator of the WCF formula is based on the assumption that the cash flow CFi occurs at the end of the day. If cash flows are assumed to occur at the beginning of the day, the numerator should be (n - ti + 1).

Example 8.6.1(g) - Portfolio Return (Modified Dietz Calculation)

A portfolio has a value of $10 million at the beginning of a month. During the month, there was a cash inflow of $100,000 on day 12 and $200,000 on day 18. At the end of the month, the portfolio value was $11 million. Using the modified Dietz formula, the return on the portfolio for the month is calculated as follows:

$$WCF = 100{,}000 \times \frac{30 - 12}{30} + 200{,}000 \times \frac{30 - 18}{30}$$

$$= 60{,}000 + 80{,}000 = 140{,}000$$

$$R_p = \frac{11{,}000{,}000 - 10{,}000{,}000 - 300{,}000}{10{,}000{,}000 + 140{,}000} = 6.90\%$$

The main advantage of the modified Dietz method is that it does not require portfolio valuation on the date of each cash flow. Its main disadvantage is that it provides a less accurate estimate of the true TWR. It can be affected by cash flows, especially if these are large relative to the size of the portfolio or when the cash flows occur during periods of high market volatility.

i. **Global Investment Performance Standards (GIPS)**

Prior to February 2005, the Association of Investment Management and Research (AIMR) issued and prescribed the minimum standards for the presentation of investment portfolio performance. On 4 February 2005, the Chartered Financial Analysts Institute Board of Governors approved the revised GIPS, which created

a single global standard of investment performance reporting and increased minimum standards worldwide. The GIPS replaced the various country-specific performance standards and are widely accepted among investment managers internationally.

Some of the fundamental principles include:

1. Total return, accounting for realized and unrealized capital gains and income, must be used to measure portfolio investment performance.

2. TWR must be used.

3. Portfolios must be valued (marked to market) at least monthly, and returns have to be geometrically linked i.e., CAGR must be used for annualized returns.

4. Performance must be calculated after deducting actual trading expenses (e.g., brokerage, commissions, etc).

5. Dividends, interest income, and other sources of time-based income have to be accrued and accounted for in portfolio performance.

6. The annual returns for all years must be shown. If the investment management company is more than 10 years old, a 10-year performance record must be presented. Otherwise, the performance record has to be presented for the number of years since its inception.

7. Disclosure of whether performance results were calculated gross or net of investment management fees.

8. The use of leverage and derivatives has to be expressly disclosed.

8.6.2 Risk-Adjusted Measures of Performance

In evaluating the performance of the portfolio manager, it is inadequate to just consider returns without considering the risk taken to achieve the returns.

There are 3 generally recognised methods of measuring risk-adjusted return, also referred to as the **composite measures of portfolio performance**:

a. Sharpe Ratio;

b. Treynor Ratio; and

c. Jensen's Alpha.

a. **Sharpe Ratio**

Developed by William Sharpe, this measure relates the portfolio's excess return, or return above the risk-free rate to the portfolio's total risk. The numerator is also referred to as the risk premium. The total risk of the portfolio is measured by its standard deviation. The Sharpe portfolio performance measure (or Sharpe ratio) is defined as:

$$\text{Sharpe ratio} = \frac{R_p - R_f}{\sigma_p}$$

where:

R_p = average rate of return for the portfolio
R_f = risk-free rate of return
σ_p = standard deviation of return for the portfolio

When comparing the Sharpe ratio between funds, the question is: "How much return above the risk-free rate am I getting for the portfolio volatility I am bearing as measured by the standard deviation of the portfolio?"

The Sharpe ratio is, thus, a measure of the excess return per unit of total risk (standard deviation). The higher the Sharpe ratio, the better the portfolio performance.

Sharpe Ratio (Risk-Adjusted Performance Measure)

Consider the performance of the following portfolios:

Portfolio	Annual Return	Standard Deviation
A	0.10	0.15
B	0.16	0.20
Market	0.13	0.18

Assuming a risk-free rate of 6%, the Sharpe measures for these portfolios are as follows:

$S_A = \frac{0.10 - 0.06}{0.15} = 0.27$ or 27% $S_B = \frac{0.16 - 0.06}{0.20} = 0.50$ or 50%

$S_M = \frac{0.13 - 0.06}{0.18} = 0.39$ or 39%

Portfolio A has the lowest excess return per unit of total risk and underperforms both portfolio B and the aggregate market portfolio. In contrast, portfolio B performs better than portfolio A and the aggregate market portfolio.

b. **Treynor Ratio**

The Treynor ratio relates the average excess return on the portfolio (exactly the same variable as in the Sharpe measure) to its systematic risk, as measured by the portfolio's beta.

$$\text{Treynor ratio} = \frac{R_p - R_f}{\beta_p}$$

where β_p = beta for the portfolio

In this case, we are calculating the excess return per unit of systematic risk. The Treynor measure implies that systematic risk is the proper measure of risk to use when evaluating portfolio performance.

When comparing the Treynor ratio between funds, the question is: "How much return above the risk-free rate am I getting for the market sensitivity that the portfolio is exposing me to, as captured by β_p?"

Treynor Ratio (Risk-Adjusted Performance Measure)

Consider the performance of the following portfolios:

Portfolio	Annual Return	Beta
A	0.14	0.9
B	0.18	1.1
Market	0.20	1.0

Based on a risk-free rate of 6 percent, the Treynor measures for these portfolios are:

$$T_A = \frac{0.14 - 0.06}{0.9} = 0.089 \text{ or } 8.9\%$$

$$T_B = \frac{0.18 - 0.06}{1.1} = 0.109 \text{ or } 10.9\%$$

$$T_M = \frac{0.20 - 0.06}{1.0} = 0.140 \text{ or } 14\%$$

c. **Sharpe Ratio versus Treynor Ratio**

The main difference between the Sharpe and Treynor performance measures lies in the definition of risk. The Sharpe performance measure considers total risk, as measured by the standard deviation of returns, while the Treynor measure considers systematic risk, as measured by the beta of the portfolio.

If portfolios are perfectly diversified - that is, the correlation coefficient between a portfolio's return and the market return is 1.0 - the two measures will provide identical portfolio performance rankings – the total risk measured by standard deviation is only systematic risk, since all unsystematic risk has been diversified away. As portfolios become less diversified, the possibility of differences in rankings increases. This leads to the following conclusion: the Sharpe measure **accounts for the extent of a portfolio's diversification during the measurement period**. Differences in rankings between the two measures can result from substantial differences in portfolio diversification. If a portfolio is inadequately diversified, its performance ranking using the Treynor ratio is likely to be higher than its ranking using the Sharpe measure, because the Treynor ratio ignores non-systematic risks, which are present in an inadequately diversified portfolio.

d. **Jensen's Alpha**

Jensen's alpha is based on the CAPM and measures the risk-adjusted return as alpha.

The Jensen measure uses the CAPM to determine the expected return on a portfolio (R_e) given the market's performance during the measurement period (R_m), the risk-free rate (R_f), and the portfolio's beta.

The expected return on a portfolio is given as:

$$\mathbf{R_e = R_f + \beta_p(R_m - R_f)}$$

The portfolio's alpha is then the difference between the portfolio's actual return during the period and the return that would be expected from the CAPM.

When the actual return is higher than the CAPM predicted return, there is positive alpha, which means that the fund manager outperformed a risk adjusted CAPM benchmark.

$$\alpha_p = R_{actual} - R_{predicted}$$

$$= R_{actual} - [R_f + \beta_p (R_m - R_f)]$$

A positive alpha reflects superior performance while a negative alpha indicates inferior performance.

Some other common forms of risk-adjusted performance measures are described below:

e. **The Information Ratio (or Appraisal Ratio)**

The information or appraisal ratio looks at the amount of excess return relative to the excess risk. When comparing two portfolios with positive alpha, the appraisal ratio compares the α_p per unit of excess risk. A higher information ratio suggest that the fund manager is generating more outperformance per unit of excess risk he is taking.

$$\text{Information ratio} = \alpha_p / \sigma_p$$

As such, the higher the information ratio, the better.

f. **The Sortino Ratio**

The Sharpe ratio is applied to portfolios with returns that are expected to be normally distributed. Both positive and negative deviation are included in the standard deviation calculation in the Sharpe ratio. However, when fund managers are able to protect their downside risk with derivatives, short securities and the market, and also enjoy the effects of leveraging to take advantage of positive volatility, they can create positively skewed portfolios. It would be unfair to penalize fund managers for having a high positive deviation and a positively skewed return distribution. Hence, with the Sortino ratio, the primary concern is downside risk, so the denominator is an average of downside volatility for the period that the fund is being evaluated. This is particularly useful in comparing the performance of hedge funds.

$$\textbf{Sortino ratio} = \frac{R_p - R_f}{\sigma_{negative}}$$

The Sortino ratio calculates the amount of portfolio return above the risk-free rate per unit of negative volatility.

8.6.3 Performance Attribution

Performance attribution seeks to determine the reasons why a portfolio manager did better or worse than a properly constructed benchmark. The purpose is to decompose the total performance of a portfolio into specific components that can be associated with specific decisions made by the portfolio manager.

Attribution analysis often begins from the broadest asset allocation choices and progressively focuses on finer details of portfolio choice. The difference between the performance of a managed portfolio and that of a benchmark portfolio may be expressed as the sum of the contributions to performance of a series of decisions made at the various levels of the portfolio construction process. For example, one common attribution system decomposes relative performance into three components:

a. Asset allocation choices across equity, fixed-income, and money markets;

b. Industry (sector) choice within each market; and

c. Security choice within each sector.

Part of the analysis process involves identifying a performance benchmark (called the bogey) for comparing the portfolio's results. The bogey is designed to measure returns if a "passive" strategy were adopted. Weights assigned to the bogey portfolio will depend on the risk tolerance of the investor. Any difference between the portfolio's results and the passive benchmark must be due to either asset allocation bets or security selection bets.

a. **Characteristics of Good Benchmarks**

Useful benchmarks should be:

i. Unambiguous – the asset class, securities, and composition should be clearly stated.

ii. Investable – the benchmark portfolio should be practically and realistically investable as a passively managed portfolio.

iii. Measurable – the returns on the benchmark portfolio can be obtained from the market and its performance measured.

iv. Consistent with the investment manager’s investment philosophy.

v. Bought in by the investment manager as an acceptable benchmark.

Example 8.6.3 - Portfolio Performance Benchmark with a Bogey Portfolio

To illustrate, consider the following hypothetical managed portfolio and bogey portfolio, which invest in equities, bonds, and money market securities:

Managed Portfolio

Asset Classes	Weight	Actual Return	Weighted Return
Equities	0.70	7.50%	5.25%
Bonds	0.10	2.00%	0.20%
Cash	0.20	0.50%	0.10%
			5.55%

Bogey Portfolio

Asset Classes	Benchmark Weight	Benchmark Return	Weighted Return
Equities	0.60	6.00%	3.60%
Bonds	0.30	1.50%	0.45%
Cash	0.10	0.50%	0.05%
			4.10%

Excess return = 5.55% - 4.10% = 1.45%

The benchmark weights have been set at 60% equities, 30% bonds, and 10% cash. The return on the bogey portfolio is 4.10%. Given a return of 5.55% on the managed portfolio, this represents an excess return of 1.45%.

The next step is to allocate the 1.45% excess return to the separate decisions that contributed to it.

b. **Asset Allocation Decision**

The effect of the manager's asset allocation choices can be computed by comparing the weights of the managed portfolio and the bogey portfolio as follows:

Asset Classes	Actual Weight	Benchmark Weight	Excess Weight	Benchmark Return	Contribution to Performance
Equities	0.70	0.60	0.10	6.00%	0.60%
Bonds	0.10	0.30	-0.20	1.50%	-0.30%
Cash	0.20	0.10	0.10	0.50%	0.05%
					0.35%

Asset allocation contributed 0.35% to the portfolio's overall excess return of 1.45%. The major factor contributing to the outperformance is the higher weighting of equities, which provided a return of 6%.

c. **Sector and Security Selection Decisions**

The remaining excess return of 1.10% is attributed to sector selection and security selection as illustrated below:

Asset Classes	Actual Return	Benchmark Return	Excess Return	Actual Weight	Contribution to Performance
Equities	7.50%	6.00%	1.50%	0.70	1.05%
Bonds	2.00%	1.50%	0.50%	0.10	0.05%
					1.10%

Equity return is 7.50% for the managed portfolio versus an equity return of 6% for the benchmark. The bond return is 2.00% for the managed portfolio versus 1.50% for the bond index. The superior performance in both equities and bonds, weighted by proportions invested in each asset class, contributed 1.10% to the overall excess return.

A summary of the portfolio attribution is as follows:

		Percentage
Asset allocation		0.35%
Sector and security selection		
- Equities	1.05%	
- Bonds	0.05%	1.10%
Excess Return of Portfolio		1.45%

8.7 Portfolio Management and Fund Sales

Depending on whether the investment representative is working in a fund management team, a bank, an insurer, or other segment in the financial industry, the role of fund or portfolio sales and after-care can be an integral part of fund management or a separate function. Operationally, there will be differences. However, there needs to be proper communication between the portfolio managers and originators of the product and services. In particular, the sales force should ensure that the right message is conveyed and that there is no misrepresentation with regards to:

1. What the portfolio is targeting to do (or what it can and cannot do).
2. The underlying assets and instruments used in the portfolio.
3. The risks that a client would be allowing the fund manager to assume.
4. The level of service and aftercare that is rendered by portfolio managers and service teams.
5. The reliance on 3rd party vendors, who receive payments (e.g., retrocessions) for promoting the fund.

Often, financial institutions have strong expertise and understanding of the product within the portfolio management team, but not within the sales force. Mis-selling can result when there is poor communication and a lack of enforcement of required practices when marketing portfolio management services.

8.8 Summary

Efficient Market Hypothesis (EMH)

There are 3 forms of the EMH:

1. **Weak Form of the EMH** - All past price information is factored into the current stock price. Past price and volume data have no relation with the future direction of stock prices. As no fixed ''patterns'' exist, technical analysis techniques will not be able to consistently produce excess returns.

2. **Semi-Strong Form of the EMH** - All available market and non-market public information have been factored into stock prices. This implies that neither fundamental analysis nor technical analysis techniques will be able to reliably produce excess returns.

3. **Strong Form of the EMH** - All market, non-market, and inside information have been factored into stock prices and no one has monopolistic access to relevant information. This implies that no one can earn excess returns.

Passive versus Active Management Strategies

1. Investors who subscribe to the EMH are likely to be inclined towards **passive management**, such as a **buy-and-hold** strategy. Passive investors believe applying such a strategy will, over the long term, produce results that are just as good as active management strategies.

2. Investors, who are non-subscribers to the EMH, generally favour **active management** strategies, which seek to outperform the market on a risk-adjusted basis. Active investors believe they have some advantages, such as superior analytical skills or better access to information, over other market participants. Examples of active strategies are security selection, sector rotation, and market timing.

Portfolio Theory

1. Studies have shown that portfolio risk is reduced as we increase the number of securities in the portfolio.

2. There are two sources of risk: systematic risk and non-systematic risk. Systematic risk is attributed to macroeconomic factors and cannot be eliminated by diversification. Non-systematic risk or firm-specific risk can be controlled and minimised through diversification.

3. Correlation between assets will change over time and, during major crises, correlations between financial asset classes rise.

4. For successful diversification, we should seek securities that are uncorrelated to each another. Otherwise, risk cannot be significantly reduced. Portfolio risk is lowest when the returns of assets are negatively correlated.

5. The capital asset pricing model (CAPM) is concerned with the equilibrium relation between the systematic risk and the expected return of assets.

6. The capital market line (CML) specifies the equilibrium relation between expected return and total risk for efficiently diversified portfolios.

7. The security market line (SML) specifies the equilibrium relation between expected return and systematic risk. This relation applies to individual securities as well as portfolios.

8. Beta is the relative measure of systematic risk for a security or portfolio. It indicates the volatility of a security or portfolio return relative to the volatility of the market portfolio's return, multiplied by the correlation coefficient.

9. The security market line (SML) is useful for identifying overvalued and undervalued securities, as well as cheap and expensive markets.

10. According to the arbitrage pricing theory (APT), security returns are affected by risk factors that represent broad economic forces.

The Portfolio Management Process

1. The portfolio management process is a dynamic, continuous, and systematic, with the following elements: (a) setting investment objectives, (b) developing and implementing strategies, (c) monitoring market conditions, (d) reviewing and adjusting the portfolio, and (e) measuring investment portfolio performance.

2. In defining the investment policy, the following objectives and constraints have to be analysed: investor's return requirement and risk tolerance, liquidity requirements, time horizon, tax considerations, legal and regulatory constraints, and client's unique needs and preferences.

3. Asset allocation involves determining the weights of each asset class in a portfolio so as to maximize return at a level of risk consistent with the investor's objectives.

4. There are two approaches to asset allocation. Strategic asset allocation constitutes the long-run asset mix and is not expected to change very often. Tactical asset allocation seeks to enhance return in the short run by shifting the asset mix of a portfolio in response to changes in economic and capital market conditions.

Performance Measurement

1. A proper measure of total return must consider both the income and capital components. Return measurements have to adjust for any deposits into, or withdrawals from, the portfolio.

2. For evaluating the performance of the portfolio manager, the time-weighted return is more appropriate than the dollar-weighted return since the former is not influenced by the size and timing of cash flows.

3. The modified Dietz method of computing returns is widely used as it does not require the valuation of the portfolio on the date of each cash flow. The modified Dietz method weights each cash flow by the amount of time it is held in the portfolio.

4. The Global Investment Performance Standards (GIPS) created a single global standard of investment performance reporting and increased minimum standards worldwide.

5. The most common composite measures of portfolio performance are:
   - Sharpe ratio, which relates the portfolio's excess return to the total risk (standard deviation) of the portfolio.
   - Treynor ratio, which relates the portfolio's excess return to the portfolio's systematic risk (beta).
   - Jensen's alpha, which measures the difference between the portfolio's actual return and the return that would be expected from the CAPM based on the portfolio's risk.

6. If a portfolio is inadequately diversified, its performance ranking using the Treynor ratio measure is likely to be higher than its ranking using the Sharpe ratio.

7. The Information or Appraisal ratio compares the active return generated per unit of active risk borne and can be used to compare the efficiency with which alpha is generated by active managers.

8. The Sortino ratio measures the return above the risk-free rate per unit of negative volatility. It is particularly useful for hedge funds when returns are positively skewed and non-normally distributed.

9. Performance attribution is concerned with why a portfolio manager did better or worse than the benchmark portfolio. The performance of a portfolio is decomposed into specific components to determine the reason for superior or inferior performance.

Chapter 9:
Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITs) and Insurance

Learning Objectives

The candidate should be able to:

- ✓ Describe exchange-traded funds (ETFs).
- ✓ Understand the role of ETFs in investments.
- ✓ Delineate the advantages and risks of investing in ETFs.
- ✓ Define a unit trust and describe its features.
- ✓ Distinguish between unit trusts and investment trusts.
- ✓ Define the advantages of investing in unit trusts.
- ✓ Describe the major types of unit trusts.
- ✓ Discuss the evaluation of unit trusts.
- ✓ Evaluate the charges and fees in unit trusts.
- ✓ Explain how the bid and offer prices of unit trusts are determined.
- ✓ Explain what a Real Estate Investment Trust (REIT) is.
- ✓ Compare REITs and business trusts and explain their differences.
- ✓ Explain what stapled securities are.
- ✓ Understand and describe the following alternative investment instruments:
    - Hedge Funds;

- Insurance;
- Annuities.

✓ Describe the CPF Investment Scheme.

9.1 Introduction

In Singapore, most pooled investments are known as **Collective Investment Schemes (CIS) or Unit Trusts**, and they are governed by the Code on Collective Investment Schemes (Code on CIS) issued by the MAS.

There are currently three main structures available in Singapore that can be set up which will allow a wide group of investors to participate in the investment management capabilities of asset managers, namely:

a. unit trusts;

b. investment holding/dealing companies; and

c. limited partnerships.

However, these structures are sometimes less than ideal as vehicles for investment funds. For example, a company can only pay dividends out of profits and there are restrictions on return of capital. Thus, the Variable Capital Companies Act was passed in October 2018 to govern a new structure called the Variable Capital Company (VCC), which is essentially a company with two key features. First, a VCC can freely issue and redeem shares and pay dividends using its capital. Second, a VCC can adopt an umbrella structure with multiple sub-funds, which increases operating efficiency.[47] This encourages funds that have structures similar to a VCC in other jurisdictions to re-domicile in Singapore and can potentially help Singapore enlarge its share of the global asset management market.

Now that we have introduced investment funds structures, we will proceed to examine the different types of pooled investment funds available. We begin by discussing conventional CIS such as exchange-traded funds (ETFs) and unit trusts, which invest in a wide range of products but, primarily, in listed equity and fixed-income securities. We then examine alternative CIS that focus on niche asset classes/strategies such as real estate investment trusts (REITs), business trusts, hedge funds, and private equity. Lastly, we will cover insurance products, which pool not only investment funds but mortality/longevity risks as well.

9.2 Exchange-Traded Funds

ETFs are funds designed to track the performance of specific indices, or fixed baskets of stocks or debt securities. They are different from most open-end funds or closed-end funds because they do not involve any active management of securities. The underlying assets may comprise local or international securities. In addition to stocks, an ETF may also be based on other asset classes such as fixed income, commodities, money market, or currency. Through ETFs, investors have access to a wide range of markets and asset classes.

Over the last two decades, there has been a proliferation of ETFs launched by different financial institutions on the major bourses around the world. Singapore has numerous SGX-listed ETFs that track the performance of major indices and selected asset classes (mostly long ETFs). Some of the more popular ETFs include the S&P

[47] See Appendix A1 for a discussion of the key features of the VCC structure.

500 SPDRs (pronounced as "Spiders") that tracks the S&P 500 Index, and the DJIA DIAMONDS that tracks the Dow Jones Industrial Average.

SPDR Straits Times Index ETF

The SPDR STI ETF is Singapore's first locally created ETF, designed to track the performance of the Straits Times Index (STI). Originally listed as "Street TRACKS STI", it was listed and traded on SGX-ST since April 2002. The ETF's investment objective is to replicate the performance of the STI as closely as possible, before expenses.

Table 9.2: Features of the SPDR STI ETF

Trading name	STI ETF 100
Pricing	About 1/100th of STI level. For example, if the STI is at 1,700, each unit of STI ETF 100 will be priced at $17.00.
Minimum bid size	$0.01
Board lot size	100 units
Dividend yield	Dividend yield is expected to approximate the yield on the STI. Investors can expect to receive dividends twice a year.
Annual costs	0.3% (includes management fee, trustee fee, and other fund expenses)

The SPDR STI ETF is like a mutual fund that tracks the performance of the STI. The ETF invests in the component stocks of the STI in proportion to their respective weights in the index. ETF units can be bought and sold by investors during trading hours like any security listed on the Exchange. This is unlike mutual funds, which can only be bought or sold at the end of the day with the mutual fund manager.

ETFs are passive investment instruments that seek to replicate the price movement of a benchmark index made up of a basket of securities. If the replication is good, the price movement of the ETF should follow the benchmark.

The ETF's Net Asset Value (NAV) reflects the value of the ETF units owned by the investor and is determined at the daily close of trading in all market(s) that the ETF is listed, using a process typically explained in the Trust Deed or Prospectus. The NAV per unit represents fair value for the ETF. However, the ETF's market price can deviate from its NAV per unit, as price is also determined by investor sentiment, which affects the supply and demand for the units.

ETFs have been designed to minimise commissions and expense ratios compared to an average mutual fund. Annual fees are lower because ETFs are passively managed funds. The main costs of investing in an ETF are the total expense ratio and the brokerage commissions when buying or selling the ETF. Instead of buying the STI ETF on the exchange, investors can also buy it from primary or participating dealers, who effectively purchase the ETF shares from the ETF sponsor and offer them to the market.

Unlike a typical unit trust that creates units for investors when they purchase units, units of the STI ETF can only be created for, or redeemed by, participating dealers in sizes of 100,000 units. To create an ETF share,

the participating dealer has to deposit a specified portfolio of stocks that closely approximates the underlying securities of the index to the trustee of the ETF. To redeem, ETF shares are delivered to the trustee of the ETF in exchange for the underlying basket of securities.

ETFs on the exchange are quoted at bid and ask prices, which track but do not necessarily always reflect the fund's NAV. If the exchange-listed price deviates sufficiently from the NAV, arbitraging will reduce the differential. For example, when the traded price of an ETF is higher than its NAV, an arbitrageur will sell the ETF shares and buy the underlying portfolio of securities. If the traded price of the ETF falls below the NAV, the arbitrageur will sell the underlying portfolio of securities and buy the ETF shares.

9.2.1 Types of ETF Structures

ETFs are either physically or synthetically structured to replicate the basket of stocks or bonds that make up the index that is being traded by the ETF. Physically replicated ETFs comprise cash holdings of the actual securities in the proportions reflected in the index. Synthetically replicated ETFs comprise statistically correlated holdings of securities which, together, mimic the price behaviour of the index. Synthetically replicated ETF may use swaps or derivatives to replicate the index.

a. **Cash or Direct Replication (Physical ETFs)**

Direct or cash replication is more common in ETFs. There are two ways of doing this:

i. **Full replication,** where the ETF purchases all of the listed stocks in the exact proportions that make up the index; and

ii. **Representative sampling,** where the ETF invests in a subset of the index, comprising the more dominant stocks in the index. This method is used when the stocks that make up the index are not liquid.

b. **Synthetic Replication (Synthetic ETFs)**

Synthetic replication uses swaps and derivatives to mimic the index. One advantage of this replication method is that it reduces the error of the ETF in tracking the index. A disadvantage is counterparty risk, as the counterparty to the derivative contract might default on its obligations.

An example of synthetic replication is an ETF that uses a total return swap, which is a contractual arrangement to swap or exchange interest or return obligations. Swap-based ETFs usually do not pay dividends. Instead, they enter into total return swap where the counterparty to the ETF will deliver periodic payments of income and capital gains return on the basket of stocks that make up the reference index. The swap-based ETF would pay a fee to the counterparty that is contractually obligated to deliver the returns of the reference index to the ETF. Hence, there is a risk to investors if the counterparty cannot fulfil its payments.

9.2.2 Investment Strategies with ETFs

ETFs may be used with the following strategies:

a. Long-term buy-and-hold strategies;

b. Short-term tactical trading because of the price transparency of ETFs and their liquidity;

c. Core-satellite allocations, where the fund is made up of a core that mimics a board-based index, and satellites of higher-risk securities that potentially provider higher return;

d. Hedging against other risks in the investment portfolio;

e. "Cash equitization", which is converting cash to equities through investing in ETFs, to minimise the cash drag[48] on a portfolio.

9.2.3 Securities Borrowing and Lending for ETFs

Securities lending is a practice where ETFs pay agents to lend out shares in their portfolios to other investors and the ETFs earn interest for lending the stocks. For example, an ETF can transfer the stocks to another investor over the lending period (e.g., 30 days) in exchange for collateral such as cash or other securities equivalent to e.g., 105% of the stocks' value. If the ETF needs to sell the lent stocks, it will do a reverse transaction with the borrower.

Usually, the stock borrowers are investors who have sold the stock short in anticipation of a fall in the stock price and need to borrow the stock for delivery, with the intention to subsequently buy back the stock at a lower price and return it to the ETF provider.

However, if the borrower is unable to fulfil its obligations under the securities lending arrangements, the ETF will face a loss equal to the value of the lent stocks. Thus, securities borrowing and lending creates counterparty risks for the ETF.

9.2.4 Advantages of Investing in ETFs

The following are some of the advantages of investing in ETFs:

a. **Diversification.** Buying ETF shares allows investors to gain exposure to a diversified portfolio of securities.

b. **Small capital outlay.** Diversification can be achieved with a small capital outlay.

c. **Low costs.** The annual management fees for ETFs are normally less than 1% compared to the 1-2% that is charged by actively managed unit trusts. In addition, there are no initial charges or redemption fees when trading ETFs.

d. **Liquidity.** ETF shares can be traded at any time during the trading hours of the exchange at prevailing prices. Unit trusts, on the other hand, can only be transacted at their daily closing NAVs.

e. **Transparency.** Investors are always aware of the basket of securities that are represented in an ETF whereas unit trusts usually provide portfolio holdings information only on a quarterly, semi-annual, or annual basis.

9.2.5 Risks of Investing in Standard ETFs

The risks in cash-replicated ETFs (physical ETFs) and synthetically replicated ETFs (synthetic ETFs) are different in the following respects:

a. **Tracking error** is greater in physical ETFs than in synthetic ETFs, principally because synthetic ETFs can achieve more precise tracking using swaps and derivatives.

[48] The underperformance of a portfolio from holding idle cash.

b. **Counterparty risk.** Synthetic ETFs have counterparty risks from the swap and derivative arrangements which they enter. Investors are therefore exposed to the risk of counterparties not fulfilling their obligations on those instruments. Physical ETFs can also have counterparty risk, which would arise if third-party custodians have securities lending arrangements with other parties.

c. **Market and sector risks.** When the prices of the underlying securities fall, the value of the ETF will suffer. Sector-related funds may be negatively impacted by the performance of the specific sectors that they track.

d. **Foreign exchange risk.** Investments in ETFs that have constituent securities that are traded in foreign currencies are subject to foreign exchange risk that may affect overall returns.

e. **Taxation.** Non-US investors holding US-established ETFs may be subject to US withholding taxes on dividends and other distributions.

9.2.6 Smart Beta ETFs

A **smart beta ETF** seeks to provide a better risk-return performance than an index of selected stocks. The selection and weighting of the index stocks is based on a set of criteria that relate to better-performing stocks (e.g., improvements to cashflow, dividends, revenue/earnings ratios, etc.).

In contrast to a standard ETF that typically follows an index's market capitalization weighting, a smart beta ETF will select, and weight securities based on some positive performance criteria. To some extent, smart beta ETFs simply entail good fundamental stock selection that weights the top performers more heavily than other stocks using a set of quantitative criteria.

Other smart beta ETFs seek to exploit market anomalies or overlay technical analysis measures such as momentum and volatility in making their selections. There is a growing number of so-called smart beta ETFs being launched on the major exchanges.

9.2.7 Volatility (VIX) Driven ETFs

A volatility index (VIX) or VIX futures ETF is another type of ETF that is becoming increasing popular. Such ETFs are meant for very short-term trading rather than for investing.

In a bullish market, volatility will generally fall. Traders might try their best to trade into long volatility ETFs on days when the volatility spikes. However, this is difficult to predict and time consistently. Holding on to a long volatility position to wait for the upward spike can be deadly as volatility trends downward, and a 1-2% fall each day will soon lead to massive losses over time. High volatility generally does not last, and the beneficiaries tend to be those that shorted volatility (i.e., invested in the inverse of the VIX). Please refer to Figure 9.2.7(a).

Figure 9.2.7(a) - Trading Volatility in the VIX using ETFs is Difficult



Table 9.2.7(b) - Volatility-Driven ETFs and ETNs in the US

CODE	POPULAR VOLATILITY ETFs AND ETNs	Total Assets ($m)	YTD Return	Avg Volume	1 July 2021 Closing Price
VXX	iPath S&P 500 VIX Short-Term Futures ETN	$1,009.7	-56.85%	17,964,400	$29.15
SVXY	ProShares Short VIX Short-Term Futures	$585.3	35.26%	2,794,200	$55.88
VXZ	iPath S&P 500 VIX Mid-Term Futures ETN	$36.3	-17.29%	12,000	$25.36
VIXM	ProShares VIX Mid-Term Futures ETF	$92.5	-17.11%	45,900	$30.39
XVZ	iPath S&P 500 Dynamic VIX ETN	$12.9	1.78%	4,800	$37.90

Table 9.2.7(b) shows a list of the EIX ETFs in the US. Under bear market conditions, volatility will be more significant and frequent. This will benefit those who have long volatility ETF positions. Even then, market conditions can change quite dramatically and rapidly. It is, therefore, important for the trader to be fleet-footed and disciplined to realise target gains and limit losses.

9.2.8 High Risk Leveraged ETFs and Long-Short ETF Pairs

One of the trends worth paying attention to is the rise of **leveraged ETFs**. A leveraged ETF uses financing and financial derivatives (such as options and futures) to drastically amplify the potential returns arising from tracking a particular underlying index. Such ETFs are usually leveraged 2x or 3x the index value and there are also those that track the inverse movement of the underlying index. Leveraged ETFs are often issued in pairs (long and short a particular index).

Large speculators with enough financial muscle will, at times, attempt to drive the index one way or the other by moving key index securities in order to make leveraged profits of 2x-3x on their ETF position. The practice is not exactly illegal unless there is proof that parties have colluded to manipulate the movement of the index. Such manipulation activity tends to be greater for the smaller stock indices (e.g., the NASDAQ Biotech Index or the NYSE Arca Gold Miners Index). In general, leveraged ETFs are not suitable for retail investors.

Leveraged ETFs typically amplify the daily percentage price change. A 2-4% daily change is normal for leveraged ETFs and a +/- 20-25% change is possible in a single trading session, so there is potential for massive gains and losses.

Singapore has several inverse ETFs (e.g., Deutsche Bank x-trackers S&P 500 Inverse Daily UCITS ETF), which traders use at specific points in the US market cycle. In Figure 9.2.8(a), the value of the DB x-trackers S&P 500 Inverse Daily UCITS ETF was at USD15.99 in November 2017, which is less than a quarter of its value of USD79.72 when it first began trading. In the near term, there may be sufficient volatility to warrant trading interest; however, betting against a bullish market is difficult and traders need to be nimble to make any gains.

Figure 9.2.8(a) - Example of Inverse ETF in Singapore (DB x-trackers S&P 500 Inverse Daily UCITS ETF)



Table 9.2.8(b) lists some examples of popular leveraged ETFs.

Table 9.2.8 (b) - Example of Popular Pairs of Leveraged ETFs

Long Position	YTD % 1 Jul 21	Short Position	YTD % 1 Jul 21	Daily Performance Of
LABU (3x)	-21.89	LABD (-3x)	-28.53	S&P Biotechnology Select Industry Index
NUGT (2x)	-16.43	DUST (-2x)	-3.62	NYSE Arca Gold Miners Index
YINN (3x)	-9.22	YANG (-3x)	-16.52	Daily performance of the FTSE China 50 Index
TQQQ (3x)	+34.83	SQQQ (-3x)	-39.79	NASDAQ 100 index

Inverse or short ETFs can be issued independently, but they are often introduced as part of a pair to long positions. This allows traders to make a profit or hedge their positions when a particular index fall. The inverse ETFs can help traders offset loss-making long positions during a trading day or over a short period.

Finally, due to the sharp movements of leveraged ETFs, issuers of leveraged ETFs often carry out reverse splits to reset prices at higher levels. Hence, you will see this regular price resetting taking place for many of the popular ETFs.

Clearly, some ETFs are not meant for the average retail investor or even institutional portfolio investments. Rather, they are intended for speculative traders who are willing to take on substantial risks and volatility and have the time and emotional demeanour to manage these positions effectively.

9.3 Unit Trusts

A **unit trust** is a CIS that pools funds from different investors to be invested and managed by one or more fund managers under the care of an independent **trustee**. In Singapore, the Code on CIS sets out the best practices on the management, operation, and marketing of schemes that trustees and managers are expected to observe. In the US, a CIS is known as a **mutual fund** and may not be issued in the form of a trust.

In the UK, the term unit trust is actively used. In Germany, it is referred to as an investment fund. In Switzerland and Luxembourg, the more common term is **Sicaf** (Société d'investissement à capital fixe) for closed-end investment company and **Sicav** (Société d'investissement à capital variable) for open-ended investment company. It is important to note this naming convention since the bulk of offshore investment funds are domiciled in Luxembourg.

In Singapore, the unit trust is the most common form of CIS. It is set up to pool the financial resources of small investors to invest the funds in securities under the management of professional fund managers. The cash and investments are registered in the name of, and held by, the trustee, whose job is to safeguard the investors and ensure that the managers carry out their job within the terms of the **trust deed**. This document, which is an agreement between the trustee and the managers, sets out the aims and objectives of the trust, the charges for management and the basic rules of operation. For some offshore funds, redemption and purchase of fund units are carried out directly by the servicing team for the asset managers.

9.3.1 Fund Domicile

The domicile of a fund is where the fund is based and set up. A fund can be registered in multiple jurisdictions, but it can only have one place of domicile.

There exists heavy competition between Singapore and other jurisdictions to be the domicile for funds. Singapore benefits from being a hub for fund management activities, having a business-friendly and well-respected regulatory framework, as well as having strong communication and transport infrastructure on top of existing tax treaties. By launching new fund structures such as the VCC, Singapore hopes to encourage a larger portion of alternative investment funds to be domiciled and managed in Singapore.

9.3.2 Local and Offshore Unit Trusts

Singapore also allows the marketing of offshore funds, regardless of whether those funds are managed by Singapore managers. Hence, funds domiciled in Singapore are also competing for investor attention with offshore funds that are marketed actively by foreign banks.

Under Singapore law, there are three ways to offer participation in offshore funds:

i. Through a public offering on the local exchange (e.g., foreign-domiciled ETFs with Singapore agents);

ii. Through a private placement; or

iii. By invoking an exception under Division 5A of Part IV of the Companies Act (Cap 50) (Division 5A — Exemptions from Divisions 1 and 5 in relation to Prospectus Requirements)[49]. This exception allows for offshore funds to be marketed to Institutional Investors, Expert Investors, and Accredited Investors[50] in Singapore.

The presence of both offshore and local-domiciled funds is beneficial for Singapore as a global wealth management and asset management hub.

9.3.3 Open-End and Closed-End Unit Trusts

The differences between open-end and closed-end unit trusts are shown in Table 9.3.3:

Table 9.3.3 – Open-End vs Closed-End Unit Trusts

Features of a Unit Trust	Types of Unit Trusts	
	Open-End	**Closed-End**
Market	Over-the-Counter	Exchange Traded
Counterparty	Issuer	Listed exchange or issuer
Exit	Sale or redemption through trustee	Sale in the market
Issued shares/units	Continuous issuance	Fixed number of units
Price determined by	Net Asset Value or NAV (performance of fund net of management fees)	Market demand and supply as well as performance of fund

In Singapore, most unit trusts are "open-end". They are not listed on the exchange[51] and are offered to retail investors. The "open-end" feature describes the availability of fresh units that the fund manager can sell to retail investors. Closed-end funds do not issue fresh units - the number of units is fixed, and these funds are usually sold to accredited investors via private placements.

A unit trust is based on the principle of dividing a fund into equal portions, which are referred to as **units**. Each unit represents the same proportion of the value of the assets held by the trust. For example, if a unit trust's investments are worth $10 million and there are 5 million units issued, each unit will be worth $2. Most unit

[49] Division 5A of Part IV of the Companies Act (Cap 50) (Division 5A — Exemptions from Divisions 1 and 5 in relation to Prospectus Requirements).

[50] For definition of the different investor types, please refer to Section 1.1 of Chapter 1.

[51] All CIS are posted on the MAS website https://www.mas.gov.sg/regulation/capital-markets/offers-of-collective-investment-schemes. Another useful website for funds in Singapore is www.fundsingapore.com.

trusts' investments are valued daily and, each day, the previous day's valuation prices are published in the newspapers and serve as the opening dealing prices for that day. The value of a unit trust is its **net asset value (NAV)** which is the value of the unit trust's assets minus its liabilities.

9.3.4 Role of Trustee

The managers neither hold the investments nor issue or redeem units. This is done by the trustee[52], who holds the trust's cash, shares, and other investment assets. In an open-end trust, the trustee will create new units when there are new purchases and redeem units when the unit holders liquidate their holdings. Hence, the capitalization of such a unit trust is "open", with the number of units outstanding changing daily.

A prospectus is typically issued with details about the unit trust, such as the investment objectives, focus and approach, its performance, fee charges, and so on. Unit trusts are more suitable as medium to long-term investments. Foreign funds (not incorporated in Singapore) may also be offered to the public if these are ''recognised funds'' for retail distribution.

9.3.5 Advantages of Investing in Unit Trusts

There are many benefits from investing in unit trusts:

a. **Diversification and divisibility.** By pooling investors' money, unit trusts enable shareholders to hold fractional shares of many different securities. A diversified portfolio of securities can shield the investors from large losses.

b. **Small capital outlay.** The initial investment outlay (e.g., $1,000 to $5,000) is affordable to individual investors. For this amount, the average investor can invest in a portfolio of securities that would otherwise be out of his reach.

c. **Professional management.** Most unit trusts have full-time staff of security analysts and portfolio managers who have experience in managing investments and will be able to respond quickly to changes in market conditions.

d. **Flexibility.** The choice of unit trust does not need to be restricted to the initial purchase. Some investment companies allow the unit trust holders to switch between a "family of funds" – i.e., the unit holder can switch from one fund to another within the same family conveniently. Switching between funds provides unit holders flexibility as they can respond to changes in their investment plans and market movements. For example, if the investment plan changes from low to moderate risk, the unit holder could switch from an income fund to an equity fund.

e. **Liquidity.** Unit holders can sell their holdings to the trustees, who must buy back the units based on the unit trust's NAV, less the relevant charges. If the investor held shares in 30 companies instead, it would be more time-consuming to sell all of them.

f. **Lower transaction costs.** By trading large blocks of securities, investment companies can achieve substantial savings on brokerage fees and commissions.

[52] Please refer to the MAS Financial Institution Directory at Financial Institutions Directory (mas.gov.sg) for list of CIS-approved trustees in Singapore.

g. **Security.** The assets of a unit trust are always legally held on the investors' behalf by an independent trustee, and not by the fund manager.

9.3.6 Disadvantages of Investing in Unit Trusts

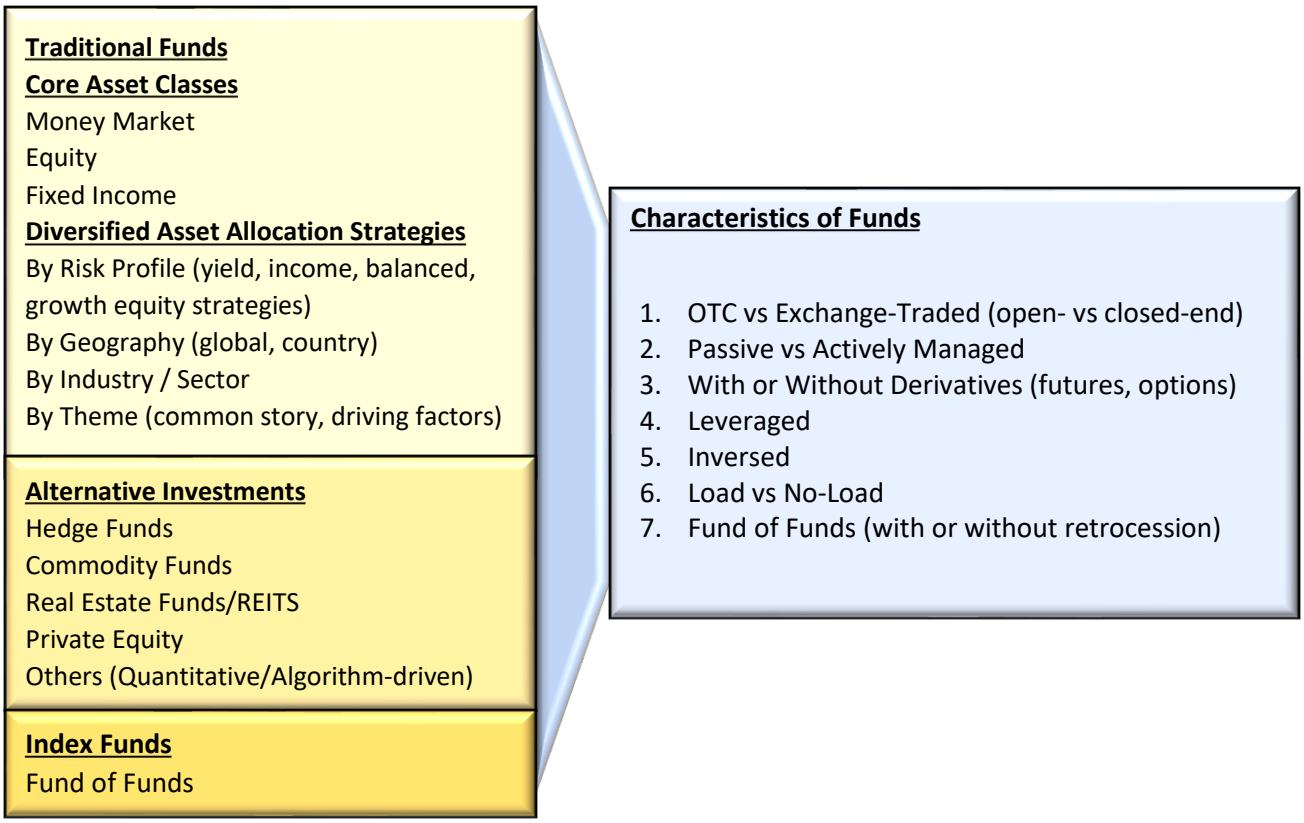
a. **Sales charges.** Relatively high sales charges are incurred when an investor buys into a unit trust, and transaction costs can be high if there is frequent buying and selling of unit trusts.

b. **Management fees.** Unit trusts pay annual management fees to the fund manager regardless the fund's performance. Investors should compare the fees among different funds as high management fees will affect the fund's return.

9.3.7 Major Types of Unit Trust Funds

Unit trusts can be classified in terms of the assets they invest in, such as money market funds, equity funds, and bond funds. Please refer to Figure 9.3.7 for the different types of unit trust funds.

Figure 9.3.7 – Types of Unit Trust Funds



The following are examples of various types of unit trust funds in Singapore:

1. **Money market funds** hold short-term (typically less than one year) fixed-income instruments, such as bank certificates of deposit, commercial paper, and Treasury bills. Because of the large denominations involved in direct dealing, these funds provide an avenue for investors seeking competitive money market rates. Interest income is earned and credited daily. Investors in money market funds normally do not pay a sales charge or a redemption charge but they do pay a management fee.

2. **Fixed-income funds** invest in fixed-income securities. Some funds specify that only particular types of bonds will be purchased. There are corporate bond funds, US government bond funds, convertible bond funds, high-yield bond funds, and so on.

3. **Equity funds** invest in equity securities. A few specialized investment companies concentrate on the securities of firms in a particular industry or sector. For example, there are chemical funds, aerospace funds, technology funds, and gold funds. Others provide a convenient means for holding the equity securities of firms in a particular country, such as Korea and Japan funds. Some funds may have a regional orientation, such as European or Asian equity funds, while others may be globally invested.

4. **Balanced funds** invest in both equity and fixed-income securities. The basic objectives of balanced funds are to generate income as well as long-term capital growth. These funds typically hold relatively constant mixes of bonds and equities. They have fairly limited potential of price increase but provide a high degree of safety and moderate to high income potential. Somewhat similar to balanced funds are **flexible income funds**. These funds seek to "provide liberal current income" by altering the proportions of equities and bonds periodically based on market expectations to rebalance the fund composition.

5. **Multi-asset funds** generally consist of more than 2 asset classes. In addition to equity and fixed income securities, multi-asset funds also include other assets like private equity and commodities. The premise of multi-asset funds is that, by diversifying into different asset classes, the funds will be able to withstand various investment conditions as each asset class contributes to returns at different points of the market cycle. Consequently, the investor would not need to rebalance his portfolio when market conditions change. In addition, the fund manager will make tactical calls. Multi-asset funds are suitable for investors who would like to be well diversified and are willing to rely on a fund manager to make the investment calls.

6. **Index funds** are designed to track the performance of a specific market index. The fund is "passively managed" in a fairly static portfolio and is always fully invested in the securities of the index that it tracks. If the overall market advances, a good index fund follows the rise and vice versa if the market declines. The management fees of index funds are significantly lower than those charged by actively managed funds.

7. **Fund of funds** are part of an investment strategy that allocates investments to the fund managers who are likely to have the “best” performance, choosing managers with different fund strategies at different periods in the market cycle in order to achieve the targeted return. Many asset managers now invest directly in funds instead of selecting securities. Others have funds taking up 70-80% of their portfolio so, in a way, many large fund managers are a “fund of funds”.

8. While such managers substitute funds instead of the underlying securities to gain scale economies and liquidity advantages, the Fund of Funds approach is different in that a set of qualitative and quantitative criteria are used to evaluate and select the “best” fund managers. There is also a deliberate attempt to overweight and underweight fund managers based on their specific approach and strategy, especially for Fund of Hedge Funds.

9. **Multi-manager funds** invest in two or more external funds managers or other unit trusts. The main selling point of a multi-manager fund is the availability of a larger set of expertise. For example, a global equity multi-manager fund may be investing in several regional mutual funds. Each of these regional unit trusts is managed by a professional who specializes in that specific region.

10. Multi-manager funds are easily confused with multi-asset funds since multi-asset funds may also engage several underlying fund managers to manage the different asset classes in the fund. As such, both use a set of subsidiary fund managers. Note that more fees are incurred when more managers are involved.

11. **Capital-guaranteed funds** It is not usual for funds to offer investors a capital guarantee. There are specific guidelines for a fund to use terms such as “guaranteed”, “assured”, “insured”, or “warranty”, either in its name or in its promotion and description. Under the Code on CIS - Appendix 4, when such terms are used, the fund must:

    a. have an eligible guarantor with a minimum rating of AA (for financial institutions) and AAA (for other companies). The guarantor should not be an issuer of transferable securities and money market instruments that constitute more than 10% of the scheme’s NAV; and

    b. have a written agreement of an unconditional guarantee provided by the guarantor to the trustee, which can be legally enforced in Singapore by the trustee on behalf of the investors. The guarantee should be in respect of not less than 100% of the capital invested by the participants, less initial sales charges or front-end loads.

    As such, the preference is to not use the terms such as “guarantee” listed above. Most capital-guaranteed funds have similar structures, allocating most of their capital to low-risk assets such as fixed-income securities and money market instruments. This portion of the fund helps to preserve the capital at maturity. The remaining balance is invested in riskier assets such as options linked to equities, funds or indices, or direct equities, which provide the upside return potential. The maturity of the funds is determined by the expiration date of the guaranteed portion. Usually, the capital guarantee will apply only if the funds are held to maturity.

12. **Capital preservation funds**. These funds are different from capital-guaranteed funds. They aim to provide some level of capital preservation. The language used is less onerous for the issuer and the fund does not require a guarantor.

13. **Hedge funds**, unlike traditionally managed funds, employ a wide variety of investment techniques such as the use of derivatives, short selling, and arbitrage to generate returns. They tend to focus on achieving absolute returns rather than relative performance against a market index. In a market-neutral fund, each long position is offset by a short position of equal value. This reduces the fund's correlation with the market. The name 'hedge fund' can be a misnomer as hedge funds are not necessarily "hedged". For example, a hedge fund manager may take directional bets on the markets (i.e.; they can be net long or short). Hedge fund managers often invest their own money in their funds and are paid a performance fee.

14. **Quant funds** use price and market modelling based on complex mathematical algorithms to seek out investing and trading opportunities in the market. Quite probably, the best known example is Long-Term Capital Management, whose board of directors included Myron S. Scholes and Robert C. Merton, who shared the 1997 Nobel Memorial Prize in Economic Sciences for a "new method to determine the value of derivatives".

15. **High frequency trading (HFT) funds** rely on mathematical algorithms and high-speed data transmission to seek out profitable investing and trading opportunities in the market. Although not all HFT funds are hedge funds, there are many hedge funds who use quantitative models and algorithms, and some have reported significant profits from using HFT.

16. A **family of funds** or **umbrella fund** is a set of funds with different investment objectives offered under a single unit trust. Umbrella funds generally permit their investors to switch from one fund to another within the family at little to no cost. This feature allows the sub-investor to inexpensively change his investment strategy as the need arises.

Investors in Singapore can tap on their CPF funds to invest in products included under the CPF Investment Scheme (CPFIS). To protect the interests of CPF members, fund managers who manage CPF monies must

report vital information quarterly to the CPF Board. In addition, a funds-tracking company is appointed to publish information on performance and risks of CPFIS-included funds to facilitate investment decision making.

9.3.8 Fund Documentation - Prospectus, Product Highlights Sheets, Monthly Fund Fact Sheets, Annual Reports

Nowadays, with access to asset managers' websites, information on a given fund can be easily obtained from its **prospectus** and **product highlights sheet**. For new fund launches, investors can look at the prospectus and product highlights sheet, which will provide details about the asset managers, investment approach adopted, types of investments selected, the risk exposures, and information on the different types of fees and how they are calculated.

The **monthly fund fact sheets** and **annual reports** provide investors with regular updates on the performance of the funds and some summary information about the funds. Summary information on the monthly fund fact sheets include:

1. Fund objective
2. Key information on fees
3. Performance data and ratios
4. Benchmark comparison
5. Top holdings
6. Country allocation
7. Risk statistics
8. Sector allocation

For fund fact sheet illustrations, please refer to www.fundsingapore.com.

9.3.9 Fund Objectives

The statement on the fund's objectives will indicate the target outcomes.

The fund objectives should also give an idea of the investment approach taken to achieve the outcomes, e.g., through a diversified portfolio with different asset classes, a diversified portfolio of global equities for long-term capital growth, a diversification within a sector only, or investing in a non-diversified portfolio.

9.3.10 Fees and Expenses

a. **Load vs No-Load Funds**

Unit trusts can be classified into **no-load funds** and **load funds**. A no-load fund neither imposes an initial sales charge nor redemption charges, i.e., the units are priced at the NAV and the bid and offer prices are the same. A load fund with a front-end load charge is sold to investors at the NAV plus a sales commission (i.e., the load). Most load funds are sold by brokers, financial planners, and employees of insurance companies. The load often serves as a commission for the services of these distributors of the unit trust and the front-end load is usually 1.5 to 5% of the NAV.

When unit holders want to sell their units, they usually receive an amount equal to the NAV of the unit trust multiplied by the number of units sold. However, some funds impose a **realization charge** or **redemption fee**, also referred to as the **back-end load**. This fee may be waived if the investor has owned the units for more than a stipulated period (e.g., 60 days), so its basic purpose is to discourage investors from selling their units right after buying them. By discouraging excessive trading, the unit trust avoids the transaction costs associated with having to sell securities frequently in order to meet investors' redemption requirements.

b. **Management Fee**

All unit trusts charge annual **management fees** to compensate their professional managers. This fee typically ranges from 0.5% to 2.0% of the NAV and is accrued daily and debited directly from the unit trust's asset value.

c. **Various Operational Costs**

Under an umbrella fund structure, a **switching fee** of about 1% may be imposed when the unit holder switches from one sub-fund to another. Frequent switching of funds can, thus, be an expensive affair for unit holders. Other charges include annual **trustee, custodian and audit fees**. These operating costs can be accrued daily, monthly, or as and when they occur and are deducted from the assets of the unit trust. These fees usually range between 0.1 to 0.15% of the NAV.

There are also other costs involved in setting up a unit trust, which would include **legal and marketing costs**. These costs can either be a one-time charge to the unit trust or amortized over a number of years. In addition to a front-end load, investment companies may charge a **distribution fee** annually, which pays for advertising, promoting, and selling the unit trust to prospective purchasers. These expenses, like the management fee, are charged directly against the NAV of the fund. Some of these costs may be seemingly "waived" but end up as expenses charged on the portfolio.

d. **Performance Fee**

In some funds, an additional **incentive charge** is paid out to the fund manager for substantially exceeding benchmark targets. Naturally, there are measures to ensure that a fund manager does not unfairly benefit from a fund that recovers after first falling sharply. For instance, a certain performance level (high watermark) is set to ensure that the fund manager generates a certain level of profits for the fund before the performance bonus can be taken out of the NAV.

The total expense ratio measures total expenses (including management fees) against the net asset value of the fund. When making comparisons between funds, this ratio is often reviewed to assess a fund manager's operational efficiency.

9.3.11 Performances and Benchmarks

Past performance is another factor that must be considered when selecting a unit trust. While "past performance is no guarantee of future performance", it is still a useful guide for investors. A unit trust is usually managed relative to a benchmark so that the investor can compare the fund's return against the benchmark. If the unit trust's return is higher than the benchmark, the fund manager has done well, and vice versa.

Typically, the performance of a fund is charted against the benchmark since the fund's inception, accompanied by a table showing fund performance over the last 3 or 5 years. It is also important to check whether the return calculations also show the fund's returns net of management fees to gain a better picture of the performance of the fund. Most fund fact sheets should indicate returns net of fees.

Where a fund has dual currency tranches, the fund's return may also be reflected in the alternate currency. For example, the fund's return in USD may be quite different when reflected in SGD or EUR given exchange rate movements over the period.

For traditional funds with a relative return target, the benchmark is usually an index. A few funds might use a weighted combination of indices from two or more asset classes. For funds with absolute return targets, an interbank rate (e.g., LIBOR+1%) might be the benchmark target to beat.

9.3.12 Net Asset Value

For funds sold over-the-counter (OTC), the value of the fund is called the Net Asset Value (NAV) which is:

$$\frac{\text{Fund Securities + Other Fund Assets - Liabilities (including mgt fees) - Debt used for leveraging (if any)}}{\text{Total fund units outstanding}}$$

For open-end funds, the outstanding units will change with additional issuances and redemptions. The NAV of the fund will rise and fall with the value of its investments and should be reflected net of management fees.

9.3.13 Bid-Offer Price

There is a difference between the NAV and the price at which an investor can buy and sell the fund. The principle is that a unit holder buying new units should pay a sum sufficient to purchase an amount of the underlying securities in the trust in proportion to his holding, including the expenses necessary for that purchase. The existing unit holders should not unfairly bear the costs associated with a transaction that they did not create.

To achieve equity between incoming and outgoing unit holders, unit trust valuations show two prices, an offer price and bid price. The **offer price** is the price at which an investor can buy the units. This price represents the cost of creating new units that are equal in value to existing units based on market prices at the latest valuation.

The steps involved in calculating the offer price are:

a. Determine the NAV per unit by assessing the market value of the unit trust's securities.

b. Add expenses related to purchase of securities such as stockbrokers' commission and clearing fees.

c. Add cash held by trust and accrued income.

d. Add manager's initial charge.

Conversely the **bid price** is the price at which the investor can sell the units. It is equivalent to the NAV of each unit should all of the securities be sold and the proceeds distributed in cash to unit holders. The steps involved in calculating the bid price are:

a. Determine the NAV per unit by assessing the market value of the unit trust's securities.

b. Deduct expenses related to the sale of securities.

c. Add cash held by trust and accrued income.

9.3.14 Top Holdings and Concentration Risk

Many funds do not allow a single security to constitute more than 5% of total asset holdings, but some managers may allow the fund's exposure to a particular security to rise to as high as 8-9% in an attempt to outperform the market.

The choices made for top holdings can significantly affect returns, concentration risk, and volatility. Under- or over-weighting certain securities that have a dominant position in the benchmark indices (e.g., Tencent, Alibaba, Apple in technology, especially) can have a material impact on the fund's performance.

9.3.15 Allocation by Asset Class, Market and Sectors

The fund fact sheet will show a breakdown of how the fund is invested by asset class (for multi-asset funds), by market and by sector.

Reviewing the fund's allocation is useful because it gives the investor an idea of how much exposure is favoured in one country or sector over others and some investors may prefer to have more exposure in one country or sector over another.

9.3.16 Risk-Adjusted Return and Other Risk Ratios

Besides looking at benchmarked return performance, we should also review the portfolio's risk-adjusted returns and risk using measures such as the following:

a. Sharpe ratio;
b. Treynor ratio;
c. Beta;
d. Jensen's alpha;
e. Information ratio; or
f. Sortino ratio (for alternative investment funds with asymmetrical return distributions).

Please refer to Chapter 8 for details on how these measures are calculated. Generally, investors should pick funds that generate higher returns above the risk-free rate per unit risk taken.

9.3.17 Evaluating Suitability of Unit Trusts and Foreign Funds Sold in Singapore

When evaluating whether a unit trust is suitable for a prospective client, all aspects in relation to understanding a client (as discussed in Chapter 8) will apply. By building a client profile using the concepts of client understanding and using information in a unit trust's fund fact sheet as highlighted in Section 9.3.8, investment representatives would be able to select funds that meet their clients' investment needs. Table 9.3.17 sets out the client profiling considerations that investment representatives must take into account when determining the suitability of investment funds for clients.

Table 9.3.17 – Client Profile and Suitability

Considerations in Client Profiling	Investment Fund (Suitability)
• Risk tolerance • Return expectations • Specific objectives • Investment time horizon • Reference functional currency	The selected fund should match the client's risk appetite and return expectation, along with all other aspects of the client's profile. Note, however, that when the fund in question only makes up a small part of the client's overall portfolio, it may have a

• Key preferences and constraints • Investment style and specific needs	dedicated objective with different risk and return characteristics that, when combined with the client's other investments, would meet the client's objectives.

a. **Investment Objectives and Time Horizon**

Investors have different investment objectives and time horizons. Knowing the investor's investment objectives and time horizon is essential to choosing the right unit trust. Generally, unit trusts are better suited for investors with medium to long-term investment horizons, the exception being certain leveraged ETFs and highly volatile commodity ETFs, which are meant more for traders.

Figure 9.3.17(a) shows the risk continuum of various types of funds, with money market funds being the least risky and specialized funds being the riskiest:

Figure 9.3.17(a) – Risk Continuum of Unit Trusts



To meet best practice standards, some fund houses provide additional statements highlighting the type of clients the product is suitable for. The following example illustrates product suitability statements for two different funds.

Product Suitability Statements – Who is the Product Suitable for?

Fund A: Conservative Fund

The fund is only suitable for investors who:

- Seek long term capital growth.
- Seek regular income distribution in respect of the Distribution Class.
- Are looking for exposure to the securities of companies which are listed on the Singapore Exchange (SGX-ST) and are comfortable with the volatility of a single country equity fund.

Fund B: Aggressive Fund

The fund is only suitable for investors who:

- Seek capital appreciation with no need for income.
- Seek access to the dynamic emerging market countries.

- Are comfortable that the portfolio may generate potentially higher returns but comes with a higher risk profile than that of a diversified Global Equity Fund.

b. **Diversification**

A common motivation for investing in a unit trust is to have a diversified portfolio through the various securities that are in the fund. Depending on the investment objectives, some funds are very narrowly focused and only invest in a very limited number of stocks. For example, a unit trust whose investment objective is to achieve growth by investing in China-related, large-cap companies will have less diversification than another unit trust which aims to achieve growth by investing in global large-cap companies.

c. **Expenses**

Another consideration is the expense ratio which indicates the cost and efficiency of the fund (refer to Section 9.3.10 on fees and expenses).

d. **Fund Size and Capacity**

A small fund size can affect the performance of the unit trust as the fixed expenses in running the fund can have a material impact on its performance, as such expenses contribute to the total expense ratio of the fund. For example, a SGD10,000 legal expense that is billed to an SGD8 million fund is less significant (0.125% of the fund) than when billed to a SGD 1 million fund (1% of the fund).

On the other hand, unit trusts that have grown very big are difficult to manage. For example, a unit trust with SGD10 billion of funds will be huge relative to the market. With such a large size, the fund manager will not be as nimble as smaller players and may face difficulty in adding value. Thus, when a unit trust reaches its optimal capacity, it will typically stop accepting fresh subscriptions.

9.3.18 Retrocessions

Retrocessions are fees received by banks and other financial institutions for distributing third-party products to clients. Hence, when recommendations of funds are made to new clients, there could be a conflict of interest and the independence of the investment advisors are compromised. Clients are often not fully aware of such practices. Additionally, the retrocession fees paid by the fund houses to the banks are a cost offset against the fund performance, thus negatively affecting the NAV of the fund for existing clients.

While this practice of receiving retrocessions from funds has been arrested by the courts in Switzerland, it is still widely practised in Asia by some financial institutions.

Some investors have expressed similar concerns about the hidden fees and retrocessions received by institutions that manage fund of funds when selecting external funds. Fund of fund houses assert that they only select the best funds available and carry out due diligence on the various funds. However, if they are receiving retrocessions, this would potentially diminish the objectivity of the fund selection process. Furthermore, investors are also concerned about the dual layer of fees they have to pay, as they would incur fees at the fund level and then again at the fund of fund level. Retrocessions may result in significant overall effective costs to clients. Some investors, however, are less concerned with paying such fees if the overall return significantly compensates for the cost.

9.4 Real Estate Investment Trusts

9.4.1 Overview of REITs

A Real Estate Investment Trust (REIT) is a CIS in which the funds of individual investors are pooled together to invest in real estate assets. In Singapore, REITs are governed by requirements set out in Appendix 6 -Property Funds (Property Funds Appendix) in the Code on CIS. A REIT is a unit trust that is predominantly invested in real estate and real estate-related assets, specifically, the Code on CIS requires a minimum of at least 75% of the fund to be invested in income-producing real estate. A simple way to understand a REIT is to see it as a passive investment vehicle that engages in real estate investments for the collection of rent.

In many respects, a REIT is like any other unit trust. However, unlike a typical unit trust, a REIT requires a wider range of specialists to manage it. Another difference is that a REIT typically has a Sponsor, usually with a stake in the REIT, that sources the properties injected into the initial portfolio and may continue to provide a pipeline of assets for the REIT. A third distinguishing factor is the availability of up-to-date valuations of the underlying assets—unlike equities and bonds in unit trusts where current market values can readily be obtained from the stock exchanges or bond dealers, the Code on CIS only requires the properties on a REIT to be valued once a year.

Unit trusts that are not REITs are typically not listed whereas REITs are listed on the SGX Mainboard in Singapore. However, unlisted property trusts, while constituted in the same way as listed REITs, are not publicly traded.

There are many different types of REITs. REITs can be sector-specific (e.g., those that invest in retail malls). Alternatively, they can consist of investments in different property sectors such as healthcare, hospitality, industrial, offices, and retail. REITs are long-term investments and are subject to fluctuations of the property cycle.

9.4.2 Advantages of REITs

Since a REIT is similar to a unit trust, the benefits are largely similar but become clear when compared to the alternative of direct investments in real estate for rental income. First, buying into a REIT is much more affordable and achieves greater diversification than purchasing a single property. Second, a REIT has significantly greater liquidity because it can be traded easily and quickly on the exchange without incurring substantial transaction costs. Third, since REITs that distribute at least 90% of their taxable income each year enjoy favourable tax treatment, many of them do; there is no corresponding tax advantage for direct investments. Lastly, REITs come with professional property management services without having to separately contract for such services.

9.4.3 Risks of Investing in REITs

The risks associated with investing in REITs vary and depend on the characteristics and features of the REIT, which includes the geographical location of the properties. Investors should consider not only the expected distribution yield, but also the quality and lease tenor of the underlying properties, as well as the tenant quality and concentration and the weighted-average lease expiry.

Some of the risks associated with investing in REITs include:

a. **Market Risk.** REITs are listed and traded on a stock exchange. Like other exchange-traded securities, its price is dependent on the supply and demand for the REIT, exposing the investor to market risk. REIT prices reflect the outlook for the specific sectors (e.g., healthcare, hospitality, industrial, office, retail etc.), the state of the property market, the economic outlook, the management of the REIT, interest rates, and other factors. Investors must be able to bear this price volatility.

b. **Income Risk**. REITs are attractive because they typically distribute at least 90% of their income each year[53].However, a REIT may not be able to make a distribution if it incurs an operating loss. This loss might come from the expiration of leases, lower occupancies, falling rentals, or even contractual failure of the tenants. The investor should evaluate if the REIT has taken any measure to mitigate these risks, such as lease tenor, quality of tenants and types of tenants.

c. **Concentration Risk.** As a unit trust, REITs afford some extent of risk diversification but may be vulnerable to concentration risk if a substantial portion of the income is derived from only a few property assets or assets within the same industry e.g., hospitality. If the REIT depends on a small number of tenants for its rental income, it is also exposed to concentration risk.

d. **Liquidity Risk.** Although REITs themselves can be bought or sold on the exchange, the underlying properties of the REITs may not be easily liquidated because the market for physical real estate is private, with buyers and sellers dealing bilaterally with each other instead of through an exchange. This exposes REITs to liquidity risk as they cannot dispose or alter the composition of the investments quickly in a falling property market.

e. **Leverage Risk.** In Singapore, REITs are subject to a single-tier leverage limit of 45%. The need to service the debt used to finance the acquisition of underlying properties implies less income for distribution and greater volatility in the equity value of the REIT.

f. **Refinancing Risk**. The aforementioned use of debt financing also exposes the REIT to the risk of increases in interest rates when refinancing the loans. As REITs typically distribute at least 90% of their income annually to unit holders, it is not easy for them to build up cash reserves for debt repayment, so they typically resort to fresh borrowing (e.g., medium term notes issue) or raising new equity (e.g., rights issue). If the REIT is unable to make these arrangements, it may be forced to liquidate some of its investments, which may result in a reduction in lease income. Due to the lack of cash reserves, REITs usually have to or raise funds in the capital markets for additional capital expenditures as well.

9.5 Business Trusts

9.5.1 Overview of Business Trusts

Business Trusts (BTs) are governed by the Business Trusts Act 2004, which was passed in the Singapore Parliament in September 2004 to regulate this new form of business structure. Like other trust entities, such as unit trusts, a BT is established under a trust deed. A BT is a hybrid structure with characteristics of both companies and trusts but, unlike a company, but a BT is not a separate legal entity. BTs have become a popular form of listing on the SGX, with participation from port infrastructures and pay television.

[53] Note that unlisted property trusts are not the same as REITs. Property trusts are not required by law to distribute any income and distribution of income under property trusts are subject to the decision of the Board of Directors.

In Singapore, BTs are a single responsible entity organization and the trustee-manager is vested with the roles of both trustee and manager of the trust assets. The trustee-manager has legal ownership of the BT's assets and is responsible for looking after the unit holders' interests and managing the business of the BT. Investors subscribe for units in the BT and are paid distributions from the BT's income.

9.5.2 Advantages of Business Trusts

BTs are an alternative form of business entity with the following advantages:

a. Limited liability of unit holders;

b. Distributions can be paid out of operating cash flow instead of accounting profits; and

c. Investment restrictions applicable to CIS do not apply to BTs.

The main attractive feature of BTs is their effectiveness in unlocking value from fixed assets that generate cash flow. Typically, businesses such as infrastructure providers or asset-backed businesses find a BT structure attractive. Such businesses usually have depreciating assets that reduce accounting profits despite strong positive cashflows, which makes BT structures attractive because dividends can be paid out of operating cash flow.

Like REITs, BTs pool investors' money and enable unit holders to have fractional ownership of underlying business assets, facilitating diversification. However, unlike REITs, BTs can actively engage in any business area whereas REITs are required by law to invest in real estate for rental income.

9.5.3 Risks in Business Trusts

BTs are subject to the same set of risks as unit trusts and REITs (i.e., market risk, liquidity risk, income risk, concentration risk, and refinancing risk). In the case of BT refinancing risk, the investor should remember that while there is no cap on its debt limit, debt could still become a problem if the BT takes this as an opportunity for indiscriminate borrowing, which would result in an unsustainable debt structure for the trust.

Given the wide range of businesses that are allowed to use this route for fund raising investors should analyse the contractual details of the BT's assets to determine the consistency and stability of its income sources. For example, existing port infrastructure providers on traditional shipping routes are potentially at risk as global warming maps out new shipping routes that are more efficient and economical, creating fresh demand for new port locations.

9.5.4 Comparison between Listed REITs and Business Trusts

Table 9.5.4 lists the differences between listed REITs and BTs.

Table 9.5.4 - Comparison between listed REITs and Business Trusts

	Listed REIT	Business Trust
Entity	Constituted by a trust deed	Constituted by a trust deed

	Listed REIT	Business Trust
Ownership	Unit holders do not have individual legal ownership of the assets but have beneficial interests	Unit holders do not have individual legal ownership of the assets but have beneficial interests
Independence of Trustee and Manager	Trustee and Manager are separate and independent	Single responsible entity is the Trustee-Manager, who is the manager of the trust assets
Trust objective	Passive investments	Permitted to actively engage in business operation
Distribution	Mandatory distribution of minimum of 90% of annual income to qualify for favorable tax treatment	No mandatory minimum. Can distribute from operating cash flow
Gearing	45% limit	No limit
Restrictions on investments	Minimum 90% to be generated by rental income	None
Regulations	Securities and Futures Act, Code on CIS	Business Trusts Act
Taxation	Tax exempt for individual unit holders	Single-tier taxation

9.6 Stapled Securities

A stapled security is created when an issuer lists two or more securities that are traded as a single instrument on the exchange. The stapled security would share the same investor. For example, several Singapore REITs in the hospitality sector is stapled to a business trust (see Chapter 9.5 - Business Trusts) for a listing. Stapled securities are popular with issuers because they combine a passive income security with an active income security, which enhances the attractiveness of the issue. From the investor's perspective, stapled securities provide some diversification for business risks.

In a stapled security, the issuer is bound by the regulations specific to each of the security being stapled. Therefore, if a REIT and a business trust are stapled together, the issuer has to pay the minimum 90% of annual income to unit holders on the REIT to qualify for favourable tax treatment but is free from paying dividends on the business trust. Also, such stapled securities cannot be marketed as REITs.

9.7 Undertakings for Collective Investment in Transferable Securities

Undertakings for Collective Investment in Transferable Securities *(*UCITS) is a European Union directive, applicable in the UK since 2002. The original UCITS objectives were to remove the barriers to cross-border distribution and marketing of funds by allowing funds to invest in a wider range of financial instruments. The UCITS provides the "passport" for funds to be sold internationally.

UCITS funds are attractive to retail investors globally because of the common standards of disclosure and the consistent investment guidelines for the funds. Another appealing feature is that, under UCITS III, funds can invest in a wide range of assets, beyond the usual equities and bonds. For example, UCITS III allows the use of derivatives and index tracking products to limit risks or increase returns.

Since 2006, MAS has allowed UCITS funds to be distributed and sold in Singapore if these meet the local regulatory conditions. The number of UCITS funds registered for sale in Singapore has increased significantly over the last 5-10 years, which reflects the trend in Asia as well.

9.8 Alternative Investments

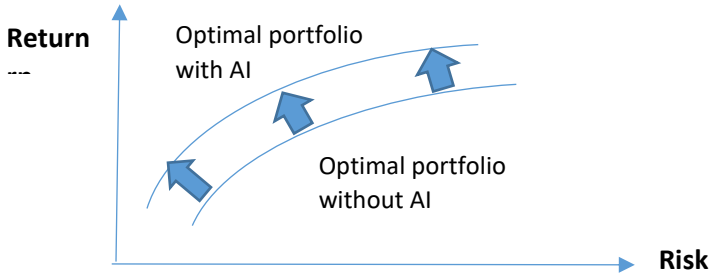
In addition to REITs and Business Trusts, other **alternative investments** include:

a. Hedge funds;
b. Private equity funds;
c. Commodity funds; and
d. Real estate funds.

The main benefit of investing in alternative investments is the ability to further diversify the portfolio using asset classes that are less correlated with the traditional asset classes of equities and fixed income. Through such diversification, the portfolio's risk-adjusted returns may potentially be increased (see Figure 9.8).

Unfortunately, during major financial crises (such as the Great Financial Crisis of 2008), investing in alternative investments do not necessarily provide much of a buffer since such investments are also affected by a widespread massive sell-off of assets. However, under normal market conditions, alternative investments have been found to be less correlated with equity and bonds.

Figure 9.8 – Optimal Portfolio with and without Alternative Investments



9.8.1 Overview of Hedge Funds

Unlike traditional funds, hedge funds can take short positions with the use of derivatives. Depending on how the policies and mandate of the hedge fund is defined, hedge funds either adopt a:

a. **single strategy approach**, where the fund manager just follows one approach; or

b. **multi-strategy approach**, which emphasizes one approach over others given the opportunities arising in a particular period.

9.8.2 Hedge Funds vs Conventional Funds

Like conventional funds, hedge funds can take on long positions to exploit positive market movements but would also be exposed to market downturns. However, unlike conventional funds, hedge funds can use short-selling or derivatives to reduce (or hedge) their portfolios' beta sensitivity to the market.

Hedge funds often have absolute return targets and can rely on leverage to significantly augment the return on an investment strategy to outperform the market (i.e., to produce alpha). However, some of these strategies require a longer time to unwind and hedge funds can be less transparent with their pricing and investment positions. Since it takes time to unwind from a position, the fund manager will only know the actual price when the trades are executed, hence the lack of pricing transparency.

Hedge fund managers often receive incentive fees on top of their higher management fees when compared to conventional fund managers. Table 9.8.2 sets out the different features of hedge funds and conventional funds.

Table 9.8.2 – Hedge Funds vs Conventional Funds

Hedge Funds	Conventional Funds
• Capture α (freedom to hedge away or reduce β) • Performance depends mostly on the fund managers' skills and their fund strategies.	• Capture $\alpha + \beta$ (limited ability to reduce β risk) • Performance depends mostly on market factors and fund managers' skills.
• Absolute return targets.	• Index-based benchmark targets.
• Can be leveraged significantly to augment impact of strategy e.g., to increase yields.	• Limited ability to use debt and leverage.
• Require longer time to unwind investment and less transparent pricing.	• More transparent with regards to investments and pricing.
• Management fees and performance incentives for fund managers.	• Management fees for fund managers.

9.8.3 Risks of Investing in Hedge Funds

As a direct consequence of features that distinguish hedge funds from conventional funds, investing in hedge funds exposes an investor to additional risks:

a. **Risky investment strategies**: Hedge funds can use speculative investment strategies which, when combined with the use of leverage, can result in outsized losses. Sometimes, leveraged positions are hedged through the use of derivatives but this comes with its associated risks, such as basis and counterparty risks. Basis risk is the risk that remains as a result of an imperfect hedge because a complete offsetting position may not exist. Counterparty risk arises when the other party to a derivative transaction does not make good on its contractual obligation.

b. **Illiquidity**: As hedge funds can invest in highly illiquid assets, they may not be able to unwind their positions at fair prices. An unanticipated market downturn can result in margin calls (for leveraged positions) and unusually large investor redemptions, which can lead to liquidation of assets at

unfavourable prices. To protect against this, many hedge funds have lock-up periods during which investors are not allowed to redeem their investment and some have gate provisions, which allow the manager to limit the withdrawals on a given withdrawal date to a percentage of the fund's net assets.

c. **Poorer investor protection**: Hedge funds are more loosely regulated than conventional funds and are not required to provide the same level of disclosures. Thus, it can be challenging to evaluate the terms of investment in a hedge fund. Given the opaque investment processes of hedge funds, there can also be agency risk, where managers do not necessarily act in the best interest of their investors and take poor leveraged bets in a misguided attempt to meet performance targets.

Even though hedge funds can use leverage and derivatives, it is incorrect to assume that all hedge funds are riskier than conventional funds. Some hedge funds with absolute return objectives actually look to protect downside risk and preserve wealth through their investments in income-oriented securities. Hedge funds can also adopt market-neutral strategies or provide downside protection with the help of options and futures. Hence, the risks of investing in hedge funds really depend on the type of instruments used and how they are used in the investment strategy.

9.8.4 Hedge Funds – Relative Value Arbitrage Strategies

Relative value arbitrage strategies look at the relative value between two securities or asset classes and aim to identify any potential mispricing through a combination of fundamental and technical analysis. Hedge fund managers take on long positions in undervalued securities while at the same time shorting overvalued securities from the same industries.

a. **Long-Short Equity Strategy**

This is the most popular and common form of the long-short strategies used by hedge funds. Basically, long-short equity strategists take long positions in stocks that are expected to appreciate and take short positions on stocks that are expected to depreciate.

Long-Short Equity – Market Neutral Strategy

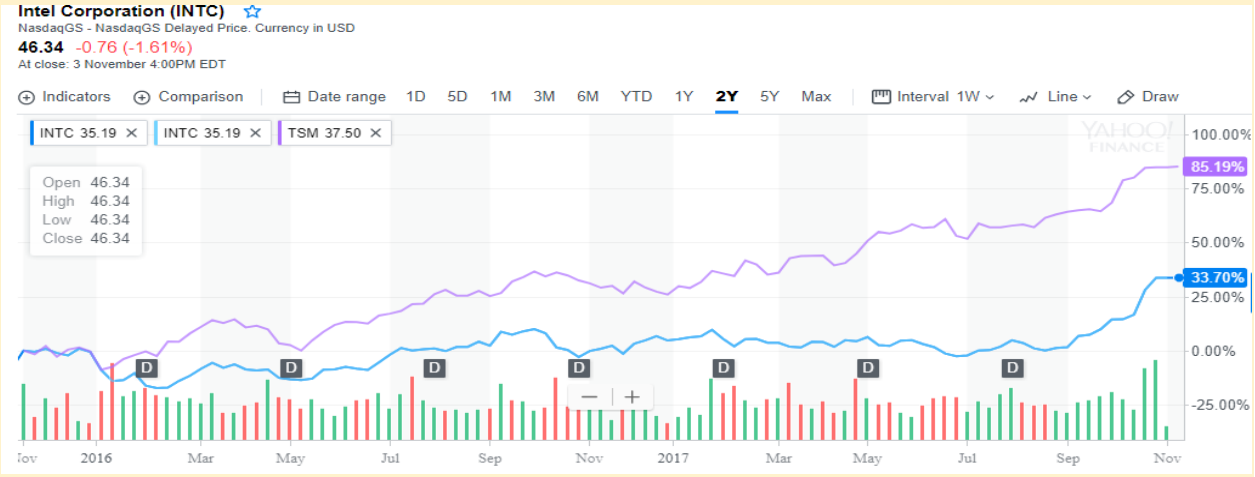
A news report in January 2017 suggested that Apple Inc. (Apple) is likely to appoint TSMC Limited (TSMC) as its semiconductor supplier for its new product. There is currently a positive view on the US market and consumer demand. Market watchers also foresee a likely rebound of semiconductor demand. However, while the view on Intel (INTEL, a rival semiconductor company) is also positive, is it not as promising as TSMC.

As such, a hedge fund manager decides to take a long position on TSMC and a short position on Intel. If he believes his stock picks are correct and wants to be market neutral, then the USD amount invested in TSMC ADRs (American Depository Receipts) should be equal to the amount of USD he uses to short sell Intel.

By doing so, as long as he is correct in his assumption that TSMC would outperform Intel (say by more than 1%), he would be able to make gains after transaction and financing costs, regardless of whether both stocks rise or fall in tandem in response to market-wide fluctuations. This would then be considered a **long-short equity market neutral strategy**.

Naturally, if TSMC rises and Intel stays flat or falls, the hedge fund manager makes gains. However, if his outlook is wrong and Intel actually outperforms TSMC, he will make losses instead.

Figure 9.8.4(a) – Example of Long-Short Strategy with TSMC and INTEL



Source: Yahoo Finance

For such strategies, it will be useful to set a target on when to close out the positions. The positions can be rolled over (i.e.; continued) but there must be justification for further widening. If there is no additional news that would cause a further widening of the premium of TSMC over INTEL, the spread position should be closed out.

If the manager has strong conviction in his positive research view, he can choose to take a net long position on this long-short strategy by investing more in TSMC than is short-sold in Intel but, if this is done, the strategy is no longer market-neutral and will have a partial exposure to a decline in the sector.

b. **Convertible Arbitrage Strategy**

Convertible arbitrage strategy is a hedge fund strategy that also uses the relative value approach. However, unlike the long-short equity strategy, the relative value analysis is done between the convertible securities and the common stock that are issued by the same company.

As such, this strategy involves the purchase of the convertible securities while shorting the underlying stock. The rationale is that, for a variety of reasons, convertible securities are usually slower in repricing to fair value as compared to the underlying stock, which leads to arbitrage opportunities. However, this strategy is not without risk—the underlying stock's price can rise and the price of the convertible security (e.g., the convertible bond) can fall.

Other relative value stategies that are less widely used include **fixed-income arbitrage** (where hedge funds seek to exploit yield differences between different types of bonds), **credit long-short,** and **volatility arbitrage**.

9.8.5 Hedge Funds – Event-Driven Strategies

Event-driven strategies are those that emerge when specific events present themselves. Hence, hedge funds seek to opportunistically take advantage of these situations. Examples of specific events that present opportunities for hedge funds include financial distress (e.g., bankruptcies and liquidation), acquisitions, capital restructuring, and consolidation.

a. **Distressed Securities**

Some hedge funds (sometimes called vulture funds) look selectively for opportunities to capitalize on bonds that are priced at distressed levels. Depending on the degree of the crisis facing a particular company, the distressed bonds can be priced at 10% - 99% of par. Hedge funds purchase distressed debt and borrowings that they believe to be undervalued.

For example, when there was a global downturn in the oil and gas sector, the bonds of many oil-related companies (including several in Singapore) were priced at distressed levels. Some investors and banks were willing to sell their distressed debt at a discount and these would be purchased by certain investors and hedge funds who were willing to bear the risk for potential gains.

Typically, hedge funds have a few options to deal with the distressed debt, such as:

i. reselling the debt at a much higher price to the market when the credit situation improves;

ii. exchanging their debt for equity, and attempting to sell the equity for a profit or combining their stake with a private equity firm and other investors to make a takeover bid; or

iii. allowing the company some time to recover on its own and pay off the full principal at maturity, which will yield substantial gains above the discounted price paid.

Investing in distressed debt can be very risky and should be managed by experts. For example, a well-known Swiss bank was forced to sell its billion-dollar portfolio to a distressed-debt specialist at a huge loss in 2016 when it eventually decided to get out of the high-risk business.

b. **Corporate Events Arbitrage**

Event-driven hedge funds also look for corporate events like mergers, acquisitions, takeovers, and liquidations. Depending on how the companies are being valued and the deal process, hedge funds could be involved in:

i. buying the target and selling the acquirer, when the acquirer is overvalued;

ii. buying both the acquirer and target, when the acquisition is believed greater value for the merged entity due to synergies; or

iii. buying shares of the target involved in a hostile takeover with competing offers.

However, not every corporate finance deal lends itself to mispricing opportunities, so hedge funds looking for such opportunities may face dry periods.

9.8.6 Hedge Funds – Global Macro Strategies

Global macro strategies take directional views on various financial products and their derivatives in anticipation of the market reaction from significant economic or political changes in one or more countries.

The fund manager will make investment decisions to long or short investments based on their view of how macroeconomic trends will affect interest rates, currencies, sovereign credit, commodities, equity indices, and equities and bonds.

Such funds could adopt a spectrum of approaches to make their decisions, ranging from very qualitative fundamental approaches to quantitative computer-based models, or a blend of both. Due to the size of their

positions and risks taken, they need to be invested in fairly liquid assets in order to unwind or hedge their positions.

Examples of Global Macro Strategies for Hedge Funds

One example of an effective global macro strategy would be to short the Pound sterling (GBP) and British stocks on the London Exchange during the Brexit Referendum.

Conversely, a global macro strategy that shorted US equities based on the expectation of a market decline following the Republican victory in the US elections would have had to reverse its position and make adjustments fairly quickly to avoid losses when the US equity markets rallied shortly after.

As illustrated by the examples above, global macro hedge funds would have to be nimble to unwind positions and cut losses.

9.8.7 Hedge Funds – Directional Computer-Based Quantitative Trading Strategies

Directional quantitative trading strategies employ a combination of technical analysis, market data, and funds flow to make investment decisions. These strategies mainly follow technical pricing trends and momentum. As long as the trend continues to be intact, the funds will continue with their position until a trend reversal is observed.

Momentum Trading Strategy

A fund following a momentum trading strategy would continue to buy options and futures contracts on various indices, in line with an upward trend momentum. The fund can be thought of as adopting the philosophy that “the trend is your friend”.

Once the model signals a slowdown in momentum or a reversal, the cumulative gains are realized at one go and the model awaits confirmation of the next trend – either up or down. If the indicators signal further correction, then the shorting of futures and options begins until the next reversal point, at which the short positions are closed out with purchases and the fund waits for the next signal.

Such momentum trading strategies benefit from clear and persistent trends over several weeks or months. When there are disruptive reversals or a high frequency of reversals over very short periods, the is likely to incur too much transaction costs to make profits.

9.8.8 Funds of Hedge Funds – Multi-Strategies

There are funds of hedge funds that adopt different strategies, allocating more to some strategies over others in a given period. The rationale is that, for any particular period, some strategies are likely to outperform others due to market conditions.

The fund of hedge funds approach attempts to provide the expertise and due diligence to ensure:

i. adequate diversification between hedge funds so that there is less concentration risk; and

ii. that monies can be actively reallocated to strategies that do well under certain market conditions.

Naturally, the service comes with additional cost of management, which is incurred on top of the performance and incentive fees paid to hedge fund managers.

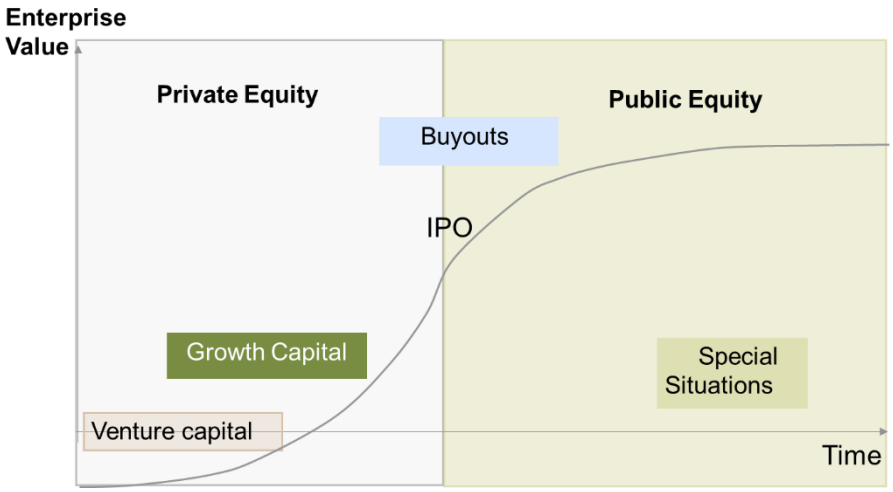
With diversification, returns will typically fall to the average return achieved by hedge funds, which may not fit the expectation that some investors have when investing in hedge funds.

9.8.9 Overview of Private Equity Funds

Private equity funds are long-term investments that are sold OTC. They are sold mainly to accredited individuals, as well as corporate clients, institutional clients, and sovereign wealth funds to complement their overall investment portfolio. These funds are typically invested in businesses at different stages of their respective growth cycles (see Figure 9.8.9).

Private equity also provides businesses with capital, which would help them grow to a particular size before they are either listed (through initial public offerings (IPOs)) or sold off to other major corporations privately. As such, private equity provides the original founders a viable exit option and investors the opportunity to realize gains.

Figure 9.8.9 – Private Equity Capital for Companies at Different Stages of Growth



9.8.10 Stages of Private Equity Investment

The various stages of private equity investment are as follows:

a. **Venture Capital** - Venture capitalists co-invest with an entrepreneur to provide seed and/or start-up capital in the early stages of a business. For example, many newly founded tech and biotech companies are typically in the conceptual or research stage where products are still not developed. These companies may have to go through several years of research and development and incur losses before providing potential revenues and profits. In return for the higher risk that venture capitalists take on with such financing, they expect a higher-than-average return on their investment.

b. **Growth Capital** – During this stage, private equity can provide funds for companies looking to scale up and take advantage of opportunities to enter during periods of high growth or expansion. When private equity funds provide growth capital, they tend to focus on later-stage investments such as companies that are

already established, but still require funds or co-sharing of expansion risks when scaling up their product range, production volume, distribution, and market reach.

c. **Buyout Funds** – These funds invest in mature companies with stable cash flows, established products, and developed personnel. The buyout could occur pre-IPO for owners and businesses who want to exit before the IPO process. Alternatively, the buyout could also be part of the IPO process with an agreement to transfer the shares between major shareholders 1-2 years after the IPO to facilitate the smooth running of the company.

d. **Special Situations:**

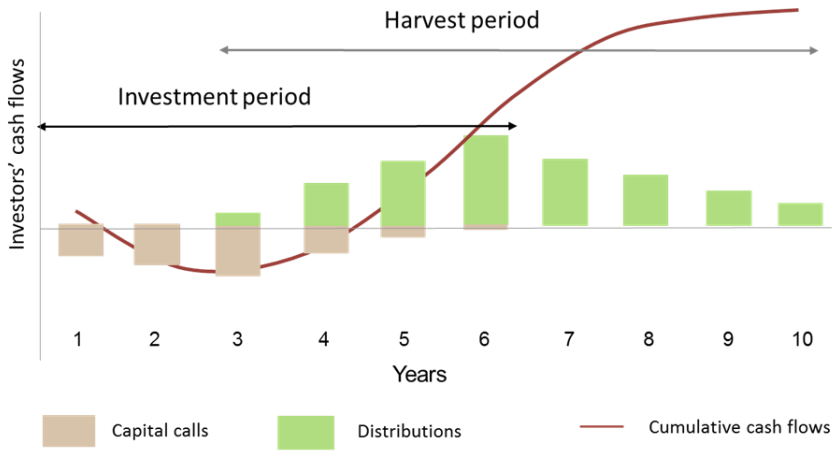
i. **Going Private** - This could occur when major shareholders of listed entities believe that the market is undervaluing the company. As such, by purchasing the company and taking it private, they can benefit by having a larger share of the profits. Private equity vehicles can be used as an intermediary for the purchase of a listed business to take the company private; alternatively, major shareholders can raise the private capital amongst friendly parties on their own to make an offer to the remaining shareholders in order to take the company private. When the time is ripe, the business can be re-listed or consolidated with other businesses for listing.

ii. **Distressed Businesses** – Private equity firms can work with various private parties to take over companies facing financial distress. During market downturns, some good businesses may succumb to financing and liquidity issues and are subsequently forced to sell their businesses. Private equity companies can work with creditor banks to take over such businesses.

9.8.11 The Private Equity J-Curve

Investors who agree to invest in private equity must be mindful of the nature of such investments. Private equity investments tend to have negative initial returns and cash flows for a relatively long period of time. As such, the cash flows for private equity typically follow a J-curve, as shown in Figure 9.8.11.

Figure 9.8.11 – The Private Equity J-Curve



The first 5-6 years represent the investment period. If an investor commits $10 million towards a private equity fund, the $10 million will not all be invested at the outset. Instead, the private equity manager will make several cash calls and draw down on the $10 million capital as and when good opportunities arise. If many good opportunities arise earlier, the draw down will naturally be faster. The harvest period typically begins from the 6th year onwards when the fund will start making payouts.

The incentive for investing in private equity is that the expected internal rate of return would typically be in the double digits (10-15%). However, the downside is that the funds would be locked up for several years and any early redemption or withdrawal before the 6th year will typically result in losses for the investor. Thus, to address suitability concerns, it must be highlighted to investors that private equity investments are long-term in nature. The investor must be willing to sacrifice liquidity in order to secure the targeted returns and the first few years will simply involve drawdowns. Returns will usually only accrue from the 6th year onward and early redemptions will result in losses.

9.8.12 Commodity Funds and Real Estate Funds

Commodity and real estate funds are largely similar to commodity ETFs and REITs, which have been covered earlier in this Study Guide. The main difference is that commodity and real estate funds are sold OTC, while commodity ETFs and REITs are traded on exchanges.

Additionally, some commodity funds and real estate funds sold OTC distinguish themselves from their exchange-traded counterparts by being more transparent about their investment approach and types of investments.

a. **Commodity Funds**

Commodity funds offer indirect exposure to a portfolio of commodity-related stocks, bonds, futures, and funds. Commodity funds can either focus on pure sectors (e.g., metals, agriculture, energy) or they can be a fund of funds that are diversified across different types of commodities. Such funds of funds could overweight one sector over others. Please refer to Table 9.8.12(a) for the underlying commodities of different commodity funds.

The more diversified the fund, the less volatile it should be. However, because of exposure to the futures market, some degree of volatility in commodity funds is unavoidable. Agriculture commodity funds suffer from the same problem as their underlying commodities – the successes in agriculture production supply often outstrip demand, causing long downward cycles of overhanging supply.

The energy, base metal, and ferrous metal sectors are more cyclical in nature. They do well when there is a strong pick-up in global economic activity, which drives the underlying demand for such commodities. Conversely, these sectors tend to do poorly when economic activity slows.

Funds invested in precious metals will typically do well in times of uncertainty and volatility in the traditional asset classes (stocks and bonds). They can be heavily influenced by the movement of major currencies and interest rates.

Table 9.8.12(a) – Underlying Commodities of Commodity Funds

Agriculture Funds	Energy Funds	Metals Funds
Grains Soft Commodities Livestock Dairy Biofuel	Crude Oil Natural Gas Refined Products Petrochemicals Electricity	Precious Metals Base Metals Ferrous Metals

b. **Real Estate Funds**

Real estate funds are not too different from REITS. Real estate funds sold OTC can cover agricultural, industrial, residential, commercial, medical, and even undeveloped land banks. Although they are focused on generating stable and high rental income, real estate funds sold OTC through private banks can be less liquid.

As with REITs, the main concern when investing in real estate funds is the mismatch between unit liquidity and the illiquidity of the underlying assets. In a property bubble situation, property asset prices can fall very sharply once the bubble has burst. This leads to redemptions of fund units by investors to stem their losses. When redemptions of fund units start to mount, funds that do not have deep pockets and that have difficulty unwind their property holdings may be caught in a liquidity crunch and, hence, be unable to meet the redemptions. By selling off the underlying property at market lows, real estate fund managers would be forced to realise losses. Hence, in such difficult situations, real estate funds will typically seek to delay redemptions if possible, instead.

9.9 Insurance

As astute in investment management as we can be, there are some events that are unpredictable as to when they occur but have severe financial repercussions when they do occur. Risk management processes must be in place to mitigate the effects of such unforeseen events.

This section covers the ways in which investment representatives can help individuals make better plans to meet their financial goals and to be prepared for life's uncertainties and eventualities. Moreover, some insurance products also serve as investment instruments, and investment representatives should have a good understanding of such products so as to provide better advice for their clients.

We will be covering basic insurance products, annuities, and the use of CPF funds for investments. The two main types of life insurance policies that can also serve as savings/investment instruments are:

a. Whole Life Insurance; and

b. Endowment Insurance.

9.9.1 Whole Life Insurance Policy

The premiums payable for a whole life insurance policy provide a combination of life cover and investment and, therefore, have cash or surrender values. In the case of a whole life insurance policy, the sum assured will be payable on the death of the life insured, whenever this takes place. Policy owners pay fixed monthly or annual premiums and beneficiaries receive fixed death benefits. In the meantime, reserves/assets backing up the future liabilities of the policies earn interest and grow, building up cash values that can be withdrawn through the surrender (cancellation) of the policy, subject to its terms.

Alternatively, if policy owners do not want to surrender their policies, they have access to accumulated policy savings. Policy owners can normally obtain loans under their policies from the life insurance company for an amount up to a certain percentage of the policy's cash value, and the limit is insurer- and policy-specific.

9.9.2 Endowment Insurance Policy

The premiums payable for an endowment insurance policy also provide a combination of life cover and investment. However, endowment insurance policies usually mature after a fixed period of time (e.g., 10, 15 or 20 years) and the sum assured is payable on the maturity date or on the life insured's death, whichever is earlier. As in a whole life insurance policy, this sum assured is fixed.

Endowment policies carry higher premiums than conventional whole life policies and term insurance but may be useful in meeting special lump-sum needs such as paying for college expenses or buying a retirement home. The guaranteed cash values received may be less than the sum of the premiums paid, because part of the premiums will pay for insurance protection while the rest is invested and subject to investment risks.

9.9.3 Using Insurance Policies for Investments

Both whole life and endowment insurance policies are, thus, effectively investments that become payable at a future date, either on death or earlier. Policies structured to provide a fixed amount of death benefit (sum assured) only, they are known as **non-profit (non-participating)** policies, in contrast to **with-profits** policies that will be explained in the sections that follow.

The return of insurance policies can be linked to the insurer's investment performance either through a with-profits policy, where benefits are indirectly affected by investment performance, or through an **investment-linked life insurance** policy, which has a direct link with investment performance.

a. **With-Profits Policies**

Every year, the insurer will carry out a valuation of the assets and liabilities of its life fund. This will normally reveal a surplus, part of which can be allocated to the with-profits policy owner in the form of an addition to the sum assured. This addition, called a bonus or profit, is usually reversionary, which means that it is payable only at the same time as the sum assured is paid.

Under a with-profits policy, the link between the policy benefits and the investment and operating (such as mortality experience and level of operating expenses) performance of the insurer is not direct and depends on the annual valuation of the fund assets and liabilities, where many other factors are taken into consideration.

The allocation of the surplus in the form of a reversionary bonus will ultimately depend on the decision of the board of directors, which are based in part on the recommendations made by the appointed actuary of the life insurance company. Therefore, the bonuses added to the policy tend to be only loosely linked to the investment and operating performance. Allowances must be made due to guarantees for the basic sum assured and so the bonus system will not directly reflect the value of the underlying assets of the life fund. In addition, bonuses are generally declared on a yearly basis. Therefore, they cannot match and reflect the daily fluctuations in the value of assets.

b. **Investment-linked Life Insurance Policies**

Investment-linked life insurance policies offer investors policies with values that are directly linked to the investment performance of the underlying instruments. This is usually done by formally linking the values of the policies to units in a fund run by the insurer or by external fund managers. Such funds usually consist of a mixture of equity funds, fixed income funds, managed funds, and other types of funds. The investment element of investment-linked life insurance policies varies according to the underlying assets of the portfolio and fluctuates daily according to the performance of those investments.

9.10 Annuities

An annuity is a series of payments guaranteed for a number of years or over the lifetime of the party that receives the payments (annuitant).

9.10.1 Types of Annuities

There are two main types of annuities:

a. An **immediate annuity** is a contract where payments to the annuitant begin as soon as it is purchased. An immediate annuity is always purchased with a lump-sum (single premium); and

b. A **deferred annuity** is a contract where the payments to the annuitant begin at some future date. The date is specified in the contract or at the annuitant's option. The amount that the annuitant will periodically receive depends on his contributions, the interest earned on the contributions, the annuitant's gender, and the annuitant's age when the payments begin. Deferred annuities may be purchased with either a single premium or periodic premiums.

Examples of non-insurance annuities include monthly payments on home mortgages and fixed-rate coupon payments on a bond.

9.10.2 Annuity Calculations

The present value of an annuity can be calculated with the formula below:

$$PV_{annuity} = C \times \left[\frac{1 - (1 + i)^{-n}}{i}\right]$$

where
C = cash flow per period
i = interest rate
n = number of payments

Example – Present Value of an Annuity

An amount of $1,000 is paid every year for 5 years. The annual discount rate is 3%. The present value of this stream of cash flows is:

$$PV_{annuity} = 1000 \times \left[\frac{1 - (1 + 0.03))^{-5}}{0.03}\right] = \$4,579.71$$

With a financial calculator:

- Input PMT = -1,000, N = 5, and I/Y = 3
- Press CPT, followed by PV to get $4,579.71

The future value of an annuity is given by the formula below:

$$FV_{annuity} = C \times \left[\frac{(1 + i)^n - 1}{i}\right]$$

where
C = cash flow per period
i = interest rate
n = number of payments

Example – Future Value of an Annuity

An amount of $1,000 is paid every year for 8 years. The annual rate of discount is 5%. The future value of this stream of cash flows is

$$FV_{annuity} = 1000 \times \left[\frac{(1 + 0.05))^8 - 1}{0.05}\right] = \$9,549.11$$

With a financial calculator:

- Input PMT = -1,000, N = 8 and I/Y = 5
- Press CPT, followed by FV to get $9,549.11

9.10.3 Uses of Annuities

a. **Insurance against the Possibility of Outliving One's Income**

In its pure form, a life annuity may be defined as a contract whereby, for a cash consideration, one party (the insurer) agrees to pay the other (the annuitant) a stipulated sum (the annuity) periodically throughout life (lifetime). However, the consideration paid for the annuity will be fully earned by the insurer immediately upon the death of the annuitant. Therefore, the purpose of an annuity is to protect against loss of income arising from excessive longevity. Hence, this is exactly the opposite of the purpose of a life insurance policy, which furnishes protection against loss of income arising out of premature death.

b. **Investment Vehicle or a Savings Instrument**

Most annuities are savings instruments designed to first accumulate funds and then systematically liquidate the funds, usually during the retirement years of an annuitant. The annuitant can put in a lump sum (single premium) or pay a series of regular premiums over a period of years. At the annuity starting date, the annuitant receives a guaranteed income for the rest of his life. The amount of the income depends on the cash value in the annuity and the payment option selected.

As such, annuities are an attractive option for an individual who has not yet accumulated an estate but wants to achieve financial independence in his old age. Professional entertainers and athletes, who enjoy a very large income for a limited period of time, will find annuities particularly attractive as a savings medium. Under such circumstances, annuities are an appropriate investment because they can be purchased through flexible periodic premiums or through a single premium, when the annuitant comes into possession of large amounts of money. Therefore, annuities can be used as a savings instrument which provides protection against loss of income arising from excessive longevity by liquidating a principal sum.

c. **Hedge against Adverse Financial Developments**

Wealthy individuals who have already accumulated an estate, either through inheritance or their own personal efforts, purchase annuities as a hedge against adverse financial developments. Large estates can be wrecked through unwise investments. As a result, individuals who were once wealthy can depend on the payments from annuities that they purchased when they were more affluent.

d. **Comparing Life Insurance and Annuities**

While life insurance products are intended to guard against living too short a life, life annuity products are intended to guard against living longer than expected, by providing support for post-retirement living expenses for as long as the annuitants live. As in the case of life insurance products, there are many different variations of life annuity products. To achieve the aim of providing sufficient retirement income, life annuity products are often structured as savings and investment products before the annuity payments begin. The savings/asset-building phase is an extremely crucial part of retirement planning and an integral part of many life annuity products, where regular premiums are paid into the retirement savings fund for investment and growth.

9.11 Risk Management for Retirement Funds

For investment of CPF funds which are ultimately meant for retirement, the CPF Board in Singapore has provided guidance that is generally aligned with risk management best practices.

For example:

a. Investors should be made aware that investments in funds with higher proportions invested in equity are generally riskier (more volatile).

b. Investment providers should highlight the focused or concentration risk associated with any investment fund. This may relate to geographical, sector, or specific investment-related concentration risk.

c. For funds to be included under CPFIS and thus eligible to take in CPF monies, they must meet the CPFIS funds admission criteria, which include being in the top quartile of global peer group and benchmarking total expense ratios to the median of CPFIS funds in the respective risk class.

Diversification and dollar-cost-averaging is recommended to help investors spread out their risks and potentially improve their overall returns.

9.11.1 Dollar Cost Averaging Versus Market Timing

a. **Market Timing**

Market timing is the technique of shifting investments in and out of risky asset classes in order to:

i. capture the upside for risky asset classes during bull markets; and
ii. exit the market before downturns (i.e., avoid the downside for risky asset classes during bear markets).

Market timing is considered an active investment approach.

b. **Dollar Cost Averaging**

Dollar cost averaging is a practice adopted by investors who do not want to undertake market-timing risks. Investors invest in steady, equal amounts over a period of time. Since equal dollar amounts are invested, fewer shares are accumulated when the price is high and more shares are accumulated when the price is low. Unfortunately, executing dollar cost averaging accurately will cause the investor to likely end up with odd lots.

Dollar Cost Averaging

An investor with $60,000 is considering buying a security currently priced at $5 and has the following choices:

i. Invest $60,000 all at the start of the year which would mean purchasing 60,000/5.00 = 12,000 units in total; or

ii. Buy $5,000 worth of the security evenly over the next 12 months.

The security's price movements over the next 12 months are shown in the table below:

Month	Price	Quantity	Cost
1	5.00	1,000	5,000.00
2	4.80	1,042	5,000.00
3	4.60	1,087	5,000.00
4	4.60	1,087	5,000.00
5	4.70	1,064	5,000.00
6	4.50	1,111	5,000.00
7	4.50	1,111	5,000.00
8	4.50	1,111	5,000.00
9	4.30	1,163	5,000.00
10	4.30	1,163	5,000.00
11	5.00	1,000	5,000.00
12	4.80	1,042	5,000.00
	4.622	12,980	60,000.00

Average cost of purchase	4.622
Average of 12 months price	4.633

Investing all at the start of the year would, not surprisingly, translate to an average price per unit of $60,000/12,000 = $5. However, using dollar cost averaging would translate to a much lower average price per unit of $60,000/12980 = $4.622. In fact, this is also always mathematically lower than the prices averaged over the twelve months ($4.6233), but this does not mean much – it is more important that the average cost of purchase be lower than the final price at the end of the 12 months, but that will depend on the trend of the stock price.

c. **Missing the Best Days**

Advocates of passive investing cite empirical evidence suggesting that most investors fail to use market timing to make a positive impact on their investments on a sustainable basis. Not only are the odds against them, the "punishments" inflicted on investment returns can be devastating if the best trading days are missed.

Let us assess this by looking at the best 25 days of the S&P from 1970-2016 (47 years). Suppose we have a young investor who started investing since he was 18 for his retirement 47 years later in 2016. Now, if he missed investing in the best 25 days of the S&P, his return falls from 1,910% to just 371%, or from 6.7% to 3.0% annually. This is even lower than the return on a 1-month US T-Bill rolled over regularly, which would generate an average yield of 4.9% over the entire period. So, it effectively suggests that passive investing using a buy-hold strategy would do much better.

There are two further considerations. The best trading days usually occur immediately after some of the worst trading days. Hence, the chances of missing out on the best trading days are high because many investors sell on panic. Furthermore, best trading days tend to occur within close proximity of each other, which further increases the risk of missing out on most of the best trading days.

It should be noted that it is almost impossible to predict the best and worst trading days in the market because the daily movements in any market are affected by many highly unpredictable events. Thus, the best and worst days are usually only identified on hindsight. In this sense, instead of trying to identify the good and bad days, a better strategy may well be to use dollar cost averaging.

d. **Missing the Best and Worst Days**

Even though market timing by investors may lead to missing out on some of the best days, such investors may miss some of the worst days as well. To get a more balanced perspective, when we remove the worst 25 days of the S&P 500 performance in our calculation for the period, we find that the return rebounds to 2,750% (7.6% annualized), which is better than the 1,910% (6.7% annualized) earned from the buy-and-hold investor for the entire period.

These results do not necessarily suggest that market timing is superior to passive management. Rather, as market-timers are not always invested in the market, their non-participation means that they are likely to miss some of the best but also the worst days. Direct evidence showing that market timers can beat a passive strategy is needed.

Unfortunately, most working adults do not have the time, expertise, or emotional temperament to be disciplined active managers of their own investment portfolio. 70-80% of active fund managers have been found to underperform the market each year. Hence, dollar cost averaging as a passive form of investing for retirement may be a good approach.

9.12 Summary

1. **Exchange traded funds (ETFs)** are open-end investment funds listed on an exchange and are designed to track the performance of specific indices or baskets of underlying assets in various asset classes. **Cash-replicated ETFs** comprise the actual securities or a representative sampling of them, held in the index proportions. **Synthetically replicated ETFs** comprise statistically correlated holdings of securities, using swaps and derivatives to mimic the price behaviour of the index. The advantages of investing in ETFs include the ability to diversify with a small capital outlay, low transaction costs, liquidity, and transparency regarding the underlying securities represented in the portfolio.

2. Securities lending is a practice in which ETFs pay agents to lend out shares in the ETF portfolios to other investors. The ETFs will earn interest for lending the stocks.

3. **Leveraged ETFs** are high-risk, volatile short-term investments that are more suited for traders and speculators than for investors.

4. A **unit trust** is a collective investment vehicle set up to pool the financial resources of investors and invest the funds in assets to be managed by professional fund managers. There are various types of unit trusts, e.g., money market funds and bond funds.

5. Investors of unit trusts transact directly with the investment company, which will redeem its own units at or near its net asset value . The capitalisation is "open", with the number of outstanding units changing as and when unit holders purchase or redeem their units.

6. The **net asset value (NAV)** is the value of the unit trust's assets less its liabilities (including management fees).

7. There are many advantages associated with investment in unit trusts, such as diversification, affordability, professional management, flexibility to switch from one sub-fund to another, liquidity, lower transaction costs, and security.

8. A **load fund** is sold to investors at NAV plus a sales charge, while a **no-load fund** does not incorporate a sales charge. Besides the sales commission, there are other fees and charges such as management fees, redemption fees, trustee and audit fees.

9. The fund factsheet will typically show or specify the:

    a. investment objective of the fund;

    b. strategy of the fund stating how the fund is managed and what it invests in;

    c. key features of its management style;

    d. key risks associated with the fund;

    e. historical performance for the last 5 years;

    f. current NAV and/or the bid-ask prices of the fund; and

    g. key performance indicators.

10. In selecting a unit trust, the prospective unit holder has to determine whether the investment objectives of the unit trust are congruent with his personal investment objectives, the amount of load charges and other fees imposed by the investment company, and the performance record of the unit trust.

11. There are a wide variety of unit trusts available, including those invested in a single asset class and those invested in multiple asset classes. Unit trusts with an **umbrella structure** provide investors with the flexibility to switch from one sub-fund to another, while index funds are **passively managed** to track the performance of a particular index such as a stock index. Other unit trusts may incorporate features such as a regular savings plan or insurance coverage.

12. A **real estate investment trust (REIT)** is an investment vehicle that pools the funds of individual investors to invest in real estate assets in order to earn rental income.

13. A **stapled security** is created when an issuer lists two or more securities that are traded as a single instrument on the exchange. For example, a REIT and a business trust can be combined to form a stapled security. The stapled security will comprise an equal share of the REIT and the business trust.

14. A **business trust** is a fund that has a hybrid structure with characteristics of a company and a trust. It uses a single responsible entity, the Trustee-Manager.

15. **UCITS** funds are attractive to retail investors because of the global common standards of disclosure and the consistent investment guidelines for the funds.

16. **Hedge funds** seek to capture alpha (α) and they have the freedom to hedge away or reduce beta (β, or sensitivity to the market). Hedge funds seek to achieve performance mostly from the fund managers' skills and their fund strategies. Popular hedge fund strategies include **relative value long-short strategies**, **event-driven strategies**, **global macro strategies**, and **computer-based directional strategies**.

17. **Private equity** are long-term investments that typically involve drawdowns on investments for the first 5-6 years before return payouts. Double-digit returns are expected but early redemptions will typically result in losses.

18. The risk in **real estate funds** arises from the illiquidity of the underlying property asset relative to the real estate fund units sold.

19. Life insurance policies are intended to protect against mortality risk, and a **whole life** policy pays out the sum assured on death whereas an **endowment policy** pays out on death or the maturity date, whichever is earlier. A **non-participating** policy only pays out the guaranteed benefit whereas a **participating (or with-profits)** policy provides both guaranteed and non-guaranteed benefits, with the latter indirectly linked to the investment performance of the insurance company. However, **investment-linked** policies link policy values directly to the values of the underlying investments.

20. Annuities are intended to protect against longevity risk but can also be used as an investment vehicle or a hedge against adverse financial developments. The **immediate** annuity starts regular payments to the annuitant as soon as it is purchased whereas the **deferred** annuity starts regular payments only at a future date.

21. **Dollar cost averaging** refers to regularly investing equal dollar amounts in a security over a period of time. This approach may be suitable for investors who have very little time to monitor their accounts.

Chapter 9: Additional Reference Materials

A1: Key Features of Variable Capital Companies

The Variable Capital Company (VCC) is a new legal entity structure for investment funds and was introduced to offer asset managers greater flexibility and lower costs, thus encouraging fund managers to co-locate fund domiciliation with their substantive fund management activities in Singapore and further deepen the asset servicing ecosystem.

The following are key features of this new structure:

- A VCC can be used for both traditional and alternative funds, as well as both open- and closed-end funds.

- A VCC can house a standalone fund or a collection of sub-funds under an umbrella structure, with segregation of assets and liabilities of the sub-fund. An umbrella structure will allow asset managers to achieve cost efficiencies by consolidating administrative functions at the umbrella fund level.

- Shares of the VCC can be redeemed without shareholders' approval, and dividends are payable from capital, thus offering greater flexibility than existing investment companies. Shares are to be redeemed and issued at the NAV.

- A VCC must be managed by a fund manager that is registered, licensed, or exempted by the MAS.

- For taxation purposes, a VCC is a company and a single tax entity.

- A VCC does not need to publicly disclose its shareholder's register and, thus, offers investor privacy.

A2: CPF Investment Risk Classification System

The CPF Board has classified unit trusts included under the CPF investment Scheme (CFPIS) under different categories of risk. The Risk Classification System splits the investment risk associated with a unit trust into two axes: Equity risk and Focus risk.

i. **Equity risk** is related to the proportion of the "riskier' types of investments in the unit trust. The greater the proportion of assets invested in stocks, the higher the equity risk. Conversely, the greater the proportion of investments in bonds and cash, the lower the equity risk. There are four categories of equity risk:

   - Higher risk – equity funds;
   - Medium to high risk – balanced funds;
   - Low to medium risk – bond funds; and
   - Lower risk – cash funds.

ii. **Focus risk** reflects how focused the investments of the unit trust are in one particular geographical region, country, industry, or individual companies. There are two categories of focus risk: broadly diversified or narrowly focused:

   - A **broadly diversified** unit trust will tend to have investments that are spread across relatively more geographical regions, countries, industries, and individual companies.
   - A **narrowly diversified** unit trust will tend to have investments that may be focused in particular geographical regions, countries, industries, or individual companies.

In general, a unit trust will have a higher level of focus risk because it is relatively less diversified. For example, a unit trust that invests in shares of companies from around the world would have lower focus risk (i.e.; more broadly diversified) than a unit trust that invests in shares of Japanese corporations only. The Japan-only unit trust would have higher focus risk because it is less diversified (invests in one foreign country only).

Within the focus risk category, CPFIS-included unit trusts have been further divided into the following sub-groups:

- **Regional** – Unit trusts that have a regional exposure e.g., Europe, Asia or Emerging Markets

- **Sector –** Unit trusts with specific industry exposure e.g., healthcare and biotechnology
- **Country –** Unit trusts that are country-specific e.g., Japan, China

These broad sub-groups have been provided to make it easier for CPF members to perform comparisons across unit trusts that are roughly similar in terms of geographic focus.

Chapter 10: Warrants

Learning Objectives

The candidate should be able to:

- ✓ Describe the features of stock warrants.
- ✓ Explain the valuation of warrants, using intrinsic value, conversion price and warrant premium.
- ✓ Describe the gearing ratio in the context of warrant pricing.
- ✓ Understand the determinants of the minimum and maximum values of warrants.
- ✓ Illustrate how the warrant premium is influenced by the underlying stock price, the life of the warrant, exercise price, price volatility of the underlying stock, dividend on the underlying stock, leverage, interest rates and size of the warrant issue.

10.1 Definition and Terminology

10.1.1 Definition of Warrants

A warrant is a derivative instrument that gives the holder (buyer) the right, but not the obligation, to buy or sell a specified number of securities (usually the issuer's common share) at a fixed price (also referred as the exercise or strike price) within a specified time period.

Warrants issued by companies are traded on the Singapore Exchange Securities Trading Ltd (SGX-ST) platform and are also known as "company warrants". Warrants may be considered as long-dated equity call options. While call options expire within a few months, warrants typically have maturities of a few years.

Warrants are generally issued by corporations as a 'sweetener', attached to an offering of bonds or rights issues. This allows the issuer to obtain financing at a lower interest rate, hence reduce its financing cost. The warrants can be detached from the host security and then traded separately. Since the warrants are issued by the company as an incentive, there is no price paid per se for the warrant by the buyer of the host security.

When the warrants are exercised, new shares will be issued, increasing the number of shares outstanding and resulting in a dilution of the issuer's earnings among its shareholders. This differs from equity call options, which do not cause earnings dilution because no new shares are issued when the options are exercised.

10.1.2 Company Warrants vs Structured Warrants

The 2 types of warrants available on SGX are:

Company Warrant

A company warrant is one issued by companies to its shareholders. Its maturity is usually measured in years from the date of issue. If exercised, the company issues new shares, so the number of shares in the market increases.

Structured Warrant

A structured warrant is a warrant issued by a bank, financial institution, or other investment entity. The entity (i.e.; usually third-party issuer, not the underlying company) is in effect selling the buyer (holder) of the warrant an option to buy or sell the underlying company's shares at a pre-determined price at a later date. The underlying company does not endorse or guarantee such warrants.

The terminology for structured warrants is more complex compared to that of company warrants. Table 10.1.2(a) illustrates the terms of a structured warrant using an example of a warrant, CapitalaMBeCW181009 (COWW), issued by Macquarie Bank.

Table 10.1.2(a) – Terms of a Structured Warrant CapitalaMBeCW181009 (COWW)

Terms/Code	Attributes	Details
Capitala	Underlying asset (security)	The underlying asset is CapitaLand Limited shares.
MB	Issuer	Macquarie Bank Limited is the issuer.
e	Exercise style	Represents a European style warrant.
CW	Type of warrant	Represents a Call Warrant (PW represents a Put Warrant).
181009	Expiry Date (in YYMMDD format)	The warrant will expire on 9 October 2018.
COWW	Warrant code or symbol	A unique 4-digit identification code ending with the letter "W" assigned by SGX.

Source: Macquarie Bank Limited, Singapore Warrants Education webpage – https://www.warrants.com.sg/eduhtml

The differences between company and structured warrants are summarised in Table 10.1.2(b).

Table 10.1.2(b) – Characteristics of Company and Structured Warrants

Company-issued Warrants	Structured Warrants
Issued by company on its own shares	Issued by third party such as financial institutions
Less liquidity	More liquidity
One direction only i.e.; call warrant	Both call and put warrants can be issued
Usually one exercise price	Can have warrants on the same underlying asset but with different exercise prices
Maturity typically measured in years	Shorter-term maturity of usually 6 – 12 months
New shares issued upon exercise	No new shares issued upon exercise

10.1.3 Basic Features of Warrants

Some of the basic features of company warrants are as follows:

a. **Underlying security**. This refers to the security that can be purchased upon conversion of the warrant.

b. **Exercise price.** This is the price at which the warrant holder can buy the underlying security. Typically, when warrants are issued, the exercise price or strike price is set above the prevailing share price.

c. **Conversion ratio.** The conversion ratio states the number of warrants required for the conversion to one share.

d. **Expiration date.** The last date that a warrant can be exercised.

When warrants are not converted into the underlying shares, their holders do not receive dividends.

Figure 10.1.3 – Example of a Company Warrant Term Sheet - Duty Free International

Warrants / Convertible Securities / Certificates

DUTY FREE INTERNATIONALW220513

Last Date/ Time to submit Subscription Notice	13 MAY 2022 05:00 PM
Listing Date	18 MAY 2017
Expiry Date	13 MAY 2022
Outstanding Balance	491,400,042
Price ($) as at 2018-16-01	0.030
High for the year ($)	0.000
Low for the year ($)	0.000
High for the month ($)	0.000
Low for the month ($)	0.000
Exercise Price	SGD 0.43000
Underlying Stock	DUTY FREE INTERNATIONALLIMITED
Underlying Stock Price ($) as at 2018-01-16	0.265
Conversion Ratio	1 warrant(s) : 1 share(s)
Cash ($)	0.000
Premium/(Discount) (%) (Cash)	73.58
Premium/(Discount) (%) (Bonds)	
Gearing (x)	8.83
Remarks	

10.2 Warrant Valuation

Let us use the warrants of Duty Free International Limited (DF Int'l) (as shown in Figure 10.1.3) as an illustrative example. One DF Int'l warrant can be converted into one DF Int'l share by paying an exercise price of $0.430. The following sections illustrate the calculations of intrinsic value, conversion price, warrant premium, and gearing ratio based on the warrants of DF Int'l. On the date that the warrants of DF Int'l were listed (i.e., 18 May 2017), its stock was trading at $0.38 as shown in Figure 10.2. By 17 January 2018, its share price had decreased to $0.26 and warrants were trading at $0.030.

Figure 10.2 – Share Price of Duty-Free International Stock on Listing Date of Warrants



Source: Yahoo Finance

10.2.1 Intrinsic Value

When the share price is greater than the exercise price, the warrant is "in-the-money" and the intrinsic value is the difference between the current share price and the exercise price. When the warrant is in-the-money,

$$\textbf{Intrinsic Value} = \frac{\textbf{(SP - EP)}}{\textbf{n}}$$

where:

SP = share price
EP = exercise price
n = conversion ratio

When the share price is below the exercise price, the warrant is "out-of-the-money" and the intrinsic value is zero.

Since the issuance of the warrant by DF Int'l, its share price has remained below the exercise price of the warrant. So, the intrinsic value is zero since the difference between SP and EP is negative.

Assuming the share price of DF Int'l rises to $0.48 today. Given this scenario, the warrant would now be in-the-money, and the intrinsic value of the warrant would be ($0.480 - $0.430) ÷ 1 = $0.05.

10.2.2 Conversion Price

This is the total price that the investor has to pay to convert warrants into one underlying share. As shown in Figure 10.1.3, the warrant price for DF Int'l is $0.03.

Conversion price = nWP + EP

= $0.030 + $0.430

= $0.46

where WP = warrant price

10.2.3 Warrant Price and Warrant Premium

The warrant premium is the additional amount that is spent by purchasing warrants and paying the exercise price to convert to one share, as opposed to buying the share outright from the market. The term "premium" in warrant premium does not have the same meaning as that used in conventional options, where the premium is in fact the option price.

Warrant premium ($) = nWP + EP – SP

The above computation gives the warrant premium in dollar terms. To calculate the warrant premium in percentage, the formula is:

$$\textbf{Warrant Premium (\%)} = \frac{\textbf{nWP + EP - SP} \times \textbf{100}}{\textbf{SP}}$$

Let us examine the warrant premium of DF Int'l under the following 2 scenarios:

Scenarios	Scenario 1 (As at 17th Jan 2018)	Scenario 2 (Today)
Underlying Stock Price	$0.26	$0.480
Intrinsic Value	Zero per warrant	($0.480 - $0.430) ÷ 1 = $0.05 per warrant
Warrant Price ($)	0.03 per warrant	$0.08 per warrant (assumed price rise)
Warrant Premium ($)	0.030 + 0.430 – 0.26 = $0.20	0.080 + 0.430 – 0.480 = $0.030
Warrant Premium (%)	$\frac{0.20}{0.26}$ X $100\% = 76.92\%$	$\frac{0.030}{0.480}$ X $100\% = 6.25\%$

Investors are interested in warrants mainly because of the leverage effect. Leverage offers the potential for greater percentage gains (and losses) than the underlying share for a given change in the price of the underlying share. For example, comparing Scenarios 1 and 2, the share price of DF Int'l increased by 85% from $0.26 to $0.480 but the warrant price increased by 167% from $0.03 to $0.08.

10.2.4 Gearing Ratio

The gearing ratio compares the price of warrants needed to convert to one share relative to the share price at a point in time. Gearing gives investors the ability to participate in the performance of the underlying share at a lower price.

Gearing ratio = SP / nWP

= 0.265 / 0.030 = 8.8x (approximately)

10.2.5 Minimum Value and Maximum Value

Warrants should trade within well-specified price bounds in an arbitrage-free market.

For a warrant with a conversion ratio of 1, the **maximum value** of a warrant is the underlying share price. Most warrant prices do not trade at their maximum value and prices fall as the warrants approach expiration.

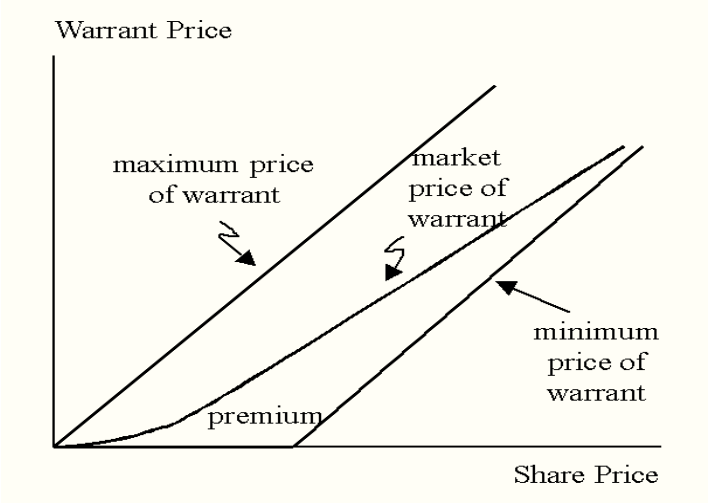
For a warrant with a conversion ratio of 1, the **minimum value** of a warrant is the intrinsic value of the warrant.

Minimum value = SP – E if SP > EP

= 0 if SP ≤ EP

In the DF Int'l example, when the share price of DF Int'l is $0.26 and the exercise price is $0.43, the share price (SP) is lower than the exercise price (EP), and the warrant is **out-of-the-money**. If the share price of DF Int'l rises to $0.48, the SP is greater that the EP, the warrant is now **in-the-money**. Should the SP be equal to the EP, the warrant is trading **at-the-money**.

Figure 10.2.5 –Warrant Price Bounds



When the share price is less than the exercise price, the minimum value of the warrant is zero as represented by the horizontal line. As the share price rises above the exercise price, the minimum price becomes positive and rises. Notice (as shown in Figure 10.2.5) that as the price of the underlying share increases beyond a point, the premium of the warrant price over the intrinsic value decreases. This is because the leverage potential of the warrant declines at some point, with a given share price increase constituting an increasingly smaller percentage of the share price. Also, note that the line representing the maximum price for the warrant is at a

45-degree angle to the x-axis, meaning that the maximum price of a warrant is equal to its underlying share price.

10.3 Factors Affecting the Price of a Company Warrant

As highlighted earlier, warrants issued by companies (i.e., company warrants) are typically attached to bonds or rights issues as a sweetener but can be detached and traded independently.

The price of a warrant when detached from its host security will be directly influenced by the following factors (see Table 10.3):

a. **Underlying share price.** Warrants are affected by the price of the underlying shares. Unlike structured warrants which may be call or put warrants, company warrants are all call warrants. As such, they are worth more if the price of the underlying shares rises.

b. **Life of the warrant.** Other things being equal, the longer the time before the warrants expire, the greater the warrant price. A warrant that is out-of-the-money may become attractive later as the share price appreciates. As the life of the warrant gets shorter, it will lose its speculative value. On the day of expiration, warrants are worth their intrinsic value, after which they are worthless.

c. **Exercise price.** The lower the exercise price, the greater the warrant price. The ability to convert warrants into a share by paying a lower exercise price increases the intrinsic value of the warrant and, hence, the warrant's price.

d. **Price volatility of the underlying share.** The more volatile the price of the underlying share, the higher the warrant price, because warrants benefit from price upside but limit price downside. Volatility can change over time.

e. **Dividend on the underlying share**. As warrant holders are not entitled to dividends, the declaration of dividends by the company will reduce the call warrant's price slightly, since call warrants "lose out".

f. **Leverage.** Warrants that have greater leverage opportunities than others are able to command higher prices.

g. **Interest rates.** The higher the interest rate, the higher the call warrant price due to interest savings from the lower capital needed to buy warrants compared to an equivalent long stock position. Even though company warrants are call warrants, for completeness, the higher the interest rate, the lower the put warrant price. This is because more interest has to be paid to buy warrants compared whereas an equivalent short stock position gains more interest on the short-sale proceeds.

h. **Size of warrant issue.** The more warrants a company has outstanding, the larger the potential earnings dilution of the existing shares. If the warrants were deep in-the-money (i.e., the share price is much higher than the exercise price), the conversion of the warrants would increase the number of outstanding shares and reduce the value of each share. Thus, there is a negative relation between the size of the warrant issue and the warrant price.

Table 10.3 - Factors Influencing Value of Company Warrants

Factors to Consider	**Warrant Price**
Remaining life of warrant ↑	↑
Exercise price ↑	↓
Underlying share price ↑	↑
Volatility ↑	↑
Dividends expectations ↑	↓
Leverage opportunities ↑	↑
Interest rates ↑	↑
Issue size of Warrant ↑	↓

10.4 Summary

1. Company warrants are call warrants and are only issued by the company. Both call and put structured warrants are issued by a third-party, usually a bank. Company warrants are settled with a delivery of shares. Structured warrants are cash-settled i.e., no delivery of shares.

2. A company warrant gives the investor an option to buy a stated number of shares of common stock at a specified price within a specified period. Unlike common shares, warrants expire, and warrant holders are not entitled to dividend payments.

3. In addition to the fundamental value of the underlying share, the warrant investor has to consider the intrinsic value, warrant premium, gearing ratio, and the expiration date of the warrant when evaluating warrants.

4. The maximum value of a warrant is the underlying share price. The minimum value of a warrant is the intrinsic value of the warrant, which is zero or the difference between the underlying share price and the exercise price, whichever is larger.

5. The price that investors are willing to pay for a warrant depends on the underlying share price of the warrant, life of the warrant, exercise price, price volatility of the underlying share, dividend expectations on the underlying share, leverage opportunity, interest rate level, and size of the warrant issue.

Chapter 11: Technical Analysis and Quantitative Analysis

Learning Objectives

The candidate should be able to:

- ✓ Explain the philosophy and assumptions in the application of technical analysis to investment decisions.
- ✓ Discuss the challenges to technical analysis.
- ✓ Describe the various technical indicators.
- ✓ Distinguish between support and resistance levels.
- ✓ Describe share price and volume analysis techniques.
- ✓ Explain chart patterns such as head and shoulders, double tops and bottoms, consolidation rectangles, triangles, flags and pennants, wedges and gaps in identifying price trends.
- ✓ Explain Elliott Wave Theory as a technical analysis approach.
- ✓ Describe the essentials of quantitative analysis.

11.1 Overview

11.1.1 What is Technical Analysis?

Besides fundamental analysis, another approach to security evaluation is technical analysis. Though controversial in nature, many traders employ technical trading rules or indicators to make investment decisions. In this Chapter, the basic philosophy and assumptions underlying technical analysis, its challenges, and common technical indicators and chart patterns are examined. The Elliott Wave Theory, a popular technical analysis approach based on sets of cycles and ratios movements, will also be described. Finally, we will discuss quantitative analysis and how it is applied in the investment process.

11.1.2 Basic Philosophy and Assumptions

Technical analysis refers to the methodology of forecasting fluctuations in securities prices through an examination of historical market data. This methodology can be applied to individual securities, the entire market, and in any asset class such as foreign exchange, gold, or bonds.

Technical analysis uses charts, technical trading rules, and indicators. It is sometimes called market or internal analysis because it studies demand and supply conditions based on market data such as price changes and trading volume. Technical analysis differs from fundamental analysis, which focuses on factors that affect a company's business profitability such as economic, industry, and company-specific factors.

Technical analysis believes that prices are determined by demand and supply which are, in turn, affected by both fundamental (such as macroeconomic forces) and non-fundamental factors (such as market sentiments and speculative forces). Technical analysts, therefore, study past price and volume behaviour to identify shifts in supply and demand forces, so as to predict future price behaviour. Technical analysts are not concerned about a security's fair value.

The underlying assumptions of technical analysis are that:

a. A security's market value is determined only by its supply and demand.

b. Supply and demand are affected by both rational and irrational factors. These factors include not only economic variables employed by fundamental analysts but also hopes, fears, opinions, moods, and guesses of potential buyers and sellers. The market constantly assesses all these factors.

c. Prices of individual securities and the overall market exhibit trends, which tend to persist over an appreciable length of time.

d. Trends develop in reaction to changes in the supply-demand balance. Such changes can be detected by studying market actions.

e. Chart patterns tend to repeat themselves.

Generally, technical analysts and non-technical analysts accept assumptions (a) and (b). Most would agree that, at any point in time, the price of a security is determined by its supply and demand. Most would also agree that supply and demand is governed by many variables. A key difference in opinion relates to the influence of irrational factors – technical analysts expect irrational factors to persist while fundamental analysts expect only a temporary effect, with rational beliefs prevailing in the long run. Certainly, many would agree that the market constantly assesses all these factors.

Opinions differ on assumption (c) about the speed of adjustment of share prices to changes in supply and demand. Technical analysts believe that new information affecting supply and demand is priced by the market over a period of time rather than at one point in time, given the fact that there are different sources of information or because some investors have earlier access to the information than others. Therefore, technical analysts believe that prices are expected to gradually adjust to reflect the gradual flow of information. As the speed of adjustment is gradual, prices are expected to exhibit trends that persist over long periods.

Assumption (d) implies that as prices move from one equilibrium value to a new equilibrium value, the price change can be detected sooner or later in charts. Technical analysts hope to benefit from this by being among the first to detect a shift in trend.

11.1.3 Challenges to Technical Analysis

The Efficient Market Hypothesis as discussed in Chapter 8 (Portfolio Management) presents the greatest challenge to technical analysis. Studies that tested weak-form market efficiency generally support the notion that prices adjust rapidly to new information coming into the market. Because markets price in all information very quickly, there is little opportunity for technical analysts to take advantage of market data.

According to the random walk theory, share prices follow a random walk, which means price movements are independent over time. Therefore, whether past price data can be used to predict future prices is questionable.

Another criticism against technical analysis is that many technical analysts are familiar with chart patterns and act on them at the same time. If a popular technical analysis trading rule is followed by many market participants, some participants will act first, causing prices to change quicker than expected, so that only those who act earliest will benefit. The popularity and the resulting competition will eventually negate the value of the technical trading rule. That said, as chart reading can be a very subjective exercise, it is not unusual to have several interpretations of the same chart pattern. There is, therefore, always an element of doubt and disagreement in interpreting chart patterns.

11.2 Price-based Technical Indicators

Technical analysts use price-based technical indicators, which attempt to mine information from current and historical market prices, in their attempt to 'beat' the market. Typically, these technical indicators ‘measure’ market momentum and the relative state of the market; whether it is ‘overbought’ or ‘oversold’.

The following are some common technical indicators.

11.2.1 Moving Average

A moving average is the average of the security’s closing price over a given period of time. Typically, each day of the moving average is given the same weight in computing the average. Hence in a 100-day moving average, each day is given a weight of 1/100. A more sophisticated moving average might give higher weights to more recent prices than distant prices. The moving average is more stable than the daily price because of the inclusion of past prices, making it a more useful measure for detecting trends. If the current market price starts to diverge directionally from the moving average, this may indicate a trend reversal. This is further discussed in Section 11.5.2.

11.2.2 Advance-Decline Line (Breadth Index)

The daily movements of an index can be misleading because the index is based on the movement of selected shares, which may not represent the overall market. The advance-decline (A/D) line, which is an index that covers all securities traded, is designed to overcome this weakness by capturing the breadth of the market. The A/D line measures, on a cumulative daily basis, the net difference between the number of shares, traded on the SGX for example, that have price advances and those that have price declines, in a particular period (usually a day or a week). The net advance or decline is calculated by subtracting the number of declines from the number of advances. The A/D line is drawn by plotting a running total of these numbers over time.

The A/D line is compared to an index, such as the Straits Times Index (STI), and normally rises and falls in line with the index. If both are rising (declining), the market is said to be technically strong (weak). In a rising market, any divergence between the index and A/D line is of concern because it means that most shares are not participating in the rising market, indicating a market peak. On the other hand, if the A/D line is rising and the index is declining, it signals a reversal of the falling trend.

11.2.3 Moving Average Convergence Divergence (MACD)

MACD is a trend following momentum indicator that shows the relationship between two moving averages of a security's price. It is designed to reveal changes in the strength, direction, momentum, and duration of a trend in a stock's price.

The MACD indicator is a collection of three time series calculated from historical price data, most often the closing price. The MACD series is the difference between a "fast" (short period) exponential moving average (EMA), and a "slow" (longer period) EMA of the price series. The 'signal' or average series is an EMA of the MACD series itself. An exponentially weighted moving average reacts more significantly to recent price changes than a simple moving average (SMA), which applies an equal weight to all observations in the period.

The MACD indicator thus depends on the three-time parameters, namely the time constants of the three EMAs. The notation "MACD (a,b,c)" usually denotes the indicator where the MACD series is the difference of EMAs with characteristic times a and b, and the average series is an EMA of the MACD series with characteristic time c. These parameters are usually measured in days. The most commonly used values are 12, 26, and 9 days, that is, MACD (12,26,9).

A fast EMA responds more quickly than a slow EMA to recent changes in a stock's price. By comparing EMAs of different periods, the MACD series can indicate changes in the trend of a security. When the MACD crosses and falls below the signal line, it is a bearish signal that indicates that it may be time to sell. Conversely, when the MACD crosses and rises above the signal line, the indicator gives a bullish signal, which suggests that the stock price is likely to experience upward momentum.

11.2.5 Bollinger Bands

Bollinger Bands are a type of price envelope developed by John Bollinger. Price envelopes define upper and lower price range levels. Bollinger Bands are envelopes plotted at a standard deviation level above and below a simple moving average of the price. Because the distance of the bands is based on standard deviation, they adjust to volatility swings in the underlying price.

The first step in calculating Bollinger Bands is to compute the simple moving average (SMA) of the security in question, typically using a 20-day SMA. A 20-day SMA would average out the closing prices for the first 20 days as the first data point. The next data point would drop the earliest price, add the price on day 21 and take the average, and so on, keeping the data set to 20. Next, the standard deviation of the security's price is obtained based on the data set.

Finally, multiply that standard deviation value by two (a commonly used multiplier) and both add and subtract that amount from each point along the SMA. Those produce the upper and lower bands.

The purpose of Bollinger Bands is to provide a relative definition of high and low prices of a market. By definition, prices are high at the upper band and low at the lower band. Many traders believe the closer the prices move to the upper band, the more overbought the market, and the closer the prices move to the lower

band, the more oversold the market. An overbought or oversold market may be primed for a trend reversal or price correction.

Individual price-based technical indicators are typically not used in isolation but rather in conjunction with other indicators and techniques. Moving averages (MA) and relative strength index (RSI), both to be discussed in section 11.5, are also 2 common indicators used in technical analysis.

11.3 Trends, Support and Resistance Levels

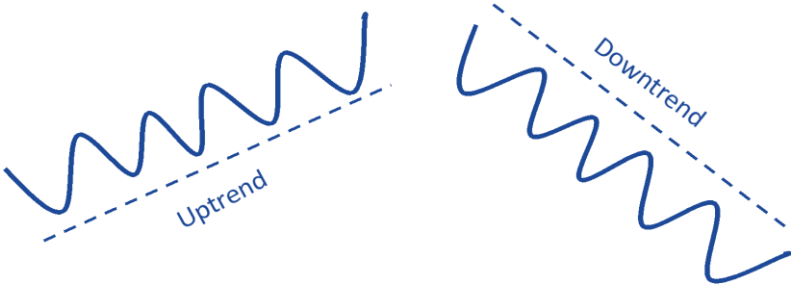
Technical analysts using charts to study the movements of the overall market and individual securities often rely on trend lines and trend channels to identify trends. An uptrend is represented by a rising trend line connecting a series of ascending lows while a downtrend is represented by a falling trend line connecting a series of descending highs.

An uptrend is in place when current highs are above previous highs and current lows are also above previous lows. Note that when current highs (or peaks) are no longer above previous highs and, more importantly, if current lows are falling below previous lows, a trend reversal may potentially be occurring.

Likewise, a downtrend occurs when current lows are lower than previous lows and current peaks are also lower than previous peaks. A potential change in the trend may occur when this pattern is no longer intact.

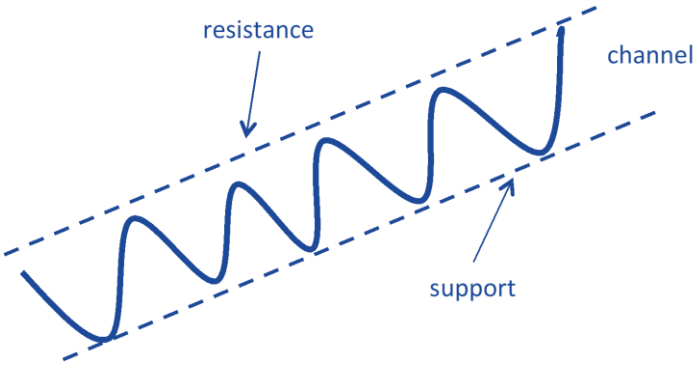
Figure 11.3(a) below illustrates both rising and falling trend lines.

Figure 11.3(a) – Trend lines



The trading range between the top and bottom trend lines is called the **trend channel**, which may be rising, flat, or declining. Figure 11.3(b) illustrates a rising trend channel. If share prices stay within this rising channel, the technician would hold the share and ride the trend.

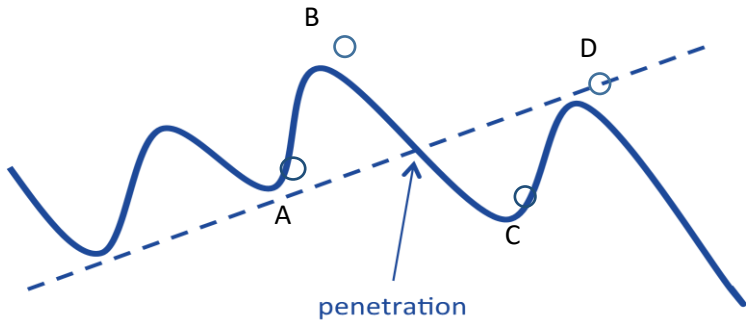
Figure 11.3(b) – Rising Trend Channel



A support level is a price range where a substantial increase in demand is expected. At this level, prices are relatively attractive for buyers who had missed the opportunity to buy earlier. As prices increase, a resistance level develops where buyers refuse to pay higher prices. At the same time, sellers may be attracted by higher prices to take profits on their holdings. This supply is called the market "overhang".

A movement through and beyond either the support or resistance level is considered a breakout and signals a strong confirmation and continuation of the primary trend. A breakout above the resistance level is a bullish signal, whereas a breakout below the support level is a bearish signal. Besides determining the trend, technical analysts are also alert to signs of reversals. A penetration below a support line as illustrated in Figure 11.3(c) would be a sell signal as it indicates a reversal of the uptrend.

Figure 11.3(c) – Trend Reversal



In the trend reversal example shown above, the price movement to point "C" has resulted in a new low that is past the previous low at point "A". Similarly, the price movement to point "D" is lower than the previous high at point "B". This confirms that the trend has reversed downwards.

11.4 Significance of Volume

Besides monitoring price movements, technical analysts also monitor trading volume. Volume, defined as the number of securities transacted, shows the intensity of excess demand or supply at a point in time and reflects changes in investors' attitudes.

For example, if prices advance on low volume, the enthusiasm implied from the price rise is not as strong as when a price rise is accompanied by very high volume. Therefore, a price increase accompanied by heavy volume relative to the security's normal trading volume is an indication of bullish activity. Similarly, a price decline with heavy volume is very bearish because it reflects a strong and widespread desire to sell the security. In a sustained bull market, if prices "correct" downward on declining volume, this may indicate a lull followed by a continuation of the rally. On the other hand, a rally that develops on contracting volume could signal a possible trend reversal.

11.5 Share Price and Volume Techniques

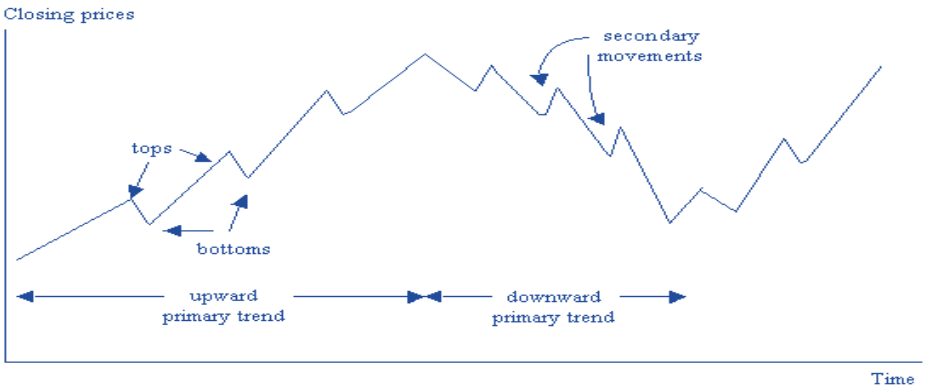
Three share price and volume techniques are discussed here, namely, the Dow Theory, the Moving Average Line, and the Relative Strength Index.

11.5.1 The Dow Theory

The Dow Theory, named after its creator Charles Dow, is the oldest and most popular method used to identify long-term trends in the equity market. The Dow Theory is based on the movements of the Dow Jones Industrial Average (DJIA) and the Dow Jones Transportation Average (DJTA).

The Dow Theory postulates 3 movements in the market (see Figure 11.5.1(a)): the primary movement, the secondary movement, and the daily fluctuations. The **primary movement** is the market trend, which lasts from several months to several years. The trend is either bullish (up) or bearish (down). Historically, bear markets last for a shorter period than bull markets. The Dow Theory attempts to determine the primary movement in the market. The **secondary movement** of the market is shorter in duration than the primary movement and opposite in direction. The secondary movement usually lasts from several weeks to months, and usually retraces one-third to two-thirds of the previous advance (decline) in a bull (bear) market. **Day-to-day fluctuations** are considered random and have no forecasting value for long-term investors. In the Dow Theory, these fluctuations are not considered to have any impact on long-term price trends.

Figure 11.5.1(a) – Line Chart of Daily Closing Prices with Dow Theory Signals



The Dow Theory forecasts are based upon the primary and secondary movements of the market. The purpose of Dow Theory is to determine where the market is and where it is going, but not the magnitude of the movements. In practice, the Dow Theory states that if the swings of the market index are successively higher and the successive lows are also higher, then the market trend is up, and investors are in a bull market. Conversely, if the successive highs and successive lows are lower, then the market direction is heading downwards, and investors are in a bear market.

Figure 11.5.1(b) shows how the Dow Theory could be used. The trend from P_1 to P_3 is bullish, because P_2 is higher than P_1 and P_3 is higher than P_2, and because L_2 is higher than L_1 and L_3 is higher than L_2.

Figure 11.5.1(b) – Upward Primary Trend

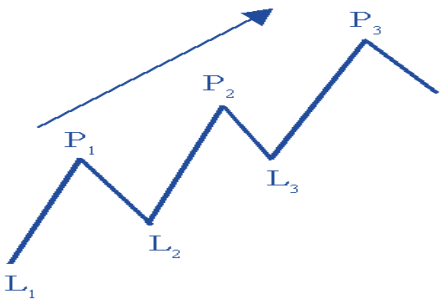
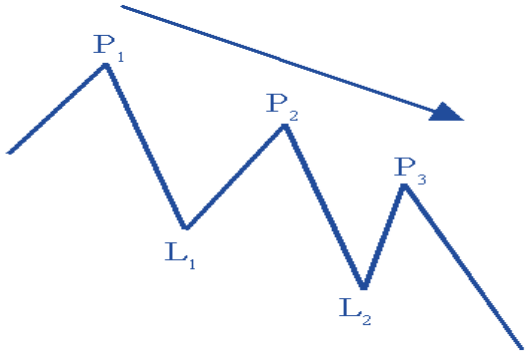


Figure 11.5.1(c) shows a primary downtrend. When the market trend changes, P_2 is lower than P_1 and P_3 is lower than P_2. Likewise, L_2 is lower than L_1.

Figure 11.5.1(c) – Downward Primary Trend



A new primary trend will only be confirmed when both the DJIA and DJTA give the same signals. When the two averages diverge from each other, the Dow Theory assumes that the prior trend is maintained.

The main weaknesses of the Dow Theory are as follows:

a. It relies too heavily on the DJIA, which contains a limited selection of shares. DJIA shares are characterised by old, large "blue chip" companies. Aggressive "high-tech" companies are under-represented in the DJIA.

b. It is slow in identifying a major trend and may miss an important price move or be "whiplashed" (i.e., be caught by a short-term price reversal even though the long-term trend is intact but not yet visible).

c. It uses the Transportation Average, which comprises mainly shares of railway companies, which no longer have a major impact on the economy.

d. It is designed to identify the major trend and, hence, is not very useful for trading the short trend.

e. It does not help to identify which shares to buy or sell.

11.5.2 Moving Average Line

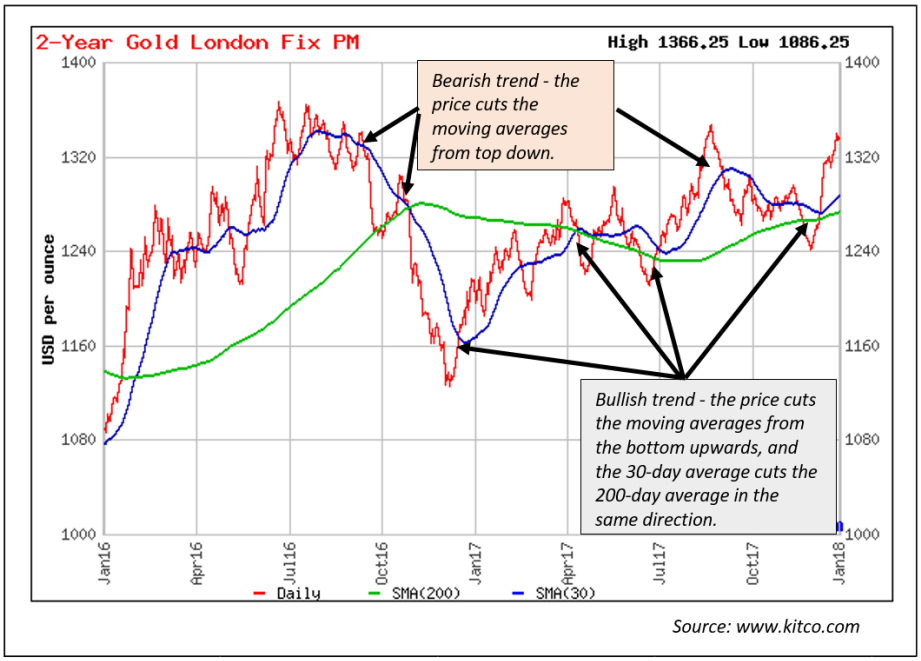
Technical analysts compute moving averages of a series so that erratic price changes are smoothened out and the general underlying trend can be clearly observed. Moving averages are used as a means of confirming the direction of price trends. Popular moving averages used in technical analysis are the 200-day and 50-day moving averages. The 200-day moving average is normally used to identify long-term trends while a 20- to 30-day moving average is used to identify short- and intermediate-term trends.

Trend changes are identified not by a change in the direction of the moving average, but by a crossover between the moving average and the index itself. If the overall price trend has been negative for some time, the moving-average price line would generally lie above current prices. If the price reverses and breaks upwards through the moving-average line from below and is accompanied by heavy trading volume, most technical analysts would consider this a very positive change and expect a reversal of the downtrend.

In contrast, in an uptrend, the moving-average line would be rising but below current prices. If current prices break downwards through the moving-average line from above and is accompanied by heavy trading volume, this would be considered a bearish pattern that would signal a reversal of the long-run uptrend.

Figure 11.5.2, shows a bullish price trend on gold, as confirmed by the 200-day and 30-day moving averages.

Figure 11.5.2 - Gold Moving Average Line



The moving average is an average of past prices. Hence, if prices continue to rise, the moving-average line would also be rising, but at a rate below that of current prices. If current prices (which lead the moving averages) are falling, the moving average line will follow the downtrend. As such, the price line will naturally cut the moving average line when a trend reversal occurs.

Typically, moments when prices cut the moving average line would be accompanied by heavy trading. This trend would be even stronger if:

a. the price cuts both short- and long-term moving averages, and

b. the short-term average cuts the long-term average in the same direction.

11.5.3 Relative Strength Index (RSI)

The Relative Strength Index (RSI) is a momentum indicator that measures the degree of buying or selling interest in a particular security. This index was developed by J. Welles Wider and measures the speed of change of price movements. The basic formula for RSI is shown below:

$$RSI = 100 - \left[\frac{100}{1 + \frac{Average\ of\ upward\ price\ change}{Average\ of\ downward\ price\ change}}\right]$$

The RSI is now a standard part of the charting tools available today and provides an easy visual measure of investment interest. The index ranges from zero to 100, and an RSI above 70 is considered overbought. Conversely, an RSI below 30 would signal an oversold condition.

A better way to use the RSI is to track the past peaks and troughs of the underlying and examine the typical RSI levels for the particular market and security in order to better calibrate the overbought and oversold RSI levels.

The RSI is useful in situations of surprises and shocks. One such scenario is the emotional selling of stocks under conditions which are considered unwarranted, resulting in oversold conditions. For the contrarian, a low RSI of 20-30 or lower could signal a potential buying opportunity. Likewise, in situations of overly optimistic demand, a strong RSI of 70-80 or higher could signal a profit taking opportunity.

The RSI is less useful when trends are steady and consistent. One example would be where constant demand for a growth stock causes its RSI to be persistently high. The RSI level should then not be viewed as an indication to sell; rather one should look for an opportunity to accumulate the stock when the RSI is not as high.

Figure 11.5.3 – Relative Strength Index Chart

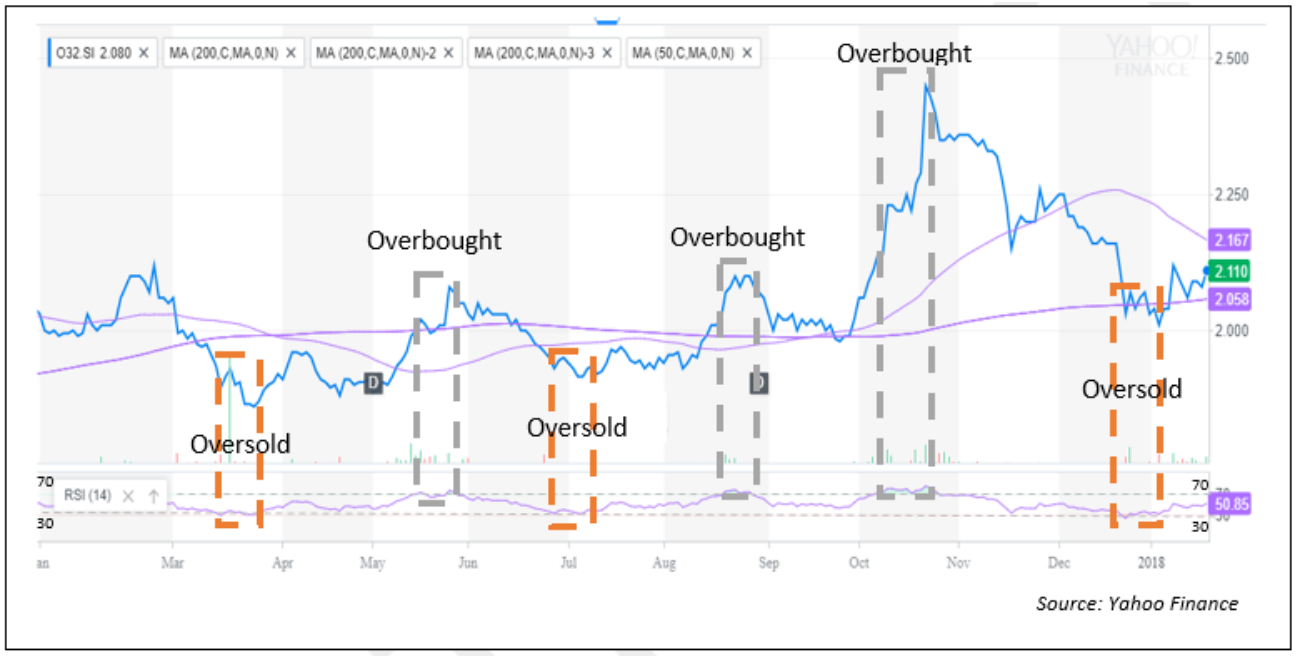


Figure 11.5.3 illustrates OLAM International, a commodity based local stock that has gone through several strong price fluctuations since 2012. Note that, in the case of this stock, periods of overselling and overbuying can last for as long as one-two weeks, during which prices can continue to rise or fall, respectively. Hence, technical analysts will usually wait for a few days of overbuying (before recommending taking profit) and overselling (before recommending accumulation).

11.6 Charts and Chart Patterns

Charts are a technical analyst's main tools. There are different charts, each suited to a different type of analysis. Some of these include:

a. Line charts;

b. Bar charts;

c. Candlestick charts; and

d. Point and figure charts.

Most of these will have chart points to show the high, low, opening and closing prices for each trading day.

Technical analysts believe that chart patterns can be used to predict future price movements. There are two types of price patterns – reversal and continuation. Reversal patterns indicate that an important reversal in trend is occurring. The continuation patterns suggest that the market is only taking a pause, possibly to correct a short term overbought or oversold condition, after which the existing trend is expected to resume.

11.6.1 Head and Shoulders

The head-and-shoulders pattern is a reversal pattern and is probably one of the more reliable indicators of major reversal patterns. The upright head-and-shoulders formation occurs at market peaks. It consists of a "left shoulder", a "head", and a "right shoulder":

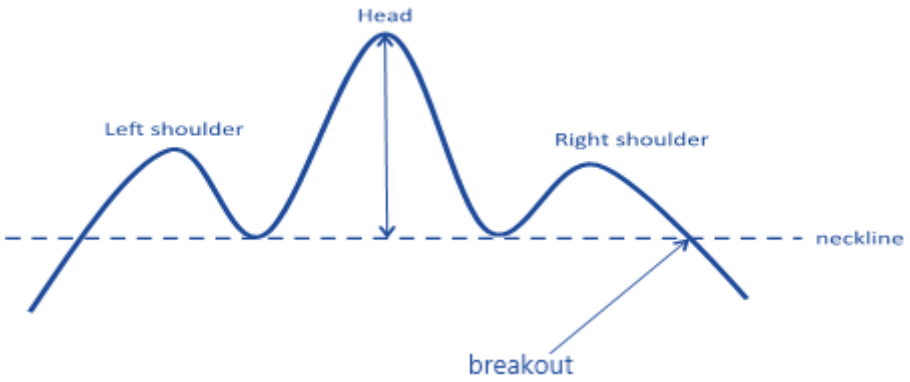
Left shoulder - 1st strong price rally normally accompanied by heavy volume, followed by a price decline with considerably less volume.

Head - Prices advance again on high volume to reach a higher level than the top of the left shoulder before falling on lower volume. Prices will fall below the top of the left shoulder but stay above or near the low of the left shoulder.

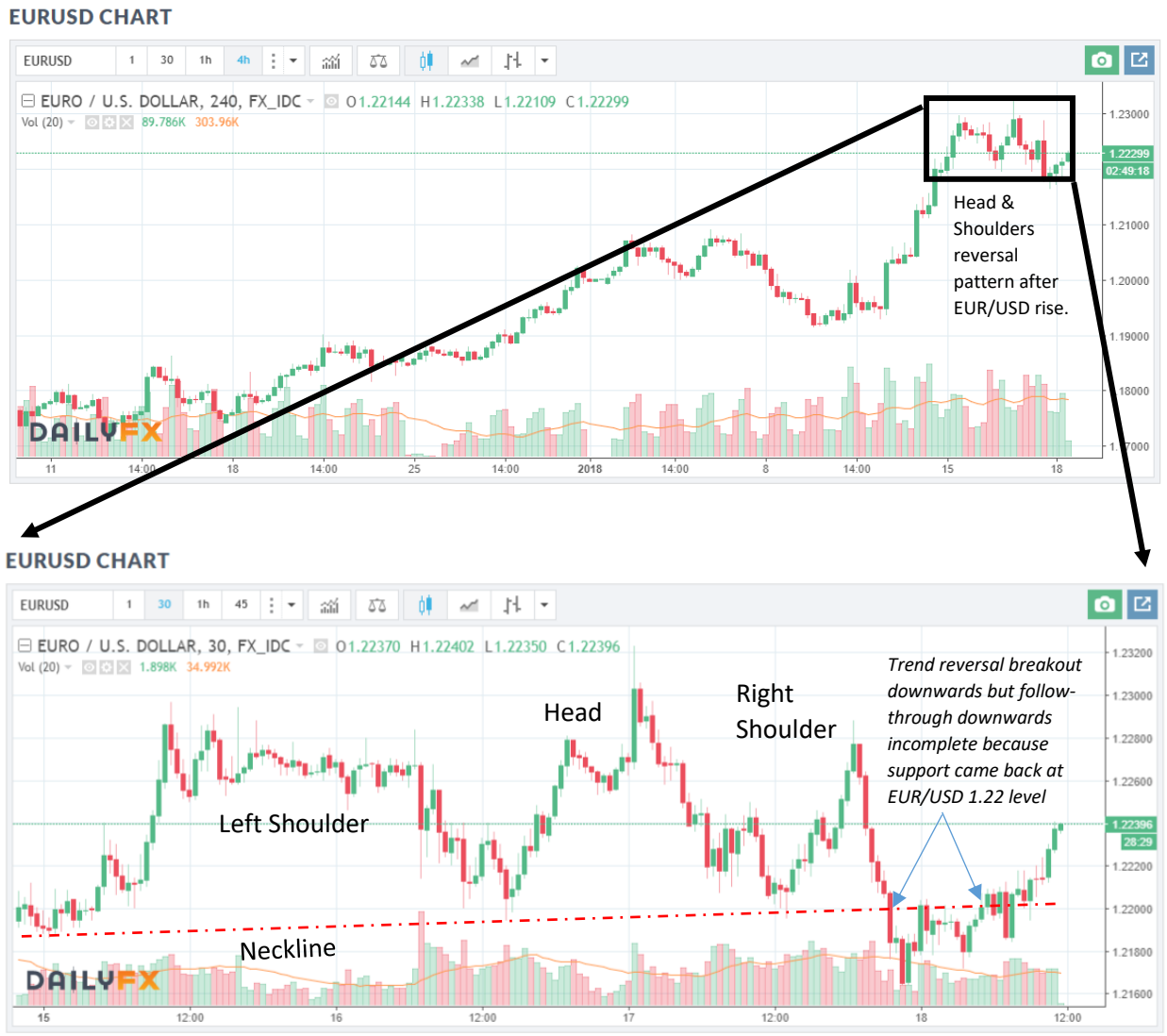
Right shoulder - A 3rd rally occurs, but this time on less volume than the volume during the formation of either the left shoulder or the head. Prices fail to reach the height of the head and fall through the neckline.

A breakout is confirmed if the price closes below the neckline by 3% of the share price. Once penetrated, the potential magnitude of decline is measured by the distance between the head and the neckline, projected downward from the neckline as shown in Figure 11.6.1(a).

Figure 11.6.1(a) – Upright Head and Shoulders



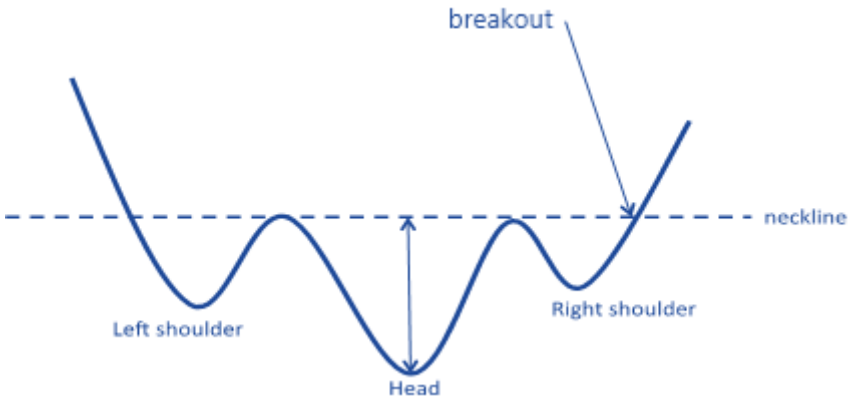
When price reactions are quite rapid, the formation of such patterns can take place over a few weeks. Otherwise, they may take place over a few months or years. The formation of such patterns tends to take a shorter time frame for currencies. However, for other products like commodities, the price pattern formation may take a long time.



Inverse Head and Shoulders

While the regular upright head and shoulders formation indicates a market peak, the inverse head and shoulders formation indicates a market bottom (see Figure 11.6.1 (b)).

Figure 11.6.1(b) – Inverse Head and Shoulders



Similarly, the inverse head-and-shoulders pattern also comprises a "left shoulder", a "head", and a "right shoulder".

Left shoulder – The pattern begins on the left with a significant selloff on high volume, followed by a brief rally that is sometimes supported by short covering. This rally is usually of lower volume.

Head - The second wave of selling involves those participating in the initial brief rally also giving up their positions. This heavy volume of selling causes the price to bottom out and form the head. The head may be extended and consolidated for a period with the selling absorbed over time by bargain hunters. The RSI would also typically indicate an oversold position at this point in time. Prices are driven up slightly due to pick-up in demand from parties who see value in the oversold position or current price level.

Right shoulder – Following this, some profit-taking for shorter-term traders and those who did not manage to sell-off earlier may occur at this stage. This round of sell-off are typically much lighter, shorter-lived, and is met with demand, which results in the right shoulder becoming a low higher than the previous low. With this, the first part of the reversal is created. The resistance at the right shoulder creates the neckline.

When demand picks up and the price breaks past the neckline, volume will typically rise as well. Once the pattern is fully formed, it implies that the prior downtrend is over, and an uptrend reversal is likely.

Figure 11.6.1(c) – Inverse Head and Shoulders



Figure 11.6.1(c) shows that crude oil prices over the three years (Oct 2014 to Dec 2017) has reflected an inverse head-and-shoulders pattern. If you look at the price charts for some of the oil companies, like Exxon as shown on figure 11.6.1(d), a similar inverse head-and-shoulders pattern is also apparent in the later part of 2017.

Figure 11.6.1(d) – Inverse Head and Shoulders

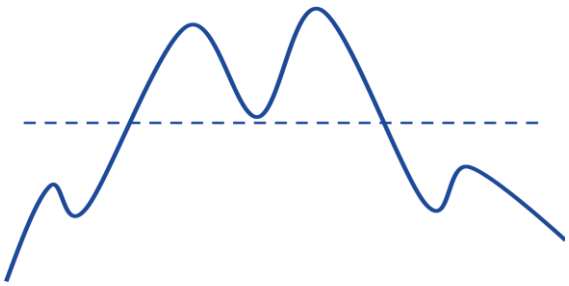


Note that in the stock price chart of Exxon in Figure 11.6.1(d), we have also reflected the two previous indicators RSI and Moving Averages, which confirms the reversal pattern. The price pattern on the right shoulder is particularly interesting because the price rests on the 50-day moving average after cutting it and the 200-day moving average twice from the bottom up previously. The price broke the neckline convincingly and trended upwards thereafter.

11.6.2 Double and Triple Tops and Bottoms

A double top is a reversal pattern indicating a market top. It consists of two "tops" separated by a valley in prices. The main feature of a double top is that the 2nd top is formed with distinctly lower volume than the 1st. A double top is confirmed when the neckline is broken on the way down, as shown in Figure 11.6.2(a).

Figure 11.6.2(a) – Double Top



There can also be more tops rising to reach a resistance level which results in a selloff. Once the resistance level is reached, the volume of each subsequent rise falls and the number of buyers pushing prices toward the peak eventually dwindles. This forces prices to correct downwards as buying interest is eventually overwhelmed by selling interest.

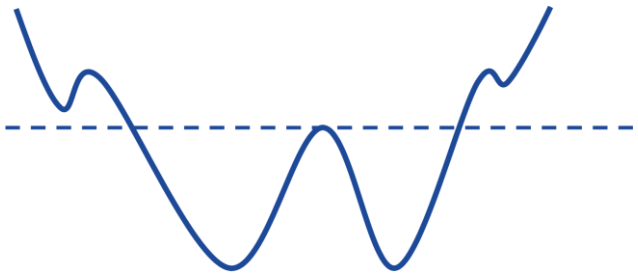
An example of double and triple tops is seen in the price of gold traded against the USD in 2017 and early 2018 (refer to Figure 11.6.2(a)).

Figure 11.6.2(a) – Double Top



A double bottom occurs in a bear market and signals a potential reversal in trend. The formation of the 1st bottom is normally accompanied by heavy volume while the 2nd bottom is usually dull with little trading volume. Volume increases significantly on the breakout. Some "double" patterns can extend to form triple tops or bottoms. As with the double top, a double bottom formation is confirmed if the neckline is broken on the way up. Refer to Figure 11.6.2 (b) for illustration.

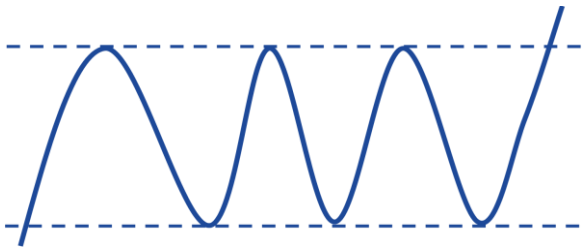
Figure 11.6.2(b) – Double Bottom



11.6.3 Consolidation Rectangles

A consolidation rectangle is formed when prices move sideways within a narrow range for an extended period of time, usually in a continuation pattern. A breakout from the consolidation rectangle would be technically significant with a break upwards indicating a bullish trend and a break downwards signalling further weakness. During the formation of the rectangle, it is difficult to predict which way the price will ultimately break. Therefore, it is assumed that the prevailing trend will continue until proven otherwise. Refer to Figure 11.6.3.

Figure 11.6.3 – Consolidation Rectangle

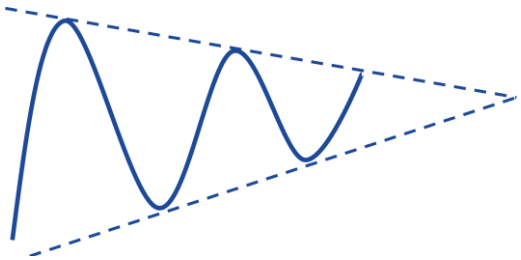


11.6.4 Triangles

Triangles are formed with four points of contact – two support and two resistance points. There are generally three types of triangle formation.

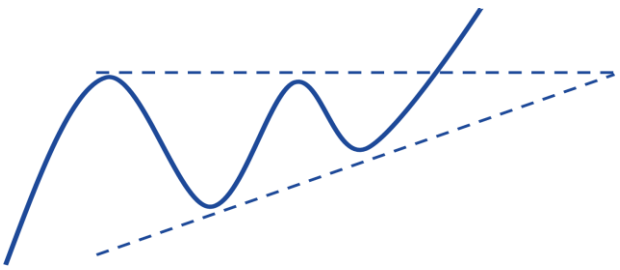
Symmetrical triangles are formed by price consolidations bounded by an upward sloping support line and a downward sloping resistance line. A breakout may occur on either side of the symmetrical triangle and is an indication of a major move in the same direction. Refer to Figure 11.6.4(a).

Figure 11.6.4(a) – Symmetrical Triangle



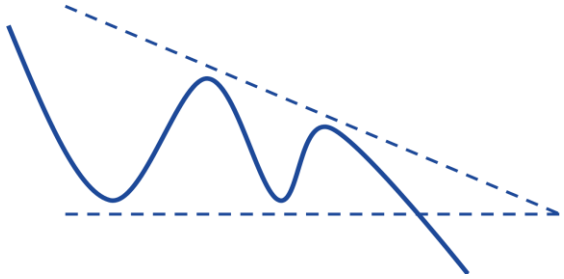
Ascending triangles are bullish indicators, formed by price consolidations bounded by an upward sloping support line and an almost horizontal resistance line. The ascending triangle indicates that buyers are more aggressive than sellers. Once the limited supply is absorbed, prices are expected to advance rapidly as shown in Figure 11.6.4(b).

Figure 11.6.4(b) – Ascending Triangle



Descending triangles are bearish indicators and are consolidations within an almost horizontal support line and a downward sloping resistance line. Sellers are more aggressive resulting in lowering rally peaks while buyers are buying at a set price. Once the limited demand is filled at the floor of the descending triangle, prices will fall sharply if there is continued selling. Refer to Figure 11.6.4(c).

Figure 11.6.4(c) – Descending Triangle

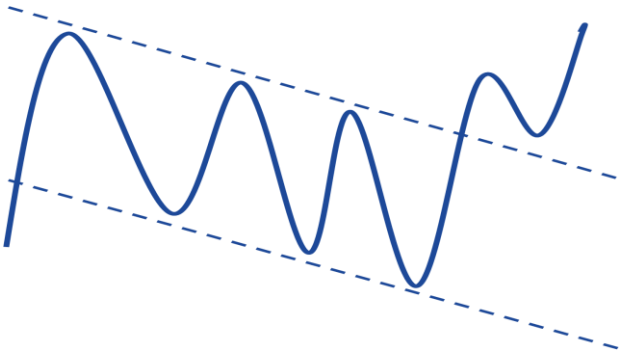


11.6.5 Flags and Pennants

The flag formation looks like a flag on the chart and may be described as a parallelogram of price fluctuations. It usually occurs after a sharp, almost vertical rise or fall in price, followed by declining volume. As the flag is completed, prices break out in the same direction that they were moving in before.

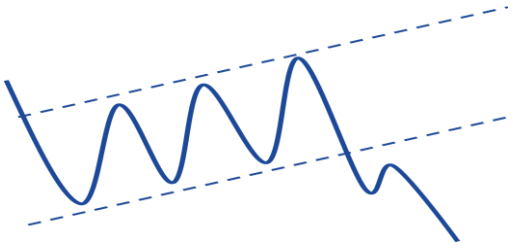
In a rising market, the flag is usually formed with a slight downtrend. Volume is normally very heavy just before the flag formation begins. As the pattern develops, volume will dry up, only to explode as the price works its way out of the formation. Flags can take as few as 5 days to as many as 3-5 weeks to form. Essentially, they represent a period of controlled profit taking in a rising market. Refer to Figure 11.6.5(a).

Figure 11.6.5(a) – Flag Formation in a Rising Market



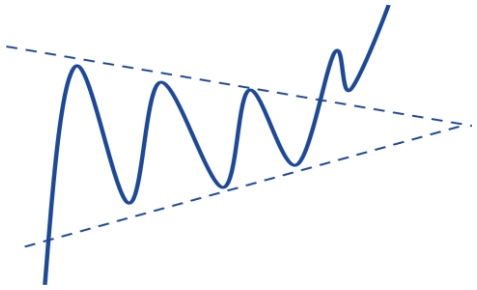
In a falling market, the flag is usually formed with a slight upward bias instead but is also accompanied by declining volume. This formation of rising prices with declining volume is essentially bearish. When the price breaks out from the flag downwards, the almost vertical slide that preceded the formation of the flag resumes. Although volume tends to pick up at this point, it does not have to be explosive, unlike the breakout from a flag formed during an upward move. Refer to Figure 11.6.5(b).

Figure 11.6.5(b) – Flag Formation in a Falling Market



A pennant is formed after a sharp rise or fall in price with lower volume during its formation. The difference between a pennant and a flag is that the pennant is formed by a set of converging, instead of parallel lines. Refer to Figure 11.6.5(c).

Figure 11.6.5(c) – Pennant

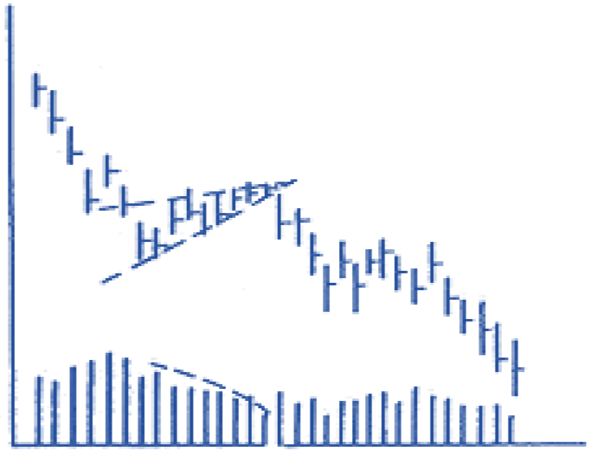


11.6.6 Wedges

A wedge is formed by a series of converging peaks and troughs. The converging lines in a wedge both move in the same direction.

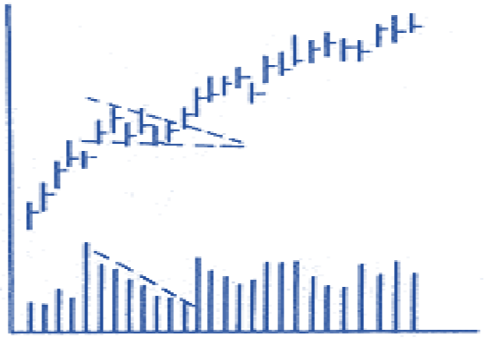
In a rising wedge, both boundary lines slant upwards. A rising wedge usually develops on a falling price trend. Following its completion, prices usually fall very sharply, especially if volume is high. Rising wedges are characteristic of bear market rallies. Refer to Figure 11.6.6 (a).

Figure 11.6.6(a) – Rising Wedge



A falling wedge is bounded by converging lines that slant downwards. It represents a temporary interruption of a rising trend. When prices move out of a falling wedge, they are likely to drift sideways before they begin to rise. Refer to Figure 11.6.6 (b).

Figure 11.6.6 (b) – Falling Wedge

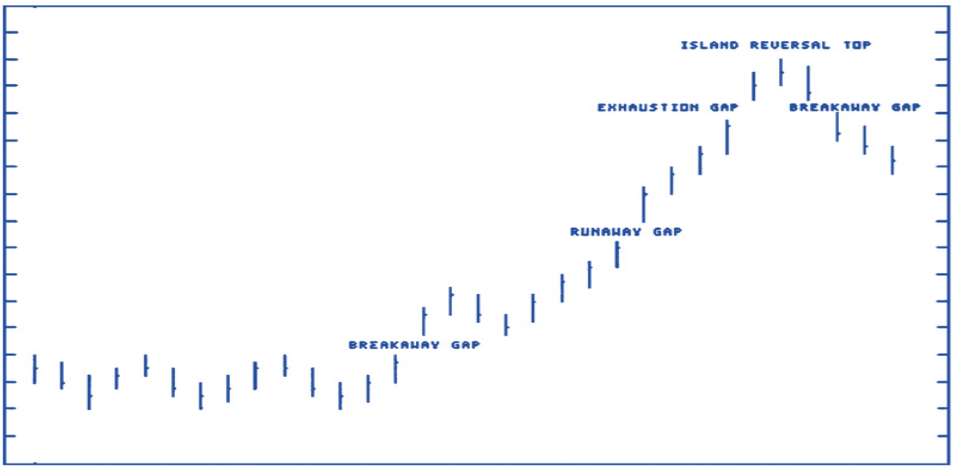


11.6.7 Gaps

A gap represents a price range where no transactions occur. Gaps occur when the lowest (highest) traded price for a given period is higher (lower) than the highest (lowest) traded price in the previous period. On a chart, a gap appears as an empty vertical space between adjacent trading periods. Daily gaps are more common than weekly or monthly gaps. A gap is closed or covered when the price retraces the entire range of the gap. A gap may be covered in a few days, weeks or months, if at all. Although the general principle is that gaps will eventually be covered, this is not always the case. Accordingly, trading strategies should not be implemented solely on the assumption that a gap will be filled in the immediate future (see Figure 11.6.7).

There are three types of gaps. A **breakaway gap** appears when a price breaks out of a price pattern. Generally, the development of a breakaway gap emphasizes the bullishness or bearishness of the breakout. To confirm a change to a bullish sentiment, an upside breakout should be accompanied by relatively heavy volume. **Runaway gaps** or **continuation gaps** occur during rapid, straight-line rises or falls when price trends accelerate, and emotions run high. **Exhaustion gaps** represent the end of a rapid rise or fall and are the last in a series of runaway gaps. It is not easy to ascertain whether a gap is another runaway gap or the exhaustion gap. Generally, breakaway gaps develop at the beginning of a move, runaway gaps in the middle of a move, and exhaustion gaps at the end of a move.

Figure 11.6.7 – Gaps

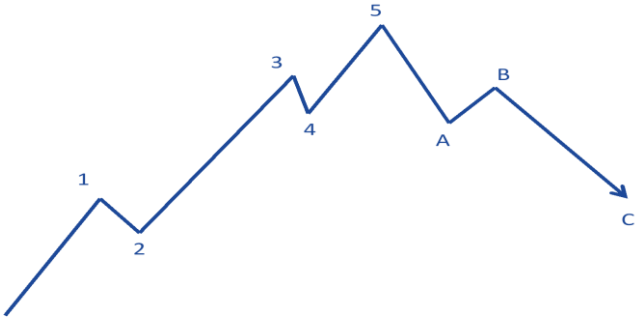


11.7 Elliot Wave Theory

In his study of equity prices in 1938, R.N. Elliott came up with the Elliott Wave Theory, which states that prices move in cycles of waves and sub-sets of waves and exhibit the Golden Ratio, 1.618[54]. Each cycle comprises a 5-wave advance and a 3-wave retreat.

In Elliot Wave charts, the markets move in 5-wave and 3-wave patterns, and the turning points that create these patterns are often at Fibonacci support and resistance levels (61.8, 161.8, etc.). See Figure 11.7.

Figure 11.7 – Elliott Waves



11.8 Quantitative Analysis

Quantitative analysis essentially involves the abstraction of market reality, which comprises the economy, financial markets and its environment, into relations that are incorporated into a financial model. The objectives of these models are to identify price anomalies and arbitrage possibilities and, lately, to even increase the speed of access to market information that will give an advantage to the investor.

The significance and importance of quantitative analysis is evidenced by the rapid growth in the use and development of quantitative techniques. Quantitative analysis uses statistics and calculus for a variety of purposes such as pricing, algorithmic trading, asset allocation, risk management, trading strategy modelling, portfolio optimization, and identification of trading and investment opportunities in the markets.

Quantitative analysts are typically called “quants” or “rocket scientists”. Within an organization, quants could be employed in the front office to develop trading strategies using complex mathematical tools, or in the risk management department to measure the risks that have been bundled into a transaction or product.

Quant funds are funds that extensively use quantitative analysis in their fund management decisions. One can look at this development as an outcome of financial markets that are becoming more sophisticated in their understanding and measurement of risks. While less sophisticated traders might just look at price changes, quants will decompose these changes so that specific risk profiles can be created to benefit from these changes. Therefore, instead of just looking at the first order of price change, i.e., delta, quants will look at price changes in their 2nd and 3rd order derivatives.

However, there is some controversy about the usefulness and fairness of using such models. For example, Long Term Capital Management was a hedge fund that relied extensively on quantitative analysis. Despite the elegance and seeming mathematical robustness of their quantitative strategies, it still incurred huge losses.

[54] Elliot Wave charts exhibit patterns that are related to Fibonacci’s series (0,1,1,2,3,5,8,13,21…..) where each number in the series is obtained by adding the two preceding consecutive numbers (e.g. 0+1=1, 1+1=2, 1+2=3, 2+3=5, and so on). As the Fibonacci series progresses, the ratio of consecutive numbers (e.g. 21÷13) will approach the Golden Ratio of 1.618.

High frequency trading, which relies on physical transmission speeds and complex price discovery algorithms, are under intense scrutiny from regulators.

Given the rapid advances in computer prowess and technology, and increasing global competition in the financial markets, quantitative analysis will continue to develop as a tool that is necessary to enhance industry standards in fund and risk management.

11.9 Summary

1. Technical analysis, also known as market or internal analysis, involves the study of past prices and volume data to predict the future price direction of individual securities or the aggregate market.

2. Technical analysis assumes that market prices discount everything and prices move in trends. Any shifts in trends can be detected by studying the charts.

3. The greatest challenge to technical analysis is the Efficient Market Hypothesis, which argues that prices adjust rapidly to new information entering the market and, therefore, technical analysts have little opportunity to take advantage of such information. In addition, popular technical analysis trading rules are followed by many market participants and the resulting competition will eventually negate the effectiveness of such rules.

4. Technical indicators used include the A/D line and moving average. The A/D line or breadth index measures the aggregate movement of all securities traded in the market. Moving averages are calculated to obtain a smooth curve so that erratic price changes are smoothed out and the trend can be more clearly observed.

5. MACD and Bollinger Bands are another two popular technical indicators. MACD is a trend following momentum indicator that shows the relationship between two moving averages of a security's price. It is designed to reveal changes in the strength, direction, momentum, and duration of a trend in a stock's price. Bollinger Bands are envelopes plotted at a standard deviation level above and below a simple moving average of the price. The closer the prices move to the upper band, the more overbought the market, and the closer the prices move to the lower band, the more oversold the market.

6. According to the Dow Theory, there are 3 movements in the equity market – the primary movement, the secondary movement, and the daily fluctuations. The objective of the Dow Theory is to identify the primary trend of the market.

7. The moving average line helps to smooth out the erratic movements in share prices so that the underlying trend is distinct. Crossovers between the moving average and the index are used to identify trend reversals.

8. The Relative Strength Index (RSI) is useful for identifying securities that are overbought or oversold by the market.

9. Various types of charts are used for technical analysis, including line chart, bar chart, candlestick chart, and point and figure chart.

10. There are two types of chart patterns – reversal and continuation. Reversal patterns, which indicate reversals in trends, include the head and shoulders formation, and double tops and bottoms. Consolidation rectangles, triangles, flags, pennants, and wedges are continuation patterns.

11. The Elliott Wave Theory posits that price movements occur in cycles of 8 waves per cycle, in which there are 5 advance waves as prices rise, and 3 retreat waves as prices fall.

12. Quantitative analysis involves the building of financial models that try to express the market and economic reality in terms of arbitrage and investment opportunities.

Chapter 12:
Case Studies

Learning Objectives

Case studies form 10% of the weightage in the CM-EIP examination. There will be a total of 3 case studies, each comprising 3 related questions.

A case study is based on elements from real-world situations that require candidates to:

- identify the key issues or problems;
- analyse the situation by applying the relevant theories and concepts acquired; and
- make recommendations based on their analysis.

These case studies challenge the ability of candidates to apply financial knowledge and concepts gained from Chapters 2 to 11 and test the efficiency of application where multiple concepts may be examined through these case questions that follow. Each case study will present adequate information for candidates to understand and identify the problem, analyse the information, and propose a solution to the problem. The following sections illustrate case studies and case questions that are similar in format to those tested in the exam.

Case Study 12.1 Individual vs Institutional Investor

John Lim is moving from stockbroking to a private bank managing accredited investor. In his current job at the brokerage, he works mostly with institutional investors. As such, he has to adjust to the differences in the client type. He has come to you, as his soon-to-be-new colleague, to understand the key differences.

Understand the definition and needs of different types of investors

Question 1

Which of the following is an accredited investor? (*Select all options that apply*)

a. An entity with S$5 million worth of net assets.

b. An university lecturer with $500,000 in net personal assets.

c. An accountant that earns S$400,000 a year.

d. An university endowment trust that has S$6 billion in investible funds.

Rationale / Workings:

Option (a) and (b) are incorrect as both the individual and entity do not qualify as accredited investors. Refer to Section 1.1.1.
Option (c) and (d) are accredited investors. The individual earns an income of more than S$300,000 per year, whilst the university endowment trust has funds exceeding S$10 million in value. Refer to Section 1.1.1.

Understand the definition and needs of different types of investors

Question 2
Which of the following is not likely to be a return objective of an institutional investor?
a. Current income to meet recurring liquidity needs.
b. Long-term capital gain to beat inflation.
c. Potential for higher yields.
d. Regular savings to fund children's education.

Rationale / Workings:
Options (a) through (c) can be return objectives of both individual and institutional investors but Option (d) is likely to be unique to an individual investor.
Refer to Figure 1.1.1.

Understand the definition and needs of different types of investors

Question 3
Which of the following shows the key differences between an accredited investor and a retail investor?
(*Select all options that apply*)
a. A retail investor is likely to be more risk averse than an accredited investor.
b. The accredited investor can withstand a higher level of monetary risk than the retail investor.
c. A retail investor has a longer investment time horizon than an accredited investor.
d. An accredited investor is able to access a wider range of investment products than a retail investor.

Rational/workings:
Option (a) is correct as retail investors usually trade in small amounts and tend to trade less frequently. Due to the small trade size of retail investors, they have little influence over corporate decisions or the capital markets.
Option (b) is correct given the accredited investor wealth, they are considered to be financially savvy and have greater ability to understand financial instruments. Thus, accredited investor can withstand a higher level of monetary risk than the retail investor.
Option (c) is incorrect because investment time horizon is dependent on the neds of the investors, not the type of investors.
Option (d) is correct given that that the accredited investor can withstand a higher level of monetary risk than the retail investor. Hence accredited investors may have access to products that are more complex and riskier, such as unrated bonds and products that require neither a prospectus nor a reasonable basis for recommendation by a representative of a capital market intermediary (CMI).
Refer to Section 1.1.1.

Case Study 12.2 Risk, Return, and Time Value Calculations

Edna Wong just turned 35 years old and would like to set aside $200,000 for 30 years in an investment with the following annual rates of return under different economic scenarios:

State of Economy	Probability	Return
Boom	0.3	12%
Normal	0.5	8%
Recession	0.2	-6%

She is concerned about the risks of the investment and how inflation might affect her eventual retirement nest egg. The average long-term inflation rate is expected to be 2%.

Compute expected return on an investment

Question 1
The expected rate of return on the investment is:
a. 5.30%
b. 6.40%
c. 8.45%
d. 9.28%

Rationale / Workings:
The expected return on an investment is the weighted average of the returns across economic states, where the weights are the probabilities of the economic state occurring. That is: E(R) = 0.3(0.12) + 0.5(0.08) + 0.2(-0.06) = 0.0640 = 6.40%.
Refer to Section 2.2.2.

Compute standard deviation and coefficient of variation as risk measures

Question 2
The coefficient of variation associated with this investment is:
a. 0.47
b. 0.68
c. 1.01
d. 1.33

Rationale / Workings:
The variance of the returns on an investment is the weighted average of the squared deviation of the returns from the mean return. Variance is the first calculation to be made to derive standard deviation. CV measures the risk per unit of expected return. That is:

Variance (σ^2) = $0.3(0.12 - 0.064)^2 + 0.5(0.08 - 0.064)^2 + 0.2(-0.06 - 0.064)^2 = 0.004144$

Standard deviation (σ) = $\sqrt{0.04144} = 0.064374 = 6.4374\%$

Coefficient of variation (CV) = 0.064374/0.064 = 1.01
Refer to Section 2.3.2.

Calculate the future value of a single cash flow

Question 3
Assuming that the expected return of the investment is the average return for the next thirty years, the size of Edna's nest egg when she turns 65, in today's dollars, would be equal to:

a. $709,264.72
b. $1,610,176.77
c. $2,279,904.66
d. $2,865,831.06

Rationale / Workings:
Since expected inflation rate is 2%, the real rate of return is:

[(1 + 0.064) / (1 + 0.02)] - 1 = 0.0431 = 4.31%
Future value investment in real terms = 200,000 $(1+0.0431)^{30}$ = $709,264.72
Refer to Section 2.5.2.

Case Study 12.3 Key Drivers of Market Movement and Assets

You are part of a team that has been tasked to produce the market outlook for US, China, and South East Asia over the next 12 months. Your boss has tasked you to make a standard list of items to cover in the report, which would serve as an outline of the key topics that the research team should provide in the research report for the sales force and clients.

Follow the economic calendar and explain the three main classes of economic indicators - leading, coincident and lagging indicators.

Question 1
One key concern of your US clients is the timing and occurrence of a future economic downturn. Which of the following suggests a potential economic downturn?
a. Average duration of unemployment is increasing.
b. The 10-year T-bond spread over the Fed Funds rate is increasing.
c. Disposable personal incomes reported by the Bureau of Economic Analysis starts to rise.
d. New orders by suppliers start to decline.

Rationale / Workings
Average duration of unemployment is a lagging indicator, which makes Option (a) incorrect.
Option (c) is also incorrect because reported disposable personal income is not only a coincident indicator, but the direction is also inconsistent with an economic downturn.
Although the bond yield spread is a leading indicator, an increase in the spread generally bodes well for economic health, which makes Option (b) incorrect.
Option (d) is correct because suppliers normally cut orders when they expected demand to drop.
Refer to Section 3.2.7.3.

Describe and explain how key economic factors affect market movement and asset values.

Question 2

At a major global conference held recently, some strategists postulated that there could be a situation where a slowdown in the Chinese economy would occur with rising interest rates. Which of the following asset classes are likely to do well in such a scenario?

a. cash, real estate

b. defensive stocks, investment grade bonds

c. cyclical stocks, high-yield bonds

d. growth stocks, precious metals

Rationale / Workings

A slowdown coupled with rising interest rates is characteristic of a late bull to early bear phase. Defensive stocks and investment grade bonds are likely to do well in such an economic environment, so Option (b) is correct.

Although an increase in cash allocation may be wise, holding real estate may be risky given that it is typically leveraged and interest rates rise, so option (a) is incorrect.

Option (c) is possibly the worst of the options because cyclical stocks will perform worse than the market when things turn south, and issuers of high-yield bonds may default.

While some allocation to precious metals may be a good idea during times of uncertainty, growth stocks are less likely achieve to the growth expectations and may experience contraction in price-earnings multiples, making Option (d) incorrect.

Refer to Section 3.2.10.4.

Describe how investment behaviour influences investment decisions, market movement, and asset prices.

Question 3

The discussion on the Chinese economy at the global conference very quickly escalated into a situation in which many uninformed fund managers reduced their positions in a given asset class out of fear of loss, simply because news had spread that many are doing likewise. Even though the likelihood of the discussed scenario is unlikely, the massive selling triggered computer-driven trading strategies. What is the root cause for the fall in prices for this asset class?

a. Herd instinct

b. Loss aversion

c. Cognitive bias

d. Algorithmic trading

Rationale / Workings:

Option (a) is correct because the uninformed fund managers simply followed the crowd without exercising independent decision-making ability and doing the due diligence.

Although the managers may have been influenced by loss aversion to reduce their holdings and algorithmic trading may have exacerbated the fall, they did not provide the original impetus for this selling, so Options (b) and (d) are incorrect.

Also, since there is no mention of how the original discussants formed their conclusions, there is insufficient information to attribute the fall for prices to cognitive bias, making Option (c) incorrect.

Refer to Section 3.4.1.

Case Study 12.4 Foreign Exchange Carry Trade

An investor is tempted to invest in IDR 100 billion worth of Indonesian government bonds. He reasons that IDR-denominated 1-year Government Bonds are currently paying interest of 6%, while banks in Singapore are charging 1.5% interest for investment related loans. The SGD/IDR exchange rate currently stands at 10,076.68 and he would like you to help him assess three exchange rate scenarios.

Demonstrate how a carry trade works

Question 1
Ignoring changes in bond prices, what is the annual gain, in Singapore dollars, from the carry trade if the currency exchange rate stays flat at current levels?
a. SGD 148,858.55
b. SGD 446,575.66
c. SGD 595,434.21
d. SGD 9,923,903.51

Rationale / Workings:
Since the currency exchange rate stays flat, we need not concern ourselves with gains/losses due to currency conversion and can simply work in SGD dollars.
The purchase of IDR 100 bn of Indonesian government bonds has to be funded by an SGD-equivalent loan of 100 bn/10,076.68 = SGD 9,923,903.51.
Since the investor would earn 6% on the bonds and pay 1.5% on the loan, and bond capital gains/losses are ignored, the annual gain from the carry trade would be (0.06 – 0.015) x 9,923,903.51 = SGD 446,575.66.
Refer to Sections 4.4.2 and 4.7.3.

Demonstrate how a carry trade works

Question 2
What is the total gain if the IDR weakens against the SGD by 10% over the next year instead?
a. – SGD 605,357.18
b. – SGD 942,771.98
c. SGD 605,357.18
d. SGD 942,771.98

Rationale / Workings:
At the current exchange rate, the purchase of IDR 100 bn of Indonesian government bonds has to be funded by an SGD-equivalent loan of 100 bn/10,076.68 = SGD 9,923,903.51.
The investor has to pay an annual interest expense of 0.015 x 9,923,903.51 = SGD 148,858.55.
However, the investor gains an annual interest income of 0.06 x 100bn = IDR 6 bn. Since the IDR weakens by 10% against the SGD, this interest income would be converted back at an exchange rate of 10,076.68/0.9 = 11,196.31. This implies an SGD-equivalent total interest income of 6 bn/11,196.31 = SGD 535,890.84.
After 1 year, the face value of the IDR 100 bn bonds would be converted back at the exchange rate of 11,196.31, which would be 100 bn / 11,196.31 = SGD 8,931,514.04.
Thus, the total gain for the carry trade would be (8,931,514.04 – 9,923,903.51) + (535,890.84 – 148,858.55) = – SGD 605,357.18. Even though interest income is greater than borrowing expenses.

The unfavourable foreign exchange rate movements have caused the investor's IDR denominated investment and income receipt to depreciate in value.
In this case, the carry trade is not profitable to the investor.

Demonstrate how a carry trade works

Question 3
Suppose in a year, the SGD/USD rate is 0.7 and the USD/IDR rate is 13,086.60. How will the gain from the carry trade compared to the first scenario, where the SGD/IDR rate stays the same at 10,076.68?
a. Lower gain.
b. Same gain.
c. Higher gain.
d. Insufficient information to conclude.

Rationale / Workings:
First, we establish the SGD/IDR cross rate, which is 0.7 x 13,086.60 = 9,160.62.
Since the SGD/IDR rate of 9,160.62 is lower than the 10,076,68 examined in Question 1, the IDR has appreciated against the SGD in this case.
The investor's IDR-denominated investment and income would appreciate in value due to the favourable exchange rate movement and cause the gain to be higher than the SGD 446,575.66 computed in Question 1.
Refer to Sections 4.4.4 and 4.7.3.

Case Study 12.5 Equities Analysis

Mr Quek received a recommendation to buy Mvidia, a semi-conductor company that has exhibited strong earnings growth. Although demand for semi-conductors in the PC industry remains sluggish, companies like Mvidia that are focused in areas of high-performance computing (HPC) (e.g., cloud computing, big data, and artificial intelligence) are better positioned to experience strong growth. However, Mr Quek has also received counter arguments that the price-earnings ratio of Mvidia is too high and has come to you for advice. He has collected information on the share price as at end 2018 and consensus EPS projections for Mvidia:

Mvidia Stock Price: USD 40

Jan	EPS%	EPS Change
FY18A	1.12	-5.0
FY19E	1.08	-3.6
FY20F	2.57	+138.0
FY21F	4.19	+63.0
FY22F	4.67	+11.4
FY23F	5.81	+24.4

A – Actual

E – Estimated

F - Forecast

Applying Price Earnings ratio (PER) in stock and explain how it is calculated and used.

Question 1

What is the prospective PER for Mvidia and why is it likely to be the case?

a. The high ratio of 35.71 is likely because the market has overvalued Mvidia.

b. The high ratio of 35.71 is likely because the market is factoring in Mvidia's growth potential.

c. The high ratio of 37.04 is likely because the market has overvalued Mvidia.

d. The high ratio of 37.04 is likely because the market is factoring in Mvidia's growth potential.

Rationale / Workings:

Prospective PER of Mvidia = 40/1.08 = 37.04x.

Since PERs can be high because of strong expected future growth, the prospective PER is high likely because investors are factoring in the high double-digit long-term growth potential of Mvidia.

Refer to Sections 6.8.1 and 6.8.1.3

Applying % growth in EPS of stock and explain it is calculated and used.

Question 2

What is the 5-year compounded annualised EPS growth rate of Mvidia?

a. 31.57%

b. 38.99%

c. 40.01%

d. 52.30%

Rationale / Workings:

$$\text{5-year EPS Growth Rate of Mvidia (\%)} = \left[\left(\frac{\text{Future EPS}}{\text{Current EPS}}\right)^{\frac{1}{\text{No. of Years}}} - 1\right]*100\%$$

$$= \left[\left(\frac{5.81}{1.12}\right)^{\frac{1}{5}} - 1\right]*100\%$$

$$= 38.99 \quad \%$$

Refer to Section 6.8.2.

Applying Price-Earnings Growth (PEG) ratio in comparing stocks and explain how it is calculated and used.

Question 3

What recommendation would you give to Mr Quek using Mvidia's prospective PEG ratio?

a. Buy Mvidia given the PEG ratio of 0.92.

b. Sell Mvidia given the PEG ratio of 0.92.

c. Buy Mvidia given the PEG ratio of 0.95.

d. Sell Mvidia given the PEG ratio of 0.95.

Rationale / Workings:

Using the two values from Questions 1 and 2, we can derive Mvidia's prospective Price-Earnings Growth (PEG) ratio as follows:

$$\text{Prospective PEG of Mvidia} = \frac{\text{Prospective PER}}{\text{\% Growth in EPS}}$$

$$= \frac{37.04}{38.99}$$

$$= 0.95$$

As the PEG ratio for Mvidia is less than 1, this indicates that Mvidia is slightly undervalued even after accounting for its strong expected future grow, and so Mr Quek can consider buying the stock.
Refer to Section 6.8.2.

Case Study 12.6 Estimating the Fair Value of a Stock

One of your new clients is curious about how your firm's research team estimates the fair value of companies. As a test to ascertain how familiar your firm is with the valuation process, the client has requested for a brief explanation of the steps undertaken when estimating the fair value of a company. You provide him with a description of your valuation process, but he wanted further clarification on some steps.

Explain the use and computation of DCF in the estimation of fair market value (intrinsic value of a stock).

Question 1
The free cash flows to a levered firm should be discounted using the firm's:
a. cost of ordinary equity only.
b. cost of preferred equity only.
c. cost of debt only.
d. weighted average cost of capital.

Rationale / Workings:
Option (d) is correct because the appropriate discount rate should take into account capital from all sources for which a return is required.
Option (a) can be correct in the special circumstance when the firm is all-equity funded but is incorrect in this case because the firm is levered, while Options (b) or (c) are incorrect under any circumstance, since a firm must have ordinary equity.

Explain the components in the estimation of free cash flow to the Firm (FCFF).

Question 2
Which of the following is NOT part of estimating free cash flows to the firm (FCFF)?
a. Interviewing company management on its revenue growth potential.
b. Consulting industry specialists on expected profitability margins.
c. Estimating the amounts of future fixed asset investments.
d. Analysing historical ratios to make interest expense projections.

Rationale / Workings:
Options (a) and (b) are incorrect because the revenue growth potential and profitability margins are needed to come up with earnings before interest and taxes, which is a key input for the FCFF computation.
Option (c) is incorrect because capital expenditures are another key input for the FCFF computation.
Option (d) is correct because interest expense is excluded from the FCFF computation.
Refer to Sections 5.11.5.2 and 6.7.3.

Computation of Enterprise Value in the estimation of fair market value (intrinsic value of a stock).

Question 3
Enterprise value can be calculated using _____. (*Select all options that apply*)
a. Sum of the present values of the FCFFs for the projection period plus the terminal value
b. Sum of the present values of the FCFFs for the projection period plus the present value of the terminal value
c. Sum of the FCFFs for the projection period plus the present value of the terminal value
d. Sum of the estimated market values of debt, common stock, and preferred stock

Rationale / Workings:
Options (b) and (d) are correct because enterprise value can be calculated using one of two ways: either in terms of thinking about the worth today of the cash flows that are generated are by the firm or the value today of the claims on those cash flows.
Options (a) and (c) are incorrect because they fail to account for time value of money for the FCFFs in the projection period and for the terminal value, respectively.
Refer to Sections 6.7.4 and 6.7.5

Case Study 12.7 Fixed-Income Investments

David was offered the following USD fixed-income instruments:

Bonds Offered	Maturity	Yield to Maturity	Rating
New Offshore Oil Corp: 10 year - 6% callable bond	5 years left	4.5%	A-
Diversified Commodities Company: 10-year 4% bond	6 years left	6.0%	BBB
Indonesian Coal Mining & Energy Company: 8% bond	4 years left	7.0%	unrated

In general, the market expects interest rates to rise over the next year as the commodities, offshore oil, coal mining, and energy industries are coming out of a slump. David does not understand bond jargon and has come to you for advice.

Explain the relationship between coupon rate, market yield, required yield, and bond price.

Question 1
David wonders why some bonds can be bought at prices below face value. You can explain to him that this is because the:

a. Indonesian Coal Mining & Energy Company bonds are unrated.
b. New Offshore Oil Corp bonds are callable.
c. YTM of the Diversified Commodities Company bonds is higher than the coupon rate.
d. YTMs of the New Offshore Oil Corp and Indonesian Coal Mining & Energy Company bonds are lower than their respective coupon rates.

Rationale / Workings:
A bond has to be priced at a discount in order for the YTM to be higher than the coupon rate, so Option (c) is correct.
Option (d) is wrong because those bonds will be priced at a premium.
Also, while investors may require higher yields for unrated compared to well-rated bonds, and callable compared to equivalent non-callable bonds, this does not necessarily mean that the yields are higher than the respective coupon rates, both Options (a) and (b) are incorrect.
Refer to Section 7.19.2.

Identify the various risks that a fixed-income investor is exposed to.

Question 2
In the event that David purchases one of the offered bonds and insists on holding his bond to maturity, what should he be most concerned with?
a. A bond is callable.
b. Interest rates are increasing.
c. Interest rates are decreasing.
d. The coupon rate of a bond.

Rationale / Workings:
A callable bond can be redeemed by the issuer before the maturity date and, if that happens, the investor is potentially exposed to reinvestment risk from having to reinvest the proceeds of the redeemed bonds at lower yields, so Option (a) is correct.

Holding the bond to maturity means that the yield to maturity rather than the coupon rate is of relevant interest, even though to enjoy the actual yield to maturity, coupons must be reinvested at the same yield. Option (a) is a much bigger concern than Option (d) for an investor who would like to hold the bond to maturity. So, Option (d) is incorrect.

Also, while changes in interest rates are likely to influence the price of a bond, this is of little concern to the investor that holds the bond to maturity since the price will converge to par on maturity, so Options (b) and (c) are incorrect.
Refer to Sections 7.12.3 & 7.19.3.

Explain the concepts of duration and convexity and the implications for a bond's price risk.

Question 3
Which of the following scenarios is likely to happen in the coming year?
a. Diversified Commodities Company bond price is likely to decrease the most in percentage terms.
b. New Offshore Oil Corp bonds are likely to be called.
c. Indonesian Coal Mining & Energy Company bonds are likely to default.
d. Indonesian Coal Mining & Energy Company bonds are likely to increase the least in percentage terms.

Rationale / Workings:
As interest rates are expected to increase, bond prices are likely to drop. Since the bonds of Diversified Commodities Company have the longest maturity and lowest coupon rate, their prices are most sensitive to changes in interest rate, so Option (a) is correct.
Although Indonesian Coal Mining & Energy Company bond prices are least sensitive to changes in interest rate due to the shortest maturity and highest coupon rate, an expected increase in interest rate is likely to decrease the price, not increase it, so Option (d) is incorrect.
Also, an expected increase in interest rate means that New Offshore Oil Corp bonds are unlikely to be called, making Option (b) incorrect.
Even though Indonesian Coal Mining & Energy Company bonds are not rated by the major credit-rating agencies, it does not necessarily mean that the company has poor credit, as the company could simply have chosen not to seek ratings from the major global rating agencies. Furthermore, as commodities, energy, and coal mining industries are expected to recover from the slump, it is a stretch to conclude that its bonds are likely to default, making Option (c) incorrect.
Refer to Section 7.17.

Case Study 12.8 Portfolio Management

Mr Walter Koh is a retiree with investable assets of $2 million, all of which is currently passively managed. However, he is willing to have $500,000 of it managed more actively and is prepared to risk $100,000 of it in full. He plans to gift about $100,000 from this actively managed fund three months from now to an art endowment fund. Also, he is not a big spender but hopes to fund at least part of his annual expenses, which are highly predictable, from his active portfolio for the next twenty years. He is otherwise not concerned about liquidity. He is considering four prospective portfolios:

Table 12.8 – Strategic Asset Allocation of Proposed Portfolios

	Portfolio			
Asset Class	**A**	**B**	**C**	**D**
Money market funds	20%	10%	20%	20%
Treasury bills	10%	0%	10%	10%
Long-term government bonds	10%	80%	10%	50%
High yield bonds	10%	0%	40%	0%
Developed-market equities	50%	10%	20%	20%

Also, Castor & Pollux shares, which have a beta (β) of 1.2 and a past return of 9.5%, have been recommended by Mr Koh's financial advisor on its alpha potential after conducting a thorough fundamental analysis of the company. The risk-free rate is 3% and the market risk premium 6%.

State the different levels of market efficiency under EMH in the context of the current market environment.

Question 1
Mr Koh does not believe one can achieve superior return through fundamental analysis. In this case, which form(s) of market efficiency must he necessarily believe in?
(*Select all options that apply*)
a. Weak form

b. Semi-weak form
c. Semi-strong form
d. Strong form

Rationale / Workings:
Conducting fundamental analysis in an attempt to identify a mispriced stock assumes that all publicly available information has not already been reflected in the current stock price. Therefore, Mr Koh must necessarily believe that the market is semi-strong form efficient, making Option (c) correct.
Since all publicly available information includes historical price and volume information, he must also believe that the market is weak form efficient, making Option (a) correct as well.
There is insufficient information on Mr Koh's views on the price impact of private information for us to conclude that he necessarily believes the market to be strong form efficient, so Option (d) is incorrect. Semi-weak form is not a standard classification of market efficiency and is also incorrect.
Refer to Section 8.2.

Explain the risk-adjusted measures of performance using Jensen's Alpha.

Question 2
As a result of Mr Koh's belief in market efficiency, he requests his advisor to calculate the risk-adjusted return of Castor & Pollux. Jensen's alpha for Castor & Pollux is:
a. – 2.9%
b. – 0.7%
c. 0.7%
d. 2.9%

Rationale / Workings:
Using the CAPM, the required return for Castor & Pollux shares is 0.03 + 1.2 (0.06) = 10.2%. Since the actual return is 9.5%, Jensen's alpha = 9.5% - 10.2% = – 0.7%.
Refer to Section 8.6.2.4.

Appreciate how investment portfolios are constructed and managed.

Question 3
Of the 4 portfolios, which is best align with Mr Koh's investment objectives?
a. Portfolio A
b. Portfolio B
c. Portfolio C
d. Portfolio D

Rationale / Workings:
Mr Koh needs $100,000, which is 20% of his $500,000 portfolio, as a gift shortly and so should allocate that to money market investments, the most liquid asset class, making Option (b) incorrect due to its low allocation to this asset class.

Option (c) is incorrect because it has a combined 60% allocation to high-yield bonds and equities, which is likely to be risky for Mr Koh given that he would only be willing to risk up to $100,000 in value.
Option (a) is incorrect for the same reason and also the fact that the allocation to long-term government bonds is likely too low to fund his annual expenses.
Option (d) is correct because it has sufficient amounts allocated to money market funds for his gift, an allocation to equity that takes into consideration his risk aversion and a sufficiently high allocation to long-term government bonds so that the coupons can be used to fund his annual expenses.
Refer to Section 8.5.6.

Case Study 12.9 Warrants and Foreign Exchange

Ms JL Tan has two questions for you, her broker. First, she is looking at the structured warrants of Poseidon Lines ("Poseidon"), a shipping company. Today, Poseidon shares are trading at $2.50 and its warrants at $1.30. The warrants have a conversion ratio of 1:1 and an exercise price of $1.50.

Second, Ms Tan has sold some of her ADRs denominated in USD and will receive USD50,000 2 business days from today. She wants to sell them for SGD.

Explain the valuation of warrants, using intrinsic value, conversion price and warrant premium

Question 1
What is the intrinsic value of Poseidon's structured warrant?
a. $0
b. $1
c. $1.50
d. $2.50

Rationale / Workings:
Since the warrant is in-the-money, the intrinsic value is (2.50 – 1.50)/1 = $1, making Option (b) the correct answer.
Refer to Section 10.2.1.

Understand the determinants of the minimum and maximum values of warrants

Question 2
You expect Poseidon's share price to rise to $4.00 and Ms Tan would like to know her minimum gain on the warrants in % terms if she buys them today. You can explain to her it is:

a. 92.31%.
b. 207.69%.
c. 307.69%.
d. impossible to determine because the future price of the structured warrant is not known with certainty.

Rationale / Workings:
If Poseidon's share price rises to $4.00, the intrinsic value of its warrant will be (4.00 – 1.50)/1 = $2.50. Since the warrant has a conversion ratio of 1, the minimum value of the warrant will be its intrinsic value of $2.50

and the minimum percentage gain on the warrant is 2.50/1.30 – 1 = 92.31%, making Option (a) correct. Although we do not have the future price of the structured warrant, we only need the future minimum price in order to compute the minimum percentage gain, making Option (d) incorrect.
Refer to Section 10.2.5.

Understand how foreign exchange rates are quoted

Question 3
Ms Tan has received two quotes as shown below:
Quote A: USD/SGD 1.2500 – 1.2550
Quote B: USD/SGD 1.2475 – 1.2515
The best quote she can sell her ADR proceeds at for SGD is:
a. USD/SGD 1.2500
b. USD/SGD 1.2550
c. USD/SGD 1.2475
d. USD/SGD 1.2515

Rationale / Workings:
All quotations refer to the number of units of SGD (the counter currency) that equal one unit of USD (the base currency). For a given quotation, the first rate is the price at which the market maker will buy USD and the second rate is the price at which the market maker will sell USD. Since Ms Tan would like to sell USD, the first rate of each quote applies and the highest one is the best quote, making Option (a) correct.
Refer to Section 4.4.2.

Case Study 12.10 Unit Trusts, REITs and Business Trusts

Mr James Lim finds the jargon associated with some collective investment schemes confusing. He approaches you to help demystify the differences between unit trusts, business trusts, and real estate investment trusts (REITs). Mr Lim has some investable funds and is deciding whether to invest them in a property or one of the aforementioned schemes. He is more interested in receiving a steady stream of income than in making large trading gains.

Distinguish between unit trusts and investment trusts

Question 1
The common features among unit trusts, business trusts and real estate investment trusts is/are that:
(*Select all options that apply*)

a. They are established under a trust deed.
b. They are exchange-traded.
c. They allow for diversification.
d. They are professionally managed.

Rationale / Workings:
All three types of trusts are established under a trust deed, so Option (a) is correct. By pooling and investing the funds, all three types of trusts also allow for diversification of risks, albeit to a different extent, so Option (c) is correct. The assets under all three types of trust are professionally managed, so Option (d) is also correct. However, while all REITs are listed, unit trusts are typically not listed and some business trusts are unlisted as well, so Option (b) is incorrect.
Refer to Sections 9.3, 9.3.5, 9.4.1, 9.4.2, 9.5.1, 9.5.2, and 9.5.4.

Compare REITs and business trusts and explain their differences.

Question 2
Mr Lim enquires about the differences between business trusts and REITs. You explain that the main difference(s) is/are that:
(*Select all options that apply*)
a. The trustee and manager are separate in a REIT but not in a business trust.
b. Business trusts do not face investment restrictions, unlike REITs.
c. Business trusts are unlisted whereas REITs are listed.
d. there is a 75% gearing limit applicable on REITs but only a 45% gearing limit on business trusts.

Rationale / Workings:
Unlike a REIT, a business trust is a single responsible entity organization with a trustee-manager, so Option (a) is correct. As business trusts can actively engage in any business area but REITs are required by law to invest only in real estate for rental income, business trusts can pursue investment strategies that are more aggressive, so Option (b) is correct. Although some business trusts are unlisted, most are listed, so Option (c) is incorrect. Finally, there is no gearing limit for business trusts but a 45% gearing limit for REITs, so Option (d) is incorrect.
Refer to Section 9.5.1, 9.5.2 & 9.5.4.

Distinguish between unit trusts and investment trusts

Question 3
Given Mr Lim's investment objectives, the collective investment scheme that is most likely to suit his need is:
a. A unit trust described as an aggressive fund.
b. A business trust comprising investments in golf course assets.
c. A REIT that focuses on heartland retail malls.
d. All of the above.

Rationale / Workings:
A unit trust focused on growth generates returns from capital gains and is unlikely to allow Mr Lim to receive a steady stream of income, so Option (a) is incorrect. Cash flows from golf course assets are likely to be highly sensitive to economic fluctuations. Further, there is no mandatory minimum distribution for business trusts. Hence, Option (b) will not provide the steady stream of income that Mr Lim desires as well and is incorrect. A REIT is often seen as an alternative to investments in physical real estate and REITs typically distribute at least 90% of their annual income, which is likely to be more stable for retail malls in the heartland. Therefore option (c) is the best answer, which also means Option (d) is incorrect.
Refer to Sections 9.3.17 & 9.5.4.

Appendix A: Review Questions

Candidates should note that the sole purpose of the Review Questions is to familiarize candidates with the scope and general nature of the examinations, and the format of the examination questions.

The Review Questions are not intended to be used as preparatory study material for the examinations, nor do the questions cover all the material tested in the examination.

The correct answer for each question is in **bold**.

Chapter 2: Risk, Return and Time Value Calculations

1. As an investment manager, you are assessing a project that has an immediate cash outflow followed by a series of cash inflows over the next 10 years. You calculate the project's net present value (NPV), which is the sum of the present values of the cash inflows minus the initial cash outflow, and also calculate the project's internal rate of return (IRR).

   Now, you discovered that you have underestimated the discount rate. What will happen when you correct for the underestimation?

   a. **There will be a reduction to NPV with no change to IRR.**
   b. There will be no change to both the NPV and IRR.
   c. There will be an increase to NPV and a reduction to IRR.
   d. Both NPV and IRR will increase.

2. A diversified investment portfolio has an expected return of 6% and a standard deviation of 12%. There is a 68% probability of the portfolio producing a return that ranges from -6% to +18% (i.e.; plus/minus one standard deviation from the expected return). The portfolio's returns follow a normal distribution and hence the probability of a loss exceeding 6% is 16%. We can expect the portfolio's loss to exceed 6% once every _____.

   a. 5 – 6 years.
   b. **6 – 7 years.**
   c. 7 – 8 years.
   d. 8 – 9 years.

3. Which of the following statements about value at risk (VaR) are TRUE?

   a. It is not an applicable measure for loss probability to assets such as bonds, shares and currencies.
   b. It measures and quantifies the minimum possible loss of an investment over a specific time and within a confidence level.
   c. It can be calculated by applying the Monte Carlo simulation or the historical method.
   d. It can be used when determinants (market factors) of future returns are different from those of past returns.

   - (a) and (d).
   - (c) and (d).
   - (a) and (c).
   - **(b) and (c).**

4. Which of the following are forms of operational risk?

   a. There is a flash crash in the financial market caused by a massive erroneous sell order.
   b. There is poor system infrastructure, leading to technical disruptions.
   c. There are human errors, resulting in erroneous trades.
   d. Inability to access funding of investments due to a credit downgrade of the borrower.

   - **(a), (b) and (c).**
   - (b) and (c).
   - (b), (c) and (d).
   - (a) and (d).

5. You have decided to purchase a brand-new car as your new job requires you to travel island wide. The bank will only provide financing for 5 years at a 10% annual rate, compounded monthly. The upfront payment to the car dealer is 20% of the purchase price of $80,000. The bank loan will be used to finance the remaining 80% of the purchase price with the 1st payment due 1 month from today. If you do not make any of the monthly loan instalments, you will end up with a loan balance of $105,299.77. What is the compound annual growth rate of the loan balance?

   a. 10.00%.
   b. 10.24%.
   c. **10.47%.**
   d. 10.84%.

Chapter 3: Key Drivers of Market Movements and Asset Values

6. Which of the following constitutes an expansionary monetary policy tool?

   a. An increase in the discount rate by the central bank.
   b. A decrease in government expenditure on infrastructure construction.
   c. A decrease in the reserve ratio of banks by the central bank.
   d. An increase in demand for domestic currency, causing a rise in exchange rate.

7. Which of the following are coincident indicators within the broad categories of economic indicators?

   a. Industrial Production Index.
   b. Unemployment Claims.
   c. Adjusted M2 Money Supply.
   d. Personal Income and Consumption Expenditure.

   - (a) and (c).
   - (c) and (d).
   - **(a) and (d).**
   - (b) and (d).

8. Price competition starts to occur at which phase of the industry life cycle?

   a. **Stabilisation.**
   b. Decline.
   c. Early Development.
   d. Mature Growth.

9. Which market-driven factor typically triggers market correction and profit taking during a bull market?

   a. Influx of hot money.
   b. Increasing bulk of dark pool trades.
   c. Influence from structured products.
   d. **Increasing liquidity shortage.**

10. If an investor believes that he has not lost until he sells his investment portfolio, the investor is exhibiting:

    a. Confirmation bias.
    b. **Loss aversion.**
    c. Herd instinct.
    d. Gambler's fallacy.

Chapter 4: Foreign Exchange

11. If there is an increase in Singapore household income, what is the most likely impact on the Singapore trade deficit and value of the Singapore dollar (SGD)?

    a. The trade deficit will decrease, and the value of SGD will not change.
    b. The trade deficit will decrease, and the SGD will appreciate.
    c. **The trade deficit will increase, and the SGD will depreciate.**
    d. The trade deficit will not change, and the SGD will depreciate.

12. According to the 2019 BIS Triennial Central Bank Survey, which of the following currency pairs are within the eight most actively traded?

    a. JPY/NZD.
    b. EUR/USD.
    c. USD/CHF.

d. USD/ZAR.

- (a) and (c).
- (b) and (d).
- (a) and (d).
- **(b) and (c).**

13. In EUR/USD you are quoted the following prices by four different banks. You are a buyer of US dollars. Which rate is the best for you?

a. 1.3625/27.
b. 1.3631/35.
c. 1.3620/22.
d. **1.3633/38.**

14. You are considering hedging the EUR exposure if you can borrow USD for 6 months at an annual rate of 1.25% and receive interest at an annual rate of 2.00% on 6-month EUR deposits. What would be the 6-month EUR/USD outright forward rate? Given the spot rate is at 1.42.

a. 1.4096.
b. 1.4200.
c. **1.4147.**
d. 1.4253.

15. Which of the following is CORRECT regarding Interest Rate Parity (IRP) and its implication?

a. IRP suggests that firms should simply borrow where interest rates are lowest.
b. The currency with a higher interest rate will be at a forward discount to the currency with the lower interest rate.
c. If IRP does not hold, there are opportunities to profit from arbitrages.
d. IRP governs the relationship between interest rates and currency exchange rates.

- (a), (c) and (d).
- (a), (b) and (c).
- **(b), (c) and (d).**
- (a), (b), (c) and (d).

Chapter 5: Company Analysis and Understanding Financial Statements

16. Kelly deposited $100,000 in a joint bank account she established for a medium-sized nail grooming salon chain that she is going to co-own and operate as Be-Dazzled Nails. By recording the deposit in the balance sheet, it will _____.

a. Increase an asset and increase a liability.
b. Decrease an asset and decrease a liability.
c. **Increase an asset and increase shareholder's equity.**
d. Decrease an asset and decrease shareholder's equity.

17. Derrick has prepared his company's financial statements for the year ended 31 December 2019, which reflects an operating profit of $280,000, current assets of $510,000 and a total assets figure of $833,000. He has since discovered that the depreciation figure he used was overstated as he double counted the depreciation cost of $55,000 in his computations. Which of the following statements is TRUE?

    a. **The revised operating profit for the year will be $335,000.**
    b. The revised depreciation expense figure will be $110,000.
    c. The revised current assets figure will be $565,000.
    d. The revised total assets figure will be $778,000.

18. The purposes of the statement of cash flows are to:

    a. Evaluate liquidity.
    b. Facilitate banking relationship.
    c. Analyse cash position.
    d. Compare investment performance.

    - (a) and (b).
    - (a), (b) and (c).
    - **(a) and (c).**
    - (a) and (d).

19. The following are financial data extracts from the books of Zeckers-Murrell Ltd (ZML) denoted in '000 USD.

Parameters	2018	2017
Average inventory	250	300
Credit Sales	864	628
Cost of Goods Sold	710	511
Average receivables	231	350
Shareholders' Equity	1,571	1,228

    Calculate the activity ratio that reflects how long it takes for ZML to collect its accounts receivable in 2018.

    a. 3.45.
    b. 3.74.
    c. **97.59.**
    d. 118.75.

20. Referring to the financial data in Q19, what is the missing information required to compute book value per share (BVPS)?

    a. Full year dividends.
    b. Weighted Average Number of Shares.
    c. Gross Number of Redeemed Shares.
    d. **Total Number of Shares Outstanding.**

Chapter 6: Equity Securities

21. Which of the following statements are TRUE about preference shares?

    a. The preference rate of dividend is usually fixed and specified.
    b. The issuer is not obliged to pay preference dividend if there are poor earnings.
    c. Preference shares can be issued as redeemable shares.
    d. The preference dividend must be paid after ordinary dividend has been paid.

    - **(a), (b) and (c).**
    - (b), (c) and (d).
    - (a), (c) and (d).
    - (a), (b), (c) and (d).

22. A price-weighted index consists of four stocks. One of the stocks is subject to a 3-for-1 stock split. What is most likely to happen to the divisor for the index?

    a. **Decrease.**
    b. Increase.
    c. Nullify.
    d. Remain unchanged.

23. MaMa Rich is a listed food and beverage firm. Stock of MaMa Rich is slightly more volatile than the market with a beta of 1.3 and the market's expected rate of return is 9.33%. With the US 5-year Treasury bill rate currently at 5.8%, what is the firm's cost of equity capital?

    a. 9.515%.
    b. 10.123%.
    c. **10.389%.**
    d. 11.761%.

24. The price-to-sales ratio (P/Sales) is a helpful valuation technique when examining firms _____.

    a. With similar profit margins.
    b. **With low, negative or zero income.**
    c. In high growth industries.
    d. That hold primarily liquid assets.

25. Which of the following are NOT advantages of American Depositary Receipts (ADRs)?

    a. ADRs may be subjected to currency risk.
    b. ADRs are a flexible mechanism for raising capital.
    c. ADRs allow for greater investor portfolio diversification.
    d. ADRs carry counterparty risk of the depository bank.

    - (a), (b) and (d).

- **(a) and (d).**
- (b), (c) and (d).
- (b) and (c).

Chapter 7: Deposits and Fixed Income Securities

26. As compared with bonds selling at par, deep discount bonds such as Zero-Coupon Bonds will have:

a. Greater reinvestment risk.
b. **Greater price volatility.**
c. Less call protection.
d. None of the above.

27. What is the main feature of a callable bond?

a. **It gives the issuer the option to redeem the bond before maturity.**
b. It gives the buyer the option to redeem the bond before maturity.
c. It gives the issuer the option to redeem the bond after maturity.
d. It gives the buyer the option to redeem the bond after maturity.

28. Which of the following are potential risks that may arise when investing in foreign government bonds?

a. Corporate sector downgrade risk.
b. Sovereign credit risk.
c. Currency risk.
d. Political risk.

- (a), (b) and (c).
- **(b), (c) and (d).**
- (c) and (d).
- (a) and (c).

29. What is the value of a bond that matures in 3 years, has an annual coupon rate of $110, and a par value of $1,000. Assume a required rate of return of 11%.

a. $330.
b. $970.
c. **$1,000.**
d. $1,330.

30. What happens to a bond's yield-to-maturity (YTM) if there is an increase in the bond price?

a. The bond's YTM rises.
b. **The bond's YTM falls.**
c. The bond's YTM remains unchanged.
d. The bond's YTM could fluctuate, depending on market conditions.

Chapter 8: Portfolio Management

31. If you wish to eliminate the risk of your single security portfolio, you will select a security that is _____ to your portfolio.

a. Perfectly positively correlated.
b. Perfectly negatively correlated.
c. Not correlated.
d. Highly correlated.

32. Which of the following will affect the expected return of a portfolio under the Capital Asset Pricing Model?

a. Risk-free rate.
b. Idiosyncratic risk of an individual security.
c. Systematic risk of portfolio.
d. Market risk premium.

- (a), (b) and (c).
- (b), (c) and (d).
- **(a), (c) and (d).**
- (a), (b), (c) and (d).

33. John recently bought a financial product with an initial investment capital of $500,000. At the end of the product's tenure, John's investment value in the product is now $550,000. He also received distributions from the investment (worth $10,000) in the interim. What was the return on John's investment?

a. 2%.
b. 8%.
c. 10%.
d. **12%.**

34. When analysing a portfolio with frequent injections of capital, what would be the best measure for evaluating the investment manager's performance?

a. Simple return.
b. Expected return.
c. **Time-Weighted return.**
a. Dollar-Weighted return.

35. The Sharpe performance measure is defined as:

a. The difference between the portfolio's excess return and the portfolio's total risk.
b. The difference between the portfolio's excess return and the portfolio's systematic risk.

c. The difference between the actual return on the portfolio and the return that would be expected from the capital asset pricing model.
d. **The portfolio's excess return divided by the portfolio's total risk.**

Chapter 9 – Exchange Traded Funds (ETFs), Unit Trusts, Real Estate Investment Trusts (REITS), and Insurance

36. Which of the following are advantages of investing in ETFs?

a. Access to superior hedge fund manager.
b. Small capital outlay.
c. Low costs.
d. Transparency.

- (a), (b) and (d).
- (b) and (c).
- (a), (c) and (d).
- **(b), (c) and (d).**

37. Which of the following is NOT a feature of real estate investment?

a. Provides security of income.
b. Provides a hedge against inflation.
c. Requires a high capital outlay.
d. **Requires annual injection of capital.**

38. Which of the following statements are CORRECT?

a. Apart from REITs, trusts that invest in real estate can be structured as Business Trusts.
b. A Business Trust has a single responsible entity that serves as both the trustee and the manager of the Business Trust.
c. A Business Trust has no gearing limit.
d. Both REITs and Business trust are required to distribute a minimum of 90% of annual income.

- (a), (c) and (d).
- **(a), (b) and (c).**
- (b), (c) and (d).
- (a), (b), (c) and (d).

39. Which of the following are main characteristics of hedge funds?

a. Use of leverage to increase potential returns.
b. Measured against absolute returns rather than relative returns.
c. Lower fees than other financial products.
d. Relatively unrestricted investment style.

- **(a), (b) and (d).**
- (b), (c) and (d).
- (a), (b) and (c).

- (a), (c) and (d).

40. Which statement best describes private equity investments?

a. Investments that are exclusive to private banking clients.
b. Investments in companies that are not publicly listed.
c. Investments in companies that operate in the private sector.
d. Investments that are not publicly disclosed to outsiders and non-clients.

Chapter 10 Warrants

41. Company A's warrant is trading at $0.35; its exercise price is $0.50 while the share price is $0.65. What is its intrinsic value?

a. $0.85.
b. $0.15.
c. $0.30.
d. $1.00.

42. A warrant with an exercise price of $1.00 is trading at $1.20. One warrant converts to one share. What is its conversion price?

a. $1.00.
b. $1.20.
c. $2.20.
d. $0.20.

43. The market price of Company B's common share is $5 and the exercise price of the warrant is $3.50. Two warrants can be converted into one common share. The warrant price is currently trading at $2. What is the warrant premium?

a. 10%.
b. **50%.**
c. 57%.
d. 100%.

44. A warrant is trading at $0.40. The exercise price is $0.80. The share price is $2.40. What is its gearing? (Assume that the warrant is company-issued; in practice they all have a conversion ratio of 1, unless otherwise stated.)

a. 2x.
b. **6x.**
c. 3x.
d. 5x.

45. Which of the following will directly affect the price of a company warrant?

a. The length of time before expiration date.
b. The coupon rate of bonds issued by the company.

c. The underlying share price.
d. The number of warrants issued by the company.

- (a), (b) and (c).
- **(a), (c) and (d).**
- (b) and (d).
- (a) and (c).

Chapter 11 – Technical Analysis and Quantitative Analysis

46. The technical analyst is one who:

a. Buys shares whose intrinsic value is higher than the market price.
b. Tends to be long-term oriented.
c. Studies the prospects of earnings growth in forecasting future price trends.
d. **Studies price and volume patterns of stocks.**

47. Which of the following indicates a buy signal to technical analysts?

a. **The stock breaks through the moving average line from below.**
b. A head and shoulders pattern is formed with a breakout below the neckline by 3% of the share price.
c. Both the Dow Jones Industrial Average and the Dow Jones Transportation Average are moving down.
d. The advance-decline line is falling in a rising market.

48. Which of the following are bullish indicators?

a. Ascending triangles.
b. Descending triangles.
c. Double bottoms.
d. Reverse heads and shoulder.

- **(a), (c) and (d).**
- (b), (c) and (d).
- (b) and (c).
- (c) and (d).

49. According to the Dow Theory, daily fluctuations are considered:

a. The secondary trend.
b. A moving average.
c. The primary trend.
d. **Random and have no long-term forecasting value.**

50. A low relative strength indicator (RSI) reading (below 20) could indicate that the stock is:

a. The market has peaked and is likely to head down.
b. Overbought.

c. **Oversold.**
d. The stock has performed well compared with other stocks in the same risk category as measured by beta.

Appendix B: Formulae Sheet

Please note that this Formulae Sheet will be provided during the Exam. The examination might not cover all formulas provided.

Risk and Return and Time Value Calculations	
1	**Measures of Return** (i) Total Return on a Security = $\frac{\text{Income + Capital Gain or Loss}}{\text{Purchase Price}}$ (ii) Expected Rate of Return = E(R) = $\sum_{i=1}^{n} r_i \, p_i$ where E(R) = expected return on a security r_i = ith possible return p_i = probability of the ith return r_i n = number of possible returns
2	**Measures of Risk** (i) Variance of Returns = $\sigma^2 = \sum p_i \, [r_i - E(R)]^2$ (ii) Standard Deviation of Returns = $\sigma = \sqrt{\sigma^2}$ (iii) Coefficient of Variation = CV = $\frac{\text{Standard Deviation of Returns}}{\text{Expected Rate of Return}}$
3	**Time Value of Money** (i) Simple Interest = $\frac{\text{Principal x Interest Rate (in \%) x Interest Period (in days)}}{36{,}500}$ (ii) Future value of an investment if compounded annually at a rate of k for n years: $FV_n = PV(1 + k)^n$ where:

	FV_n = future value of the investment at the end of n years PV = present value or original amount invested at the beginning of the first year n = number of years during which the compounding occurs k = annual interest rate (iii) Compounded annual growth rate (CAGR), the effective rate that is earned on the initial cash flow: $CAGR = \left[\frac{FV}{PV}\right]^{\frac{1}{n}} - 1$ where: CAGR = the compounded annual growth rate FV = future value PV = present value n = number of years in the time period (iv) Future value of an investment for which interest is compounded more than once a year: $FV_n = PV(1 + k/m)^{mn}$ where m = number of interest periods in the year
4	**Present Value of a Single Cash Flow** Present value of an investment: $PV = \frac{FV_n}{(1 + k)^n}$ where: PV = present value of the future sum of money FV_n = future value of the investment at the end of n years n = number of years to the receipt of payment k = annual discount (or interest) rate
Key Drivers of Market Movements and Asset Values	
5	**Quantity Theory of Money** (i) $M \times V = P \times Q$ where: M = stock of money (currency outside banks plus demand deposits) V = velocity of money circulation (number of times per period that

	an average unit of money changes hands/accounts) P = general price level or a price index Q = the number of transactions made in an economy during a year
Foreign Exchange	
6	**(i) The Forward Foreign Exchange Contract:** Forward Points = $\frac{S \times (R_c - R_b) \times n/360}{1 + (R_b \times n/360)}$ where S = spot rate R_c = annualised interest rate of counter-currency R_b = annualised interest rate of base currency n = number of days from spot date **(ii) Interest Rate Parity:** $F = S\left[\frac{1 + R_c(\frac{n}{360})}{1 + R_b(\frac{n}{360})}\right]$ where F = forward rate S = spot rate R_c = annualised interest rate of counter-currency R_b = annualised interest rate of base currency n = number of days
Company Analysis and Understanding Financial Statements	
7	**Per Share Ratios** (i) Book Value Per Share = $\frac{\text{Shareholders' Equity}}{\text{Total Number of Shares Outstanding}}$ (ii) Fully Diluted Earnings Per Share = $\frac{\text{Net Earnings}}{\text{Fully-Diluted Number of Shares}}$ (iii) Dividend Per Share = $\frac{\text{Dividends Paid}}{\text{Total Number of Shares Outstanding}}$
8	**Liquidity Ratios** (i) Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$ (ii) Quick Ratio = $\frac{\text{Cash + Marketable Securities + Accounts Receivable}}{\text{Current Liabilities}}$ (iii) Cash Ratio = $\frac{\text{Cash + Near-Cash}}{\text{Current Liabilities}}$ (iv) Cash flow Ratio = $\frac{\text{Cash Flow From Operating Activities}}{\text{Current Liabilities}}$

9	**Leverage Ratios** (i) Debt/Equity Ratio = $\frac{\text{Total Debt}}{\text{Total Shareholders' Equity}}$ (ii) Debt-to-Total Assets Ratio = $\frac{\text{Total Debt}}{\text{Total Assets}}$ (iii) Total Liabilities-to-Total Assets Ratio = $\frac{\text{Total Liabilities}}{\text{Total Assets}}$ (iv) Interest Coverage = $\frac{\text{Earnings Before Interest and Taxes}}{\text{Interest Expense}}$ (v) EBITDA Interest Coverage = $\frac{\text{EBITDA}}{\text{Interest Expense}}$ (vi) Total Fixed Charge Coverage = $\frac{\text{Earnings Before Interest, Taxes and Lease Payments}}{\text{Interest Expense + Lease Payments + Preferred Dividend / (1-Tax Rate)}}$
10	**Operating Efficiency / Activity Ratios** (i) Total Asset Turnover = $\frac{\text{Sales}}{\text{Average Total Assets}}$ (ii) Inventory Turnover = $\frac{\text{Sales}}{\text{Average Inventory}}$ (iii) Days to Sell Inventory = $\frac{\text{365 days}}{\text{Inventory Turnover}}$ Or $= \frac{\text{Average Inventory}}{\text{Sales/365}}$ (iv) Receivables Turnover = $\frac{\text{Sales}}{\text{Average Accounts Receivables}}$ (v) Collection Period = $\frac{\text{365}}{\text{Receivables Turnover}}$ Or $= \frac{\text{Average Accounts Receivables}}{\text{Sales / 365}}$
11	**Profitability Ratios** (i) Gross Profit Margin = $\frac{\text{Gross Profit}}{\text{Sales}}$ (ii) Operating Profit Margin = $\frac{\text{Earnings Before Interest and Tax}}{\text{Sales}}$ (iii) Net Profit Margin = $\frac{\text{Net Income}}{\text{Sales}}$ (iv) Return on Equity = $\frac{\text{Net Income}}{\text{Average Common Equity}}$ (v) Pre-tax Return on Assets = $\frac{\text{Pre-Tax Income + Interest Expense}}{\text{Average Total Assets}}$

	(vi) After-tax Return on Assets = $\frac{\text{Net Income + Interest Expense (1-Tax Rate)}}{\text{Average Total Assets}}$ (vii) Earnings Per Share = $\frac{\text{Net Income Available to Common Shareholders}}{\text{Average Number of Common Shares Outstanding}}$
Equity Securities	
12	RNAVPS = (Revised Asset Value – Market Value of Liabilities)/(Total Outstanding Number of Shares)
13	**Valuation Ratios** (i) Price-Earnings Ratio (P/E) = $\frac{\text{Market Price Per Share}}{\text{Earnings Per Share}}$ (ii) Price-Earnings Growth Ratio (PEG) = $\frac{\text{P/E}}{\text{\% Growth in Earnings}}$ (iii) Price-to-Book Value Ratio (P/BV) = $\frac{\text{Market Price Per Share}}{\text{Book Value Per Share}}$ (iv) Price-to-Revised Net Asset Value Ratio (P/RNAV) = $\frac{\text{Market Price Per Share}}{\text{(Market Value of Assets - Liabilities)/Shares Outstanding}}$ (v) Price-to-Sales Ratio = $\frac{\text{Current Market Price Per Share}}{\text{Sales Per Share}}$ (vi) Enterprise Value-to-Cash Flow Ratio = $\frac{\text{Market Cap + Debt + Preferred Shares - Cash}}{\text{Cash Flow from Operations}}$ (vii) Enterprise Value-to-EBITDA Ratio = $\frac{\text{Market Cap + Debt + Preferred Shares + Minority Interest + Unfunded Pension - Associates - Cash}}{\text{Earnings Before Interest, Taxes, Depreciation and Amortisation}}$
14	**Dividend Yield** Dividend Yield = $\frac{\text{Dividend Per Share}}{\text{Market Price Per Share}}$ x 100
15	**Dividend Discount Model:** $P_o = \frac{D_1}{(1+k)} + \frac{D_2}{(1+k)^2} + \frac{D_3}{(1+k)^3} + \ldots + \frac{D_\infty}{(1+k)^\infty}$ where: P_0 = intrinsic value or theoretical value of the share today D_t = dividends expected to be received at the end of period *t*

	k = required rate of return (discount rate) on the share **Dividend Discount Model with subsequent sale of the share:** $P_o = \frac{D_1}{(1+k)} + \frac{D_2}{(1+k)^2} + \ldots + \frac{D_n}{(1+k)^n} + \frac{P_n}{(1+k)^n}$ where: n = holding period P_n = expected selling price of the share at the end of period *n*
16	**Constant Growth Dividend Discount Model** $P_0 = \frac{D_0(1+g)}{(1+k)} + \frac{D_0(1+g)^2}{(1+k)^2} + \frac{D_0(1+g)^3}{(1+k)^3} + \ldots + \frac{D_0(1+g)^\infty}{(1+k)^\infty}$ where: P_0 = intrinsic value or theoretical value of the share today D_0 = current dividend payment g = constant growth rate of dividends k = required rate of return (discount rate) on the share The above equation can be simplified to: $P_0 = \frac{D_1}{k-g}$ where D_1 = dividend expected to be received at the end of period 1 Under the Constant Growth Dividend Discount Model: Required Rate of Return (k)= $\frac{D_1}{P_0} + g$ The growth rate of dividends (g) can be estimated as a function of the return on equity (ROE) and the retention rate (RR): *g = Retention Rate x Return on Equity = RR x ROE* **Zero-Growth Model** $P_0 = \frac{\text{Dividend}}{\text{Discount Rate}}$

17	**Multi-Stage Growth Dividend Discount Model** For two-stage dividend growth: $P_0 = \sum_{t=1}^{n} \frac{D_0 (1 + g_1)^t}{(1 + k)^t} + \frac{D_n(1 + g_c)^t}{k - g_c} \frac{1}{(1 + k)^n}$ where: P_0 = intrinsic value of the share today D_0 = current dividend g_1 = growth rate when dividends are more predictable g_c = constant-growth rate when dividends are less predictable k = required rate of return n = number of periods of supernormal (or subnormal) growth D_n = dividend at the end of the abnormal growth period
Deposits and Fixed Income Securities	
18	T-bill Yield, $Y_d = \frac{D}{F} \times \frac{360}{t}$ where Y_d = annualised yield on a bank discount basis (expressed as a decimal) D = dollar discount, which is equal to the difference between the face value and price F = face value t = number of days remaining to maturity
19	**Bond Valuation** (i) Value of Bond (With Annual Interest): $P = \sum_{t=1}^{n} \frac{C_t}{(1 + k)^t} + \frac{M}{(1 + k)^n}$ where P = present value of the bond today (t = 0) C_t = coupon payments for period t M = maturity value n = number of years to maturity *k* = required rate of return

	(ii) Value of Bond (With Semi-Annual Interest): $P = \sum_{t=1}^{2n} \frac{C_t}{(1 + k/2)^t} + \frac{M}{(1 + k/2)^{2n}}$ (iii) Value of Zero-Coupon Bond: $P = \frac{M}{(1 + k/2)^{2n}}$
20	**Yield Measures** (i) Current Yield = $\frac{\text{Annual Interest Payment}}{\text{Bond Price}}$ (ii) Yield-to-Maturity (k) can be determined by solving the following equation for k: $P = \frac{C_1}{1 + k/2} + \frac{C_2}{(1 + k/2)^2} + \frac{C_3}{(1 + k/2)^3} + \ldots + \frac{C_{2n}}{(1 + k/2)^{2n}} + \frac{M}{(1 + k/2)^{2n}}$ (iii) Effective Annual Yield = $(1 + \text{Semi-Annual Interest Rate})^2 - 1$ (iv) Approximate Yield-to-Maturity = $\frac{\text{Annual Interest Payment} \pm \text{Amortised Gain or Loss}}{(\text{Current Market Price} + \text{Par Value})/2}$ (v) Yield to Maturity (k) for a Zero-Coupon Bond can be determined by solving the following equation for k: $\frac{k}{2} = \left[\frac{\text{Maturity Value}}{\text{Price}}\right]^{1/2n} - 1$ (vi) Taxable Equivalent Yield = $\frac{\text{Tax Exempt Yield}}{1 - \text{Marginal Tax Rate}}$
21	**Analysis of Convertible Bonds** (i) Conversion Value = Conversion Ratio x Share Price (ii) Conversion Ratio = $\frac{\text{Par Value of Convertible Bond}}{\text{Conversion Price}}$ (iii) Conversion Parity Price = $\frac{\text{Market Price of Convertible Bond}}{\text{Conversion Ratio}}$ (iv) Conversion Premium ($) = Market Value of Convertible Bond – Conversion Value

	(v) Conversion Premium (%) = $\frac{\text{Market Value of Convertible Bond - Conversion Value}}{\text{Conversion Value}}$ (vi) Breakeven Time/Payback = $\frac{\text{Conversion Premium}}{\text{Bond Income - Dividend Income}}$ (vii) Premium Over Straight Value (%) = $\frac{\text{Market Price of Convertible Bond}}{\text{Investment Value}} - 1$
Portfolio Management	
22	**Calculating Portfolio Return and Risk** (i) Expected Return of a Portfolio = $R_p = \sum_{i=1}^{n} w_i \ r_i$ where R_p = expected return on the portfolio w_i = portfolio weight for the ith security r_i = expected return on i^{th} security n = number of different securities in the portfolio $\sum w_i = 1.0$ (ii) Portfolio Risk (for two-asset portfolio) = $\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2}$ where w_1 and w_2 = portfolio weights of security 1 and security 2, respectively σ_1 and σ_2 = standard deviations of security 1 and security 2, respectively $\rho_{1,2}$ = correlation coefficient of security 1 and 2
23	**(i) Capital Market Line (CML):** $E(R_p) = R_f + \frac{E(R_m) - R_f}{\sigma_m} \sigma_p$ where $E(R_p)$ = expected return on any efficient portfolio, P, on the CML R_f = rate of return on the risk-free asset $E(R_m)$ = expected return on market portfolio, M σ_m = standard deviation of the market portfolio returns σ_p = standard deviation of the efficient portfolio P's returns **(ii) Capital Asset Pricing Model (CAPM):** $E(R_i)$ = Risk-free Rate + Risk Premium $= R_f + \beta_i(R_m - R_f)$ where $E(R_i)$ = expected return on security or portfolio i R_f = risk-free rate R_m = expected return on the market portfolio β_i = systematic risk of security or portfolio i relative to the market portfolio $R_m - R_f$ = market risk premium

24 Arbitrage Pricing Theory

$R_i = E(R_i) + b_{i1}f_1 + b_{i2}f_2 + \ldots + b_{in}f_n + e_i$

where R_i = actual rate of return on security i in a given period
$E(R_i)$ = expected return on security i in a given period
f = deviation of a systematic factor f from its expected value
b_i = sensitivity of security i's returns to a factor
e_i = random error term, unique to security i

25 Measuring Investment Returns

(i) Return on the Portfolio (No Deposits or Withdrawals):

$$R_p = \frac{V_E - V_B}{V_B}$$

where V_E = end value of the portfolio
V_B = beginning value of the portfolio

(ii) Return on the Portfolio (with Deposits or Withdrawals at the end of the measurement period):

$$R_p = \frac{(V_E + \text{Withdrawals} - \text{Deposits}) - V_B}{V_B}$$

(iii) Return on the Portfolio (with Deposits or Withdrawals at the beginning of measurement period):

$$R_p = \frac{V_E - (V_B + \text{Deposits} - \text{Withdrawals})}{V_B + \text{Deposits} - \text{Withdrawals}}$$

(iv) Annualizing Returns:

Annual Return = $[(1+ r_1)(1+ r_2)(1+ r_3)(1+ r_4)] - 1$

where r_1, r_2, r_3, r_4 are quarterly returns.

(v) Estimating the TWR using the modified Dietz method:

$$R_p = \frac{V_E - V_B - CF}{V_B + WCF}$$

where:
CF = net cash flow over the period
WCF = weighted cash flow

	The formula for WCF is: $WCF = \sum_i CF_i \frac{n - t_i}{n}$ where: CF_i = cash flow on day t_i t_i = number of days from beginning of period until CF_i n = number of days in the period
26	**Risk-Adjusted Measures of Performance** **(i) Sharpe Ratio** Sharpe ratio = $\frac{R_p - R_f}{\sigma_p}$ where: R_p = average rate of return for the portfolio R_f = risk-free rate of return σ_p = standard deviation of return for the portfolio **(ii) Treynor Ratio** Treynor ratio = $\frac{R_p - R_f}{\beta_p}$ where β_p = beta for the portfolio **(iii) Jensen's Alpha** The expected return as given by the CAPM is: $R_e = R_f + \beta_p(R_m - R_f)$ The portfolio's alpha is then the difference between the portfolio's actual return during the period, R_a, and the return that would be expected from the CAPM. $\alpha_p = Ra - R_e$ $= Ra - [R_f + \beta_p (R_m - R_f)]$ **(iv) Information Ratio** Information ratio = a_p / ε_p where: α_p = Jensen's alpha ε_p = Excess risk

	(v) Sortino Ratio Sortino ratio = $\frac{R_p - R_f}{\sigma_{negative}}$ where $\sigma_{negative}$ = average downside volatility
Exchange Traded Funds (ETFs), , Unit Trusts, Real Estate Investment Trusts (REITS), and Insurance	
27	Net Asset Value = [Fund Securities + Other Fund Assets – Liabilities (including management fees) – Debt used for leveraging (if any)]/Total fund units outstanding
28	**Annuity Calculations** (i) Present value, $PV_{annuity} = C \times \left[\frac{1-(1+i)^{-n}}{i}\right]$ (ii) Future value, $FV_{annuity} = C \times \left[\frac{(1+i)^{n}-1}{i}\right]$ where: C = cash flow per period i = interest rate n = number of payments
Warrants	
29	**Warrant Valuation** (i) Intrinsic Value = $\frac{(SP - EP)}{n}$ where SP = share price EP = exercise price n = conversion ratio (ii) Conversion Price = nWP + EP where WP = warrant price (iii) Warrant Premium ($) = nWP + EP – SP (iv) Warrant Premium (%) = $\frac{nWP + EP - SP \times 100}{SP}$ (v) Gearing Ratio = $\frac{SP}{nWP}$

	(vi) Minimum Value of Warrant	=	SP – EP	if SP > EP
		=	0	if SP ≤ EP

Appendix C: Essential Readings

1. **Analysis of Investments and Management of Portfolios** / Frank K. Reilly, Keith C. Brown
2. **Fundamentals of Investment Management** / Geoffrey A. Hirt, Stanley B. Block
3. **Essentials of Investments** / Zvi Bodie, Alex Kane, Alan J. Marcus
4. **Fundamentals of investments: Valuation and Management** / Bradford D. Jordan, Thoman W. Miller, Steven D. Dolvin
5. **Investments: Principles of Portfolio and Equity Analysis** / Michael G. McMillan, et al.
6. **Technical Analysis plain and simple: Charting the markets in your language** / Michael N. Kahn
7. **Managing Investment Portfolios: A Dynamic Process** / John L. Maginn, Donald L. Tuttle, Jerald E. Pinto, Dennis W. McLeavey

Appendix D: Glossary

A

Absolute returns The returns achieved over a certain time period. Absolute returns do not consider the risk of the investment or the returns achieved by similar investments.

Accounts payable Money owed by a company to suppliers that have extended the company credit.

Accounts receivable Money owed to a company by customers who purchase on credit.

Accredited Investor Is a person or a business entity considered to be financially savvy and has higher risk tolerance with privileged access to complex and risky products.

Alpha Outperformance relative to a relevant market benchmark.

Alternative investments A diverse asset class that typically includes private equity, real estate, and commodities. It provides an alternative to traditional investments, such as debt and equity securities.

Amortisation The process of expensing the costs of intangible assets over their useful lives.

Annualizing Returns A geometric average to show what an investor would earn over a period of time if the annual return was compounded. An annualized total return provides only a snapshot of an investment's performance.

Annuity The exchange of an initial amount for a fixed number of future payments of a certain amount.

Appraisal Assessment or estimation of the value of an asset, such as real estate, that is subject to certain assumptions, which may not always be realistic.

Appreciation A situation in which a currency is getting stronger relative to other currencies.

Arbitrage An opportunity to make money by taking advantage of a price difference between two markets.

Arbitrage pricing theory (APT) A multi-factor asset pricing model based on the idea that an asset's returns can be predicted using the linear relationship between the asset's expected return and a number of macroeconomic variables that capture systematic risk. It is a useful tool for analysing portfolios from a value investing perspective, in order to identify securities that may be temporarily mispriced.

Ask prices Prices at which a dealer is willing to sell an asset or a security, typically qualified by a maximum quantity (ask size). Also called *offer price*.

Assets Items that have value and include real assets and financial assets.

Asset allocation The process to determine the proportion of a portfolio to hold in various asset classes or the proportion of a portfolio held in various asset classes.

Asset-backed securities Financial securities created by securitisation whose associated payments and value are backed by a pool of underlying assets, such as car loans, credit card receivables, bank loans, or airplane leases.

Asset class A broad grouping of similar types of investments, such as shares, bonds, real estate, and commodities.

Asset managers See *investment managers*.

Asset turnover A measure that indicates the volume of revenues being generated by the assets used in the business, or how effectively the company uses its assets to generate revenue.

B

Balance of payments Record that tracks transactions between residents of one country and residents of the rest of the world over a period of time, usually a year.

Balance sheet A statement of the company's financial position at a specified point in time; essentially, it shows the company's assets, liabilities or debt, and owner-supplied capital or equity.

Banks Financial institutions that collect deposits from savers and transform them into loans to borrowers.

Banker's acceptance An instrument representing a promised future payment by a bank. The payment is accepted and guaranteed by the bank as a time draft to be drawn on a deposit.

Barriers to entry Obstacles, such as licences, brand loyalty, or control of natural resources, that prevent competitors from entering the market.

Basis point A measure equal to 0.01% or 0.0001.

Benchmark A comparison portfolio (e.g., the S&P 500 Index).

Beta A generic term for market risk, systematic risk, or non-diversifiable risk.

Bid prices Prices at which a dealer is willing to buy an asset or a security, typically qualified by a maximum quantity (bid size).

*Blue chip C*ompanies known for quality, reliability and the ability to weather downturns and operate profitably in the face of adverse economic conditions, which helps to contribute to their long record of stable and reliable growth

Board of directors A group of people whose job is to monitor the company's business activities on behalf of its shareholders.

Bond A formal contract that represents a loan from an investor (bondholder) to an issuer. The contract describes the key terms of the debt obligation, such as the interest rate and the maturity.

Bonus Issue An issue of additional shares to shareholders instead of a dividend, in proportion to the shares already held.

Book values Balance sheet values of a company's assets, liabilities, and equity.

Breakeven point The number of units produced and sold at which the company's profit is zero—that is, revenues exactly cover total costs. If the company sells less units than the breakeven point, it will suffer a loss. In contrast, if it sells more units than the breakeven point, it will make a profit.

Brokerage services Trading services provided to clients who want to buy and sell securities; they include not only execution services (that is, processing orders on behalf of clients) but also investment advice and research.

Brokers Trading services providers who act as agents and, in exchange for a commission, arrange trades by finding sellers for their clients who want to buy and buyers for their clients who want to sell.

Business cycles Economy-wide fluctuations in economic activity.

Business trust An unincorporated commercial organization in which assets are held in trust by trustees, who have been appointed to manage the assets of one or more beneficiaries.

Buyouts funds Private equity investment strategy that consists of financing established companies that require capital to restructure and facilitating a change of ownership.

C

Call market Market in which trades can be arranged only when the market is called, which is usually once a day.

Call option The right (but not the obligation) to buy an underlying at the exercise price until the option expires.

Callable bond A bond that provides the issuer with the right to buy back (call) the bond from bondholders prior to the maturity date at a pre-specified price.

Capital account A component of the balance of payments that reports capital transfers between domestic entities and foreign entities, such as debt forgiveness or the transfer of assets by migrants entering or leaving the country.

Capital Asset Pricing Model (CAPM) It describes the relationship between systematic risk and expected return for assets, particularly stocks. CAPM is widely used throughout finance for pricing risky securities and generating expected returns for assets given the risk of those assets and cost of capital.

Capital markets See *financial markets*.

Capitalisation-weighted indices Indices for which the weight assigned to each security depends on the security's market capitalisation—that is, the market price of the security multiplied by the number of units outstanding of the security. Also called *cap-weighted*, *market-weighted*, or *value-weighted indices*.

Capitalised Classifying a cost as generating long-term economic benefits and reporting it as an asset rather than charging it as an expense to current operations.

Cash Flow Statement A financial statement that summarizes the amount of cash and cash equivalents the go in and out of a company.

Caveat Emptor The onus is on the buyer to do the due diligence check in a buy and sell transaction.

Central Provident Fund (CPF) A compulsory comprehensive social security scheme for working Singaporeans and permanent residents primarily to fund their retirement, healthcare, and housing needs.

Certificate of Deposit (CD) A bank-issued investment instrument where a bank typically agrees to repay the principal plus interest on a fixed maturity date. CDs are typically negotiable, i.e.; they can be sold on the secondary market, allowing the principal and accrued interest to be redeemed before maturity.

Clearing All activities that occur from the arrangement of the trade to its settlement.

Clearing houses Trading services providers that arrange for final settlement of trades.

Closed-end funds Pooled investment vehicles that have a fixed number of shares and thus do not issue or redeem shares on demand. Investors who want to buy or sell closed-end funds must trade with investors willing to sell or buy these funds.

Coincident indicators Measures of economic activity that are intended to measure the current state of the economy rather than the past or to predict the future. Coincident indicators have a tendency to change at the same time as the economy measured as a whole.

Collateral Specific assets (generally a tangible asset) that a borrower pledges to a lender to secure a loan.

Collateralised debt obligation (CDO) A structured financial product that pools together cash flow-generating assets and repackages the asset pool into discrete tranches that can be sold to investors.

Collective Investment Scheme Is an investment scheme wherein several individuals come together to pool their money for investing in a particular asset(s) and for sharing the returns arising from that investment.

Commercial paper (CP) A short-term, unsecured promissory note usually issued in bearer form, i.e.; it is a negotiable instrument. By issuing the paper, the issuer promises to pay the bearer the face value of the paper on a fixed maturity date. CP is usually issued at a discount.

Commodity funds Mutual funds that seek to track the underlying price of various commodities and natural resources. This can include assets such as corn, copper, oil, wheat and gold.

Commodities Physical products, such as precious and base metals (e.g., gold, copper), energy products (e.g., oil), and agricultural products that are typically consumed (e.g., corn, cattle, wheat) or used in the manufacture of goods (e.g., lumber, cotton, sugar).

Common shares Also known as common stocks, ordinary shares, or voting shares, it is the main type of equity security issued by a company. It represents an ownership stake in the company.

Compound interest Interest that is calculated on principal and interest; it assumes reinvestment of interest received. Compound interest is often referred to as interest on interest.

Conflict of interest When either the employee's personal interests or the employer's interests conflict with the interests of the client. Conflicts of interest can also arise when employee's and employer's interests conflict.

Consumer price index Constructed by determining the weight (or relative importance) of each product and service in a typical household's spending in a particular base year and then measuring the overall price of that basket of goods from year to year.

Convertible bond A bond that offers the bondholder the right to convert the bond into a pre-specified number of shares of common stock of the issuing company.

Correlation A measure of the strength of a relationship between two variables; essentially, two variables are correlated when a change in one variable is always accompanied by a change in another variable. Variables can be positively or negatively correlated.

Correlation coefficient A number between –1 and +1 that measures the consistency or tendency for two variables to move in tandem with each other.

Counterparty risk Risk that one of the parties to a contract will fail to honour the terms of the contract.

Coupon rate The interest rate for a bond. The bond's coupon rate multiplied by its par value equals the annual interest owed to the bondholders.

Credit default swap (CDS) A contract that protects the buyer against a loss of value in a debt security or index of debt securities. The contract will specify under what conditions the other party has to make payment to the buyer of the CDS.

Credit rating Assessment of the credit quality of a bond based on the creditworthiness of the issuer.

Credit rating agencies Investment research service providers that specialise in providing opinions about the credit quality of bonds and of their issuers.

Credit risk For a lender, the risk of loss caused by a borrower's failure to honour the contract and make a promised payment in a timely manner.

Credit spread The difference between a risky bond's yield and the yield on a government bond with the same maturity.

Currency risk The risk associated with the fluctuation of exchange rates; also called foreign exchange risk.

Currency swap The exchange of debt service obligations denominated in different currencies.

Current account A component of the balance of payments that indicates how much a country consumes and invests (outflows) with how much it receives (inflows). It includes three components, the goods and services account, the income account, and the current transfers account.

Current account balance The sum of the goods and services, income, and current transfers accounts.

Current account deficit A negative current account balance.

Current account surplus A positive current account balance.

Current assets Short-term assets that are expected to be converted into cash, used up, or sold within the current operating period (usually one year), such as cash, inventories, and accounts receivable.

Current liabilities Short-term liabilities that must be repaid within the next year.

Current ratio A liquidity ratio calculated as current assets divided by current liabilities.

Current yield The annual coupon payment divided by the current market price.

Custodians Typically, banks and brokerage firms that hold money and securities for safekeeping on behalf of their clients.

D

Dark pools Type of alternative trading venue that does not display orders from clients to other market participants.

Dealers Trading services providers who participate in their clients' trades and stand ready to buy or sell when their clients want to sell or buy, providing liquidity and profiting when they can buy securities for less than they sell them. Also called *market makers*.

Debenture *A* type of debt instrument that is not secured by collateral and usually has a term greater than 10 years.

Debt-to-equity ratio A measure of financial leverage that indicates the extent to which debt is used in the financing of the company.

Debt-to-total assets ratio An indicator of a company's financial leverage where the company's total assets are financed by creditors.

Debt security A debt instrument, such as a government bond, corporate bond, certificate of deposit, municipal bond or preferred stock, that can be bought or sold between two parties and has basic terms defined.

Default A situation in which the bond issuer fails to make the promised payments.

Deflation A persistent and pronounced decrease in prices across most products and services in an economy.

Demand The quantity of a product or service buyers are willing and able to buy at a given price.

Deposit Insurance A scheme that protects depositors' money with a member bank for up to a certain threshold per depositor per bank. All full banks and finance companies in Singapore are members of the scheme.

Depository Receipt A security issued by a financial institution that represents an economic interest in a foreign company. The financial institution holds the foreign company's shares in custody and issues depositary receipts against the shares held. These depositary receipts trade like common stock on the local stock exchange.

Depreciation The process of allocating the cost of an asset over the asset's estimated useful life; a situation in which a currency is getting weaker relative to other currencies.

Depreciation expense The amount of depreciation allocated each year and reported on the income statement as an expense.

Derivatives Contracts (agreements to do something in the future) that derive their value from the performance of an underlying asset, event, or outcome.

Direct investments Purchase of securities issued by companies, governments, and individuals and of real assets, such as real estate, art, or timber.

Discount rate The rate used to calculate the present value of some future amount.

Discounted Cash Flow (DCF) Analysis A valuation method used to estimate the value of an investment based on its future cash flows. In a DCF, future cash flows are estimated and discounted to their respective present values. They are then added to the discounted terminal value to reach an estimated company value.

Discrete data Data that show observations only as distinct values.

Discretionary relationships Relationships that permit the service provider to act on behalf of the client.

Distressed Private equity investment strategy that focuses on purchasing the debt of troubled companies that may have defaulted or are on the brink of defaulting.

Distribution The set of values that a variable can take, showing their observed or theoretical frequency of occurrence.

Diversification The practice of combining assets and types of assets with different characteristics in a portfolio for the purpose of reducing risk.

Dividend per share The amount of cash dividends the company pays for each share outstanding.

Dividend Discount Model (DDM) A mode of determining whether a stock is over- or undervalued. According to the DDM, when dividend is assumed to grow at a constant rate, the price of a stock can be estimated by dividing the annual dividend by the value of the required rate of return less the growth rate.

Dividend Payout Ratio A measure of the portion of the earnings that are paid out in dividends.

Dividend yield The ratio of a company's annual dividend compared to its share price.

Document A piece of written, printed, or electronic matter that provides information or evidence or that serves as an official record.

Dollar-weighted Rate of Return A measure of the performance of an investment. The dollar-weighted rate of return is calculated by finding the rate of return that will set the present values of all cash flows equal to the value of the initial investment.

E

Earnings per share The amount of income earned during a period per share of common stock; net income divided by the number of shares outstanding.

Earnings yield (E/P) It is the percentage of how much a company earned per share. This is the inverse of P/E ratio.

EBITDA Earnings before interest, taxes, depreciation, and amortization, is a measure of a company's overall financial performance and is used as an alternative to simple earnings or net income in some circumstances.

Economic growth The percentage change in real output (real GDP) for an economy.

Economic indicator A measure that offers insight regarding economic activity.

Economics The study of production, distribution, and consumption or the study of choices in the presence of scarce resources.

Economies of scale Cost savings arising from a significant increase in output without a simultaneous increase in fixed costs.

Efficient Frontier A set of optimal portfolios that offer the highest expected return for a defined level of risk or the lowest risk for a given level of expected return.
Efficient Market Hypothesis (EMH) An investment theory that that hypothesise that share prices reflect all information at any given time and in a liquid market.

Electronic Communication Network Is a computerized system that automatically matches buy and sell orders for securities in the market and connects major brokerages and individual traders for direct trades without a middleman.

Elliott Wave Theory A form of technical analysis that finance traders use to analyse financial market cycles and forecast market trends by identifying extremes in investor psychology, highs and lows in prices, and other collective factors.

Enterprise value (EV) is a measure of a company's total *value*, often used as a more comprehensive alternative to equity market capitalization. EV includes in its calculation the market capitalization of a company and short-term and long-term debt as well as any cash on the company's balance sheet.

Enterprise Value to Sales (EV/S) ratio A financial ratio that compares the total value of the company to its sales. Some analysts regard EV/S as a superior metric to Price-to-sales.

Equity Assets minus liabilities; the shareholders' (owners') investment in the company.

Equity security Represents ownership interest held by shareholders in an entity (a company, partnership or trust), realized in the form of shares of capital stock.

Eurobond International long-term debt securities with maturities over one year denominated in any Eurocurrency. International distribution is a key feature and they are usually in bearer form, but the bonds can be issued in any currency or any interest basis.

Exchange rate The rate at which one currency can be exchanged for another.

Exchange-traded funds (ETFs) Pooled investment vehicles that are typically passively managed to track a particular index or sector and that trade continuously as common stocks on exchanges or through dealers.

Exercise price Specified in an options contract, the price to trade the underlying in the future. Also called the *strike price*.

Expense ratio **(ER),** also sometimes known as the management *expense ratio* (MER), measures how much of a fund's assets are used for administrative and other operating *expenses.*

Expenses The cost of using up company resources to earn revenues. Typical expenses include operating expenses (such as cost of sales; selling, general, and administrative expenses; and depreciation expenses); interest expenses; and income taxes.

Exports Products and services that are produced within a country's borders and then transported to another country.

F

Fair value Value that reflects the amount for which an asset could be sold in an arm's length transaction between willing and unrelated parties.

Fed Fund Rate The interest rate at which banks and other depository institutions lend money to each other, usually on an overnight basis.

Federal Reserve aka The Fed, is the central banking system of the United States of America

Financial Advisers Act (FAA) Provides guidance to financial advisers about FAA regime and covers topics including licensing requirements, exemptions, representative notification framework, business conduct and reporting requirements. The Monetary Authority of Singapore is the regulatory body responsible for the administration of the FAA.

Financial assets Claims on other assets; for example, a share of stock represents ownership in a company or a claim to some of the company's assets and earnings.

Financial capital Money provided to individuals, companies, and governments to finance their needs.

Financial institutions Financial intermediaries, such as banks and insurance companies, whose role is to collect money from savers and to invest it in financial assets.

Financial markets Places where buyers and sellers can trade securities; also called securities markets.

Fiscal policy The use of taxes and government spending to affect the level of aggregate expenditures.

Fixed costs Costs that do not fluctuate with the level of output of the company.

Fixed rate bonds A bond with a finite life that offers a coupon rate that does not change over the life of the bond. Also known as *straight bonds*.

Floating rate bonds A bond with a finite life that offers a coupon rate that changes over time. Also known as *variable-rate bonds*.

Foreign exchange market A highly integrated decentralised network in which currencies are traded.

Foreign exchange risk See *currency risk*.

Forward contract An agreement between two parties in which one party agrees to buy from the seller an underlying at a later date for a price established at the start of the contract.

Forward market Foreign exchange market in which currencies are traded now but delivered at some future date.

Forward rate The exchange rate for forward market transactions.

Foundations Grant-making institutions funded by gifts and by the investment income that they produce.

Fraud Intentional deception, such as deliberately causing or falsely reporting losses to collect insurance settlements.

Free Cash Flow to the Firm (FCFF) The amount of cash flow from operations available for distribution after depreciation expenses, taxes, working capital, and investments are accounted for and paid. FCFF is essentially a measurement of a company's profitability after all expenses and reinvestments.

Front office Client-facing activities that provide direct revenue generation, such as sales, marketing, and customer service activities.

Funds of funds Investment vehicles that invest in other funds.

Future value The amount to which a payment or series of payments will grow by a stated future date.

Futures contract An agreement that obligates the seller, at a specified future date, to deliver to the buyer a specified underlying in exchange for the specified futures price.

G

GDP per capita Gross domestic product divided by the population; a measure of average output or income per person.

Generally Accepted Accounting Principles aka **GAAP** A common set of accepted accounting principles, standards, and procedures that companies and their accountants must follow when they compile their financial statements.

Geometric average The average compounded return for each period; the average return for each period assuming that returns are compounding.

Global Investment Performance Standards (GIPS) A set of voluntary standards used by investment managers to ensure the full disclosure and fair representation of their investment performance.

Goodwill An intangible asset that arises when a company purchases another company and pays more than the fair value of the net assets (assets minus liabilities) of the purchased company.

Gross domestic product The total value of all final products and services produced within an economy in a given period of time (output definition), or equivalently, the aggregate income earned by all households, all companies, and the government within an economy in a given period of time (income definition). Nominal GDP uses current market values. Real GDP adjusts for changes in price levels.

Gross profit Sales minus the cost of sales.

Growth capital Private equity investment strategy that usually focuses on financing companies with proven business models, good customer bases, and positive cash flows and profits but that need additional capital to support their growth plans.

H

Hedge To reduce or eliminate risk caused by fluctuations in the prices of commodities, securities, or currencies.

Hedge funds Private investment pools that investment managers organise and manage. They are characterised by their availability to only a limited number of investors, agreements that lock up the investors' capital for fixed periods, and performance-based managerial compensation.

High-yield bonds See *non-investment-grade bonds*.

Hyperinflation Price increases so large and rapid that people find it difficult to purchase products and services.

I

Imports Products and services that are produced outside a country's borders and then brought into the country.

Income statement A financial statement that identifies the profit or loss of a company during a given time period, such as one year.

Index fund A portfolio of securities structured to track the returns of a specific index called the benchmark index.

Industrial production A measure of economic output by the following three segments of an economy: manufacturing, mining, and utilities.

Inflation The percentage increase in the general price level from one period to the next; a sustained rise in the overall level of prices for products and services.

Inflation risk The risk associated with inflation.

Information ratio A reward-to-risk ratio defined as the portfolio's mean active return (the difference in average return between the portfolio and its benchmark) over its active risk (tracking error).

Initial public offering (IPO) The first issuance of common shares to the public by a formerly private corporation.

Insider trading Trading while in possession of material non-public information.

Institutional investor A nonbank person or organization that trades securities in large quantities or dollar amounts that it qualifies for preferential treatment and lower commissions.

Insurance companies Financial institutions that help individuals and companies offset the risks they face; also among the largest investors.

Insurance company A company that sells insurance contracts (policies) that provide payments in the event that losses occur.

Intangible assets Assets lacking physical substance, such as patents and trademarks.

Interest Payment for the use of borrowed money.

Interest rate risk The risk associated with decreases in bond prices resulting from increases in interest rates.

International Financial Reporting Standards (**IFRS)** A set of common rules so that financial statements can be consistent, transparent and comparable around the world. IFRS are issued by the International Accounting Standards Board (IASB).

Interest coverage ratio A debt ratio and profitability ratio used to determine how easily a company can pay interest on its outstanding debt.

Inventory turnover A ratio showing how many times a company has sold and replaced inventory in a given period.

Intrinsic value The perceived or calculated value of an asset, an investment, or a company.

Inventories The unsold units of production on hand.

Investment banks Financial intermediaries that assist companies and governments raise capital and can provide other services, such as strategic advisory services, brokerage and dealing services, and research services; also known as merchant banks.

Investment companies Companies that exist solely to hold investments on behalf of their shareholders, partners, or unitholders, including mutual funds, hedge funds, venture capital funds, and investment trusts.

Investment-grade bonds Bonds rated BBB– or higher by Standard & Poor's and Fitch or Baa3 or higher by Moody's.

Investment managers Investment professionals who receive authority from their clients to trade securities and assets on their behalf. Also called *asset managers*.

Investment policy statement (IPS) A written planning document describing a client's investment objectives—return requirements and risk tolerance—over a relevant time horizon, along with constraints that apply to the client's portfolio. The IPS serves as a guide to what is required and acceptable in the investment portfolio.

Investment risk The risk associated with investing that arises from the fluctuation in the value of investments.

Investment vehicles Assets, such as securities and real assets, created by the investment industry to help investors move money from the present to the future, with the hope of increasing the value of their money.

Issuers Typically companies and governments that sell securities to investors in exchange for cash to raise money.

J

J curve The graphical representation of net cash flow (that is, the cash distributions net of carried interest minus the sum of the capital calls and management fees) for limited partners. It shows that net cash flows are negative in the early years of a fund but turn positive toward the end of a fund's life.

Jensen's Alpha A risk-adjusted performance measure that represents the average return on a portfolio or investment, above or below that predicted by the capital asset pricing model (CAPM), given the portfolio's or investment's beta and the average market return.

Junk bonds See *non-investment-grade bonds*.

L

Lagging indicators Turning points that signal a change in economic activity after output has already changed.

Leading indicators Turning points that signal changes in the economy in the future, and thus are considered useful for economic prediction and policy formulation.

Legal risk The risk that an external party could sue an organisation for breach of contract or other violations.

Leverage ratio Measurements of a company's abilities to service its debts.

Liability A monetary obligation of a company as a result of previous events.

Limit price Ceiling price for a buy order and floor price for a sell order. A trade cannot be arranged at a price higher than the specified limit price when buying or a price lower than the specified limit price when selling.

Limited liability Liability that does not exceed an investor's initial contribution of capital. For example, shareholders are protected by limited liability, which means that higher claimants—particularly debt investors—cannot recover money from the personal assets of the shareholders if the company's assets are insufficient to fully cover their claims.

Liquidity risk The risk that an asset or security cannot be bought or sold quickly without a significant concession in price.

London Interbank Offered Rate (LIBOR) The most widely used reference rate, defined as the average interest rate that banks charge each other in the London interbank market. Also called Libor.

Long positions Positions for which investors own assets or securities. These positions increase in value when prices rise.

Long-term debt Money borrowed from banks or other lenders that is to be repaid during periods of greater than one year.

M

Macroeconomics The branch of economics that studies the economy as a whole.

Management fees Fees that limited partners must pay general partners to compensate them for managing investments. Management fees are typically set as a percentage of the amount the limited partners have committed rather than the amount that has been invested.

Market equilibrium The price where quantity demanded equals quantity supplied. At the equilibrium price, demand and supply in the market are balanced, and neither buyers nor sellers have an incentive to try to change the price, all other factors remaining unchanged.

Market makers See *dealers*.

Market risk The risk caused by changes in market conditions affecting prices.

Maturity date Date when the borrower must repay the amount borrowed.

Microeconomics The branch of economics that studies how individuals and companies make decisions to allocate scarce resources.

Model risk The risk arising from the use of models and associated with inappropriate underlying assumptions, the unavailability or inaccuracy of historical data, data errors, and misapplication of models.

Modified Dietz Method A measure of the ex-post performance of an investment portfolio in the presence of external flows.

Monetary Authority of Singapore (MAS) Acts as Singapore's central bank and is responsible for the conduct of monetary policy, the issuance of currency, the oversight of payment systems and serving as banker to and financial agent of the Government of Singapore.

Monetary policy Actions taken by a nation's central bank to affect aggregate output and prices through the money supply or credit.

Money laundering A process in which criminals use financial services to transfer money from illegal operations to other legal activities; the money becomes "clean" in the process.

Money market The part of financial market that deals in the borrowing and lending of short-term loans for less than a year.

Money market funds Special class of open-end mutual funds that investors view as uninsured interest-paying bank accounts. Unlike other open-end mutual funds, regulators permit money market funds to accept deposits and satisfy redemptions at a constant price per share if they meet certain conditions.

Multiplier effect An initial increase (decrease) in spending produces an increase (decrease) in GDP and consumption greater than the initial change in spending.

N

Net asset value (NAV) Total net value of a fund (the value of all assets minus the value of all liabilities) divided by the fund's current total number of shares outstanding.

Net book value Calculated as the gross value of an asset minus accumulated depreciation, where accumulated depreciation is the sum of all reported depreciation expenses for the particular asset.

Net exports The difference between exports and imports of goods and services; also called balance of trade or trade balance.

Net income The difference between revenue and expenses; income available to distribute to shareholders.

Net present value The present value of future cash flows minus the cost of the investment, or the present value of cash inflows minus the present value of cash outflows.

Net profit margin A profitability ratio that indicates how much (percentage) of each monetary unit of revenue is left after all costs and expenses are covered.

Non-investment-grade bonds Bonds rated BB+ or lower by Standard & Poor's and Fitch and Ba1 or lower by Moody's. Also called high-yield bonds or junk bonds.

Non-tariff barriers A range of measures, such as certification, licensing, sanctions, or embargoes, that make it more difficult and expensive for foreign producers to compete with domestic producers.

Normal distribution A symmetrical distribution in which the mean, median, and mode are the same value. The distribution is completely described by its mean and variance (or standard deviation).

O

Offer prices See *ask prices*.

Open-end mutual funds Pooled investment vehicles that have the ability to issue or redeem (repurchase) shares on demand. When investors want to invest in a mutual fund, the fund issues new shares in exchange for cash that the investors deposit. When existing investors want to withdraw money, the fund redeems the investors' shares and pays them cash.

Open market operations Activities that involve the purchase and sale of government bonds by a central bank.

Operating income Income generated by the company from its usual business activities before taking into account financing costs and taxes. It is often referred to as earnings before interest and taxes (**EBIT**).

Operating profit margin A profitability ratio calculated as operating income divided by revenue.

Operational risk The risk of losses from inadequate or failed people, systems, and internal policies and procedures, as well as from external events that are beyond the control of the organisation but that affect its operations.

Opportunity cost The cost of any activity measured in terms of the value of the best alternative that is not chosen; the value that investors forgo by choosing a particular course of action. For example, the cost of not having money to invest, spend, or hold; the cost of giving up opportunities to use money.

Option contract An agreement in which the buyer of the option has the right, but not the obligation, to trade the underlying.

Option premium The amount paid by the option buyer to the option seller, at the initiation of the option contract, to compensate option sellers for their risk.

Orders Instructions that people who want to trade give brokers or trading venues and that specify what security to trade, whether to buy or sell, and how much should be bought or sold. They also have other instructions attached to them, such as order execution, exposure, and time-in-force instructions.

Over-the-counter (OTC) markets Another name for quote-driven markets dating from when securities were literally traded over a counter in the dealer's office.

P

Par value The stated value or face value of a security; the amount the investor would be entitled to receive in a liquidation scenario, which also serves as the principal value on which coupon payments are calculated.

Passive investment managers Managers who follow a buy-and-hold approach and seek to match the return and risk of a benchmark.

Penny Stock A common stock valued at less than one dollar, and therefore highly speculative.

Performance attribution The process of identifying the source of a fund manager's return. Potential sources include asset allocation, sector selection, stock selection, and currency exposure.

Performance measurement The process of measuring the performance of investments, including the calculation of reward-to-risk ratios.

Perpetual bond A bond with no maturity date

Physical capital The means of production; tangible goods such as equipment, tools, and buildings.

Policies Principles of action adopted by a company.

Political risk The risk of a change in the ruling political party of a country leads to changes in policies that can affect everything from monetary and fiscal policies to investment incentives, public investments, and procurement.

Pooled investments Indirect investment vehicles in which investors pool their money together to gain the advantages of being part of a large group. The resulting economies of scale can significantly improve investment returns.

Portfolio The group of assets in which savings are invested.

Portfolio Theory A mathematical framework for assembling a portfolio of assets such that the expected return is maximized for a given level of risk.

Position Quantity of an asset or security that a person or institution owns or owes.

Preferred stock aka **preference shares**; a type of equity security that ranks between debt securities and common stock. It typically does not carry voting rights but has priority over common stock in the receipt of dividends.

Present value The present discounted value of future cash flows.

Price-earnings Growth ratio A valuation metric that factors in growth expectations and helps determine whether a stock is under- or overvalued given its expected growth rate.

Price-to-book value ratio The ratio of a company's share price to its book value per share to determine whether a stock is appropriately priced relative to the value of its assets.

Price-to-earnings ratio The ratio of a company's share (market) price to its earnings per share.

Price-to-cash flow ratio A metric to determine whether a stock is appropriately priced using the market value per share divided by the operating cash flow per share.

Price-to-sales ratio A metric to determine whether a stock is appropriately priced. It is calculated by dividing market value per share by revenue per share.

Price-weighted index Indices for which the weight assigned to each security is determined by dividing the price of the security by the sum of all the prices of the securities.

Primary market Markets in which issuers, such as companies and governments, sell their securities to investors.

Principal The money initially lent on which interest is paid.

Private equity A type of alternative investment that consists of investing in private companies—that is, companies that are not listed on a stock exchange.

Private placement Type of primary market transaction in which companies sell securities directly to a small group of qualified investors, usually with the assistance of an investment bank.

Procedures What the company must do to achieve a desired outcome.

Processes Individual steps that a company must take, from start to finish, to achieve a desired outcome. They divide procedures into manageable actions.

Proprietary trading Direct trading by traders with their clients rather than by arranging trades with others on behalf of their clients.

Prospectus Document that discloses the investment policies, deposit and redemption procedures, fees and expenses, and past performance statistics associated with an investment vehicle.

Put option The right (but not the obligation) to sell the underlying at the exercise price until expiration.

Q

Quick ratio A liquidity ratio that indicates a company's ability to satisfy current liabilities with its most liquid assets.

R

Real assets Physical assets, such as land, buildings, machinery, cattle, and gold.

Real estate Land and buildings. Also a type of alternative investment that consists of direct and indirect investments in land and buildings.

Real estate equity funds Often open-end funds that hold investments in hundreds of commercial properties.

Real estate investment trusts (REITs) Public companies that mainly own, and in most cases operate, income-producing real estate.

Rebalancing The process of adjusting the weights of the constituent securities in an index or the weights of assets in a portfolio.

Recession A period during which real GDP decreases (i.e., negative growth) for at least two successive quarters, or a period of significant decline in total output, income, employment, and sales lasting more than a few months.

Receivables turnover ratio An accounting measure used to quantify a company's effectiveness in collecting its receivables or money owed by clients.

Regulations Rules that set standards for conduct and carry the force of law.

Reinvestment risk Risk that in a period of falling interest rates, the periodic coupon payments received during the life of a bond and/or the principal payment received from a bond that is called early must be reinvested at a lower interest rate than the bond's original coupon rate.

Relative returns The difference between holding-period returns (absolute returns) and returns on a benchmark over the same holding period.

Repurchase Agreements Also known as a repo, or a sale and repurchase agreement, is the sale of securities together with an agreement to buy them back at a future date, in return for a short-term loan. The repo seller, who is receiving the loan, is offering securities as collateral against the loan.

Reserve currency A currency held in significant quantities by many governments and institutions as part of their foreign exchange reserves.

Reserve requirement The requirement for banks to hold reserves in proportion to the size of deposits.

Residual claimants Investors whose claims rank last. Common shareholders are residual claimants. In the event of a company's liquidation, common shareholders share proportionately in the remaining company assets after all other claimants have been satisfied.

Retail Investor An individual investor who purchases securities for their own personal account rather than on behalf of organization. They typically trade in much smaller amounts than accredited or institutional investors.

Retained earnings The accumulated net income that is retained by the company rather than distributed to its owners (shareholders) as dividends.

Return on assets (ROA) A profitability ratio that indicates a company's net income generated per monetary unit invested in total assets.

Return on equity (ROE) A profitability ratio calculated as net income divided by average shareholders' equity.

Revenues The amount charged (and expected to be received) for the delivery of products or services in the ordinary activities of a business.

Reverse Stock Split A process by which shares of corporate stock are effectively merged to form a smaller number of proportionally more valuable shares.

Rights Issue An issue of shares offered at a special price by a company to its existing shareholders in proportion to their holding of old shares

Risk The effect of uncertain future events on an organisation or on the outcomes the organisation achieves.

Risk appetite An organisation's willingness to take on risk, which depends on its attitude toward risk and on its risk culture.

Risk management An iterative process used by organisations to support the identification and management of risk (or uncertainty) and reduce the chances and/or effects of adverse events while enhancing the realisation of opportunities and the ability to achieve company objectives.

Risk tolerance The level of risk an organisation or person is able and willing to take on.

S

Secondary market Market in which investors trade securities and contracts with each other but not with the original security issuer.

Securities Financial assets that can be traded, such as shares and bonds.

Securities and Futures Act (SFA) The primary source of securities law that regulates activities and institutions in the securities, futures and derivatives industry in Singapore, including dealing in capital markets products, clearing facilities, and related matters. The Monetary Authority of Singapore is the regulatory body responsible for the administration of the SFA.

Securitisation Creation of new financial products by buying and repackaging securities or other assets; the creation and issuance of new debt securities that are backed by a pool of other debt securities.

Settlement Activity that consists of the final exchange of cash for securities following a trade.

Settlement risk The risk that when settling a transaction, a firm performs one side of the deal but the counterparty does not complete its side of the deal as agreed, often because it has declared bankruptcy.

SGX Catalist The first sponsor-supervised listing platform in Asia and caters to the needs of smaller or fast-growing companies.

SGX Mainboard A listing platform that caters to the needs of more established companies with higher entry and listing requirements in the Singapore Exchange.

Share buyback See *share repurchase*.

Share repurchase A transaction in which a company uses its cash to buy back its own shares from existing shareholders. This transaction reduces the number of shares outstanding.

Shareholders The owners of shares (stock) of a corporation.

Sharpe ratio A reward-to-risk ratio defined as the excess portfolio return (portfolio return minus risk-free return) over the standard deviation of portfolio returns.

Short positions Positions for which investors sell assets or securities that they do not own, a process that involves borrowing the assets or securities, selling them, and repurchasing them later to return them to their owner. These positions increase in value when prices fall.

Sinking Fund Provision A provision in some bond indentures requiring the issuer to put money aside to repay bondholders at maturity.

Skewed A distribution is skewed when the bulk (majority) of the values (possibly including the median) lie either to the right or to the left of the mean; the distribution is not symmetrical.

Sovereign risk The risk that a foreign government will not repay its debt because it does not have either the ability or the willingness to do so.

Sovereign wealth funds Funds created by governments to invest surpluses for the benefit of current and future generations of their citizens.

Spot market Foreign exchange market in which currencies are traded now and delivered immediately.

Spot rate The exchange rate for spot market transactions.

Standard deviation A measure, in the same units as the original data, of the variability, volatility, or dispersion of a data set around the average value of that data set (i.e., the arithmetic mean). It is the positive square root of the variance.

Statement of cash flows A financial statement that identifies the sources and uses of cash over a time period and explains the change in the company's reported cash balance during the period.

Stock dividend A transaction in which a company distributes additional shares of its common stock to shareholders instead of cash. This transaction increases the number of shares outstanding but does not affect the company's value because the stock price decreases accordingly.

Stock split A transaction in which a company increases the number of shares outstanding. For example, in a two-for-one stock split, the company doubles the number of shares outstanding and the stock price is halved, but the company's value is unaffected.

Strategic asset allocation The long-term mix of assets that is expected to achieve the client's long-term objectives, given the client's investment constraints.

Strike price See *exercise price*.

Structured product, also known as a market-linked investment, is a pre-packaged structured finance investment strategy based on a single security, a basket of securities, options, indices, commodities, debt issuance or foreign currencies, and to a lesser extent, derivatives.

Structured warrant A form of structured investment products issued by a third-party financial institution over a wide range of assets, including the shares of an un-related listed company, a basket of companies' shares or an index.

Supply The quantity of a product or service sellers are willing and able to sell at a given price.

Swaps Derivatives in which two parties swap cash flows or other financial instruments over multiple periods (months or years) for mutual benefit, usually to manage risk.

Systematic risk Also known as market risk, it is the risk created by general economic conditions that affect all risky investments. Systematic risk factors include changes in macroeconomic conditions, interest rate risk, and political risk, among others.

Systemic failure Failure of the financial system as a whole, including loss of access to credit and collapse of capital markets.

T

Tactical asset allocation The decision to deliberately deviate from the strategic asset allocation in an attempt to add value based on forecasts of the short-term relative performance of asset classes.

Tariffs Taxes (duties) levied on imported goods and services, which allow governments not only to establish trade barriers but also to raise revenue.

Technical Analysis A trading discipline applied to evaluate investments and identify trading opportunities in price trends and patterns on charts.

Term structure of interest rates The relationship between the yields to maturity offered by government bonds and the maturities of these government bonds.

Terminal value, or horizon value, determines the value of a business or project beyond the forecast period when future cash flows can be estimated**.**

Time-weighted rate of return A measure of investment performance in which the overall measurement period is divided into sub-periods. The timing of each individual cash flow is identified and then defines the beginning of the sub-period in which it occurs.

Total asset turnover ratio A measure on the value of a company's sales or revenues relative to the value of its assets. It can be used as an indicator of the efficiency with which a company is using its assets to generate revenue.

Tracking error The standard deviation of the differences between the deviation over time of the returns on a portfolio and the returns on its benchmark; a synonym of active risk.

Trade balance See *net exports*.

Trade bills A bill of exchange that is used to pay for goods.

Trade deficits When the value of exports is lower than the value of imports.

Trade surplus When the value of exports is higher than the value of imports.

Transaction costs Costs associated with trading, which include explicit costs (mainly brokerage commissions) and implicit costs (bid–ask spreads, price impact, and opportunity costs).

Treasury Bill (**T-Bill**) A short-term debt obligation backed by the U.S. Treasury Department with a maturity of one year or less. T-bills are usually sold in denominations of $1,000.

Treasury bond (T-bond) A marketable, fixed-interest US government debt security with a maturity of more than 10 years. T-bonds make interest payments semi-annually, and the income received is only taxed at the federal level.

Treynor ratio A reward-to-risk ratio defined as the excess portfolio return (portfolio return minus risk-free return) over the beta of portfolio returns.

Trust Deed A legal document that defines the *trust* such as the trustee, beneficiaries, settlor and appointer, and the objectives and terms and conditions of the trust.

U

Underwriters The (lead) investment bank or brokerages guarantee the sale of the securities at an offering price that is negotiated with the issuer of a public offering.

Unit trust An unincorporated fund comprising investors' money, which is invested in a variety of financial assets to meet the unit trust's objectives set by a fund manager.

V

Value at risk An estimate of the minimum loss of value that can be expected for a given period within a given level of probability.

Variable costs Costs that fluctuate with the level of output of the company.

Variable-rate bonds See *floating-rate bonds*.

Variance A measure of dispersion that is equal to the standard deviation squared (i.e., the standard deviation multiplied by itself).

Venture capital Private equity investment strategy that consists of financing "start-up" companies that exist merely as an idea or a business plan.

Voting rights The rights of shareholders to vote—for example, to elect the members of the board of directors.

W

Warrant An equity-like security that entitles the holder to buy a pre-specified amount of common stock of the issuing company at a pre-specified stock price prior to a pre-specified expiration date.

Weighted Average Cost of Capital (WACC) The rate that a company is expected to pay on average to all its security holders to finance its assets.

Y

Yield curve Term structure of interest rates presented in graphical form.

Yield to maturity The discount rate that equates the present value of a bond's promised cash flows to its market price.

Z

Zero-coupon bonds Bonds that do not offer periodic interest payments during the life of the bond. The only cash flow offered by a zero-coupon bond is a single payment equal to the bond's par value to be paid on the bond's maturity date.



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